

Semantic Audit: A Lean-Verified 51/50 Cutoff for Four-Arc Minimizers

Independent semantic review completed and accepted by
openai-codex/gpt-6-astra

Verdict and scope

Equivalent to the frozen modeled claim. No critical mathematical error identified in the main proof, supplement, indexed declaration translations, generated-family descriptions, or adopted external citation. This is a review of the current author-05 submission and locked source, not an endorsement of a previous review or package-status banner. No previous reviewer report or generated SemanticAudit manuscript was used as evidence.

The conclusion is deliberately scoped: for every real $\lambda \geq 51/50$, every modeled FourArcCandidate satisfying the formal CMV incidence hypotheses is nonminimal against the defined equal-area admissible competitors. It is not a proof of the full source-paper classification, a general BV/frontier equivalence, a sharp threshold, or a conclusion below 51/50. The main proof expressly distinguishes those stronger statements.

Coverage and integrity evidence

I compared all 850 nongenerated author statements and their mathematical explanations with the indexed Lean contracts, their surrounding constructor/section conventions, and the relevant proof dependencies. Each receives exactly one review below; the source_sha256 field intentionally carries the inventory's **declaration code_sha256**, not the whole-file hash. Exact script checks recomputed all 4,433 indexed declaration hashes and all 181 source-file hashes, with no mismatch. All 850 nongenerated supplement statements, explanations and embedded source blocks match the current author/source records. The supplement also reproduces all eight author-selected generated representatives correctly. Its exhaustive generated tables have all 3,583 expected unique names, locations and declaration hashes.

The two family reviews cover every indexed generated entry, not just the representatives. Exact-rational parsing checked all compact boxes 2..1023 and middle bricks 0..112, all 1,135 internal interval seams after including explicit compact boxes 0 and 1, and the advertised endpoint/call coverage of all 65 compact batch/prefix theorems. Normalized source comparison found one compact local candidate contract, one middle local candidate contract, and the expected common condition/certificate schemas across their respective families. The 1,022 compact Conditions proofs and 113 middle CellCertificate/Conditions proofs supply the analytic obligations rather than assuming them.

A full frozen-source search found no occurrences of sorry, admit, axiom, native_decide, implemented_by, unsafe or extern. This textual check is not a substitute for kernel checking. I did not use compilation success as a semantic oracle and do not claim a fresh full Lean build.

Geometry and meaning of minimality

The metric issue is handled explicitly. PlanePoint is a real coordinate product, while EuclideanPlane is WithLp 2 of that product. FrontierMeasure transports Euclidean H^1 on the complete topological frontier back to the coordinate plane. Thus lengths are not measured in the product sup norm. StripDensity assigns density 1 on both interfaces $|y|=1$. WeightedArea is the Lebesgue integral on the carrier. The low-level WeightedPerimeter integral has a measure parameter, but the admissible competitor perimeter always uses the canonical FrontierMeasure of its own carrier; no competitor can choose a cheap auxiliary boundary measure.

FinitePerimeterRegion requires measurability and the two integrability properties for this complete-frontier model. The paper instead works with reduced-boundary/relaxed perimeter. The manuscript makes that distinction and does not assert an unproved equality for every measurable/BV set. Type-(iii) competitors enter the general constructor through a constructed integrable carrier; they are not assumed scalar competitors. For the explicit carriers, the package proves the geometric area and frontier identities it needs, including Euclidean arc lengths, null intersections at junctions, and removal of glued internal chords.

The candidate has $0 < h \leq 1$ and $0 < \alpha < \pi/2$; its CMV hypotheses are exactly $\lambda > 1$ and $\lambda \cos(\alpha) = h$. No minimization, stationary-fold, certificate, or desired comparison is hidden in that structure. The endpoint $h=1$ remains in the model and final theorem. TypeThreeAssembly has the full regular domain $0 < h < 1$, not just $h > 1/2$. Its major exterior caps below one half and the zero-width/singleton lower boundary at one half are explicitly treated. The realization interface determines the competitor by its carrier; it does not independently prescribe a convenient perimeter.

The arc parameterization deserves particular care: $\text{unitCircleArc}(r,s) = \text{rexp}(is/r)$, with signed arclength parameter s , not an angle parameter on a radius-one circle. The author correctly records the radius and span hypotheses, and the totalized zero-radius/zero-partition conventions where relevant. General cap half-angles range over $(0,\pi)$, so doubling a cap does not silently exclude the major-cap case.

Scalar formulas and all-curvature reduction

With $q=2h^3-1$, the manuscript's B3 includes $\pi/2$, whereas LeanSuffixAnalytic.typeThreeAngle does not. CMVSuffixModel explicitly proves $\text{angleTerm} = \text{typeThreeAngle} + \pi/2$ before identifying the carrier area and perimeter. The resulting $A_3 = (B_3 + (2h^3+1)D_3)/h^3$, $P_3 = 2(B_3 + D_3)/h^3$, $A_4 = 2(B_4 + h^4D_4)/h^4$ and $P_4 = 4B_4/h^4$ agree with the geometric formulas and the cited source equations after translating the angles. The normalizations, density factors and signs are not interchanged.

On the regular domain, $A_i' = K_i/h^3$ and $P_i' = K_i/h^2 = h * A_i'$. The radical-cube gap is positive on the full interval $-1 < x < 1$ for $\lambda > 1$, including zero and negative x ; this matters for small type-(iii) curvature. The fold monotonicity and type-(iv) stationary minimum supply the global area comparison needed by the envelope, including the endpoint via continuity.

StationaryEqualAreaPair records equal area and the type-(iv) stationary fold with regular curvatures; negative K3 and strict perimeter improvement are separate, proved certificate consequences. They are not smuggled into the meaning of a stationary pair. A3 tends to infinity at zero, and the negative-fold branch gives the selected descending root. Uniqueness assertions are used with their actual scope: uniqueness on a compact trapping set in the continuity lemma, and uniqueness on the negative-fold branch when that sign is supplied. The manuscript does not infer that there is only one regular root of every equal-area level.

The same-curvature inequality is $A_3(h) < A_4(h)$, and decreasing-branch reasoning gives $\text{root}(h) < h$, the sign required by the envelope derivative. The proof derives continuity using local compact trapping and closed-graph uniqueness, derives simplicity from $K_3 < 0$, and differentiates the equal-area identity. Consequently $\Delta' = (\text{root}-h) * A_4'$. The fold is the strict maximum of the perimeter gap; a negative gap there gives negativity on every $0 < h \leq 1$. At $h=1$ the argument uses one-sided continuity, not a nonexistent regular derivative or control of an arbitrary total root function above one. These are derived analytic properties, not hypotheses containing the all-curvature conclusion.

Certificate soundness and generated coverage

QInterval is an ordered rational interval with closed real containment. Multiplication encloses all four endpoint products, and reciprocal/division retain positivity hypotheses. nsmul means scaling by a nonnegative rational, including zero and fractions, not only a natural-number multiplier. The Taylor sine/cosine approximations and their remainder bounds apply

on their stated full rational domain; inverse trigonometric uses additionally establish principal branches. The centered mean-value estimates keep the required common intermediate point. The IVT lemma permits weak endpoint signs and a degenerate interval and returns a closed-interval zero, as the author states.

The middle-face map is exactly $(-F, C4, E, C3)$, with coupled affine coordinates. The simultaneous-zero argument is not an unsupported invocation of a general multivariate Miranda theorem. In `MiddleFaceAssembly's Conditions.exists_zero`, the incidence equations produce explicit continuous arccos root projections, trapped by the incidence face signs. A one-dimensional IVT solves F along the ξ projection; a second solves E along the ρ projection. F and $C4$ are independent of the later ρ/σ choice, preserving the earlier zeros. Certified branches then identify the analytic curvatures and justify denominator clearing, while $K3$ and G give the descending branch and strict perimeter sign.

The compact-cell Conditions impose strict perimeter negativity on the whole $h3/h4$ slab, whereas their area-face signs are conditioned on a stationary $h4$. The author preserves this quantifier distinction. The local candidate boundaries apply to arbitrary modeled candidate curvature at the certified density; they do not restrict the candidate itself to a stationary slab.

Exact coverage is $[51/50, 102001/100000]$ for the first prototype, $[102001/100000, 25781/25000]$ for the 113 middle bricks, $[25781/25000, 33/32]$ for the right prototype, and $[33/32, 9/7]$ for the 1,024 compact boxes including the two explicit ones. All interfaces are retained. The generated compact family contains 3,131 indexed declarations: three entries per box $2..1023$ and 65 coverage theorems. The generated middle family contains 452: four entries per brick $0..112$. Each family's aliases are definitionally the advertised generic structures, not independent altered models.

High-density splice

The normalized area $a(\theta)$ is strictly increasing from $(0, \pi)$ onto positive reals. `thetaOf` chooses this inverse on positive inputs and explicitly returns $\pi/2$ on nonpositive inputs; geometric theorems use only positive inputs. The normalized saving $g(X) = 2L(X) - L(2X)$ has its global positive-domain minimum at $xstar$, with $g(xstar) = 3\cos(\theta_{star})$, where $3\theta_{star} - 3\sin(\theta_{star}) * \cos(\theta_{star}) - \pi = 0$. The complementary doubled-cap angle is $\pi - \theta_{star}$, so the proof retains the correct branch.

The replacement difference is the division-free identity $P_{rep} - P_{candidate} = c(1 - \lambda \text{dag}(X))$, valid for arbitrary real λ . The manuscript's saving factorization $\lambda \text{dac}(g(X) - 1/\lambda)$ is used only for $\lambda > 1$. It proves strict improvement for the exact rational threshold 1.2581840884 ; since this is below $9/7$, the final case $\lambda > 9/7$ is covered with overlap. The lower and compact branches cover their closed upper endpoints. The final theorem therefore has precisely the stated $\lambda \geq 51/50$ and $0 < h \leq 1$ domains.

External source and independent cross-check

I checked the adopted CMV citation against the frozen Canete2010.pdf and the stable arXiv:0906.1256v1 record. The authors/title match. In that version, Section 3.2 contains the strip setup; Propositions 3.5/3.6, Lemma 3.8 and Proposition 3.9 have the cited roles; Conjecture 3.12 and equations (26)-(27) are on printed page 19. Theorem 3.16 runs through the cited printed pages 22-24, with the cap-replacement comparison and Remark 3.17 on page 24. The manuscript treats the source classification as motivation for its explicit model, not as a silently imported formal axiom. Incidental literature comments inside Lean are expressly not adopted citations, and the normalized arc construction and derivative are proved internally. No citation issue remains.

As an independent numerical smoke check, I evaluated the displayed profiles and geometric horizontal-slice cap integrals at 60-decimal working precision. At $\lambda = 51/50$, $33/32$ and $9/7$, I solved for the stationary type-(iv) fold and descending equal-area roots, then checked strict perimeter improvement at $h4 = 0.01, 0.5$ and 1 . Additional checks at $h3 = 0.25, 0.5$ and 0.75 covered major-cap, degenerate-width and minor-cap geometry and $P3' = h3 * A3'$. All 18 scenarios passed. The stationary gaps were approximately -0.00818522727888 , -0.0152033510607 and -0.317139378278 , respectively. These observations are supplementary sanity evidence only; they do not replace the exact rational certificates or establish a new threshold.

Disposition

All 850 nongenerated translations and both exhaustive generated-family descriptions are equivalent in their explicitly stated constructor/section context. The main proof supplies the geometric realization, analytic reduction, exact certificate coverage and high-density splice needed for the frozen modeled exclusion. No conditional fragment is substituted for that final theorem. All issue lists are empty because no specific semantic defect was identified, not because a prior status or successful compilation was accepted as proof.

Package-level findings

Relation: equivalent

No package-level semantic errors were reported.

Declaration-by-declaration audit

AdmissibleCompetitor

Source hash: 52c3c4597213ca5cc7d4e12b21ad852ff148bcb8179b3f52a9c03b7d3665311b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3952-3960, the source contract is inductive AdmissibleCompetitor (lam : $\mathbb{R}$) where | empty | general (region : FinitePerimeterRegion lam) | fourArc (a : FourArcAssembly) | replacement (a : ReplacementAssembly). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

AdmissibleCompetitor.WeightedArea

Source hash: d58b51a28690771d1b404e9048905bb6eae43f08f6798b27166b1234f3251f5e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3968-3975, the source contract is def WeightedArea {lam : $\mathbb{R}$ } : AdmissibleCompetitor lam $\rightarrow$ $\mathbb{R}$ | .empty => 0 | .general region => region.weightedArea | .fourArc a => a.weightedArea lam | .replacement a => a.weightedArea lam. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

AdmissibleCompetitor.WeightedPerimeter

Source hash: 17b5c5914d38bc0479d502c48cde9715c303aa7070585ade87ba56d307321437

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3976-3980, the source contract is def WeightedPerimeter {lam : $\mathbb{R}$ } (r : AdmissibleCompetitor lam) : $\mathbb{R}$:= _root_.WeightedPerimeter lam (FrontierMeasure r.carrier). Although area is dispatched by competitor

constructor, perimeter is canonically evaluated on the complete frontier of that constructor's carrier. A type-(iii) competitor enters through the general constructor; the quantification is not restricted to four-arc competitors.

Findings: none.

AdmissibleCompetitor.carrier

Source hash: b57d4c4c33776a05ffbbbc22f1b363a7f356aa7cd089d7c2b3335ead1bb65644

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3961-3967, the source contract is `def carrier {lam : ℝ} : AdmissibleCompetitor lam → Set PlanePoint | .empty => ∅ | .general region => region.carrier | .fourArc a => a.carrier | .replacement a => a.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

AdmissibleCompetitor.weightedArea_eq_carrier

Source hash: bf49ad5d7ddb4e090057dafec5dba684f6e36be017290e55451b4a9c6cfabce1

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3981-3990, the source contract is `theorem weightedArea_eq_carrier {lam : ℝ} (r : AdmissibleCompetitor lam) : r.WeightedArea = _root_.WeightedArea lam r.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

CMVPublishedCutoff51_50.candidate_not_isWeightedPerimeterMinimizer_from_51_50

Source hash: fe8c3d4297822263e0764fc31d2cba27c4c202e21a8868a614e1c8b94789b37f

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVPublishedCutoff51_50.lean:22-39, the source contract is `theorem candidate_not_isWeightedPerimeterMinimizer_from_51_50 {lam : ℝ} (hlower : (51 / 50 : ℝ) ≤ lam) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. The only external inputs are $\lambda \geq 51/50$, a FourArcCandidate λ , and its SatisfiesCMVTypeIVHypotheses. The proof splits at 33/32 and 9/7, using the lower and compact closed cover and then the cap-replacement theorem. Candidate curvature remains $0 < h \leq 1$, including $h = 1$; no sharpness assertion or conclusion below 51/50 is added.

Findings: none.

CMVSuffixModel.FourArcCandidate.endpoint_weightedArea_eq_typeFourArea

Source hash: 1d3c440482f713868366732790a87ef12add8058ecc71af2e6fb07df04bb3b83

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:159-166, the source contract is theorem `endpoint_weightedArea_eq_typeFourArea` (`hcandidate : candidate.SatisfiesCMVTypeIVHypotheses`) (`_hendpoint : candidate.h = 1`) : `candidate.WeightedArea = LeanSuffixAnalytic.typeFourArea lam candidate.h`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.endpoint_weightedPerimeter_eq_typeFourPerimeter

Source hash: 9a9144c9e0869ee21b16e09b6aad1d31570ea82a7d5d876263c80c006e5e5661

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:167-176, the source contract is theorem `endpoint_weightedPerimeter_eq_typeFourPerimeter` (`hcandidate : candidate.SatisfiesCMVTypeIVHypotheses`) (`_hendpoint : candidate.h = 1`) : `candidate.WeightedPerimeter = LeanSuffixAnalytic.typeFourPerimeter lam candidate.h`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_analytic_typeThree

Source hash: 99c923b45cbc09e9ba012e383efbf5a498fe0a194180526cbe54c89d50a0f151

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:201-224, the source contract is theorem `not_isWeightedPerimeterMinimizer_of_analytic_typeThree` (`hcandidate : candidate.SatisfiesCMVTypeIVHypotheses`) (`h3 : ℝ`) (`hh3 : 0 < h3`) (`hh3_one : h3 < 1`) (`equalArea : LeanSuffixAnalytic.typeThreeArea h3 = LeanSuffixAnalytic.typeFourArea lam candidate.h`) (`strictPerimeter : LeanSuffixAnalytic.typeThreePerimeter lam h3 < LeanSuffixAnalytic.typeFourPerimeter lam candidate.h`) : `¬ candidate.IsWeightedPerimeterMinimizer`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair

Source hash: d6670f69ae0dd1a579233b44583ff3b5ab90f13ac210ff1ddaabbb1088c5b5e2

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:225-248, the source contract is `theorem not_isWeightedPerimeterMinimizer_of_stationaryEqualAreaPair (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) (pair : LeanSuffixAnalytic.StationaryEqualAreaPair lam) (hfoldThree : LeanSuffixAnalytic.typeThreeFold lam pair.h3 < 0) (hperimeter : LeanSuffixAnalytic.typeThreePerimeter lam pair.h3 < LeanSuffixAnalytic.typeFourPerimeter lam pair.h4) : ¬ candidate.IsWeightedPerimeterMinimizer`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.regular_weightedArea_eq_typeFourArea

Source hash: 48c010f5756d8d71b74c4c11f50c789619d79d3e32a79f62433bedf4290ea164

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:142-149, the source contract is `theorem regular_weightedArea_eq_typeFourArea (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) (_hregular : candidate.h < 1) : candidate.WeightedArea = LeanSuffixAnalytic.typeFourArea lam candidate.h`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.regular_weightedPerimeter_eq_typeFourPerimeter

Source hash: 7c7abd9d4131a70966d7ee06757ae9000eb6c0370ad6826014e6939680589328

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:150-158, the source contract is `theorem regular_weightedPerimeter_eq_typeFourPerimeter (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) (_hregular : candidate.h < 1) : candidate.WeightedPerimeter = LeanSuffixAnalytic.typeFourPerimeter lam candidate.h`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.weightedArea_eq_typeFourArea

Source hash: e8473928799276d8ba026ded04a5483fef2171ee9679b0cfa48b5117289324b3

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:177-186, the source contract is **theorem weightedArea_eq_typeFourArea** (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : candidate.WeightedArea = LeanSuffixAnalytic.typeFourArea lam candidate.h. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is typeThreeAngle + $\pi/2$; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.FourArcCandidate.weightedPerimeter_eq_typeFourPerimeter

Source hash: c47649486a6e39478b2697659276a1423a248517e94449da31be1eb9b63be5ce

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:187-200, the source contract is **theorem weightedPerimeter_eq_typeFourPerimeter** (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : candidate.WeightedPerimeter = LeanSuffixAnalytic.typeFourPerimeter lam candidate.h. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is typeThreeAngle + $\pi/2$; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.angleTerm_eq_typeThreeAngle

Source hash: fe9ab761a6d7819237c18cc5b6d5b714cc699cb312b84007d27510fce813ca50

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:42-52, the source contract is **theorem angleTerm_eq_typeThreeAngle** : a.angleTerm = LeanSuffixAnalytic.typeThreeAngle lam a.h + $\pi / 2$. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is typeThreeAngle + $\pi/2$; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.scalarWeightedArea_eq_typeThreeArea

Source hash: 09eff0e6af5f2cf4916a82b585443109aad4f3e8bb020859b5a5f6526f9c2f0d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:78-86, the source contract is **theorem scalarWeightedArea_eq_typeThreeArea** : a.scalarWeightedArea = LeanSuffixAnalytic.typeThreeArea lam a.h. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is typeThreeAngle + $\pi/2$; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.scalarWeightedPerimeter_eq_typeThreePerimeter

Source hash: 1cbaa2764b23e5bf11163fb498e9653b454e16f1f8f6b84d752362e7a842718b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:87-99, the source contract is `theorem scalarWeightedPerimeter_eq_typeThreePerimeter : a.scalarWeightedPerimeter = LeanSuffixAnalytic.typeThreePerimeter lam a.h`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CMVSuffixModel.TypeThreeAssembly.sineGap_eq_typeThreeDelta

Source hash: 0b54dd8dd926b7670949326ff7f7fb7d21250d76068f1da2eb207447216f1aac

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVSuffixModel.lean:53-77, the source contract is `theorem sineGap_eq_typeThreeDelta : a.sineGap = LeanSuffixAnalytic.typeThreeDelta lam a.h`. This bridge identifies actual carrier integrals with the scalar profiles. In particular the geometric type-(iii) angle term is `typeThreeAngle + pi/2`; regular-only wrappers are not substituted for the separate endpoint-inclusive candidate bridge.

Findings: none.

CapSide

Source hash: 6863fffc025769eb8169e9582a3d673fe180025902f766c2557257f6937cc7a7e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:92-99, the source contract is `inductive CapSide where | upper | lower deriving DecidableEq`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

CompactCellAssembly.Conditions

Source hash: 5c3b2aa26c0b8b3d8801158e0abfe33b06aca94b7f9fdd276d46c23ec0373f44

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellAssembly.lean:25-62, the source contract is `structure Conditions (lambdaInterval h3Interval h4Interval : QInterval) : Prop where
lambda_one_lt : ∀ {lam : ℝ}, lambdaInterval.RealContains lam → 1 < lam
h3_slab_regular : ∀ {h : ℝ}, h3Interval.RealContains h → h ∈ Ioo (0 : ℝ) 1
h3_slab_above_half : ∀ {h : ℝ}, h3Interval.RealContains h → 1 / 2 < h
h4_slab_regular : ∀ {h : ℝ}, h4Interval.RealContains h → h ∈ Ioo (0 : ℝ) 1
typeFourFold_lowerFace_neg : ∀ {lam : ℝ}, lambdaInterval.RealContains lam → LeanSuffixAnalytic.typeFourFold lam (h4Interval.lo : ℝ) < 0
typeFourFold_upperFace_pos : ∀ {lam : ℝ}, lambdaInterval.RealContains lam → 0 < LeanSuffixAnalytic.typeFourFold lam (h4Interval.hi : ℝ)
typeThreeFold_neg_on_slab : ∀ {lam h : ℝ}, lambdaInterval.RealContains lam → h3Interval.RealContains h → LeanSuffixAnalytic.typeThreeFold lam h < 0
typeThreeArea_lowerFace_gap_pos : ∀ {lam h4`

$\mathbb{R}$ }, $\text{lambdaInterval.RealContains } \text{lam} \rightarrow \text{h4Interval.RealContains } h_4 \rightarrow$
 $\text{LeanSuffixAnalytic.typeFourFold } \text{lam } h_4 = 0 \rightarrow 0 < \text{LeanSuffixAnalytic.typeThreeArea } \text{lam}$
 $(\text{h3Interval.lo} : \mathbb{R}) - \text{LeanSuffixAnalytic.typeFourArea } \text{lam } h_4$
 $\text{typeThreeArea_upperFace_gap_neg} : \forall \{ \text{lam } h_4 : \mathbb{R} \}, \text{lambdaInterval.RealContains } \text{lam} \rightarrow$
 $\text{h4Interval.RealContains } h_4 \rightarrow \text{LeanSuffixAnalytic.typeFourFold } \text{lam } h_4 = 0 \rightarrow$
 $\text{LeanSuffixAnalytic.typeThreeArea } \text{lam } (\text{h3Interval.hi} : \mathbb{R}) -$
 $\text{LeanSuffixAnalytic.typeFourArea } \text{lam } h_4 < 0 \text{ typeThreePerimeter_gap_neg} : \forall \{ \text{lam } h_3 h_4 :$
 $\mathbb{R} \}, \text{lambdaInterval.RealContains } \text{lam} \rightarrow \text{h3Interval.RealContains } h_3 \rightarrow$
 $\text{h4Interval.RealContains } h_4 \rightarrow \text{LeanSuffixAnalytic.typeThreePerimeter } \text{lam } h_3 -$
 $\text{LeanSuffixAnalytic.typeFourPerimeter } \text{lam } h_4 < 0$. The condition record is analytic data, not the
conclusion in disguise: density jump, regular h3/h4 slabs with h3 above one half, K4 opposite signs, K3 negativity,
conditional equal-area face signs at a K4 zero, and a strict perimeter sign on the entire Cartesian slab. The author preserves
that last unconditional slab quantifier.

Findings: none.

CompactCellAssembly.Conditions.candidate_not_isWeightedPerimeterMinimizer

Source hash: 6281e20f4e6b992e95255e50221808300ba4ec7b1c545ab00d1868e256be0776

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellAssembly.lean:312-338, the source contract is
theorem candidate_not_isWeightedPerimeterMinimizer (conditions : Conditions
lambdaInterval h3Interval h4Interval) {lam : $\mathbb{R}$ } (hlam : lambdaInterval.RealContains
lam) (candidate : _root_.FourArcCandidate lam) (hcandidate :
candidate.SatisfiesCMVTypeIVHypotheses) : $\neg$ candidate.IsWeightedPerimeterMinimizer. The
Conditions record contains regular root slabs, opposite stationary-fold and area-face signs, negative type-(iii) fold, and a
strict perimeter sign. The generic assembly derives a stationary pair and then uses the all-curvature envelope; candidate
curvature is not confined to the certificate slab.

Findings: none.

CompactCellAssembly.Conditions.stationaryEqualArea_improvement_exists

Source hash: e63a75b5ca3c02a2700397ede0d62a135c22960b730b6191885ca6432b43a9fe

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellAssembly.lean:244-264, the source contract is
theorem stationaryEqualArea_improvement_exists (conditions : Conditions lambdaInterval
h3Interval h4Interval) {lam : $\mathbb{R}$ } (hlam : lambdaInterval.RealContains lam) : $\exists h_4 h_3 :$
 $\mathbb{R}, \text{h4Interval.RealContains } h_4 \wedge \text{LeanSuffixAnalytic.typeFourFold } \text{lam } h_4 = 0 \wedge$
 $\text{h3Interval.RealContains } h_3 \wedge \text{LeanSuffixAnalytic.typeThreeArea } \text{lam } h_3 =$
 $\text{LeanSuffixAnalytic.typeFourArea } \text{lam } h_4 \wedge \text{LeanSuffixAnalytic.typeThreePerimeter } \text{lam } h_3$
 $< \text{LeanSuffixAnalytic.typeFourPerimeter } \text{lam } h_4$. The Conditions record contains regular root slabs,
opposite stationary-fold and area-face signs, negative type-(iii) fold, and a strict perimeter sign. The generic assembly
derives a stationary pair and then uses the all-curvature envelope; candidate curvature is not confined to the certificate
slab.

Findings: none.

CompactCellAssembly.Conditions.typeFourFold_existsUnique_in_slab

Source hash: c2359e743fc358937d19c1ac3887ef019ddb92c97b2b12e00651546f5b742979

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellAssembly.lean:76-106, the source contract is theorem typeFourFold_existsUnique_in_slab (conditions : Conditions lambdaInterval h3Interval h4Interval) {lam : $\mathbb{R}$ } (hlam : lambdaInterval.RealContains lam) : $\exists! h_4 : \mathbb{R}, h4Interval.RealContains h_4 \wedge \text{LeanSuffixAnalytic.typeFourFold lam } h_4 = 0$. The Conditions record contains regular root slabs, opposite stationary-fold and area-face signs, negative type-(iii) fold, and a strict perimeter sign. The generic assembly derives a stationary pair and then uses the all-curvature envelope; candidate curvature is not confined to the certificate slab.

Findings: none.

CompactCellAssembly.Conditions.typeThreeEqualArea_existsUnique_in_slab

Source hash: 79c92a7e4df192f65dc043cccd8bea4a018e8826ba49c09591fd046f371a56f7

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellAssembly.lean:107-177, the source contract is theorem typeThreeEqualArea_existsUnique_in_slab (conditions : Conditions lambdaInterval h3Interval h4Interval) {lam h₄ : $\mathbb{R}$ } (hlam : lambdaInterval.RealContains lam) (hh₄ : h4Interval.RealContains h₄) (hstationary : LeanSuffixAnalytic.typeFourFold lam h₄ = 0) : $\exists! h_3 : \mathbb{R}, h3Interval.RealContains h_3 \wedge \text{LeanSuffixAnalytic.typeThreeArea lam } h_3 = \text{LeanSuffixAnalytic.typeFourArea lam } h_4$. The Conditions record contains regular root slabs, opposite stationary-fold and area-face signs, negative type-(iii) fold, and a strict perimeter sign. The generic assembly derives a stationary pair and then uses the all-curvature envelope; candidate curvature is not confined to the certificate slab.

Findings: none.

CompactCellAssembly.QInterval

Source hash: bf869b1f520d1cedce20479248192ec784229f327d43ecc88c3fd90b71146250

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellAssembly.lean:20-24, the source contract is abbrev QInterval := ScalarSuffixCertificate.QInterval. The Conditions record contains regular root slabs, opposite stationary-fold and area-face signs, negative type-(iii) fold, and a strict perimeter sign. The generic assembly derives a stationary pair and then uses the all-curvature envelope; candidate curvature is not confined to the certificate slab.

Findings: none.

CompactCellCertificate.QInterval

Source hash: baf33f14dafc66f34a92b548a53e5c251fc54f718db15c2e52f89ec02947b54d

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:19-23, the source contract is abbrev QInterval := ScalarSuffixCertificate.QInterval. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.QInterval.realContains_arcsin_of_taylor

Source hash: 4923f85862130adb8d9efbeee24dfca326699114fa32f80b790af2693e065f1b

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:24-44, the source contract is theorem `realContains_arcsin_of_taylor` (`a : QInterval`) (`x : ℚ`) (`hx : (-1 : ℚ) ≤ x ∧ x ≤ 1`) (`hbranch : (a.lo : ℝ) ∈ Icc (-(π / 2)) (π / 2) ∧ (a.hi : ℝ) ∈ Icc (-(π / 2)) (π / 2)`) (`hlo : (ScalarSuffixCertificate.sinTaylor27Interval a.lo).hi ≤ x`) (`hhi : x ≤ (ScalarSuffixCertificate.sinTaylor27Interval a.hi).lo`) : `a.RealContains (arcsin (x : ℝ))`. `QInterval` containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.QInterval.realContains_arcsin_of_taylor_interval

Source hash: 72d1f4184baa0c96534f19a83e2653ab69774d487022002946ed89c5dcb4c89d

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:45-68, the source contract is theorem `realContains_arcsin_of_taylor_interval` (`a xInterval : QInterval`) (`x : ℝ`) (`hx : xInterval.RealContains x`) (`hxUnit : (-1 : ℝ) ≤ x ∧ x ≤ 1`) (`hbranch : (a.lo : ℝ) ∈ Icc (-(π / 2)) (π / 2) ∧ (a.hi : ℝ) ∈ Icc (-(π / 2)) (π / 2)`) (`hlo : (ScalarSuffixCertificate.sinTaylor27Interval a.lo).hi ≤ xInterval.lo`) (`hhi : xInterval.hi ≤ (ScalarSuffixCertificate.sinTaylor27Interval a.hi).lo`) : `a.RealContains (arcsin x)`. `QInterval` containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.halfPiBox

Source hash: b24b5a3000649c97c552eff6f175f59f2e5aa6602a533417871ec37dfc3c128c

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:193-201, the source contract is `def halfPiBox : QInterval := ScalarSuffixCertificate.piInterval.nsmul (1 / 2) (by norm_num) @[simp] theorem halfPiBox_lo : halfPiBox.lo = 392699 / 250000 := by norm_num [halfPiBox, ScalarSuffixCertificate.piInterval, LeanSuffixReflective.QInterval.nsmul]`. `QInterval` containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.halfPiBox_sound

Source hash: 32569ba98b96e5c623e6b4690e061422a54fc3e99bc88a52ad664b5adadcdbf6

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:202-210, the source contract is theorem halfPiBox_sound : halfPiBox.RealContains $(\pi / 2)$. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.principalAsinBranch_of_nonneg_of_hi_le_halfPiBox

Source hash: 5d213510a122acf90e70e2f5809f269336785657fb7f1298c72b17c1dfca9aca

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:211-223, the source contract is theorem principalAsinBranch_of_nonneg_of_hi_le_halfPiBox (a : QInterval) (h : $0 \leq a.lo \wedge a.hi \leq \text{halfPiBox}.lo$) : (a.lo : $\mathbb{R}$) $\in \text{Icc } (-\pi / 2) (\pi / 2) \wedge (a.hi : \mathbb{R}) \in \text{Icc } (-\pi / 2) (\pi / 2)$. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.realContains_of_mem_uIcc

Source hash: ebce25efd40e9df8676596e53d18f779f851450dc32118fcfd8ff8a4a80eceb

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:69-76, the source contract is theorem realContains_of_mem_uIcc {i : QInterval} {c h x : $\mathbb{R}$ } (hc : i.RealContains c) (hh : i.RealContains h) (hx : $x \in \text{uIcc } c \ h$) : i.RealContains x. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFourAngleBox

Source hash: 0938a788ac4433df3e54045185685f79b05098a7ac19313f95d43a2d5b337139

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:582-584, the source contract is def typeFourAngleBox (lambdaInterval asinHI asinRatioI : QInterval) : QInterval := (lambdaInterval.mul (halfPiBox.sub asinRatioI)).add asinHI. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFourAreaBox

Source hash: d6bce36771e12b0ae40beb52da88876842d09c71cb25452547718f2a63131dfa

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:589-596, the source contract is `def typeFourAreaBox (lambdaInterval hI uI dI asinHI asinRatioI : QInterval) (hlambda : 0 < lambdaInterval.lo) (hhSq : 0 < (hI.mul hI).lo) : QInterval :=`

`ScalarSuffixCertificate.QInterval.divPos (LeanSuffixReflective.QInterval.nsmul 2 (by norm_num) ((typeFourAngleBox lambdaInterval asinHI asinRatioI).add (hI.mul (typeFourDeltaBox lambdaInterval uI dI hlambda)))) (hI.mul hI) hhSq`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFourArea_centered_enclosure

Source hash: 5c97a83974bdea22e535e6c42e2259805e5f45e9f57a2175a11ade991b773562

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:77-105, the source contract is `theorem typeFourArea_centered_enclosure (lambdaInterval hInterval : QInterval) (center : ℚ) {lam h : ℝ} (_hlam : lambdaInterval.RealContains lam) (hh : hInterval.RealContains h) (hcenter : hInterval.RealContains (center : ℝ)) (hlambda : 1 < lam) (hregular : ∀ {x : ℝ}, hInterval.RealContains x → x ∈ Ioo (0 : ℝ) 1) (base slope displacement : QInterval) (hbase : base.RealContains (LeanSuffixAnalytic.typeFourArea lam (center : ℝ))) (hslope : ∀ x, hInterval.RealContains x → slope.RealContains (LeanSuffixAnalytic.typeFourFold lam x / x ^ 3)) (hdisplacement : displacement.RealContains (h - (center : ℝ))) : (base.add (slope.mul displacement)).RealContains (LeanSuffixAnalytic.typeFourArea lam h)`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFourDeltaBox

Source hash: ab7d789a19d88d75ee1647ab012fc735ddecd9dc900020900794197c4a8a7372

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:585-588, the source contract is `def typeFourDeltaBox (lambdaInterval uI dI : QInterval) (hlambda : 0 < lambdaInterval.lo) : QInterval := (dI.divPos lambdaInterval hlambda).sub uI`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFourFoldBox

Source hash: c03c54c4b04b12af79ff4aba9a1baf5c6d0801c18971d3619c3c5a83e175fdfd

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:602-607, the source contract is `def typeFourFoldBox (lambdaInterval hI uI dI asinHI asinRatioI : QInterval) (_hlambda : 0 < lambdaInterval.lo) (hu : 0 < uI.lo) (hd : 0 < dI.lo) : QInterval := ((hI.mul ((uI.recipPos hu).sub (lambdaInterval.divPos dI hd))).sub (typeFourAngleBox lambdaInterval asinHI asinRatioI)).nsmul 4 (by norm_num). QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.`

Findings: none.

CompactCellCertificate.typeFourPerimeterBox

Source hash: 9acb08313f7038b05bcb4ead730aaeb55230fdc38d1e200f1e4b75f2584d5063

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:597-601, the source contract is `def typeFourPerimeterBox (lambdaInterval hI asinHI asinRatioI : QInterval) (hh : 0 < hI.lo) : QInterval := (ScalarSuffixCertificate.QInterval.divPos ((typeFourAngleBox lambdaInterval asinHI asinRatioI).nsmul 4 (by norm_num)) hI hh). QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.`

Findings: none.

CompactCellCertificate.typeFourPerimeter_centered_enclosure

Source hash: b6de403709bd1fd67e6a59e747fa0828b9b2000aa26dc332ce4736414ad4a315

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:106-134, the source contract is `theorem typeFourPerimeter_centered_enclosure (lambdaInterval hInterval : QInterval) (center : Q) {lam h : ℝ} (_hlam : lambdaInterval.RealContains lam) (hh : hInterval.RealContains h) (hcenter : hInterval.RealContains (center : ℝ)) (hlambda : 1 < lam) (hregular : ∀ {x : ℝ}, hInterval.RealContains x → x ∈ Ioo (0 : ℝ) 1) (base slope displacement : QInterval) (hbase : base.RealContains (LeanSuffixAnalytic.typeFourPerimeter lam (center : ℝ))) (hslope : ∀ x, hInterval.RealContains x → slope.RealContains (LeanSuffixAnalytic.typeFourFold lam x / x ^ 2)) (hdisplacement : displacement.RealContains (h - (center : ℝ))) : (base.add (slope.mul displacement)).RealContains (LeanSuffixAnalytic.typeFourPerimeter lam h). QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.`

Findings: none.

CompactCellCertificate.typeFourShiftedInterval

Source hash: c59ab8bb2b04642fd5ba4e5bffd2f3c42cf4105e98a359d7aa614a9c22b9d322

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:361-368, the source contract is `def typeFourShiftedInterval (hI : QInterval) : QInterval := (hI.add (LeanSuffixReflective.QInterval.point 1)).nsmul (1 / 2) (by norm_num)`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFour_atom_enclosures

Source hash: 6e810d9ecc043611b95444c9f9834f7abe166e7b764027bba215f104fde2bca9

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:608-665, the source contract is `theorem typeFour_atom_enclosures (lambdaInterval hI uI dI asinHI asinRatioI : QInterval) {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : hI.RealContains h) (hatoms : uI.RealContains (sqrt (1 - h ^ 2)) ^ dI.RealContains (sqrt (lam ^ 2 - h ^ 2)) ^ asinHI.RealContains (arcsin h) ^ asinRatioI.RealContains (arcsin (h / lam))) (hlambda : 0 < lambdaInterval.lo) (hhPos : 0 < hI.lo) (hhSq : 0 < (hI.mul hI).lo) (huPos : 0 < uI.lo) (hdPos : 0 < dI.lo) : (typeFourAreaBox lambdaInterval hI uI dI asinHI asinRatioI hlamba hhSq).RealContains (LeanSuffixAnalytic.typeFourArea lam h) ^ (typeFourPerimeterBox lambdaInterval hI asinHI asinRatioI hhPos).RealContains (LeanSuffixAnalytic.typeFourPerimeter lam h) ^ (typeFourFoldBox lambdaInterval hI uI dI asinHI asinRatioI hlamba huPos hdPos).RealContains (LeanSuffixAnalytic.typeFourFold lam h)`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeFour_atoms_of_exact_bounds

Source hash: 782e78df1d670aee0c342670496e6eb38d375401e162a64061d13d6eb4739dd2

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:369-432, the source contract is `theorem typeFour_atoms_of_exact_bounds (lambdaInterval hI uI dI asinHI asinRI : QInterval) {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : hI.RealContains h) (hlambaLoPos : 0 < lambdaInterval.lo) (hlambaOne : 1 < lam) (hexact : 0 ≤ (((typeFourShiftedInterval hI).nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).lo ^ (((typeFourShiftedInterval hI).nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).hi ≤ 1 ^ 0 ≤ uI.lo ^ uI.lo ^ 2 ≤ 1 - (((typeFourShiftedInterval hI).nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).hi ^ 2 ^ 1 - (((typeFourShiftedInterval hI).nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).lo ^ 2 ≤ uI.hi ^ 2 ^ 0 ≤ dI.lo ^ dI.lo ^ 2 ≤ lambdaInterval.lo ^ 2 - (((typeFourShiftedInterval`

$$\begin{aligned}
& hI).nsmul\ 2\ (by\ norm_num)).sub\ (LeanSuffixReflective.QInterval.point\ 1)).hi\ ^\ 2\ \wedge \\
& \lambdaInterval.hi\ ^\ 2\ -\ (((typeFourShiftedInterval\ hI).nsmul\ 2\ (by\ norm_num)).sub \\
& (LeanSuffixReflective.QInterval.point\ 1)).lo\ ^\ 2\ \leq\ dI.hi\ ^\ 2\ \wedge\ (0\ \leq\ asinHI.lo\ \wedge \\
& asinHI.hi\ \leq\ halfPiBox.lo)\ \wedge\ (ScalarSuffixCertificate.sinTaylor27Interval\ asinHI.lo).hi \\
& \leq\ (((typeFourShiftedInterval\ hI).nsmul\ 2\ (by\ norm_num)).sub \\
& (LeanSuffixReflective.QInterval.point\ 1)).lo\ \wedge\ (((typeFourShiftedInterval\ hI).nsmul\ 2 \\
& (by\ norm_num)).sub\ (LeanSuffixReflective.QInterval.point\ 1)).hi\ \leq \\
& (ScalarSuffixCertificate.sinTaylor27Interval\ asinHI.hi).lo\ \wedge\ (0\ \leq\ asinRI.lo\ \wedge \\
& asinRI.hi\ \leq\ halfPiBox.lo)\ \wedge\ (ScalarSuffixCertificate.sinTaylor27Interval\ asinRI.lo).hi \\
& \leq\ (ScalarSuffixCertificate.QInterval.divPos\ (((typeFourShiftedInterval\ hI).nsmul\ 2\ (by \\
& norm_num)).sub\ (LeanSuffixReflective.QInterval.point\ 1))\ \lambdaInterval \\
& h\lambdaLoPos).lo\ \wedge\ (ScalarSuffixCertificate.QInterval.divPos \\
& (((typeFourShiftedInterval\ hI).nsmul\ 2\ (by\ norm_num)).sub \\
& (LeanSuffixReflective.QInterval.point\ 1))\ \lambdaInterval\ h\lambdaLoPos).hi\ \leq \\
& (ScalarSuffixCertificate.sinTaylor27Interval\ asinRI.hi).lo) : uI.RealContains\ (sqrt\ (1 \\
& -\ h\ ^\ 2))\ \wedge\ dI.RealContains\ (sqrt\ (lam\ ^\ 2\ -\ h\ ^\ 2))\ \wedge\ asinHI.RealContains\ (arcsin\ h) \\
& \wedge\ asinRI.RealContains\ (arcsin\ (h\ /\ lam)).
\end{aligned}$$
 QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreeAngleBox

Source hash: 07cf7ee9a93efff6da0c0dde5b35b60c96d4a49a9bfb27da2b96e97cee62677e

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:436-438, the source contract is `def typeThreeAngleBox (lambdaInterval asinQI asinRatioI : QInterval) : QInterval := (lambdaInterval.mul (halfPiBox.sub asinRatioI)).add asinQI`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreeAreaBox

Source hash: ef992989a954766b4ace02b65812003aa8c6c32c075d2783aff3bdaa521effd7

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:443-450, the source contract is `def typeThreeAreaBox (lambdaInterval hI uI dI asinQI asinRatioI : QInterval) (hlambda : 0 < lambdaInterval.lo) (hhSq : 0 < (hI.mul hI).lo) : QInterval := ScalarSuffixCertificate.QInterval.divPos ((typeThreeAngleBox lambdaInterval asinQI asinRatioI).add halfPiBox |>.add (ScalarSuffixCertificate.QInterval.mul ((typeThreeQBox hI).add (LeanSuffixReflective.QInterval.point 2)) (typeThreeDeltaBox lambdaInterval uI dI hlambda))) (hI.mul hI) hhSq`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreeDeltaBox

Source hash: 1e88f30488cceb14eb20e3a119a4f1d4e5d4e9a2623b803c1f4033ff8844eadb

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:439-442, the source contract is `def typeThreeDeltaBox (lambdaInterval uI dI : QInterval) (hlambda : 0 < lambdaInterval.lo) : QInterval := (dI.divPos lambdaInterval hlambda).sub uI`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreeFoldBox

Source hash: 134abfc2384eaab1be42cb464447703ca4b8c8946a097fc0f4846639ef17f9e4

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:458-472, the source contract is `def typeThreeFoldBox (lambdaInterval hI uI dI asinQI asinRatioI : QInterval) (hlambda : 0 < lambdaInterval.lo) (hu : 0 < uI.lo) (hlambdaD : 0 < (lambdaInterval.mul dI).lo) : QInterval := let qI := typeThreeQBox hI let angleI := typeThreeAngleBox lambdaInterval asinQI asinRatioI let deltaI := typeThreeDeltaBox lambdaInterval uI dI hlambda let first := ScalarSuffixCertificate.QInterval.divPos ((LeanSuffixReflective.QInterval.point 1).add qI) uI hu let second := ScalarSuffixCertificate.QInterval.divPos ((lambdaInterval.mul lambdaInterval).add qI) (lambdaInterval.mul dI) hlambdaD ((hI.mul (first.sub second)).nsmul 4 (by norm_num)).sub (((angleI.add halfPiBox).add deltaI).nsmul 2 (by norm_num)).QInterval`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreePerimeterBox

Source hash: 5a67f5bcbe9b9ff1470b4fe9640a7fcc5189a1c047837c4c7b6352a695bd3349

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:451-457, the source contract is `def typeThreePerimeterBox (lambdaInterval hI uI dI asinQI asinRatioI : QInterval) (hlambda : 0 < lambdaInterval.lo) (hh : 0 < hI.lo) : QInterval := (ScalarSuffixCertificate.QInterval.divPos ((typeThreeAngleBox lambdaInterval asinQI asinRatioI).add halfPiBox |>.add (typeThreeDeltaBox lambdaInterval uI dI hlambda)) hI hh).nsmul 2 (by norm_num)`. QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreePerimeter_centered_enclosure

Source hash: 716e11bc278bf7afe71f82cb84a51dd8766c11970a07ccc39643388de278cc03

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:164-192, the source contract is theorem typeThreePerimeter_centered_enclosure (lambdaInterval hInterval : QInterval) (center : Q) {lam h : ℝ} (_hlam : lambdaInterval.RealContains lam) (hh : hInterval.RealContains h) (hcenter : hInterval.RealContains (center : ℝ)) (hlambda : 1 < lam) (hregular : ∀ {x : ℝ}, hInterval.RealContains x → x ∈ Ioo (0 : ℝ) 1) (base slope displacement : QInterval) (hbase : base.RealContains (LeanSuffixAnalytic.typeThreePerimeter lam (center : ℝ))) (hslope : ∀ x, hInterval.RealContains x → slope.RealContains (LeanSuffixAnalytic.typeThreeFold lam x / x ^ 2)) (hdisplacement : displacement.RealContains (h - (center : ℝ))) : (base.add (slope.mul displacement)).RealContains (LeanSuffixAnalytic.typeThreePerimeter lam h). QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThreeQBox

Source hash: 9902aca80d47ee5ab1219050853cf20b2a80b983bed70658848ab626cc339f9f

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:433-435, the source contract is def typeThreeQBox (hI : QInterval) : QInterval := (hI.nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1). QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThree_area_perimeter_atom_enclosure

Source hash: b6ef0ef761987f79259cc0d419f8a32fbed4691f7270b539e416653bb41f839b

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:529-581, the source contract is theorem typeThree_area_perimeter_atom_enclosure (lambdaInterval hI uI dI asinQI asinRatioI : QInterval) {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : hI.RealContains h) (hatoms : uI.RealContains (sqrt (1 - LeanSuffixAnalytic.typeThreeShape h ^ 2)) ^ dI.RealContains (sqrt (lam ^ 2 - LeanSuffixAnalytic.typeThreeShape h ^ 2)) ^ asinQI.RealContains (arcsin (LeanSuffixAnalytic.typeThreeShape h)) ^ asinRatioI.RealContains (arcsin (LeanSuffixAnalytic.typeThreeShape h / lam))) (hlambda : 0 < lambdaInterval.lo) (hhPos : 0 < hI.lo) (hhSq : 0 < (hI.mul hI).lo) : (typeThreeAreaBox lambdaInterval hI uI dI asinQI asinRatioI hlambda hhSq).RealContains (LeanSuffixAnalytic.typeThreeArea lam h) ^ (typeThreePerimeterBox lambdaInterval hI uI dI asinQI asinRatioI hlambda

$hhPos$).`RealContains` (`LeanSuffixAnalytic.typeThreePerimeter lam h`). `QInterval` containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactCellCertificate.typeThree_atoms_of_exact_bounds

Source hash: c51a322c3566880879bc58cb2483ec639ff97dddc9cd3e9ac9a18afac5600b84

Relation: equivalent

Equivalent in the declared constructor/section context. At `CompactCellCertificate.lean:224-360`, the source contract is

```
theorem typeThree_atoms_of_exact_bounds (lambdaInterval hI uI dI asinQI asinRatioI :
QInterval) {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : hI.RealContains
h) (hlambdaLoPos : 0 < lambdaInterval.lo) (hlambdaOne : 1 < lam) (hqNonneg : 0 ≤
((hI.nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).lo) (hqLeOne
: ((hI.nsmul 2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).hi ≤ 1)
(huNonneg : 0 ≤ uI.lo) (huLower : uI.lo ^ 2 ≤ 1 - ((hI.nsmul 2 (by norm_num)).sub
(LeanSuffixReflective.QInterval.point 1)).hi ^ 2) (huUpper : 1 - ((hI.nsmul 2 (by
norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).lo ^ 2 ≤ uI.hi ^ 2) (hdNonneg
: 0 ≤ dI.lo) (hdLower : dI.lo ^ 2 ≤ lambdaInterval.lo ^ 2 - ((hI.nsmul 2 (by
norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).hi ^ 2) (hdUpper :
lambdaInterval.hi ^ 2 - ((hI.nsmul 2 (by norm_num)).sub
(LeanSuffixReflective.QInterval.point 1)).lo ^ 2 ≤ dI.hi ^ 2) (hasinQBranch : 0 ≤
asinQI.lo ^ asinQI.hi ≤ halfPiBox.lo) (hasinQLower :
(ScalarSuffixCertificate.sinTaylor27Interval asinQI.lo).hi ≤ ((hI.nsmul 2 (by
norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).lo) (hasinQUpper : ((hI.nsmul
2 (by norm_num)).sub (LeanSuffixReflective.QInterval.point 1)).hi ≤
(ScalarSuffixCertificate.sinTaylor27Interval asinQI.hi).lo) (hasinRatioBranch : 0 ≤
asinRatioI.lo ^ asinRatioI.hi ≤ halfPiBox.lo) (hasinRatioLower :
(ScalarSuffixCertificate.sinTaylor27Interval asinRatioI.lo).hi ≤
(ScalarSuffixCertificate.QInterval.divPos ((hI.nsmul 2 (by norm_num)).sub
(LeanSuffixReflective.QInterval.point 1)) lambdaInterval hlambaLoPos).lo)
(hasinRatioUpper : (ScalarSuffixCertificate.QInterval.divPos ((hI.nsmul 2 (by
norm_num)).sub (LeanSuffixReflective.QInterval.point 1)) lambdaInterval
hlambaLoPos).hi ≤ (ScalarSuffixCertificate.sinTaylor27Interval asinRatioI.hi).lo) :
uI.RealContains (sqrt (1 - LeanSuffixAnalytic.typeThreeShape h ^ 2)) ^ dI.RealContains
(sqrt (lam ^ 2 - LeanSuffixAnalytic.typeThreeShape h ^ 2)) ^ asinQI.RealContains
(arcsin (LeanSuffixAnalytic.typeThreeShape h)) ^ asinRatioI.RealContains (arcsin
(LeanSuffixAnalytic.typeThreeShape h / lam)). QInterval containment is closed real containment of
rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch
hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.
```

Findings: none.

CompactCellCertificate.typeThree_fold_atom_enclosure

Source hash: e0e739ea56f950a43598c09273d8fde5874e105be082082bb67c5da1055a66a6

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactCellCertificate.lean:473-528, the source contract is theorem typeThree_fold_atom_enclosure (lambdaInterval hI uI dI asinQI asinRatioI : QInterval) {lam h : $\mathbb{R}$ } (hlam : lambdaInterval.RealContains lam) (hh : hI.RealContains h) (hatoms : uI.RealContains (sqrt (1 - LeanSuffixAnalytic.typeThreeShape h ^ 2)) ^ dI.RealContains (sqrt (lam ^ 2 - LeanSuffixAnalytic.typeThreeShape h ^ 2)) ^ asinQI.RealContains (arcsin (LeanSuffixAnalytic.typeThreeShape h)) ^ asinRatioI.RealContains (arcsin (LeanSuffixAnalytic.typeThreeShape h / lam))) (hlambda : 0 < lambdaInterval.lo) (huPos : 0 < uI.lo) (hlambaD : 0 < (lambdaInterval.mul dI).lo) : (typeThreeFoldBox lambdaInterval hI uI dI asinQI asinRatioI hlambda huPos hlambaD).RealContains (LeanSuffixAnalytic.typeThreeFold lam h).QInterval containment is closed real containment of rational endpoints. The radical and inverse-trigonometric enclosures retain their positivity and principal-branch hypotheses, and the computed bounds enclose the stated real expressions, not sampled approximations.

Findings: none.

CompactFirstCell.QInterval

Source hash: c522fb7a23b8535f69b64bd9659ad4aed8fc8dc5b89426617a6ca4e04ee665a3

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:19-23, the source contract is abbrev QInterval := ScalarSuffixCertificate.QInterval. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.QInterval.realContains_arcsin_of_taylor

Source hash: 89a8b20d50a5cd5fff8193c5fcee59fcf1459ec24caa265aff4f074622f22d58

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:24-38, the source contract is theorem realContains_arcsin_of_taylor (a : QInterval) (x : $\mathbb{Q}$) (hx : (-1 : $\mathbb{Q}$) ≤ x ∧ x ≤ 1) (hbranch : (a.lo : $\mathbb{R}$) ∈ Icc (-(π / 2)) (π / 2) ∧ (a.hi : $\mathbb{R}$) ∈ Icc (-(π / 2)) (π / 2)) (hlo : (ScalarSuffixCertificate.sinTaylor27Interval a.lo).hi ≤ x) (hhi : x ≤ (ScalarSuffixCertificate.sinTaylor27Interval a.hi).lo) : a.RealContains (arcsin (x : $\mathbb{R}$)). This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.compactCandidate_not_isWeightedPerimeterMinimizer

Source hash: 0cf4dd7a93e6150799254c4879293eeac9af163fdbd193bd279b77559f43d237

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:1749-1757, the source contract is `theorem compactCandidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (h1lam : lambdaInterval.RealContains lam) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.h3Center

Source hash: 56aacc195857950b6ae7b136dff6ad4341c307ca690bae0f998846b98084779a

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:70-71, the source contract is `def h3Center : ℚ := (h3Interval.lo + h3Interval.hi) / 2`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.h3Interval

Source hash: cfb76a49841952dce830375124f9a01e8a81c790ef5f2c6d14a5bb739e437d32

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:62-66, the source contract is `def h3Interval : QInterval := ⟨132572791017099 / 140737488355328, 133157808072187 / 140737488355328, by norm_num⟩`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.h4Center

Source hash: fcc3fc93d1f6d9a14b3c9f8cd836a4da459d6e2e9d5879a858b078d451152165

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:67-69, the source contract is `def h4Center : ℚ := (h4Interval.lo + h4Interval.hi) / 2`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.h4Interval

Source hash: 21261a7072ad9b4e7e79f4c2420b3ea2c0b04219a9a6039f38c82d6959dee705

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:57-61, the source contract is `def h4Interval : QInterval := ⟨34445267662685 / 35184372088832, 275599301332823 / 281474976710656, by norm_num⟩`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.lambdaInterval

Source hash: 33288724289e17e6b58f251168fcac1ba8bb868b76aed0ce4b2d7a7825c0065c

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:53-56, the source contract is `def lambdaInterval : QInterval := ⟨33 / 32, 236601 / 229376, by norm_num⟩`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.realContains_arcsin_of_taylor_interval

Source hash: 04625f4bdbd8d2aa4776f4b5c665d47c58bb244e2be98c92835ce4982a969a1c

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:39-52, the source contract is `theorem realContains_arcsin_of_taylor_interval (a xInterval : QInterval) (x : ℝ) (hx : xInterval.RealContains x) (hxUnit : (-1 : ℝ) ≤ x ∧ x ≤ 1) (hbranch : (a.lo : ℝ) ∈ Icc (-(π / 2)) (π / 2) ∧ (a.hi : ℝ) ∈ Icc (-(π / 2)) (π / 2)) (hlo : (ScalarSuffixCertificate.sinTaylor27Interval a.lo).hi ≤ xInterval.lo) (hhi : xInterval.hi ≤ (ScalarSuffixCertificate.sinTaylor27Interval a.hi).lo) : a.RealContains (arcsin x)`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeFourArea_centered_enclosure

Source hash: 23fd914cbcc4badfdce7ba2492c48ade47c8348ae2e3255286932df9b5b0b656

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:422-439, the source contract is `theorem typeFourArea_centered_enclosure {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : h4Interval.RealContains h) (base slope displacement : QInterval) (hbase : base.RealContains (LeanSuffixAnalytic.typeFourArea lam (h4Center : ℝ))) (hslope : ∀ x, h4Interval.RealContains x → slope.RealContains (LeanSuffixAnalytic.typeFourFold lam x / x ^ 3)) (hdisplacement : displacement.RealContains (h - (h4Center : ℝ))) : (base.add`

$(\text{slope} \cdot \text{displacement})$). `RealContains (LeanSuffixAnalytic.typeFourArea lam h)`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeFourFold_lowerFace_neg

Source hash: dcd660b276bba4f208c5ded09cca5851f937ad4f80478890332c45cce47b8a0

Relation: equivalent

Equivalent in the declared constructor/section context. At `CompactFirstCell.lean:182-300`, the source contract is `theorem typeFourFold_lowerFace_neg {lam : ℝ} (h1am : lambdaInterval.RealContains lam) : LeanSuffixAnalytic.typeFourFold lam (h4Interval.lo : ℝ) < 0`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeFourFold_upperFace_pos

Source hash: 2a5ed0fdf81ba4e8918a25fc36a2303fc5b7fea92508d672753fd14471fc1b6f

Relation: equivalent

Equivalent in the declared constructor/section context. At `CompactFirstCell.lean:301-421`, the source contract is `theorem typeFourFold_upperFace_pos {lam : ℝ} (h1am : lambdaInterval.RealContains lam) : 0 < LeanSuffixAnalytic.typeFourFold lam (h4Interval.hi : ℝ)`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeFourPerimeter_centered_enclosure

Source hash: af4df9fa49b636ddab2f85af4f4c2bbf86e908c96cd4cc740bea472cf83b8977

Relation: equivalent

Equivalent in the declared constructor/section context. At `CompactFirstCell.lean:440-458`, the source contract is `theorem typeFourPerimeter_centered_enclosure {lam h : ℝ} (h1am : lambdaInterval.RealContains lam) (hh : h4Interval.RealContains h) (base slope displacement : QInterval) (hbase : base.RealContains (LeanSuffixAnalytic.typeFourPerimeter lam (h4Center : ℝ))) (hslope : ∀ x, h4Interval.RealContains x → slope.RealContains (LeanSuffixAnalytic.typeFourFold lam x / x ^ 2)) (hdisplacement : displacement.RealContains (h - (h4Center : ℝ))) : (base.add (slope.mul displacement)).RealContains (LeanSuffixAnalytic.typeFourPerimeter lam h)`. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeThreeArea_lowerFace_minus_stationaryTypeFour_pos

Source hash: 38de632d8808d27f117549af1dc1f71c9d50996e4ee9a767cd56ea761fcaac99

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:1527-1550, the source contract is theorem typeThreeArea_lowerFace_minus_stationaryTypeFour_pos {lam h₄ : ℝ} (hlam : lambdaInterval.RealContains lam) (hh₄ : h4Interval.RealContains h₄) (_hstationary : LeanSuffixAnalytic.typeFourFold lam h₄ = 0) : 0 < LeanSuffixAnalytic.typeThreeArea lam (h3Interval.lo : ℝ) - LeanSuffixAnalytic.typeFourArea lam h₄. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeThreeArea_upperFace_minus_stationaryTypeFour_neg

Source hash: 7ae9709ef08dafd6acf18487ff86ef2f78b6f5204c9d53ab45605c720440adb8

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:1551-1574, the source contract is theorem typeThreeArea_upperFace_minus_stationaryTypeFour_neg {lam h₄ : ℝ} (hlam : lambdaInterval.RealContains lam) (hh₄ : h4Interval.RealContains h₄) (_hstationary : LeanSuffixAnalytic.typeFourFold lam h₄ = 0) : LeanSuffixAnalytic.typeThreeArea lam (h3Interval.hi : ℝ) - LeanSuffixAnalytic.typeFourArea lam h₄ < 0. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeThreeFold_cell_enclosure

Source hash: c9da53af261130db3061bf11c0e74ee8196fae038a2277ad282922335d64dc0a

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:1018-1122, the source contract is theorem typeThreeFold_cell_enclosure {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : h3Interval.RealContains h) : typeThreeFoldCellInterval.RealContains (LeanSuffixAnalytic.typeThreeFold lam h). This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeThreeFold_neg_on_slab

Source hash: 708577dd88ded1a5fe8468072b9f381b014a4ba74d97e65eac8e88ea179dff0e

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:1123-1130, the source contract is theorem typeThreeFold_neg_on_slab {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : h3Interval.RealContains h) : LeanSuffixAnalytic.typeThreeFold lam h < 0. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.

Findings: none.

CompactFirstCell.typeThreePerimeter_centered_enclosure

Source hash: 0acb33ec361f95f1dd6ecbaa5f29b078d12d8e006095b8006b41890c9e85efd5

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:477-495, the source contract is `theorem typeThreePerimeter_centered_enclosure {lam h : ℝ} (hlam : lambdaInterval.RealContains lam) (hh : h3Interval.RealContains h) (base slope displacement : QInterval) (hbase : base.RealContains (LeanSuffixAnalytic.typeThreePerimeter lam (h3Center : ℝ))) (hslope : ∀ x, h3Interval.RealContains x → slope.RealContains (LeanSuffixAnalytic.typeThreeFold lam x / x ^ 2)) (hdisplacement : displacement.RealContains (h - (h3Center : ℝ))) : (base.add (slope.mul displacement)).RealContains (LeanSuffixAnalytic.typeThreePerimeter lam h)). This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.`

Findings: none.

CompactFirstCell.typeThreePerimeter_minus_typeFour_neg

Source hash: 131bce34e1b590459900c2653104e4d2af2313f086bde253fa5a38a3822148fb

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactFirstCell.lean:1575-1598, the source contract is `theorem typeThreePerimeter_minus_typeFour_neg {lam h3 h4 : ℝ} (hlam : lambdaInterval.RealContains lam) (hh3 : h3Interval.RealContains h3) (hh4 : h4Interval.RealContains h4) : LeanSuffixAnalytic.typeThreePerimeter lam h3 - LeanSuffixAnalytic.typeFourPerimeter lam h4 < 0. This is the explicit zero-based first compact box. Its exact density and curvature intervals are certificate domains; the candidate-facing conclusion, where stated, applies to every modeled candidate at that density rather than only the stationary root.`

Findings: none.

CompactSecondCell.QInterval

Source hash: 1a69616e446623c478186f47d8f25ae8cb966c1989db56c2e34bb70bf61b9a3c

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:19-21, the source contract is `abbrev QInterval := ScalarSuffixCertificate.QInterval`. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.checkedConditions

Source hash: c9b3f9f7a265a22f128e05df51ad2c73f46eedace1983e930b75a5cccf6ea3ba

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:639-659, the source contract is theorem checkedConditions : CompactCellAssembly.Conditions lambdaInterval h3Interval h4Interval where lambda_one_lt. The theorem supplies every field of CompactCellAssembly.Conditions for the concrete second box with no certificate premise. In particular the perimeter sign is on the whole h3/h4 slab; the equal-area face hypotheses remain restricted to stationary h4 as specified by the generic record.

Findings: none.

CompactSecondCell.compactCandidate_not_isWeightedPerimeterMinimizer

Source hash: bf9eb3ed65e181f0e0de3991de3f82fd1169d1b5acecfb53b060a331f19ad1ae

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:672-681, the source contract is theorem compactCandidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlam : lambdaInterval.RealContains lam) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.firstTwoBoxes_candidate_not_isWeightedPerimeterMinimizer

Source hash: 1f54334812872c86ebe3794a932ab059b0b9f0ae3518130a266bed3d7f94e321

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:682-702, the source contract is theorem firstTwoBoxes_candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (33 / 32 : ℝ) ≤ lam) (hupper : lam ≤ (118329 / 114688 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.h3Center

Source hash: 571f803f2356f6d9d1aef243b2c46e837820a4eb7a7aa3d5322e7d084660f89b

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:39-40, the source contract is def h3Center : ℚ := 8301881813483 / 8796093022208. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.h3Interval

Source hash: bee35652f5b82397fd6b36e73591556e7291b9c519843c28a1442d7cd4f919a5

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:31-35, the source contract is `def h3Interval : QInterval := ⟨265075342400937 / 281474976710656, 266245093661975 / 281474976710656, by norm_num⟩`. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.h4Center

Source hash: 666f145a9e19df37e5fddc56423955a5afdef7d5d8965955b503afe70df65af5

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:36-38, the source contract is `def h4Center : ℚ := 275560807394631 / 281474976710656`. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.h4Interval

Source hash: af804f12d01723b2d7a86305f48af3d27bbb99fca40d12171d80e480fb2a6639

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:26-30, the source contract is `def h4Interval : QInterval := ⟨275542293618597 / 281474976710656, 275579321170665 / 281474976710656, by norm_num⟩`. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

CompactSecondCell.lambdaInterval

Source hash: 69be0395e97b77b018ac7b8ae43d4c795e095c167e0e517d474ad3537fb6b8ec

Relation: equivalent

Equivalent in the declared constructor/section context. At CompactSecondCell.lean:22-25, the source contract is `def lambdaInterval : QInterval := ⟨236601 / 229376, 118329 / 114688, by norm_num⟩`. This is the explicit second compact box and its splice to the first. Closed interval membership retains the common rational endpoint; local root-slab conditions do not become extra hypotheses on an arbitrary candidate.

Findings: none.

EuclideanPlane

Source hash: dfa2db26da624a2b5333b886cff8c9cd3151218ea8aa3e3cb39c78359b32470e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:34-37, the source contract is `abbrev EuclideanPlane := WithLp 2 (ℝ × ℝ)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

F

Source hash: 5b2bb966fff4872f29347307b668a0e2d316ac47616a7d86bd1b9925aa189c42

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:43-45, the source contract is `noncomputable def F (θ : ℝ) : ℝ := 3 * θ - 3 * (sin θ * cos θ) - π`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

F_continuous

Source hash: 944215df53ab5a1f8d80d30f68581f0d2605e8a8c87867772cbaa9697833f26f

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:57-61, the source contract is `lemma F_continuous : Continuous F`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

F_pi_div_two

Source hash: 1bd957238f9de023b62b37f505e2ceebe0cf17d7fa9db27be00533881b095aac

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:76-81, the source contract is `lemma F_pi_div_two : F (π / 2) = π / 2`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

F_strictMonoOn

Source hash: 159fcb80f1fa4e926941eda01dd71a676dd66fef614b9045e53e04469f3d4efe

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:62-71, the source contract is `lemma F_strictMonoOn : StrictMonoOn F (Icc 0 π)`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$,

including the major-cap branch.

Findings: none.

F_zero

Source hash: 0054c59a03a184000fc9f7673875bd1fa77b8d5a1989b4d2de26bcf5f101a9b0

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:72-75, the source contract is `lemma F_zero :  $F\ 0 = -\pi$ .  $F(\theta) = 3\theta - 3\sin(\theta) \cdot \cos(\theta) - \pi$  has its unique relevant root in  $(0, \pi/2)$ . The chosen normalized area is positive, and the supplementary angle  $\pi - \theta$  remains in  $(0, \pi)$ , including the major-cap branch.`

Findings: none.

FinitePerimeterRegion

Source hash: 2efb7176d17b946809f4cd07b78237ddd03cd4197c405c38d08e97342106a1a0

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:74-82, the source contract is `structure FinitePerimeterRegion (lam :  $\mathbb{R}$ ) where carrier : Set PlanePoint measurable_carrier : MeasurableSet carrier finite_weighted_area : IntegrableOn (StripDensity lam) carrier finite_weighted_perimeter : Integrable (StripDensity lam) (FrontierMeasure carrier)`. The record requires a measurable carrier, area-density integrability, and density integrability against its canonical complete-frontier measure. It is not a theorem characterizing all BV finite-perimeter sets; the author explicitly limits the name to this regular-carrier model.

Findings: none.

FinitePerimeterRegion.weightedArea

Source hash: 3892310b68bd795c4ddff8f1d276f58be0efcb96c1e8c4fad96ec7d73deea20a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:83-85, the source contract is `def weightedArea {lam :  $\mathbb{R}$ } (r : FinitePerimeterRegion lam) :  $\mathbb{R}$  := WeightedArea lam r.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly

Source hash: 1ac9185bddaf1e6a0074fb39d1e29417262cc431ccce861d3c65ec2e5216fdb1

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2491-2498, the source contract is `structure FourArcAssembly where core : StripCore outerAngle :  $\mathbb{R}$  outerAngle_pos :  $0 < \text{outerAngle}$  outerAngle_lt_pi_div_two :  $\text{outerAngle} < \pi / 2$` . The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.boundaryMeasure

Source hash: a07e8d5b98aab97a737c50d9ee9eae871142e1adb2b58a3132be8c4b6a093cce

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2525-2529, the source contract is `def boundaryMeasure (a : FourArcAssembly) : Measure PlanePoint := leftCoreBoundaryMeasure a.core + rightCoreBoundaryMeasure a.core + capBoundaryMeasure a.upperCap + capBoundaryMeasure a.lowerCap`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.boundaryTrace

Source hash: d8944f364e38eaa5c7b77ce182d417bec91ac425efc229739d886be00ad658c6

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2625-2630, the source contract is `def boundaryTrace (a : FourArcAssembly) : Set PlanePoint := StripCore.leftArcTrace a.core ∪ (StripCore.rightArcTrace a.core ∪ (OneSidedCircularCap.arcTrace a.upperCap ∪ OneSidedCircularCap.arcTrace a.lowerCap))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.boundaryTrace_subset_frontier_carrier

Source hash: 160385ee99b171abc63b7cc7cb5ade687af8d5249c1ab26a5150be49b8c4654b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3244-3251, the source contract is `theorem boundaryTrace_subset_frontier_carrier (a : FourArcAssembly) : a.boundaryTrace ⊆ frontier a.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.carrier

Source hash: a55616796db22264c4fe11d48e20ab173ecb661c3bdced26752617ea90d1f816

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2519-2524, the source contract is `def carrier (a : FourArcAssembly) : Set PlanePoint := a.core.carrier ∪ a.upperCap.carrier ∪ a.lowerCap.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.frontier_carrier

Source hash: 7da383d241cb3cc30fc9da784ef7200d9bb87d82e25006745acf6f7f7d86a78f

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3252-3259, the source contract is theorem `frontier_carrier (a : FourArcAssembly) : frontier a.carrier = a.boundaryTrace`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.frontier_carrier_subset_boundaryTrace

Source hash: 0d72e7b21e3528564863313aa2366a2e4806e01c00a4772872911d424abf3cc5

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2795-2819, the source contract is theorem `frontier_carrier_subset_boundaryTrace (a : FourArcAssembly) : frontier a.carrier  $\subseteq$  a.boundaryTrace`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.lowerCap

Source hash: a60a9393b363872f08f12dbab41380b51afefa746b9504c5841f28d389091bfe

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2509-2518, the source contract is `def lowerCap (a : FourArcAssembly) : OneSidedCircularCap where chord := a.core.chord theta := a.outerAngle midpointX := 0 baseY := -1 side := .lower chord_pos := a.core.chord_pos theta_pos := a.outerAngle_pos theta_lt_pi := lt_trans a.outerAngle_lt_pi_div_two (by linarith [Real.pi_pos])`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.upperCap

Source hash: 3f084f0dbec6bee8992e556fe2ca304aae5eb8981815636bf15454a3b8e613f9

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2499-2508, the source contract is `def upperCap (a : FourArcAssembly) : OneSidedCircularCap where chord := a.core.chord theta := a.outerAngle midpointX := 0 baseY := 1 side := .upper chord_pos := a.core.chord_pos theta_pos := a.outerAngle_pos theta_lt_pi := lt_trans a.outerAngle_lt_pi_div_two (by`

`linarith [Real.pi_pos]`). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.weightedArea

Source hash: ad43970e4ac79b935baed41701092f4d901053e35bb88583dcbd5a9003db13a3

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2541-2545, the source contract is `def weightedArea (lam : ℝ) (a : FourArcAssembly) : ℝ := WeightedArea lam a.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.weightedArea_formula

Source hash: 839e3c1d7a83b5801ac118ef77feb57b415f9e5a71dfa154b8987ba3b91118a0

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2554-2602, the source contract is `theorem weightedArea_formula (lam : ℝ) (a : FourArcAssembly) : a.weightedArea lam = a.core.euclideanArea + lam * (a.upperCap.euclideanArea + a.lowerCap.euclideanArea)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.weightedPerimeter

Source hash: df8ef218db7076c4681d4506265ae824cfa4b1a2ec872c320aeb4f283b25493b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2546-2548, the source contract is `def weightedPerimeter (lam : ℝ) (a : FourArcAssembly) : ℝ := WeightedPerimeter lam a.boundaryMeasure`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcAssembly.weightedPerimeter_formula

Source hash: 43ff4cddb6774032399c0e7d7f5767a18500986a522e0ace1e470e2035290acd

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2603-2624, the source contract is `theorem weightedPerimeter_formula (lam : ℝ) (a : FourArcAssembly) : a.weightedPerimeter lam = a.core.boundaryArcLength + lam * (a.upperCap.arcLength + a.lowerCap.arcLength)`. The

coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

FourArcCandidate

Source hash: 24dc0ff22e0918bdb2c7f16fab0da52bc08326d88e5cb8e1d2e8a8f7ec2bee62

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:23-35, the source contract is `structure FourArcCandidate (lam : ℝ) where h : ℝ alpha : ℝ h_pos : 0 < h h_le_one : h ≤ 1 alpha_pos : 0 < alpha alpha_lt_pi_div_two : alpha < π / 2`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.CapReplacement

Source hash: e1ad18311ba23ee8f10b78f8ffe05ca7f5160069b4c67ed9620f490f9f00dfa9

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:51-58, the source contract is `def CapReplacement : ReplacementAssembly where core := candidate.stripCore replacementAngle := candidate.replacementAngle replacementAngle_pos := candidate.replacementAngle_pos replacementAngle_lt_pi := candidate.replacementAngle_lt_pi`. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.IsWeightedPerimeterMinimizer

Source hash: f04b5c158c1a510cd5a8af38df8f99424aeed5e376f986e1d437f95e6a4cbe17

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:90-94, the source contract is `def IsWeightedPerimeterMinimizer : Prop := ∀ competitor : AdmissibleCompetitor lam, competitor.WeightedArea = candidate.WeightedArea → candidate.WeightedPerimeter ≤ competitor.WeightedPerimeter`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.SatisfiesCMVTypeIVHypotheses

Source hash: c09b9c7180b6131040333c0b3c2623f3c89af7f54fe8ab0b9f6a41be873ca0dc

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:84-89, the source contract is `structure SatisfiesCMVTypeIVHypotheses : Prop where density_jump : 1 < lam
snell_incidence : lam * cos candidate.alpha = candidate.h`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.WeightedArea

Source hash: 8256dd516e32d853aaed6a85d09fdb0b62411bd0bf67f1df7ffd123e47baed99

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:76-78, the source contract is `def WeightedArea : ℝ := candidate.region.WeightedArea`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.WeightedPerimeter

Source hash: d649676f2fd717389a6e9a17a385fe854bc399e17ecfdad24bd8a7bcc9a36521

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:79-83, the source contract is `def WeightedPerimeter : ℝ := candidate.region.WeightedPerimeter`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.alpha_eq_arccos

Source hash: 46219b8ad0fc6b679968f8972242a6ad511174ce969ce84539a4054206de6595

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:276-292, the source contract is `theorem alpha_eq_arccos (h : candidate.SatisfiesCMVTypeIVHypotheses) : candidate.alpha = arccos (candidate.h / lam)`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.assembly

Source hash: ca67fddb2d167afab63544438387902bb645349ebfde9d83c7a2719a2277c941

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:66-72, the source contract is `def assembly : FourArcAssembly` where `core := candidate.stripCore` `outerAngle := candidate.alpha` `outerAngle_pos := candidate.alpha_pos` `outerAngle_lt_pi_div_two := candidate.alpha_lt_pi_div_two`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.beta

Source hash: f3520601e601026585d5024a59c3bee3be514d265897b65e66919a9d3ea7fd1e

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:39-41, the source contract is `def beta : ℝ := π / 2 - arcsin candidate.h`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.candidate_area_formula

Source hash: b6d944ecc5d4bab77cadf263520f194142a9d6855088a9b98e5873be144ab8cc

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:410-430, the source contract is `theorem candidate_area_formula (h : candidate.SatisfiesCMVTypeIVHypotheses) :`
`candidate.WeightedArea = 4 * sin candidate.alpha / candidate.h + 2 * (lam * candidate.alpha - sin candidate.alpha * cos candidate.beta) / candidate.h ^ 2 + 2 * (arcsin candidate.h - candidate.h * √(1 - candidate.h ^ 2)) / candidate.h ^ 2`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.candidate_perimeter_cap_components

Source hash: 5aef3e70645e6bc7cb387847d94c3bc5e794e60aba832cf7143171cce0f52718

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:98-105, the source contract is `theorem candidate_perimeter_cap_components : candidate.WeightedPerimeter = candidate.stripCore.boundaryArcLength + 2 * lam * candidate.capChord * arc candidate.capX`. The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.candidate_perimeter_formula

Source hash: b4949bdc7019f429151f2faf734db8466461943a33be006f4c104caebb493fe0

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:391-409, the source contract is `theorem candidate_perimeter_formula (_h : candidate.SatisfiesCMVTypeIVHypotheses) : candidate.WeightedPerimeter = 4 * lam * candidate.alpha / candidate.h + 4 * arcsin candidate.h / candidate.h`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capArea

Source hash: eb6c25ab8d5e1e4d2ce589b98fc3af2d04e58bd682ad273d1b07834426ee23d0

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:45-49, the source contract is `def capArea : ℝ := (candidate.alpha - sin candidate.alpha * cos candidate.alpha) / candidate.h ^ 2`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capArea_eq_chord_sq_mul_area

Source hash: 0e9545ae1662d69d5550dc9264e42c4c5b03b78571f6473a85822aca53c6073

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:104-109, the source contract is `theorem capArea_eq_chord_sq_mul_area : candidate.capArea = candidate.capChord ^ 2 * area candidate.alpha`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capChord

Source hash: c09aa141e8bbabbe7279bce2473dd47aa931f875442fe1fc79d759685536fb1e

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:42-44, the source contract is `def capChord : ℝ := 2 * sin candidate.alpha / candidate.h`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capChord_ne

Source hash: 58b855530ffa4fa2baf00c3fc6982a6cafa75c224fae7347ad43ffd8cbb2b782

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:102-103, the source contract is `theorem capChord_ne : candidate.capChord ≠ 0`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capChord_pos

Source hash: f52b3f7ad66a860b3cbec5075902f43e4ec10a9d92c573767d420b9a7f6a16c6

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:99-101, the source contract is `theorem capChord_pos : 0 < candidate.capChord`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capReplacementRegion

Source hash: c2d6906ef5b69db6b314e7ac4d82d54432092c90cdf06b81f1a87bed1432f7c3

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:69-78, the source contract is `def capReplacementRegion : AdmissibleCompetitor lam := .replacement`
`candidate.CapReplacement @[simp] theorem capReplacement_core :`
`candidate.CapReplacement.core = candidate.stripCore := rfl @[simp] theorem`
`capReplacement_angle : candidate.CapReplacement.replacementAngle =`
`candidate.replacementAngle := rfl`. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.capX

Source hash: a20fa39233b648b55a696be0bec2aefc6c5d3bf3467f9a8163b812924a00e64

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:50-52, the source contract is `def capX : ℝ := candidate.capArea / candidate.capChord ^ 2`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capX_eq_area

Source hash: ef35b1cb5934a4532e1255f444a69fb5e100682afde4a7e8de56f9cb84bae537

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:116-119, the source contract is **theorem capX_eq_area** : candidate.capX = area candidate.alpha. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.capX_pos

Source hash: be506fe40b7a525a5e22df3e7a8d9312d3e56af70a1f9488dad065e7eeb4fa1f

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:120-126, the source contract is **theorem capX_pos** : $0 < \text{candidate.capX}$. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.cap_replacement_perimeter_difference

Source hash: cef0cb1c446892a6b4545fed077efd78b2014155506ae9d18d92702507d05712

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:124-133, the source contract is **theorem cap_replacement_perimeter_difference** : candidate.capReplacementRegion.WeightedPerimeter - candidate.WeightedPerimeter = candidate.capChord * (1 - lam * g candidate.capX). The exact identity is $P(\text{replacement}) - P(\text{candidate}) = c(1 - \lambda \cos(\alpha))$, which is division-free and valid for arbitrary real λ . The main proof's opposite-sign saving $\lambda \cos(\alpha) - 1/\lambda$ is used only when $\lambda > 1$. Thus there is no illicit division at $\lambda = 0$ or reversal of the improvement sign.

Findings: none.

FourArcCandidate.cap_replacement_preserves_area

Source hash: 40dbfacc980c0caa8be7bb02e41318ad70f3b454256d042f4cd60d664125314b

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:90-97, the source contract is **theorem cap_replacement_preserves_area** : candidate.capReplacementRegion.WeightedArea = candidate.WeightedArea. The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.cap_replacement_strictly_improves

Source hash: c953f49d8456f94f9248f042331d7c54782ee3366628a4a57c9db7cc82ecdd4a

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:157-176, the source contract is `theorem cap_replacement_strictly_improves (h $\lambda$  : (1.2581840884 :  $\mathbb{R}$ )  $\leq$   $\lambda$ ) : candidate.capReplacementRegion.WeightedPerimeter < candidate.WeightedPerimeter`. The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.density_pos_of_campaign_threshold

Source hash: cc9e8590326c21e2d3058fbcef78da468bb9c4210d5a293f7f28898b6488b124

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:150-156, the source contract is `theorem density_pos_of_campaign_threshold (h $\lambda$  : (1.2581840884 :  $\mathbb{R}$ )  $\leq$   $\lambda$ ) :  $0 < \lambda$` . The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.exposed_lower_chord_length

Source hash: f04f252743e50cee8e8705384217c84f1c347cf4e87d10a166f74cf59b5b7fd6

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:95-99, the source contract is `theorem exposed_lower_chord_length : candidate.CapReplacement.exposedLowerChord.euclideanLength = candidate.capChord`. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.lowerCap_euclideanArea

Source hash: 14a860cf99daaf759c3fa7838aed86edfb2879de1a5a3f6f7d84f9996c7da155

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:159-166, the source contract is `theorem lowerCap_euclideanArea : candidate.assembly.lowerCap.euclideanArea = candidate.capArea`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.not_isWeightedPerimeterMinimizer_of_typeThree

Source hash: 6807359794d5b2629c85ed715f93a0b06f1e0f5dc3b504c99a54401226562f55

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3101-3118, the source contract is theorem not_isWeightedPerimeterMinimizer_of_typeThree (a : TypeThreeAssembly lam) (realization : TypeThreeAssembly.ModeledCompetitorRealization a) (equalArea : a.scalarWeightedArea = candidate.WeightedArea) (strictPerimeter : a.scalarWeightedPerimeter < candidate.WeightedPerimeter) : ¬ candidate.IsWeightedPerimeterMinimizer. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

FourArcCandidate.original_arc_eq_ell

Source hash: b932d1e6553b24d46c1b0161df0040ebd64f34413f6c447a3da4b2b0f95c234a

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:27-33, the source contract is theorem original_arc_eq_ell : arc candidate.capX = ell candidate.alpha. The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.region

Source hash: 8ab9f84c197a1a6dbe90236469e9ff5cc2696f04e42133db44cdf34546558b75

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:73-75, the source contract is def region : AdmissibleCompetitor lam := .fourArc candidate.assembly. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.replacementAngle

Source hash: e82e3e1d3191a88d1c944b28e4f02eaa8d736356029ed2fa74ba08f57ef344d7

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:27-28, the source contract is def replacementAngle : $\mathbb{R}$:= 00f (2 * candidate.capX). The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacementAngle_area

Source hash: c942d60f253b362e97e724222a6abdc2af0e9c796571a40aeeb1a4c26cb3d0d7

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:39-43, the source contract is **theorem replacementAngle_area** : $\text{area candidate.replacementAngle} = 2 * \text{candidate.capX}$. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacementAngle_lt_pi

Source hash: 5c7cddce688eec8cbb1b3197a2fe6a78bfdc4de573e3f9b6cf8479b677800d6f

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:35-38, the source contract is **theorem replacementAngle_lt_pi** : $\text{candidate.replacementAngle} < \pi$. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacementAngle_mem

Source hash: 01b8487a31a244058e7174a351228b05206d647acdd690d9902ffea504fec4b6

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:29-31, the source contract is **theorem replacementAngle_mem** : $\text{candidate.replacementAngle} \in \text{Ioo } 0 \ \pi$. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacementAngle_pos

Source hash: 9919f7c2d00e4667c39a44e5eadcdd729471eff6713284062fbdbad13333572

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:32-34, the source contract is **theorem replacementAngle_pos** : $0 < \text{candidate.replacementAngle}$. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacement_cap_arc_length

Source hash: 15d3b8bc212201f87fa01ba9f302a81f8ee2f8ead5c18096e4f073110885dea8

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:89-94, the source contract is **theorem replacement_cap_arc_length** : $\text{candidate.CapReplacement.upperCap.arcLength} = \text{candidate.capChord} * \text{arc}(2 * \text{candidate.capX})$. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacement_cap_area

Source hash: 5517f888e5fc2a04dba3edea18ff03637b20f17abc45bd337950afb694709ec8

Relation: equivalent

Equivalent in the declared constructor/section context. At CapReplacement.lean:79-88, the source contract is **theorem replacement_cap_area** : $\text{candidate.CapReplacement.upperCap.euclideanArea} = 2 * \text{candidate.capArea}$. The replacement retains the strip core, doubles the upper cap area, and exposes the lower chord. Its new half-angle is in $(0, \pi)$, so a major cap is not silently discarded; the statement uses the candidate constructor inequalities.

Findings: none.

FourArcCandidate.replacement_perimeter_cap_components

Source hash: ad5283b2c2158e01ae2eb9db43ef15a8cb068a20feefa5923311c31d1fd30f36

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:106-123, the source contract is **theorem replacement_perimeter_cap_components** : $\text{candidate.capReplacementRegion.WeightedPerimeter} = \text{candidate.stripCore.boundaryArcLength} + \text{lam} * \text{candidate.capChord} * \text{arc}(2 * \text{candidate.capX}) + \text{candidate.capChord}$. The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.replacement_weightedArea_components

Source hash: f60c2f169a3606748cb65977bfe0e42bbb44dc76480dad9bc42ac52ed335dcbb

Relation: equivalent

Equivalent in the declared constructor/section context. At GeometricBridge.lean:79-89, the source contract is **theorem replacement_weightedArea_components** : $\text{candidate.capReplacementRegion.WeightedArea} = \text{candidate.stripCore.euclideanArea} + \text{lam} * (2 * \text{candidate.capArea})$. The bridge uses the actual cap area, chord length and canonical perimeter formulas. The geometric improvement has a strict sign and equal weighted area; an arbitrary scalar record is not accepted as a competitor.

Findings: none.

FourArcCandidate.side_caps_area_formula

Source hash: a639bd8c9eef40f2c4f87fdb62b671cf7db0ce41013e7d923721f6f472bc8cf1

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:351-363, the source contract is `theorem side_caps_area_formula : 8 * area candidate.stripCore.sideAngle = 2 * (arcsin candidate.h - candidate.h *  $\sqrt{1 - \text{candidate.h}^2}$ ) / candidate.h ^ 2`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.sin_alpha_pos

Source hash: b92e14d3743fcaa1d6c447d96c6334ba3f5f129152aa26ff47b97d6fd8de2e28

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:95-98, the source contract is `theorem sin_alpha_pos : 0 < sin candidate.alpha`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.stripCore

Source hash: 66aff9e1eb934bbbeeb8dd09a1a68a4c23aef3ee09735d3ba9bc12eadb4664ee

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:53-65, the source contract is `def stripCore : StripCore where chord := candidate.capChord curvature := candidate.h chord_pos := by exact div_pos (mul_pos (by norm_num) (sin_pos_of_pos_of_lt_pi candidate.alpha_pos (lt_trans candidate.alpha_lt_pi_div_two (by linarith [Real.pi_pos]))) candidate.h_pos curvature_pos := candidate.h_pos curvature_le_one := candidate.h_le_one`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.upperCap_euclideanArea

Source hash: 63eb34459de3e09e8c932883edd9446c50efc2cc1a148ef60c5c77cbd56d7496

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:153-158, the source contract is `theorem upperCap_euclideanArea : candidate.assembly.upperCap.euclideanArea = candidate.capArea`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.weightedArea_components

Source hash: 654554fba4e2f4847eae4f853550061afa29cf158da02de52109cffffacefd702e

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:364-373, the source contract is `theorem weightedArea_components : candidate.WeightedArea = candidate.stripCore.euclideanArea + 2 * lam * candidate.capArea`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FourArcCandidate.weightedPerimeter_components

Source hash: d02953cec6a51d470b1b940fa2318bbd5c86354d0e420d179322c86356c9434a

Relation: equivalent

Equivalent in the declared constructor/section context. At FourArcCandidate.lean:374-390, the source contract is `theorem weightedPerimeter_components : candidate.WeightedPerimeter = candidate.stripCore.boundaryArcLength + 2 * lam * candidate.capChord * ell candidate.alpha`. The candidate constructor has $0 < h \leq 1$ and $0 < \alpha < \pi/2$. CMV incidence additionally means $\lambda > 1$ and $\lambda \cos(\alpha) = h$. These conditions are not a hidden minimality or strict-improvement assumption.

Findings: none.

FrontierMeasure

Source hash: 7323bb80c0a565daf410d27d199e2143ee33a0ecca0460c52504f2e7e50b48e2

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:61-66, the source contract is `def FrontierMeasure (carrier : Set PlanePoint) : Measure PlanePoint := Measure.map planeEuclideanHomeomorph.symm ((μH[1] : Measure EuclideanPlane).restrict (frontier (planeEuclideanHomeomorph ' ' carrier)))`. FrontierMeasure is the pullback, through the coordinate-preserving Euclidean homeomorphism, of H^1 restricted to the complete topological frontier. Neither the product sup norm nor an independently chosen boundary measure is used.

Findings: none.

HorizontalSegment

Source hash: e7bebc91d2e392d80b21e5e8b5487591a5477fe10be16627e5b6180dd120506

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:392-399, the source contract is `structure HorizontalSegment where chord : ℝ midpointX : ℝ baseY : ℝ chord_pos : 0 < chord`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

HorizontalSegment.carrier

Source hash: 094b96b3006204081e6d336250f56e63f6b337cc71255210b9bae8ae3b894041

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:406-408, the source contract is `def carrier (s : HorizontalSegment) : Set PlanePoint := {p | p.2 = s.baseY ^ |p.1 - s.midpointX| ≤ s.chord / 2}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

HorizontalSegment.euclideanLength

Source hash: c965c7e2ff7089171e9318c88e3733eb9629da0051af6a9aad082bf1ad5a5aef

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:409-415, the source contract is `def euclideanLength (s : HorizontalSegment) : ℝ := s.chord`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

HorizontalSegment.leftEndpoint

Source hash: d64f9e0ea48cfbbcb1b06bed874068659f0011957eca2b8226c738430113a0615

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:400-402, the source contract is `def leftEndpoint (s : HorizontalSegment) : PlanePoint := (s.midpointX - s.chord / 2, s.baseY)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

HorizontalSegment.rightEndpoint

Source hash: a14791633639531ab0cb8801c8e0e16383dd7712b7d5730aa7f0d9178c83e5da

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:403-405, the source contract is `def rightEndpoint (s : HorizontalSegment) : PlanePoint := (s.midpointX + s.chord / 2, s.baseY)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

LeanSuffixAnalytic.DescendingEqualAreaEnvelope

Source hash: ee2db888cf8e5e37251e350d2425fb86808e5c1f7d808208572e20c0e3ea40a9

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:279-300, the source contract is
structure DescendingEqualAreaEnvelope (lam : $\mathbb{R}$) where fold : $\mathbb{R}$ root : $\mathbb{R} \rightarrow \mathbb{R}$
density_jump : $1 < \text{lam}$ fold_pos : $0 < \text{fold}$ fold_lt_one : $\text{fold} < 1$ fold_stationary :
typeFourFold lam fold = 0 root_pos : $\forall \{h_4 : \mathbb{R}\}, 0 < h_4 \rightarrow h_4 \leq 1 \rightarrow 0 < \text{root } h_4$
root_lt_one : $\forall \{h_4 : \mathbb{R}\}, 0 < h_4 \rightarrow h_4 \leq 1 \rightarrow \text{root } h_4 < 1$ equalArea : $\forall \{h_4 : \mathbb{R}\}, 0 < h_4$
 $\rightarrow h_4 \leq 1 \rightarrow \text{typeThreeArea lam (root } h_4) = \text{typeFourArea lam } h_4$ root_fold_neg : $\forall \{h_4 : \mathbb{R}\}, 0 < h_4 \rightarrow h_4 \leq 1 \rightarrow \text{typeThreeFold lam (root } h_4) < 0$. Equal area and the negative-fold branch are
distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The
main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.FoldGapNegative

Source hash: e0a51358d0910c8fb2dd4ffa9d95125b8210328ea93cd0771ca9e9c1138b2ced

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:577-581, the source contract is **def**
FoldGapNegative {lam : $\mathbb{R}$ } (env : DescendingEqualAreaEnvelope lam) : Prop :=
equalAreaPerimeterGap lam env.root env.fold < 0. Equal area and the negative-fold branch are distinct
requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main
proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.InRetainedSuffix

Source hash: 18acd275369860c6ce3970cef1307119f694180bde02eaf9f16412e8cd68698

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1159-1161, the source contract is **def**
InRetainedSuffix (lam : $\mathbb{R}$) : Prop := (51 / 50 : $\mathbb{R}$) $\leq$ lam $\wedge$ lam $\leq$ 9 / 7. The profiles and fold
numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint
statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.StationaryEqualAreaPair

Source hash: 865710967f267b9ec6707c5d925cd799ff4862bc66ec64cf0a22afce00862eef

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1177-1188, the source contract is
structure StationaryEqualAreaPair (lam : $\mathbb{R}$) where h₃ : $\mathbb{R}$ h₄ : $\mathbb{R}$ h₃_gt_half : $1 / 2 <$
h₃ h₃_lt_one : h₃ < 1 h₄_pos : $0 < h_4$ h₄_lt_one : h₄ < 1 equalArea : typeThreeArea lam
h₃ = typeFourArea lam h₄ stationary : typeFourFold lam h₄ = 0. StationaryEqualAreaPair stores
regular h₄, h₃ strictly above one half, equal area, and K₄(h₄)=0. It does not itself store K₃(h₃)<0 or a perimeter inequality.
Those are separate inputs to the envelope/improvement theorem, exactly as the surrounding explanation distinguishes.

Findings: none.

LeanSuffixAnalytic.allCurvature_typeThreeImprovement_of_envelope

Source hash: c4f7d83c3e3a4e728eb9be863cb4e20b5d04c133005f9a44bb6548808441f1ae

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:645-658, the source contract is theorem allCurvature_typeThreeImprovement_of_envelope {lam h₄ : ℝ} (env : DescendingEqualAreaEnvelope lam) (hfoldGap : FoldGapNegative env) (hh₄ : 0 < h₄) (hh_{4_one} : h₄ ≤ 1) : ∃ h₃ : ℝ, 0 < h₃ ∧ h₃ < 1 ∧ typeThreeArea lam h₃ = typeFourArea lam h₄ ∧ typeThreePerimeter lam h₃ < typeFourPerimeter lam h₄. Equal area and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.centered_meanValue_exists

Source hash: 9e16c64246998ba49ad719b4074913f0337c8612efed42e98f32935fcd55a3e5

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:93-129, the source contract is theorem centered_meanValue_exists (f f' : ℝ → ℝ) {c h : ℝ} (hderiv : ∀ x ∈ Set.uIcc c h, HasDerivAt f (f' x) x) : ∃ ξ ∈ Set.uIcc c h, f h = f c + f' ξ * (h - c). The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap

Source hash: e8643a1549a791c31f1d0a91f7cb9b302d6832ce9c0d1d8e354d9c3e22351fcb

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:34-39, the source contract is def equalAreaPerimeterGap (lam : ℝ) (root : ℝ → ℝ) (h₄ : ℝ) : ℝ := typeThreePerimeter lam (root h₄) - typeFourPerimeter lam h₄. Equal area and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap_hasDerivAt

Source hash: 42f42f4f683c9b8ff6e876958c9ae3b87d6bfd1ebf63072c7e612401d7b5ee93

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:184-225, the source contract is theorem equalAreaPerimeterGap_hasDerivAt {lam h : ℝ} {root : ℝ → ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) (hroot_pos : 0 < root h) (hroot_lt_one : root h < 1) (hroot_differentiable : DifferentiableAt ℝ root h) (hequalArea : ∀ {x : ℝ}, 0 < x → x ≤ 1 → typeThreeArea lam (root x) = typeFourArea lam x) : HasDerivAt (equalAreaPerimeterGap lam root) ((root h - h) * deriv (typeFourArea lam) h) h. Equal area

and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap_le_fold

Source hash: 0bde810fe503972e30290f54b6517ae65288b7e84131b161bebac58ed0796e33

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:632-644, the source contract is theorem equalAreaPerimeterGap_le_fold {lam : $\mathbb{R}$ } (env : DescendingEqualAreaEnvelope lam) {h₄ : $\mathbb{R}$ } (hh₄ : 0 < h₄) (hh₄_one : h₄ ≤ 1) : equalAreaPerimeterGap lam env.root h₄ ≤ equalAreaPerimeterGap lam env.root env.fold. Equal area and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.equalAreaPerimeterGap_lt_fold

Source hash: 1a0b26ad0e4af7fdd933118930486f792fc517471c75bd1495b190e7f97f3420

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:582-631, the source contract is theorem equalAreaPerimeterGap_lt_fold {lam : $\mathbb{R}$ } (env : DescendingEqualAreaEnvelope lam) {h₄ : $\mathbb{R}$ } (hh₄ : 0 < h₄) (hh₄_one : h₄ ≤ 1) (hne : h₄ ≠ env.fold) : equalAreaPerimeterGap lam env.root h₄ < equalAreaPerimeterGap lam env.root env.fold. Equal area and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.equalAreaRoot_continuousOn_of_compact_unique

Source hash: 58b73f54ea82a27e264112e64332402373c869c542bc7277160adca9dcc99fdd

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:71-107, the source contract is theorem equalAreaRoot_continuousOn_of_compact_unique {lam : $\mathbb{R}$ } {s K : Set $\mathbb{R}$ } {root : $\mathbb{R}$ → $\mathbb{R}$ } (hlam : 1 < lam) (hK : IsCompact K) (hs_domain : s ⊆ Ioc 0 1) (hK_regular : K ⊆ Ioo 0 1) (hroot_mem : MapsTo root s K) (hequalArea : ∀ h ∈ s, typeThreeArea lam (root h) = typeFourArea lam h) (hunique : ∀ h ∈ s, ∀ y ∈ K, typeThreeArea lam y = typeFourArea lam h → y = root h) : ContinuousOn root s. The theorem allows an arbitrary compact K contained in (0,1) and an arbitrary S contained in (0,1], not necessarily intervals or connected sets. A selected root is trapped in K, satisfies equal area, and is unique within K. Closed-graph compactness gives continuity on S; no global uniqueness outside K is inferred.

Findings: none.

LeanSuffixAnalytic.equalAreaRoot_hasDerivAt

Source hash: 4240cf112f1c537233001e48d7956d1d7de811e741deeb8b94332f375fdbb1f0

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:165-183, the source contract is theorem equalAreaRoot_hasDerivAt {lam h : ℝ} {root : ℝ → ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) (hroot_pos : 0 < root h) (hroot_lt_one : root h < 1) (hroot_continuous : ContinuousAt root h) (hareaThree_ne : deriv (typeThreeArea lam) (root h) ≠ 0) (hequalArea : ∀ {x : ℝ}, 0 < x → x ≤ 1 → typeThreeArea lam (root x) = typeFourArea lam x) : HasDerivAt root (deriv (typeFourArea lam) h / deriv (typeThreeArea lam) (root h)) h. The implicit derivative assertion assumes interior h₄, a regular selected h₃, continuity at that point, nonvanishing A₃ derivative, and the equal-area identity on the modeled domain. The denominator need only be nonzero here, not negative; the negative-fold application proves that stronger sign separately.

Findings: none.

LeanSuffixAnalytic.equalArea_gap_identity

Source hash: 7a449a0d73201ece195743b096dbde5032c8598ab1748985129132f301a8ba9d

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1208-1220, the source contract is theorem equalArea_gap_identity {lam h₃ h₄ : ℝ} (hh₃ : h₃ ≠ 0) (hh₄ : h₄ ≠ 0) (harea : typeThreeArea lam h₃ = typeFourArea lam h₄) : typeThreePerimeter lam h₃ - typeFourPerimeter lam h₄ = 4 * (((h₃ - h₄) / h₄ ^ 2) * typeFourAngle lam h₄ + h₃ / h₄ * typeFourDelta lam h₄ - typeThreeDelta lam h₃). The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourAngle

Source hash: ed3b0f3804a218b0cd68be6333d68c33832835819c8ca12dd83eb68bae07439d

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:241-278, the source contract is theorem hasDerivAt_typeFourAngle {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeFourAngle lam) (1 / sqrt (1 - h ^ 2) - lam / sqrt (lam ^ 2 - h ^ 2)) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourArea

Source hash: 406477265a826e873557d3950f02d636becf1dd68c773947e69154fec9631e9e

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:496-553, the source contract is theorem hasDerivAt_typeFourArea {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeFourArea lam) (typeFourFold lam h / h ^ 3) h. The profiles and fold

numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourDelta

Source hash: 71475be52cec7e059c20046cff749d550fce2a4275677924c383af205dd192c8

Relation: equivalent

Equivalent in the declared constructor/section context. At `LeanSuffixAnalytic.lean:279-305`, the source contract is `theorem hasDerivAt_typeFourDelta {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeFourDelta lam) (h / sqrt (1 - h ^ 2) - h / (lam * sqrt (lam ^ 2 - h ^ 2))) h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourFold

Source hash: df765bf2b71951f20ce86d7897c71befd43183478a79e7706b6061a82ade97f0

Relation: equivalent

Equivalent in the declared constructor/section context. At `LeanSuffixAnalytic.lean:457-495`, the source contract is `theorem hasDerivAt_typeFourFold {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeFourFold lam) (4 * h ^ 2 * (1 / sqrt (1 - h ^ 2)) ^ 3 - lam / sqrt (lam ^ 2 - h ^ 2) ^ 3) h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeFourPerimeter

Source hash: fe6311de3122c760c199f2555194ef9855f7218b10bcc5afe2c6f598e18071ca

Relation: equivalent

Equivalent in the declared constructor/section context. At `LeanSuffixAnalytic.lean:761-767`, the source contract is `theorem hasDerivAt_typeFourPerimeter {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeFourPerimeter lam) (typeFourFold lam h / h ^ 2) h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeAngle

Source hash: 6ec918819525fbc766cbbf40969985e480000f0bcbcf23d536b033e9e80b6380

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:154-205, the source contract is theorem hasDerivAt_typeThreeAngle {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeThreeAngle lam) (2 * (1 / sqrt (1 - typeThreeShape h ^ 2) - lam / sqrt (lam ^ 2 - typeThreeShape h ^ 2))) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeArea

Source hash: 7e0c0da90901c546a8edb152b4ed9a40769e6bd51ef8318f50390135bcace455

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:700-724, the source contract is theorem hasDerivAt_typeThreeArea {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeThreeArea lam) (typeThreeFold lam h / h ^ 3) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeDelta

Source hash: 6a259e36542b424c2657b96b59d66f9d38d054281073fe5c590c3110a33522c8

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:206-240, the source contract is theorem hasDerivAt_typeThreeDelta {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeThreeDelta lam) (2 * typeThreeShape h * (1 / sqrt (1 - typeThreeShape h ^ 2) - 1 / (lam * sqrt (lam ^ 2 - typeThreeShape h ^ 2)))) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeFold

Source hash: c8c79d60a2ba5a7f202a48aa5b5851e2b132f448f609c9f5925a2a8edab62ee5

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:375-456, the source contract is theorem hasDerivAt_typeThreeFold {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeThreeFold lam) (16 * h ^ 2 * (1 / sqrt (1 - typeThreeShape h ^ 2) ^ 3 - lam / sqrt (lam ^ 2 - typeThreeShape h ^ 2) ^ 3)) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreePerimeter

Source hash: bd5364506b45967c0ea0097c8997e2b79343b02b6ddca131541b965a63dfb88f

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:725-732, the source contract is theorem hasDerivAt_typeThreePerimeter {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeThreePerimeter lam) (typeThreeFold lam h / h ^ 2) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.hasDerivAt_typeThreeShape

Source hash: fc5b59d714869fde00b69f1950c83bd7a32c47524e01ad8d04455ff346c39d58

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:146-153, the source contract is theorem hasDerivAt_typeThreeShape (h : ℝ) : HasDerivAt typeThreeShape 2 h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.radicalCubeGap_pos

Source hash: ed04fbfe0d2a5c8c68ae2429d8da8593a273f2471a8fcc329e85c97a6780176d

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:349-374, the source contract is theorem radicalCubeGap_pos {lam x : ℝ} (hlam : 1 < lam) (hx : -1 < x) (hx1 : x < 1) : 0 < 1 / sqrt (1 - x ^ 2) ^ 3 - lam / sqrt (lam ^ 2 - x ^ 2) ^ 3. The radical cube gap is strictly positive for $\lambda > 1$ and every $-1 < x < 1$, including $x=0$ and negative x . This full domain is needed when $x=2\lambda^3-1$; the author does not narrow it to positive x .

Findings: none.

LeanSuffixAnalytic.reducedFoldGap

Source hash: 90c9f564cc40a2e05814e95c52b8ea113ecc45335e828ff2a54c8e9f2caf08b3

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:83-92, the source contract is def reducedFoldGap (lam h₃ h₄ : ℝ) : ℝ := (h₃ - h₄) / h₄ * (1 / sqrt (1 - h₄ ^ 2) - lam / sqrt (lam ^ 2 - h₄ ^ 2)) + h₃ / h₄ * typeFourDelta lam h₄ - typeThreeDelta lam h₃. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.stationaryEqualArea_gap_identity

Source hash: 7ca190ef8d3e84eb702882d4666162a0576db626556bb2591b0ba437f522132f

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1221-1238, the source contract is theorem stationaryEqualArea_gap_identity {lam h₃ h₄ : ℝ} (hh₃ : h₃ ≠ 0) (hh₄ : h₄ ≠ 0) (harea : typeThreeArea lam h₃ = typeFourArea lam h₄) (hfold : typeFourFold lam h₄ = 0) : typeThreePerimeter lam h₃ - typeFourPerimeter lam h₄ = 4 * reducedFoldGap lam h₃ h₄. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.stationaryEqualArea_typeThree_improves_iff

Source hash: d313a9a9b483a140c805abe01de0a03576005db4ae69b2c417e95c5e91300b63

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1239-1252, the source contract is theorem stationaryEqualArea_typeThree_improves_iff {lam h₃ h₄ : ℝ} (hh₃ : h₃ ≠ 0) (hh₄ : h₄ ≠ 0) (harea : typeThreeArea lam h₃ = typeFourArea lam h₄) (hfold : typeFourFold lam h₄ = 0) : typeThreePerimeter lam h₃ < typeFourPerimeter lam h₄ ↔ reducedFoldGap lam h₃ h₄ < 0. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourAngle

Source hash: 94e6a9f5b491cb7ca4802d89fba1fa7ccb864bbe777d6e41d89b1d6bf892bdd0

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:53-56, the source contract is def typeFourAngle (lam h : ℝ) : ℝ := lam * arccos (h / lam) + arcsin h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourAngle_eq_checker

Source hash: d6ea741d7af136e9d0263d84561c8171caafa7b1249aceaa9a355bce3a415957

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:139-145, the source contract is theorem typeFourAngle_eq_checker (lam h : ℝ) : typeFourAngle lam h = lam * arccos (h / lam) + π / 2 - arccos h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use lambda > 1 and 0 < h < 1; endpoint statements have separate continuity support. The type-(iii) profile includes the required pi/2 offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourArea

Source hash: 775f7921b66716563dd5e0e4c35562d880f5cef2a71b40cef616cd10eb9488d2

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:61-64, the source contract is `def typeFourArea (lam h : ℝ) : ℝ := 2 * (typeFourAngle lam h + h * typeFourDelta lam h) / h ^ 2`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourArea_deriv_neg_before_zero

Source hash: e504f043a30eb739ae5f3ba9174d0ca2cd6aa51665b8ee6815ad3b1b3ea9932b

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1071-1082, the source contract is `theorem typeFourArea_deriv_neg_before_zero {lam fold h : ℝ} (hlam : 1 < lam) (hfold : fold ∈ Set.Ioo (0 : ℝ) 1) (hzero : typeFourFold lam fold = 0) (hh : h ∈ Set.Ioo (0 : ℝ) 1) (hlt : h < fold) : deriv (typeFourArea lam) h < 0`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourArea_deriv_pos_after_zero

Source hash: 574cf814295ee9aa4cb9406582955f50506dd66296979a4f903f27b9ae8e2667

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1083-1094, the source contract is `theorem typeFourArea_deriv_pos_after_zero {lam fold h : ℝ} (hlam : 1 < lam) (hfold : fold ∈ Set.Ioo (0 : ℝ) 1) (hzero : typeFourFold lam fold = 0) (hh : h ∈ Set.Ioo (0 : ℝ) 1) (hlt : fold < h) : 0 < deriv (typeFourArea lam) h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourArea_fold_le

Source hash: 00808b809e3e67ecf06ff39be8e712a367908c043a9c526b98df7d97ff33901d

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1095-1129, the source contract is `theorem typeFourArea_fold_le {lam fold h : ℝ} (hlam : 1 < lam) (hfold : fold ∈ Set.Ioo (0 : ℝ) 1) (hzero : typeFourFold lam fold = 0) (hh : h ∈ Set.Ioc (0 : ℝ) 1) : typeFourArea lam fold ≤ typeFourArea lam h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFourDelta

Source hash: a4c8b30971f59f8f04626dbddf49af92c98dbfa9126bf3766c0be8b6472cbfb3

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:57-60, the source contract is `def typeFourDelta (lam h : ℝ) : ℝ := sqrt (lam ^ 2 - h ^ 2) / lam - sqrt (1 - h ^ 2)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeFourFold

Source hash: e492724315b735482cab607f768694fcd028864cb336800b7469d841099293ee

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:78-82, the source contract is `def typeFourFold (lam h : ℝ) : ℝ := 4 * (h * (1 / sqrt (1 - h ^ 2) - lam / sqrt (lam ^ 2 - h ^ 2)) - typeFourAngle lam h)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeFourFold_strictMonoOn

Source hash: 16af12c38f3bb1c629d7204f781995879e693ae4c65a0b5d7c3475b2043d0222

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1056-1070, the source contract is `theorem typeFourFold_strictMonoOn {lam : ℝ} (hlam : 1 < lam) : StrictMonoOn (typeFourFold lam) (Set.Ioo 0 1)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeFourFold_zero_unique

Source hash: bc70631d3da63af952fb4321a7d0a57845b40dd8854182d958b30d4e8971577c

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1150-1158, the source contract is `theorem typeFourFold_zero_unique {lam h1 h2 : ℝ} (hlam : 1 < lam) (hh1 : h1 ∈ Set.Ioo (0 : ℝ) 1) (hh2 : h2 ∈ Set.Ioo (0 : ℝ) 1) (hz1 : typeFourFold lam h1 = 0) (hz2 : typeFourFold lam h2 = 0) : h1 = h2`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeFourPerimeter

Source hash: ad0bebbd79c17351c0f12ce6a7fdae3e930dba33de0077415a6116df1d1e6e25

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:65-70, the source contract is `def typeFourPerimeter (lam h : ℝ) : ℝ := 4 * typeFourAngle lam h / h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeFour_variational_hasDerivAt

Source hash: be216789e3804b1c530f202d3470319a386e5c80dc03952cfaf3ebc685f27297

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:733-753, the source contract is `theorem typeFour_variational_hasDerivAt {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeFourArea lam) (deriv (typeFourArea lam) h) h ^ HasDerivAt (typeFourPerimeter lam) (h * deriv (typeFourArea lam) h) h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThreeAngle

Source hash: fb7617b8b745792c6f019f3cbcf3c26d5e54f1fc7dd47bb69cec919c712d110a

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:35-38, the source contract is `def typeThreeAngle (lam h : ℝ) : ℝ := lam * arccos (typeThreeShape h / lam) + arcsin (typeThreeShape h)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThreeAngle_add_halfPi

Source hash: 309f9c361ad038b7eae75200b2b081130c71a155a43f2f07c9ad2ae282ce68ff

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:130-138, the source contract is `theorem typeThreeAngle_add_halfPi (lam h : ℝ) : typeThreeAngle lam h +  $\pi / 2$  = lam * arccos (typeThreeShape h / lam) +  $\pi$  - arccos (typeThreeShape h)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThreeArea

Source hash: 63d13d5879252e16ac6f53a2d4449b4f511e353d6016e98158793484dcfaa8f7

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:44-48, the source contract is `def typeThreeArea (lam h : ℝ) : ℝ := (typeThreeAngle lam h + π / 2 + (typeThreeShape h + 2) * typeThreeDelta lam h) / h ^ 2`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_deriv_eq_explicit

Source hash: ec96d74eb7c231919df47f4c5a7bcce391a0bf68733b2ba3385beb5c882b9ca6

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:649-699, the source contract is `theorem typeThreeArea_deriv_eq_explicit {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : deriv (typeThreeArea lam) h = ((2 * (2 * (1 / sqrt (1 - typeThreeShape h ^ 2) - lam / sqrt (lam ^ 2 - typeThreeShape h ^ 2)) + 2 * typeThreeShape h * (1 / sqrt (1 - typeThreeShape h ^ 2) - 1 / (lam * sqrt (lam ^ 2 - typeThreeShape h ^ 2)))) * h - 2 * (typeThreeAngle lam h + π / 2 + typeThreeDelta lam h)) / h ^ 3`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use  $\lambda > 1$  and  $0 < h < 1$ ; endpoint statements have separate continuity support. The type-(iii) profile includes the required  $\pi/2$  offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique

Source hash: 15660b04d9615f11056408ab4ae99cb03c7fde3d8d26f5056c89e742987b7edb

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1013-1039, the source contract is `theorem typeThreeArea_descending_root_existsUnique {lam anchor target : ℝ} (hlam : 1 < lam) (hanchor : anchor ∈ Set.Ioo (0 : ℝ) 1) (hfold : typeThreeFold lam anchor < 0) (htarget : typeThreeArea lam anchor ≤ target) : ∃! root : ℝ, root ∈ Set.Ioo (0 : ℝ) 1 ∧ typeThreeArea lam root = target ∧ typeThreeFold lam root < 0`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use  $\lambda > 1$  and  $0 < h < 1$ ; endpoint statements have separate continuity support. The type-(iii) profile includes the required  $\pi/2$  offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_descending_root_existsUnique_of_stationary_pair

Source hash: f368bc947db4d0b119fefc4c63c0de6d0d14cb14db7e5a28faa26c535eb880b8

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:1130-1149, the source contract is `theorem typeThreeArea_descending_root_existsUnique_of_stationary_pair {lam h3 fold h4 fold h4 : ℝ} (hlam : 1 < lam) (hh3 fold : h3 fold ∈ Set.Ioo (0 : ℝ) 1) (hh4 fold : h4 fold ∈ Set.Ioo (0 : ℝ) 1) (hh4 : h4 ∈ Set.Ioc (0 : ℝ) 1) (hfoldThree : typeThreeFold`

$\text{lam } h_3 \text{fold} < 0$) ($\text{hfoldFour} : \text{typeFourFold } \text{lam } h_4 \text{fold} = 0$) ($\text{hequalFold} : \text{typeThreeArea } \text{lam } h_3 \text{fold} = \text{typeFourArea } \text{lam } h_4 \text{fold}$) : $\exists! \text{root} : \mathbb{R}, \text{root} \in \text{Set.Ioo } (0 : \mathbb{R}) 1 \wedge \text{typeThreeArea } \text{lam } \text{root} = \text{typeFourArea } \text{lam } h_4 \wedge \text{typeThreeFold } \text{lam } \text{root} < 0$. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\text{lambda} > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle .

Findings: none.

LeanSuffixAnalytic.typeThreeArea_descending_root_lt

Source hash: f0218743e3c66d2b3ba1e5bdde4d44d6f7c58ed79dc9416fb8fb18d852a14824

Relation: equivalent

Equivalent in the declared constructor/section context. At SameCurvatureArea.lean:586-622, the source contract is $\text{theorem typeThreeArea_descending_root_lt } \{\text{lam } \text{root } h : \mathbb{R}\} (\text{hlam} : 1 < \text{lam}) (\text{hroot} : \text{root} \in \text{Set.Ioo } (0 : \mathbb{R}) 1) (\text{hh} : 0 < h) (\text{hh1} : h \leq 1) (\text{hfold} : \text{typeThreeFold } \text{lam } \text{root} < 0) (\text{harea} : \text{typeThreeArea } \text{lam } \text{root} = \text{typeFourArea } \text{lam } h) : \text{root} < h$. The strict same-curvature area inequality is proved for $\text{lambda} > 1$ and $0 < h \leq 1$. Combining it with monotonicity of the negative-fold branch gives a descending equal-area root strictly below the four-arc curvature, not above it.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_lt_typeFourArea

Source hash: 292616e61180882b74b7405ba97fd168e8ed5d3653115a72ad0fa3e78e338f65

Relation: equivalent

Equivalent in the declared constructor/section context. At SameCurvatureArea.lean:225-265, the source contract is $\text{theorem typeThreeArea_lt_typeFourArea } \{\text{lam } h : \mathbb{R}\} (\text{hlam} : 1 < \text{lam}) (\text{hh} : 0 < h) (\text{hh1} : h \leq 1) : \text{typeThreeArea } \text{lam } h < \text{typeFourArea } \text{lam } h$. The strict same-curvature area inequality is proved for $\text{lambda} > 1$ and $0 < h \leq 1$. Combining it with monotonicity of the negative-fold branch gives a descending equal-area root strictly below the four-arc curvature, not above it.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_root_le_of_fold_neg

Source hash: a9a71e3bf8960b5c96a8fc59bf3fe2a372f70aa365bd91817e7ca128cb8520b9

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:954-973, the source contract is $\text{theorem typeThreeArea_root_le_of_fold_neg } \{\text{lam } \text{root } y : \mathbb{R}\} (\text{hlam} : 1 < \text{lam}) (\text{hroot} : \text{root} \in \text{Set.Ioo } (0 : \mathbb{R}) 1) (\text{hy} : y \in \text{Set.Ioc } (0 : \mathbb{R}) 1) (\text{hfold} : \text{typeThreeFold } \text{lam } \text{root} < 0) (\text{harea} : \text{typeThreeArea } \text{lam } \text{root} = \text{typeThreeArea } \text{lam } y) : \text{root} \leq y$. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\text{lambda} > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle .

Findings: none.

LeanSuffixAnalytic.typeThreeArea_strictAntiOn_Icc

Source hash: fd210f8a463759cd45da345eae2826fb3a7bdf59d29239656d68c8efc64fb7fc

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:108-123, the source contract is theorem typeThreeArea_strictAntiOn_Icc {lam a b : ℝ} (hlam : 1 < lam) (ha : 0 < a) (hb : b < 1) (hderiv : ∀ y ∈ Ioo a b, deriv (typeThreeArea lam) y < 0) : StrictAntiOn (typeThreeArea lam) (Icc a b). Equal area and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_strictAntiOn_Icc_of_fold_neg_right

Source hash: 1b1d7870499bdaf74cbcd036814fc1cce139b4cf2463cce4ba39b71ae55024b7

Relation: equivalent

Equivalent in the declared constructor/section context. At EqualAreaEnvelope.lean:124-138, the source contract is theorem typeThreeArea_strictAntiOn_Icc_of_fold_neg_right {lam a b : ℝ} (hlam : 1 < lam) (ha : 0 < a) (hb : b < 1) (hb_pos : 0 < b) (hfoldb : typeThreeFold lam b < 0) : StrictAntiOn (typeThreeArea lam) (Icc a b). Equal area and the negative-fold branch are distinct requirements. Interior differentiation requires a simple root; the closure endpoint uses one-sided continuity. The main proof derives these properties rather than assuming the all-curvature perimeter comparison.

Findings: none.

LeanSuffixAnalytic.typeThreeArea_tendsto_atRight_zero

Source hash: e82ffa97f750ea7c64d3d22eebba58e03fe2fc0f731297b9787ae711ded5a49a

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:828-871, the source contract is theorem typeThreeArea_tendsto_atRight_zero {lam : ℝ} (hlam : 1 < lam) : Tendsto (typeThreeArea lam) (ℕ[>] 0) atTop. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThreeDelta

Source hash: f96722ed089f4e5911c6013a0aa601eb4080090036491702cd0fa78e3e31113f

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:39-43, the source contract is def typeThreeDelta (lam h : ℝ) : ℝ := sqrt (lam ^ 2 - typeThreeShape h ^ 2) / lam - sqrt (1 - typeThreeShape h ^ 2). The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThreeFold

Source hash: de466d8508a0def3484419e5a98402f15940017ff9d116432fb67c21c1d22c86

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:71-77, the source contract is `def typeThreeFold (lam h : ℝ) : ℝ := 4 * h * ((1 + typeThreeShape h) / sqrt (1 - typeThreeShape h ^ 2) - (lam ^ 2 + typeThreeShape h) / (lam * sqrt (lam ^ 2 - typeThreeShape h ^ 2))) - 2 * (typeThreeAngle lam h + π / 2 + typeThreeDelta lam h)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreeFold_strictMonoOn

Source hash: 548db5145f9bcf2011e54d3648e735633b97c51f2514a21b4368c3304bb06a32

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:872-893, the source contract is `theorem typeThreeFold_strictMonoOn {lam : ℝ} (hlam : 1 < lam) : StrictMonoOn (typeThreeFold lam) (Set.Ioo 0 1)`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreeFold_zero_between_of_area_eq

Source hash: 314dbeb5db22bb018c3bb2b871eb2f76a4711df985ead7c117e1cc5036bed173

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:905-925, the source contract is `theorem typeThreeFold_zero_between_of_area_eq {lam a b : ℝ} (hlam : 1 < lam) (ha : a ∈ Set.Ioo (0 : ℝ) 1) (hb : b ∈ Set.Ioc (0 : ℝ) 1) (hab : a < b) (harea : typeThreeArea lam a = typeThreeArea lam b) : ∃ c ∈ Set.Ioo a b, typeThreeFold lam c = 0`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreePerimeter

Source hash: 22e35e36f9c0930c9f9233f0004c686c48f6f282298a578def5a6e6bba11f34e

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:49-52, the source contract is `def typeThreePerimeter (lam h : ℝ) : ℝ := 2 * (typeThreeAngle lam h + π / 2 + typeThreeDelta lam h) / h`. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside `typeThreeAngle`.

Findings: none.

LeanSuffixAnalytic.typeThreePerimeter_deriv_eq_height_mul_area_deriv

Source hash: 7607f8124dad06702d520d2b8e20b8eab5117f85a2007d935f98d72479aad4e2

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:640-648, the source contract is theorem typeThreePerimeter_deriv_eq_height_mul_area_deriv {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : deriv (typeThreePerimeter lam) h = h * deriv (typeThreeArea lam) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThreeShape

Source hash: 9ef91461acc972465b98decfe31d601788f1354eb0ffbb8d094be6f5c46ca593

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:32-34, the source contract is def typeThreeShape (h : ℝ) : ℝ := 2 * h - 1. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixAnalytic.typeThree_variational_hasDerivAt

Source hash: 727f5f60431aed129e76617edde1e3ccdf8d18a1ae71fbf80ca33147d09b9cfa

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixAnalytic.lean:554-639, the source contract is theorem typeThree_variational_hasDerivAt {lam h : ℝ} (hlam : 1 < lam) (hh : 0 < h) (hh1 : h < 1) : HasDerivAt (typeThreeArea lam) (deriv (typeThreeArea lam) h) h ∧ HasDerivAt (typeThreePerimeter lam) (h * deriv (typeThreeArea lam) h) h. The profiles and fold numerators are the displayed exact real formulas. Regular derivative arguments use $\lambda > 1$ and $0 < h < 1$; endpoint statements have separate continuity support. The type-(iii) profile includes the required $\pi/2$ offset outside typeThreeAngle.

Findings: none.

LeanSuffixReflective.QInterval

Source hash: aa9fdae3c31a981fd9f3e3c84090b4a600570661da326e67bfa152cc3da18001

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:19-26, the source contract is structure QInterval where lo : ℚ hi : ℚ ordered : lo ≤ hi. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.RealContains

Source hash: 82c7a510d68058d7816af0b7c59da61b39569db9b49cfeb125a24aaafc395c508

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:30-33, the source contract is `def RealContains (i : QInterval) (x : ℝ) : Prop := (i.lo : ℝ) ≤ x ∧ x ≤ (i.hi : ℝ)`. RealContains is the conjunction of the two weak real endpoint inequalities. Neither a positive-width condition nor strict interior containment is smuggled into the notation.

Findings: none.

LeanSuffixReflective.QInterval.add

Source hash: bdc18247a80ffa4a9b050a8d17f0cc87f5a29f49311d73038324983c96603c22

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:37-40, the source contract is `def add (i j : QInterval) : QInterval := ⟨i.lo + j.lo, i.hi + j.hi, add_le_add i.ordered j.ordered⟩`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.neg

Source hash: 81fd6f63aa66c1f67c60e066b58e12cde99767a33f0cf5e8a79a120837332892

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:41-44, the source contract is `def neg (i : QInterval) : QInterval := ⟨-i.hi, -i.lo, neg_le_neg i.ordered⟩`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.nsmul

Source hash: aee3f878b71376e5cd4779ff5df12d1e44522ca38190dd8fec337d507ede104

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:48-59, the source contract is `def nsmul (q : ℚ) (hq : 0 ≤ q) (i : QInterval) : QInterval := ⟨q * i.lo, q * i.hi, mul_le_mul_of_nonneg_left i.ordered hq⟩ @[simp] theorem realContains_point (x : ℚ) (y : ℝ) : (point x).RealContains y ↔ y = (x : ℝ) := by constructor · intro h exact le_antisymm h.2 h.1 · rintro rfl exact ⟨le_rfl, le_rfl⟩`. Despite its name, nsmul takes a nonnegative rational multiplier with a proof of nonnegativity. It is not restricted to natural numbers or positive integers; zero and fractional multipliers are covered by the author's formulation.

Findings: none.

LeanSuffixReflective.QInterval.point

Source hash: 35a71df9981ec16d7188d1e717bf2773425dff09af794e6cabfae2678adb7fb9

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:34-36, the source contract is `def point (x : ℚ) : QInterval := ⟨x, x, le_refl⟩`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.realContains_add

Source hash: 29bd8dc11237ddddd13e05a341310e25a85723026439683a6ff469efc9539f38

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:65-71, the source contract is `theorem realContains_add {i j : QInterval} {x y : ℝ} (hx : i.RealContains x) (hy : j.RealContains y) : (i.add j).RealContains (x + y)`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.realContains_neg

Source hash: c819be23ad835c80289fa7a60e6d94935c325c424a832a1d3dc0780c7f34f5fd

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:72-77, the source contract is `theorem realContains_neg {i : QInterval} {x : ℝ} (hx : i.RealContains x) : i.neg.RealContains (-x)`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.realContains_sub

Source hash: 8bd28ee4b73b928fde4081285e5ca72b48a1703fbb61327a22c0b7e93e0fc99c

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:78-85, the source contract is `theorem realContains_sub {i j : QInterval} {x y : ℝ} (hx : i.RealContains x) (hy : j.RealContains y) : (i.sub j).RealContains (x - y)`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

LeanSuffixReflective.QInterval.sub

Source hash: edd84d3089fe87cd8f669b75895897b3124604659a9914bc80d5d6266ee85fd2

Relation: equivalent

Equivalent in the declared constructor/section context. At LeanSuffixReflective.lean:45-47, the source contract is `def sub (i j : QInterval) : QInterval := i.add j.neg`. Intervals have rational ordered endpoints and closed containment. Arithmetic enclosure statements concern the real-valued operations after rational casting; exact checks do not stand for floating-point evidence.

Findings: none.

MiddleFaceAssembly.Brick

Source hash: 0941d7d9eb0eaebceb5359d2c2683a5354c44a25e853225ca2ccbf95f4460178

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:27-37, the source contract is `structure Brick where lamLo : $\mathbb{Q}$ lamHi : $\mathbb{Q}$ centers : $\text{Fin } 4 \rightarrow \mathbb{Q}$ slopes : $\text{Fin } 4 \rightarrow \mathbb{Q}$ yXiSlope : $\mathbb{Q}$ vRhoSlope : $\mathbb{Q}$ radii : $\text{Fin } 4 \rightarrow \mathbb{Q}$ lamOrdered : $\text{lamLo} < \text{lamHi}$ radiiPos : $\forall j, 0 < \text{radii } j$` . The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.derivRho

Source hash: c1e483238adef1a69bda282637a3ed4325a085039bf7cd8b957f9f3fdb1c5e6e

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1190-1197, the source contract is `def derivRho (t rho sigma : $\mathbb{R}$) : $\mathbb{R}$:= let ww := (b.centers 2 : $\mathbb{R}$) + b.slopes 2 * t + rho let vv := (b.centers 3 : $\mathbb{R}$) + b.slopes 3 * t + b.vRhoSlope * rho + sigma let ll := (lamMid b : $\mathbb{R}$) + t (-sin ww + (b.vRhoSlope : $\mathbb{R}$) * (ll * sin vv))`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.derivSigma

Source hash: 36bccfa8df42ee44f7ce2efdeb4df70c3f66a17d14d767e920177bbf1afa9876

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1198-1202, the source contract is `def derivSigma (t rho sigma :  $\mathbb{R}$ ) :  $\mathbb{R}$  := let vv := (b.centers 3 :  $\mathbb{R}$ ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma ((lamMid b :  $\mathbb{R}$ ) + t) * sin vv`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.derivT

Source hash: c4e519ef7409cb708317db418241213f452ef4b4aa73c4ae26a3279d8dcb518d

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1181-1189, the source contract is `def derivT (t rho sigma : ℝ) : ℝ := let ww := (b.centers 2 : ℝ) + b.slopes 2 * t + rho let vv := (b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma let ll := (lamMid b : ℝ) + t * (-(b.slopes 2 : ℝ) * sin ww - cos vv + (b.slopes 3 : ℝ) * (ll * sin vv))`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.reduced

Source hash: 4b58f212985cabd9417414d5466594d9a780e38a8e0c1797d384042e1674c68b

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1170-1175, the source contract is `def reduced (t rho sigma : ℝ) : ℝ := cos ((b.centers 2 : ℝ) + b.slopes 2 * t + rho) - ((lamMid b : ℝ) + t) * cos ((b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.reduced_hasDerivAt_rho

Source hash: 78960ca166459039fcd13d2b254e90bc61da69bb828a41a82ae24260c24dc81e

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1229-1250, the source contract is `theorem reduced_hasDerivAt_rho (t rho sigma : ℝ) : HasDerivAt (fun u => reduced b t u sigma) (derivRho b t rho sigma) rho`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.reduced_hasDerivAt_sigma

Source hash: 4b0f795e660ad8612256b86b0826502532784fc2e8dabb3224b9d16a883aa33e

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1251-1270, the source contract is `theorem reduced_hasDerivAt_sigma (t rho sigma : ℝ) : HasDerivAt (fun u => reduced b t rho u) (derivSigma b t rho sigma) sigma`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C3Residual.reduced_hasDerivAt_t

Source hash: 7dbe2b5150934c034dcc9733f2ecaa267ad4409fadfc1f5c1ad22c9ee3309dc2

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1203-1228, the source contract is theorem reduced_hasDerivAt_t (t rho sigma : $\mathbb{R}$) : HasDerivAt (fun u => reduced b u rho sigma) (derivT b t rho sigma) t. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.derivEta

Source hash: 1a437771f514f75dbb66dda75e523e87384fcd4e7f50961082bd2649fc01f800

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:707-710, the source contract is def derivEta (t xi eta : $\mathbb{R}$) : $\mathbb{R}$:= ((lamMid b : $\mathbb{R}$) + t) * sin ((b.centers 1 : $\mathbb{R}$) + b.slopes 1 * t + b.yXiSlope * xi + eta). The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.derivT

Source hash: 4215795dac7a2debf79984f7d7c608299b079841931e2bdd1d8a1d5e697b0802

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:693-700, the source contract is def derivT (t xi eta : $\mathbb{R}$) : $\mathbb{R}$:= -(b.slopes 0 : $\mathbb{R}$) * sin ((b.centers 0 : $\mathbb{R}$) + b.slopes 0 * t + xi) - cos ((b.centers 1 : $\mathbb{R}$) + b.slopes 1 * t + b.yXiSlope * xi + eta) + (b.slopes 1 : $\mathbb{R}$) * (((lamMid b : $\mathbb{R}$) + t) * sin ((b.centers 1 : $\mathbb{R}$) + b.slopes 1 * t + b.yXiSlope * xi + eta)). The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.derivXi

Source hash: ed53afb5e433a1f19562ef174d9cbdc3d2d47f108aac2160d30aaf8712b1aa82

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:701-706, the source contract is def derivXi (t xi eta : $\mathbb{R}$) : $\mathbb{R}$:= -sin ((b.centers 0 : $\mathbb{R}$) + b.slopes 0 * t + xi) + (b.yXiSlope : $\mathbb{R}$) * (((lamMid b : $\mathbb{R}$) + t) * sin ((b.centers 1 : $\mathbb{R}$) + b.slopes 1 * t + b.yXiSlope * xi + eta)). The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.reduced

Source hash: 6ad85092e6bdf9c53b20e583c6a631e4f162d95899defb77f3673f60e1415681

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:683-687, the source contract is `def reduced (t xi eta : ℝ) : ℝ := cos ((b.centers 0 : ℝ) + b.slopes 0 * t + xi) - ((lamMid b : ℝ) + t) * cos ((b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.reduced_hasDerivAt_eta

Source hash: 3ee50bbc4e028a15db6fda4f71c882933ff623e2cf7891f6aa4ee3930451e70b

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:758-775, the source contract is `theorem reduced_hasDerivAt_eta (t xi eta : ℝ) : HasDerivAt (fun u => reduced b t xi u) (derivEta b t xi eta) eta`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.reduced_hasDerivAt_t

Source hash: 57a105b0af38480d894e98b27a5a27170b796d726656e5e9828ac63318bb34f4

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:711-736, the source contract is `theorem reduced_hasDerivAt_t (t xi eta : ℝ) : HasDerivAt (fun u => reduced b u xi eta) (derivT b t xi eta) t`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.C4Residual.reduced_hasDerivAt_xi

Source hash: 5c5aed8fcfa67c76068e264e5cfde03a674d4b1ee2a4b7f25289608330e62d6c

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:737-757, the source contract is `theorem reduced_hasDerivAt_xi (t xi eta : ℝ) : HasDerivAt (fun u => reduced b t u eta) (derivXi b t xi eta) xi`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.CellCertificate

Source hash: c5e1550765346522815e726fec39295f689896e3151daad7810df51dd4df337f

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1444-1484, the source contract is structure CellCertificate (b : Brick) : Prop where lambda_one_lt : $\forall t, (tInterval\ b).RealContains\ t \rightarrow 1 < lam\ b\ (pointOf\ t\ 0)$ branches : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow let\ p := pointOf\ t\ z\ 0 < x\ b\ p \wedge x\ b\ p < \pi / 4 \wedge 0 < y\ b\ p \wedge y\ b\ p < \pi / 4 \wedge 0 < w\ b\ p \wedge w\ b\ p < \pi / 4 \wedge 0 < v\ b\ p \wedge v\ b\ p < \pi / 4$ F_xi_low_pos : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 0 = -radiiReal\ b\ 0 \rightarrow 0 < F\ b\ (pointOf\ t\ z)$ F_xi_high_neg : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 0 = radiiReal\ b\ 0 \rightarrow F\ b\ (pointOf\ t\ z) < 0$ C4_eta_low_neg : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 1 = -radiiReal\ b\ 1 \rightarrow C4\ b\ (pointOf\ t\ z) < 0$ C4_eta_high_pos : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 1 = radiiReal\ b\ 1 \rightarrow 0 < C4\ b\ (pointOf\ t\ z)$ E_rho_low_neg : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 2 = -radiiReal\ b\ 2 \rightarrow E\ b\ (pointOf\ t\ z) < 0$ E_rho_high_pos : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 2 = radiiReal\ b\ 2 \rightarrow 0 < E\ b\ (pointOf\ t\ z)$ C3_sigma_low_neg : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 3 = -radiiReal\ b\ 3 \rightarrow C3\ b\ (pointOf\ t\ z) < 0$ C3_sigma_high_pos : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow z\ 3 = radiiReal\ b\ 3 \rightarrow 0 < C3\ b\ (pointOf\ t\ z)$ K3_neg : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow K3\ b\ (pointOf\ t\ z) < 0$ G_neg : $\forall t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow G\ b\ (pointOf\ t\ z) < 0$. The record contains exact branch bounds, four pairs of strict opposite-face signs for the specified residuals, K3 negativity and G negativity throughout the brick. These are sufficient analytic certificate obligations, not assumed roots or preexisting nonminimality.

Findings: none.

MiddleFaceAssembly.CellCertificate.sound

Source hash: 3fff65f4ffa94f40ec377aeb7e5b713831d3e0a0f69889bb8c84ffd5cf5fd19a

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1510-1536, the source contract is theorem sound {b : Brick} (certificate : CellCertificate b) : Conditions b where lambda_one_lt. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Conditions

Source hash: 5c36f9057f1d62e8c0bcb35e1c262b414fda26bcb9bf4a32565a3c1e70dc5568

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1485-1509, the source contract is structure Conditions (b : Brick) : Prop where lambda_one_lt : $\forall t, (tInterval\ b).RealContains\ t \rightarrow 1 < lam\ b\ (pointOf\ t\ 0)$ branches : $\forall t\ z, (tInterval$

$b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow let\ p := pointOf\ t\ z\ 0 < x\ b\ p \wedge x\ b\ p < \pi / 4 \wedge 0 < y\ b\ p \wedge y\ b\ p < \pi / 4 \wedge 0 < w\ b\ p \wedge w\ b\ p < \pi / 4 \wedge 0 < v\ b\ p \wedge v\ b\ p < \pi / 4$ lowFaces : $\forall\ t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow \forall\ j, z\ j = -radiiReal\ b\ j \rightarrow rootMap\ b\ t\ z\ j < 0$ highFaces : $\forall\ t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow \forall\ j, z\ j = radiiReal\ b\ j \rightarrow 0 < rootMap\ b\ t\ z\ j$ K3_neg : $\forall\ t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow K3\ b\ (pointOf\ t\ z) < 0$ G_neg : $\forall\ t\ z, (tInterval\ b).RealContains\ t \rightarrow InSymmetricBox\ (radiiReal\ b)\ z \rightarrow G\ b\ (pointOf\ t\ z) < 0$. The Conditions interface has the same brick and residual geometry as the certificate soundness theorem. Its face-sign and branch requirements lead to actual common zeros via the incidence-root projections and two one-dimensional IVTs; a general multivariate existence principle is not silently postulated.

Findings: none.

MiddleFaceAssembly.Conditions.RootProjection

Source hash: 2f76fbe7f8dbd852f7dc66c768c04d395e099b30fab88a0cc5ace08e7978fcf4

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1993-2008, the source contract is
`structure RootProjection (b : Brick) (t : ℝ) (z : Vec4) : Prop where mem_box : InSymmetricBox (radiiReal b) z equations : $\forall\ j, rootMap\ b\ t\ z\ j = 0$ branches : let p := pointOf t z 0 < x b p $\wedge$ x b p < $\pi / 4$ $\wedge$ 0 < y b p $\wedge$ y b p < $\pi / 4$ $\wedge$ 0 < w b p $\wedge$ w b p < $\pi / 4$ $\wedge$ 0 < v b p $\wedge$ v b p < $\pi / 4$ K3_neg : K3 b (pointOf t z) < 0 G_neg : G b (pointOf t z) < 0 typeThreeHeight_regular : $(1 + \cos (w\ b\ (pointOf\ t\ z))) / 2 \in Ioo\ (0 : \mathbb{R})\ 1$ typeFourHeight_regular : $\cos (x\ b\ (pointOf\ t\ z)) \in Ioo\ (0 : \mathbb{R})\ 1$. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.`

Findings: none.

MiddleFaceAssembly.Conditions.candidate_not_isWeightedPerimeterMinimizer

Source hash: ceb5515954483428fb5ae9be3255392f99e773ed979053baf3c18aa42dc8d6cd

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:2424-2438, the source contract is
`theorem candidate_not_isWeightedPerimeterMinimizer (conditions : Conditions b) {t lam0 : ℝ} (ht : (tInterval b).RealContains t) (hlam : (lamMid b : ℝ) + t = lam0) (candidate : _root_.FourArcCandidate lam0) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : $\neg$ candidate.IsWeightedPerimeterMinimizer. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.`

Findings: none.

MiddleFaceAssembly.Conditions.rootProjection

Source hash: b0b149b1edc9d2dd81f962164ee26472b0cb945a7e8f0c2a208f3fb617b6fe84

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:2009-2035, the source contract is theorem rootProjection (conditions : Conditions b) (t : $\mathbb{R}$) (ht : (tInterval b).RealContains t) : $\exists z, \text{RootProjection } b \ t \ z$. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Conditions.root_exists

Source hash: beb4b56e77f5562892d76e328ff8aac16683506f9b4a5b897368a7309f80cb5d

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1914-1992, the source contract is theorem root_exists (conditions : Conditions b) (t : $\mathbb{R}$) (ht : (tInterval b).RealContains t) : $\exists z, \text{InSymmetricBox } (\text{radiiReal } b) \ z \wedge \forall j, \text{rootMap } b \ t \ z \ j = 0$. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Conditions.stationaryEqualAreaPair_exists

Source hash: c9edc4aa9422891e0d25fbd6875c005c42dad37c4605fbf65a3d8fdd6fa0a683

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:2349-2423, the source contract is theorem stationaryEqualAreaPair_exists (conditions : Conditions b) {t : $\mathbb{R}$ } (ht : (tInterval b).RealContains t) : $\exists \text{pair} : \text{LeanSuffixAnalytic.StationaryEqualAreaPair } ((\text{lamMid } b : \mathbb{R}) + t), \text{LeanSuffixAnalytic.typeThreeFold } ((\text{lamMid } b : \mathbb{R}) + t) \text{ pair.h}_3 < 0 \wedge \text{LeanSuffixAnalytic.typeThreePerimeter } ((\text{lamMid } b : \mathbb{R}) + t) \text{ pair.h}_3 < \text{LeanSuffixAnalytic.typeFourPerimeter } ((\text{lamMid } b : \mathbb{R}) + t) \text{ pair.h}_4$. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.abs_image_sub_le_of_hasDerivAt

Source hash: b7f0e88d7d03b44d4326633ce2286459e2f413f783621bf0b54b9742012c3366

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:435-447, the source contract is theorem abs_image_sub_le_of_hasDerivAt {f f' : $\mathbb{R} \rightarrow \mathbb{R}$ } {R M u : $\mathbb{R}$ } (hR : $0 \leq R$) (hu : $u \in \text{Icc } (-R) \ R$) (hderiv : $\forall q \in \text{Icc } (-R) \ R, \text{HasDerivAt } f \ (f' \ q) \ q$) (hbound : $\forall q \in \text{Icc } (-R) \ R, |f' \ q| \leq M$) : $|f \ u - f \ 0| \leq M * |u|$. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.derivEta

Source hash: 3f40afdbac974dd2d7b928190653d4044cf2a42449bc6353b0884369f5086079

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:239-246, the source contract is `def derivEta (t xi eta rho sigma : ℝ) : ℝ := formulaDeriv ((lamMid b : ℝ) + t) ((b.centers 0 : ℝ) + b.slopes 0 * t + xi) ((b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta) ((b.centers 2 : ℝ) + b.slopes 2 * t + rho) ((b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma) 0 0 1 0 0`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.derivRho

Source hash: 3de77d285680f5b628dc186984128f3143ab98fcad154680680ec2be723c01a7

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:247-254, the source contract is `def derivRho (t xi eta rho sigma : ℝ) : ℝ := formulaDeriv ((lamMid b : ℝ) + t) ((b.centers 0 : ℝ) + b.slopes 0 * t + xi) ((b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta) ((b.centers 2 : ℝ) + b.slopes 2 * t + rho) ((b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma) 0 0 0 1 b.vRhoSlope`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.derivSigma

Source hash: c526b5c951f77329daf2d0b5ffbf1fbd6d8a07576e3b33918a14d59b930c31ad

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:255-262, the source contract is `def derivSigma (t xi eta rho sigma : ℝ) : ℝ := formulaDeriv ((lamMid b : ℝ) + t) ((b.centers 0 : ℝ) + b.slopes 0 * t + xi) ((b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta) ((b.centers 2 : ℝ) + b.slopes 2 * t + rho) ((b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma) 0 0 0 0 1`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.derivT

Source hash: 63d0ee648f437cabdbbba64daa954d7e00b41424d9d35808ce7891e2d6f80a5

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:223-230, the source contract is `def derivT (t xi eta rho sigma : ℝ) : ℝ := formulaDeriv ((lamMid b : ℝ) + t) ((b.centers 0 : ℝ) + b.slopes 0 * t + xi) ((b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta) ((b.centers 2 : ℝ) + b.slopes 2 * t + rho) ((b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma) 1 (b.slopes 0) (b.slopes 1) (b.slopes 2) (b.slopes 3)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.derivXi

Source hash: d1a73b3463380014d7f1b81b1d3b02ebd95fa6a7e6e2d4551638737c0c79430f

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:231-238, the source contract is `def derivXi (t xi eta rho sigma : ℝ) : ℝ := formulaDeriv ((lamMid b : ℝ) + t) ((b.centers 0 : ℝ) + b.slopes 0 * t + xi) ((b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta) ((b.centers 2 : ℝ) + b.slopes 2 * t + rho) ((b.centers 3 : ℝ) + b.slopes 3 * t + b.vRhoSlope * rho + sigma) 0 1 b.yXiSlope 0 0`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.formulaDeriv

Source hash: 7bb93e09ba7a6620ea0d2d9c467898bc5f608acecb169dcb1185ff20303362eb

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:151-164, the source contract is `def formulaDeriv (ll xx yy ww vv dl dx dy dw dv : ℝ) : ℝ := let nn4 := ll * yy + π / 2 - xx + cos xx * (sin yy - sin xx) let nn3 := ll * vv + π - ww + (cos ww + 2) * (sin vv - sin ww) let dn4 := dl * yy + ll * dy - dx + (-sin xx * dx) * (sin yy - sin xx) + cos xx * (cos yy * dy - cos xx * dx) let dn3 := dl * vv + ll * dv - dw + (-sin ww * dw) * (sin vv - sin ww) + (cos ww + 2) * (cos vv * dv - cos ww * dw) (-4 : ℝ) * cos xx * sin xx * dx * nn3 + 2 * cos xx ^ 2 * dn3 + 2 * (1 + cos ww) * sin ww * dw * nn4 - (1 + cos ww) ^ 2 * dn4`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.formulaDerivExpr

Source hash: c2d0ab98cf3dfe4dcf08e2029ce80163a7ccf657f84dbd95007f146980c85f86

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:165-222, the source contract is `def formulaDerivExpr (ell ex ey ew ev esx ecx esy ecy esw ecw esv ecv ep : CertificateExpr) (dl dx dy dw dv : $\mathbb{Q}$) : CertificateExpr := let nn4 := .add (.add (.add (.mul ell ey) (.mul (.rational (1 / 2)) ep)) (.neg ex)) (.mul ecx (.add esy (.neg esx))) let nn3 := .add (.add (.add (.mul ell ev) ep) (.neg ew)) (.mul (.add ecw (.rational 2)) (.add esv (.neg esw))) let dn4 := .add (.add (.add (.add (.mul (.rational dl) ey) (.mul ell (.rational dy))) (.neg (.rational dx))) (.mul (.mul (.neg esx) (.rational dx)) (.add esy (.neg esx)))) (.mul ecx (.add (.mul ecy (.rational dy)) (.neg (.mul ecx (.rational dx))))) let dn3 := .add (.add (.add (.add (.mul (.rational dl) ev) (.mul ell (.rational dv))) (.neg (.rational dw))) (.mul (.mul (.neg esw) (.rational dw)) (.add esv (.neg esw)))) (.mul (.add ecw (.rational 2)) (.add (.mul ecv (.rational dv)) (.neg (.mul ecw (.rational dw))))) .add (.add (.mul (.mul (.mul (.mul (.rational (-4)) ecx) esx) (.rational dx) nn3) (.mul (.mul (.rational 2) (.mul ecx ecx)) dn3)) (.add (.mul (.mul (.mul (.mul (.rational 2) (.add (.rational 1) ecw)) esw) (.rational dw) nn4) (.neg (.mul (.mul (.add (.rational 1) ecw) (.add (.rational 1) ecw)) dn4))))). The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.`

Findings: none.

MiddleFaceAssembly.EResidual.reduced

Source hash: ccfc5bbc3e7c269d91f270f90b97e975e0915b24935827f4200f4750d122e6b7c

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:135-144, the source contract is `def reduced (t xi eta rho sigma : $\mathbb{R}$) : $\mathbb{R}$:= let ll := (lamMid b : $\mathbb{R}$) + t let xx := (b.centers 0 : $\mathbb{R}$) + b.slopes 0 * t + xi let yy := (b.centers 1 : $\mathbb{R}$) + b.slopes 1 * t + b.yXiSlope * xi + eta let ww := (b.centers 2 : $\mathbb{R}$) + b.slopes 2 * t + rho let vv := (b.centers 3 : $\mathbb{R}$) + b.slopes 3 * t + b.vRhoSlope * rho + sigma let nn4 := ll * yy + π / 2 - xx + cos xx * (sin yy - sin xx) let nn3 := ll * vv + π - ww + (cos ww + 2) * (sin vv - sin ww) 2 * cos xx ^ 2 * nn3 - (1 + cos ww) ^ 2 * nn4. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.`

Findings: none.

MiddleFaceAssembly.EResidual.reduced_hasDerivAt_eta

Source hash: 6360f8f0c5e7362adbbd3906be241fc3112c0fa4aed9c65f6b00b6c0506d2a0e

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:358-375, the source contract is `theorem reduced_hasDerivAt_eta (t xi eta rho sigma : $\mathbb{R}$) : HasDerivAt (fun u => reduced b t xi u rho sigma) (derivEta b t xi eta rho sigma) eta. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.`

Findings: none.

MiddleFaceAssembly.EResidual.reduced_hasDerivAt_rho

Source hash: 5cd6c45099afc8ddab65369fcdbbfab3f07c85f8b5c029f56ce626072c6d9e4f

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:376-397, the source contract is theorem reduced_hasDerivAt_rho (t xi eta rho sigma : $\mathbb{R}$) : HasDerivAt (fun u => reduced b t xi eta u sigma) (derivRho b t xi eta rho sigma) rho. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.reduced_hasDerivAt_sigma

Source hash: bd919b6d5eb11263919e208acc2f1dc4dcb2acf50a28cc2e3a87fcc770f90e3b

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:398-418, the source contract is theorem reduced_hasDerivAt_sigma (t xi eta rho sigma : $\mathbb{R}$) : HasDerivAt (fun u => reduced b t xi eta rho u) (derivSigma b t xi eta rho sigma) sigma. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.reduced_hasDerivAt_t

Source hash: 14eb220aacc6bd2935a6edcbfbb7db3a21ce8b05aba5a15bc8f61479beb9b277

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:311-336, the source contract is theorem reduced_hasDerivAt_t (t xi eta rho sigma : $\mathbb{R}$) : HasDerivAt (fun u => reduced b u xi eta rho sigma) (derivT b t xi eta rho sigma) t. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.EResidual.reduced_hasDerivAt_xi

Source hash: 9be284052453dd5776bd813cbdef160106e0042f33d5c122d353506cebfd8680

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:337-357, the source contract is theorem reduced_hasDerivAt_xi (t xi eta rho sigma : $\mathbb{R}$) : HasDerivAt (fun u => reduced b t u eta rho sigma) (derivXi b t xi eta rho sigma) xi. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.FResidual.derivEta

Source hash: 074c2a2bce7a0d615a389c52439e114479e4ad4b8be810f92b43ae51021af99c

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:971-977, the source contract is `def derivEta (t xi eta : ℝ) : ℝ := let xx := (b.centers 0 : ℝ) + b.slopes 0 * t + xi let yy := (b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta let ll := (lamMid b : ℝ) + t let bb := ll * yy + π / 2 - xx cos xx * cos yy - sin xx * cos yy * bb - sin xx * sin yy * ll`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.FResidual.derivT

Source hash: 88426adb6d025457273a686218437b5b2fa20ed7c46488942a2e65430d12f11b

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:958-970, the source contract is `def derivT (t xi eta : ℝ) : ℝ := let xx := (b.centers 0 : ℝ) + b.slopes 0 * t + xi let yy := (b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta let ll := (lamMid b : ℝ) + t let bb := ll * yy + π / 2 - xx let dx := (b.slopes 0 : ℝ) let dy := (b.slopes 1 : ℝ) (-dx) * sin xx * (sin yy - sin xx) + cos xx * (dy * cos yy - dx * cos xx) - (dx * cos xx * sin yy * bb + sin xx * dy * cos yy * bb + sin xx * sin yy * (yy + ll * dy - dx))`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.FResidual.reduced

Source hash: 0e2d8d4b1eb446b26168d3d1be2a5c6a59a042bb296a5fe1b9a5da048c481863

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:946-952, the source contract is `def reduced (t xi eta : ℝ) : ℝ := let xx := (b.centers 0 : ℝ) + b.slopes 0 * t + xi let yy := (b.centers 1 : ℝ) + b.slopes 1 * t + b.yXiSlope * xi + eta let ll := (lamMid b : ℝ) + t cos xx * (sin yy - sin xx) - sin xx * sin yy * (ll * yy + π / 2 - xx)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.FResidual.reduced_hasDerivAt_eta

Source hash: 3fc54562c683d5fdcf05f0c5a400e333c95946017d61238476c9d5e2fb027ad3

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1009-1036, the source contract is `theorem reduced_hasDerivAt_eta (t xi eta : ℝ) : HasDerivAt (fun u => reduced b t xi u) (derivEta b t xi eta) eta`. The brick uses affine, coupled angle predictors and a closed error box. The residual

order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.FResidual.reduced_hasDerivAt_t

Source hash: e3ee85a81e52b7d5809d62785536115b9fa172b7ee957d2ee76c2badb09aafc0

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:978-1008, the source contract is theorem `reduced_hasDerivAt_t (t xi eta : ℝ) : HasDerivAt (fun u => reduced b u xi eta) (derivT b t xi eta) t`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.InSymmetricBox

Source hash: 7113db679549bacbee758495cf87555b49cdbc440060902d1105828bd48ba194

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:24-26, the source contract is `abbrev InSymmetricBox := ScalarSuffixCertificate.InSymmetricBox`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.B3

Source hash: 89bf9f374fc276401ac90ed32d58d6789681d13bdd40586c5140a98d2d570b68

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:71-71, the source contract is `def B3 (p : Point) : ℝ := lam b p * v b p + π - w b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.B4

Source hash: 4571608d53f0160a6715950590485e849c7587509687cead6580411b7968f1d6

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:68-68, the source contract is `def B4 (p : Point) : ℝ := lam b p * y b p + π / 2 - x b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is $(-F, C4, E, C3)$; branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.C3

Source hash: 2033f5e817a7ce4ac881490411dee667ce9a476eadf4a060349976ec81a234e9

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:78-78, the source contract is `def C3 (p : Point) : ℝ := cos (w b p) - lam b p * cos (v b p)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.C4

Source hash: 60b08b58076f301aaff633a26cf94d71e808af4d3ade0beea2f7a2c0dba972c5

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:75-75, the source contract is `def C4 (p : Point) : ℝ := cos (x b p) - lam b p * cos (y b p)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.D3

Source hash: 68fc1a2e05e1a65f1772f92a4660582f4e3c321b287cc3594448b912c6e7c624

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:72-72, the source contract is `def D3 (p : Point) : ℝ := sin (v b p) - sin (w b p)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.D4

Source hash: b20a244421f72357884724349d2944bd92ce6795e69082a93fe34e5d606f3a73

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:69-69, the source contract is `def D4 (p : Point) : ℝ := sin (y b p) - sin (x b p)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.E

Source hash: 7b79ccc631128297b34feba1be2bb747e1b7da7d54a858df51a0cb1a25f13437

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:79-80, the source contract is `def E (p : Point) : ℝ := 2 * cos (x b p) ^ 2 * N3 b p - (1 + cos (w b p)) ^ 2 * N4 b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face

hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.F

Source hash: 6d400bd716c439d9d2d74f6bd49be9dc19b55b45847d35d694074dc09fbf9777

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:76-77, the source contract is `def F (p : Point) : ℝ := cos (x b p) * D4 b p - sin (x b p) * sin (y b p) * B4 b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.G

Source hash: cd83bee785b1a55d445030d234df204752ee92464928a982e6215564d1ecad3b

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:81-83, the source contract is `def G (p : Point) : ℝ := cos (x b p) * (B3 b p + D3 b p) - (1 + cos (w b p)) * B4 b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.K3

Source hash: ced932efb916edd340f60f22ba0266b2b91a924f0bdc75b35e1b476fed409a5e

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:84-89, the source contract is `def K3 (p : Point) : ℝ := let h3 := (1 + cos (w b p)) / 2 * h3 * ((1 + cos (w b p)) / sin (w b p) - (lam b p ^ 2 + cos (w b p)) / (lam b p * (lam b p * sin (v b p)))) - 2 * (B3 b p + D3 b p)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.N3

Source hash: 558a63c2734e558c548a2f1daef67adb206ef229c2adc0edc698dc7c3f00e2c9

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:73-74, the source contract is `def N3 (p : Point) : ℝ := B3 b p + (cos (w b p) + 2) * D3 b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.N4

Source hash: 471118e366d83932c15b82b181a3387515c42d96f12d08eec3d20f86b701eec8

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:70-70, the source contract is `def N4 (p : Point) : ℝ := B4 b p + cos (x b p) * D4 b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.lam

Source hash: 09c8d628c82b18ad9565a08f3c76eeb7195571504223cb2dac220bc1b4b12c5a

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:62-62, the source contract is `def lam (p : Point) : ℝ := (lamMid b : ℝ) + p.t`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.pointOf

Source hash: 123eaac2727784b857a8a1d07bab6e08557c023e8c73b6b961970da83630664

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:92-93, the source contract is `def pointOf (t : ℝ) (z : Vec4) : Point := ⟨t, z 0, z 1, z 2, z 3⟩`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.radiiReal

Source hash: 1a51bc7fbed7528bb90b612010e93b671df7d5284f9acc9f65a81a654617ee53

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:91-91, the source contract is `def radiiReal : Vec4 := fun j => (b.radii j : ℝ)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.rootMap

Source hash: bbd8ee6a24e6b32afa58618b3b175c83e9f669c90a7f1ad7d90339051f50892f

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:94-97, the source contract is `def rootMap (t : ℝ) (z : Vec4) : Vec4 := let p := pointOf t z ![-F b p, C4 b p, E b p, C3 b p]`. The map has four coordinates in the order (-F,C4,E,C3). The sign on F is essential: it converts positive F on the low xi face to the negative-low convention. The incidence and equal-area coordinates are not permuted.

Findings: none.

MiddleFaceAssembly.Model.rootMap_continuous

Source hash: 6092edc775a5fa26e9773ae9ff53a2555f68e09f5126315e8d8440fbed6260b3

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:114-134, the source contract is `theorem rootMap_continuous (t : ℝ) : Continuous (rootMap b t)`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.v

Source hash: 384327e74651ca26a49098121b5edc0d2aec2203e7adb9b9c2f01e9786a1b96b

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:66-67, the source contract is `def v (p : Point) : ℝ := b.centers 3 + b.slopes 3 * p.t + b.vRhoSlope * p.rho + p.sigma`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.w

Source hash: e0c3c0b0f31f78a2f6c2d6cd4c7b667830ebe97b8269a630415448ba726152aa

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:65-65, the source contract is `def w (p : Point) : ℝ := b.centers 2 + b.slopes 2 * p.t + p.rho`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.x

Source hash: 4080f6cd78cee5e77729c1d21f119ce966fec918b9c6006a3798cb8dc3bc93f

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:63-63, the source contract is `def x (p : Point) : ℝ := b.centers 0 + b.slopes 0 * p.t + p.xi`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable

with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Model.y

Source hash: e9aa09104ad8189cc4fa9f32855b42af7f901b88654816cd3974a6e8c5d751d4

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:64-64, the source contract is `def y (p : Point) : ℝ := b.centers 1 + b.slopes 1 * p.t + b.yXiSlope * p.xi + p.eta`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Point

Source hash: f6b6070cff3d67fea5f3a1bb07944579342b7bc314fcbc32fbf625a57ff72fc6

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:50-61, the source contract is `structure Point where t : ℝ xi : ℝ eta : ℝ rho : ℝ sigma : ℝ`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.PointEquationSemantics

Source hash: 478344dbc781da1c2b9f6b07701b41f4488a93fff689eda8b3059bdb2ad9f499

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1537-1551, the source contract is `structure PointEquationSemantics (b : Brick) (p : Point) : Prop where typeThreeHeight_gt_half : 1 / 2 < (1 + cos (w b p)) / 2 typeThreeHeight_lt_one : (1 + cos (w b p)) / 2 < 1 typeFourHeight_pos : 0 < cos (x b p) typeFourHeight_lt_one : cos (x b p) < 1 equalArea_of_E_zero : E b p = 0 → LeanSuffixAnalytic.typeThreeArea (lam b p) ((1 + cos (w b p)) / 2) = LeanSuffixAnalytic.typeFourArea (lam b p) (cos (x b p)) stationary_of_F_zero : F b p = 0 → LeanSuffixAnalytic.typeFourFold (lam b p) (cos (x b p)) = 0 perimeter_lt_of_G_neg : G b p < 0 → LeanSuffixAnalytic.typeThreePerimeter (lam b p) ((1 + cos (w b p)) / 2) < LeanSuffixAnalytic.typeFourPerimeter (lam b p) (cos (x b p))`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.QInterval

Source hash: ab557efab6ff5d67c61f860d37b3645b971e70e7a32bfb18825874961bbbc432

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:22-22, the source contract is `abbrev QInterval := ScalarSuffixCertificate.QInterval`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.Vec4

Source hash: 26b6384e4234f00c2040b7fed071a12c3e6f967500dfcaee15eb69b43a2cf5e3

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:23-23, the source contract is `abbrev Vec4 := ScalarSuffixCertificate.Vec4`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.lamMid

Source hash: 2d4db33efad481c4328c1c7dfb050b5a976c6575fe3896f4791861bbf4743632

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:38-38, the source contract is `abbrev lamMid (b : Brick) :  $\mathbb{Q}$  := (b.lamLo + b.lamHi) / 2`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.pointEquationSemantics

Source hash: c757c9ef8ed58ec827195f2ddcc9f974e6ab47521674ab7d055018fa79f178cf

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1552-1700, the source contract is `theorem pointEquationSemantics (b : Brick) (p : Point) (hlam : $1 < \text{lam } b \text{ } p$) (hbranches : $0 < x \text{ } b \text{ } p \wedge x \text{ } b \text{ } p < \pi / 4 \wedge 0 < y \text{ } b \text{ } p \wedge y \text{ } b \text{ } p < \pi / 4 \wedge 0 < w \text{ } b \text{ } p \wedge w \text{ } b \text{ } p < \pi / 4 \wedge 0 < v \text{ } b \text{ } p \wedge v \text{ } b \text{ } p < \pi / 4$) (hC4 : $C4 \text{ } b \text{ } p = 0$) (hC3 : $C3 \text{ } b \text{ } p = 0$) : PointEquationSemantics b p`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.tInterval

Source hash: f0133991a8583c976d09c8410dc69827234287519ce36ced5deb237d26a4d63

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:41-45, the source contract is `def tInterval (b : Brick) : QInterval := ⟨-tRadius b, tRadius b, by dsimp [tRadius] linarith [b.lamOrdered]⟩`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.tRadius

Source hash: 18f0ba106ace07af0a9a6397628ed7aba1c4508d8c6f08edfb081b20e3385643

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:39-40, the source contract is `abbrev tRadius (b : Brick) : ℚ := (b.lamHi - b.lamLo) / 2`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceAssembly.typeThreeArea_deriv_eq_K3

Source hash: a51562aaac4e05b4f1f95d421868e79ec4eae70d59c9a31981d5897af5fec042

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceAssembly.lean:1701-1774, the source contract is `theorem typeThreeArea_deriv_eq_K3 (b : Brick) (p : Point) (hlam : 1 < lam b p) (hbranches : 0 < w b p ∧ w b p < π / 4 ∧ 0 < v b p ∧ v b p < π / 4) (hC3 : C3 b p = 0) : deriv (LeanSuffixAnalytic.typeThreeArea (lam b p)) ((1 + cos (w b p)) / 2) = K3 b p / ((1 + cos (w b p)) / 2) ^ 3`. The brick uses affine, coupled angle predictors and a closed error box. The residual order is (-F,C4,E,C3); branch and face hypotheses are not interchangeable with the resulting common-zero or strict-perimeter conclusions.

Findings: none.

MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer

Source hash: 65047cf47373d68e2f54c04db5fee6610caf78e9eb55ac3e2777e49e93efeff1

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells.lean:29-74, the source contract is `theorem candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (102001 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (25781 / 25000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. The density intervals are closed and joined at their exact rational endpoints. The exported theorem takes only density bounds, a candidate, and its CMV hypotheses; neither stationary curvature nor externally supplied analytic data is added.

Findings: none.

MiddleFaceCells.candidate_not_isWeightedPerimeterMinimizer_from_51_50

Source hash: 4e139d2067a97ded95a1f47d09a5d5b68e8fd03114a4a32d5431bb047f26e9f1

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells.lean:75-102, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer_from_51_50 {lam : ℝ} (hlower : (51 / 50 : ℝ) ≤ lam) (hupper : lam ≤ (33 / 32 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. The density intervals are closed and joined at their exact rational endpoints. The exported theorem takes only density bounds, a candidate, and its CMV hypotheses; neither stationary curvature nor externally supplied analytic data is added.

Findings: none.

MiddleFaceCells0To19.candidate_not_isWeightedPerimeterMinimizer

Source hash: b28637e44e18b6358362a61f82cb49563ab505b8a06a5dc1be41f5b00b1477fb

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells0To19.lean:40-265, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (102001 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (102201 / 100000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells104To112.candidate_not_isWeightedPerimeterMinimizer

Source hash: 4570e7b8cef789aedef57b10eef603ac2c0b313f81a77a3d704cb2879ca1fd52d

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells104To112.lean:29-133, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (103041 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (25781 / 25000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells20To31.candidate_not_isWeightedPerimeterMinimizer

Source hash: a93747927463ab3765198fcfbb919eb2715b5ee65fc623d802fbd018e733d4eb

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells20To31.lean:32-169, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (102201 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (102321 / 100000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬`

`candidate.IsWeightedPerimeterMinimizer`. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells32To43.candidate_not_isWeightedPerimeterMinimizer

Source hash: 2e75386747807f06bb44919c15005abd3135f1cb638447100a8d72debc73bacf

Relation: equivalent

Equivalent in the declared constructor/section context. At `MiddleFaceCells32To43.lean:32-169`, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (102321 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (102441 / 100000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells44To55.candidate_not_isWeightedPerimeterMinimizer

Source hash: 3d231e37509eafaa66d654d02538cb90db625899f8572fbb742c157a20888d4a

Relation: equivalent

Equivalent in the declared constructor/section context. At `MiddleFaceCells44To55.lean:32-169`, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (102441 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (102561 / 100000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells56To67.candidate_not_isWeightedPerimeterMinimizer

Source hash: b26f20f880618fce0adba5633b5b3fe601d543d544685cd5c5e6e541816f591e

Relation: equivalent

Equivalent in the declared constructor/section context. At `MiddleFaceCells56To67.lean:32-169`, the source contract is theorem `candidate_not_isWeightedPerimeterMinimizer {lam : ℝ} (hlower : (102561 / 100000 : ℝ) ≤ lam) (hupper : lam ≤ (102681 / 100000 : ℝ)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells68To79.candidate_not_isWeightedPerimeterMinimizer

Source hash: 64c898dd26ac2e9b8857bb1839001b7ae6426e285d6390e32038f347657f972f

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells68To79.lean:32-169, the source contract is theorem candidate_not_isWeightedPerimeterMinimizer {lam : $\mathbb{R}$ } (hlower : (102681 / 100000 : $\mathbb{R}$) $\leq$ lam) (hupper : lam $\leq$ (102801 / 100000 : $\mathbb{R}$)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : $\neg$ candidate.IsWeightedPerimeterMinimizer. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells80To91.candidate_not_isWeightedPerimeterMinimizer

Source hash: 506c0e3dda4c848a77bd896e79f1ba0ef204016d55e8b8f1dccd1e1a06f429f4

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells80To91.lean:32-169, the source contract is theorem candidate_not_isWeightedPerimeterMinimizer {lam : $\mathbb{R}$ } (hlower : (102801 / 100000 : $\mathbb{R}$) $\leq$ lam) (hupper : lam $\leq$ (102921 / 100000 : $\mathbb{R}$)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : $\neg$ candidate.IsWeightedPerimeterMinimizer. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

MiddleFaceCells92To103.candidate_not_isWeightedPerimeterMinimizer

Source hash: eb8d5eea5007ec9d93249eb58e5ea168ebe71c7c90ef757bae9ec70ea6beb8ed

Relation: equivalent

Equivalent in the declared constructor/section context. At MiddleFaceCells92To103.lean:32-169, the source contract is theorem candidate_not_isWeightedPerimeterMinimizer {lam : $\mathbb{R}$ } (hlower : (102921 / 100000 : $\mathbb{R}$) $\leq$ lam) (hupper : lam $\leq$ (103041 / 100000 : $\mathbb{R}$)) (candidate : _root_.FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : $\neg$ candidate.IsWeightedPerimeterMinimizer. This batch uses only the closed endpoint inequalities and candidate CMV hypotheses. Its case splits cover the stated consecutive middle bricks without excluding shared endpoints; the output is actual candidate nonminimality, not just a stationary scalar inequality.

Findings: none.

OneSidedCircularCap

Source hash: a9dd2ba7ee653be367b6a2072de1384c3362b0214117e3aecceae763bc6f8403

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:100-112, the source contract is structure OneSidedCircularCap where chord : $\mathbb{R}$ theta : $\mathbb{R}$ midpointX : $\mathbb{R}$ baseY : $\mathbb{R}$ side : CapSide chord_pos : $0 < \text{chord}$ theta_pos : $0 < \text{theta}$ theta_lt_pi : $\text{theta} < \pi$. The coordinate carriers

live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcLength

Source hash: 60daa6fa20eb20410aecf4ceebd5a41e50c48a0eb958e6b55f1bcb20cd85ff25

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:167-170, the source contract is `def arcLength (c : OneSidedCircularCap) : ℝ := c.chord * ell c.theta`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcLength_pos

Source hash: 615464c0ac73967d872be72b9e5d931fb0644d2db70b6e9168ad1dd7c81aa184

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:220-222, the source contract is `theorem arcLength_pos (c : OneSidedCircularCap) : 0 < c.arcLength`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcPoint

Source hash: 2f12b44987266410d1b21333c27225fca467817287b4028245b465856f5540c3

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:124-133, the source contract is `def arcPoint (c : OneSidedCircularCap) (t : ℝ) : PlanePoint := match c.side with | .upper => (c.midpointX + c.radius * sin t, c.baseY + c.radius * (cos t - cos c.theta)) | .lower => (c.midpointX + c.radius * sin t, c.baseY - c.radius * (cos t - cos c.theta))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcPoint_in_closed_side

Source hash: 4692c99be4beede3a58f1e164acd1cfe21db7a9160da63340de9490acd46c184

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:240-252, the source contract is `theorem arcPoint_in_closed_side (c : OneSidedCircularCap) {t : ℝ} (ht : |t| ≤ c.theta) : match c.side with | .upper => c.baseY ≤ (c.arcPoint t).2 | .lower => (c.arcPoint t).2 ≤ c.baseY`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcPoint_in_open_side

Source hash: 3dbf6a865d1d6331f9c3406c05af249acf00d7c487349d982b588394a814d697

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:253-266, the source contract is `theorem arcPoint_in_open_side (c : OneSidedCircularCap) {t : ℝ} (ht : |t| < c.theta) : match c.side with | .upper => c.baseY < (c.arcPoint t).2 | .lower => (c.arcPoint t).2 < c.baseY`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcPoint_left

Source hash: 4e22ae8901c6f895322d9f1591e370b613671c823882adf707a5fac1ae750287

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:228-232, the source contract is `theorem arcPoint_left (c : OneSidedCircularCap) : c.arcPoint (-c.theta) = c.leftEndpoint`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcPoint_right

Source hash: ba639553cc75debbcac47cb7f164b0cdee0e14e9586cdc6d1cf857f7fa27f21b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:233-239, the source contract is `theorem arcPoint_right (c : OneSidedCircularCap) : c.arcPoint c.theta = c.rightEndpoint`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.arcTrace

Source hash: 5e3edb81f4f16ef8dbff7c15d5756a176b1f99c3bd5fb86ce703134a0a727a89

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:656-661, the source contract is `def arcTrace (c : OneSidedCircularCap) : Set PlanePoint := {p | c.radiusSquaredAt p = c.radius ^ 2 ^ match c.side with | .upper => c.baseY ≤ p.2 | .lower => p.2 ≤ c.baseY}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.carrier

Source hash: e0644dd0b5f330958b79efbb3810eb1cc2f4356e5db251a95736b532b75bfcf1

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:146-152, the source contract is `def carrier (c : OneSidedCircularCap) : Set PlanePoint := {p | c.radiusSquaredAt p ≤ c.radius ^ 2 ^ match c.side with | .upper => c.baseY ≤ p.2 | .lower => p.2 ≤ c.baseY}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.center

Source hash: 58a018a3af71f1eb6c5aebb7ee395b60aab79f7b55a9d384d43694baace0d346

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:117-123, the source contract is `def center (c : OneSidedCircularCap) : PlanePoint := match c.side with | .upper => (c.midpointX, c.baseY - c.radius * cos c.theta) | .lower => (c.midpointX, c.baseY + c.radius * cos c.theta)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.chordCarrier

Source hash: 0705736836c9203b120221cb0770c365bada15ab749e72d0791d55f58eba4c1e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:160-163, the source contract is `def chordCarrier (c : OneSidedCircularCap) : Set PlanePoint := {p | p.2 = c.baseY ^ |p.1 - c.midpointX| ≤ c.chord / 2}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.chordCarrier_subset_carrier

Source hash: a35dcecd8fd218da748131deaec9b02f5328c1034952ad269b6ba1f5b686d947

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:311-334, the source contract is `theorem chordCarrier_subset_carrier (c : OneSidedCircularCap) : c.chordCarrier  $\subseteq$  c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.ell_theta_pos

Source hash: 88c4230a8ce8dbe3ed5e64385c18de0588039e6c242ead1e2df6299d087f0ad8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:216-219, the source contract is `theorem ell_theta_pos (c : OneSidedCircularCap) :  $0 < \text{ell } c.\text{theta}$` . The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.euclideanArea

Source hash: 5c476b88f35c9e9f6afc55ed7ed9ef6be1395a7b5a3003452ff8691443c49da

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:164-166, the source contract is `def euclideanArea (c : OneSidedCircularCap) :  $\mathbb{R} := c.\text{chord}^2 * \text{area } c.\text{theta}$` . The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.euclideanArea_pos

Source hash: 3902ccdc9eb3694014c5a47aad3179ebe09a3fd3cddb066b9c4c1ca933463218

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:213-215, the source contract is `theorem euclideanArea_pos (c : OneSidedCircularCap) :  $0 < c.\text{euclideanArea}$` . The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.frontier_carrier

Source hash: d9cb2c749721b51457faae17a95033cf908527d3fdc652ebedac fbb2292d579d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:697-761, the source contract is `theorem frontier_carrier (c : OneSidedCircularCap) : frontier c.carrier = arcTrace c ∪ c.chordCarrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.interiorCarrier

Source hash: d9986ab2dd520dfd41d6ccb99ca9878df085185d6c10c136685b43549b4c4dc1

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:153-159, the source contract is `def interiorCarrier (c : OneSidedCircularCap) : Set PlanePoint := {p | c.radiusSquaredAt p < c.radius ^ 2 ∧ match c.side with | .upper => c.baseY < p.2 | .lower => p.2 < c.baseY}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.interiorCarrier_subset_interior

Source hash: e055c7441e3e2230b6bd3c3989fc3043eb631862e9cdc4e7fe3a7963b0242ddc

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:687-696, the source contract is `theorem interiorCarrier_subset_interior (c : OneSidedCircularCap) : c.interiorCarrier ⊆ interior c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.isClosed_carrier

Source hash: d757ff8cbf988a97323a1f1c27dc667eab9b6475f2eb7d1c16beffa82e3009a4

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:364-391, the source contract is `theorem isClosed_carrier (c : OneSidedCircularCap) : IsClosed c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.isOpen_interiorCarrier

Source hash: ec8c11c26c05148d770698c167cdec56869de41dc90864a365cf2514de87bc1

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:662-686, the source contract is `theorem isOpen_interiorCarrier (c : OneSidedCircularCap) : IsOpen c.interiorCarrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.leftEndpoint

Source hash: a193f337e9021d7e081f4666c9f784092558b2268ec0e9060f11b3d18dfc3c8d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:134-137, the source contract is `def leftEndpoint (c : OneSidedCircularCap) : PlanePoint := (c.midpointX - c.chord / 2, c.baseY)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.measurableSet_carrier

Source hash: e0f551f525fbf0409babb736205ec484e0a0c2ff7039ce55de1e7a7fc5595a27

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:335-363, the source contract is `theorem measurableSet_carrier (c : OneSidedCircularCap) : MeasurableSet c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.mem_chordCarrier_of_mem_carrier_of_eq_base

Source hash: 424261710f758c71ebe7e57dd13ebd11ffc59cb51b840ff8fc86319890ddae19

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:291-310, the source contract is `theorem mem_chordCarrier_of_mem_carrier_of_eq_base (c : OneSidedCircularCap) {p : PlanePoint} (hp : p ∈ c.carrier) (hy : p.2 = c.baseY) : p ∈ c.chordCarrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.radius

Source hash: e898cc7066a75a606fdc3b1d1e563939ad569bdb0f66fd289569e58c7f4c44b8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:113-116, the source contract is `def radius (c : OneSidedCircularCap) : ℝ := c.chord / (2 * sin c.theta)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.radiusSquaredAt

Source hash: a1c0f970b426c7f0e6a013d8d8bbb613fbba3b5c253445dacecdb9b47cb6785a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:142-145, the source contract is `def radiusSquaredAt (c : OneSidedCircularCap) (p : PlanePoint) : ℝ := (p.1 - c.center.1) ^ 2 + (p.2 - c.center.2) ^ 2`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.radius_mul_sin

Source hash: 4b800b6287f8bc78c34b40f39fff02ab7d25631ad2fd22f8819095ceda60143e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:223-227, the source contract is `theorem radius_mul_sin (c : OneSidedCircularCap) : c.radius * sin c.theta = c.chord / 2`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.radius_pos

Source hash: 7bcdcf3411946f45cb3c854e76728bf4e1477fc16c0bd8673b5d0f6f904d6adf

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:210-212, the source contract is `theorem radius_pos (c : OneSidedCircularCap) : 0 < c.radius`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.rightEndpoint

Source hash: e83d4942af5a95dd72bd914bc15f416e046e239fc23caff7a851807f832f26a3

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:138-141, the source contract is `def rightEndpoint (c : OneSidedCircularCap) : PlanePoint := (c.midpointX + c.chord / 2, c.baseY)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.sin_theta_pos

Source hash: 8c22795350624813eb7b9aec2fdcda7bfc438d144ebb38b607db3cac77688583

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:207-209, the source contract is `theorem sin_theta_pos (c : OneSidedCircularCap) : 0 < sin c.theta`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

OneSidedCircularCap.volume_carrier

Source hash: 0d51f9d06b45af9eac50b2062d616aee51a260200ad5a2e96d2ecbe8e98be85c

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1996-2028, the source contract is `theorem volume_carrier (c : OneSidedCircularCap) : volume c.carrier = ENNReal.ofReal c.euclideanArea`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

PlanePoint

Source hash: b6c33ab7d9e6652b4833188977c0694ee665f31ebd07e572ea7a67ee7a8a0183

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:30-33, the source contract is `abbrev PlanePoint := ℝ × ℝ`. The abbreviation is exactly the Cartesian product of two real lines. Euclidean length is deliberately introduced separately; this declaration alone does not endow the product norm with Euclidean length semantics.

Findings: none.

ReplacementAssembly

Source hash: e0b73790946058b82b8d6a4bdde47ad7560092e26acf3d58cb0c91dc0cfa445f

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3281-3288, the source contract is structure ReplacementAssembly where core : StripCore replacementAngle : $\mathbb{R}$ replacementAngle_pos : $0 < \text{replacementAngle}$ replacementAngle_lt_pi : replacementAngle $< \pi$. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.below_exposedLowerChord_not_mem

Source hash: 3316f6365842786c2f5614825bd4efdba288774b79cdd05c569f08a4717428cd

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3442-3459, the source contract is theorem below_exposedLowerChord_not_mem (a : ReplacementAssembly) {p : PlanePoint} (hp : p ∈ a.exposedLowerChord.carrier) {epsilon : $\mathbb{R}$ } (hepsilon : $0 < \text{epsilon}$) : (p.1, p.2 - epsilon) $\notin$ a.carrier. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.boundaryMeasure

Source hash: 75dfabe9f5e9f8aa75724a2e5f410dd1e358abaf0454833ef91226757c6630cb

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3310-3314, the source contract is def boundaryMeasure (a : ReplacementAssembly) : Measure PlanePoint := leftCoreBoundaryMeasure a.core + rightCoreBoundaryMeasure a.core + capBoundaryMeasure a.upperCap + segmentBoundaryMeasure a.exposedLowerChord. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.boundaryTrace

Source hash: c14f256441519af3907ade461995fd9aa134a8b3a94b950059f337d7edd4be97

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3487-3492, the source contract is def boundaryTrace (a : ReplacementAssembly) : Set PlanePoint := StripCore.leftArcTrace a.core ∪ (StripCore.rightArcTrace a.core ∪ (OneSidedCircularCap.arcTrace a.upperCap ∪ a.exposedLowerChord.carrier)). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.boundaryTrace_subset_frontier_carrier

Source hash: 6c897595e7667c418f2a95e4221cabe00a508a6ab4dac16eff0c51ccb2b4d8a9

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3933-3940, the source contract is `theorem boundaryTrace_subset_frontier_carrier (a : ReplacementAssembly) : a.boundaryTrace  $\subseteq$  frontier a.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.carrier

Source hash: 36b80491dc0fecbf358d87c6bfe7194faf4bcad4ac91372874e8af60a30fd23e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3305-3309, the source contract is `def carrier (a : ReplacementAssembly) : Set PlanePoint := a.core.carrier  $\cup$  a.upperCap.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.exposedLowerChord

Source hash: f5b2acb16292464686c2dc7d4da985f73189845b89d73512b7e850cb9f2f8b26

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3299-3304, the source contract is `def exposedLowerChord (a : ReplacementAssembly) : HorizontalSegment where chord := a.core.chord midpointX := 0 baseY := -1 chord_pos := a.core.chord_pos`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.exposedLowerChord_subset_carrier

Source hash: 4d73b08bb97103964040bbdf24b1dbc10e728893bf174f22b32f933c62dcef7f

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3429-3441, the source contract is `theorem exposedLowerChord_subset_carrier (a : ReplacementAssembly) : a.exposedLowerChord.carrier  $\subseteq$  a.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.exposedLowerChord_subset_frontier

Source hash: f8eb7dd82f960c22028fc1d5bb6797612d50aa3f62628a630157cdc32d61c499

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3460-3481, the source contract is **theorem exposedLowerChord_subset_frontier** ($a : \text{ReplacementAssembly}$) :

$a.\text{exposedLowerChord}.\text{carrier} \subseteq \text{frontier } a.\text{carrier}$. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.frontier_carrier

Source hash: 5eb2c9bf9d9bf270c9c126108b477ade628695b91787f859535adefef6f41836a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3941-3951, the source contract is **theorem frontier_carrier** ($a : \text{ReplacementAssembly}$) : $\text{frontier } a.\text{carrier} = a.\text{boundaryTrace}$. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.frontier_carrier_subset_boundaryTrace

Source hash: 690416c38a1e91ad4ec069f59dc7a6046805e39bbae1ba5d6d8c4028b6ea7f3e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3595-3613, the source contract is **theorem frontier_carrier_subset_boundaryTrace** ($a : \text{ReplacementAssembly}$) : $\text{frontier } a.\text{carrier} \subseteq a.\text{boundaryTrace}$. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.isClosed_carrier

Source hash: 7d390bd6162828599bf858ab60165e80e65e5c04a4677e586ff23aba97d10394

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3579-3580, the source contract is **theorem isClosed_carrier** ($a : \text{ReplacementAssembly}$) : $\text{IsClosed } a.\text{carrier}$. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.upperCap

Source hash: b30d86506333a29d0fea0a0c213d5ab5e404eb8902dd90c6019d57dad3b8f56f

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3289-3298, the source contract is `def upperCap (a : ReplacementAssembly) : OneSidedCircularCap` where `chord := a.core.chord` `theta := a.replacementAngle` `midpointX := 0` `baseY := 1` `side := .upper` `chord_pos := a.core.chord_pos` `theta_pos := a.replacementAngle_pos` `theta_lt_pi := a.replacementAngle_lt_pi`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.weightedArea

Source hash: 30517c93164485b2476e45ea547b19187ad53db44a33af2bf70144ca4c93c662

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3326-3329, the source contract is `def weightedArea (lam : ℝ) (a : ReplacementAssembly) : ℝ := WeightedArea lam a.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.weightedArea_formula

Source hash: f1ba4256fd2f5c3e87624e59042924594647c2f49dcba6ed06cd14dc8d287789

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3338-3362, the source contract is `theorem weightedArea_formula (lam : ℝ) (a : ReplacementAssembly) : a.weightedArea lam = a.core.euclideanArea + lam * a.upperCap.euclideanArea`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.weightedPerimeter

Source hash: ff8986acb5666758cd2b4555262842f122d66f79d9aaa241c37692769a5c2fbc

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3330-3332, the source contract is `def weightedPerimeter (lam : ℝ) (a : ReplacementAssembly) : ℝ := WeightedPerimeter lam a.boundaryMeasure`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

ReplacementAssembly.weightedPerimeter_formula

Source hash: 0ec330da63b1183620a69c25b9b52f69e166e1fbc55093558ecffbfbdc73ed619

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:3363-3387, the source contract is `theorem weightedPerimeter_formula (lam : ℝ) (a : ReplacementAssembly) : a.weightedPerimeter lam = a.core.boundaryArcLength + lam * a.upperCap.arcLength + a.exposedLowerChord.euclideanLength`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

RightPrototypeCell.B3

Source hash: fbc773e40d79a1d6d31822cdbc56d9a9607c6a7f8abc1ba0f47027df7b631eae

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:122-122, the source contract is `def B3 (p : Point) : ℝ := lam p * v p + π - w p`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.B4

Source hash: 30e079ec002735e04de6acf4de75c1ef255bd47fc8db540544074fad80b3a6bc

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:119-119, the source contract is `def B4 (p : Point) : ℝ := lam p * y p + π / 2 - x p`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.Brick

Source hash: c2ed826f9f797c84fdf65a35205275ba3596085d3c93941f0721f070655d9db0

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:20-22, the source contract is `abbrev Brick := MiddleFaceAssembly.Brick`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.C3

Source hash: bb334eb31664c80305d156013a66d4b3847b5f60bc740d68b3db43dd6612439b

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:128-128, the source contract is `def C3 (p : Point) : ℝ := cos (w p) - lam p * cos (v p)`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.C3_sigma_high_face_margin

Source hash: 5b3bf1ce51b69bccdd8bfe9c8c7de37ad57b8cb9ed698111103e6ab5d416bdcdb

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:1818-1844, the source contract is `theorem C3_sigma_high_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 3 = radiiReal 3 → (1 / 10000000000 : ℝ) < C3 (pointOf t z)`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.C3_sigma_low_face_margin

Source hash: 6fb3362f5b47edfd5c864f48fb4f2f4eb79f2e4e22847db0995b5bffe22e621e

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:1791-1817, the source contract is `theorem C3_sigma_low_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 3 = -radiiReal 3 → C3 (pointOf t z) < -(1 / 10000000000 : ℝ)`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.C4

Source hash: 0c76dd8701b0777d1f4e908247b8a477e6ed59455006924fb2e56c23786a4bd0

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:126-126, the source contract is `def C4 (p : Point) : ℝ := cos (x p) - lam p * cos (y p)`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.C4_eta_high_face_margin

Source hash: 0c746a428e5a281f55bd8109c26b40469bd8fb48b26367a9a7fa6a1522dfd978

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:1215-1244, the source contract is theorem C4_eta_high_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \ 1 = \text{radiiReal } 1 \rightarrow (11 / 10000000000 : \mathbb{R}) < C4 \text{ (pointOf } t \ z)$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.C4_eta_low_face_margin

Source hash: b33b3e4037ad423afeac36d4ddac306b90781f3f172e1644c7e6268fd9b3e55b

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:1188-1214, the source contract is theorem C4_eta_low_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \ 1 = -\text{radiiReal } 1 \rightarrow C4 \text{ (pointOf } t \ z) < -(11 / 10000000000 : \mathbb{R})$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.D3

Source hash: 4496291242cc1a371596577f5bdbec61dd292b63c92e9b87cf551b140d32a368

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:123-123, the source contract is def D3 (p : Point) : $\mathbb{R} := \sin (v \ p) - \sin (w \ p)$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.D4

Source hash: 458c17f197940395eb065d606c56a29538dee5cfc382d507f31979fa9b14e3b0

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:120-120, the source contract is def D4 (p : Point) : $\mathbb{R} := \sin (y \ p) - \sin (x \ p)$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.E

Source hash: 473eec5a0b00f868ac4f7ad090edf435fe47f07ca5bd2b59758512636541fbfa

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:129-130, the source contract is `def E (p : Point) : ℝ := 2 * cos (x p) ^ 2 * N3 p - (1 + cos (w p)) ^ 2 * N4 p`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.E_rho_high_face_margin

Source hash: eb3eec8dea60a8df64d88b6b4837361d3d879b92fece643a52712269468a17a0

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:2056-2088, the source contract is `theorem E_rho_high_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 2 = radiiReal 2 → (14 / 10000000000 : ℝ) < E (pointOf t z)`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.E_rho_low_face_margin

Source hash: 6272cbb2e67c611a239abd88e60ef57c905b209d0a231e982ffd7c0dc0e89972

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:2028-2055, the source contract is `theorem E_rho_low_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 2 = -radiiReal 2 → E (pointOf t z) < -(14 / 10000000000 : ℝ)`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.F

Source hash: ff764cf036982bdc0d990c94070c7ae41de884ec397b4b250e1096fedd89ff34

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:127-127, the source contract is `def F (p : Point) : ℝ := cos (x p) * D4 p - sin (x p) * sin (y p) * B4 p`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.F_xi_high_face_margin

Source hash: ffe50f040a2dc348cba7032d1ad7f3678a00d83ce29c2e6aded2308288cab74e

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:1572-1597, the source contract is theorem F_xi_high_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \neq 0 = \text{radiiReal } 0 \rightarrow F(\text{pointOf } t z) < -(86 / 1000000000 : \mathbb{R})$. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.F_xi_low_face_margin

Source hash: 09956ba2c2677ff4925e529c6578299ab10ba3f2d08cee3e462aa50640f59f58

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:1546-1571, the source contract is theorem F_xi_low_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \neq 0 = -\text{radiiReal } 0 \rightarrow (88 / 1000000000 : \mathbb{R}) < F(\text{pointOf } t z)$. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.G

Source hash: 818854a9d2ac34b27a374a124d11eeffc488199875161c552aa19dcc6709e1ea

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:131-137, the source contract is def G (p : Point) : $\mathbb{R} := \cos(x p) * (B3 p + D3 p) - (1 + \cos(w p)) * B4 p$. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.K3

Source hash: dded0a92a5475f561e4ec09b53d31fd672246c1e3a18679f1d82548595f59954

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:238-245, the source contract is def K3 (p : Point) : $\mathbb{R} := \text{let } h3 := (1 + \cos(w p)) / 2^4 * h3 * ((1 + \cos(w p)) / \sin(w p) - (\text{lam } p^2 + \cos(w p)) / (\text{lam } p * (\text{lam } p * \sin(v p)))) - 2 * (B3 p + D3 p)$. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.N3

Source hash: 79fd22b0e88fADF424caa5c32cd0621ba146426141946780c13923b4adb4108c

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:124-125, the source contract is `def N3 (p : Point) : ℝ := B3 p + (cos (w p) + 2) * D3 p`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.N4

Source hash: 8317292c5d87208707548fa6a07d20553105db710dba94daee68266201ce16ef

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:121-121, the source contract is `def N4 (p : Point) : ℝ := B4 p + cos (x p) * D4 p`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.Point

Source hash: cff2e93e4339fbf01155e35df8506905eb069e6d43c06718e6b5df7c41ed32d5

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:109-110, the source contract is `abbrev Point := MiddleFaceAssembly.Point`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.affine_branches

Source hash: 9a77fd11c6c2c0e0a916f823adc4aea4c608cadccacf0dbe2100c6ac6d1bce78

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:707-741, the source contract is `theorem affine_branches : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → let p := pointOf t z 0 < x p ∧ x p < π / 4 ∧ 0 < y p ∧ y p < π / 4 ∧ 0 < w p ∧ w p < π / 4 ∧ 0 < v p ∧ v p < π / 4`. After the let-bound pointOf substitution, all eight inequalities are present: x,y,w,v are each strictly between zero and pi/4. The universal quantifiers range over the prototype's closed density interval and error box, not just over a face or a selected zero.

Findings: none.

RightPrototypeCell.brick

Source hash: 46529d69eb028da84cb1e1cd48a1286b5a20bbf7043954e20190c659fde37b2d

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:23-39, the source contract is `def brick : Brick` where `lamLo := 25781 / 25000` `lamHi := 33 / 32` `centers := ! [2048263294902610889 / 1000000000000000000, 3193765780936307979 / 1000000000000000000, 4769698287901120439 / 1000000000000000000, 2663161348942592843 / 5000000000000000000] slopes := ! [140927032410729353 / 1000000000000000000, 954393643324404653 / 2500000000000000000, 4401546318656929969 / 1000000000000000000, 5503812760533025527 / 1000000000000000000] yXiSlope := 13 / 20 vRhoSlope := 219 / 250 radii := ! [1 / 10000000, 1 / 10000000, 1 / 10000000, 1 / 10000000] lamOrdered := by norm_num radiiPos := by intro j; fin_cases j <;> norm_num. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.`

Findings: none.

RightPrototypeCell.candidate_not_isWeightedPerimeterMinimizer

Source hash: 5c6a266d071e9cf7ce0214b6d5870976fd096dbfbf585b5acb5f6cd350590871

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:2222-2240, the source contract is `theorem candidate_not_isWeightedPerimeterMinimizer {lam0 : ℝ} (hlam : lamInterval.RealContains lam0) (candidate : _root_.FourArcCandidate lam0) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : ¬ candidate.IsWeightedPerimeterMinimizer`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.cellCertificate

Source hash: 37a6b153f1c09f5b0bd0af4e9abeec7bf284e6e510faf353cac76a90f727cac4

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:2089-2179, the source contract is `theorem cellCertificate : MiddleFaceAssembly.CellCertificate brick where lambda_one_lt`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.checkedConditions

Source hash: 907b696193f7401fd5d37dde529193bc1f7dd81a2b600bbb7b76d63e5238086e

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:2180-2184, the source contract is `theorem checkedConditions : MiddleFaceAssembly.Conditions brick`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.descendingK3

Source hash: e9afb66c8c2aa5658ce6fd84c67f02d2de09068f00ce68b48aceab8eeae79c86

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:766-934, the source contract is theorem descendingK3 : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow K3 (pointOf t z) < 0$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.gap_upper

Source hash: bfd6569809511ed8dd358788886ef4c9bc9af352f129411e514e6363ef806197

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:935-1049, the source contract is theorem gap_upper : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow G (pointOf t z) < -(437 / 1000000 : \mathbb{R})$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.lam

Source hash: 80d4ea0b007cc38c9cef09318804f92bb1d2031dffde45077e78cad062961da3

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:111-111, the source contract is def lam (p : Point) : $\mathbb{R} := (lamMid : \mathbb{R}) + p.t$. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.lamInterval

Source hash: f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:79-80, the source contract is def lamInterval : QInterval := ⟨brick.lamLo, brick.lamHi, brick.lamOrdered.le⟩. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.lamMid

Source hash: 6872199d838a1fa3e543ff329eb7ed7979df53e9ab7df478a48be79baedaf0cf

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:75-75, the source contract is `abbrev lamMid :  $\mathbb{Q}$  := (brick.lamLo + brick.lamHi) / 2`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.lambda_iff_centered

Source hash: 8f9d4d2567e466a05274e0772e055d5bcc7985aa0cebd52f3a01300950766dca

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:81-93, the source contract is `theorem lambda_iff_centered {lam :  $\mathbb{R}$ } : lamInterval.RealContains lam  $\leftrightarrow$  tInterval.RealContains (lam - (lamMid :  $\mathbb{R}$ ))`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.lambda_in_retainedSuffix

Source hash: d64b377e3281e7e0b8cb140bc4c8494ca73e2578a3420d0547192d8dcd31aa70

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:94-100, the source contract is `theorem lambda_in_retainedSuffix {lam :  $\mathbb{R}$ } (h : lamInterval.RealContains lam) : LeanSuffixAnalytic.InRetainedSuffix lam`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.pointOf

Source hash: 123eaac2727784b857a8a1d07babc6e08557c023e8c73b6b961970da83630664

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:268-269, the source contract is `def pointOf (t :  $\mathbb{R}$ ) (z : Vec4) : Point := ⟨t, z 0, z 1, z 2, z 3⟩`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.radiiReal

Source hash: ffbda00bb8ae49e94f67d9ad5dd16012ff60af58d9c1760a6abfb99b1bb96909

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:267-267, the source contract is `def radiiReal : Vec4 := fun j => (brick.radii j : ℝ)`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.tInterval

Source hash: 4051c1639ae05eb9e136816b4912782037642807485ecc541b8a4729aa41a839

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:78-78, the source contract is `def tInterval : QInterval := ⟨-tRadius, tRadius, by norm_num [tRadius, brick]⟩`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.tRadius

Source hash: 7ec2e049acded7f92cfecfef90c8523058c254046f34ddfc61af74c906f331c0

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:76-77, the source contract is `abbrev tRadius : ℚ := (brick.lamHi - brick.lamLo) / 2`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.v

Source hash: 7b17efff67b8a9b22200a9061ba58fb57a901753e3d49bff62bf8ea7f6d2a8ea

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:116-118, the source contract is `def v (p : Point) : ℝ := brick.centers 3 + brick.slopes 3 * p.t + brick.vRhoSlope * p.rho + p.sigma`. These statements use the retained upper prototype density interval $[25781/25000, 33/32]$ and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.w

Source hash: 52d839481355b169feb0cd07c321cec9f11b8bdc2b69f116735e48df02da7c3d

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:115-115, the source contract is `def w (p : Point) : ℝ := brick.centers 2 + brick.slopes 2 * p.t + p.rho`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.x

Source hash: 3f1552bfdd08e7586bfffec5b6dc6b1a410691f11d8196031e1e7ca9cbf58889a

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:112-112, the source contract is `def x (p : Point) : ℝ := brick.centers 0 + brick.slopes 0 * p.t + p.xi`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

RightPrototypeCell.y

Source hash: 9e7642719b786de830ac8e4bdac8c25ca9f746737432d174122c9b0d847d92b9

Relation: equivalent

Equivalent in the declared constructor/section context. At RightPrototypeCell.lean:113-114, the source contract is `def y (p : Point) : ℝ := brick.centers 1 + brick.slopes 1 * p.t + brick.yXiSlope * p.xi + p.eta`. These statements use the retained upper prototype density interval [25781/25000,33/32] and its recorded closed error box. Affine angle bounds, inverse branches and face signs are quantified over that box, not over arbitrary real angles.

Findings: none.

ScalarSuffixCertificate.CertificateExpr

Source hash: b7255be61ccd2c3030071696969cf641da72d2598375adad83bc3d9088d769f2

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:167-175, the source contract is inductive CertificateExpr where | rational (q : ℚ) | atom (x : ℝ) (i : QInterval) (sound : i.RealContains x) | add (a b : CertificateExpr) | neg (a : CertificateExpr) | mul (a b : CertificateExpr). The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.CertificateExpr.enclosure

Source hash: 93def124f5286c57b1f3cae76f5c9b5b01ebe4c4e710fe712c5ca0db00b71550

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:183-190, the source contract is `def enclosure : CertificateExpr → QInterval | rational q => LeanSuffixReflective.QInterval.point q | atom _ i _ => i | add a b => a.enclosure.add b.enclosure | neg a => a.enclosure.neg | mul a b => a.enclosure.mul b.enclosure`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.CertificateExpr.sound

Source hash: 7903a28236dd3b57076df8e33c74ea38e4b9e0dfa4dd8e31c52c47cb44f7b8cc

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:191-200, the source contract is `theorem sound (e : CertificateExpr) : e.enclosure.RealContains e.value`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.CertificateExpr.value

Source hash: a0149812d66bc28eb03b0052f9694933e7c6247aec19b155763c96f72897c502

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:176-182, the source contract is `def value : CertificateExpr → ℝ | rational q => q | atom x _ _ => x | add a b => a.value + b.value | neg a => -a.value | mul a b => a.value * b.value`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.CertificateExpr.value_neg_of_hi_neg

Source hash: db3beb0c3287dfb56bc535077265688e9bbbd719c9e044e825c88f873d5dd3ed

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:201-205, the source contract is `theorem value_neg_of_hi_neg (e : CertificateExpr) (h : e.enclosure.hi < 0) : e.value < 0`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.AnalyticFaceBounds

Source hash: 55d57a38a3bc2fde4d82f77d73cfb2e9ed05adcef3d85713a2b783119493f

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3244-3249, the source contract is structure AnalyticFaceBounds : Prop where lowFaces : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow \forall j, z j = -radiiReal j \rightarrow rootMap t z j \leq 0$ highFaces : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow \forall j, z j = radiiReal j \rightarrow 0 \leq rootMap t z j$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.B3

Source hash: fbc773e40d79a1d6d31822cbbd56d9a9607c6a7f8abc1ba0f47027df7b631eae

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:506-506, the source contract is def B3 (p : Point) : $\mathbb{R} := \lambda m p * v p + \pi - w p$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.B4

Source hash: 30e079ec002735e04de6acf4de75c1ef255bd47fc8db540544074fad80b3a6bc

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:503-503, the source contract is def B4 (p : Point) : $\mathbb{R} := \lambda m p * y p + \pi / 2 - x p$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.Brick

Source hash: fe061415d65cea6f77fb8c663f249825b26067221bd27165829e3e0950822743

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:425-436, the source contract is structure Brick where lamLo : $\mathbb{Q}$ lamHi : $\mathbb{Q}$ centers : $Fin\ 4 \rightarrow \mathbb{Q}$ slopes : $Fin\ 4 \rightarrow \mathbb{Q}$ yXiSlope : $\mathbb{Q}$ vRhoSlope : $\mathbb{Q}$ radii : $Fin\ 4 \rightarrow \mathbb{Q}$ lamOrdered : $lamLo < lamHi$ radiiPos : $\forall j, 0 < radii j$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.C3

Source hash: bb334eb31664c80305d156013a66d4b3847b5f60bc740d68b3db43dd6612439b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:512-512, the source contract is `def C3 (p : Point) : ℝ := cos (w p) - lam p * cos (v p)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.C3_sigma_high_face_margin

Source hash: d14003647889eccbbfb10aaee4768200f899b458693a4ef6f074530360518a08

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:2949-2970, the source contract is `theorem C3_sigma_high_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 3 = radiiReal 3 → (1 / 10000000000 : ℝ) < C3 (pointOf t z)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.C3_sigma_low_face_margin

Source hash: 33c17a00a6d5f6981f6c2016b520d75f7bdfba5da4ccd3f18d97796ac3d45fac

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:2927-2948, the source contract is `theorem C3_sigma_low_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 3 = -radiiReal 3 → C3 (pointOf t z) < -(1 / 10000000000 : ℝ)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.C4

Source hash: 0c76dd8701b0777d1f4e908247b8a477e6ed59455006924fb2e56c23786a4bd0

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:510-510, the source contract is `def C4 (p : Point) : ℝ := cos (x p) - lam p * cos (y p)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.C4_eta_high_face_margin

Source hash: 4dda50fdf98d96d0c6d5298b4b08eca53dc73079c6357bc4d6c0fad14a84b1e3

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:2212-2232, the source contract is theorem C4_eta_high_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \ 1 = \text{radiiReal } 1 \rightarrow (11 / 10000000000 : \mathbb{R}) < C4 (\text{pointOf } t \ z)$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.C4_eta_low_face_margin

Source hash: 9d228ca47508753791f14fd62b25ef0aa8eb7814e1715715afaf3fe27d1a441b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:2190-2211, the source contract is theorem C4_eta_low_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \ 1 = -\text{radiiReal } 1 \rightarrow C4 (\text{pointOf } t \ z) < -(11 / 10000000000 : \mathbb{R})$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.D3

Source hash: 4496291242cc1a371596577f5bdbec61dd292b63c92e9b87cf551b140d32a368

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:507-507, the source contract is def D3 (p : Point) : $\mathbb{R} := \sin (v \ p) - \sin (w \ p)$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.D4

Source hash: 458c17f197940395eb065d606c56a29538dee5cfc382d507f31979fa9b14e3b0

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:504-504, the source contract is def D4 (p : Point) : $\mathbb{R} := \sin (y \ p) - \sin (x \ p)$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.E

Source hash: 473eec5a0b00f868ac4f7ad090edf435fe47f07ca5bd2b59758512636541fbfa

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:513-514, the source contract is `def E (p : Point) : ℝ := 2 * cos (x p) ^ 2 * N3 p - (1 + cos (w p)) ^ 2 * N4 p`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.E_rho_high_face_margin

Source hash: a325916020527c8ce3fd863cee68df30e9b7de45125ec6369905d4e21da5a392

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3219-3243, the source contract is `theorem E_rho_high_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 2 = radiiReal 2 → (14 / 1000000000 : ℝ) < E (pointOf t z)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.E_rho_low_face_margin

Source hash: fbca51390968494bba3f952ecf1e30c03e3575007068f814db8a31c612e35640

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3196-3218, the source contract is `theorem E_rho_low_face_margin : ∀ t z, tInterval.RealContains t → InSymmetricBox radiiReal z → z 2 = -radiiReal 2 → E (pointOf t z) < -(14 / 1000000000 : ℝ)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.F

Source hash: ff764cf036982bdc0d990c94070c7ae41de884ec397b4b250e1096fedd89ff34

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:511-511, the source contract is `def F (p : Point) : ℝ := cos (x p) * D4 p - sin (x p) * sin (y p) * B4 p`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.F_xi_high_face_margin

Source hash: ab944f08473f17ea4d7ed3fe1083b1fc50813a95437c8cb991f719dd9c66f87b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:2639-2663, the source contract is theorem F_xi_high_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \neq 0 = \text{radiiReal } 0 \rightarrow F(\text{pointOf } t z) < -(86 / 10000000000 : \mathbb{R})$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.F_xi_low_face_margin

Source hash: 3a89e7a208c35818420fe325762c3418f78b64fba4a4d38be2bc20f08e1bb47d

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:2616-2638, the source contract is theorem F_xi_low_face_margin : $\forall t z, t \text{Interval.RealContains } t \rightarrow \text{InSymmetricBox radiiReal } z \rightarrow z \neq 0 = -\text{radiiReal } 0 \rightarrow (88 / 10000000000 : \mathbb{R}) < F(\text{pointOf } t z)$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.G

Source hash: 818854a9d2ac34b27a374a124d11eefc488199875161c552aa19dcc6709e1ea

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:515-521, the source contract is def G (p : Point) : $\mathbb{R} := \cos(x p) * (B3 p + D3 p) - (1 + \cos(w p)) * B4 p$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.K3

Source hash: dded0a92a5475f561e4ec09b53d31fd672246c1e3a18679f1d82548595f59954

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:720-727, the source contract is def K3 (p : Point) : $\mathbb{R} := \text{let } h3 := (1 + \cos(w p)) / 24 * h3 * ((1 + \cos(w p)) / \sin(w p) - (\text{lam } p^2 + \cos(w p)) / (\text{lam } p * (\text{lam } p * \sin(v p)))) - 2 * (B3 p + D3 p)$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.N3

Source hash: 79fd22b0e88fadf424caa5c32cd0621ba146426141946780c13923b4adb4108c

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:508-509, the source contract is `def N3 (p : Point) : ℝ := B3 p + (cos (w p) + 2) * D3 p`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.N4

Source hash: 8317292c5d87208707548fa6a07d20553105db710dba94daee68266201ce16ef

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:505-505, the source contract is `def N4 (p : Point) : ℝ := B4 p + cos (x p) * D4 p`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.Point

Source hash: faab379d3c93e3fdbeaacb3e4e673d5518c5a8dbdfc4b2177da8137b5ec11ed2

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:488-494, the source contract is `structure Point` where `t : ℝ xi : ℝ eta : ℝ rho : ℝ sigma : ℝ`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.PointEquationSemantics

Source hash: 57c3cf6c23b8d0095e640676ca4732afaae3f1e4c4e649a7f2e21c72d97bbd64

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:522-537, the source contract is `structure PointEquationSemantics (p : Point) : Prop` where `typeThreeHeight_gt_half : 1 / 2 < (1 + cos (w p)) / 2` `typeThreeHeight_lt_one : (1 + cos (w p)) / 2 < 1` `typeFourHeight_pos : 0 < cos (x p)` `typeFourHeight_lt_one : cos (x p) < 1` `equalArea_of_E_zero : E p = 0 → LeanSuffixAnalytic.typeThreeArea (lam p) ((1 + cos (w p)) / 2) = LeanSuffixAnalytic.typeFourArea (lam p) (cos (x p))` `stationary_of_F_zero : F p = 0 → LeanSuffixAnalytic.typeFourFold (lam p) (cos (x p)) = 0` `perimeter_lt_of_G_neg : G p < 0 → LeanSuffixAnalytic.typeThreePerimeter (lam p) ((1 + cos (w p)) / 2) < LeanSuffixAnalytic.typeFourPerimeter (lam p) (cos (x p))`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.affine_branches

Source hash: e4d28d57906a58d6905767ae59f9a6cca32fef94fecf64598fa3b381471d528f

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:1590-1624, the source contract is theorem affine_branches : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow \text{let } p := \text{pointOf } t z \ 0 < x p \wedge x p < \pi / 4 \wedge 0 < y p \wedge y p < \pi / 4 \wedge 0 < w p \wedge w p < \pi / 4 \wedge 0 < v p \wedge v p < \pi / 4$. The complete conclusion includes positivity and strict pi/4 upper bounds for each of x,y,w,v, after pointOf t z is introduced. Both t and the full error vector are universally quantified subject to the recorded prototype box constraints.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.analyticFaceBounds

Source hash: f2c7f2a169ba9c55d2865d26a4898b9b5bc1410e5acd882f9326aa435e600409

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3250-3280, the source contract is theorem analyticFaceBounds : AnalyticFaceBounds. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.brick

Source hash: 688841e317d3df26432b03f8a47b552ed700304ce27b849b8c43cf708ddb321b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:437-453, the source contract is def brick : Brick where lamLo := 51 / 50 lamHi := 102001 / 100000 centers := ![925629104183851933 / 5000000000000000000, 2705068892345424047 / 10000000000000000000, 1046818865074353649 / 2500000000000000000, 4608185511215091997 / 10000000000000000000] slopes := ![439944824343995293 / 2000000000000000000, 2510476247114540243 / 5000000000000000000, 6186129361874951931 / 10000000000000000000, 3760062975752449919 / 5000000000000000000] yXiSlope := 67 / 100 vRhoSlope := 112 / 125 radii := ![1 / 10000000, 1 / 100000000, 1 / 10000000, 1 / 100000000] lamOrdered := by norm_num radiiPos := by intro j; fin_cases j <=> norm_num. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.candidate_not_isWeightedPerimeterMinimizer

Source hash: addf63fb8e32b74a68ab723aad485618435e5da3c6625d624ab901eb3e183553

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:4190-4203, the source contract is theorem candidate_not_isWeightedPerimeterMinimizer {lam0 : $\mathbb{R}$ } (hlam : lamInterval.RealContains lam0) (candidate : _root_.FourArcCandidate lam0) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) : $\neg$ candidate.IsWeightedPerimeterMinimizer. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.descendingK3

Source hash: 77707dff8f9a285fbbcbce0fee4560bb9495981e335a6a037b5c4be7daa5c9da3

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:1646-1814, the source contract is theorem descendingK3 : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow K3 (pointOf t z) < 0$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.gap_upper

Source hash: bfd0e938dc561a18e65fa20cc38b1be820b9d6c633f0485b08fc114b41463547

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:1815-1930, the source contract is theorem gap_upper : $\forall t z, tInterval.RealContains t \rightarrow InSymmetricBox radiiReal z \rightarrow G (pointOf t z) < -(437 / 1000000 : \mathbb{R})$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.lam

Source hash: 80d4ea0b007cc38c9cef09318804f92bb1d2031dffde45077e78cad062961da3

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:495-495, the source contract is def lam (p : Point) : $\mathbb{R} := (lamMid : \mathbb{R}) + p.t$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.lamInterval

Source hash: f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:458-459, the source contract is `def lamInterval : QInterval := ⟨brick.lamLo, brick.lamHi, brick.lamOrdered.le⟩`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.lamMid

Source hash: 6872199d838a1fa3e543ff329eb7ed7979df53e9ab7df478a48be79baedaf0cf

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:454-454, the source contract is `abbrev lamMid : ℚ := (brick.lamLo + brick.lamHi) / 2`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.lambda_iff_centered

Source hash: 80585faf74c765a727b4d57aad40e1258e6fc487898dae74ceaba8d89c1df2a3

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:460-472, the source contract is `theorem lambda_iff_centered {lam : ℝ} : lamInterval.RealContains lam ↔ tInterval.RealContains (lam - (lamMid : ℝ))`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.lambda_in_retainedSuffix

Source hash: 7d841cce60699559819b45f6ee022939aef986b15a3732746d752fc33d94e2fa

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:473-479, the source contract is `theorem lambda_in_retainedSuffix {lam : ℝ} (h : lamInterval.RealContains lam) : LeanSuffixAnalytic.InRetainedSuffix lam`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.pointEquationSemantics

Source hash: 55d2eea611fa72c52ef509b0c8b4a1aa98765b6f9627ee1108103f1668f29bc9

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:538-687, the source contract is theorem pointEquationSemantics (p : Point) (h_{lam} : 1 < lam p) (hbranches : 0 < x p ∧ x p < π / 4 ∧ 0 < y p ∧ y p < π / 4 ∧ 0 < w p ∧ w p < π / 4 ∧ 0 < v p ∧ v p < π / 4) (hC4 : C4 p = 0) (hC3 : C3 p = 0) : PointEquationSemantics p. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.pointOf

Source hash: 123eaac2727784b857a8a1d07babc6e08557c023e8c73b6b961970da83630664

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:801-802, the source contract is def pointOf (t : ℝ) (z : Vec4) : Point := ⟨t, z 0, z 1, z 2, z 3⟩. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.point_equations_to_scalar_improvement

Source hash: db5cd193da271e760511f0742066e9fa2712d960f6dbb5c7963df325913da93

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:688-719, the source contract is theorem point_equations_to_scalar_improvement (p : Point) (h_{lam} : 1 < lam p) (hbranches : 0 < x p ∧ x p < π / 4 ∧ 0 < y p ∧ y p < π / 4 ∧ 0 < w p ∧ w p < π / 4 ∧ 0 < v p ∧ v p < π / 4) (hC4 : C4 p = 0) (hF : F p = 0) (hC3 : C3 p = 0) (hE : E p = 0) (hG : G p < 0) : let h₃ := (1 + cos (w p)) / 2 let h₄ := cos (x p) 1 / 2 < h₃ ∧ h₃ < 1 ∧ 0 < h₄ ∧ h₄ < 1 ∧ LeanSuffixAnalytic.typeThreeArea (lam p) h₃ = LeanSuffixAnalytic.typeFourArea (lam p) h₄ ∧ LeanSuffixAnalytic.typeFourFold (lam p) h₄ = 0 ∧ LeanSuffixAnalytic.typeThreePerimeter (lam p) h₃ < LeanSuffixAnalytic.typeFourPerimeter (lam p) h₄ ∧ LeanSuffixAnalytic.reducedFoldGap (lam p) h₃ h₄ < 0. The complete let-bound conclusion yields h₃ in (1/2,1), h₄ in (0,1), equal area, K₄=0, strict perimeter improvement and a negative reducedFoldGap from the incidence/residual zero equations, G<0, lambda>1 and the recorded branch inequalities. It does not conclude K₃<0; that separate descending-fold sign is supplied later by the uniform certificate. The pi/2 shift and positive denominator clearing are retained.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.radiiReal

Source hash: ffbda00bb8ae49e94f67d9ad5dd16012ff60af58d9c1760a6abfb99b1bb96909

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:800-800, the source contract is `def radiiReal : Vec4 := fun j => (brick.radii j : ℝ)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.realConclusion

Source hash: dc3fbce7b832643e59d10760d43a70cfab194d45c7293f0a25257849be954285

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3950-3975, the source contract is `theorem realConclusion {lam0 : ℝ} (hlam : lamInterval.RealContains lam0) : ∃ p : Point, lam p = lam0 ∧ InSymmetricBox radiiReal (vars p) ∧ C4 p = 0 ∧ F p = 0 ∧ C3 p = 0 ∧ E p = 0 ∧ G p < 0`. For each lambda in the prototype interval, the theorem constructs one point at that same lambda, with its vars in the closed error box, C4=F=C3=E=0 and G<0. The proof obtains it as pointOf at the triangular-root witness and uses the uniform gap bound. It neither assumes the common zero nor claims root uniqueness or a K3 sign as part of this conclusion.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.realConclusion_scalarImprovement

Source hash: bb2d746cbd9e64bb1870debe196a905e1574b27a24542c427057bd40fc2db5ef

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3976-4031, the source contract is `theorem realConclusion_scalarImprovement {lam0 : ℝ} (hlam : lamInterval.RealContains lam0) : ∃ p : Point, lam p = lam0 ∧ InSymmetricBox radiiReal (vars p) ∧ let h3 := (1 + cos (w p)) / 2 let h4 := cos (x p) 1 / 2 < h3 ∧ h3 < 1 ∧ 0 < h4 ∧ h4 < 1 ∧ LeanSuffixAnalytic.typeThreeArea (lam p) h3 = LeanSuffixAnalytic.typeFourArea (lam p) h4 ∧ LeanSuffixAnalytic.typeFourFold (lam p) h4 = 0 ∧ LeanSuffixAnalytic.typeThreePerimeter (lam p) h3 < LeanSuffixAnalytic.typeFourPerimeter (lam p) h4 ∧ LeanSuffixAnalytic.reducedFoldGap (lam p) h3 h4 < 0 ∧ deriv (LeanSuffixAnalytic.typeThreeArea (lam p)) h3 < 0`. The implication extracts the same certified point from realConclusion and converts it to the stated scalar improvement through point_equations_to_scalar_improvement. The root and gap information are obtained from the conclusion's witnesses rather than inserted as extra premises.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.regularStationaryFold_existsUnique

Source hash: 0524e42b6946a1eef76cb933934bd9181ca74751bcf4f0c445f7b35510a5a81c

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:4082-4103, the source contract is theorem regularStationaryFold_existsUnique {lam0 : $\mathbb{R}$ } (hlam : lamInterval.RealContains lam0) : $\exists! h_4 : \mathbb{R}, h_4 \in \text{Ioo } (0 : \mathbb{R}) 1 \wedge \text{LeanSuffixAnalytic.typeFourFold lam0 } h_4 = 0$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.regularStationaryFold_scalarImprovement

Source hash: 7fefcab04ab1793e87d3e44914f4cea3510747aca02a4b68cef1c2b92278828d

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:4124-4157, the source contract is theorem regularStationaryFold_scalarImprovement {lam0 h4 : $\mathbb{R}$ } (hlam : lamInterval.RealContains lam0) (hh4 : h4 $\in$ Ioo (0 : $\mathbb{R}$) 1) (hstationary : LeanSuffixAnalytic.typeFourFold lam0 h4 = 0) : $\exists h_3 : \mathbb{R}, 1/2 < h_3 \wedge h_3 < 1 \wedge \text{LeanSuffixAnalytic.typeThreeArea lam0 } h_3 = \text{LeanSuffixAnalytic.typeFourArea lam0 } h_4 \wedge \text{LeanSuffixAnalytic.typeThreePerimeter lam0 } h_3 < \text{LeanSuffixAnalytic.typeFourPerimeter lam0 } h_4 \wedge \text{LeanSuffixAnalytic.reducedFoldGap lam0 } h_3 h_4 < 0 \wedge \text{deriv } (\text{LeanSuffixAnalytic.typeThreeArea lam0}) h_3 < 0$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.rootMap

Source hash: 9341d8d5e2fed305c16e751079c6bca9d9e015e2672d8c238a0e48e3da9b8c83

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:803-806, the source contract is def rootMap (t : $\mathbb{R}$) (z : Vec4) : Vec4 := let p := pointOf t z ![-F p, C4 p, E p, C3 p]. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.rootMap_continuous

Source hash: 2df03b7ca4b2fad1bc9d6c65609dfe1a30f18068ea4f5844ffe3335f9f65c08b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:807-815, the source contract is theorem rootMap_continuous (t : $\mathbb{R}$) : Continuous (rootMap t). The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals

are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.stationaryEqualAreaPair_exists

Source hash: 48939925c6e8499402fbc852be5c65cf7f2c3959981d591edde1fff923f7115e

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:4158-4189, the source contract is theorem stationaryEqualAreaPair_exists {lam0 : $\mathbb{R}$ } (hlam : lamInterval.RealContains lam0) : $\exists$ pair : LeanSuffixAnalytic.StationaryEqualAreaPair lam0, LeanSuffixAnalytic.typeThreeFold lam0 pair.h₃ < 0 $\wedge$ LeanSuffixAnalytic.typeThreePerimeter lam0 pair.h₃ < LeanSuffixAnalytic.typeFourPerimeter lam0 pair.h₄. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.tInterval

Source hash: 4051c1639ae05eb9e136816b4912782037642807485ecc541b8a4729aa41a839

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:457-457, the source contract is def tInterval : QInterval := ⟨-tRadius, tRadius, by norm_num [tRadius, brick]⟩. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.tRadius

Source hash: 7ec2e049acded7f92cfecfef90c8523058c254046f34ddfc61af74c906f331c0

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:455-456, the source contract is abbrev tRadius : $\mathbb{Q}$:= (brick.lamHi - brick.lamLo) / 2. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.triangular_root

Source hash: 84268649affbe380ad5f3767cb23b92f1867804e119cb1a35ce8cf1fd4dbb9a5

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:3856-3942, the source contract is theorem triangular_root (t : $\mathbb{R}$) (ht : tInterval.RealContains t) : $\exists$ z, InSymmetricBox radiiReal z $\wedge$ $\forall$ j, rootMap t z j = 0. The exact rational enclosures and Taylor remainders have their stated

domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.typeThreeArea_deriv_eq_K3

Source hash: 8a9cde70bef0dfad1eb697f78cb39f5a363eb9e41679d7d14dd8c12231f3ec28

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:728-798, the source contract is theorem typeThreeArea_deriv_eq_K3 (p : Point) (hlam : 1 < lam p) (hbranches : 0 < w p ^ w p < pi / 4 ^ 0 < v p ^ v p < pi / 4) (hC3 : C3 p = 0) : deriv (LeanSuffixAnalytic.typeThreeArea (lam p)) ((1 + cos (w p)) / 2) = K3 p / ((1 + cos (w p)) / 2) ^ 3. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.v

Source hash: 7b17efff67b8a9b22200a9061ba58fb57a901753e3d49bff62bf8ea7f6d2a8ea

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:500-502, the source contract is def v (p : Point) : ℝ := brick.centers 3 + brick.slopes 3 * p.t + brick.vRhoSlope * p.rho + p.sigma. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.vars

Source hash: d389f5673106602bb401f76b3d594d64ec6f00b7fcff209bdfc571267bc953d5

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:799-799, the source contract is def vars (p : Point) : Vec4 := ![p.xi, p.eta, p.rho, p.sigma]. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.w

Source hash: 52d839481355b169feb0cd07c321cec9f11b8bdc2b69f116735e48df02da7c3d

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:499-499, the source contract is def w (p : Point) : ℝ := brick.centers 2 + brick.slopes 2 * p.t + p.rho. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box;

scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.x

Source hash: 3f1552bfdd08e7586bfffec5b6dc6b1a410691f11d8196031e1e7ca9cbf58889a

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:496-496, the source contract is `def x (p : Point) : ℝ := brick.centers 0 + brick.slopes 0 * p.t + p.xi`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.FirstPrototypeCell.y

Source hash: 9e7642719b786de830ac8e4bdac8c25ca9f746737432d174122c9b0d847d92b9

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:497-498, the source contract is `def y (p : Point) : ℝ := brick.centers 1 + brick.slopes 1 * p.t + brick.yXiSlope * p.xi + p.eta`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.InSymmetricBox

Source hash: 5499419d9ab0a5d1357ae99f00c375e3edfdb9ed0f97f8657ce5edf639f1f5d0

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:420-424, the source contract is `def InSymmetricBox (r z : Vec4) : Prop := ∀ j, -r j ≤ z j ∧ z j ≤ r j`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval

Source hash: 9fe7a736ba4c430250485562ba236783155c010cef8e4943fa7819a28f9aa406

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:24-29, the source contract is `abbrev QInterval := LeanSuffixReflective.QInterval`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.divPos

Source hash: 2642acecc562d12111d17e260928f80af5d34794a1e8e1d3abc20a33a085768a

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:50-53, the source contract is `def divPos (i j : QInterval) (hj : 0 < j.lo) : QInterval := i.mul (j.recipPos hj)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.mul

Source hash: 0d7a9188e958ff3c0238d78f3d3a0728ab82af07abbb6a56e4a749efa8520624

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:30-44, the source contract is `def mul (i j : QInterval) : QInterval where lo := min (min (i.lo * j.lo) (i.lo * j.hi)) (min (i.hi * j.lo) (i.hi * j.hi)) hi := max (max (i.lo * j.lo) (i.lo * j.hi)) (max (i.hi * j.lo) (i.hi * j.hi)) ordered := by calc min (min (i.lo * j.lo) (i.lo * j.hi)) (min (i.hi * j.lo) (i.hi * j.hi)) ≤ i.lo * j.lo := (min_le_left _ _).trans (min_le_left _ _ _ ≤ max (max (i.lo * j.lo) (i.lo * j.hi)) (max (i.hi * j.lo) (i.hi * j.hi)) := (le_max_left _ _).trans (le_max_left _ _)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.negative_of_realContains

Source hash: dd92d9ea0b74bda9075f7d8e3782bf5b435ea264bc0d9f911316c7d07b50aa26

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:153-158, the source contract is `theorem negative_of_realContains {i : QInterval} {x : ℝ} (hx : i.RealContains x) (hi : i.hi < 0) : x < 0`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.positive_of_realContains

Source hash: 4196ef90c8a23949f2eeda7be784e128a7fc085f82c49c1113a0dbdd35e4d445

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:159-166, the source contract is `theorem positive_of_realContains {i : QInterval} {x : ℝ} (hx : i.RealContains x) (hi : 0 < i.lo) : 0 < x`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.realContains_centered_meanValue

Source hash: 0ed29af5eec0bee2a0714dc6eef9fd9127d30190d983af8320b3b10560c11496

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:131-145, the source contract is theorem `realContains_centered_meanValue (f f' : $\mathbb{R} \rightarrow \mathbb{R}$) {c h : $\mathbb{R}$ } (base slope displacement : QInterval) (hderiv : $\forall x \in \text{Set.uIcc } c \ h, \text{HasDerivAt } f (f' \ x) \ x)$ (hbase : base.RealContains (f c)) (hslope : $\forall x \in \text{Set.uIcc } c \ h, \text{slope.RealContains } (f' \ x)$) (hdisplacement : displacement.RealContains (h - c)) : (base.add (slope.mul displacement)).RealContains (f h))`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.realContains_divPos

Source hash: f39233850c94c2277651049ffd4664a24cd1dbc0d60498abc5472f08a163d72f

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:120-130, the source contract is theorem `realContains_divPos {i j : QInterval} {x y :  $\mathbb{R}$ } (hj :  $0 < j.\text{lo}$ ) (hx : i.RealContains x) (hy : j.RealContains y) : (i.divPos j hj).RealContains (x / y)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.realContains_mul

Source hash: bcb27cff58a492c7470bc52a00dd315717a6bfbc49a464e152ee6b024b200dfd

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:54-97, the source contract is theorem `realContains_mul {i j : QInterval} {x y :  $\mathbb{R}$ } (hx : i.RealContains x) (hy : j.RealContains y) : (i.mul j).RealContains (x * y)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.realContains_nsmul

Source hash: 230f7166aba334e53ba446c86429b3ff8b9f0576ec86363de14a61caa8a2624d

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:98-108, the source contract is theorem `realContains_nsmul {i : QInterval} {x :  $\mathbb{R}$ } {q :  $\mathbb{Q}$ } (hq :  $0 \leq q$ ) (hx : i.RealContains x) : (i.nsmul q hq).RealContains ((q :  $\mathbb{R}$ ) * x)`. The enclosure theorem applies to

every nonnegative rational scaling factor, including zero and fractions, and transports the interval endpoints through the rational-to-real cast. This matches the deliberately broader meaning of `nsmul` used in the author's explanation.

Findings: none.

ScalarSuffixCertificate.QInterval.realContains_recipPos

Source hash: 6bebb0eb9a5daccfb0e032373488c963884df078d4bcae42d816f15559377e3f

Relation: equivalent

Equivalent in the declared constructor/section context. At `ScalarSuffixCertificate.lean:109-119`, the source contract is `theorem realContains_recipPos {i : QInterval} {x : ℝ} (hi : 0 < i.lo) (hx : i.RealContains x) : (i.recipPos hi).RealContains (1 / x)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.realContains_sqrt

Source hash: 78d08ac2e456c76c8cc84e056f4a63de293d6015fbff06b8f125a80a09ec3dd5

Relation: equivalent

Equivalent in the declared constructor/section context. At `ScalarSuffixCertificate.lean:146-152`, the source contract is `theorem realContains_sqrt {i : QInterval} {x : ℝ} (hlo : 0 ≤ i.lo) (hloSq : (i.lo : ℝ) ^ 2 ≤ x) (hhiSq : x ≤ (i.hi : ℝ) ^ 2) : i.RealContains (sqrt x)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.QInterval.recipPos

Source hash: 90ef275967dd2ae91683ae88e032a660973bb176bd69a1f9f83086d63e1c2bed

Relation: equivalent

Equivalent in the declared constructor/section context. At `ScalarSuffixCertificate.lean:45-49`, the source contract is `def recipPos (i : QInterval) (hi : 0 < i.lo) : QInterval where lo := 1 / i.hi hi := 1 / i.lo ordered := one_div_le_one_div_of_le hi i.ordered`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.Vec4

Source hash: 5221a72a407ada42251ae04520dcef5209887e8aa5988ed22eb9dbd06e5ba073

Relation: equivalent

Equivalent in the declared constructor/section context. At `ScalarSuffixCertificate.lean:418-419`, the source contract is `abbrev Vec4 := Fin 4 → ℝ`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.cosTaylor26

Source hash: 46d8e337f6347f1173e5244107f4620ec9e73c77dbe96ffed5fdcf32a93ba782

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:219-223, the source contract is `def cosTaylor26 (x : ℚ) : ℚ := ∑ k ∈ Finset.range 14, (-1 : ℚ) ^ k * x ^ (2 * k) / (2 * k).factorial`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.cosTaylor26Interval

Source hash: 97a38531dd63947f237378c23320eb2947ffd81d0b5929b322bafd6a94ec891e

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:230-234, the source contract is `def cosTaylor26Interval (x : ℚ) : QInterval := let p := cosTaylor26 x let r := |x| ^ 27 / (Nat.factorial 27 : ℚ) ⟨p - r, p + r, by linarith [show 0 ≤ r by positivity]⟩`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.cosTaylor26Interval_sound

Source hash: 527e561394d27648791691ec45e120c58633f8289f9131a5419b3b4150dbd9b7

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:311-325, the source contract is `theorem cosTaylor26Interval_sound (x : ℚ) : (cosTaylor26Interval x).RealContains (cos (x : ℝ))`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.oppositeFace_zero

Source hash: 77378bc472ea0851732650cda76cddb51de35a2f8573fa992dd84ae00b26c15

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:411-417, the source contract is `theorem oppositeFace_zero {a b : ℝ} (hab : a ≤ b) {f : ℝ → ℝ} (hf : ContinuousOn f (Icc a b)) (ha : f a ≤ 0) (hb : 0 ≤ f b) : ∃ x ∈ Icc a b, f x = 0`. The IVT uses $a \leq b$, continuity on $[a,b]$, $f(a) \leq 0$ and $0 \leq f(b)$, and returns a zero in the closed interval. Equality at an endpoint and the degenerate case $a=b$ are allowed; the author does not claim strict interior existence.

Findings: none.

ScalarSuffixCertificate.piInterval

Source hash: da867cd8e471ad99a2b0bc5bbf7e433df88cee3a9989f797f883fa79eab1b795

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:206-208, the source contract is `def piInterval : QInterval := ⟨3.141592, 3.141593, by norm_num⟩`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.piInterval_sound

Source hash: 0e2096e4d4fa24f53f2ff61998135a39f6cbe4314d1809b974cfbb5eccd9fe5b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:209-213, the source contract is `theorem piInterval_sound : piInterval.RealContains π`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.realContains_arccos_of_arcsin

Source hash: 6b1989559f420605398bd419fe5c29784f5a3ffc9b8acebbd0bee1ba892adc65

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:403-410, the source contract is `theorem realContains_arccos_of_arcsin {p a : QInterval} {x : ℝ} (hp : p.RealContains π) (ha : a.RealContains (arcsin x)) : ((p.nsmul (1 / 2) (by norm_num)).sub a).RealContains (arccos x)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.realContains_arcsin_of_sin_bounds

Source hash: 1ba394185e599a9b82bbb3c3e941ce7390ce6cfc17342f097ef698f1fe9212b

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:393-402, the source contract is `theorem realContains_arcsin_of_sin_bounds {i : QInterval} {x : ℝ} (hx : x ∈ Icc (-1 : ℝ) 1) (hbranch : (i.lo : ℝ) ∈ Icc (-(π / 2)) (π / 2) ∧ (i.hi : ℝ) ∈ Icc (-(π / 2)) (π / 2)) (hlo : sin (i.lo : ℝ) ≤ x) (hhi : x ≤ sin (i.hi : ℝ)) : i.RealContains (arcsin x)`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.realContains_cos_interval

Source hash: eedd5e166794798d683fb5483e2b9b97111a653338c01d010cd7592bb93d98f1

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:339-352, the source contract is theorem realContains_cos_interval {input output : QInterval} {x : ℝ} (hx : input.RealContains x) (hbranch : $0 \leq (\text{input.lo} : \mathbb{R}) \wedge (\text{input.hi} : \mathbb{R}) \leq \pi$) (hlo : $(\text{output.lo} : \mathbb{R}) \leq \cos (\text{input.hi} : \mathbb{R})$) (hhi : $\cos (\text{input.lo} : \mathbb{R}) \leq (\text{output.hi} : \mathbb{R})$) : output.RealContains (cos x). The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.realContains_sin_interval

Source hash: 21959c159381deb1e01f48cd4776b6a292e590b4e5bd4cd351519ff872a0aa25

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:326-338, the source contract is theorem realContains_sin_interval {input output : QInterval} {x : ℝ} (hx : input.RealContains x) (hbranch : $-(\pi / 2) \leq (\text{input.lo} : \mathbb{R}) \wedge (\text{input.hi} : \mathbb{R}) \leq \pi / 2$) (hlo : $(\text{output.lo} : \mathbb{R}) \leq \sin (\text{input.lo} : \mathbb{R})$) (hhi : $\sin (\text{input.hi} : \mathbb{R}) \leq (\text{output.hi} : \mathbb{R})$) : output.RealContains (sin x). The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.sinTaylor27

Source hash: cad29fad0ca5f0f5a021df9a703aaac6f17eda67757816193f0541445c680252

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:214-218, the source contract is def sinTaylor27 (x : ℚ) : ℚ := $\sum k \in \text{Finset.range } 14, (-1 : \mathbb{Q})^k * x^{(2 * k + 1)} / (2 * k + 1)!. \text{factorial}$. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.sinTaylor27Interval

Source hash: 9588051f8a2ff67152b4c9b894d6f120c8732ea5b8195eebd297f8f7e92a5437

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:224-229, the source contract is def sinTaylor27Interval (x : ℚ) : QInterval := let p := sinTaylor27 x let r := $|x|^{28} / (\text{Nat.factorial } 28 : \mathbb{Q})$ <p - r, p + r, by linarith [show $0 \leq r$ by positivity]]. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

ScalarSuffixCertificate.sinTaylor27Interval_sound

Source hash: cccc0e43a547af36c7ebd29335f5601a5c6d0913d8691b65f34e008faa2bf0e0

Relation: equivalent

Equivalent in the declared constructor/section context. At ScalarSuffixCertificate.lean:296-310, the source contract is theorem `sinTaylor27Interval_sound (x :  $\mathbb{Q}$ ) : (sinTaylor27Interval x).RealContains (sin (x :  $\mathbb{R}$ ))`. The exact rational enclosures and Taylor remainders have their stated domains. Local prototype claims inherit the recorded density/error box; scalar residuals are converted to geometric equal-area improvement only after incidence and angle-branch conditions have been established.

Findings: none.

StripCore

Source hash: a32ba501747725a0a1307cf7bd2abbffd387c8bc9d810f24cf348c7c04513aad

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:416-425, the source contract is `structure StripCore` where `chord :  $\mathbb{R}$  curvature :  $\mathbb{R}$  chord_pos :  $0 < \text{chord}$  curvature_pos :  $0 < \text{curvature}$  curvature_le_one :  $\text{curvature} \leq 1$` . The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.boundaryArcLength

Source hash: 74e634700740d7651d4ed773b1cb3cb71b11c7eedacd0c71b6a84a0e6ec43665

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:457-459, the source contract is `def boundaryArcLength (c : StripCore) :  $\mathbb{R}$  := 4 * ell c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.carrier

Source hash: 05f714bedd5ca3f09fd1dac168964dc190e4830b8099d994e590458d8167e259

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:449-452, the source contract is `def carrier (c : StripCore) : Set PlanePoint := c.rectangleCarrier  $\cup$  c.leftCapCarrier  $\cup$  c.rightCapCarrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.carrier_y_bounds

Source hash: dfd3510aaac02f0e3230984bb1d2516c15f954b7d416fa7e30c71ad44efa3ab8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:482-490, the source contract is `theorem carrier_y_bounds (c : StripCore) {p : PlanePoint} (hp : p ∈ c.carrier) : |p.2| ≤ 1`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.cos_sideAngle

Source hash: 5c26f39a6dccf790fede705362af68051955e479d2108518ff90a5d9985148ed

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:478-481, the source contract is `theorem cos_sideAngle (c : StripCore) : cos c.sideAngle = √(1 - c.curvature ^ 2)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.euclideanArea

Source hash: d05ac0be663f49bfb5425917ab3d2dcdd3367e8ef25ec467ccc3d1ef4ccf07a7

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:453-456, the source contract is `def euclideanArea (c : StripCore) : ℝ := 2 * c.chord + 8 * area c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.frontier_carrier

Source hash: 34bdf246812f32ba3d6167bf3b3ff5fea490b0816170740d97a7d77e703585bf

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1088-1199, the source contract is `theorem frontier_carrier (c : StripCore) : frontier c.carrier = leftArcTrace c ∪ (rightArcTrace c ∪ (upperChordTrace c ∪ lowerChordTrace c))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.horizontalChord_subset_carrier

Source hash: 334fc66a21b600efd108f9065db91fc12f64adadeb2b25ecfa3e64aa16686847

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:491-499, the source contract is `theorem horizontalChord_subset_carrier (c : StripCore) {y : ℝ} (hy : |y| ≤ 1) : {p : PlanePoint | p.2 = y ∧ |p.1| ≤ c.chord / 2} ⊆ c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.isClosed_carrier

Source hash: d28fca8c993d5d2b9aa0b2ebe7f3f0372ebdd951a5cda90d26adf748bface0e4

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:805-843, the source contract is `theorem isClosed_carrier (c : StripCore) : IsClosed c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.leftArcTrace

Source hash: 702872b40533619867ab566579d52d3f6a44e812910796d96106d1c50acbbfc2

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:762-766, the source contract is `def leftArcTrace (c : StripCore) : Set PlanePoint := {p | (p.1 - c.leftCenterX) ^ 2 + p.2 ^ 2 = c.radius ^ 2 ∧ p.1 ≤ -c.chord / 2 ∧ |p.2| ≤ 1}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.leftCapCarrier

Source hash: 2ca301cf56be587995c60fad814a6021c666b20d0f45e1fa3f2964b23f678cbc

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:440-443, the source contract is `def leftCapCarrier (c : StripCore) : Set PlanePoint := {p | (p.1 - c.leftCenterX) ^ 2 + p.2 ^ 2 ≤ c.radius ^ 2 ∧ p.1 ≤ -c.chord / 2 ∧ |p.2| ≤ 1}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.leftCenterX

Source hash: 01fd3b14ca7e8a9d341e71e2fb86b773a9bea051016f7f6499489c3b65bab474

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:431-433, the source contract is `def leftCenterX (c : StripCore) : ℝ := -c.chord / 2 + c.radius * cos c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.lowerChordTrace

Source hash: faa702d9afdf2eed044fad4474692be94c38bb3a9868779993853eb74445596

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:774-776, the source contract is `def lowerChordTrace (c : StripCore) : Set PlanePoint := {p | p.2 = -1 ∧ |p.1| ≤ c.chord / 2}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.measurableSet_carrier

Source hash: c4a110b00efc328d9afc36bfb523f064a5cb69170025121ec86c0022fc798171

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:500-551, the source contract is `theorem measurableSet_carrier (c : StripCore) : MeasurableSet c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.radius

Source hash: f23d45618cc8f0a402155f6b0b9c9bed86df54007d8b97c4f4027753ebfcd373

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:426-428, the source contract is `def radius (c : StripCore) : ℝ := 1 / c.curvature`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.radius_pos

Source hash: 754f25da7343b067aca1ccf07fea956e09bfd461a16a072435b6b938ea98f462

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:460-462, the source contract is `theorem radius_pos (c : StripCore) : 0 < c.radius`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.rectangleCarrier

Source hash: 5204ba429abcba2c9a7bfd26b44272d1068d5b57b8a97a88331831661e9408f9

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:437-439, the source contract is `def rectangleCarrier (c : StripCore) : Set PlanePoint := {p | -c.chord / 2 ≤ p.1 ∧ p.1 ≤ c.chord / 2 ∧ |p.2| ≤ 1}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.rightArcTrace

Source hash: 059982ba84229cdd58242cd7f98720be73ad274e207529f914e522d73ba5cbeb

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:767-770, the source contract is `def rightArcTrace (c : StripCore) : Set PlanePoint := {p | (p.1 - c.rightCenterX) ^ 2 + p.2 ^ 2 = c.radius ^ 2 ∧ c.chord / 2 ≤ p.1 ∧ |p.2| ≤ 1}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.rightCapCarrier

Source hash: 0f7231f5e97656f30e8e5131e2afc098206a3ff9953e2cd02ae8207ac4d54b12

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:444-448, the source contract is `def rightCapCarrier (c : StripCore) : Set PlanePoint := {p | (p.1 - c.rightCenterX) ^ 2 + p.2 ^ 2 ≤ c.radius ^ 2 ∧ c.chord / 2 ≤ p.1 ∧ |p.2| ≤ 1}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.rightCenterX

Source hash: ff1acc9e9b3ef170758d23460b0f614f6eb1c034ebd024b460f189e9f303f923

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:434-436, the source contract is `def rightCenterX (c : StripCore) : ℝ := c.chord / 2 - c.radius * cos c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.sideAngle

Source hash: 4f6b6599037149db7ce42f881c45d0eb7f35fbc43f124df7507d6a83a6d6e15b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:429-430, the source contract is `def sideAngle (c : StripCore) : ℝ := arcsin c.curvature`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.sideAngle_le_pi_div_two

Source hash: 0c0857ab96c23693b4448efce2e70a6069dda6f460cf0118b7e365f0f1074987

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:466-469, the source contract is `theorem sideAngle_le_pi_div_two (c : StripCore) : c.sideAngle ≤ π / 2`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.sideAngle_lt_pi

Source hash: 4d165924e20daf34cd0cd3a93af12e6637e356347b723990beb9d1a85af59a9e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:470-473, the source contract is `theorem sideAngle_lt_pi (c : StripCore) : c.sideAngle < π`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.sideAngle_pos

Source hash: 20886da5ba412dbc0666239218b14d32a1bad68cf7a0f4837b3f6216a46aefbd

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:463-465, the source contract is `theorem sideAngle_pos (c : StripCore) : 0 < c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.sin_sideAngle

Source hash: e85f9cb3c0c2b22c1ce74fdc60660033205e7a0821e76cfdd7248571352ea184

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:474-477, the source contract is `theorem sin_sideAngle (c : StripCore) : sin c.sideAngle = c.curvature`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.upperChordTrace

Source hash: 71db3f396f38ac0c4ed32b75273f08b0092359d4c0fdd7bc77976000e7791fb0

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:771-773, the source contract is `def upperChordTrace (c : StripCore) : Set PlanePoint := {p | p.2 = 1 ∧ |p.1| ≤ c.chord / 2}`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.volume_carrier

Source hash: a426f0908aec7b73455f4a0093170232e841cf46ab8f580ee2fe442ff66a7997

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2202-2242, the source contract is `lemma volume_carrier (c : StripCore) : volume c.carrier = ENNReal.ofReal c.euclideanArea`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.volume_leftCapCarrier

Source hash: ee7dbe308fa5b9feabca0b7d1f31d2c91cdf7e530a43dcc64a52dce7563fb7b8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2076-2138, the source contract is lemma `volume_leftCapCarrier (c : StripCore) : volume c.leftCapCarrier = ENNReal.ofReal (4 * area c.sideAngle)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.volume_rectangleCarrier

Source hash: b2c1d8716422a2e0169b6aaca489790c14b0594bc75f1e77b6bbec1349d5110

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2060-2075, the source contract is lemma `volume_rectangleCarrier (c : StripCore) : volume c.rectangleCarrier = ENNReal.ofReal (2 * c.chord)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.volume_rightCapCarrier

Source hash: 7bc214af5d8d445665e416853a1fab80112ca5fe1116500de0e9b2b6285dfdf7

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2139-2201, the source contract is lemma `volume_rightCapCarrier (c : StripCore) : volume c.rightCapCarrier = ENNReal.ofReal (4 * area c.sideAngle)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripCore.weightedArea_formula

Source hash: 22039501f3884648956188fd011ab629e4d948f733db9c19194b8b310134e20a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2315-2333, the source contract is lemma `StripCore.weightedArea_formula (lam : ℝ) (c : StripCore) : WeightedArea lam c.carrier = c.euclideanArea`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

StripDensity

Source hash: 51e629a8115629e4652f417b9dcfc46beea6a584d8ebcb351e746df181f97ab1

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:43-54, the source contract is `def StripDensity (lam : ℝ) (p : PlanePoint) : ℝ := if |p.2| ≤ 1 then 1 else lam @[simp] theorem stripDensity_upper_interface (lam x : ℝ) : StripDensity lam (x, 1) = 1 := by simp [StripDensity] @[simp] theorem stripDensity_lower_interface (lam x : ℝ) : StripDensity lam (x, -1) = 1 := by simp [StripDensity]`. The test is $|p.2| \leq 1$, so both interfaces $y = \pm 1$ receive density 1, not lambda. The definition is total for real lambda; the geometric exclusion theorem supplies $\lambda \geq 51/50$.

Findings: none.

TypeThreeAssembly

Source hash: ba80fb031079587a301c08eeb69ef793b83ab9e2bcde30a061c829264b7e1959

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:29-35, the source contract is `structure TypeThreeAssembly (lam : ℝ) where h : ℝ density_jump : 1 < lam h_pos : 0 < h h_lt_one : h < 1`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.HasModeledCompetitor

Source hash: 6f62c793e9d706d4618e398ebe39871915cbb898f90f26305f1f8dad0a496a66

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3076-3080, the source contract is `def HasModeledCompetitor : Prop := Nonempty (ModeledCompetitorRealization a)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization

Source hash: 7461e543281be1bed3a7575ad4267ecf1122dd3b3edd989aadf4754226aea410

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3043-3052, the source contract is `structure ModeledCompetitorRealization where competitor : AdmissibleCompetitor lam carrier_eq : competitor.carrier = a.carrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization.asCompetitor

Source hash: 6c62dc13fbd8bcc95bc9815a8c9dace231a72bb0614d3ccc231b26790b97baab

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3053-3054, the source contract is `def asCompetitor : AdmissibleCompetitor lam := realization.competitor`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization.asCompetitor_weightedArea

Source hash: ac19f7d26c088671b4dd0552d598cd1124637e2815ea1238d0d5daf917f4ca9b

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3059-3064, the source contract is `theorem asCompetitor_weightedArea : realization.asCompetitor.WeightedArea = a.scalarWeightedArea`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.ModeledCompetitorRealization.asCompetitor_weightedPerimeter

Source hash: c73f289c45ecd5870b19789b66c484f95d8c8aad15bb42c8e172d87d107523d8

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3065-3075, the source contract is `theorem asCompetitor_weightedPerimeter : realization.asCompetitor.WeightedPerimeter = a.scalarWeightedPerimeter`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.WeightedArea

Source hash: 5d054747c494e793da416a19e2d895420eb9e121366a5645e79b7ab53a0c3c7f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:265-268, the source contract is `def WeightedArea : ℝ := _root_.WeightedArea lam a.carrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.WeightedPerimeter

Source hash: ac2df4f2daf804e358c901d1a6b496eaadc35076745e389cd399733fc5dead10

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:269-272, the source contract is `def WeightedPerimeter : ℝ := _root_.WeightedPerimeter lam (FrontierMeasure a.carrier)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.angleTerm

Source hash: 8089feb5fd033f48e23e1ab4579cce22afbbbfdede128400123f3b8b2b2a245332

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:289-291, the source contract is `def angleTerm : ℝ := lam * a.outerAngle + π - a.innerAngle`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.bottomChord

Source hash: 08cc63d3aedc396b5efa9b27fd1bac931beb8b7d14cbce558e5fde942be277c2

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:136-139, the source contract is `def bottomChord : ℝ := 2 * a.sideHalfWidth`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.bottomChord_eq_zero_iff

Source hash: 1b82dffbe772075bcb5095c9dc075644068e9232dc927ae0cd23919b863062dd

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:355-358, the source contract is `theorem bottomChord_eq_zero_iff : a.bottomChord = 0 ↔ a.h = 1 / 2`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.bottomChord_nonneg

Source hash: bbe5a0b68c27b1eab70889a99796728aff04d58c47eda0333c16191c78a8f12a

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:352-354, the source contract is theorem bottomChord_nonneg : $0 \leq a.\text{bottomChord}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.bottomChord_pos

Source hash: 5deb1da8ce95e59329d432d4f1484bad3b915566f05bd8a9e6b37a69ad4de177

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:359-362, the source contract is theorem bottomChord_pos (hh : $a.h \neq 1/2$) : $0 < a.\text{bottomChord}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.bottomSegmentCarrier

Source hash: 18e70891a1386b33220d27cc96af45d0ffdcc88b970fd467c64035c04da589f5

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:132-135, the source contract is def bottomSegmentCarrier : Set PlanePoint := {p | p.2 = -1} ∩ {p | |p.1| ≤ a.sideHalfWidth}. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.bottomSegmentCarrier_subset_frontier

Source hash: 82d466e4484f4553cdf0fe29df6958b297f06ff314377fa09237775f676763e1

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1058-1079, the source contract is theorem bottomSegmentCarrier_subset_frontier : $a.\text{bottomSegmentCarrier} \subseteq \text{frontier } a.\text{carrier}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.boundaryJoinSet

Source hash: 4d107bd731d106706191e0cc66c1eb6e52b0ad83df797a9379d71e46e9ea68c7

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1130-1135, the source contract is `def boundaryJoinSet : Set PlanePoint := {(-a.radius * sin a.outerAngle, 1), (a.radius * sin a.outerAngle, 1), (-a.sideHalfWidth, -1), (a.sideHalfWidth, -1)}`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.boundaryJoinSet_finite

Source hash: fb8b2d3307f8905b1616f7f8934468dae8dc2d1df098e8fa72f92f5a3759db5d

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1136-1141, the source contract is `theorem boundaryJoinSet_finite : a.boundaryJoinSet.Finite`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.boundaryTrace

Source hash: 77ded3af26dffe034b843162e20202cd8e868c69330a0cb95819f9e3b2a0bbf5

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1595-1599, the source contract is `def boundaryTrace : Set PlanePoint := a.leftArcTrace ∪ (a.rightArcTrace ∪ (OneSidedCircularCap.arcTrace a.outerCap ∪ a.bottomSegmentCarrier))`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.boundaryTrace_subset_frontier_carrier

Source hash: daeda01e39ff76764527644f12850660eec76e1d4031dd8579c24a3b55dda6ba

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2840-2847, the source contract is `theorem boundaryTrace_subset_frontier_carrier : a.boundaryTrace ⊆ frontier a.carrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.carrier

Source hash: 0849c97e398a59f917657f18205fab9ffd6aa8e524ee751befd78f1c4735a4ff

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:262-264, the source contract is `def carrier : Set PlanePoint := a.coreCarrier ∪ a.outerCap.carrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.continuous_realizedLeftArcParam

Source hash: 5aa13a8e0a54ece98427e269f77eefa81d89485ad9d1f32dde5c0e6f344ef9b4

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:690-699, the source contract is `lemma continuous_realizedLeftArcParam : Continuous a.realizedLeftArcParam`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.continuous_realizedRightArcParam

Source hash: 3d9c0740b38c1fa009799b8c35785a0cb767211b94d4ca0bc37637a119a58afd

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:700-709, the source contract is `lemma continuous_realizedRightArcParam : Continuous a.realizedRightArcParam`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.coreCarrier

Source hash: 5e82c9d38bf8a70ad9c2ae143c293bbf0238ee48148c3dd2d922c95471160878

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:99-103, the source contract is `def coreCarrier : Set PlanePoint := a.rectangleCarrier ∪ a.leftSegmentCarrier ∪ a.rightSegmentCarrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.coreCarrier_y_bounds

Source hash: a0a8c181bd45290d862033f8524f84e76d590a21a370fbbc0214aab783d9df85

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1447-1455, the source contract is theorem coreCarrier_y_bounds {p : PlanePoint} (hp : p ∈ a.coreCarrier) : |p.2| ≤ 1. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.core_weightedArea_formula

Source hash: 5a80fac760bbad4e64452fa4d575b54c924b85ffbad6442cb8801547dabcbb7

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1530-1550, the source contract is theorem core_weightedArea_formula : _root_.WeightedArea lam a.coreCarrier = a.rectangleArea + a.sideCap.euclideanArea. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.cos_innerAngle

Source hash: e774d0a2578a71402d098d5ac027f2d6d1dbb90425ca111832168cfb9f013412

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:178-183, the source contract is theorem cos_innerAngle : cos a.innerAngle = a.shapeParameter. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.cos_outerAngle

Source hash: 3eb066bbadef1d5d0722d2c322b7cfe8bce8e1eb62204f2872b1f8f1bbb41a24

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:184-188, the source contract is theorem cos_outerAngle : cos a.outerAngle = a.outerArgument. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.cos_sideArcAngle

Source hash: a8f9a8c72dcee4cc4dc59b4da10f2ae6d2741d7865a4b2faca10168fa17cc76a

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:231-237, the source contract is theorem cos_sideArcAngle : cos a.sideArcAngle = -a.shapeParameter. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.euclidean_frontier

Source hash: 5e03c2ded7f93ebce1b77460290c9d90f4e5be41990287351d94b69c624d60ea

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2937-2949, the source contract is theorem euclidean_frontier : frontier (planeEuclideanHomeomorph '' a.carrier) = (planeEuclideanHomeomorph '' a.leftArcTrace) $\cup$ ((planeEuclideanHomeomorph '' a.rightArcTrace) $\cup$ ((planeEuclideanHomeomorph '' OneSidedCircularCap.arcTrace a.outerCap) $\cup$ (planeEuclideanHomeomorph '' a.bottomSegmentCarrier))). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.exact_euclidean_bottomSegmentCarrier_hausdorffMeasure

Source hash: 56949577898a4e9f1638a1a8bab2c43a8263ea9e6cf0976fce47fbff56b3fe00

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1101-1129, the source contract is theorem exact_euclidean_bottomSegmentCarrier_hausdorffMeasure : ($\mu_H[1]$: Measure EuclideanPlane) (planeEuclideanHomeomorph '' a.bottomSegmentCarrier) = ENNReal.ofReal a.bottomChord. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.exact_euclidean_leftArcParam_hausdorffMeasure

Source hash: c6dc496bb93e612df391bdcd13229ada6ddb03a05f7a4a85318498972631a8fc

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:742-778, the source contract is theorem exact_euclidean_leftArcParam_hausdorffMeasure : ($\mu_H[1]$: Measure EuclideanPlane) (a.realizedLeftArcParam '' Icc (-a.radius * a.sideArcAngle) 0) = ENNReal.ofReal (a.radius * a.sideArcAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.exact_euclidean_leftArcTrace_hausdorffMeasure

Source hash: 9b484f3710815971495ad59b2a292d470a94ac2f3cad3edb96b8870979733b3

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:817-824, the source contract is theorem exact_euclidean_leftArcTrace_hausdorffMeasure : ($\mu H[1]$: Measure EuclideanPlane) (planeEuclideanHomeomorph ' ' a.leftArcTrace) = ENNReal.ofReal (a.radius * a.sideArcAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.exact_euclidean_rightArcParam_hausdorffMeasure

Source hash: f1f573c6c931308699d4611129bc237b3c89fe21d3a82fbb640a2bc4f452f054

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:779-816, the source contract is theorem exact_euclidean_rightArcParam_hausdorffMeasure : ($\mu H[1]$: Measure EuclideanPlane) (a.realizedRightArcParam ' ' Icc 0 (a.radius * a.sideArcAngle)) = ENNReal.ofReal (a.radius * a.sideArcAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.exact_euclidean_rightArcTrace_hausdorffMeasure

Source hash: 82deac992e3b3149a73b5a9867d35c880f6af8171bd8434cc5a883788eaac184

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:825-832, the source contract is theorem exact_euclidean_rightArcTrace_hausdorffMeasure : ($\mu H[1]$: Measure EuclideanPlane) (planeEuclideanHomeomorph ' ' a.rightArcTrace) = ENNReal.ofReal (a.radius * a.sideArcAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.frontier_carrier

Source hash: 0ce2b73ef2aba31ce763d170404d962fb774d3c494bf5afe10757388adbfbdb30

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2848-2854, the source contract is theorem frontier_carrier : frontier a.carrier = a.boundaryTrace. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.frontier_carrier_subset_boundaryTrace

Source hash: e4a9f2900fa53b9204f0a28ca553872a35240978918b3cf32712160073f759cb

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2443-2458, the source contract is theorem frontier_carrier_subset_boundaryTrace : frontier a.carrier $\subseteq$ a.boundaryTrace. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.hasModeledCompetitor

Source hash: 6010bc1b92d07ee7af37566e925a969f227d520e7144371bb334d772b94b8386

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3081-3100, the source contract is theorem hasModeledCompetitor : a.HasModeledCompetitor. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.innerAngle

Source hash: bb9acabc2bc9f5427d7041305c401b4e2793720759dd063e3153448db0793dd9

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:53-55, the source contract is def innerAngle : $\mathbb{R}$:= arccos a.shapeParameter. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.innerAngle_lt_pi

Source hash: 31385b2dc48409bb1698d05b2b8cedcea6151f9ce4bedc38534228b77f838b5c

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:168-170, the source contract is theorem innerAngle_lt_pi : a.innerAngle $< \pi$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.innerAngle_pos

Source hash: e78599ffc8793999ca239c142df39a815df5b9220a19bd2bd882d169b85658b5

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:165-167, the source contract is theorem innerAngle_pos : $0 < a.\text{innerAngle}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.innerAngle_semicircular

Source hash: 136b95db3cfbe6ffba49a9d377637e6fd44e92cfd37653b368ff453e7d5af642

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:370-373, the source contract is theorem innerAngle_semicircular (hh : $a.h = 1 / 2$) : $a.\text{innerAngle} = \pi / 2$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.innerSin_pos

Source hash: 88694b992a38f1dff3c67f32d082f4eb742c98f01874e62941887bfa3c13f04d

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:216-218, the source contract is theorem innerSin_pos : $0 < \sin a.\text{innerAngle}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.integrableOn_carrier

Source hash: f50bce5009ef96b4081db29960a7e72186fb76edbfedf9fa6fd538bc59ab8c22

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3000-3011, the source contract is theorem integrableOn_carrier : IntegrableOn (StripDensity lam) a.carrier. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.integrable_frontierMeasure

Source hash: 242285e9e93302048576853114a4bc7f6ca16cdc917f22c33a2fe025def880a8

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:3012-3042, the source contract is theorem integrable_frontierMeasure : Integrable (StripDensity lam) (FrontierMeasure a.carrier). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major

exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isClosed_bottomSegmentCarrier

Source hash: 4502cfa8cbb9cfeeb9921da9cbccfe0b4a2a7c969d9473aede4beb6ee9ada554

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1589-1594, the source contract is theorem `isClosed_bottomSegmentCarrier : IsClosed a.bottomSegmentCarrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isClosed_carrier

Source hash: 86d49c14fed38ff9d123ec7002dc853d7ab13d112817308f4d1ef5ac7d463980

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1307-1309, the source contract is theorem `isClosed_carrier : IsClosed a.carrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isClosed_coreCarrier

Source hash: c084bdc57b0172b2f03efa14200c62e0c7ef9b7e0265bac3ea60cec5c75fc390

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1303-1306, the source contract is theorem `isClosed_coreCarrier : IsClosed a.coreCarrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isClosed_leftSegmentCarrier

Source hash: 9a06aa24438ad23f7c7f01c4c87ea6fc01e48b3ecec4d358799e726bea34abb1

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1289-1295, the source contract is theorem `isClosed_leftSegmentCarrier : IsClosed a.leftSegmentCarrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isClosed_rectangleCarrier

Source hash: a040fbd7853bc5d56529ed0d45def53a17db2084e9e6f891db5067b23f32a560

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1282-1288, the source contract is theorem `isClosed_rectangleCarrier : IsClosed a.rectangleCarrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isClosed_rightSegmentCarrier

Source hash: 97ec89af687cf23451a62fdd3fb4816f90150533b2e6c5aca3fc7cb1fe8e300f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1296-1302, the source contract is theorem `isClosed_rightSegmentCarrier : IsClosed a.rightSegmentCarrier`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.isometry_sideArcComplexIsometryMap

Source hash: 5f9f98564eebaed09c3cd958295e743ceabd92f2b665793225e0fb64c4812ba4

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:653-664, the source contract is theorem `isometry_sideArcComplexIsometryMap (center : PlanePoint) : Isometry (sideArcComplexIsometryMap center)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.leftArcParam

Source hash: 4f29c060ae66edf060cbefb1375e17d2584dc8c159343c7bce4a1ff009d7632c

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:120-126, the source contract is `def leftArcParam (s : ℝ) : PlanePoint := (a.leftCenter.1 + a.radius * sin (s / a.radius), a.leftCenter.2 - a.radius * cos (s / a.radius))`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.leftArcParam_image_Icc

Source hash: a53d3bd58a5cf0ca117ca4045079faea921dc67e709030e1bc93b3bc04963479

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:446-546, the source contract is `theorem leftArcParam_image_Icc : a.leftArcParam '' Icc (-a.radius * a.sideArcAngle) 0 = a.leftArcTrace`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.leftArcTrace

Source hash: cfb74d8a54830ee4e695b37318a8d8c8a086063cc569f3e8d4fa167da3d5d0b9

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:104-110, the source contract is `def leftArcTrace : Set PlanePoint := {p | (p.1 - a.leftCenter.1) ^ 2 + (p.2 - a.leftCenter.2) ^ 2 = a.radius ^ 2 ^ p.1 ≤ a.leftCenter.1 ^ (-1 : ℝ) ≤ p.2 ^ p.2 ≤ 1}`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.leftCenter

Source hash: ae634544d50a625f2da0ab83dddef536c2004aefe8a002e19ec3ff792534bdf2

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:70-72, the source contract is `def leftCenter : PlanePoint := (-a.sideHalfWidth, a.radius - 1)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.leftSegmentCarrier

Source hash: 7a6b1fb5a536fc1614b926138e279c79369045c2b292e00c145e1a0e699dbced

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:85-91, the source contract is `def leftSegmentCarrier : Set PlanePoint := {p | (p.1 - a.leftCenter.1) ^ 2 + (p.2 - a.leftCenter.2) ^ 2 ≤ a.radius ^ 2} ∩ ({p | p.1 ≤ a.leftCenter.1} ∩ ({p | (-1 : ℝ) ≤ p.2} ∩ {p | p.2 ≤ 1}))`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.leftUpperJunction

Source hash: a660a79d351317f739116ce079a065c0c8bcd856fae9b73442ad8d1339dd7ec

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:946-957, the source contract is theorem leftUpperJunction : (a.leftCenter.1 - a.radius * sin a.innerAngle, 1) = a.outerCap.leftEndpoint. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.measurableSet_carrier

Source hash: 25696aa660c5c71e8ce3e6a179a069d25633bc99ff664ad8e93ea6e319cb501f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1310-1316, the source contract is theorem measurableSet_carrier : MeasurableSet a.carrier. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_bottomSegmentCarrier

Source hash: ac873340c9031353c9d26e2404b771c8a0ff1b638ea75b1a24ed627ed5bc802d

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2855-2860, the source contract is theorem measurableSet_euclidean_bottomSegmentCarrier : MeasurableSet (planeEuclideanHomeomorph ' ' a.bottomSegmentCarrier). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_leftArcTrace

Source hash: 791dc1e2fe6e2d0cb10c51a3bd4f4018c5ee53a25eeef24b22bc0d4e6f0f1192

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:732-736, the source contract is theorem measurableSet_euclidean_leftArcTrace : MeasurableSet (planeEuclideanHomeomorph ' ' a.leftArcTrace). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.measurableSet_euclidean_rightArcTrace

Source hash: 64197e0a340a1643ad8bc1525fe961f5b8792a0e7cb9321824fa2436e9d02bb4

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:737-741, the source contract is theorem measurableSet_euclidean_rightArcTrace : MeasurableSet (planeEuclideanHomeomorph ' ' a.rightArcTrace). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.modeledWeightedArea

Source hash: 35499fdca0a3bbbed11909d6dec0bbb1c83123d6ea5cb0599717ee289f567e87e

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:278-284, the source contract is def modeledWeightedArea : $\mathbb{R}$:= a.rectangleArea + a.sideCap.euclideanArea + lam * a.outerCap.euclideanArea. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.modeledWeightedArea_formula

Source hash: b0e70e8aeaefad1e18dbaff1d3d6364f544c6042f48ce97bebd3fa5c6cc8e74e

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:885-895, the source contract is theorem modeledWeightedArea_formula : a.modeledWeightedArea = a.scalarWeightedArea. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.modeledWeightedPerimeter

Source hash: 76198be6eb5b8790a4af34449bf29c4b963314e1e4bd268b4d3c9cdb2e69ea0d

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:285-288, the source contract is def modeledWeightedPerimeter : $\mathbb{R}$:= a.sideCap.arclength + lam * a.outerCap.arclength + a.bottomChord. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.modeledWeightedPerimeter_formula

Source hash: bb4c05a2ca2fd9d5b63be1b6ad16da716cc43497abcad228054765e6eaa2e39e

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:896-903, the source contract is theorem modeledWeightedPerimeter_formula : a.modeledWeightedPerimeter = a.scalarWeightedPerimeter. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.one_lt_radius

Source hash: 4e892eef63bc8252d5a243e2b5495e748b55fa956d516b3510c9e5289ec264d9

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:148-152, the source contract is theorem one_lt_radius : $1 < a.radius$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerAngle

Source hash: 0e2ff2070394fdda5b4ec09824f47ed543dabdfef101c076ad378e0c92be2ff47

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:59-63, the source contract is def outerAngle : $\mathbb{R} := \arccos a.outerArgument$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerAngle_lt_pi

Source hash: ce135b6a3dc191fb33efd487aab9da31365b8ec038647a624b62a436ba911d34

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:174-177, the source contract is theorem outerAngle_lt_pi : $a.outerAngle < \pi$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerAngle_pos

Source hash: 760f9b90fd549c278aa760178ec027d7c643f8b5bd6d4bf942c86f6a65bb0e28

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:171-173, the source contract is theorem outerAngle_pos : $0 < a.outerAngle$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerAngle_semicircular

Source hash: 9ff26a1092e7089058ecf98f24fecce78972031b475870c0e4d30009e831f8ee

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:374-377, the source contract is theorem outerAngle_semicircular (hh : $a.h = 1 / 2$) : $a.outerAngle = \pi / 2$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerArgument

Source hash: d6cfc025aa1368c77ccefa6b050dbf8bfa113c35569b4434625ec2d735b91f41

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:56-58, the source contract is def outerArgument : $\mathbb{R} := a.shapeParameter / \lambda$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerArgument_mem

Source hash: fc2a8b1032448c01ece41eff0bcd5d748f70ee556d5721867bb36b59d56ce63a

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:156-164, the source contract is theorem outerArgument_mem : $-1 < a.outerArgument \wedge a.outerArgument < 1$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap

Source hash: 5482055613f2ec0454ddd9c71d722bea04494543bc659e447732870c662eb86f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:251-261, the source contract is def outerCap : OneSidedCircularCap where chord := $2 * a.radius * \sin a.outerAngle$ theta := $a.outerAngle$ midpointX := 0 baseY := 1 side := .upper chord_pos := mul_pos (mul_pos

(by norm_num) a.radius_pos) a.outerSin_pos theta_pos := a.outerAngle_pos theta_lt_pi := a.outerAngle_lt_pi. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap_arcLength

Source hash: e4c6a09e492a4c9395262e1c277f2cd0b185e72ae34042a7f14138b12c6b4c19

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:875-884, the source contract is theorem outerCap_arcLength : a.outerCap.arcLength = 2 * a.radius * a.outerAngle. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap_center

Source hash: 1057a7a30ad6e7558c1c0e586fc0147d20f1fc01e6c3e508acda901e55191c4c

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:929-933, the source contract is theorem outerCap_center : a.outerCap.center = a.upperCenter. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap_euclideanArea

Source hash: 3c7709918bd2e08d7c7ea431b339d946f2eb6c3c9d1138f4304b6afe64a153c2

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:851-864, the source contract is theorem outerCap_euclideanArea : a.outerCap.euclideanArea = a.radius ^ 2 * (a.outerAngle - sin a.outerAngle * cos a.outerAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap_leftEndpoint

Source hash: 369e532ae6bbe742ee13471a1a8cc638a185e8c74c4f29cdde432417763b0fdb

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:934-938, the source contract is theorem outerCap_leftEndpoint : a.outerCap.leftEndpoint = (-a.radius * sin a.outerAngle, 1). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap_radius

Source hash: f107696210fe89f5a746d54584303c70da482f05587bc229f2d37aaa9494b57f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:419-422, the source contract is `theorem outerCap_radius : a.outerCap.radius = a.radius`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerCap_rightEndpoint

Source hash: 0bf0b00b58f0396d9b9b8bc9e08617bcd76cf3f0b76937e23b52dec1f9b5fe12

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:939-945, the source contract is `theorem outerCap_rightEndpoint : a.outerCap.rightEndpoint = (a.radius * sin a.outerAngle, 1)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.outerSin_pos

Source hash: feedd827e6203dedcb21f4f474b9fea36ae3ea708b33b644b944093c5f958959

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:213-215, the source contract is `theorem outerSin_pos : 0 < sin a.outerAngle`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.pairwiseOverlap_h1_null

Source hash: 545b8d3e4922237126e7364ee9bb07b444cf8cada72b02e8afca89501d2f79b3

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1258-1273, the source contract is `theorem pairwiseOverlap_h1_null : (μH[1] : Measure EuclideanPlane) (planeEuclideanHomeomorph '' pairwiseOverlap a.leftArcTrace a.rightArcTrace (OneSidedCircularCap.arcTrace a.outerCap) a.bottomSegmentCarrier) = 0`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.pairwiseOverlap_subset_boundaryJoinSet

Source hash: f8736f4dd5a67fc4149e15b1ed3e3f21030a49e12643cd517b41158566d4664f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1142-1257, the source contract is theorem pairwiseOverlap_subset_boundaryJoinSet : pairwiseOverlap a.leftArcTrace a.rightArcTrace (OneSidedCircularCap.arcTrace a.outerCap) a.bottomSegmentCarrier $\subseteq$ a.boundaryJoinSet. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.radius

Source hash: 44bdb144ee33a0fe634670e914d151b3f9a50f2b2f7db8da7c3fad23bdfbf70e

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:47-49, the source contract is def radius : $\mathbb{R} := 1 / a.h$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.radius_mul_h

Source hash: e90d4095fef5e1e089527a179a53736b5dbf462b6149abecb5375e96b7380078

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:197-200, the source contract is theorem radius_mul_h : a.radius * a.h = 1. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.radius_pos

Source hash: a610fc617d9a92f1154e4b9627bb16a0d909464945f9e22ccb900c975acb2c7d

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:145-147, the source contract is theorem radius_pos : $0 < a.radius$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam

Source hash: 51fbf9517aa10415de53dc106db635c5f70b9c39860ba75778e696cdebf2d6c3

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:665-668, the source contract is `noncomputable def realizedLeftArcParam (s : ℝ) : EuclideanPlane := planeEuclideanHomeomorph (a.leftArcParam s)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam_eq

Source hash: 538a7925bd9a6977d5e74933bd5f951885abb82dc534e3d102db01bf6aa01a1e

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:672-680, the source contract is `lemma realizedLeftArcParam_eq (s : ℝ) : a.realizedLeftArcParam s = sideArcComplexIsometryMap a.leftCenter (unitCircleArc a.radius s)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.realizedLeftArcParam_image_Icc

Source hash: 06e6662a39cd836487840418c7224db9c1b8779c37aa295aab4e1109503a76c7

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:710-720, the source contract is `theorem realizedLeftArcParam_image_Icc : a.realizedLeftArcParam ' ' Icc (-a.radius * a.sideArcAngle) 0 = planeEuclideanHomeomorph ' ' a.leftArcTrace`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.realizedRightArcParam

Source hash: 8a963395af5bf096a9c5748471f90364728d8fe511c4e2cc0591bd2068d9e016

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:669-671, the source contract is `noncomputable def realizedRightArcParam (s : ℝ) : EuclideanPlane := planeEuclideanHomeomorph (a.rightArcParam s)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.realizedRightArcParam_eq

Source hash: 3339fa9296e8c62289cd021c3334658bb86b73401d06cffbaa790d7423ffd93e

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:681-689, the source contract is lemma realizedRightArcParam_eq (s : ℝ) : a.realizedRightArcParam s = sideArcComplexIsometryMap a.rightCenter (unitCircleArc a.radius s). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.realizedRightArcParam_image_Icc

Source hash: 9469dc298ecad88323016b9907111e322a864f0831a991b637ca11824881af16

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:721-731, the source contract is theorem realizedRightArcParam_image_Icc : a.realizedRightArcParam '' Icc 0 (a.radius * a.sideArcAngle) = planeEuclideanHomeomorph '' a.rightArcTrace. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rectangleArea

Source hash: 3314187b7b48ab958a7d8415bcf10afdb2dd85317d354dfd049f5344c2569e90

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:273-277, the source contract is def rectangleArea : ℝ := 4 * a.sideHalfWidth. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rectangleCarrier

Source hash: d451775e4018b5bae444c6a8ed6911c5adcc23db4a0efc68ab71c655f2933477

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:80-84, the source contract is def rectangleCarrier : Set PlanePoint := ({p | -a.sideHalfWidth ≤ p.1} ∩ {p | p.1 ≤ a.sideHalfWidth}) ∩ ({p | (-1 : ℝ) ≤ p.2} ∩ {p | p.2 ≤ 1}). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rightArcParam

Source hash: 74515bd4fc474dfa2c52a14662b8cc34aa0f40ef9d8185819e05d1de94ccf588

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:127-131, the source contract is `def rightArcParam (s : ℝ) : PlanePoint := (a.rightCenter.1 + a.radius * sin (s / a.radius), a.rightCenter.2 - a.radius * cos (s / a.radius))`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rightArcParam_image_Icc

Source hash: d47f88fb5eadc69e4f4bc7ce3bc201dc34d60ad570f34d3d29c4d0184ef0f8f1

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:547-648, the source contract is `theorem rightArcParam_image_Icc : a.rightArcParam '' Icc 0 (a.radius * a.sideArcAngle) = a.rightArcTrace`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rightArcTrace

Source hash: 828cebab0c56e67a2a3cfaae9e912a8a76f497287dcea6b589e1a22a40edd616

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:111-119, the source contract is `def rightArcTrace : Set PlanePoint := {p | (p.1 - a.rightCenter.1) ^ 2 + (p.2 - a.rightCenter.2) ^ 2 = a.radius ^ 2 ∧ a.rightCenter.1 ≤ p.1 ∧ (-1 : ℝ) ≤ p.2 ∧ p.2 ≤ 1}`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rightCenter

Source hash: c1d50740a1dfb76a9c424f19f6e78589f8a8ab930ab1727da5282aecd42c3f25

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:73-75, the source contract is `def rightCenter : PlanePoint := (a.sideHalfWidth, a.radius - 1)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rightSegmentCarrier

Source hash: 889596cedc4d7d957005581540478e3a458f9f3d1f37c9af6b1a713eee05b078

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:92-98, the source contract is `def rightSegmentCarrier : Set PlanePoint := {p | (p.1 - a.rightCenter.1) ^ 2 + (p.2 - a.rightCenter.2) ^ 2 ≤ a.radius ^ 2} ∩ ({p | a.rightCenter.1 ≤ p.1} ∩ ({p | (-1 : ℝ) ≤ p.2} ∩ {p | p.2 ≤ 1}))`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.rightUpperJunction

Source hash: ae6c6a14cc5b76c0f3aa60155a1fb16c629e4d67ae173c7593aa69c135bbff39

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:958-968, the source contract is `theorem rightUpperJunction : (a.rightCenter.1 + a.radius * sin a.innerAngle, 1) = a.outerCap.rightEndpoint`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.scalarWeightedArea

Source hash: 195fdb382725116907cc721bab689c5c6f6f34b356a442f900ba649d89b62b9c

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:295-299, the source contract is `def scalarWeightedArea : ℝ := (a.angleTerm + (a.shapeParameter + 2) * a.sineGap) / a.h ^ 2`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.scalarWeightedPerimeter

Source hash: fcbac6602a2b133a8e714c37e1ef51bfb698ab0ae754dcd7eca0cb6e9c49204d

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:300-302, the source contract is `def scalarWeightedPerimeter : ℝ := 2 * (a.angleTerm + a.sineGap) / a.h`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.shapeParameter

Source hash: 29190b0ec0676db3ee8da889fc7b1e078bce75188f6b4271945e3fee045f2920

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:50-52, the source contract is `def shapeParameter : ℝ := 2 * a.h - 1`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.shapeParameter_mem

Source hash: 018fe2d07fb99b9420ec19a75327dc616f0119968b1c22fb83265de47b1261df

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:153-155, the source contract is `theorem shapeParameter_mem : -1 < a.shapeParameter ∧ a.shapeParameter < 1`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideArcAngle

Source hash: c68849312591ff361810874588f817eab8327b61ad9b815244ad1ab61a55597a

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:64-65, the source contract is `def sideArcAngle : ℝ := π - a.innerAngle`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideArcAngle_lt_pi

Source hash: 27c4b01d9c8710ab675792367c9a16f3b7952720c452f471c97b46d487e08c08

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:223-226, the source contract is `theorem sideArcAngle_lt_pi : a.sideArcAngle < π`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideArcAngle_pos

Source hash: fb3e4cdd6ff568be2285a238c6293705a5d18ec17ca0000cf9fc5fd3be6f53e2

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:219-222, the source contract is `theorem sideArcAngle_pos : 0 < a.sideArcAngle`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at

$h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideArcComplexIsometryMap

Source hash: 2d2b0ae6442b1aeb7779e0e2c761b53071a3b1948418645b546ba443dcc3b322

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:649-652, the source contract is `noncomputable def sideArcComplexIsometryMap (center : PlanePoint) (z : ℂ) : EuclideanPlane := WithLp.toLp 2 (center.1 + z.im, center.2 - z.re)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideCap

Source hash: 62a5fb2c6003ebc71ae1cea156d819929dd2061bdfd35d03ac33c3bf896e8fd0

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:238-250, the source contract is `def sideCap : OneSidedCircularCap where chord := 2 * a.radius * sin a.innerAngle theta := a.sideArcAngle midpointX := 0 baseY := 1 side := .lower chord_pos := mul_pos (mul_pos (by norm_num) a.radius_pos) a.innerSin_pos theta_pos := a.sideArcAngle_pos theta_lt_pi := a.sideArcAngle_lt_pi`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideCap_arcLength

Source hash: 0735084a84605f457abd29f126bfc3b41f13fe2e361355b34befa71a89121ae9

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:865-874, the source contract is `theorem sideCap_arcLength : a.sideCap.arcLength = 2 * a.radius * a.sideArcAngle`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideCap_center

Source hash: 5c32cc16475c4c5293c8fcbce19421c65fe608e9916172122a7cb5ff16d6b389

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:430-438, the source contract is `theorem sideCap_center : a.sideCap.center = (0, a.radius - 1)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideCap_euclideanArea

Source hash: b6b94a03ab653ee46e011e7fa757125fdbdfe26f1473cd6911d9100fe92583e4

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:833-850, the source contract is theorem sideCap_euclideanArea : a.sideCap.euclideanArea = a.radius ^ 2 * (a.sideArcAngle - sin a.sideArcAngle * cos a.sideArcAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideCap_radius

Source hash: 860b391d38da56536b3934963761766a28238619240605477033823fc900d1af

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:423-429, the source contract is theorem sideCap_radius : a.sideCap.radius = a.radius. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideHalfWidth

Source hash: c1a6374b950f6fdf60b58062fd5753e6cafd953a8fbac46cb6164191b66af1b3

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:66-69, the source contract is def sideHalfWidth : $\mathbb{R}$:= a.radius * (sin a.outerAngle - sin a.innerAngle). The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideHalfWidth_eq_zero_iff

Source hash: aa31042682af8c9fc65e3a19a4a9ce7dc71ed7e2ad3e807732dcfcc2da35644f

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:344-351, the source contract is theorem sideHalfWidth_eq_zero_iff : a.sideHalfWidth = 0 $\leftrightarrow$ a.h = 1 / 2. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideHalfWidth_nonneg

Source hash: be70cb71b25ed2f43754461915097af9183cc7aac3504655dda98d8a67b605ec

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:338-340, the source contract is theorem sideHalfWidth_nonneg : $0 \leq a.\text{sideHalfWidth}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sideHalfWidth_pos

Source hash: a769284707d6bf25198d747cd5846bae22f2f394cc9d09e601cfcab854579d7a

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:341-343, the source contract is theorem sideHalfWidth_pos (hh : $a.h \neq 1/2$) : $0 < a.\text{sideHalfWidth}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sin_inner_le_sin_outer

Source hash: def468198882d033570072298a83ae1d41fc003f0ecb9d0b4ee3a6b94e6a1d10

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:303-311, the source contract is theorem sin_inner_le_sin_outer : $\sin a.\text{innerAngle} \leq \sin a.\text{outerAngle}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sin_inner_lt_sin_outer

Source hash: b3a6348544f17baf0cf9a8d51663f5b8b6187359294117163fc5d2afba78dd58

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:312-337, the source contract is theorem sin_inner_lt_sin_outer (hh : $a.h \neq 1/2$) : $\sin a.\text{innerAngle} < \sin a.\text{outerAngle}$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sin_sideArcAngle

Source hash: 41165bd4a88985079154632e1304282b8868bb98cc6b1d97bb3ea19fae0dd

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:227-230, the source contract is theorem `sin_sideArcAngle` : `sin a.sideArcAngle = sin a.innerAngle`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.sineGap

Source hash: e9d7a7074e27fe28a23aa0db9c4cf9b0ef4e5f2181f1cfab6098f7cc2e240ecb

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:292-294, the source contract is `def sineGap` : $\mathbb{R} := \sin a.outterAngle - \sin a.innerAngle$. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.upperCenter

Source hash: 6dabb1e2d2dfef1af758b047dfb6728a97191f4c64742b75a35f102c81c1c3c1

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:76-79, the source contract is `def upperCenter` : `PlanePoint := (0, 1 - a.radius * cos a.outterAngle)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.volume_coreCarrier

Source hash: e422839d8a82e578aa8619bf8a6d240c5dc840fd4d4e1880ecc09a0307558380

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1481-1529, the source contract is theorem `volume_coreCarrier` : `volume a.coreCarrier = ENNReal.ofReal (a.rectangleArea + a.sideCap.euclideanArea)`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.weightedArea_eq_modeledWeightedArea

Source hash: 5f7913f0171c133683a735330062b186a060d0e518987c5488549de7bfd06347

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1562-1583, the source contract is theorem `weightedArea_eq_modeledWeightedArea` : `a.WeightedArea = a.modeledWeightedArea`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.weightedArea_formula

Source hash: 53616c2c25615a95257aa4773c87d4c69464dc905650e6ed5b53696dc5f08570

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:1584-1588, the source contract is theorem `weightedArea_formula` : `a.WeightedArea = a.scalarWeightedArea`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.weightedPerimeter_eq_modeledWeightedPerimeter

Source hash: 3dd334c8f3cff9146f23a5d90e6da2520151c567d49c8604446b90aa7a9f4d2b

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2950-2993, the source contract is theorem `weightedPerimeter_eq_modeledWeightedPerimeter` : `a.WeightedPerimeter = a.modeledWeightedPerimeter`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

TypeThreeAssembly.weightedPerimeter_formula

Source hash: c6f51109b06f9f61cbbcb63e9d6e1d950e5a79c8decdf76b0e2fc3f366f8eca

Relation: equivalent

Equivalent in the declared constructor/section context. At TypeThreeAssembly.lean:2994-2999, the source contract is theorem `weightedPerimeter_formula` : `a.WeightedPerimeter = a.scalarWeightedPerimeter`. The assembly constructor has $\lambda > 1$ and the full regular range $0 < h < 1$. This includes a major exterior cap below $h=1/2$ and the zero-width/singleton lower boundary at $h=1/2$. Any extra branch restriction in this particular declaration remains explicit.

Findings: none.

WeightedArea

Source hash: 959b5ddd7363861d4d0d79c211781b3b19e09df50087472690ae0581725b8c6d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:55-60, the source contract is `def WeightedArea (lam : ℝ) (carrier : Set PlanePoint) : ℝ := ∫ p in carrier, StripDensity lam p`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

WeightedPerimeter

Source hash: 8d1f1a60a6d4c5ce03ccda31c1bc34ec4f837b6ce55275e4b0c06fe04e566dba

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:67-73, the source contract is `def WeightedPerimeter (lam : ℝ) (boundaryMeasure : Measure PlanePoint) : ℝ := ∫ p, StripDensity lam p ∂boundaryMeasure`. This is the low-level weighted integral against the explicitly supplied boundary measure, as the author says. The canonical choice is imposed at the admissible-competitor perimeter definition, so this auxiliary measure parameter is not a freedom in the minimization problem.

Findings: none.

arc

Source hash: d1b33dc05b11585801ca5ae18eed6d15a605f3d852a3120f8561f8ec87bdc3a5

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:267-268, the source contract is `noncomputable def arc (x : ℝ) : ℝ := ell (00f x)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

arc_eq

Source hash: 3b036fd44214b60f3d2baaf90be272e48c10ba5e310883f73c7eb20af846505a

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:269-271, the source contract is `lemma arc_eq (x : ℝ) : arc x = ell (00f x)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area

Source hash: 8c44e8dfff416b2208afe91aad54bf1678ad6f2299c75592fe1fed4f6cbf55dc2

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:92-93, the source contract is `noncomputable def area (θ : ℝ) : ℝ := (θ - sin θ * cos θ) / (4 * sin θ ^ 2)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_continuous0n

Source hash: 7d3a7659000f0acd2289cd80ef6ad9d2f860c6244018818d478eae78e42ab6d

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:212-216, the source contract is `lemma area_continuous0n : Continuous0n area (Ioo 0 π)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_deriv_pos

Source hash: 9b40e28d2efa8873c77b49eab27397ba54c7d20988e9930377fea092b5b1fc66

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:125-130, the source contract is `lemma area_deriv_pos {θ : ℝ} (h0 : 0 < θ) (hπ : θ < π) : 0 < (sin θ - θ * cos θ) / (2 * sin θ ^ 3)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_lt_self

Source hash: 505f4cfdbccff59a35c0181afbb77947295aa04ad4c13ed1bce28a912b3e1042

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:158-172, the source contract is `lemma area_lt_self {θ : ℝ} (h0 : 0 < θ) (h1 : θ ≤ 1) : area θ < θ`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_pi_div_two

Source hash: 038ce6e93459592481f7a2c059cf51404c3244d96fa3828e4004b771a35871c4

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:66-69, the source contract is `lemma area_pi_div_two : area (π / 2) = π / 8`. The normalized saving is $2L(X) - L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to

nonpositive cap area.

Findings: none.

area_pi_sub

Source hash: 6ddb90a2a474720c53dc6e87015838175f9560adcb74d22d3aa99ec3da8673f

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:134-140, the source contract is `lemma area_pi_sub {θ : ℝ} (_hθ : θ ∈ Ioo 0 π) : area (π - θ) = (π - θ + sin θ * cos θ) / (4 * sin θ ^ 2)`. $F(\theta)=3\theta-3\sin(\theta)*\cos(\theta)-\pi$ has its unique relevant root in $(0,\pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi-\theta$ remains in $(0,\pi)$, including the major-cap branch.

Findings: none.

area_pi_sub_eq_two_mul

Source hash: a8e24b12489f7a578d21b419b549e47a8c62b05b8eaac09a11261e5958d93945

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:141-153, the source contract is `lemma area_pi_sub_eq_two_mul {θ : ℝ} (hθ : θ ∈ Ioo 0 π) (hF : F θ = 0) : area (π - θ) = 2 * area θ`. $F(\theta)=3\theta-3\sin(\theta)*\cos(\theta)-\pi$ has its unique relevant root in $(0,\pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi-\theta$ remains in $(0,\pi)$, including the major-cap branch.

Findings: none.

area_pos

Source hash: 0af41b50988d0bace92fc2e638571302802d8538245a2c254bedb9f164463ede

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:148-157, the source contract is `lemma area_pos {θ : ℝ} (h0 : 0 < θ) (hπ : θ < π) : 0 < area θ`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_strictMonoOn

Source hash: 5637c9200e5eb02be872785f063a999d67c4275f1b2e3906ef15f278ad2dd469

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:131-147, the source contract is `lemma area_strictMonoOn : StrictMonoOn area (Ioo 0 π)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_surj0n

Source hash: fd1b58a2c230b6612a3da5e4c5ae32d43aeae3c716f031eb3ab7a09dd77ecd3b

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:217-246, the source contract is `lemma area_surj0n : Surj0n area (Ioo 0 π) (Ioi 0)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_tendsto_atTop

Source hash: f69fc5f4e4ff813e6ef3d3de0124d162050bb67630aeab29d98accb32509b368

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:188-211, the source contract is `lemma area_tendsto_atTop : Tendsto area (ℳ[<] π) atTop`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_tendsto_zero

Source hash: 6f401985b37aaf23b22e61608890d9398b8c4c885ef124fa9b1095661462feec

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:173-187, the source contract is `lemma area_tendsto_zero : Tendsto area (ℳ[>] 0) (ℳ 0)`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

area_00f

Source hash: 18eb44e54860ffad838e65be5e17f4d170f11530a4f7b414a16d834abbb9e2c1

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:254-258, the source contract is `lemma area_00f {x : ℝ} (hx : 0 < x) : area (00f x) = x`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

capBoundaryMeasure

Source hash: 5dbd12f9bc57860ad2ce92a638514e0e8d15006685aca7388e2fa2e1689504cd

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1249-1251, the source contract is `def capBoundaryMeasure (c : OneSidedCircularCap) : Measure PlanePoint := intervalPushforward (capParam c) (capStart c) (capEnd c)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

capBoundaryMeasure_integrable

Source hash: 0065afd47de261467dd963dc3f68bd9fc5e9eab2c424cfffbd00f2e7c2919a7

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2260-2267, the source contract is `lemma capBoundaryMeasure_integrable (lam : ℝ) (c : OneSidedCircularCap) : Integrable (StripDensity lam) (capBoundaryMeasure c)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

capComplexIsometryMap

Source hash: 3ed78deaf1597a91c6218ca8fe10519ba693355e7329e0301f17fcce226f5a43

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:390-397, the source contract is `noncomputable def capComplexIsometryMap (c : OneSidedCircularCap) (z : ℂ) : EuclideanPlane := match c.side with | .upper => WithLp.toLp 2 (c.midpointX + z.im, c.baseY - c.radius * Real.cos c.theta + z.re) | .lower => WithLp.toLp 2 (c.midpointX + z.im, c.baseY + c.radius * Real.cos c.theta - z.re)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capEnd

Source hash: a140ca726638df619aef507f8019eb9f7ca9503d3231f8bfa8f0a64feb377161

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1246-1248, the source contract is `def capEnd (c : OneSidedCircularCap) : ℝ := c.radius * c.theta`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

capParam

Source hash: d80d6cd5283176fe8b29770f039d1135a8892539144575a4bc1d9962ad427168

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1241-1243, the source contract is `def capParam (c : OneSidedCircularCap) (s : ℝ) : PlanePoint := c.arcPoint (s / c.radius)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

capParam_image_Icc_eq_arcTrace

Source hash: f412e9294ab946fde4ef494447fa75b0b57b0600c2bd61007639df1bb35c0e9e

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:947-1070, the source contract is `theorem capParam_image_Icc_eq_arcTrace (c : OneSidedCircularCap) : capParam c '' Icc (capStart c) (capEnd c) = c.arcTrace`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capPartitionPoint

Source hash: 49ac6ecbe67b9040843eeee672d7b0f08055016781814c3ce0fa4e7d31ff805a

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:154-156, the source contract is `def capPartitionPoint (r theta : ℝ) (N i : ℕ) : ℝ := -r * theta + (i : ℝ) * (2 * r * theta / (N : ℝ))`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capPartitionPoint_dist

Source hash: 6b90645e53e16e3855618b966b04af3ee0f15439043926ba1133c583e65b5a00

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:185-207, the source contract is `lemma capPartitionPoint_dist {r theta : ℝ} (hr : 0 < r) (htheta : theta ∈ Ioo 0 Real.pi) {N i : ℕ} (hN : 0 < N) : dist (unitCircleArc r (capPartitionPoint r theta N i)) (unitCircleArc r (capPartitionPoint r theta N (i + 1))) = 2 * r * Real.sin (theta / (N : ℝ))`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capPartitionPoint_end

Source hash: b5d0e986af4c921461bd13ce89490a41fdc02991eaadda7d26f70db465413519

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:161-166, the source contract is lemma `capPartitionPoint_end {r theta : ℝ} {N : ℕ} (hN : 0 < N) : capPartitionPoint r theta N = r * theta`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capPartitionPoint_mem

Source hash: 8f26d09621c071ac8a6a1935a21253dfb29cb269e0ec21dc71e59a40f0529e9f

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:175-184, the source contract is lemma `capPartitionPoint_mem {r theta : ℝ} (hr : 0 < r) (htheta : 0 < theta) {N i : ℕ} (hN : 0 < N) (hi : i ≤ N) : capPartitionPoint r theta N i ∈ Icc (-r * theta) (r * theta)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capPartitionPoint_zero

Source hash: 1714782661300a21ee2d4243c7e4fbe79e1e97ce545ea187699c13d709db2db9

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:157-160, the source contract is lemma `capPartitionPoint_zero (r theta : ℝ) (N : ℕ) : capPartitionPoint r theta N 0 = -r * theta`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

capStart

Source hash: 594eb6e2bbf3cbc7a2397b6af7a93506027168fa75ecf7f70b7012b2fc6e208a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1244-1245, the source contract is `def capStart (c : OneSidedCircularCap) : ℝ := -c.radius * c.theta`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_arclength_eq_two_radius_theta

Source hash: edad6e24948a1499ce1a66a5efdc84d9c8f7899f5c52d6551716312e95163fb3

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:451-456, the source contract is lemma `cap_arclength_eq_two_radius_theta (c : OneSidedCircularCap) : c.arclength = 2 * c.radius * c.theta`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

cap_integrableOn

Source hash: bbc4a1a9af0d45bee6aa1c1152d956e0e945a869f3550ca289ef7022556292ef

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2303-2308, the source contract is lemma `cap_integrableOn (lam : ℝ) (c : OneSidedCircularCap) : IntegrableOn (StripDensity lam) c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_lower_external_density

Source hash: baaf83c56acf8edc18268269454b4946af9c7098831da004f1da856c3f80d9b2

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1341-1352, the source contract is theorem `cap_lower_external_density (lam : ℝ) (c : OneSidedCircularCap) (hside : c.side = .lower) (hbase : c.baseY = -1) {s : ℝ} (hs : s ∈ Ioo (capStart c) (capEnd c)) : StripDensity lam (capParam c s) = lam`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_lower_weightedArea

Source hash: af4bcab3c0cdacd24a9ba22f365ac164c96fd2be59faf0a944ca162de8e4ada2

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2360-2389, the source contract is lemma `cap_lower_weightedArea (lam : ℝ) (c : OneSidedCircularCap) (hside : c.side = .lower) (hbase : c.baseY = -1) : WeightedArea lam c.carrier = lam * c.euclideanArea`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_lower_weightedIntegral

Source hash: 5575ee03ce1e095004c86962659c2bcf1dde6c0b1ad90b439220173f4e11e7fc

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1373-1394, the source contract is `theorem cap_lower_weightedIntegral (lam : ℝ) (c : OneSidedCircularCap) (hside : c.side = .lower) (hbase : c.baseY = -1) : ∫ p, StripDensity lam p ∂capBoundaryMeasure c = lam * c.arcLength`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_partition_polygon_lower

Source hash: 4dc7fcd936b6c56a4643f00c8d67c38ce24d6351f625a8873960fc070d61a446

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:208-268, the source contract is `lemma cap_partition_polygon_lower {r theta : ℝ} (hr : 0 < r) (htheta : theta ∈ Ioo 0 Real.pi) (N : ℕ) (hN : 0 < N) : ENNReal.ofReal ((N : ℝ) * (2 * r * Real.sin (theta / (N : ℝ)))) ≤ (μH[1] : Measure ℂ) (unitCircleArc r ' Icc (-r * theta) (r * theta))`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

cap_upper_external_density

Source hash: 0d4d779bde08a636e8536f594b78dfa9705a31925f9d2e9248d0d62fdf5aa11a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1329-1340, the source contract is `theorem cap_upper_external_density (lam : ℝ) (c : OneSidedCircularCap) (hside : c.side = .upper) (hbase : c.baseY = 1) {s : ℝ} (hs : s ∈ Ioo (capStart c) (capEnd c)) : StripDensity lam (capParam c s) = lam`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_upper_weightedArea

Source hash: 2f2165b544886739ec40b854c29666e67f06c7a423bd046c04eb224a5c6ee0ce

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2334-2359, the source contract is `lemma cap_upper_weightedArea (lam : ℝ) (c : OneSidedCircularCap) (hside : c.side = .upper) (hbase : c.baseY = 1) : WeightedArea lam c.carrier = lam * c.euclideanArea`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cap_upper_weightedIntegral

Source hash: 45644ff66dcb2be9e3cb6c626527848bf909f532df9be34ffa1977e24d25b8f4

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1353-1372, the source contract is `theorem cap_upper_weightedIntegral (lam : ℝ) (c : OneSidedCircularCap) (hside : c.side = .upper) (hbase : c.baseY = 1) : ∫ p, StripDensity lam p ∂capBoundaryMeasure c = lam * c.arcLength`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cmv_comparison_bound_optimized

Source hash: 507439948a37eb806f48ea9934d720941448689e7e9b24960cacb11d308a3d9b

Relation: equivalent

Equivalent in the declared constructor/section context. At Certificate.lean:746-753, the source contract is `theorem cmv_comparison_bound_optimized {lam x : ℝ} (hlam : (1.2581840884 : ℝ) ≤ lam) (hx : 0 < x) : 1 / lam < g x`. The exact threshold is the rational decimal 1.2581840884. The conclusion is the strict inequality $1/\lambda < g(X)$ for every $X > 0$ under the displayed threshold hypothesis. The argument uses the global minimum, not a finite set of positive-area samples.

Findings: none.

cmv_type_four_not_minimizing

Source hash: 6d0ea3949f7dbdcb13da7efae9522f57a17102179e82571502380037275e9220

Relation: equivalent

Equivalent in the declared constructor/section context. At CandidateExclusion.lean:18-37, the source contract is `theorem cmv_type_four_not_minimizing {lam : ℝ} (hlam : (1.2581840884 : ℝ) ≤ lam) (candidate : FourArcCandidate lam) (hcandidate : candidate.SatisfiesCMVTypeIVHypotheses) (hmin : candidate.IsWeightedPerimeterMinimizer) : False`. This is exclusion against the defined equal-area admissible class. The proof obtains an admissible replacement with strictly smaller canonical weighted perimeter; it does not redefine minimality as a scalar sign.

Findings: none.

continuous_realizedCapParam

Source hash: 439602ca8a442d23dec8f858626a27ca71e9723ce90b04b56ee6e94cd89590f1

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:442-450, the source contract is `lemma continuous_realizedCapParam (c : OneSidedCircularCap) : Continuous (realizedCapParam c)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

continuous_realizedLeftCoreParam

Source hash: 4dc1fbf3a43e8117f791c26e28bd5a2a2466716abc2324d51a2522841ccc46bf

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:727-735, the source contract is `lemma continuous_realizedLeftCoreParam (c : StripCore) : Continuous (realizedLeftCoreParam c)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

continuous_realizedRightCoreParam

Source hash: db06971d2e0a6e54b3ce1b2cb1ffe296cb5ed7ac07495118f10c8f5f82bfe132

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:736-744, the source contract is `lemma continuous_realizedRightCoreParam (c : StripCore) : Continuous (realizedRightCoreParam c)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

continuous_realizedSegmentParam

Source hash: 5ba4505531fe6afcb59f3956aaf9fe6e352c8d17f3326d5e955e4afc3e17c5df

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:868-872, the source contract is `lemma continuous_realizedSegmentParam (seg : HorizontalSegment) : Continuous (realizedSegmentParam seg)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

continuous_unitCircleArc

Source hash: 343bc4472a64687ca829ebbf897c060cb6c7c3cad3716259e96fb84cf091470c

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:14-17, the source contract is `lemma continuous_unitCircleArc (r : ℝ) : Continuous (unitCircleArc r)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

coreArcEnd

Source hash: 7ee06e3535f2fd49529132649f006dc8125adff906e2bb1c8e325a0ca4b835a4

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1474-1476, the source contract is `def coreArcEnd (c : StripCore) : ℝ := c.radius * c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

coreArcStart

Source hash: 4c395d23f09d7608737696a9d4043f61933ddaffbb62953e4470c1085d950b4e

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1472-1473, the source contract is `def coreArcStart (c : StripCore) : ℝ := -c.radius * c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

core_integrableOn

Source hash: ebecbc727d05931a9afed5e77aa787703ada65169aa70e3f372bec6e04bddcac

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2309-2314, the source contract is `lemma core_integrableOn (lam : ℝ) (c : StripCore) : IntegrableOn (StripDensity lam) c.carrier`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

cos_two_mul_sin_sq

Source hash: 457c64175a22d2778423d351b2d6d75e23c9d3a388b2439675e854dd25957517

Relation: equivalent

Equivalent in the declared constructor/section context. At Certificate.lean:117-122, the source contract is `lemma cos_two_mul_sin_sq (u : ℝ) : cos (2 * u) = 1 - 2 * sin u ^ 2`. The numerical constants here denote exact rationals in the reals. The normalized-cap comparison is strict and applies to positive normalized area, not just to the point attaining the minimum.

Findings: none.

ell

Source hash: 9adb32c6bf9324acec1e791c0ec46f0e17df835ffee7b3549cab07bdc3a760c9

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:89-91, the source contract is `noncomputable def ell (θ : ℝ) : ℝ := θ / sin θ`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

euclideanJoinSet

Source hash: 96d51d86f6c32a7f1c31ecb0290be17bec48e5d48a1aa38fdfbae4457025d389

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1257-1259, the source contract is `def euclideanJoinSet (q : ℝ) : Set EuclideanPlane := planeEuclideanHomeomorph '' joinSet q`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

euclideanJoinSet_finite

Source hash: 9539ffd4336468eab518f7e951c9dc4db7ee2d6bd08f0fa98bbda9dcf0817f38

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1260-1262, the source contract is `theorem euclideanJoinSet_finite (q : ℝ) : (euclideanJoinSet q).Finite`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

euclideanPairwiseOverlap

Source hash: c1559aa1737fb404d99a3376b70fbd4f0ff514e71a947a7d30b01a1ce4c1177e

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1591-1597, the source contract is `def euclideanPairwiseOverlap {α : Type*} (A B C D : Set α) : Set α := (A ∩ B) ∪ ((A ∩ C) ∪ ((A ∩ D) ∪ ((B ∩ C) ∪ ((B ∩ D) ∪ (C ∩ D))))`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

euclideanStripDensity

Source hash: 3d074997fd7c2f4c47fc25c7e8a9a39259d6fd4628b103b9b3e4da77e1679848

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1389-1391, the source contract is `def euclideanStripDensity (lam : ℝ) (z : EuclideanPlane) : ℝ := StripDensity lam (planeEuclideanHomeomorph.symm z)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

euclideanStripDensity_integrableOn_of_measure_ne_top

Source hash: 509b7e5c9b45ab61b5e958fce825f92bda813640616ddf5c7591ac420818c5d2

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1404-1411, the source contract is `lemma euclideanStripDensity_integrableOn_of_measure_ne_top (lam : ℝ) {s : Set EuclideanPlane} (hs : (μH[1] : Measure EuclideanPlane) s ≠ ∞) : IntegrableOn (euclideanStripDensity lam) s (μH[1] : Measure EuclideanPlane)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_capParam_hausdorffMeasure

Source hash: ce3a2c8ff8b380f7da5859563d117de2848052560e457ef5f88e25aa88808f43

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:457-483, the source contract is `theorem exact_euclidean_capParam_hausdorffMeasure (c : OneSidedCircularCap) : (μH[1] : Measure EuclideanPlane) (realizedCapParam c '' Icc (capStart c) (capEnd c)) = ENNReal.ofReal c.arcLength`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_capTrace_hausdorffMeasure

Source hash: 8a0602f46d010a0d38a019d5e1052cfe4c5d69fc3db5988eaac8c5dbcc5c5e1c

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1081-1088, the source contract is `theorem exact_euclidean_capTrace_hausdorffMeasure (c : OneSidedCircularCap) : (μH[1] : Measure EuclideanPlane) (planeEuclideanHomeomorph '' c.arcTrace) = ENNReal.ofReal c.arcLength`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_leftArcTrace_hausdorffMeasure

Source hash: f6b756b038d2dd53ff9e214034534494d6216a6823376bb73c389a754a1ad0d9

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:821-827, the source contract is **theorem exact_euclidean_leftArcTrace_hausdorffMeasure** ($c : \text{StripCore}$) : ($\mu H[1] : \text{Measure EuclideanPlane}$) ($\text{planeEuclideanHomeomorph } ' ' c.\text{leftArcTrace}$) = $\text{ENNReal.ofReal } (\text{coreArcEnd } c - \text{coreArcStart } c)$. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_leftCoreParam_hausdorffMeasure

Source hash: e0211c9e8b0941bc0bbeb3bd977afe28d78688025726aac98a4d738bf7f180e9

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:745-772, the source contract is **theorem exact_euclidean_leftCoreParam_hausdorffMeasure** ($c : \text{StripCore}$) : ($\mu H[1] : \text{Measure EuclideanPlane}$) ($\text{realizedLeftCoreParam } c ' ' \text{Icc } (\text{coreArcStart } c) (\text{coreArcEnd } c)$) = $\text{ENNReal.ofReal } (\text{coreArcEnd } c - \text{coreArcStart } c)$. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_rightArcTrace_hausdorffMeasure

Source hash: 47e43c0cab0cb015904a2fe672eeec0720a25537d058b25dd2b767b05bcf1d39

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:828-834, the source contract is **theorem exact_euclidean_rightArcTrace_hausdorffMeasure** ($c : \text{StripCore}$) : ($\mu H[1] : \text{Measure EuclideanPlane}$) ($\text{planeEuclideanHomeomorph } ' ' c.\text{rightArcTrace}$) = $\text{ENNReal.ofReal } (\text{coreArcEnd } c - \text{coreArcStart } c)$. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_rightCoreParam_hausdorffMeasure

Source hash: 5b9d17b9e5921b7fba027be4ebfa3845e68c3c43a92a67ce9d4854bf1b244db0

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:773-800, the source contract is **theorem exact_euclidean_rightCoreParam_hausdorffMeasure** ($c : \text{StripCore}$) : ($\mu H[1] : \text{Measure EuclideanPlane}$) ($\text{realizedRightCoreParam } c ' ' \text{Icc } (\text{coreArcStart } c) (\text{coreArcEnd } c)$) = $\text{ENNReal.ofReal } (\text{coreArcEnd } c - \text{coreArcStart } c)$. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exact_euclidean_segmentCarrier_hausdorffMeasure

Source hash: f2e6ec79942c7df8c258fab2922138cb782506dbaa0184c81a94b12e6e9b9896

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:937-946, the source contract is **theorem exact_euclidean_segmentCarrier_hausdorffMeasure** (seg : HorizontalSegment) : (μ H[1] : Measure EuclideanPlane) (planeEuclideanHomeomorph '' seg.carrier) = ENNReal.ofReal seg.euclideanLength. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

exists_F_root

Source hash: 95a62157307a9e9688c52df86b638986f0064730d81ed190bb31889a2f44b580

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:82-91, the source contract is **lemma exists_F_root** : $\exists \theta \in \text{Icc } 0 (\pi / 2), F \theta = 0$. $F(\theta) = 3\theta - 3\sin(\theta) \cdot \cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

fourArc_euclidean_frontier

Source hash: 748ff0ec3e12377aa952eabaeafb59718df5bc069c8651bdcd45a9c687a1d41d

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1338-1348, the source contract is **theorem fourArc_euclidean_frontier** (a : FourArcAssembly) : frontier (planeEuclideanHomeomorph '' a.carrier) = (planeEuclideanHomeomorph '' StripCore.leftArcTrace a.core) $\cup$ ((planeEuclideanHomeomorph '' StripCore.rightArcTrace a.core) $\cup$ ((planeEuclideanHomeomorph '' OneSidedCircularCap.arcTrace a.upperCap) $\cup$ (planeEuclideanHomeomorph '' OneSidedCircularCap.arcTrace a.lowerCap))). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

fourArc_frontier_weightedPerimeter_eq

Source hash: e95ee1864c7bc7d9bc3a3ec2659625a510c6cefcddafe0cd2efab0e2d93e1067

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1794-1844, the source contract is **theorem fourArc_frontier_weightedPerimeter_eq** (lam : $\mathbb{R}$) (assembly : FourArcAssembly) : assembly.weightedPerimeter lam = WeightedPerimeter lam (FrontierMeasure

`assembly.carrier`). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

fourArc_pairwiseOverlap_h1_null

Source hash: 01eb58d468da4075ed5a4b3c4808bf8deddca98be0f05ef720c74c5a64fd9f4f

Relation: equivalent

Equivalent in the declared constructor/section context. At `FrontierPerimeter.lean:1269-1279`, the source contract is `theorem fourArc_pairwiseOverlap_h1_null (a : FourArcAssembly) : (μH[1] : Measure EuclideanPlane) (planeEuclideanHomeomorph ' ' pairwiseOverlap (StripCore.leftArcTrace a.core) (StripCore.rightArcTrace a.core) (OneSidedCircularCap.arcTrace a.upperCap) (OneSidedCircularCap.arcTrace a.lowerCap)) = 0`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

fourArc_pairwiseOverlap_subset_joinSet

Source hash: eea91f9cc72f6aabac972107df3b4a4acd7547fe5fa392689f59302f202356a4

Relation: equivalent

Equivalent in the declared constructor/section context. At `FrontierPerimeter.lean:1142-1195`, the source contract is `theorem fourArc_pairwiseOverlap_subset_joinSet (a : FourArcAssembly) : pairwiseOverlap (StripCore.leftArcTrace a.core) (StripCore.rightArcTrace a.core) (OneSidedCircularCap.arcTrace a.upperCap) (OneSidedCircularCap.arcTrace a.lowerCap) ⊆ joinSet a.core.chord`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

g

Source hash: 84e86ef24e01fc8948ba482190efcf999cb2f81449ac1865f6de67d08ad6760b

Relation: equivalent

Equivalent in the declared constructor/section context. At `GFunction.lean:115-118`, the source contract is `noncomputable def g (x : ℝ) : ℝ := 2 * arc x - arc (2 * x)`. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

g_deriv_neg

Source hash: 89dcdcf54ccfabbb31e9736dc2f8394ba49f8c59134f8b039132ae14f9404c9bd9

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:156-166, the source contract is **theorem** `g_deriv_neg {x : ℝ} (hx : 0 < x) (h : x ≤ π / 16) : 4 * sin (θ of x) - 4 * sin (θ of (2 * x)) < 0`. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

g_deriv_pos

Source hash: 4d597413573bb96cb2e13a079780c61d7a55d50fcebe5b68b68cbf58a0cbfa39

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:167-193, the source contract is **theorem** `g_deriv_pos {x : ℝ} (h : π / 8 ≤ x) : 0 < 4 * sin (θ of x) - 4 * sin (θ of (2 * x))`. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

g_deriv_xstar

Source hash: 1eebac5648246e00a782c35d06dac92c6d57fafb232dc20b208113c4fcc2af00

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:173-182, the source contract is **theorem** `g_deriv_xstar : 4 * sin (θ of xstar) - 4 * sin (θ of (2 * xstar)) = 0`. $F(\theta)=3\theta-3\sin(\theta)\cos(\theta)-\pi$ has its unique relevant root in $(0,\pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi-\theta$ remains in $(0,\pi)$, including the major-cap branch.

Findings: none.

g_strictAntiOn

Source hash: 9a778ff550b85fbf2a285e9d241dc6696a5778402499a6962e2ebab67dde9834

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:110-119, the source contract is **lemma** `g_strictAntiOn : StrictAntiOn g (Ioc 0 xstar)`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\thetaStar)$ is identified on the actual inverse-cap branch.

Findings: none.

g_strictMono0n_Ici

Source hash: bbc00bca790c91cdaec1fe6d2fa43a3abe6e68cd5e0920d9ebd378f169cf5602

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:120-131, the source contract is **lemma** `g_strictMono0n_Ici : StrictMono0n g (Ici xstar)`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\theta star)$ is identified on the actual inverse-cap branch.

Findings: none.

g_xstar_le

Source hash: b1c43512db9300ddb51ff2adbde05838f6b0dd21a49572dbdd5454f8a1f394a2

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:132-147, the source contract is **theorem** `g_xstar_le {x : ℝ} (hx : 0 < x) : g xstar ≤ g x`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\theta star)$ is identified on the actual inverse-cap branch.

Findings: none.

g_θstar

Source hash: b7fb48fb7e90dd4fccd7f3be6e957eefae58ef3ae08e86762f2e1b503b38585a

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:183-206, the source contract is **theorem** `g_θstar : g xstar = 3 * cos θstar`. $F(\theta)=3\theta-3\sin(\theta)*\cos(\theta)-\pi$ has its unique relevant root in $(0,\pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi-\theta$ remains in $(0,\pi)$, including the major-cap branch.

Findings: none.

gd

Source hash: b105248ee05b475037dd209f9b23e6a5119f0585d6ae1093bea44c7ceded4050

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:33-34, the source contract is **noncomputable def** `gd (x : ℝ) : ℝ := 4 * sin (θ0f x) - 4 * sin (θ0f (2 * x))`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\theta star)$ is identified on the actual inverse-cap branch.

Findings: none.

gd_neg_of_lt_xstar

Source hash: f2df0e64632660e264e46b7db88395768f82d48e3e7f2345a8a2f9c75cc667c9

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:90-98, the source contract is **lemma** `gd_neg_of_lt_xstar {x : ℝ} (hx : 0 < x) (h : x < xstar) : gd x < 0`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value

$g(x_{\text{star}})=3*\cos(\theta_{\text{star}})$ is identified on the actual inverse-cap branch.

Findings: none.

gd_pos_of_xstar_lt

Source hash: aa9cc07829ae7f90ff1904fd8a3bd95114f930be277c8de2503c494dfa26a0dc

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:99-109, the source contract is `lemma gd_pos_of_xstar_lt {x : ℝ} (h : xstar < x) : 0 < gd x`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(x_{\text{star}})=3*\cos(\theta_{\text{star}})$ is identified on the actual inverse-cap branch.

Findings: none.

gd_strictMonoOn

Source hash: 882d961320b58e7e7b5a68f1c9b6da8b8cd9f145ae2152f7eeda7c84659a8f5c

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:69-89, the source contract is `lemma gd_strictMonoOn : StrictMonoOn gd (Ioo (π / 16) (π / 8))`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(x_{\text{star}})=3*\cos(\theta_{\text{star}})$ is identified on the actual inverse-cap branch.

Findings: none.

gd_xstar

Source hash: 36e1f834bfffffc0c34e5ca4b702e9f19b5aa6de3e168944b257119d1ee48362

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:40-41, the source contract is `lemma gd_xstar : gd xstar = 0`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(x_{\text{star}})=3*\cos(\theta_{\text{star}})$ is identified on the actual inverse-cap branch.

Findings: none.

hasDerivAt_F

Source hash: dcdd81a18d4f4a1ae529be33b11aab6304bb984338d093081cc21b62e19958fa

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:46-56, the source contract is `lemma hasDerivAt_F (θ : ℝ) : HasDerivAt F (6 * sin θ ^ 2) θ`. $F(\theta)=3\theta-3\sin(\theta)*\cos(\theta)$ -pi has its unique relevant root in $(0,\pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi-\theta$ remains in $(0,\pi)$, including the major-cap branch.

Findings: none.

hasDerivAt_arc

Source hash: eb72e29cac89c3bb9db09a43c638398f57ad6c7fbce7c36f45c192d4285ebfc2

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:312-333, the source contract is theorem hasDerivAt_arc $\{x : \mathbb{R}\} (hx : 0 < x) : \text{HasDerivAt arc } (2 * \sin (\theta f x)) x$. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

hasDerivAt_arc_two_mul

Source hash: 68db7ce1cccf8d83346e37f43b7e16bbcad82e0649185eda0b8b1c7290d5df1

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:119-129, the source contract is lemma hasDerivAt_arc_two_mul $\{x : \mathbb{R}\} (hx : 0 < x) : \text{HasDerivAt } (\text{fun } t \Rightarrow \text{arc } (2 * t)) (4 * \sin (\theta f (2 * x))) x$. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

hasDerivAt_area

Source hash: 77cb2c9579c9e12eb25a7bd63dc9a791629029d7c8ad37f065713a7afa870b98

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:104-124, the source contract is lemma hasDerivAt_area $\{\theta : \mathbb{R}\} (h\theta : 0 < \theta) (h\pi : \theta < \pi) : \text{HasDerivAt area } ((\sin \theta - \theta * \cos \theta) / (2 * \sin \theta ^ 3)) \theta$. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

hasDerivAt_ell

Source hash: cae8d71b9bdc0c10b243d112a9ada6dbc5d44d1c5f710020d8f577d13d56023c

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:94-103, the source contract is lemma hasDerivAt_ell $\{\theta : \mathbb{R}\} (h\theta : 0 < \theta) (h\pi : \theta < \pi) : \text{HasDerivAt ell } ((\sin \theta - \theta * \cos \theta) / \sin \theta ^ 2) \theta$. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

hasDerivAt_g

Source hash: decc0673d8052a837dfd48ad4a92fcf973c064c1bdb19f47382d7d73801df7cf

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:130-150, the source contract is `theorem hasDerivAt_g {x : ℝ} (hx : 0 < x) : HasDerivAt g (4 * sin (θ of x) - 4 * sin (θ of (2 * x))) x`. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

hasDerivAt_g'

Source hash: 8f0141bd9b2ca2ed0af5052050d85c875e33a71481490dd5e2250b9a2886c5ba

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:35-39, the source contract is `lemma hasDerivAt_g' {x : ℝ} (hx : 0 < x) : HasDerivAt g (gd x) x`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(x_{\text{star}})=3*\cos(\theta_{\text{star}})$ is identified on the actual inverse-cap branch.

Findings: none.

hausdorffMeasure_image_Ico_eq_Icc

Source hash: 1c850c9b7ca6e169808ebfb87cac4902c44102258f4c3ba0c9b6433987f2a5de

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:142-153, the source contract is `lemma hausdorffMeasure_image_Ico_eq_Icc {E : Type*} [EMetricSpace E] [MeasurableSpace E] [BorelSpace E] {f : ℝ → E} {S : Set ℝ} (hf : Set.InjOn f S) {u v : ℝ} (huv : u ≤ v) (hsub : Icc u v ⊆ S) : (μH[1] : Measure E) (f '' Ico u v) = (μH[1] : Measure E) (f '' Icc u v)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

hausdorffMeasure_one_euclideanJoinSet

Source hash: 7ebe8fae03ad531268ff769afa248cd1067ee674eba1a7f8bc77f36d3ab49ebc

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1263-1268, the source contract is `theorem hausdorffMeasure_one_euclideanJoinSet (q : ℝ) : (μH[1] : Measure EuclideanPlane) (euclideanJoinSet q) = 0`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

hausdorffMeasure_unitCircleArc_image_eq

Source hash: f4746d8c9b2c6ef1416bccca1f814590cdef08c393e10b7189ff167eb0402ab3

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:294-311, the source contract is `theorem hausdorffMeasure_unitCircleArc_image_eq {r theta : ℝ} (hr : 0 < r) (htheta : theta ∈ Ioo 0 Real.pi) : (μH[1] : Measure ℂ) (unitCircleArc r '' Icc (-r * theta) (r * theta)) = ENNReal.ofReal (2 * r * theta)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

hausdorffMeasure_unitCircleArc_image_eq_of_span_lt_two_pi

Source hash: a5a68294ffad2a36064599c08079161b28ea03171c66437e00e8f24eba632c26

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:350-389, the source contract is `theorem hausdorffMeasure_unitCircleArc_image_eq_of_span_lt_two_pi {r u v : ℝ} (hr : 0 < r) (huv : u < v) (hspan : v - u < 2 * Real.pi * r) : (μH[1] : Measure ℂ) (unitCircleArc r '' Icc u v) = ENNReal.ofReal (v - u)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

hausdorffMeasure_unitCircleArc_image_le

Source hash: 07ee598762aa4a7964c4db549c22833ebf32ee2d325dcc190dd9a72212015319

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:93-104, the source contract is `lemma hausdorffMeasure_unitCircleArc_image_le {r a b : ℝ} (hr : 0 < r) : (μH[1] : Measure ℂ) (unitCircleArc r '' Icc a b) ≤ ENNReal.ofReal (b - a)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

image_Ico_eq_image_Icc_diff_endpoint

Source hash: c4fb0e64c0405d8f9fa3cb2429d9a9bc01ac95be8aba1e08560d1aeae3f939e9

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:124-141, the source contract is `lemma image_Ico_eq_image_Icc_diff_endpoint {α : Type*} {f : ℝ → α} {S : Set ℝ} (hf : Set.InjOn f S) {u v : ℝ} (huv : u ≤ v) (hsub : Icc u v ⊆ S) : f '' Ico u v = f '' Icc u v \ {f v}`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

integral_four_union

Source hash: 8b3fbc02eae74d84636263a1b0bedcfff91c58bcc3410ac92807b51d54cc9ea4

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1562-1590, the source contract is theorem integral_four_union {f : EuclideanPlane → ℝ} {A B C D : Set EuclideanPlane} (hAB : AEDisjoint (μH[1] : Measure EuclideanPlane) A B) (hAC : AEDisjoint (μH[1] : Measure EuclideanPlane) A C) (hAD : AEDisjoint (μH[1] : Measure EuclideanPlane) A D) (hBC : AEDisjoint (μH[1] : Measure EuclideanPlane) B C) (hBD : AEDisjoint (μH[1] : Measure EuclideanPlane) B D) (hCD : AEDisjoint (μH[1] : Measure EuclideanPlane) C D) (hB : MeasurableSet B) (hC : MeasurableSet C) (hD : MeasurableSet D) (hfA : IntegrableOn f A (μH[1] : Measure EuclideanPlane)) (hfB : IntegrableOn f B (μH[1] : Measure EuclideanPlane)) (hfC : IntegrableOn f C (μH[1] : Measure EuclideanPlane)) (hfD : IntegrableOn f D (μH[1] : Measure EuclideanPlane)) : (∫ z in A ∪ (B ∪ (C ∪ D)), f z ∂(μH[1] : Measure EuclideanPlane)) = (∫ z in A, f z ∂(μH[1] : Measure EuclideanPlane)) + ((∫ z in B, f z ∂(μH[1] : Measure EuclideanPlane)) + ((∫ z in C, f z ∂(μH[1] : Measure EuclideanPlane)) + ∫ z in D, f z ∂(μH[1] : Measure EuclideanPlane))). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

integral_semicircle_scaled

Source hash: fe66bbeed922271b7cf0e4043f58e6b4f6651310c0d02aa776383ac42b18240d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1759-1791, the source contract is lemma integral_semicircle_scaled {r θ : ℝ} (hr : 0 < r) (hθ0 : 0 ≤ θ) (hθπ : θ ≤ π) : (∫ y in r * cos θ..r, 2 * √(r ^ 2 - y ^ 2)) = r ^ 2 * (θ - sin θ * cos θ). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

integral_semicircle_translated

Source hash: 06721e5dc911d180205234ff69b820e369a8d82b66f142e17c0a1377b9b27f8d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1792-1800, the source contract is lemma integral_semicircle_translated {a r θ : ℝ} (hr : 0 < r) (hθ0 : 0 ≤ θ) (hθπ : θ ≤ π) : (∫ y in a + r * cos θ..a + r, 2 * √(r ^ 2 - (y - a) ^ 2)) = r ^ 2 * (θ - sin θ * cos θ). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

integral_sqrt_one_sub_sq_from_cos

Source hash: e4a969c6c1b67c6f267db8fb9582c7cd96174fa24489265c840c646c3085a52a

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1729-1758, the source contract is lemma `integral_sqrt_one_sub_sq_from_cos {θ : ℝ} (hθ0 : 0 ≤ θ) (hθπ : θ ≤ π) : (∫ x in cos θ..1, √(1 - x ^ 2)) = (θ - sin θ * cos θ) / 2`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

intervalPushforward

Source hash: 366d70f2307ddf912f8fe8519c486284fdacdab2c9bb76f371f9a65dd8ee807d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1200-1202, the source contract is `def intervalPushforward (curve : ℝ → PlanePoint) (a b : ℝ) : Measure PlanePoint := Measure.map curve (volume.restrict (Icc a b))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

isometry_capComplexIsometryMap

Source hash: 5d4136606591a26f6abf24f8ba286ddba7154e77e9f96c81673196c1854009b0

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:398-408, the source contract is lemma `isometry_capComplexIsometryMap (c : OneSidedCircularCap) : Isometry (capComplexIsometryMap c)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

isometry_leftCoreComplexIsometryMap

Source hash: a61c4ab4c222e88c43376bf46386ccf7697c506b7a23f81458ff102288a9e14a

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:685-695, the source contract is lemma `isometry_leftCoreComplexIsometryMap (c : StripCore) : Isometry (leftCoreComplexIsometryMap c)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

isometry_rightCoreComplexIsometryMap

Source hash: 2857926ac4225e60a51ede217224fdFDA63879d202e0be14411ea5c13b2fc6bf

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:696-704, the source contract is lemma `isometry_rightCoreComplexIsometryMap (c : StripCore) : Isometry (rightCoreComplexIsometryMap c)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

isometry_unitCircleArcRotation

Source hash: 0390b164e94132c53e908a7960b31f68dc786e7272a9a9f6482fa79cade88362

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:315-322, the source contract is lemma `isometry_unitCircleArcRotation (r m : ℝ) : Isometry (unitCircleArcRotation r m)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

joinSet

Source hash: 613dcfda543c5255d80175237d7db38b3386ce55bc3215682252569ce26c42d5

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1115-1117, the source contract is `def joinSet (q : ℝ) : Set PlanePoint := {(-q / 2, 1), (q / 2, 1), (-q / 2, -1), (q / 2, -1)}`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

joinSet_finite

Source hash: c95df1795c640672a88890bc658373f829154b7fa3a2361c967c14e17f22b551

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1253-1256, the source contract is `theorem joinSet_finite (q : ℝ) : (joinSet q).Finite`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

leftCoreBoundaryMeasure

Source hash: e243bfd8c440c79be3e701f3e1ba7f0b4af716d3ad6d1c3de2eb88023b1a30c8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1487-1490, the source contract is `def leftCoreBoundaryMeasure (c : StripCore) : Measure PlanePoint := intervalPushforward (leftCoreParam c) (coreArcStart c) (coreArcEnd c)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

leftCoreBoundaryMeasure_integrable

Source hash: efdd79d96b8f43a7c478608fb7242070e102fd586af27d2826a566f07542e0a9

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2275-2281, the source contract is `lemma leftCoreBoundaryMeasure_integrable (lam : ℝ) (c : StripCore) : Integrable (StripDensity lam) (leftCoreBoundaryMeasure c)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

leftCoreComplexIsometryMap

Source hash: c0385e604fc62ea1f5b752f17a4538ba2edeedf20e52cbfb6848b3b966b2ab57

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:677-680, the source contract is `noncomputable def leftCoreComplexIsometryMap (c : StripCore) (z : ℂ) : EuclideanPlane := WithLp.toLp 2 (c.leftCenterX - z.re, z.im)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

leftCoreParam

Source hash: 80f940fe52c510e69a283605275f10b49a7ca369836a80a40ece896dbbf606db

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1477-1481, the source contract is `def leftCoreParam (c : StripCore) (s : ℝ) : PlanePoint := (c.leftCenterX - c.radius * cos (s / c.radius), c.radius * sin (s / c.radius))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

leftCoreParam_image_Icc

Source hash: 01c2af95d70e04d0bac530bc681fda34196e33e841f34e26466b4a3a9c6ecd18

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:517-596, the source contract is `theorem leftCoreParam_image_Icc (c : StripCore) : leftCoreParam c '' Icc (coreArcStart c) (coreArcEnd c) = c.leftArcTrace`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

leftCore_strip_density

Source hash: 3222d64a655ce3af15a638606821b923e89c31ae75558031c08763aeb76b72b6

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1662-1667, the source contract is `theorem leftCore_strip_density (lam : ℝ) (c : StripCore) {s : ℝ} (hs : s ∈ Icc (coreArcStart c) (coreArcEnd c)) : StripDensity lam (leftCoreParam c s) = 1`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

leftCore_weightedIntegral

Source hash: e333f475432d30c40456ab56a9f5187cebdb61288c7e07d5eec293804ac77f49

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1675-1694, the source contract is `theorem leftCore_weightedIntegral (lam : ℝ) (c : StripCore) : ∫ p, StripDensity lam p ∂leftCoreBoundaryMeasure c = 2 * ell c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

lintegral_semicircle_translated

Source hash: 3c8e90cd635d133e832b64fe2731d46db4f68d0a1a341d292df3904cf66e67e8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1801-1819, the source contract is `lemma lintegral_semicircle_translated {a r θ : ℝ} (hr : 0 < r) (hθ0 : 0 ≤ θ) (hθπ : θ ≤ π) : (∫ y in Icc (a + r * cos θ) (a + r), ENNReal.ofReal (2 * √(r ^ 2 - (y - a) ^ 2))) = ENNReal.ofReal (r ^ 2 * (θ - sin θ * cos θ))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

lipschitzWith_unitCircleArc

Source hash: 6ddc026dd017d7b20d15e385a48bc34f5bcbfcad3796885aedaeb1fd06730313

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:33-54, the source contract is lemma lipschitzWith_unitCircleArc {r : ℝ} (hr : 0 < r) : LipschitzWith 1 (unitCircleArc r). The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and N=0 conventions.

Findings: none.

measurableSet_euclidean_capTrace

Source hash: 345ff49d99187e89c9b6a7558ffa41e164108dfd32b3dc525b699de014a17b5a

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1104-1109, the source contract is theorem measurableSet_euclidean_capTrace (c : OneSidedCircularCap) : MeasurableSet (planeEuclideanHomeomorph '' c.arcTrace). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

measurableSet_euclidean_leftArcTrace

Source hash: a31ed55013e95ae690ffcc5bab6587c2933302f4970b8b673a167e6c53e1b7f8

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1089-1093, the source contract is theorem measurableSet_euclidean_leftArcTrace (c : StripCore) : MeasurableSet (planeEuclideanHomeomorph '' c.leftArcTrace). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

measurableSet_euclidean_rightArcTrace

Source hash: 771198ca90f5054f065ccd16a46e41b5fce0886173f7b3c654e6b61824be845e

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1094-1098, the source contract is theorem measurableSet_euclidean_rightArcTrace (c : StripCore) : MeasurableSet (planeEuclideanHomeomorph '' c.rightArcTrace). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

measurableSet_euclidean_segmentCarrier

Source hash: 82897ecf0c0191705cb160c8a089d85dbe55e0d4633da94f20d433c8b2862b71

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1099-1103, the source contract is theorem measurableSet_euclidean_segmentCarrier (seg : HorizontalSegment) : MeasurableSet (planeEuclideanHomeomorph ' ' seg.carrier). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

measurable_stripDensity

Source hash: b6fc908d8bba478440a5c35f7c75a94366dc691bdf7d2125848e2a8f512284af

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1234-1240, the source contract is theorem measurable_stripDensity (lam : $\mathbb{R}$) : Measurable (StripDensity lam). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

min_g_eq_three_mul_cos_thetaStar

Source hash: 76e47f07f4c6982bfe46b74b225bb1e09621a21e59136f8d33d5ed15a7efc3b0

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:148-153, the source contract is theorem min_g_eq_three_mul_cos_thetaStar {x : $\mathbb{R}$ } (hx : $0 < x$) : $3 * \cos \theta_{\text{star}} \leq g \ x$. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(x_{\text{star}}) = 3 * \cos(\theta_{\text{star}})$ is identified on the actual inverse-cap branch.

Findings: none.

monotone_capPartitionPoint

Source hash: 62e77199a2eb7f24e5f93307536675c49f368eab2d84be96574f544f2152a857

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:167-174, the source contract is lemma monotone_capPartitionPoint {r theta : $\mathbb{R}$ } (hr : $0 < r$) (htheta : $0 < \theta$) {N : $\mathbb{N}$ } (hN : $0 < N$) : Monotone (capPartitionPoint r theta N). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

ofReal_dist_le_hausdorffMeasure_image_Icc

Source hash: 272ec4c298bc0f3582f8dd04d91ee8113b10ec4ec1c37538e792ec5391cc0936

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:105-123, the source contract is lemma ofReal_dist_le_hausdorffMeasure_image_Icc {E : Type*} [MetricSpace E] [MeasurableSpace E] [BorelSpace E] {f : $\mathbb{R} \rightarrow E$ } (hf : Continuous f) {u v : $\mathbb{R}$ } (huv : $u \leq v$) : ENNReal.ofReal (dist (f u) (f v)) $\leq$ ($\mu_H[1]$: Measure E) (f '' Icc u v). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

pairwiseOverlap

Source hash: f2051dcd764bb55c1f4c980a58c56ba3f51d928d6141df3262dbdd919f1b5fbd

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1110-1114, the source contract is def pairwiseOverlap (A B C D : Set PlanePoint) : Set PlanePoint := (A $\cap$ B) $\cup$ ((A $\cap$ C) $\cup$ ((A $\cap$ D) $\cup$ ((B $\cap$ C) $\cup$ ((B $\cap$ D) $\cup$ (C $\cap$ D)))). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

pairwise_aedisjoint_of_overlap_null

Source hash: 4b46ea8bfc070cb38c17e38f03c2486a1eb119281a4706f7a55c5da843e21d49

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1598-1616, the source contract is theorem pairwise_aedisjoint_of_overlap_null {A B C D : Set EuclideanPlane} (hnull : ($\mu_H[1]$: Measure EuclideanPlane) (euclideanPairwiseOverlap A B C D) = 0) : AEDisjoint ($\mu_H[1]$: Measure EuclideanPlane) A B $\wedge$ AEDisjoint ($\mu_H[1]$: Measure EuclideanPlane) A C $\wedge$ AEDisjoint ($\mu_H[1]$: Measure EuclideanPlane) A D $\wedge$ AEDisjoint ($\mu_H[1]$: Measure EuclideanPlane) B C $\wedge$ AEDisjoint ($\mu_H[1]$: Measure EuclideanPlane) B D $\wedge$ AEDisjoint ($\mu_H[1]$: Measure EuclideanPlane) C D. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

pi_div_eight_pos

Source hash: 69407ed5917d0d3fc3922aeeb807cf1b44f3f653f3ebf198c99ff7ac4ce4eba5

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:64-65, the source contract is lemma pi_div_eight_pos : (0 : $\mathbb{R}$) < $\pi / 8$. The normalized saving is $2L(X) - L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

pi_div_sixteen_lt_xstar

Source hash: 57b6eb878ba5fce5bf94c8fc02b54dbda28d42d21c498f918d5255530a944bbc

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:42-48, the source contract is `lemma pi_div_sixteen_lt_xstar :  $\pi / 16 < xstar$` . The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\theta star)$ is identified on the actual inverse-cap branch.

Findings: none.

pi_div_two_le_00f

Source hash: 00f928404b32c84bb29cfcfab41ea9d1b3d396a41398ba66120f2ba22a120cbe

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:103-114, the source contract is `lemma pi_div_two_le_00f {x :  $\mathbb{R}$ } (h :  $\pi / 8 \leq x$ ) :  $\pi / 2 \leq 00f\ x$` . The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

pi_div_two_le_00f_two_mul

Source hash: 84550d9c9294f46b93923eb76ddfec68dbdc45510cbd269c2f4b8188b9c71e4

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:65-68, the source contract is `lemma pi_div_two_le_00f_two_mul {x :  $\mathbb{R}$ } (hx :  $x \in I_{00}(\pi / 16)(\pi / 8)$ ) :  $\pi / 2 \leq 00f(2 * x)$` . The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\theta star)$ is identified on the actual inverse-cap branch.

Findings: none.

pi_sub_0star_mem

Source hash: 2bfa7c68cdb66f6586eaf8c23f7f915842c521244d8070de93397cda7a77e466

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:161-164, the source contract is `lemma pi_sub_0star_mem :  $\pi - \theta star \in I_{00} 0 \pi$` . $F(\theta)=3\theta-3\sin(\theta)*\cos(\theta)-\pi$ has its unique relevant root in $(0,\pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi-\theta$ remains in $(0,\pi)$, including the major-cap branch.

Findings: none.

planeEuclideanHomeomorph

Source hash: 0cae97f48ab4c061d7f08a907381bf52e9560477df33ffb993c23c2cbbf11049

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:38-42, the source contract is `noncomputable def planeEuclideanHomeomorph : PlanePoint  $\approx_t$  EuclideanPlane := (WithLp.homeomorphProd 2  $\mathbb{R}$   $\mathbb{R}$ ).symm`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

realizedCapParam

Source hash: 95b1661a30d7b5d3f17854c6eabbe1a4da9d944d9ad542e596607c6cb4fd12b0

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:430-433, the source contract is `noncomputable def realizedCapParam (c : OneSidedCircularCap) (s :  $\mathbb{R}$ ) : EuclideanPlane := planeEuclideanHomeomorph (capParam c s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedCapParam_eq

Source hash: 9c421cce21d93c49e9d419ecc30c8d31de995d391da25f034d29cf3f0bd85599

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:434-441, the source contract is `lemma realizedCapParam_eq (c : OneSidedCircularCap) (s :  $\mathbb{R}$ ) : realizedCapParam c s = capComplexIsometryMap c (unitCircleArc c.radius s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedCapParam_image_Icc

Source hash: a37b40496e1e5c2ae0b778fe01e7fee7a91557ce9473bf6e0b7755afebbd941b

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1071-1080, the source contract is `theorem realizedCapParam_image_Icc (c : OneSidedCircularCap) : realizedCapParam c '' Icc (capStart c) (capEnd c) = planeEuclideanHomeomorph '' c.arcTrace`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedLeftCoreParam

Source hash: 854e6d1ea17e2b2d6576f273610fd43051ac61c0bb7e75c185b3d3aa821f7da4

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:705-708, the source contract is `noncomputable def realizedLeftCoreParam (c : StripCore) (s : ℝ) : EuclideanPlane := planeEuclideanHomeomorph (leftCoreParam c s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedLeftCoreParam_eq

Source hash: e2c0026193d6fa0dd58c252b75c6df1a108616049c60130059ddb93311c0b2a0

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:713-719, the source contract is `lemma realizedLeftCoreParam_eq (c : StripCore) (s : ℝ) : realizedLeftCoreParam c s = leftCoreComplexIsometryMap c (unitCircleArc c.radius s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedLeftCoreParam_image_Icc

Source hash: 43b7188db3ec73adee2747ff0dc65e5368084e5e17965fa7fdaa8f3c9301ccd7

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:801-810, the source contract is `theorem realizedLeftCoreParam_image_Icc (c : StripCore) : realizedLeftCoreParam c '' Icc (coreArcStart c) (coreArcEnd c) = planeEuclideanHomeomorph '' c.leftArcTrace`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedRightCoreParam

Source hash: 4f8de565538cdbe799e2faf0a2d6452b4030fcfacd5b43a7347015b8e46924a9

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:709-712, the source contract is `noncomputable def realizedRightCoreParam (c : StripCore) (s : ℝ) : EuclideanPlane := planeEuclideanHomeomorph (rightCoreParam c s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedRightCoreParam_eq

Source hash: 4aab5f14bc5465771d4021be2ff0ba59e74b69eeaba0b9a9c834bd0bf72ed625

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:720-726, the source contract is lemma `realizedRightCoreParam_eq (c : StripCore) (s : ℝ) : realizedRightCoreParam c s = rightCoreComplexIsometryMap c (unitCircleArc c.radius s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedRightCoreParam_image_Icc

Source hash: e28432904f0d4f1c849ed6ed56e30bdae00185a8e05f01ccceab8ef89d16f29b

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:811-820, the source contract is theorem `realizedRightCoreParam_image_Icc (c : StripCore) : realizedRightCoreParam c '' Icc (coreArcStart c) (coreArcEnd c) = planeEuclideanHomeomorph '' c.rightArcTrace`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedSegmentParam

Source hash: 4c1728c18c2df9a1da871e357fce4f6439c6307fe654c71c273fe06b8caaa0fd

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:864-867, the source contract is noncomputable def `realizedSegmentParam (seg : HorizontalSegment) (s : ℝ) : EuclideanPlane := planeEuclideanHomeomorph (segmentParam seg s)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedSegmentParam_image_affineSegment

Source hash: c3be9a5bc2d447d5c5665690a46411881b4f0933eb5c3cf50deaa41a88946c72

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:873-912, the source contract is theorem `realizedSegmentParam_image_affineSegment (seg : HorizontalSegment) : realizedSegmentParam seg '' Icc (segmentStart seg) (segmentEnd seg) = affineSegment ℝ (planeEuclideanHomeomorph seg.leftEndpoint) (planeEuclideanHomeomorph seg.rightEndpoint)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

realizedSegmentParam_image_carrier

Source hash: fc21a65690cb2d00a27cb6aebde20046f1dffbc15b7f30d3caf58ac1aefd1725

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:913-922, the source contract is theorem realizedSegmentParam_image_carrier (seg : HorizontalSegment) : realizedSegmentParam seg '' Icc (segmentStart seg) (segmentEnd seg) = planeEuclideanHomeomorph '' seg.carrier. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

replacement_euclidean_frontier

Source hash: e7e5168c04a70cc7a8c8b17e4b2ad4201ee6c71706067d29f52984410e3fd5d4

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1349-1358, the source contract is theorem replacement_euclidean_frontier (a : ReplacementAssembly) : frontier (planeEuclideanHomeomorph '' a.carrier) = (planeEuclideanHomeomorph '' StripCore.leftArcTrace a.core) $\cup$ ((planeEuclideanHomeomorph '' StripCore.rightArcTrace a.core) $\cup$ ((planeEuclideanHomeomorph '' OneSidedCircularCap.arcTrace a.upperCap) $\cup$ (planeEuclideanHomeomorph '' a.exposedLowerChord.carrier))). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

replacement_frontier_weightedPerimeter_eq

Source hash: 4689c717c4c51ee163362ad12dd8fdcb9855d6e5f3a329efb232be3060d7d4cf

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1845-1898, the source contract is theorem replacement_frontier_weightedPerimeter_eq (lam : $\mathbb{R}$) (assembly : ReplacementAssembly) : assembly.weightedPerimeter lam = WeightedPerimeter lam (FrontierMeasure assembly.carrier). The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

replacement_pairwiseOverlap_h1_null

Source hash: 757396ecdaece07d8adc70b78cb910b95f948774e1e20f4dcb7710270834cc6c

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1280-1290, the source contract is theorem replacement_pairwiseOverlap_h1_null (a : ReplacementAssembly) : ($\mu H[1]$: Measure EuclideanPlane) (planeEuclideanHomeomorph '' pairwiseOverlap (StripCore.leftArcTrace a.core) (StripCore.rightArcTrace a.core))

$(\text{OneSidedCircularCap.arcTrace } a.\text{upperCap}) \ a.\text{exposedLowerChord.carrier} = \emptyset$. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

replacement_pairwiseOverlap_subset_joinSet

Source hash: aade619b0e98a3d318437558f408789f4a791fd841078c5489c3fcd97b06ed64

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1204-1252, the source contract is theorem replacement_pairwiseOverlap_subset_joinSet (a : ReplacementAssembly) : pairwiseOverlap (StripCore.leftArcTrace a.core) (StripCore.rightArcTrace a.core) (OneSidedCircularCap.arcTrace a.upperCap) a.exposedLowerChord.carrier $\subseteq$ joinSet a.core.chord. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

rightCoreBoundaryMeasure

Source hash: 33cbc13fa6fcd71207ac7e5dfb0579966744c8887bb1c9af1b95e6c688f55ea8

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1491-1493, the source contract is def rightCoreBoundaryMeasure (c : StripCore) : Measure PlanePoint := intervalPushforward (rightCoreParam c) (coreArcStart c) (coreArcEnd c). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

rightCoreBoundaryMeasure_integrable

Source hash: 63cd5ea28e72c7cf39aef8bfd2e698b16940d7ae74acc20f074d60c55994913c

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2282-2288, the source contract is lemma rightCoreBoundaryMeasure_integrable (lam : $\mathbb{R}$) (c : StripCore) : Integrable (StripDensity lam) (rightCoreBoundaryMeasure c). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

rightCoreComplexIsometryMap

Source hash: 24206b125ec8e6de86a054126e34b32568c5cb9c365728de42d394ff9cc0ede6

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:681-684, the source contract is `noncomputable def rightCoreComplexIsometryMap (c : StripCore) (z : ℂ) : EuclideanPlane := WithLp.toLp 2 (c.rightCenterX + z.re, z.im)`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

rightCoreParam

Source hash: 38ea0d845e704b5d320adc3936a1b6a5fba52e17ce5fb320da7c20200afa9e4d

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1482-1486, the source contract is `def rightCoreParam (c : StripCore) (s : ℝ) : PlanePoint := (c.rightCenterX + c.radius * cos (s / c.radius), c.radius * sin (s / c.radius))`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

rightCoreParam_image_Icc

Source hash: 78bb53ade5c270538e82ae914ea314e16857498de0ee3dfc73dde7468e65c773

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:597-676, the source contract is `theorem rightCoreParam_image_Icc (c : StripCore) : rightCoreParam c '' Icc (coreArcStart c) (coreArcEnd c) = c.rightArcTrace`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

rightCore_strip_density

Source hash: d879e18591ac7ff00f0a228fad33d85d1e290fb3c77820aaf80c672ec9411662

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1668-1674, the source contract is `theorem rightCore_strip_density (lam : ℝ) (c : StripCore) {s : ℝ} (hs : s ∈ Icc (coreArcStart c) (coreArcEnd c)) : StripDensity lam (rightCoreParam c s) = 1`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

rightCore_weightedIntegral

Source hash: 12c65212e5417b38acb8560f789f61c31ede351c3de6efe2b847438f7d6cf718

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1695-1714, the source contract is `theorem rightCore_weightedIntegral (lam : ℝ) (c : StripCore) : ∫ p, StripDensity lam p ∂rightCoreBoundaryMeasure c = 2 * ell c.sideAngle`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segmentBoundaryMeasure

Source hash: f71d719edefabe163afbd5014b1ba6aa8df0398ddf3cc380ddcc9ee628163feb

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1403-1405, the source contract is `def segmentBoundaryMeasure (seg : HorizontalSegment) : Measure PlanePoint := intervalPushforward (segmentParam seg) (segmentStart seg) (segmentEnd seg)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segmentBoundaryMeasure_integrable

Source hash: ea2176915a90a62781ded65b3f02aa8993ae5482d22cc82a781ae3b26e8f8120

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2268-2274, the source contract is `lemma segmentBoundaryMeasure_integrable (lam : ℝ) (seg : HorizontalSegment) : Integrable (StripDensity lam) (segmentBoundaryMeasure seg)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segmentEnd

Source hash: 161abc8655ce59212c511d028e178416823adf64c523c73dae6b889ae03ed1f4

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1400-1402, the source contract is `def segmentEnd (seg : HorizontalSegment) : ℝ := seg.chord / 2`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segmentEndpoint_edist

Source hash: 667e4fe55a19b902ccd7cc29782bd67f737d862a1bddf625b46565ed0a83a1cd

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:923-936, the source contract is `lemma segmentEndpoint_edist (seg : HorizontalSegment) : edist (planeEuclideanHomeomorph seg.leftEndpoint) (planeEuclideanHomeomorph seg.rightEndpoint) = ENNReal.ofReal seg.euclideanLength`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

segmentParam

Source hash: ba8ed6bbc5cd19b9c405036bea7c221ae24e5c76ea0f6ef5cc2d62c327ae1dc5

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1395-1397, the source contract is `def segmentParam (seg : HorizontalSegment) (s : ℝ) : PlanePoint := (seg.midpointX + s, seg.baseY)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segmentParam_image_Icc

Source hash: 78cfdac8328c8d0308b83fec45a7d72c3a0a830edb6a3f41f9bd2da54db0f76

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:835-863, the source contract is `theorem segmentParam_image_Icc (seg : HorizontalSegment) : segmentParam seg '' Icc (segmentStart seg) (segmentEnd seg) = seg.carrier`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

segmentStart

Source hash: e90adf884523aab9d2e3af449adbc9d655d375051a612c0b4970a212a5dca61f

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1398-1399, the source contract is `def segmentStart (seg : HorizontalSegment) : ℝ := -seg.chord / 2`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segment_strip_density

Source hash: 085d6e1c928eaa3445cf1a074627690d3998f7b411d2733a06ce5de0f2e168c0

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1451-1456, the source contract is theorem `segment_strip_density (lam : ℝ) (seg : HorizontalSegment) (hbase : |seg.baseY| ≤ 1) (s : ℝ) : StripDensity lam (segmentParam seg s) = 1`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

segment_weightedIntegral

Source hash: c5e9908e4c946fc4cb2e86548a94c923ba1b55cd20dcedb1e28930ac0fd78876

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1457-1471, the source contract is theorem `segment_weightedIntegral (lam : ℝ) (seg : HorizontalSegment) (hbase : |seg.baseY| ≤ 1) : ∫ p, StripDensity lam p ∂segmentBoundaryMeasure seg = seg.euclideanLength`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

sin_lt_sin_of_le_pi_div_two

Source hash: 8163503cd18ad5473edae576484d993037d91228ee5ab16856af467dd4e699d7

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:44-48, the source contract is lemma `sin_lt_sin_of_le_pi_div_two {a b : ℝ} (ha : 0 ≤ a) (hab : a < b) (hb : b ≤ π / 2) : sin a < sin b`. The normalized saving is $2L(X) - L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

sin_lt_sin_of_pi_div_two_le

Source hash: 7a9054c5f615d11a8d495c714ceecbf95cfd8c6004ddbd3d2a23ef7f883cd55e

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:49-63, the source contract is lemma `sin_lt_sin_of_pi_div_two_le {a b : ℝ} (ha : π / 2 ≤ a) (hab : a < b) (hb : b ≤ π) : sin b < sin a`. The normalized saving is $2L(X) - L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

sin_sub_mul_cos_pos

Source hash: 0b2cec2b87fd3b3f54034a8d9f968aaacb6a3fa0df4915dc18cf123aab5092e4

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:65-88, the source contract is lemma `sin_sub_mul_cos_pos {θ : ℝ} (h0 : 0 < θ) (hπ : θ < π) : 0 < sin θ - θ * cos θ`. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

sin_θstar_pos

Source hash: 4e8cec70a1403c2eeb99e9733f21156d53cc03eb293ae991df42fda32054bfd6

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:119-122, the source contract is lemma `sin_θstar_pos : 0 < sin θstar`. $F(\theta) = 3\theta - 3\sin(\theta) \cdot \cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

stripDensity_integrableOn_of_volume_ne_top

Source hash: d26957a576fdcc692079b92ac5054cbbeda2a914af15d77e6d70ad1d418d20fd

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2289-2302, the source contract is lemma `stripDensity_integrableOn_of_volume_ne_top (lam : ℝ) {s : Set PlanePoint} (hs : volume s ≠ 0) : IntegrableOn (StripDensity lam) s`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

stripDensity_integrable_intervalPushforward

Source hash: 6315c9471ea2031d23e0cb4552b23f2812fe90938358c371a246f04d7f2d8095

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2243-2259, the source contract is lemma `stripDensity_integrable_intervalPushforward (lam : ℝ) {curve : ℝ → PlanePoint} {a b : ℝ} (hcurve : Measurable curve) : Integrable (StripDensity lam) (intervalPushforward curve a b)`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

tendsto_cap_polygon_length

Source hash: 32d7803e3d290677cdac6cd7bd331241d0a1e99888d3a8665e326de2ca241a11

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:269-293, the source contract is `lemma tendsto_cap_polygon_length {r theta : ℝ} (htheta : 0 < theta) : Tendsto (fun n : ℕ => ((n + 1 : ℕ) : ℝ) * (2 * r * Real.sin (theta / ((n + 1 : ℕ) : ℝ)))) atTop (λ (2 * r * theta))`. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

three_cos_lt_of_θstar_lt

Source hash: c9da56286228487769dd40657ed749a3741fda3b807926be4434af7975f1b3b8

Relation: equivalent

Equivalent in the declared constructor/section context. At Certificate.lean:100-106, the source contract is `lemma three_cos_lt_of_θstar_lt {t : ℝ} (ht : t ≤ π) (h : θstar < t) : 3 * cos t < 3 * cos θstar`. The numerical constants here denote exact rationals in the reals. The normalized-cap comparison is strict and applies to positive normalized area, not just to the point attaining the minimum.

Findings: none.

unitCircleArc

Source hash: 6865b0b68c07403ee39252efa35feb9b50850a4644c2bf915fb506723c7a07c7

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:11-13, the source contract is `def unitCircleArc (r s : ℝ) : ℂ := (r : ℂ) * Complex.exp (Complex.I * (s / r : ℝ))`. The circle path is $\exp(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArcRotation

Source hash: cfd45400ed0d08a69b42a215854d1f81999a500771f48a7f18476aafefe48ef

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:312-314, the source contract is `noncomputable def unitCircleArcRotation (r m : ℝ) (z : ℂ) : ℂ := z * Complex.exp (Complex.I * (m / r : ℝ))`. The circle path is $\exp(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArcRotation_apply

Source hash: dcedc37c9428c0a722c5bf130023e2389a83c7faa4036432847247c3401f6b9b

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:323-331, the source contract is lemma `unitCircleArcRotation_apply {r : ℝ} (hr : r ≠ 0) (m s : ℝ) : unitCircleArcRotation r m (unitCircleArc r s) = unitCircleArc r (s + m)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArcRotation_image_Icc

Source hash: 6bbfe5bd6af2f20e278e2e30ff5adc9fe7c0761359113d169ea5baae1f20d6a2

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:332-349, the source contract is lemma `unitCircleArcRotation_image_Icc {r : ℝ} (hr : r ≠ 0) (m u v : ℝ) : unitCircleArcRotation r m '' (unitCircleArc r '' Icc u v) = unitCircleArc r '' Icc (u + m) (v + m)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArc_dist

Source hash: 1c583965a8663dcd3d0483716b4801199d867b2ff4cbe96f04d4cbeb57ff9c08

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:55-66, the source contract is lemma `unitCircleArc_dist {r s t : ℝ} (hr : 0 < r) : dist (unitCircleArc r s) (unitCircleArc r t) = r * |2 * Real.sin (((s - t) / r) / 2)|`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArc_eq

Source hash: 9342d6a8ed8b7dcd1fbce36251ddf6b271aeea6ec0298f5d467e178c8fbf9cec

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:409-415, the source contract is lemma `unitCircleArc_eq (r s : ℝ) : unitCircleArc r s = (r : ℂ) * ((Real.cos (s / r) : ℂ) + (Real.sin (s / r) : ℂ) * Complex.I)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArc_im

Source hash: 0666e8d860656cd1bc24ade42af99acdafd9034497a0ba0f27acf44cddcd77e5

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:423-429, the source contract is `lemma unitCircleArc_im (r s : ℝ) : (unitCircleArc r s).im = r * Real.sin (s / r)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArc_injOn

Source hash: ceabbb1bd00ed7b3d5a248b0d9d8b4d41f8f1fc0af8194bd56a8092deed637e9

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:67-92, the source contract is `lemma unitCircleArc_injOn {r theta : ℝ} (hr : 0 < r) (htheta : theta ∈ Ioo 0 Real.pi) : Set.InjOn (unitCircleArc r) (Icc (-r * theta) (r * theta))`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArc_re

Source hash: 238aa74976048276575d89eb69760ceb968a224b5358a1775e61818ec60df916

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:416-422, the source contract is `lemma unitCircleArc_re (r s : ℝ) : (unitCircleArc r s).re = r * Real.cos (s / r)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

unitCircleArc_sub_factor

Source hash: d6c59e7f498e1352e21969a1c8084fa6600038a792db1172f4e5bd3b2836eece

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:18-32, the source contract is `lemma unitCircleArc_sub_factor {r s t : ℝ} (_hr : r ≠ 0) : unitCircleArc r s - unitCircleArc r t = (r : ℂ) * Complex.exp (Complex.I * (t / r : ℝ)) * (Complex.exp (Complex.I * ((s - t) / r : ℝ)) - 1)`. The circle path is $\text{rexp}(is/r)$: s is the signed arclength parameter, not the central angle, and r is not fixed to one. The author preserves the stated radius/span assumptions and, for partition formulas, the stated index and $N=0$ conventions.

Findings: none.

upper_cap_integral

Source hash: 447beffafa9faf9c0047e982aa709b3b243852770c1fde70cc3c6676215458bf

Relation: equivalent

Equivalent in the declared constructor/section context. At FrontierPerimeter.lean:1545-1552, the source contract is theorem upper_cap_integral (lam : $\mathbb{R}$) (c : OneSidedCircularCap) (hside : c.side = .upper) (hbase : c.baseY = 1) : ($\int z$ in planeEuclideanHomeomorph '' c.arcTrace, euclideanStripDensity lam z $\partial(\mu H[1] : \text{Measure EuclideanPlane})$) = lam * c.arcLength. The length measure is Euclidean one-dimensional Hausdorff measure on the complete frontier. Arc/segment unions are combined using their proved null overlaps; glued internal chords are not charged as exterior boundary.

Findings: none.

volume_horizontalLine

Source hash: ea745db5e4b12e1d05fe0934ca3c81eb24b84ea87dd522e3aa6824c86d1a2dce

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2029-2036, the source contract is lemma volume_horizontalLine (b : $\mathbb{R}$) : volume {p : PlanePoint | p.2 = b} = 0. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

volume_lower_circular_segment

Source hash: 2301e25d7ced8d1d9e2c23a9b63e02ba7c6e8f9ea615112d4c41d15aeb777045

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1945-1995, the source contract is lemma volume_lower_circular_segment (mx cy r θ : $\mathbb{R}$) (hr : $0 < r$) (h00 : $0 \leq \theta$) (h0 π : $\theta \leq \pi$) : volume {p : $\mathbb{R} \times \mathbb{R}$ | (p.1 - mx) 2 + (p.2 - cy) $^2 \leq r^2 \wedge p.2 \leq cy - r * \cos \theta$ } = ENNReal.ofReal (r $^2 * (\theta - \sin \theta * \cos \theta)$). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

volume_upper_circular_segment

Source hash: d23f6efcc432aaa58959c64bfa3f6f589362c33546093657b1717ec5dbfb1b34

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:1820-1944, the source contract is lemma volume_upper_circular_segment (mx cy r θ : $\mathbb{R}$) (hr : $0 < r$) (h00 : $0 \leq \theta$) (h0 π : $\theta \leq \pi$) : volume {p : $\mathbb{R} \times \mathbb{R}$ | (p.1 - mx) 2 + (p.2 - cy) $^2 \leq r^2 \wedge cy + r * \cos \theta \leq p.2$ } = ENNReal.ofReal (r $^2 * (\theta - \sin \theta * \cos \theta)$). The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

volume_verticalLine

Source hash: b32e9d96eca7ded1e8b5354b6d117e61bcda7b34c1fe056587cd14e1871da61b

Relation: equivalent

Equivalent in the declared constructor/section context. At CMVGeometry.lean:2037-2046, the source contract is `lemma volume_verticalLine (a : ℝ) : volume {p : PlanePoint | p.1 = a} = 0`. The coordinate carriers live in the product plane, but geometric boundary length is transported from the Euclidean WithLp realization. Constructor inequalities and the stated measurable/integrable hypotheses are part of the comparison; no reduced-boundary identification is being asserted.

Findings: none.

xstar

Source hash: 4590a64da97090d3af07ee66b4ba81cd8d175d4ce5739ecba3f1415fc4cba3d8

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:154-155, the source contract is `noncomputable def xstar : ℝ := area θstar`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

xstar_lt_pi_div_eight

Source hash: cf3af1a7b133df1071b52ee9fcdf29abd10638feaf9c104776dea7ea70a191e

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:49-55, the source contract is `lemma xstar_lt_pi_div_eight : xstar < π / 8`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar) = 3\cos(\thetaStar)$ is identified on the actual inverse-cap branch.

Findings: none.

xstar_mem_Ioo

Source hash: 720519699b0830892e8a5d92169d8b53b8ea57339507bc01453bc6fff96e87b0

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:56-60, the source contract is `lemma xstar_mem_Ioo : xstar ∈ Ioo (π / 16) (π / 8)`. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar) = 3\cos(\thetaStar)$ is identified on the actual inverse-cap branch.

Findings: none.

xstar_pos

Source hash: 285916d9a661ffdef997baeb58111dd2f99ed6b3837830effc46ed8c96b2914d

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:156-157, the source contract is `lemma xstar_pos :  $0 < xstar$ .  $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$  has its unique relevant root in  $(0, \pi/2)$ . The chosen normalized area is positive, and the supplementary angle  $\pi - \theta$  remains in  $(0, \pi)$ , including the major-cap branch.`

Findings: none.

00f

Source hash: ae683fbf0fd357d5b8a59395fa6c08ffe184192942daf192789541eb610f89e3

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:247-249, the source contract is `noncomputable def 00f (x : $\mathbb{R}$) : $\mathbb{R}$:= if h : $0 < x$ then (area_surjOn (mem_Ioi.mpr h)).choose else $\pi / 2$. For $X > 0$ the definition chooses the unique θ in $(0, \pi)$ with normalized area X ; for $X \leq 0$ it returns $\pi/2$. The author explicitly records that fallback, so total Lean division/inverse conventions do not create an unstated positive-area extension.`

Findings: none.

00f_area

Source hash: f8dabe04d85804b4ae48d760060d4f14fcddf7d4245871a8f753f3695333c49b

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:259-266, the source contract is `lemma 00f_area { θ : $\mathbb{R}$ } (h θ : $\theta \in \text{Ioo } 0 \pi$) : 00f (area θ) = θ . The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.`

Findings: none.

00f_continuousAt

Source hash: efcdf76351169996ba87c9b4ec14eb080fd4c43d9ee67eed507a1d6cc3182dbda

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:283-291, the source contract is `lemma 00f_continuousAt {x : $\mathbb{R}$ } (hx : $0 < x$) : ContinuousAt 00f x. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.`

Findings: none.

00f_image

Source hash: d81b765da2f8cf7cb10a3e3b057fc5bc93f2a0c62b4ee7c5334eaaaf98f1bd323

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:279-282, the source contract is `lemma 00f_image :  $\text{Ioo } 0 \pi \subseteq \text{00f '' (Ioi } 0)$ . The unit-chord normalized area and arc length are constructed on  $0 < \theta < \pi$ . The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse`

outside that domain.

Findings: none.

00f_le_pi_div_two

Source hash: e9cfe5c6a9cd5cf538683dfa1919d265b38f9b8ab0aa4ba7f8c7d2c013301928

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:98-102, the source contract is **lemma** $00f_le_pi_div_two \{x : \mathbb{R}\} (hx : 0 < x) (h : x \leq \pi / 8) : 00f \ x \leq \pi / 2$. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

00f_lt_pi_div_two

Source hash: 5baeed1a6d4a81b4780dc3116c27179937e00e0901298d63c17f2f452433e21c

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:94-97, the source contract is **lemma** $00f_lt_pi_div_two \{x : \mathbb{R}\} (hx : 0 < x) (h : x < \pi / 8) : 00f \ x < \pi / 2$. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

00f_lt_pi_div_two_of_mem

Source hash: 80218f0a09a02137efcf79005b6fbcec3641c36ec9134e0575885f8ad1c23afb

Relation: equivalent

Equivalent in the declared constructor/section context. At MinimumValue.lean:61-64, the source contract is **lemma** $00f_lt_pi_div_two_of_mem \{x : \mathbb{R}\} (hx : x \in I_{oo} (\pi / 16) (\pi / 8)) : 00f \ x < \pi / 2$. The derivative sign analysis concerns $X > 0$ and establishes a global normalized-saving minimum, not just a local stationary value. The value $g(xstar)=3*\cos(\theta_{star})$ is identified on the actual inverse-cap branch.

Findings: none.

00f_lt_00f_two_mul

Source hash: 095143100fd32388bdf4e1d180bcb0168b99a36a86a64f445ac9b5b1dadd6573

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:151-155, the source contract is **lemma** $00f_lt_00f_two_mul \{x : \mathbb{R}\} (hx : 0 < x) : 00f \ x < 00f \ (2 * x)$. The normalized saving is $2L(X)-L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

00f_mem

Source hash: cffddd09f082d4b6674b40f5793e1c255fa99c0cb678be3643150d9630636943

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:250-253, the source contract is lemma $00f_mem \{x : \mathbb{R}\} (hx : 0 < x) : 00f\ x \in I_{00}\ 0\ \pi$. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

00f_pi_div_eight

Source hash: 787022501b978041cc5dda7d507ec1898466d85828bc2384ff3ced42461fa420

Relation: equivalent

Equivalent in the declared constructor/section context. At GFunction.lean:73-78, the source contract is lemma $00f_pi_div_eight : 00f\ (\pi / 8) = \pi / 2$. The normalized saving is $2L(X) - L(2X)$ on $X > 0$. The factor of two in the chain rule and the principal inverse-cap-angle domain are retained; none of the geometric inequalities is extended to nonpositive cap area.

Findings: none.

00f_strictMonoOn

Source hash: 27129f3095ed7521a354aa523bb8840a0881e0fcc6c3fb3346c9443b50265957

Relation: equivalent

Equivalent in the declared constructor/section context. At MorgansArcFunction.lean:272-278, the source contract is lemma $00f_strictMonoOn : StrictMonoOn\ 00f\ (I_{oi}\ 0)$. The unit-chord normalized area and arc length are constructed on $0 < \theta < \pi$. The inverse is used geometrically only at positive area; its explicit totalization is not confused with an inverse outside that domain.

Findings: none.

00f_two_mul_xstar

Source hash: 9e384c7c73738ac8b54bf413e27f2a7a3ad3afa8254d6f30bca551f7d7a2b751

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:165-172, the source contract is lemma $00f_two_mul_xstar : 00f\ (2 * xstar) = \pi - \thetastar$. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

00f_xstar

Source hash: 5f96bbbf1ba25c54aa03e1adcc89a5374c79ae31eab08da194eece7875964d2e

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:158-160, the source contract is lemma $00f_xstar : 00f\ xstar = \thetastar$. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap

branch.

Findings: none.

θ_{star}

Source hash: 10465817c5977e25bde5deaa090eec0b969594d7d01505755dd8387cec5029e1

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:92-93, the source contract is `noncomputable def  $\theta_{\text{star}}$  :  $\mathbb{R}$  := exists_F_root.choose`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

$\theta_{\text{star_lt_of_F_pos}}$

Source hash: 8aed32b6dfdba14ba36157e57562cf1fa218ff121a66877337d2141e16cb6a9b

Relation: equivalent

Equivalent in the declared constructor/section context. At Certificate.lean:87-99, the source contract is `lemma  $\theta_{\text{star\_lt\_of\_F\_pos}}$  {t :  $\mathbb{R}$ } (ht : t ∈ Icc 0  $\pi$ ) (h : 0 < F t) :  $\theta_{\text{star}}$  < t`. The numerical constants here denote exact rationals in the reals. The normalized-cap comparison is strict and applies to positive normalized area, not just to the point attaining the minimum.

Findings: none.

$\theta_{\text{star_lt_pi}}$

Source hash: 00d6146ca546d963629c29e057f27680f814201f70ca0ff861a828ca133ab65c

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:117-118, the source contract is `lemma  $\theta_{\text{star\_lt\_pi}}$  :  $\theta_{\text{star}}$  <  $\pi$` . $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

$\theta_{\text{star_mem}}$

Source hash: c88631345f244802cc83acea97a1e4d6db39816cf66b6157c7b1a365c1b58e68

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:99-114, the source contract is `lemma  $\theta_{\text{star\_mem}}$  :  $\theta_{\text{star}}$  ∈ Ioo 0 ( $\pi / 2$ )`. $F(\theta) = 3\theta - 3\sin(\theta)\cos(\theta) - \pi$ has its unique relevant root in $(0, \pi/2)$. The chosen normalized area is positive, and the supplementary angle $\pi - \theta$ remains in $(0, \pi)$, including the major-cap branch.

Findings: none.

θstar_mem_Icc

Source hash: 60a56c312e094733952f7446cf434fefca8186b21010cd17ab60b2b44864dfffb

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:96-98, the source contract is `lemma θstar_mem_Icc : θstar ∈ Icc 0 (π / 2). F(theta)=3theta-3sin(theta)*cos(theta)-pi` has its unique relevant root in (0,pi/2). The chosen normalized area is positive, and the supplementary angle pi-theta remains in (0,pi), including the major-cap branch.

Findings: none.

θstar_pos

Source hash: 1d24d5264a96edc189b17b575a738900ce2888e0222ccbf2e0c850b185522fca

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:115-116, the source contract is `lemma θstar_pos : 0 < θstar. F(theta)=3theta-3sin(theta)*cos(theta)-pi` has its unique relevant root in (0,pi/2). The chosen normalized area is positive, and the supplementary angle pi-theta remains in (0,pi), including the major-cap branch.

Findings: none.

θstar_spec

Source hash: a1f729b0df782b1b7a5288e2aba2fab9def7f4caa3e0a74817fc2fd6b33f74fd

Relation: equivalent

Equivalent in the declared constructor/section context. At Minimiser.lean:94-95, the source contract is `lemma θstar_spec : F θstar = 0. F(theta)=3theta-3sin(theta)*cos(theta)-pi` has its unique relevant root in (0,pi/2). The chosen normalized area is positive, and the supplementary angle pi-theta remains in (0,pi), including the major-cap branch.

Findings: none.

Generated declaration audit ledger

CompactCells*.lean

Reviewed all 3,131 indexed entries: `QInterval`, `lambdaInterval` and the local candidate boundary for every box 2..1023, plus all 65 batch/prefix coverage theorems. Exact fractions show the local intervals join the explicit boxes 0 and 1 into the closed union [33/32,9/7]; every advertised batch endpoint and every included local/prefix call agrees with its stated range. All 1,022 local candidate contracts have only closed density containment, a modeled candidate and its CMV hypotheses, and use a supplied theorem `checkedConditions` rather than assuming certificate data or restricting candidate curvature. The common `Conditions` schema proves branch regularity, stationary fold signs, negative K3, appropriate equal-area face signs and strict perimeter negativity; the generic envelope/geometry bridge supplies actual nonminimality. The author-selected `Box1003` density/boundary, `Box1023` boundary and `boxes0To1023` aggregate were inspected in their full frozen blocks and agree with the family description. The aliases remain the existing `QInterval` type. No floating-point sample replaces an exact box proof.

Declaration	Source hash	Audit relation
CompactCells1003To1023.Box1003.QInterval	0aef3ddb1349e43cd56d63408011934a740f7e602452e59270f0bca0f41f74	equivalent

Declaration	Source hash	Audit relation
CompactCells1003To1023.Box1003.lamb daInterval	7954f593f751ecc31594321a2fefcd18ceb 37ff87af3e760efe036665448a30c	equivalent
CompactCells1003To1023.Box1003.comp actCandidate_not_isWeightedPerimete rMinimizer	9266c46b5eaf9a61ee35268398061782200 f1890d01f8abf70f7ead86b3f2f99	equivalent
CompactCells1003To1023.Box1004.QInt erval	1a43a246eefb6c50a81ef9733e489315752 3d94377df538d1fc2dab7847673e7	equivalent
CompactCells1003To1023.Box1004.lamb daInterval	3c0fccac63260140d2d2cad8e8094d541458 9bc30188dad2544feed361432aeff	equivalent
CompactCells1003To1023.Box1004.comp actCandidate_not_isWeightedPerimete rMinimizer	ccafaed6dc01e22df9c10d9dbee848d5ac5 0f656e30706ad79e37be76d4367b6	equivalent
CompactCells1003To1023.Box1005.QInt erval	28f18eb223623f4896f731377948f5570a4 2a7d33b0b7f32c10ae233383fcc18	equivalent
CompactCells1003To1023.Box1005.lamb daInterval	8a1dc2a70ad8a50a185714446fd6705ce11 e757f2127ff5da46c4b5df950f6d7	equivalent
CompactCells1003To1023.Box1005.comp actCandidate_not_isWeightedPerimete rMinimizer	28dc15d7ee00d56557ba18ba2b60b89a755 3da5770a61afcd20cfbd4b11ce133e	equivalent
CompactCells1003To1023.Box1006.QInt erval	a5f119ffec426f5ab5a050693081c58f27c c9424aa9e120872579e9614f0e28a	equivalent
CompactCells1003To1023.Box1006.lamb daInterval	cde8901753aee32cbe93e6f042e0b3493e9 97c8d1b981da5be266b87a90d0e35	equivalent
CompactCells1003To1023.Box1006.comp actCandidate_not_isWeightedPerimete rMinimizer	2a1a1947a524adc6c1d18c96ef2381e0475 0e391e58893084248a550082a84f7	equivalent
CompactCells1003To1023.Box1007.QInt erval	1182d581448cab66f2e6410ccf404309985 f257821d192bfdae177eb6d0e3f9b	equivalent
CompactCells1003To1023.Box1007.lamb daInterval	a33c38139e8d46e1461f881822aac58f900 46f33177e34402f9a1ff97cf5ff5a	equivalent
CompactCells1003To1023.Box1007.comp actCandidate_not_isWeightedPerimete rMinimizer	ad937a4c9b26ca0e0956de7fac92be794d8 d67fa715d82baab5db857b8c2a74b	equivalent
CompactCells1003To1023.Box1008.QInt erval	9d125dc28535496c0f8cc16349e12786fa7 e73626641d6fd1599ff592b5808ae	equivalent
CompactCells1003To1023.Box1008.lamb daInterval	2535d335a00e480a4431c68f670d33f2c19 e4361f267a21d59605cb72bc0c1ec	equivalent
CompactCells1003To1023.Box1008.comp actCandidate_not_isWeightedPerimete rMinimizer	e6214ad116ae5df988976937e4fbae694dd ba4f26896227aa690c77eb2ad6bcf	equivalent
CompactCells1003To1023.Box1009.QInt erval	95d65cfb4207522b49e995dc27a369c5c41 8f8c1faaf9d8f083340a957974ec4	equivalent
CompactCells1003To1023.Box1009.lamb daInterval	d8bcd30ee7afb5b62a0776cfc4c813b0b2a 6003aa68013fcff326bf7bb5f2ea7	equivalent
CompactCells1003To1023.Box1009.comp actCandidate_not_isWeightedPerimete rMinimizer	e92c04dc3bdea8599e90fbde215572b51bf e82d22bcb55502b84b76420efa83c	equivalent
CompactCells1003To1023.Box1010.QInt erval	8bd7d96ab0c00f196605ab439c8f85d7c85 67c4173c7d648cfe9e2841c5334d8	equivalent
CompactCells1003To1023.Box1010.lamb daInterval	0ea3fb52d192d71ae711915519aa674b5fa ee224a7f0ae1c3f664c6ba4eee82a	equivalent

Declaration	Source hash	Audit relation
CompactCells1003To1023.Box1010.compactCandidate_not_isWeightedPerimeterMinimizer	83b37f94d9672bda8fabad74e73ea5e6ddd96b225d6265fb8a2267288040b300	equivalent
CompactCells1003To1023.Box1011.QInterval	31d29a9f7cc8f9f491331bfedfa25102ab9921a068e2920a67dea55a3872ff35	equivalent
CompactCells1003To1023.Box1011.lambdaInterval	f13fa3e49f4ea2862ef61e5739d5561f6927d729a4d0d3f93fa2e45bfa763d98	equivalent
CompactCells1003To1023.Box1011.compactCandidate_not_isWeightedPerimeterMinimizer	e6c94787f7584891c53fb0cbcc50900bd40b8de2b16c5a0a13a6baa01ff897f6	equivalent
CompactCells1003To1023.Box1012.QInterval	5bd7901405f16301c1091d0c85ccb409eead3817f51b900421c2a285bbe60ebd	equivalent
CompactCells1003To1023.Box1012.lambdaInterval	4e9e546e5be03a9226056cbaf605558741fcae849ef67c1e1f1ccc00d80f0b99	equivalent
CompactCells1003To1023.Box1012.compactCandidate_not_isWeightedPerimeterMinimizer	c79527c69c02ce67a1128e442e52ca9ae20da0912e3823e4b4630307e12ff46a	equivalent
CompactCells1003To1023.Box1013.QInterval	4063f1ead2817760bc617b85317f31e90b7a59c40d9c824d5901c4a15ac8a670	equivalent
CompactCells1003To1023.Box1013.lambdaInterval	741eed6722855869eeaa45f4353ca1c4d25607115974f8eb7780800e17447fa88	equivalent
CompactCells1003To1023.Box1013.compactCandidate_not_isWeightedPerimeterMinimizer	7d0873dfd3ce4273369f1149a4f3c5da05534202483e7d22c0eceb2409847380	equivalent
CompactCells1003To1023.Box1014.QInterval	ad142bffc1586fabacd10979731e608943777f20ed176797895b53592dd5a163	equivalent
CompactCells1003To1023.Box1014.lambdaInterval	d0f3b0b115bfde671e58f6f8c357e8a45006bf2051602064080eaf7b25b9616f	equivalent
CompactCells1003To1023.Box1014.compactCandidate_not_isWeightedPerimeterMinimizer	294708d1bae36afb9ffb93cf123e85a1defaf7bf2ec2b2534c647ba3a3314de3	equivalent
CompactCells1003To1023.Box1015.QInterval	8b44b31c87f08aef997f4424f65a7a822ced66045861d2b52d86c0d21e45007c	equivalent
CompactCells1003To1023.Box1015.lambdaInterval	867e6813f99d38a785dfdd7fe121ada05edfb9e8922089f830c24804ee0338e9	equivalent
CompactCells1003To1023.Box1015.compactCandidate_not_isWeightedPerimeterMinimizer	659d47039b09be2d0a1270bfb78e195672ba710b284a1b659d8b9fbbe979441a	equivalent
CompactCells1003To1023.Box1016.QInterval	63c272a9c2c0b950e401ca707d1478510ac05380832d9d12a5120cec171d2aec	equivalent
CompactCells1003To1023.Box1016.lambdaInterval	b0997fbb9c290a01c9cd97ebe58d27c8be978415ff1722a85ac418faa242a17e	equivalent
CompactCells1003To1023.Box1016.compactCandidate_not_isWeightedPerimeterMinimizer	e3ea4685ec24cc960d213a2a451590fd3a02bd61488ac6a981c61c2280ac1473	equivalent
CompactCells1003To1023.Box1017.QInterval	00eac8527cf5985ebc4760a7a98ebf672e3b36950bd46b5ad0f04b383211233e	equivalent
CompactCells1003To1023.Box1017.lambdaInterval	44b7020558eb8ff294dbe26a29630b983312e56ed8a9803e85aad347de2d4a43	equivalent
CompactCells1003To1023.Box1017.compactCandidate_not_isWeightedPerimeterMinimizer	31b8f0f05bcea5f1222b889fb7bbe3974e02a6c596d5b87a011aff14d40cfaf4	equivalent

Declaration	Source hash	Audit relation
CompactCells1003To1023.Box1018.QInterval	4518d7ac6fd604b046e73b0f5c281663adc4327f70e0d6922c4954ccbb26701e	equivalent
CompactCells1003To1023.Box1018.lambdaInterval	2478df3775575e7c681d307e0b8e59c4f8038f80b17a481ccfddfeaacc6ea12f	equivalent
CompactCells1003To1023.Box1018.compactCandidate_not_isWeightedPerimeterMinimizer	c6f689c4446a97f22b3433ac67082410fbb15387a8d32688c18c0e6c60f4eb27	equivalent
CompactCells1003To1023.Box1019.QInterval	b3f49473524decab07888e467370c44ae9092e08d5727e9def03d84b277b20ad	equivalent
CompactCells1003To1023.Box1019.lambdaInterval	d685b2279c31a2d937fbe24f348a9527e5f315fb5d6cec7a3304fe9e5e0d7f6e	equivalent
CompactCells1003To1023.Box1019.compactCandidate_not_isWeightedPerimeterMinimizer	44859e81753d2655f85543492bcf3fd1091cb1de41fb8d7903c28bc9c1c2ee4d	equivalent
CompactCells1003To1023.Box1020.QInterval	7e9c4b0378b605932a5284215019fe28d85ca5563d4f3c36143d710cfd0e1c7a	equivalent
CompactCells1003To1023.Box1020.lambdaInterval	89a28ccc0a4771132b224330899df1d03913328356f7d1ae854103ae4cae6e25	equivalent
CompactCells1003To1023.Box1020.compactCandidate_not_isWeightedPerimeterMinimizer	315a2ec244f07531469fced2e69aa1638c1f927a83772350e31838899625eadb	equivalent
CompactCells1003To1023.Box1021.QInterval	877f2b0a740007b4a4c81257054a3f46884409000faf1d7b0c53eafa2e6d556e	equivalent
CompactCells1003To1023.Box1021.lambdaInterval	e6daec92bfe1fe96d7626e9692216bbac722129207e05f7f7666a90f85f6fb7e	equivalent
CompactCells1003To1023.Box1021.compactCandidate_not_isWeightedPerimeterMinimizer	da071604bf01fb6cac878ffdc25881c15cbca32d98197713c0727a6272509420	equivalent
CompactCells1003To1023.Box1022.QInterval	d1ea96bb28afdaddc07e3e0f7a388dd90d52c0f2060f313379c79517a635dba4	equivalent
CompactCells1003To1023.Box1022.lambdaInterval	3824f8af50dfd32c681a36f24c59a2aa1517cf11996c4f73a5fe08f1c9b67d5a	equivalent
CompactCells1003To1023.Box1022.compactCandidate_not_isWeightedPerimeterMinimizer	23cdc580d5edbbb1be74d223b2fc8eeee557aa1f51d3eea78ccd5cc2263eb57d	equivalent
CompactCells1003To1023.Box1023.QInterval	f91b3cb3d3bc591728f8b29f921a230fffb675a60da347e9a9a81590063648f71	equivalent
CompactCells1003To1023.Box1023.lambdaInterval	0bae30c198eab1d06be6d8c202438c3ceb56ac62eec7bf2af16380dd627567b8	equivalent
CompactCells1003To1023.Box1023.compactCandidate_not_isWeightedPerimeterMinimizer	4d50f38ea945f7e34f4d467709359e74bf299766fdd152d61f0f523dacb8b78c	equivalent
CompactCells1003To1023.bboxes1003To1023_candidate_not_isWeightedPerimeterMinimizer	68581cbf6c2cd53b0b398a9c517f4e9be204c435157aa95bd03c1c34523a8049	equivalent
CompactCells1003To1023.bboxes0To1023_candidate_not_isWeightedPerimeterMinimizer	a84497877b925fe28bd82df50e6461419f03e32cc7770a6672102746fe9c6b10	equivalent
CompactCells107To138.Box107.QInterval	e667e63a3eee216c3392ab1cf8779ac1d44631d32fc820e30e1b9ceb464d5016	equivalent
CompactCells107To138.Box107.lambdaInterval	45e99273f001945e452396917862956752774a252317c428eb1635882e65b66f	equivalent

Declaration	Source hash	Audit relation
CompactCells107To138.Box107.compactCandidate_not_isWeightedPerimeterMinimizer	0ca0ea6090e6cb065dce2f787a4e0e91dcacf9c82b3e8fb66460c96f0f1995844	equivalent
CompactCells107To138.Box108.QInterval	4755fff945cab80eaf9ad77aa677d0cfa0f10cec1e9ac23c91303a8a3dea3503	equivalent
CompactCells107To138.Box108.lambdaInterval	9838b082918b9c77a81e48965754e34bf60bbb7db492975617c5fb519b5fbed2	equivalent
CompactCells107To138.Box108.compactCandidate_not_isWeightedPerimeterMinimizer	19961a78bd33aea336173701b025bdcf860cae1be3c87a7ae60f47aa8c130b93	equivalent
CompactCells107To138.Box109.QInterval	f62177849be0c8c2e3d0631acc5de30b0bb53c8d8759083ae593531424a3b642	equivalent
CompactCells107To138.Box109.lambdaInterval	57264cfae9c7a21afc133ad01a7c4abb3d51218f8c0b8d1495230976114e656c	equivalent
CompactCells107To138.Box109.compactCandidate_not_isWeightedPerimeterMinimizer	5620710b85e323945f84ead91a451d8ab2c3f2e3acc0c0c2b3656c0b22bc2944	equivalent
CompactCells107To138.Box110.QInterval	a9a119ef1024e526062283615281f064afd8de6e00c24683e750e2f7d7765a44	equivalent
CompactCells107To138.Box110.lambdaInterval	a90707b248fdb52a3f651de6423ffcbf1a3ea55890f217283b0bfcd279668785	equivalent
CompactCells107To138.Box110.compactCandidate_not_isWeightedPerimeterMinimizer	63d0e11c528523d49f0a88d2293a8204fbe68c419289d3593b781b570398a9f6	equivalent
CompactCells107To138.Box111.QInterval	81705fa5e0544cb077800ef741ec3faecffbbf1083ca73318a81657f2a333b4c	equivalent
CompactCells107To138.Box111.lambdaInterval	1f4464e597a1061a6b77b8cadac099ff7a2407e0d40374784df846440c8af74c	equivalent
CompactCells107To138.Box111.compactCandidate_not_isWeightedPerimeterMinimizer	61c8f849e2b3bf48024fbf2000198c47b4ba2ae7a7978acfe19ff165ec191f44	equivalent
CompactCells107To138.Box112.QInterval	ed34d111c029b5366ad458004b7cb330e2de9ad843b3957444d0eb147bc3111f	equivalent
CompactCells107To138.Box112.lambdaInterval	cbd41b5a778e5c7bc7299b3170926afecaa cd0173aa50f5f9d9974af03945e30	equivalent
CompactCells107To138.Box112.compactCandidate_not_isWeightedPerimeterMinimizer	3a9cd6789a0b668b9e4ef7228ce3067ea7245f71f00f09f5f1283511b8c38108	equivalent
CompactCells107To138.Box113.QInterval	748a622a97ddcf0fd24f3713cbdd4d8f1c1e9ad735d7f94431c138aa24ed595a	equivalent
CompactCells107To138.Box113.lambdaInterval	31d125d981c5781883548cf18715012b9d58c7f9d9972b11c832d11aa5af2137	equivalent
CompactCells107To138.Box113.compactCandidate_not_isWeightedPerimeterMinimizer	6cf3753a45ed8aef6d1788c64e1d218afa635feb01e73e4059fe25ca2719de01	equivalent
CompactCells107To138.Box114.QInterval	d5d96a42dac6417edef434f1606c1c8789144cad93034c4d129c5fd8c0545863	equivalent
CompactCells107To138.Box114.lambdaInterval	1900f057587833c3a2a62d323cace993939f7c3167f4741a9b95a1086d80915a	equivalent
CompactCells107To138.Box114.compactCandidate_not_isWeightedPerimeterMinimizer	79ae99a0c222c41ff1a9249a60ead613f64488a9fef106d22fb8c7e3f4422919	equivalent

Declaration	Source hash	Audit relation
CompactCells107To138.Box115.QInterval	82acc9e99eeee70d8f66c5d0636388f616f02dd9813982a56d04d09e4144eab8	equivalent
CompactCells107To138.Box115.lambdaInterval	f4263df21fba15fce3d5794aefbbc7a90168fd44e2e756be1132ef1b55929646	equivalent
CompactCells107To138.Box115.compactCandidate_not_isWeightedPerimeterMinimizer	aec94a0efe0bb2956fce2631c039759143a5251f312c60120b5d55b1ae1c92e1	equivalent
CompactCells107To138.Box116.QInterval	c3281681d3850f6674e15287efabd0a405f98a90f3c958e592c508deb0164012	equivalent
CompactCells107To138.Box116.lambdaInterval	5f75f8baddeee85d6ba7db854dc399db7626fd7fec254a30f0fe9fe09f18e0c5	equivalent
CompactCells107To138.Box116.compactCandidate_not_isWeightedPerimeterMinimizer	195466ad701f5e3615da8fe573e8ea598bc988c1d0ecc522779e416c980abccf	equivalent
CompactCells107To138.Box117.QInterval	3934f74b5682ea3b2599994dac31687f97f6d282ccb7d45a86736047dc1a4d7b	equivalent
CompactCells107To138.Box117.lambdaInterval	25e045f7d62cf36fb735d85f381759c265b514dd7f18f0f5769cb957972722b0	equivalent
CompactCells107To138.Box117.compactCandidate_not_isWeightedPerimeterMinimizer	d5be3ca025f09c632b40d2996bfb92a26d1f54fb9482f65627fe1f5abe288161	equivalent
CompactCells107To138.Box118.QInterval	1305db4b75a492be05780d2bef19023193cba5fea1155543abb825a5dda09dde	equivalent
CompactCells107To138.Box118.lambdaInterval	2bcf787fe6fcdcf31fcebdb95de48f7b367fa06c16ecadc0c7330bea0046041f7	equivalent
CompactCells107To138.Box118.compactCandidate_not_isWeightedPerimeterMinimizer	8c100461773542c25e3195721dd7d95d68dac52f50a43d5e5e652976b5c12fc3	equivalent
CompactCells107To138.Box119.QInterval	d2d2cc3b6434d470830cb519e902fd9b96e9dbcb3500c8c28c472e9049a82b2	equivalent
CompactCells107To138.Box119.lambdaInterval	2fe4c7dab54dbe7504aa3943fb2d47249d6227118898f8ecde6a0b329d31cd40	equivalent
CompactCells107To138.Box119.compactCandidate_not_isWeightedPerimeterMinimizer	058fb3171e312f24b4749feeedf41790798cbecda7dfc69c6b2ffd8bb32da7a6	equivalent
CompactCells107To138.Box120.QInterval	103b533a81b2e646db03b4ad427c0b8cf1b2223b8e3a6c09a5566d938246b665	equivalent
CompactCells107To138.Box120.lambdaInterval	d61a4c01ac3f11004d3efe7c820ada78be6b57a1214f4891d73b2b7586fe0ce3	equivalent
CompactCells107To138.Box120.compactCandidate_not_isWeightedPerimeterMinimizer	bea186bc0e425adcd6913c65f9d1c99a9099cdb00d49e9634140939deee7e71c	equivalent
CompactCells107To138.Box121.QInterval	eeceb48aa4efa2d80f41189f1bbef7aba7d9f3718d9bcaeb733e8672391d27	equivalent
CompactCells107To138.Box121.lambdaInterval	63e2b656cc82abe4633b048393401388694a26d0097c0d373f25296f30b965a4	equivalent
CompactCells107To138.Box121.compactCandidate_not_isWeightedPerimeterMinimizer	996882e693ffd0be49534227f061dde3e7f18a3d7d34fde94f97a6094a1e9040	equivalent
CompactCells107To138.Box122.QInterval	35e3792a60b80dacfc16ff79c8f795d658236be89e93d0077333e7e487d3fb29	equivalent
CompactCells107To138.Box122.lambdaInterval	1c69c53a94321ccfacb37b3a03ef50e6c83da61fa2f6557ff558ac4f1f28576d	equivalent

Declaration	Source hash	Audit relation
CompactCells107To138.Box122.compactCandidate_not_isWeightedPerimeterMinimizer	ca80a2a08de62d36dde09def39689d83e1ab8217bd9e2bc63375df5985ac3ba	equivalent
CompactCells107To138.Box123.QInterval	19c72373c73a5898bdaafdc315e9ef16a823e86a8fb06f66d61400eaab658a6e	equivalent
CompactCells107To138.Box123.lambdaInterval	5c4791ca74b4df1472d86458b31a4869339f95629b2856a4cf96702faa2b1ccc	equivalent
CompactCells107To138.Box123.compactCandidate_not_isWeightedPerimeterMinimizer	29da32bc22f308b61f0582c08c56fcdcc16bb268bc70499cba7428a3277293c9	equivalent
CompactCells107To138.Box124.QInterval	87baf1dc00778cb04474c082c17f4c3712cd68ce367875707f16a1884c68928f	equivalent
CompactCells107To138.Box124.lambdaInterval	4f9b3897f26da44eeb9f2801d7045b4bd78ac8caa6563474d5f5fc925d957a8c	equivalent
CompactCells107To138.Box124.compactCandidate_not_isWeightedPerimeterMinimizer	b580f6ea51497ea001b0cf0a005793bb7527df4d5f01ca0b50e0b91658db2e11	equivalent
CompactCells107To138.Box125.QInterval	3bb0688ac6c09819422eeb5b03074eaeff1b1d2815a4f14710acfff66b67178d	equivalent
CompactCells107To138.Box125.lambdaInterval	d75724a7fd42668a349affa0dbf9e46a93f0ae8fe0c925e8c16c0f820b643c90	equivalent
CompactCells107To138.Box125.compactCandidate_not_isWeightedPerimeterMinimizer	0bb1c93a2685c852cdd1c3864e0000737ac109603618884fc2cf0029317c7ae1	equivalent
CompactCells107To138.Box126.QInterval	65dee7bf0f0c3314ba53210981e947c77d80c6ca4e235f0fd007da792b3e675e	equivalent
CompactCells107To138.Box126.lambdaInterval	37572a7a1fa32e92ad27a7448d903b36826df3d430dcecff860cacb9927cb1d5	equivalent
CompactCells107To138.Box126.compactCandidate_not_isWeightedPerimeterMinimizer	8972d2b5e263f16574992fbbba6dcf4570cf2e5312aecf37ab044a74d4c11a3ac	equivalent
CompactCells107To138.Box127.QInterval	6f2417566dcff20547c9d5102c027c4655f0365c69254bbfea9c43897f815af5	equivalent
CompactCells107To138.Box127.lambdaInterval	b51d044230b38c6f466650541d8eeeff6c33bfd80c44b1d95cb9c6115bb41fca	equivalent
CompactCells107To138.Box127.compactCandidate_not_isWeightedPerimeterMinimizer	9ec5488204315d2fcd4369ea6bfd5f0ac8a80c49ecd284f55c2908298ff17a61	equivalent
CompactCells107To138.Box128.QInterval	8cf07bbafa97ae35b24e0fc110e0e95be4432d46dd20e84ebdb84d5b73c5f0c	equivalent
CompactCells107To138.Box128.lambdaInterval	fb3d49e365c561884234e70998a748ace0129d8cb91de3476b0eb522d886dce8	equivalent
CompactCells107To138.Box128.compactCandidate_not_isWeightedPerimeterMinimizer	234686df430073286237b44cee801e495b720e190df844d75f0765206fa210ee	equivalent
CompactCells107To138.Box129.QInterval	c1bf979b33461da9e80649051f552352b338c4a51cfc303dba16e27c3d099961	equivalent
CompactCells107To138.Box129.lambdaInterval	28324986036051daef5fc64c44bbf48e1d0c7cbffcbd91da47e1ff62e99c3905	equivalent
CompactCells107To138.Box129.compactCandidate_not_isWeightedPerimeterMinimizer	f7d81a771e9c79f128aeab371c7b57a95fe5824462a8af09a32e7f337697015b	equivalent

Declaration	Source hash	Audit relation
CompactCells107To138.Box130.QInterval	faaf66d32121c4867fd8587b54234bdff78f85a36304216a117dc28a37d5076e	equivalent
CompactCells107To138.Box130.lambdInterval	b8d94c45a41c91330f6ddeec159fe7ee2925ef3f283168f4fff88435f7159241	equivalent
CompactCells107To138.Box130.compactCandidate_not_isWeightedPerimeterMinimizer	6ac942981ae41b4e13758a82e908c4780ac40187ec10c3929796a8bc98603d0e	equivalent
CompactCells107To138.Box131.QInterval	570caf3ff4493d7e8dd9c9891e513683c92c1da854f58de6ee819c57640e58b1	equivalent
CompactCells107To138.Box131.lambdInterval	e2801375490500dea91ccb37dbaf01ab1ecf2d17e34af1f7a383da10b2dcd65c	equivalent
CompactCells107To138.Box131.compactCandidate_not_isWeightedPerimeterMinimizer	98e5d302e1985d3d9f75eea4ed0ebeabee6a4e9c9318c497a774b9dc0f8f77fc	equivalent
CompactCells107To138.Box132.QInterval	ebfea2b76f6f4417a6215c8b721d5c03417271d3f627b198adb4e9173cc8ea67	equivalent
CompactCells107To138.Box132.lambdInterval	aaf982e20f6b7796b9b3722729ad678bccd64f77e7a872260313c1abccc1f2b9	equivalent
CompactCells107To138.Box132.compactCandidate_not_isWeightedPerimeterMinimizer	1a7ff74d7fc44e335f8d1a8f1fccb0c8f0b9b92306f012e8696ad50155e5aed4	equivalent
CompactCells107To138.Box133.QInterval	32fb2d89a006e5e60d72410b0ec5d50a926c094c69117b8ff976e5cf18d6703d	equivalent
CompactCells107To138.Box133.lambdInterval	645314099b27c2288c39c4d0cb392b463cc3349381d6aac28a5946f2d19b4e9	equivalent
CompactCells107To138.Box133.compactCandidate_not_isWeightedPerimeterMinimizer	abbf978d1e632a7a8a5e2c59ecf2d95d8d91477a5b8711f87bc590f2caed1774	equivalent
CompactCells107To138.Box134.QInterval	26565a0bd2a432463bbd9ac8fd3cd52509ec23b25f31297f10b8ceca9fe8e4fe	equivalent
CompactCells107To138.Box134.lambdInterval	dc27deac50d93b90835e114f7ab6be048e142370a35be5c88e445d7457e89269	equivalent
CompactCells107To138.Box134.compactCandidate_not_isWeightedPerimeterMinimizer	274e4bd501cbbdd4ce60205454386c56f9cb42a56f31897b1f347dc36fd0cf99	equivalent
CompactCells107To138.Box135.QInterval	a2028aa99b5739d5e1f5da386a8a3122d85508341952765c1c270676c7c73e8e	equivalent
CompactCells107To138.Box135.lambdInterval	4adba7b09f31eb35eafd8c10e47dd7ef966c661a0edf55ed6276c9294caf061c	equivalent
CompactCells107To138.Box135.compactCandidate_not_isWeightedPerimeterMinimizer	7be5920f912c23898ee1ff7008ec1599c7d254730400af6c9184fae2f0cae11c	equivalent
CompactCells107To138.Box136.QInterval	fa0f031c6bf7e80abc0e7d6d85ad4661427f8f9bd4349e592343c5f45a16953b	equivalent
CompactCells107To138.Box136.lambdInterval	850e68baf3057d94dcfece81ec726425e3fec1ec1da127b18fb276f2fd0e434f	equivalent
CompactCells107To138.Box136.compactCandidate_not_isWeightedPerimeterMinimizer	b590a5469c02d52f082c91aec19ce6eec8b17d510b293062f1f82eb6157eb856	equivalent
CompactCells107To138.Box137.QInterval	ac375d4f35e7d48fb0d5f05cdd05ed25fe73c6c92dae7c1edbd8df18fe349a80	equivalent
CompactCells107To138.Box137.lambdInterval	6c7ef894dd9b440266ce4854cd6f2762306d57b453861d1e5f685f6a9a5c45e0	equivalent

Declaration	Source hash	Audit relation
CompactCells107To138.Box137.compactCandidate_not_isWeightedPerimeterMinimizer	cd37b0388a2fa3d88f69ad1a783cfd5d79af2fd7fc3002d94c1a788ee3aae4e5	equivalent
CompactCells107To138.Box138.QInterval	93279139f29ace3d5c509f8081cf08bf3818bd7b24ab161c210be0b7187e8571	equivalent
CompactCells107To138.Box138.lambdaInterval	7434ed749fbc1611f4808238533569f1d23c07d4153856855d87385bec39ea62	equivalent
CompactCells107To138.Box138.compactCandidate_not_isWeightedPerimeterMinimizer	1a8e9b21f7fb48f02c6f6710362b345b29fbf1bf53f6bbdc7957ddb3704ce95e	equivalent
CompactCells107To138.boxes107To138_candidate_not_isWeightedPerimeterMinimizer	ff0f8614b90de0bf68ac6a45a4b51826c0d9819aebbb752569923f30cb3d197d	equivalent
CompactCells107To138.boxes0To138_candidate_not_isWeightedPerimeterMinimizer	6be709e753a98660172ec022123c80f08dc86c5c6dbaaa2e116440ee8e221a25	equivalent
CompactCells11To42.Box11.QInterval	5dadf750260546df002a0918e752ce4f8c1000017a18593d0fe9112eb9365954	equivalent
CompactCells11To42.Box11.lambdaInterval	35ed6c0ef6f230d169239aca3e04fc09c00308405cd703142032adc28c44ba76	equivalent
CompactCells11To42.Box11.compactCandidate_not_isWeightedPerimeterMinimizer	e25ed488330a5af7dfe27ae3e94cdb50b49ae35f3ef0a4a11d315bc4e2b43693	equivalent
CompactCells11To42.Box12.QInterval	6b5cb01da16d94ba8578d164b9d138021aa84004533177fd1aed33b426ffcba9	equivalent
CompactCells11To42.Box12.lambdaInterval	fa2ece58f06362739b1482c3f942480e2e4dc4722efb7d590cafbbe42b82e3d5	equivalent
CompactCells11To42.Box12.compactCandidate_not_isWeightedPerimeterMinimizer	d2687ea347f469495c814ce5582189853cb5921076e09a9146c3b10f64f65a26	equivalent
CompactCells11To42.Box13.QInterval	6c899b3819fdeb1f9aa07349283eb22e7b0e151a6cc9196e2c4d1da4a2198f6a	equivalent
CompactCells11To42.Box13.lambdaInterval	0607af4ba6a5fa8832b97a7f513cc12d266733dd7ef34533b974841135cbeec3	equivalent
CompactCells11To42.Box13.compactCandidate_not_isWeightedPerimeterMinimizer	c0dbe2d801c0fd15851a4a3b35021c4751cc44fe9bb3ab0e7210f115bdbbf94	equivalent
CompactCells11To42.Box14.QInterval	c9ec122c69cffedebe9936c2d91f7f1c14b19b24528db3ebe5ed93d18963359d	equivalent
CompactCells11To42.Box14.lambdaInterval	c1d5aa6d3b5b6a86a032b59712e5f8375242032dc6ce91827856370bc86e3044	equivalent
CompactCells11To42.Box14.compactCandidate_not_isWeightedPerimeterMinimizer	a3d548248bc69d9116bcc089592b0ca2ca80d55257768cc508af62ddba5ea920	equivalent
CompactCells11To42.Box15.QInterval	28fd1c3c8854c403d3966bf52d5c3405a027ca08afad17fe8d3ab2e144e88672	equivalent
CompactCells11To42.Box15.lambdaInterval	9a9e1f2e3c3f5d830f510a19cc063a108cef158c5a4a0053bbefe4bc8e601f67	equivalent
CompactCells11To42.Box15.compactCandidate_not_isWeightedPerimeterMinimizer	26b45d6f69c5132412e85e23c7e25fa2c0c84e2ff4f2a78db700e62df8795740	equivalent
CompactCells11To42.Box16.QInterval	bfbear0f46f4e56081432f35964870229724878407ca8329c357eee6fe7afbc27	equivalent

Declaration	Source hash	Audit relation
CompactCells11To42.Box16.lambdainterval	7c142e04fd01aa37a68b37c851e84e1cd50192b24e51b10857285bf123f33125	equivalent
CompactCells11To42.Box16.compactCandidate_not_isWeightedPerimeterMinimizer	853796148a1f370c1343d2da62354af455085efa7bb86d3f54f1adffefe5f3c8	equivalent
CompactCells11To42.Box17.QInterval	a706028b21c48f0a047e789de82ab0e189552846a35320f6ac83e6fb98b2afb2	equivalent
CompactCells11To42.Box17.lambdainterval	f8f9fd00f2203e3795e23de641dbd2a44b0b46a5ce50a2db12f43ddf0f821176	equivalent
CompactCells11To42.Box17.compactCandidate_not_isWeightedPerimeterMinimizer	3f5988125a896f23e484b8f98d6d958d45969bc1a2a02a82965d22bc3019b721	equivalent
CompactCells11To42.Box18.QInterval	421939a3e0dade57935a14330b9dc80f68414b42fb63b21b1d489750c5c5e710	equivalent
CompactCells11To42.Box18.lambdainterval	393de9eb26f4a13c47cb6abc31d73cdc1afe0c8773b1179b6b8333ec2911a92	equivalent
CompactCells11To42.Box18.compactCandidate_not_isWeightedPerimeterMinimizer	8de32dc221efa0f365dfbb56cb3ae8f7cbfe628aba560bb697be1f00dc4d6358	equivalent
CompactCells11To42.Box19.QInterval	72252640172414a82d8565298063464f9db5fcab1a67ee3aecc9d3070367f74e	equivalent
CompactCells11To42.Box19.lambdainterval	95bcd6716496c3af1683ace25bbcae84def2a1a82475f91382dd2bd6f450ec40	equivalent
CompactCells11To42.Box19.compactCandidate_not_isWeightedPerimeterMinimizer	494fd37b380b5741b2fea7c2fc7a398e0e2de5b9cff332decebebbba062971f68	equivalent
CompactCells11To42.Box20.QInterval	5cd563653114f2859636ebf98a9b97dd18c4e19fb145a032e9aa4caa88b62c3c	equivalent
CompactCells11To42.Box20.lambdainterval	af736e1cb590a5f2ae1205aca4a1955fe25d0d5c849c09d10ce875f867a8999a	equivalent
CompactCells11To42.Box20.compactCandidate_not_isWeightedPerimeterMinimizer	2cd462bc5d6c1462c9d3fe5fc44f3e51fe4afb467fc39c1276a47a92e42b5aa6	equivalent
CompactCells11To42.Box21.QInterval	8fa3d95272438ee8a6d7a4d1277c977d44d15fa7172a869f0fab78ca3b8ec6cb	equivalent
CompactCells11To42.Box21.lambdainterval	d3ea57470f81dc170630c50ff54a13283c352da3435e3f8cbff5426fe815ff7d	equivalent
CompactCells11To42.Box21.compactCandidate_not_isWeightedPerimeterMinimizer	c5adde95b1ac6d16b334604f3f667603c8286809d12d25c354fdc9e96ec19f65	equivalent
CompactCells11To42.Box22.QInterval	275894b802bf1e39f33583fb5c03182ad08036c18feee34733a85510b9087aae	equivalent
CompactCells11To42.Box22.lambdainterval	6f49fb55096f4c54947e65a27514664ce05e255b22efcdb30680b15a59ceef5f	equivalent
CompactCells11To42.Box22.compactCandidate_not_isWeightedPerimeterMinimizer	4194c5da3e9170b039ceb10ba2934731ac7d6c32a862e9f5e2cb4ce461500c37	equivalent
CompactCells11To42.Box23.QInterval	a300e02dca2aea8e6e9827285162da1cf623e509ccde6bac256b08da727d9021	equivalent
CompactCells11To42.Box23.lambdainterval	61d864af315c910404f363ec07754596cfbbd0e130a37cc2ab7ba3b914321e40	equivalent

Declaration	Source hash	Audit relation
CompactCells11To42.Box23.compactCandidate_not_isWeightedPerimeterMinimizer	2b2ea6bc327b386b73f1d9a29ece89b4a0920be8c1ed1018b9060575c7f776e1	equivalent
CompactCells11To42.Box24.QInterval	3ce85c9d3b1423764457217a9271ff0c32ed7a623e287024043e0b4af11925a4	equivalent
CompactCells11To42.Box24.lambdaInterval	53a2313e578ddc6d222950836b3de59d5398f0da6f98ee2619abdda9c9c1157e	equivalent
CompactCells11To42.Box24.compactCandidate_not_isWeightedPerimeterMinimizer	f3c337be480a73960fd98a4520aed8c79df9571c25c5d4ffc1cdd67fe0d066a5	equivalent
CompactCells11To42.Box25.QInterval	171b410c63d124de25758b7a9f2347576e90dc75f16378caba82094285face18	equivalent
CompactCells11To42.Box25.lambdaInterval	28c680502c80440791ceed14bc4ddbc649c394986c8a82c3a312c56b4eb7cd2d	equivalent
CompactCells11To42.Box25.compactCandidate_not_isWeightedPerimeterMinimizer	800b063236ec4bbf34e7dcc65f28b04046e8e93e7e58e96d610da31817777e1	equivalent
CompactCells11To42.Box26.QInterval	6b726f333664ba5e415c25d4e75af36f8f1b67e74d3ce12ee4479db0b8353d5f	equivalent
CompactCells11To42.Box26.lambdaInterval	e046af7b919fb2b5c4b42b11f03be39c237f858dc88b66ba0ad053aca45d931	equivalent
CompactCells11To42.Box26.compactCandidate_not_isWeightedPerimeterMinimizer	36103e4a1fdc17c3539886190d961e240c5c6dbbdfa3bc5f84dfe97d7480af04	equivalent
CompactCells11To42.Box27.QInterval	802bfda20aed2d8b5f8a5480029299eb99c2ceec4d6de9fc733ba17beefa7e24	equivalent
CompactCells11To42.Box27.lambdaInterval	d92092aabe34d1bb13e25cff39f4812e2d56e1d5807fe80a5b95d3bd896a86bc	equivalent
CompactCells11To42.Box27.compactCandidate_not_isWeightedPerimeterMinimizer	70bd6d8a894646d61b390eb3a5e19a933622cace6ead5c9e179fb1bd98d24ed6	equivalent
CompactCells11To42.Box28.QInterval	02fc6391d60f5f7f7700a22f41a5ce06c41b39e67846c35015881b5642a03ad1	equivalent
CompactCells11To42.Box28.lambdaInterval	e429588ce38d887e1b92b96b096085976369647573ee1fcae7dd6139a6dc8750	equivalent
CompactCells11To42.Box28.compactCandidate_not_isWeightedPerimeterMinimizer	96a38b510e97976752ced1cad8fc92612973e544a444581d711dd6c0214958dc	equivalent
CompactCells11To42.Box29.QInterval	0ac331b20d112ba40707de26cdc849de1f24c461eac5ba433506b5c5898f163d	equivalent
CompactCells11To42.Box29.lambdaInterval	57e18f665520f6df6c4b31f0f4a72b18bf6e0e04d9e868ac83064d94b0be1c88	equivalent
CompactCells11To42.Box29.compactCandidate_not_isWeightedPerimeterMinimizer	57f6bb2f9c56cae0cd5289b3c726da9698dfae81dc6a15435718670c060dd2ee	equivalent
CompactCells11To42.Box30.QInterval	2c5cd86da62c9e9dd2d366d35a92834bcb8d884f831c058cc69a0f3d16df980d	equivalent
CompactCells11To42.Box30.lambdaInterval	3070b83a0add6e0dda42ba58b00dcd3c7db563a57cd7a7956ae6bd074b1cd3bd	equivalent
CompactCells11To42.Box30.compactCandidate_not_isWeightedPerimeterMinimizer	0c8285084099205dabbdff10bc9b65ef26cb67ca3579abb01a62fcec2e0931a4	equivalent

Declaration	Source hash	Audit relation
CompactCells11To42.Box31.QInterval	edd7a64e3a8968e2daa7888d695da7eb5c6abe8609692b49ad55754ee7f98cfe	equivalent
CompactCells11To42.Box31.lambdaInterval	7373155b15a6ff6c4a671d4bb1e9e181f996da8ab2e5de50a87dafa8721faa35	equivalent
CompactCells11To42.Box31.compactCandidate_not_isWeightedPerimeterMinimizer	aaca620af0b65b715b7cebb4045eea6c6588af48a92c2d198b9d7491cf4fe464	equivalent
CompactCells11To42.Box32.QInterval	992f208ba2c47e2eea338d0bd5dca59eabf7d4efb062a93618227f1812b20663	equivalent
CompactCells11To42.Box32.lambdaInterval	a99d2e5f0a44f58d3e54b030483795d9a369431039029e42b17116f2ef4eedb3	equivalent
CompactCells11To42.Box32.compactCandidate_not_isWeightedPerimeterMinimizer	9aadfb095a6f5f8ea8894f9a2ce6701be657036c8148a3d488b87b33bd6b8518	equivalent
CompactCells11To42.Box33.QInterval	29cb5a3a238c0021a974e2ac0c9c45405ad39e71bbf71ee295080814740db7a7	equivalent
CompactCells11To42.Box33.lambdaInterval	fe3c59207b513cca74326544fb7993cb37040ac55ede8a93fb7d49865d538844	equivalent
CompactCells11To42.Box33.compactCandidate_not_isWeightedPerimeterMinimizer	ba354e595fbec3d31ea420648d6e2a15276e1fecdc8182035139c406b75e18b5	equivalent
CompactCells11To42.Box34.QInterval	e91e68addcd010ca7c18cf528d434d1c9a62cfbae5989807cbfcfd805c62f4ad	equivalent
CompactCells11To42.Box34.lambdaInterval	a31c325d88d7e6420026520f8cf4803a02e09756f2da6dfe522ef0db9cedcfec	equivalent
CompactCells11To42.Box34.compactCandidate_not_isWeightedPerimeterMinimizer	5a998dd16fb8ea1b54a2a3a0ce2ccc5aad36971ef785f10b9e7d8c42735f316d	equivalent
CompactCells11To42.Box35.QInterval	67d421fff5b0f7d23625bcfe706945aeb347acb0c1cea5cf4e3871dd74fa581a	equivalent
CompactCells11To42.Box35.lambdaInterval	2be8f367fb1f6de59d489c19dcc084d8abe24c2ee2b02f7a7a105231dd66251	equivalent
CompactCells11To42.Box35.compactCandidate_not_isWeightedPerimeterMinimizer	e472b8b45f7a87a43a42500b93702a016df024a7dbfe4f108615e29eb2ec147c	equivalent
CompactCells11To42.Box36.QInterval	703b8c1b92e07f6185b60b8f29c34c09d491094a1f243e5c708e03dd164be7a2	equivalent
CompactCells11To42.Box36.lambdaInterval	c43bedfdd6219641b741f341fa67517dc080c6f62fcdfae7e894bc94f241770	equivalent
CompactCells11To42.Box36.compactCandidate_not_isWeightedPerimeterMinimizer	bd31e85eb354f834b0785141f4ece915272818929427fa06cb621c830d6384e6	equivalent
CompactCells11To42.Box37.QInterval	1d1860ad9f15ffa72c1b9a46b962004426831802a35a0380682af33151aa2601	equivalent
CompactCells11To42.Box37.lambdaInterval	b41c9694e7b50b0a52f549ff178cd10628887aafa299526a3a3b7f7dcb6143df	equivalent
CompactCells11To42.Box37.compactCandidate_not_isWeightedPerimeterMinimizer	bdf39f7c66d9fce9f2656ac21bff9546c4f0f9cfa0575d393b12aafeb828d95a	equivalent
CompactCells11To42.Box38.QInterval	259350e6febd8af14c376b67fe99dd162667feda06519bd2ecee1c6b432f7d45	equivalent
CompactCells11To42.Box38.lambdaInterval	ca19fcb23c5101a837477321e27457ac658b40aef82c8ea7da82d75d5c1e63db	equivalent

Declaration	Source hash	Audit relation
CompactCells11To42.Box38.compactCandidate_not_isWeightedPerimeterMinimizer	76f774c500c659737d38bc4ca7f4d7ceef2ef4e85c0b3d04a1895d3b84ec0df	equivalent
CompactCells11To42.Box39.QInterval	fd76225421a3059a0e53bebe8b123dc84dafd68e33b8e46022106dbce6e9b921	equivalent
CompactCells11To42.Box39.lambdaInterval	df83554a8d6953f3e48e40278e29011ba10aee297eff9eda50bf6f50e7d358f9	equivalent
CompactCells11To42.Box39.compactCandidate_not_isWeightedPerimeterMinimizer	499d458b3a3224f0a18221c08bdb8bc01fae88be2eb59c864fec3e78b5e75d9e	equivalent
CompactCells11To42.Box40.QInterval	112f63a36097d132d1ec31a58c60face95208a28da5fed16567c49515211c19a	equivalent
CompactCells11To42.Box40.lambdaInterval	52dd9987434b230240fb2e9720069e8e76cca4b27a5b3c34f88ae9b5b7bca142	equivalent
CompactCells11To42.Box40.compactCandidate_not_isWeightedPerimeterMinimizer	f5398866846c93fb5701e62e7dcba8680a9acebf47da299afcef802c003b73cc	equivalent
CompactCells11To42.Box41.QInterval	8ebe75de31823f535dd39b562794e1e4c9eafadc30aacf46e8797030eb99f874	equivalent
CompactCells11To42.Box41.lambdaInterval	fe35a12b986e353f59b9fcbaaffa15a6952620a3f6de90e1c9d785601422d2a15	equivalent
CompactCells11To42.Box41.compactCandidate_not_isWeightedPerimeterMinimizer	3fcc77976b61018da6d95a9fd96b8dbec48ecf1000fab4323d96dd0884692522	equivalent
CompactCells11To42.Box42.QInterval	7bb6e46eed87a28c3c04468260bc9dd51fc9992e94ec24ef01454c93c3cbdaa9	equivalent
CompactCells11To42.Box42.lambdaInterval	6995331f0ff2bb4bde2a9d33e219a52e79b70146cb518c3d3fbf64c3e2c02f53	equivalent
CompactCells11To42.Box42.compactCandidate_not_isWeightedPerimeterMinimizer	4c0cea44e7d218c0242ee5603e92680468f3b73e3466b4c45e5939a9b03b340b	equivalent
CompactCells11To42.bboxes11To42_candidate_not_isWeightedPerimeterMinimizer	34866d24b391b999bc0cc35b5a2dc9f2cd750ba2fb0cd34020cc9d50bdc54709	equivalent
CompactCells11To42.bboxes0To42_candidate_not_isWeightedPerimeterMinimizer	87b0dd8c2e42a30635934bd4d240b4f0fe3dfce987f1fc99b5dd1e03c56305df	equivalent
CompactCells139To170.Box139.QInterval	96fbfd325d197d6ac810ce3d1ba477d88ad46e03accc7f4a2779aca04473ea51	equivalent
CompactCells139To170.Box139.lambdaInterval	7eab737548930554c94700a6bd3f19b48954f39c55693c40a36e763a8353c589	equivalent
CompactCells139To170.Box139.compactCandidate_not_isWeightedPerimeterMinimizer	4f5e7898571d4646013f0bb623106dde8df0b482a338c6f60586847e7ed336f6	equivalent
CompactCells139To170.Box140.QInterval	b6b8d85ac67ffa5032c2e60b9ab88530165267185734d20bd1a61f57afe6997c	equivalent
CompactCells139To170.Box140.lambdaInterval	bf502e34c46af08c2482b3a01056f55e3dfd8769d00f4ec6a846f040ace9770b	equivalent
CompactCells139To170.Box140.compactCandidate_not_isWeightedPerimeterMinimizer	e71b7add99315d28a5c049bb17d3088bb2cd90e286c17148f459167d9af682d5	equivalent
CompactCells139To170.Box141.QInterval	c5aa8ad0981d3ef46cec79273fd099fcbd4a03e9a773d9edf8330cab642c35b7	equivalent

Declaration	Source hash	Audit relation
CompactCells139To170.Box141.lambdaInterval	ebd8ed0ef46e0036bdca3713f42250161f2d86f5350cdbc466baaa37e2276e30	equivalent
CompactCells139To170.Box141.compactCandidate_not_isWeightedPerimeterMinimizer	b31722ff31480f7fc901c7da37eb4f9683a279bae3035f22f682a6db6a3d0c76	equivalent
CompactCells139To170.Box142.QInterval	d00d668e1df1ae7297272fe1d0e8fd86bed14c05228ab0dac25041167db0b070	equivalent
CompactCells139To170.Box142.lambdaInterval	d68bcc1322a6fc800486747b125a739ee4212111cf0d7ae485f6d573309d7837	equivalent
CompactCells139To170.Box142.compactCandidate_not_isWeightedPerimeterMinimizer	29b76b05a3229ced675fee51bdcdf45e7d204d27297a5c419195180201321026	equivalent
CompactCells139To170.Box143.QInterval	ed30b07baa3c46a50eb8668d465d6fcfc2565f2858556b8178232a4437302ee8	equivalent
CompactCells139To170.Box143.lambdaInterval	c123239135618db5be267158194bad92e0fa0b4330b28b7289d435ea3dbca077	equivalent
CompactCells139To170.Box143.compactCandidate_not_isWeightedPerimeterMinimizer	397c14585c813b89621b075e4f6449702f5fde9c8b4ee21d6c5b9f8fbb5a29ef	equivalent
CompactCells139To170.Box144.QInterval	6bfc7a1dda5830f2d341b6a18028e435bfd2bdc7fd44ee371deca812a7586827	equivalent
CompactCells139To170.Box144.lambdaInterval	8398f543591fdfe314c223c2970ac04d10d5662b124f568cee9cada28c791f24	equivalent
CompactCells139To170.Box144.compactCandidate_not_isWeightedPerimeterMinimizer	9bb352137f9a2a0e4d615c72bf27879518f52ea8936a8205f09fd0b48891607d	equivalent
CompactCells139To170.Box145.QInterval	1bfae45ac8f410e7fb5a5998768a19534eaddf23621c1858959ddfe8108b33c66	equivalent
CompactCells139To170.Box145.lambdaInterval	7ab0695cff91aaae3318e2145cb5d8187fd047b8bb8beec4d75da505f8e461d8	equivalent
CompactCells139To170.Box145.compactCandidate_not_isWeightedPerimeterMinimizer	0715a3792092b3d36751b417a37518cb5f9a4e4b2808b97f82bb9ad9ab408e48	equivalent
CompactCells139To170.Box146.QInterval	bc6d6bfaaaeaf2ba77285e075c714dbfbb848557c2972750a40c1a650b922e4	equivalent
CompactCells139To170.Box146.lambdaInterval	2b0d5f86657dd88e42cf3210cfc8eb2af30b53d13cb7086757dd932139c35a7a	equivalent
CompactCells139To170.Box146.compactCandidate_not_isWeightedPerimeterMinimizer	e45304aff9eaf5c1b73503d91b8eaaa2c874bf1b25097eb65632a0949128c966	equivalent
CompactCells139To170.Box147.QInterval	7af9fcc8eab40bb8479f9addb2444baaf0fc8805a6c2a91a2942927727ab7085	equivalent
CompactCells139To170.Box147.lambdaInterval	bbf2b5157b5ff77dab6146d347f70982c30e079046b5059d6cf4550ed4587956	equivalent
CompactCells139To170.Box147.compactCandidate_not_isWeightedPerimeterMinimizer	8d809cae1df66956fcd857f1f7bbf019fcfa4385e01cc911d36f6975bc5df5d8	equivalent
CompactCells139To170.Box148.QInterval	b7cd65884e1e7cca0b52f3f125606f997b34baa596a7dc5b36f7c4a798c0eaf5	equivalent
CompactCells139To170.Box148.lambdaInterval	57bcd16aa829beb5ec2abfd22f637a7082a265f2f4e10b24d97e9d46d7652f7	equivalent

Declaration	Source hash	Audit relation
CompactCells139To170.Box148.compactCandidate_not_isWeightedPerimeterMinimizer	a8d52d8a284546a50e6257162f3f2cae8978851788fccde04ad4bc966e80348c	equivalent
CompactCells139To170.Box149.QInterval	d31dbbb514b07d095ad3954bd7fa4758e03b2464076831c4f52cc36f8c4d714e	equivalent
CompactCells139To170.Box149.lambdaInterval	7af84fb070cf753864d8bae4454d9255c6512a0c772cf9f48dc989b9fb02ef67	equivalent
CompactCells139To170.Box149.compactCandidate_not_isWeightedPerimeterMinimizer	5f60dbd0dc2b57cbdf52414917411d58f8c0288632b67ab8eaecadcee62a404e	equivalent
CompactCells139To170.Box150.QInterval	428aa4aa9aa1115e0a715dbb34b0063fb9259c38a187b01b48dfd331c208b126	equivalent
CompactCells139To170.Box150.lambdaInterval	83f044f64ab5966922908a622789f6a34aef7f72131578caa88124b9f198f7d	equivalent
CompactCells139To170.Box150.compactCandidate_not_isWeightedPerimeterMinimizer	532256cad8d9c73e2ae2d33ea21cd6ca85cac8cac03727a651fb93b329f12364	equivalent
CompactCells139To170.Box151.QInterval	ebbd1e4c6b42c0c0948eebdf45a3f5b6a761ea9553879deb143a05b821b8a23e	equivalent
CompactCells139To170.Box151.lambdaInterval	cd1e575732ac08e433cee2324d067fedd96ae45e4d726f28ce4fe3f5f83300b1	equivalent
CompactCells139To170.Box151.compactCandidate_not_isWeightedPerimeterMinimizer	f1a1a5c0f41a12bfb2719cb790ca995d2756b2aad3129de7a062b206d3b25c98	equivalent
CompactCells139To170.Box152.QInterval	84359d97b5bb1fdae316eccc8cae31711b23271fd2d0d1c1602e9e5b3b254e00	equivalent
CompactCells139To170.Box152.lambdaInterval	db58c358ac30702efe2c2c046bd8a9c7010dfeee648457fc9a5a5e2d3002b319	equivalent
CompactCells139To170.Box152.compactCandidate_not_isWeightedPerimeterMinimizer	cf5678e65642fa8747487fed1d548adecab a73ffcab0508a757df72152547dda	equivalent
CompactCells139To170.Box153.QInterval	3656bb60eece130a026d50f451c2352271044d8934940b252cdc61fbf9cd7ff4	equivalent
CompactCells139To170.Box153.lambdaInterval	ca8aaa0b32065f870548c88b663ec49a02fc85423c5e3ae44028dc9888cee12d	equivalent
CompactCells139To170.Box153.compactCandidate_not_isWeightedPerimeterMinimizer	2a2b6f4104975e168d8955bfcd7c1a097f6c652d2287a8cb7f1f0c8e45938e8	equivalent
CompactCells139To170.Box154.QInterval	b0349e56abcfc41d38931f05c574ec2acc9c5083a387ef823c70e11361418f99	equivalent
CompactCells139To170.Box154.lambdaInterval	4d001d151a391e50a1630624c9f7321f2be867107ec93f82540b2e827b1c64a9	equivalent
CompactCells139To170.Box154.compactCandidate_not_isWeightedPerimeterMinimizer	6fdb6e4e5b60d8b7f14ed40b99a21681af5afd1229292e415caf79ac47db2eee2	equivalent
CompactCells139To170.Box155.QInterval	786256132228c7e3f6fb3c7c05e051e5db78c211cefbddb20601ebee12f2c81f	equivalent
CompactCells139To170.Box155.lambdaInterval	872574aab1510c2fcf24894ec7eb20b7c4a77c062166a1acd330cac3662cd226	equivalent
CompactCells139To170.Box155.compactCandidate_not_isWeightedPerimeterMinimizer	d6a7e3a64b3db65d346ad6e30b157b19590f303ea91ea2c8cd5fca7e3ce88edb	equivalent

Declaration	Source hash	Audit relation
CompactCells139To170.Box156.QInterval	b59abd5489edda5d581c52bc0534da07ee8f7640d83e1457a513a1e67f2d03e1	equivalent
CompactCells139To170.Box156.lambdaInterval	ff8a4629b354b48088bd5c135f5a3e4f88ab1b9cc8421776e16d6b52fbc93a7e	equivalent
CompactCells139To170.Box156.compactCandidate_not_isWeightedPerimeterMinimizer	c031085e133f05e678f977b24fd3ff0fce8aaf40926e40587a722cc648c01b7e	equivalent
CompactCells139To170.Box157.QInterval	1072c84956ae5c6e6e63d6316667385915019c039aba30ba0f8c43ea6491c9ef	equivalent
CompactCells139To170.Box157.lambdaInterval	b7441e33b28c7f68937b84e0ed3017f3e06221b66622472d8cd6479c86a9bf60	equivalent
CompactCells139To170.Box157.compactCandidate_not_isWeightedPerimeterMinimizer	6c471a7cf6db3acc0e7140a30475fdb2fca6d72045e80c23337d7d7070d79360	equivalent
CompactCells139To170.Box158.QInterval	4170f0210c59bf23f0af5e75260894d21b1f09d0c40e801f73ae7a172fff7b46	equivalent
CompactCells139To170.Box158.lambdaInterval	675951be7744b306d89104e3ef52bf60077844a56262a6abb37602f640dfba02	equivalent
CompactCells139To170.Box158.compactCandidate_not_isWeightedPerimeterMinimizer	81c711b29998fe0b1d32ed77cd4868a9b54d63046b48e114dd13ef6a93879744	equivalent
CompactCells139To170.Box159.QInterval	b5f83583aa63cd030ee5b66a047d060b340d718241a0dde79893bd75b5f25f18	equivalent
CompactCells139To170.Box159.lambdaInterval	154787958732c67136de84837579632b0ef191eb171754c0b6a607faac20785e	equivalent
CompactCells139To170.Box159.compactCandidate_not_isWeightedPerimeterMinimizer	b10e33942d5df013c912349d3e90308b022419002f6a8c3c36fd00083504ab44	equivalent
CompactCells139To170.Box160.QInterval	a248a5453d1f64a153fd7b3ebbd5cb1c23a0480fa655cf52a26717e8d867e6c5	equivalent
CompactCells139To170.Box160.lambdaInterval	dc3f1b16f82e1fbf5514e5365db01da55cc446670e7100fc79f7e342256ad38e	equivalent
CompactCells139To170.Box160.compactCandidate_not_isWeightedPerimeterMinimizer	7faba6ffe2ded59e54de97c54fe50dd900eab2c85eacd2b6ff301777ce53e418	equivalent
CompactCells139To170.Box161.QInterval	0b868224b4e3cf71f051b5e21d9025f92c162ce3693039523c5f2029114257c3	equivalent
CompactCells139To170.Box161.lambdaInterval	59217c8be0b9d9a40b5ed563ea49c03823b1d0db9f0f5612836cf38af2e80e3b	equivalent
CompactCells139To170.Box161.compactCandidate_not_isWeightedPerimeterMinimizer	13f9281598ef76402a2b07cd4f1a91ffe65ef772bbdfb817dae40df210cf960e	equivalent
CompactCells139To170.Box162.QInterval	73e9a5199d8acfbcf10a5b10307806c306867bc471d2b402381a921f1fe6384e	equivalent
CompactCells139To170.Box162.lambdaInterval	50f6d335ee696f81fa7c7decbb88df5159bb975837bf5cdcc1a4ce6a69c4fc11	equivalent
CompactCells139To170.Box162.compactCandidate_not_isWeightedPerimeterMinimizer	21cd06207b13f62c155c26b95c9b87a1daf35125c040fe380e9af7ca06aafb0	equivalent
CompactCells139To170.Box163.QInterval	c597558af7ba4c6d70c741f1a1b87283fd574d77a916a20a92f6239df9ef5d70	equivalent
CompactCells139To170.Box163.lambdaInterval	1ecf263a553326757585f4d13c5157d8e1572cbb68dc709830a14b29e92b1244	equivalent

Declaration	Source hash	Audit relation
CompactCells139To170.Box163.compactCandidate_not_isWeightedPerimeterMinimizer	939a5d950f24e668fdc952381fe941594af8907edaba92c7721b37c584bf6fc7	equivalent
CompactCells139To170.Box164.QInterval	2ef2e086693e73ee7f2a6293992e6157a4ee26edf990811d1fcf5d9ecb606635	equivalent
CompactCells139To170.Box164.lambdaInterval	9bf15763fdce3e736f94eea05e0d4743bcd488379e0a571586d4fa5f4fedfcb1	equivalent
CompactCells139To170.Box164.compactCandidate_not_isWeightedPerimeterMinimizer	755ece5bb320913a87f02706f8ce0c23f07fd14b634f8b76349825ccebcb694ef	equivalent
CompactCells139To170.Box165.QInterval	4d292416cf671e10650928d606bf02fa167581d9de1ae63cf433e6cbb611ec59	equivalent
CompactCells139To170.Box165.lambdaInterval	0a20fa40b4603e2bbafb385c3b1c871938d88571e846d8ad53a68266c5849d0f	equivalent
CompactCells139To170.Box165.compactCandidate_not_isWeightedPerimeterMinimizer	1b395a1a0da198da1c9d29b0d24a0d62406227dcd9bb9bcb4619bc19be93e85f2	equivalent
CompactCells139To170.Box166.QInterval	08adce2617d58429e5f3a677d0499e15f0b038883fc3036f105dc04abfc60be5	equivalent
CompactCells139To170.Box166.lambdaInterval	99641535aa3b99610b480375b976757909b1cc5e2ea482f463dbed4508255cc8	equivalent
CompactCells139To170.Box166.compactCandidate_not_isWeightedPerimeterMinimizer	4d9e348715fe1241552bae5990fbb01a4d15353436d4c6c9bf131a0ae92ff625	equivalent
CompactCells139To170.Box167.QInterval	b3851bcf7cfbcf71375f4e271840c1cff4e959f7952f07001f8b596b3cddf1a8	equivalent
CompactCells139To170.Box167.lambdaInterval	28d1004ea52cb61b37530169ded081ce361696b6041117243313ed8b7ab77150	equivalent
CompactCells139To170.Box167.compactCandidate_not_isWeightedPerimeterMinimizer	d37728a35925d3db83e9b72db6c213fb512854171eeda4a6b7626c22d753acff	equivalent
CompactCells139To170.Box168.QInterval	2fb7b958e0ef583325bb9dd5b5d52a6fb770bcc4d9145c087f96fe029524578	equivalent
CompactCells139To170.Box168.lambdaInterval	be0b2d905bcea63c1eb3b1e063accbd53b5448acfa54e76be485f09db9218d21	equivalent
CompactCells139To170.Box168.compactCandidate_not_isWeightedPerimeterMinimizer	446b691b40a817af6c2dab9347632c21aa28595d1fe46cfc08d2583c50c53f80	equivalent
CompactCells139To170.Box169.QInterval	b644880ef549c7ad05fb3d14ef53ba35ad5728d7afe746c6c9293441991fae1e	equivalent
CompactCells139To170.Box169.lambdaInterval	142086ae5d45cb59637c66c8512a9f7c372d7e126bb005288a8769c3a7b96469	equivalent
CompactCells139To170.Box169.compactCandidate_not_isWeightedPerimeterMinimizer	e1e8d60116f3eaed082262a1681c72ae4de73c89fffaefca9e59f9adee7275e0	equivalent
CompactCells139To170.Box170.QInterval	97af062d9d011be9569336cf1c50d619d9edcf8358fcdbe8f0e1d4f07c562ee8	equivalent
CompactCells139To170.Box170.lambdaInterval	5eff4036e6437022f43eb05e532f6c328713f15923f2bfe43b944dc1c35976d2	equivalent
CompactCells139To170.Box170.compactCandidate_not_isWeightedPerimeterMinimizer	844f7dc793168913a6ee65e0ff62afe936831b8adddf2f346442893b219018	equivalent

Declaration	Source hash	Audit relation
CompactCells139To170.Boxes139To170_Candidate_not_isWeightedPerimeterMinimizer	94931ac5e5e6c85ad9a8f787727c31546ddf22bfc85fe062061f5b3fc7bdfef4	equivalent
CompactCells139To170.Boxes0To170_Candidate_not_isWeightedPerimeterMinimizer	190d0bd7e48a18fcfc9203cec84f4d67ae2e885027442d1acda7c15f1a7661d4	equivalent
CompactCells171To202.Box171.QInterval	abdf6daf4306e8dbe17760c37a0eb592c79f60105b6799e73e35322c41ddc9de	equivalent
CompactCells171To202.Box171.LambdaInterval	217966c5de3deb703c0cf6c405487487578727fe7b5e6f82d2d66acd884203f9	equivalent
CompactCells171To202.Box171.CompactCandidate_not_isWeightedPerimeterMinimizer	0cbde0d4f3aac1755d5dfdc5be35ab8015544d57b85e32e51fab2afd95c28c5a	equivalent
CompactCells171To202.Box172.QInterval	fd4a578d7f1cabd64b05e2d2511a704a25b4e589d5b02e302ea6db56b40ce80f	equivalent
CompactCells171To202.Box172.LambdaInterval	c07cc4c548f3016afcb820186c2d764f7c6db303fc882b3f96b83aeb54e1ca9c	equivalent
CompactCells171To202.Box172.CompactCandidate_not_isWeightedPerimeterMinimizer	03291cdf1e0e6c0f8fce3c3f085d9a23a774eff54e53eb99c25617ea0785004a	equivalent
CompactCells171To202.Box173.QInterval	ba076c5f73980fb15ae96c30560b3e19316cb306219ff3168bb11787094c2f9d	equivalent
CompactCells171To202.Box173.LambdaInterval	a82e3dd248e1c3c740ec79964ee31264258e5e60ddc6e6438160a0c20274917	equivalent
CompactCells171To202.Box173.CompactCandidate_not_isWeightedPerimeterMinimizer	3aa65a93bf48e9632e85c58f9a71273bb42eb562984c52c79d84474d1314235f	equivalent
CompactCells171To202.Box174.QInterval	b1e3d35036b2087650afa153158c250941a1cc0632ce802cca5f5542c6cf451e	equivalent
CompactCells171To202.Box174.LambdaInterval	58cc4fa8d0f9a464acd3fbbbb7ee8bc6d972fe0ed3b498d6b55ba3058f2573c5	equivalent
CompactCells171To202.Box174.CompactCandidate_not_isWeightedPerimeterMinimizer	ede88503652d4c079799da58072a5605dc326c044d5ded584a544500ccb78179	equivalent
CompactCells171To202.Box175.QInterval	e03bd14d3d3b319927801907ffb498890a5a50db06f169bd5dc91ea4af5bd24f	equivalent
CompactCells171To202.Box175.LambdaInterval	14e092a2c22c02b90ec28e0d4300c2ef11df1b5957640a2ddcf22c796ca9002	equivalent
CompactCells171To202.Box175.CompactCandidate_not_isWeightedPerimeterMinimizer	7cb966fe63090d6f45978e9f94ed8ca8d50d142d5bdb6f36acd32ad07d843360	equivalent
CompactCells171To202.Box176.QInterval	ed3ef9d714027b0c0a774a91ec5279f88ca491ff850900b84cb7d2497bc33001	equivalent
CompactCells171To202.Box176.LambdaInterval	bd6cd3e034cc4e3ca9236a4f34b0de056a01d7cc37d1e225449ce4a90c6fd075	equivalent
CompactCells171To202.Box176.CompactCandidate_not_isWeightedPerimeterMinimizer	b690b3de394280e615037736b79a57cac8da0526c063393798989a0053fe145e	equivalent
CompactCells171To202.Box177.QInterval	d235f98a00234fa91ab3dfd4f63e39186cf534de600d018fc9ae13ff9179abeb	equivalent
CompactCells171To202.Box177.LambdaInterval	a5e86465e5677733e626983e271dde038d1bddde39c06794aa142cff5a71f397	equivalent

Declaration	Source hash	Audit relation
CompactCells171To202.Box177.compactCandidate_not_isWeightedPerimeterMinimizer	a42a1edba64fef88853b4846342f40d75f096fa4d417924903f6088e5bb05b5e	equivalent
CompactCells171To202.Box178.QInterval	6b6d04c46d571cecc5ad752bc3632f39953a185fcc4b9370633af4b73b7ee1c0	equivalent
CompactCells171To202.Box178.lambdaInterval	72190f04fd89f72a5433c5ae7f9982e6eed41c565984f97db3a46207e071935a	equivalent
CompactCells171To202.Box178.compactCandidate_not_isWeightedPerimeterMinimizer	5e9ebe0bac33326de2d90205f0af66bc72fee9efaef56df7c5ec5f74360001a5	equivalent
CompactCells171To202.Box179.QInterval	264596259ab6bd21a5e62e25445100335686ebf38b1e977e563563dc2632af1a	equivalent
CompactCells171To202.Box179.lambdaInterval	3eececee7fa457c375192064beebd3317618ab90965428873d56e2023da8d228	equivalent
CompactCells171To202.Box179.compactCandidate_not_isWeightedPerimeterMinimizer	8e60636c5dae2c981eb0adf00d63672faa31538d7c0b21d1a9b1c0ca8a317a	equivalent
CompactCells171To202.Box180.QInterval	a8fc77d9750168ce6e73b939748540545112d036ee58019b32083de1d783261d	equivalent
CompactCells171To202.Box180.lambdaInterval	008d0581bce7391c71c36443fee54b14c1be27fd14c23a7162088c511e54e79c	equivalent
CompactCells171To202.Box180.compactCandidate_not_isWeightedPerimeterMinimizer	dadc00e991342874c64d9956764ef5a83e46b911764572b5d8d91cfd6859ac2d	equivalent
CompactCells171To202.Box181.QInterval	6cf7a752aebf4cbf7acba08c6a87979f4008e6f6aa1039949b187c69a2b06820	equivalent
CompactCells171To202.Box181.lambdaInterval	838ad6c948757b9dd33f2028b94db61f39c50ae1333e1e4bdf24d36cc6e0caf0	equivalent
CompactCells171To202.Box181.compactCandidate_not_isWeightedPerimeterMinimizer	aa866a9c9336e83fb2e9d0a90673786bf999436636f2e1d58783956449c1c8d9	equivalent
CompactCells171To202.Box182.QInterval	1f247d2e39243386bd4599e7b41903c391c0c9301ddc7dbd578232ea94b8e3e7	equivalent
CompactCells171To202.Box182.lambdaInterval	10208da5f1b245b045d97af7e9859217fa7cf65c9afadfe82b42a6fdca0088ef	equivalent
CompactCells171To202.Box182.compactCandidate_not_isWeightedPerimeterMinimizer	69e3721053dfdd26f9512307d7c101213b5db5b5df05cd6f0a85e44a8950664f	equivalent
CompactCells171To202.Box183.QInterval	085a310a7d7110fd8331a5f805af4f32e66a6b02a6171e13345a8e38a7736844	equivalent
CompactCells171To202.Box183.lambdaInterval	db7fc94f2dcf2c8b0148156ada416f30fd8d036fcb123a866f25df5ac0366a	equivalent
CompactCells171To202.Box183.compactCandidate_not_isWeightedPerimeterMinimizer	807af68267ec0f798e12ecbe4b83314af57ecaba020c24843745b2d90311833e	equivalent
CompactCells171To202.Box184.QInterval	39a4040539e3fe2c3b94557d772d6da609c432459caf54d9f359132e7811aed4	equivalent
CompactCells171To202.Box184.lambdaInterval	99e70ae3816174022ba0dc25de6daadce5bc05fba6d0d07189b642aa0cde04b8	equivalent
CompactCells171To202.Box184.compactCandidate_not_isWeightedPerimeterMinimizer	cae5da5624fe76f0e1e4ae44f39f60de33e89610b6f5952e85d0f520bdbc1735	equivalent

Declaration	Source hash	Audit relation
CompactCells171To202.Box185.QInterval	c61946891f4031ce35c75bbdf5ed1ac0c0bc6b501db9311949e90527fcbfa25c	equivalent
CompactCells171To202.Box185.lambdaiInterval	8003b3fd01163d323a525d4c83a23d890f8003e7d20aad6e19c7e801b2a23982	equivalent
CompactCells171To202.Box185.compactCandidate_not_isWeightedPerimeterMinimizer	9af3defe89cad69cce53f8c8064ae51b8287ebf983c8de8d805ca5ca3c027ce3	equivalent
CompactCells171To202.Box186.QInterval	67b9676306452f8939c73cae6ce1149691e85df949bea4bf8309d3f82a035164	equivalent
CompactCells171To202.Box186.lambdaiInterval	94740ccddceb632c5aca963996705aa178f15b92db2f548bcf1f5927a3c97d27	equivalent
CompactCells171To202.Box186.compactCandidate_not_isWeightedPerimeterMinimizer	e5e7e7c6fb7576d8e8b4249eebf4ecb6c3fdfed98b9a201dedcf7712e07d4acc	equivalent
CompactCells171To202.Box187.QInterval	c25c444edfd7fb4c348e30f65e0b0318b4e93fc62b831d9851469c8981e17116	equivalent
CompactCells171To202.Box187.lambdaiInterval	93a35e5d14b40364a92b8d7205260f65faaf3c fb8b162a3bf47d1bc24491e305	equivalent
CompactCells171To202.Box187.compactCandidate_not_isWeightedPerimeterMinimizer	10e9f96cb85c6096b0bfe39f19fed3f575b61bbcc4f03dd12e24d7431746ed73	equivalent
CompactCells171To202.Box188.QInterval	f0f9b2fc73402e05a82b39c0bed3fcacf6e924f5c51f715ffaf3e2e3c7e5d57eb	equivalent
CompactCells171To202.Box188.lambdaiInterval	59a9cf81c922fa901edf898650074b708a86f353ead5a8b072ac44f24b461d95	equivalent
CompactCells171To202.Box188.compactCandidate_not_isWeightedPerimeterMinimizer	c60e843d1aea3b60c7f59dae03d0dd61b57cd9b5c6b0070eab0e886a0fe8c46f	equivalent
CompactCells171To202.Box189.QInterval	8b1d62053ea29d11dbe54564d794ef2771892c4f478fee53c580b16d0e8b7082	equivalent
CompactCells171To202.Box189.lambdaiInterval	2a768b5e287c4213d8e70a6c31a11f2557e3d90171dddb8c883de1663ce1ac98	equivalent
CompactCells171To202.Box189.compactCandidate_not_isWeightedPerimeterMinimizer	88b2e442fa1c500edb28cbe210f2f3b69eb2016180f7acd133f22a55c37069d3	equivalent
CompactCells171To202.Box190.QInterval	c53cda5b8476a866e9f1c64851effae329eada4a79072d33904644a1bad98b2b	equivalent
CompactCells171To202.Box190.lambdaiInterval	505125c9057a286c782ff34c63fbbd5dbf837db84e76d223fe67381005eef7e5	equivalent
CompactCells171To202.Box190.compactCandidate_not_isWeightedPerimeterMinimizer	023acb5a8d0e61fa3bfaa96a5705e8d112dc49ff8d1c8d0525c47aea424c751d	equivalent
CompactCells171To202.Box191.QInterval	a2c4988a019be040bda3518cd09c4f4eee46bd2f5967f304dd051c712d860150	equivalent
CompactCells171To202.Box191.lambdaiInterval	04a26479b2ac5fe7025e36908c3280a2e48683f0ccd805bbf5a782d3389f35a7	equivalent
CompactCells171To202.Box191.compactCandidate_not_isWeightedPerimeterMinimizer	933b4af793b5aec123f596edffa833f8138288a88e685e0a90d26c3bd5305f39	equivalent
CompactCells171To202.Box192.QInterval	b8e9b5b74f9ad0249bdca22bc978ac54c350db898e84f784278354e704ace7ef	equivalent
CompactCells171To202.Box192.lambdaiInterval	24b124e12453ed81bec065bef2c7ea90877d33b29ec8e42962e332ef37b0b695	equivalent

Declaration	Source hash	Audit relation
CompactCells171To202.Box192.compactCandidate_not_isWeightedPerimeterMinimizer	eae4044e5b388bbb3c4a16053ce9e0b0ba6947b0e3d9f9a7f9b908affe2fd625	equivalent
CompactCells171To202.Box193.QInterval	425c2c5c08b987117fbd0ac774e3d2a1c29faab7d63327821b0b9ce055fc4f14	equivalent
CompactCells171To202.Box193.lambdaInterval	39cf5cb140fa4919ea57e4f89901e3d5b6035c5adc fadf194b27d2114125c387	equivalent
CompactCells171To202.Box193.compactCandidate_not_isWeightedPerimeterMinimizer	80bb227ae81ae11a2b308cd7593cc43f3db4f21de1a8fc1e2db86e2ccfaad4f7	equivalent
CompactCells171To202.Box194.QInterval	acf232527ff0bdb53abbafbf06ddb705d4dd00572df8095939fa2c25baf33bff	equivalent
CompactCells171To202.Box194.lambdaInterval	fb0d24feb1d7e59b54b16955d1f082e7c5329c4b5f09daaa349fccd67192acb9	equivalent
CompactCells171To202.Box194.compactCandidate_not_isWeightedPerimeterMinimizer	cdd8ac3f7723566b6c756e74d789e48b7950d80c93593723bc84f5c3ddd4112d	equivalent
CompactCells171To202.Box195.QInterval	40e90156b9c97501124e8c3fc58a20e47568787db12359259504410d9bf31cbb	equivalent
CompactCells171To202.Box195.lambdaInterval	25cbbee3a0adbc79b1319836f3c22ea57de8d1ef4be6a550dcff8a961f5aa80	equivalent
CompactCells171To202.Box195.compactCandidate_not_isWeightedPerimeterMinimizer	1f830e67807d5ffa4844c81d68e23a1ebd2b47b7c49ca7c722e864db3c8223bd	equivalent
CompactCells171To202.Box196.QInterval	ba4bf33336a50a7c30b17210de4b5aee1d9620df287231db640fbdf59fd54b64	equivalent
CompactCells171To202.Box196.lambdaInterval	c606d462c9254303385a3ffa06819ea2adc8592f5b05869a4f16c2891e4f811a	equivalent
CompactCells171To202.Box196.compactCandidate_not_isWeightedPerimeterMinimizer	ff5dfcc41395f2b361c6be59812cde460c539cb371fc91e1d58e968b183f94f	equivalent
CompactCells171To202.Box197.QInterval	930da11cd7622dcca87fd380d700ce8668b4b945d69840b1f15f647da3a73bb1	equivalent
CompactCells171To202.Box197.lambdaInterval	295a6aa735928f2bdf225cbc447381aa8e3ea6c30d5b1a11517f84bef960357d	equivalent
CompactCells171To202.Box197.compactCandidate_not_isWeightedPerimeterMinimizer	1cd5cc0a831220fc2b04334145bb93037d468e809da3f3dd9d1526859f2d9067	equivalent
CompactCells171To202.Box198.QInterval	8127a70294999d405508e22904029eee35624c47fcd37db247095d7670fdbdfc	equivalent
CompactCells171To202.Box198.lambdaInterval	bb51166cbc50731c0acb59b47a228d8ce9f917c376ce7169b9c850ab4d79e47f	equivalent
CompactCells171To202.Box198.compactCandidate_not_isWeightedPerimeterMinimizer	40c98b4decf579087a745315d56fa0d4014fd7e4b5dd20a8b079ee805d8dbf58	equivalent
CompactCells171To202.Box199.QInterval	418768080a3f40a4b4680a00826e30ebaaf291ba82e1bcbdf32a8ca70444394a	equivalent
CompactCells171To202.Box199.lambdaInterval	5b246aa92fa97705bb9ab39240679ee4edd5d414dbaf9ab5a33dd5f8ae38c6ef	equivalent
CompactCells171To202.Box199.compactCandidate_not_isWeightedPerimeterMinimizer	5945a229d91ec4be28fb1bbbf426dd811092327224ef706bc0b14b8459fc2e88	equivalent

Declaration	Source hash	Audit relation
CompactCells171To202.Box200.QInterval	032d490ecbae24a1c0a9e4080d86424295ba990bad9eec638409cdd5f6a245cf	equivalent
CompactCells171To202.Box200.lambdaiInterval	2963cca907ca00b971b902930aea931d10d4a61994286e607a3cfb5bc01c2ed5	equivalent
CompactCells171To202.Box200.compactCandidate_not_isWeightedPerimeterMinimizer	da54748806ba18ae70a68df1e2121cd77bfe390c57c407fc92e3c43dfaabc15	equivalent
CompactCells171To202.Box201.QInterval	73be23fab32915a0abe2f03364c4acb7d101c163dd704f3274ed37e73078e6a5	equivalent
CompactCells171To202.Box201.lambdaiInterval	6ee2e1933cf132d0879e52e2691a94596a37ee09ea2d72afdc66873297070f74	equivalent
CompactCells171To202.Box201.compactCandidate_not_isWeightedPerimeterMinimizer	c731aee8844d9eacd1cee6db7cbb38c479be29c13e440ca4897a4ae09c4ac05c	equivalent
CompactCells171To202.Box202.QInterval	df6398c5069205afbad5c5fcea06655913a80782ffd1db230cf29baac93fdd05	equivalent
CompactCells171To202.Box202.lambdaiInterval	2493c97a875d6de07045f8dc19b0d187ba4954da75a93c47d4eafad7c1dafb5	equivalent
CompactCells171To202.Box202.compactCandidate_not_isWeightedPerimeterMinimizer	936d79892f3aa5d16e12fe2c10a18ff6761742398b379097834c544f4d34382e	equivalent
CompactCells171To202.bboxes171To202_candidate_not_isWeightedPerimeterMinimizer	ab8d83a63d780118a9f83af1e615d1d98eafd659431145a94e93f24d17943b8e	equivalent
CompactCells171To202.bboxes0To202_candidate_not_isWeightedPerimeterMinimizer	58b72b25ef150d1ddf276a3bbc1bb8d5402633158bf87eb42c64040df4198456	equivalent
CompactCells203To234.Box203.QInterval	ddce544332912a4a72665523dd0e67ec931e451fdbf97948d2e1d6a6d02166cf	equivalent
CompactCells203To234.Box203.lambdaiInterval	3bb41c54e0da945a0d3deb41a89cc2248907aba7909628914132530a29efb387	equivalent
CompactCells203To234.Box203.compactCandidate_not_isWeightedPerimeterMinimizer	e9c05345c404f829a0a985a4195006df4853fd4a1738a86c442e066a1ea54fb6	equivalent
CompactCells203To234.Box204.QInterval	7c578b4282f58f02fd3c8d86633e7ec6ff4aa87368adfc3d2029e183d08e5d3f	equivalent
CompactCells203To234.Box204.lambdaiInterval	8f89ef3a0c832c3cd32c8c6f212a1bc8bcc a8371adcad56bbb882697b4f9661	equivalent
CompactCells203To234.Box204.compactCandidate_not_isWeightedPerimeterMinimizer	2f7632a5600fd4fe5cc8545085948b02279c8627e3412d681094af069aebd277	equivalent
CompactCells203To234.Box205.QInterval	330e501bf852c19b654189cd59d2099952d9d2294d259f444c6879fcc6d2addd	equivalent
CompactCells203To234.Box205.lambdaiInterval	6582c4462b63a2b8cf8b706c04a450c855f3118ddb066fb7372046e8db8d8e8f	equivalent
CompactCells203To234.Box205.compactCandidate_not_isWeightedPerimeterMinimizer	2b010fd04ba612070eaa0c04d94abeea23784d1f02230b86a3f663afb6125b89	equivalent
CompactCells203To234.Box206.QInterval	a709ba842fdd4301c2c00b3c453a726f82c3ecf9382b07f7787bc3d5822120ee	equivalent
CompactCells203To234.Box206.lambdaiInterval	43a3aa70dd09ffd4745add3e59eb85fbdc8108c4308a09de5e9a07cca8d7013b	equivalent

Declaration	Source hash	Audit relation
CompactCells203To234.Box206.compactCandidate_not_isWeightedPerimeterMinimizer	16a4152b0f1d332699872ed3206fc9cd303732d9292d0233c0c9424e15b48a77	equivalent
CompactCells203To234.Box207.QInterval	f9b330e9aefb977723b8359cc6e5f09250a2bbe0a3e4239520caed09ae748b6a	equivalent
CompactCells203To234.Box207.lambdaInterval	cd5bcfd40e16347c4b4ef5872856daec7a36a304a71a1a095bd8fe5869f19214	equivalent
CompactCells203To234.Box207.compactCandidate_not_isWeightedPerimeterMinimizer	2a84344e7a3172ab4a8d0783e426b91acd0c072e10e72662409d4de0671e60b2	equivalent
CompactCells203To234.Box208.QInterval	8047e71a12c464ec50a8282761697742a636f4d2f458d8de9b63e392ebafcab7	equivalent
CompactCells203To234.Box208.lambdaInterval	867ddccb9ce2952fce22acd8194ee904cb8c0eead3e369c55d2a2858bf872ff1	equivalent
CompactCells203To234.Box208.compactCandidate_not_isWeightedPerimeterMinimizer	a432ef3f4d5296fabe7a6bd2f8d5ca0130191fd511e662c11f52997194b62782	equivalent
CompactCells203To234.Box209.QInterval	758ff204adc5d5a17d3f91143100af3a89c4151e05a96e3d39bcf5b19866b555	equivalent
CompactCells203To234.Box209.lambdaInterval	b58e9462b1f78b36722069d02b0726e583cf6a002317e04f6c939e5627ed05b5	equivalent
CompactCells203To234.Box209.compactCandidate_not_isWeightedPerimeterMinimizer	66869b479e01f6b8662f6801cfef938005f610290aaf51b5afe4770959092686	equivalent
CompactCells203To234.Box210.QInterval	afe1c02438b93264dec215c8f818fa6a4c653e036ddcae9bd3a067f2dcc92e2b	equivalent
CompactCells203To234.Box210.lambdaInterval	532864466236d7e99dc003c52da0c107f1d6dacd9cd57c33079788ea890919eb	equivalent
CompactCells203To234.Box210.compactCandidate_not_isWeightedPerimeterMinimizer	e1b0fd2b2a36b6ab4ce5f893bcbe7879faf3421c8a81af9c3017bbb2ca2cde65	equivalent
CompactCells203To234.Box211.QInterval	d05a795e1a514839cba5cf21dbe0cae06308c74fdd2cd4d9f8c1439bfba0453	equivalent
CompactCells203To234.Box211.lambdaInterval	95561fbe2fc6af353ac7a0513a640249d74a8db8d498d4fb9bb6ad313f9130c2	equivalent
CompactCells203To234.Box211.compactCandidate_not_isWeightedPerimeterMinimizer	729daede3ff9959fbfc0ff64715f15735c65babb46b86edfa50f715558cecef	equivalent
CompactCells203To234.Box212.QInterval	4a419397fbe6a2ffb0709054c4809d7d6cab14a2d9bdd5185fa8693ccc8b138	equivalent
CompactCells203To234.Box212.lambdaInterval	cd83bd16fd0cc168b95c19b42876973bfda9a5b7f1c888807fd14b50e21ddf0f	equivalent
CompactCells203To234.Box212.compactCandidate_not_isWeightedPerimeterMinimizer	f20dcac18ef3274ffd797801c2b295f0b4691d44acceeb86b697fc57f4cd6945	equivalent
CompactCells203To234.Box213.QInterval	baf2b07d7595285e7211185627ca5f18e12a235fa20b36777977dcc2a7a63b44	equivalent
CompactCells203To234.Box213.lambdaInterval	4aa531dc5558fbcdf57c5981e5b742a25fc8f6ba36edaa7a64886b4be034604f	equivalent
CompactCells203To234.Box213.compactCandidate_not_isWeightedPerimeterMinimizer	1defb8f58dfdf29425a17e23b2bafbb7617db531a3b53703b3fd2814750a7426	equivalent

Declaration	Source hash	Audit relation
CompactCells203To234.Box214.QInterval	774eec7e8f4a4d81a8f565112278ce172697edf351d9ef30923a13fc1652af55	equivalent
CompactCells203To234.Box214.lambdaInterval	9f368b9da6428cab01fb026393917c00d4aec2304dfc9b3c4cb843cb3b3c7e2d	equivalent
CompactCells203To234.Box214.compactCandidate_not_isWeightedPerimeterMinimizer	a2796df42d8e34a6dbc1568e7bbe3d42f9fa30afdd0486ddbbf875f2e21fe2f4	equivalent
CompactCells203To234.Box215.QInterval	8ead679749f38e7e9144a6be4b7a341a7da4a6f22d9a49122f72b092f29761ea	equivalent
CompactCells203To234.Box215.lambdaInterval	22f4bbf56e7a15171cd473ffaba18ef957c6841fdff74691489585429674c524	equivalent
CompactCells203To234.Box215.compactCandidate_not_isWeightedPerimeterMinimizer	45b69f2d062467a4ec04ddc5506549eaaa738718b5d754d5582d3146e39af9fa	equivalent
CompactCells203To234.Box216.QInterval	0aa1742c71029e51b09dae9c320e189130ffff49603bb3508c25d1070a5879346	equivalent
CompactCells203To234.Box216.lambdaInterval	bdf17f96c203f2512cf543bb7e31e73df574fef8e93e245fc342dba7d1a35cbf	equivalent
CompactCells203To234.Box216.compactCandidate_not_isWeightedPerimeterMinimizer	3044021ddb671cb6459abdea6d9d36a9b9631341d36cc377d9e7d7eb06257473	equivalent
CompactCells203To234.Box217.QInterval	52b3baeee032135eebdc36e759d221bacde97facfc38c3f39c1ac42f2e1819cc	equivalent
CompactCells203To234.Box217.lambdaInterval	803c7d86a1794c3863572efb04b3644f7eb106a389e3843a2f9b64899f515f83	equivalent
CompactCells203To234.Box217.compactCandidate_not_isWeightedPerimeterMinimizer	3aec8461e134361fbbd684d60e50b66fba1d334c5795cbcb8bad2648cb026308	equivalent
CompactCells203To234.Box218.QInterval	78bb2d4525006b10599ce057c2ac0bd54aae981e9be3e3618257afb4187fdb57	equivalent
CompactCells203To234.Box218.lambdaInterval	c12b6b66767949fce840dffeb6b1dbdf49832ecdacce92949fe5094337e59e18	equivalent
CompactCells203To234.Box218.compactCandidate_not_isWeightedPerimeterMinimizer	d470b7ab5cb97c74358acd8805c3c9bb8f50bb2a362bd57fababb80718041849	equivalent
CompactCells203To234.Box219.QInterval	952e0fc2495aa973f923822625fc63c265cc35abcf3aadb952709a87ad935478	equivalent
CompactCells203To234.Box219.lambdaInterval	3233bcccd9a3336c169e852700990bb34d84e871c473af14b5758c9e7c4f98ea	equivalent
CompactCells203To234.Box219.compactCandidate_not_isWeightedPerimeterMinimizer	814499edcf3567c879b3db2ccfd4a28d08006a01dc8f8e4d37a8f825393009db	equivalent
CompactCells203To234.Box220.QInterval	af1a2d011ff13c4f83af935286b5bf477d8df6c2fc45d44cc45c4d233c7ba217	equivalent
CompactCells203To234.Box220.lambdaInterval	4ba8b5e42e46c630f8477322af5d6cbef33733d3c059c7a3ba3da4d1edf2a4f7	equivalent
CompactCells203To234.Box220.compactCandidate_not_isWeightedPerimeterMinimizer	647a63ce34f8c68be69be0d0952252813712e5478522bb79c3458d36d850d5ac	equivalent
CompactCells203To234.Box221.QInterval	f9420771d2c588cd1aeb798644712dcd09839ea67f583994f1b4d6a2da4b7b67	equivalent
CompactCells203To234.Box221.lambdaInterval	a53daf3a6f19ab85f615e2f31b2a6dd12cb62d71ef18708e16cc7a6379fc2803	equivalent

Declaration	Source hash	Audit relation
CompactCells203To234.Box221.compactCandidate_not_isWeightedPerimeterMinimizer	67de096a5aaf8726f98d9b094dd39f58fcd38d58701cd2ca8167547534401804	equivalent
CompactCells203To234.Box222.QInterval	9e8a51dbde1a2a07175d8076d58625d5e81ca15566bec8830f36dfb28d490564	equivalent
CompactCells203To234.Box222.lambdaInterval	48a16dd5f7a52a03de9c5c498bfae5823a4a95685170e45799e8d2dd28dab224	equivalent
CompactCells203To234.Box222.compactCandidate_not_isWeightedPerimeterMinimizer	b48d7d5727430688a9b8008450dde62a15fde577130ddc88d13d1d712dc16a5c	equivalent
CompactCells203To234.Box223.QInterval	065b5f7998275f05d24c0031e2bf015b87dae985c8bdaf8a135a93a89c6699c6	equivalent
CompactCells203To234.Box223.lambdaInterval	ec51b82778b9b16f09f86e22f3efd79e444f2716745b5a96c7b9ec6e87ff0d6e	equivalent
CompactCells203To234.Box223.compactCandidate_not_isWeightedPerimeterMinimizer	2252fc71783cffe651565bc937e1b17a935396876feb1a72f5d5782202f13b45	equivalent
CompactCells203To234.Box224.QInterval	b00d3d719c67ac3a3f6aa9940433e19ae2bdd8181c66b57e2e8c6c13c298af1f	equivalent
CompactCells203To234.Box224.lambdaInterval	65b85a95609f931eaea0b503edaff59c9c45c3ddccf05a526c03740081e86676	equivalent
CompactCells203To234.Box224.compactCandidate_not_isWeightedPerimeterMinimizer	1409a792ee6c19892fed61eca6e55121e7dc62d1947b711fb2032efe52c67336	equivalent
CompactCells203To234.Box225.QInterval	9a43d0e4a5769785965b4d8305373ead2cfd4a2560ac18ae234db00c70f43e5a	equivalent
CompactCells203To234.Box225.lambdaInterval	9c35eaf6df9287b4c483d0250c6cdfc9a5c9594755ff8f419c72eca351932e75	equivalent
CompactCells203To234.Box225.compactCandidate_not_isWeightedPerimeterMinimizer	6d80c7ce235fca76c0d76f822aaf0e30be1d3155e0b96c5eb25cc133c9cb9fdc	equivalent
CompactCells203To234.Box226.QInterval	0c8f738a62d609f054eb328b000b86ee4808443f29ac03884ebf8291f8c10010	equivalent
CompactCells203To234.Box226.lambdaInterval	5d507edef3254a082dacd7eeb17ea6129225477881f4996b4f47db1ba9ca1e42	equivalent
CompactCells203To234.Box226.compactCandidate_not_isWeightedPerimeterMinimizer	c42343026c45ad2a567ea25ac14e1e6df220a7c369382fc837ffebc9399605aa	equivalent
CompactCells203To234.Box227.QInterval	0c1c2b4573c87567c9c423563bdd12f452efa00ad5d588bdb8709590a4327b9f	equivalent
CompactCells203To234.Box227.lambdaInterval	9615b00f1c44f64167866087f0afc1e4ecc001d1a6e075114619087469ffea38	equivalent
CompactCells203To234.Box227.compactCandidate_not_isWeightedPerimeterMinimizer	03c41543a8ca73c3960ca1deeddd103c4ee0459b24703a530c54a997ced5df1	equivalent
CompactCells203To234.Box228.QInterval	a51646b5b9369a9b250fe401b368ad230e1709860e1cdc817a2dad382b05e1cf	equivalent
CompactCells203To234.Box228.lambdaInterval	eee27ed63870ce6b45f5424884324a2ef5a71aa2f06d15e86b5753c9a70ca555	equivalent
CompactCells203To234.Box228.compactCandidate_not_isWeightedPerimeterMinimizer	f0a45aad90e60e129c9217ccab6910665b0a74bc9004684004981bffc82d19	equivalent

Declaration	Source hash	Audit relation
CompactCells203To234.Box229.QInterval	342b809eb2073412430655910d96550d18193f614309ca807f59d09c0f90f368	equivalent
CompactCells203To234.Box229.lambdaInterval	8b41d67e5f4e55610f9b1d762cdc60812ff4abe85e5062693305fd3c8b7136f4	equivalent
CompactCells203To234.Box229.compactCandidate_not_isWeightedPerimeterMinimizer	8f4c541980c4348bac1ee24a17e2986fdf9368b6d1da9ee5d5dbcfb7249c0b19	equivalent
CompactCells203To234.Box230.QInterval	440e3be4353fa5e8384a8dda80718922d8a4484e64704e04a14fb19052016956	equivalent
CompactCells203To234.Box230.lambdaInterval	556389d01ee71412a68b0b88e42c5b12db30619dcda3f1cd2d1dd2eed75bcfc2	equivalent
CompactCells203To234.Box230.compactCandidate_not_isWeightedPerimeterMinimizer	e7c1bc78e4858b21032c371020bb3170d78b0272c189881e044c1ee8496a67c0	equivalent
CompactCells203To234.Box231.QInterval	d461635bbef5cc35adb9da9f4161aa2230301ea118346a9baaad32e9edeff3c9	equivalent
CompactCells203To234.Box231.lambdaInterval	e38d8ede96612b86c7807394cbbf2c88009b36b3f937d1ab34aced40217acb1c	equivalent
CompactCells203To234.Box231.compactCandidate_not_isWeightedPerimeterMinimizer	2968fd9f60c188264fae7f2a3ccdf9094948da02d96e048f46ef430c9d026b29	equivalent
CompactCells203To234.Box232.QInterval	88a23288993f9403d32e5fc8df8a620aefc72f0913ba20753aecc30a299a77c	equivalent
CompactCells203To234.Box232.lambdaInterval	a7d4c329474cff6fadd7d147df6f1b90f5ab986c480566d4f1a1bc48b15a2281	equivalent
CompactCells203To234.Box232.compactCandidate_not_isWeightedPerimeterMinimizer	1d81d5985986079f6a2ddac5ae9d9897cc74454dd17232a9e084292282c168c5	equivalent
CompactCells203To234.Box233.QInterval	568dd7ffb575fa3fb046aa27f225f5607083d12ae5d50f2ddb91538088572581	equivalent
CompactCells203To234.Box233.lambdaInterval	778edc65040c82eb1975b6c874904f9975f71ffebf7ae9e557eabe197631b64f	equivalent
CompactCells203To234.Box233.compactCandidate_not_isWeightedPerimeterMinimizer	858d71e58983a3b73200663656c42c53cd55e252b98c7a1bc4c28518980c987d	equivalent
CompactCells203To234.Box234.QInterval	b24b5922cdbbe0d09d2a24dab8229698c20953c37286492cdbfd4ec395b41eb8	equivalent
CompactCells203To234.Box234.lambdaInterval	57c93fbbcf786964ac5676abb82c32fb8ea4a744d88fe9bbff1534b827a20cb	equivalent
CompactCells203To234.Box234.compactCandidate_not_isWeightedPerimeterMinimizer	6a942bef8d64c00ef7d6fdacc3f8041980f017fe235a00277c27c3075fa5db8a	equivalent
CompactCells203To234.bboxes203To234_candidate_not_isWeightedPerimeterMinimizer	b16b14c19b511f5808a209c1d2d8262bf301e9fc16ad48af8408347d2ac82315	equivalent
CompactCells203To234.bboxes0To234_candidate_not_isWeightedPerimeterMinimizer	2a4333f6da31eb87bd878ee83d151871a4f7be9ec0959079544fc17aac891bf2	equivalent
CompactCells235To266.Box235.QInterval	c8f0e458527f0057dada886e1ae5cc848dd39d75ddba5ab0072173ff5207367	equivalent
CompactCells235To266.Box235.lambdaInterval	cf31069303d2679e0c6d8e942faca147e3047a2fa08c67d40fea6660eb3aa7e1	equivalent

Declaration	Source hash	Audit relation
CompactCells235To266.Box235.compactCandidate_not_isWeightedPerimeterMinimizer	963badaabe151307d63e8ca8c6ebabc6cb5c093d0baecd5f0d77640606e53933	equivalent
CompactCells235To266.Box236.QInterval	e6acb588ecd6396e1e4fdcd7dedb11ea521e9a93d3850f3c82a193a85e606bba	equivalent
CompactCells235To266.Box236.lambdaInterval	205bb40e563fe0358fb904d92abaaf40815288afc44a6ac9551b3850a0621a91	equivalent
CompactCells235To266.Box236.compactCandidate_not_isWeightedPerimeterMinimizer	8f98e0eda6ab9f08ef42bc538bfd7719adfb98c75cd08c14d5b2deb53939486	equivalent
CompactCells235To266.Box237.QInterval	3ee23102b7c955ccc8b3c99e70ee404b8195b1866381c34a6a3bdbb37defc588	equivalent
CompactCells235To266.Box237.lambdaInterval	434c9830956f293ec25872f1f5d4faf8e2b113500f81dae14901bc6636ed9861	equivalent
CompactCells235To266.Box237.compactCandidate_not_isWeightedPerimeterMinimizer	ca605fa50545608e508438922f76123c8035f2dabd14f45f877fe021f69f718c	equivalent
CompactCells235To266.Box238.QInterval	fd1e212adc0faa25cff6bc15572602aaf0085f5ad56c129e7657b233479b79e5	equivalent
CompactCells235To266.Box238.lambdaInterval	ad52069a7bd60df6c642f60bd266b1475b36f329e872821c044ba53d70f04df8	equivalent
CompactCells235To266.Box238.compactCandidate_not_isWeightedPerimeterMinimizer	689df6f7ee64ead3a7ebe84c60b627174330ce46381627e55cfdce9783f54653	equivalent
CompactCells235To266.Box239.QInterval	ab92d8c753b9eb23549c78fc6c51b686aaf261dbb36c558f595af85ce874d593	equivalent
CompactCells235To266.Box239.lambdaInterval	19aad87e907d1ba348bbf749fb0cfe1bb0cc80ecd97a192a6f10d9ef397830bc	equivalent
CompactCells235To266.Box239.compactCandidate_not_isWeightedPerimeterMinimizer	fe2b995c1db709bb3de7d8fc5663151330492a7fec1d0e4e9a8f16c6d5e85622	equivalent
CompactCells235To266.Box240.QInterval	f2368bbd55af4d84a366cbeea3f9a5e380d1901ae4868c2e715dd84f3333e273	equivalent
CompactCells235To266.Box240.lambdaInterval	48fea21c6001bd99af6b1653cf2937dc40fadab34af735d2ad7c4e1d4557c5c7	equivalent
CompactCells235To266.Box240.compactCandidate_not_isWeightedPerimeterMinimizer	0f2258d05b22ce75c927b4dc4db283478a7c8e6d92df3f8114587ee806641d0b	equivalent
CompactCells235To266.Box241.QInterval	456847d7f1a3a4a6e4020d8217421fa33b048bf4059e1236217fc068e6d5b18c	equivalent
CompactCells235To266.Box241.lambdaInterval	ec14b4f3ea3ee994c94738fa51dfdcfb079f99783280a1910215756cef5d66d7	equivalent
CompactCells235To266.Box241.compactCandidate_not_isWeightedPerimeterMinimizer	a3b0cd79f9611daccf8f34431f9c440b0bec6b73b248db1c9a0be2b6cc1b8b84	equivalent
CompactCells235To266.Box242.QInterval	f8825c94abdfd2e8f81c0681495ba7538e3d3a302faeddae26722318e80eaca8	equivalent
CompactCells235To266.Box242.lambdaInterval	39099eb120387dd35a8ebdbadf5a6a396f06a688439181dd0d4533b014011b63	equivalent
CompactCells235To266.Box242.compactCandidate_not_isWeightedPerimeterMinimizer	04d3d58ba3475bf629aac4cb7ec187820e8e8c9b921938221e3c758befe909c0	equivalent

Declaration	Source hash	Audit relation
CompactCells235To266.Box243.QInterval	33c9236b02098fd09aaed35e201533ad62642649b167c81c13bf067a1d789d4d	equivalent
CompactCells235To266.Box243.lambdaiInterval	b25f1384d6a694a2a0396980a46b2241c64f74f0b9cb502f1cfbb3fa6f576c5d	equivalent
CompactCells235To266.Box243.compactCandidate_not_isWeightedPerimeterMinimizer	60d60dc0df599dc3f0fb2f0e3efc988bfd6891159afd0f7e526bcce1521ecd11	equivalent
CompactCells235To266.Box244.QInterval	050829c65de050eb03a6ec85b0727860e453db26fcbba87fb429f0ae3a63b255	equivalent
CompactCells235To266.Box244.lambdaiInterval	ab7c379ee21aa96e8b7e9b44c8f22bf64544868d38a7a4d0f68ab83440a209dd	equivalent
CompactCells235To266.Box244.compactCandidate_not_isWeightedPerimeterMinimizer	48d2fb8e44802852044cb484fb5c2db8992510af7d0a4488e21106f3812e6dd1	equivalent
CompactCells235To266.Box245.QInterval	776c87b0c4944a30856969db0c98d151b5fcd0a2a99f2881b791d5d316d9ea18	equivalent
CompactCells235To266.Box245.lambdaiInterval	ffcd6dc3c70d1b46ce08ba4dfbadbe79e1dbc3ea681495090db617ad9cbce4b5	equivalent
CompactCells235To266.Box245.compactCandidate_not_isWeightedPerimeterMinimizer	47918f8bb5cf372084be4be2cad8a43d9596d118f773a3dc91b5870f0d6aaa14	equivalent
CompactCells235To266.Box246.QInterval	3b3d4946216c8e5c09f7bbaba259994b60caea1aa8e9159ea0e226571860b8094	equivalent
CompactCells235To266.Box246.lambdaiInterval	2f72ec96053d14e45695902fb84a1ee41758abf94546e1fb53666178c9f5bc25	equivalent
CompactCells235To266.Box246.compactCandidate_not_isWeightedPerimeterMinimizer	f64b977dad1505f9fe6518fc923dc9f820166d2db270aeeb8f8f2fe27f28c84e	equivalent
CompactCells235To266.Box247.QInterval	ab9ea36c2fa411ab85aba040ee04caa95616cace8368b3f2765b9640f073aa96	equivalent
CompactCells235To266.Box247.lambdaiInterval	d329aa6edf28cf487d556aa684d40be967aed4b74dece854e3e817b9ff8ddcf6	equivalent
CompactCells235To266.Box247.compactCandidate_not_isWeightedPerimeterMinimizer	afd4867661b8d6eee4ba937f532ae098d1aa3f3f746e8166add81ac22c9f393a	equivalent
CompactCells235To266.Box248.QInterval	cf8bd0ea71ffda1ff5f93b072f6c606c78242eece3153cff70569854eaea3166	equivalent
CompactCells235To266.Box248.lambdaiInterval	ec57521aed607461112ac74f47694f46bc89437b1e8a6413c42c05c74468db54	equivalent
CompactCells235To266.Box248.compactCandidate_not_isWeightedPerimeterMinimizer	408e4b078ef3f20785b85fa3ca46c4b272e5ac3b341bc042188a573544a97708	equivalent
CompactCells235To266.Box249.QInterval	36cf637dbaa9c6e4023d31fbd75328f6ec30d6e427528ad0603edeb247d9e75a	equivalent
CompactCells235To266.Box249.lambdaiInterval	b911a07ba969b65a54e42a9b37a02fdd4ba3b34ea5b223c3f37de3d0d13c0a46	equivalent
CompactCells235To266.Box249.compactCandidate_not_isWeightedPerimeterMinimizer	f6843cc9d8f056bef0e17488c38b1741d0aea6e5d7bc4998612ca6b7456de043	equivalent
CompactCells235To266.Box250.QInterval	04eb6c2e2a3d2690b706c4b7e0b531a3f9a48c35ac300cfeff065d149ebf417f	equivalent
CompactCells235To266.Box250.lambdaiInterval	f06ce2ae2820bafb0aeb37d3413a79aa9a8c4502dba8f64e9db1d559040ab16c	equivalent

Declaration	Source hash	Audit relation
CompactCells235To266.Box250.compactCandidate_not_isWeightedPerimeterMinimizer	29a9225aad35b433fdecdd24261209bab26d5173720ea1cf3369584bc137aae9c	equivalent
CompactCells235To266.Box251.QInterval	963e4d61f782271b93835c6d68109a19561be3cc4d7dee32344917cb9034cba6	equivalent
CompactCells235To266.Box251.lambdaInterval	2b8b22db3bc5bdc4cc68852e5d56115a254c5ea54cab16ff8c77cbbc6ae0cc81	equivalent
CompactCells235To266.Box251.compactCandidate_not_isWeightedPerimeterMinimizer	937043be68165fc254692d4c54e20a5306bb33d67956d2719755f1db147ecb3	equivalent
CompactCells235To266.Box252.QInterval	02d911263f16331eb4476da32774a78ca05cecedfdd00283d816e3b2514b554	equivalent
CompactCells235To266.Box252.lambdaInterval	38446238a1362cd9beac1945c7b6efaeaf573c6e12665c6829014df0982a5c72	equivalent
CompactCells235To266.Box252.compactCandidate_not_isWeightedPerimeterMinimizer	4690c2c7dea4929c4edfda1feb96f47c9fef63a4b5402744c9176c77de515e6	equivalent
CompactCells235To266.Box253.QInterval	34214e31b0dc2e6a3f16cfc256ec0ddcac535eb0047adfb99545fce5f7c49669	equivalent
CompactCells235To266.Box253.lambdaInterval	a248cec8784933488982830084cc2de054c4565b80f5682e0e581f334eec7735	equivalent
CompactCells235To266.Box253.compactCandidate_not_isWeightedPerimeterMinimizer	066f796b0a6c144db341655e4c695a5064b47b4f2e2a8a37cdc075baa4fe43ba	equivalent
CompactCells235To266.Box254.QInterval	cafae28de3c1dea42b550d8c2671c5b426c039a1c4407ec971735b03f8c55665	equivalent
CompactCells235To266.Box254.lambdaInterval	d4405eb07e12f124496d90411280cfd999eddcc23e7ad0d0dc7cac848f357aab5	equivalent
CompactCells235To266.Box254.compactCandidate_not_isWeightedPerimeterMinimizer	77b9f1ede1afb3534a73b621daf62c046034e1c2a9dc4245e04db2fe7f1634de	equivalent
CompactCells235To266.Box255.QInterval	3460f460c896c74317a25ba4ad2c1984620d2f298b905535b5fcb29c27f0b287	equivalent
CompactCells235To266.Box255.lambdaInterval	246cc866e43dcbcd09bc564bce8d39be246517fb893ff3733ffb51681a16387	equivalent
CompactCells235To266.Box255.compactCandidate_not_isWeightedPerimeterMinimizer	8967fbd729cdffee978370371907edf98ca46ea6ffd49e7091c813004a364e3	equivalent
CompactCells235To266.Box256.QInterval	62ad9a4dd419195d0bbeaf8da7a49e7715c5ee2f081d9ddf907270e111ea0378	equivalent
CompactCells235To266.Box256.lambdaInterval	91440e30cc62ad7c7ebd2d2ea8759f97c898c21181afe83c78a55cb1d98c2121	equivalent
CompactCells235To266.Box256.compactCandidate_not_isWeightedPerimeterMinimizer	d5a660076dea68a67d88f7757cfd50ce65fb2824f4d940684229fe9c20057b8	equivalent
CompactCells235To266.Box257.QInterval	cbfbdb8c209d592ab8bc4947e15572fbf6c21fb76fbb7691eedb3d724df122fa	equivalent
CompactCells235To266.Box257.lambdaInterval	88edf7bb5eaac340b25140ce77fb44e2756e8e36601642b491f54cf0cc556175	equivalent
CompactCells235To266.Box257.compactCandidate_not_isWeightedPerimeterMinimizer	984593caa9e9d53328dc8839cf7d987218e6e004d67438bc8ecb08e7d445656a	equivalent

Declaration	Source hash	Audit relation
CompactCells235To266.Box258.QInterval	a14ab39461dca75e08dc10adddbc34f306654072ac0d0bf646a8996d62f38918f	equivalent
CompactCells235To266.Box258.lambdaInterval	74f0c94cf035ebf9e75a9db94565239a1392252ce5f4bb778fe1d89951178b89	equivalent
CompactCells235To266.Box258.compactCandidate_not_isWeightedPerimeterMinimizer	253c46cdfdee3cef57c3aa120124c5dbb2f068f526c313d59f152c841c602b89	equivalent
CompactCells235To266.Box259.QInterval	7aaa195550a112f044c9b2aa0cc7416d8f3055a14a3e168143ed0439f8e8303d	equivalent
CompactCells235To266.Box259.lambdaInterval	62524bf0f420b4b77c582ac6704ff3a920f800cc0bfbc50ee2d2212126207bb	equivalent
CompactCells235To266.Box259.compactCandidate_not_isWeightedPerimeterMinimizer	3509846fb2893069f376328027b45ca57b1daf7d0db39992e33919d20f86f7ba	equivalent
CompactCells235To266.Box260.QInterval	07fc1dc1dcc80470f00a28c0a364b61c8a8ef7411e21c0e4fc5dea5e27b763aa	equivalent
CompactCells235To266.Box260.lambdaInterval	6bd8324a222b4f379cc42f4fd2110124ea7434b826d3865a27b13d4423a422c8	equivalent
CompactCells235To266.Box260.compactCandidate_not_isWeightedPerimeterMinimizer	e73c48896e63548d3c90b8359a57dc38fff45c95698bdacbf434d63fe627260	equivalent
CompactCells235To266.Box261.QInterval	1f5b6b0248c2f98f82bcbba326cac840e35c0e6490c9608e3b0cbbb8c5b57e74	equivalent
CompactCells235To266.Box261.lambdaInterval	689c61a9f25e38439267beaf7880cee2a733b9f2fa6d0a912e0d1058a7cebbd6	equivalent
CompactCells235To266.Box261.compactCandidate_not_isWeightedPerimeterMinimizer	e1b3b1997d9b52116495fc14a32cd0dc1910ecceb04ac6f2855b62f7242a148a	equivalent
CompactCells235To266.Box262.QInterval	66f156b99662e37b2a9f26c9ba811d62adaa34e90abd4ca93914e524fa428683	equivalent
CompactCells235To266.Box262.lambdaInterval	96b7920f1aad2f3fc3bfb1370ac80a217d6481b69560d7bd033b6b9d9fb2eacf	equivalent
CompactCells235To266.Box262.compactCandidate_not_isWeightedPerimeterMinimizer	b61e86c44888ca43a5d0d6553bcafe4466d2496b1aa836285a4e4384e5eb1375	equivalent
CompactCells235To266.Box263.QInterval	e10cabb796e60ee5b44cb0f7b94cdc898f35271ef38169675fbbe8976a22789f	equivalent
CompactCells235To266.Box263.lambdaInterval	14ce025f97e0f80fcd024bb7be928972bd74329524a77d03942eb9214d5a7e1a	equivalent
CompactCells235To266.Box263.compactCandidate_not_isWeightedPerimeterMinimizer	30ad3be735194944e9cd3c963c5becf37b560242b339a2fa0d63c395fc3de861	equivalent
CompactCells235To266.Box264.QInterval	f9a304198d9aaf6b1c472c831fe530552ce5a9256bc516584e4e90ea6fa01e33	equivalent
CompactCells235To266.Box264.lambdaInterval	b9f6e9c598b8f3931aeacdf11d68ebfc70cc41b5775e3684dea80f9504ec171f	equivalent
CompactCells235To266.Box264.compactCandidate_not_isWeightedPerimeterMinimizer	77727887b400c7eedf05e3ab0a83bcc49bf5fee0bc29243b77859b605e14cad0	equivalent
CompactCells235To266.Box265.QInterval	dc1b442fc58f932d4d6d8cf6f6c3429c545fdbaaeec19493a2b9555e0134298	equivalent
CompactCells235To266.Box265.lambdaInterval	23d6186a9bc4d0a842bb8565c2fa9399f960dec398a5fe3b491ae7cb0b259452	equivalent

Declaration	Source hash	Audit relation
CompactCells235To266.Box265.compactCandidate_not_isWeightedPerimeterMinimizer	0a560b24c9d9984d43ce5e09bd7964d7432d96a29c817224149e63d14a8997a6	equivalent
CompactCells235To266.Box266.QInterval	60d7a0de30313d19dbac697671a5e2aa1be1a7e35bcec2e1b9e0ba2840d88682	equivalent
CompactCells235To266.Box266.lambdaInterval	ecdda9cf690a2f6be0d71cad3f5c852544f782a0d309ea25aef807fc49a3c0d5	equivalent
CompactCells235To266.Box266.compactCandidate_not_isWeightedPerimeterMinimizer	7a595a97aab78f2be70a9b8ca234ab8d29791a7f9fdbeb1b8573b34931ab92e2	equivalent
CompactCells235To266.boxes235To266_candidate_not_isWeightedPerimeterMinimizer	72052237e751d3960bf550f49b68f0e6c2bb777b4a746d382ddadc77a213a3ce	equivalent
CompactCells235To266.boxes0To266_candidate_not_isWeightedPerimeterMinimizer	fb420fdf20c9ea9177f9b327a5cbfed2623f6dc31d2e170786825355c74000c9	equivalent
CompactCells267To298.Box267.QInterval	7f09bbfb20e9be6b754b17153ed084189233e6c9a6f3de2991f09af1ce83cc16	equivalent
CompactCells267To298.Box267.lambdaInterval	43a688415160f0f9a511472c53951d668c186333bbade5286436ea2de02b36f3	equivalent
CompactCells267To298.Box267.compactCandidate_not_isWeightedPerimeterMinimizer	04e60e6cf641a8b0d15a23606805622954f313bbf7682ebd18f0c62f591c9943	equivalent
CompactCells267To298.Box268.QInterval	9712f9dba2f62028d225f83597e10b2b901837ad7a81cf2e4073ad1922b8febd	equivalent
CompactCells267To298.Box268.lambdaInterval	3b06d2811cb059204e0e8172759eeec3ed8843c359e088d7513a8d8d431a8f43	equivalent
CompactCells267To298.Box268.compactCandidate_not_isWeightedPerimeterMinimizer	b95c6541ff6876c57adc07be9564400aca3ca1fa0c178d74ab2987212f1b227e	equivalent
CompactCells267To298.Box269.QInterval	5aa8d0df59b24a34e6214599c507f825e2c903b082e5a086661d87e053916e04	equivalent
CompactCells267To298.Box269.lambdaInterval	143454ddc1f39f8edbf26f6107827f80ad0a86f9326a998a33fbb61b27f90b53	equivalent
CompactCells267To298.Box269.compactCandidate_not_isWeightedPerimeterMinimizer	7500e9be8534ffac0b7ab256f10b735f57d59b35f5eab472d7e6c35733ca7d0e	equivalent
CompactCells267To298.Box270.QInterval	68b8d2704b021f79419c4fd3d0a1f0834e76a18c4c7b9f713dd60a2726347fc0	equivalent
CompactCells267To298.Box270.lambdaInterval	1b3b0849838b87e6d28ce14a376780b4a04d497d1bc7d8a98af4b4d9231fe293	equivalent
CompactCells267To298.Box270.compactCandidate_not_isWeightedPerimeterMinimizer	7244cd3942ecb0a3dd1c71a8b718e6c06b83f12cbd008457764fd1efefac55fa	equivalent
CompactCells267To298.Box271.QInterval	e83800beaa1fa5d2fba33bceabad995aa7f86dad568c01a1c74fdd868d009bae	equivalent
CompactCells267To298.Box271.lambdaInterval	f8de82ed4b95651b7426846c926d736f7f82bd6624ff58eed98fef8d7eb1b2e0	equivalent
CompactCells267To298.Box271.compactCandidate_not_isWeightedPerimeterMinimizer	b7de13f0d85954877a1adee67668d1f4ab0eba7e2815a67e549c6873ef3146da	equivalent
CompactCells267To298.Box272.QInterval	8c7caf6a72e0283fe4799e698dfc576d9c50b75155c8c0d26ad9d5c9ba32797e	equivalent

Declaration	Source hash	Audit relation
CompactCells267To298.Box272.lambdInterval	cdc1709ca9649731fbd47f887aa4c3e4c2c7ad24da8ef1364af8f72c7c15ec0e	equivalent
CompactCells267To298.Box272.compactCandidate_not_isWeightedPerimeterMinimizer	489c872f7481cc88f26970567d03ce77e36c3156d5efa4b6fd060e84b27f4972	equivalent
CompactCells267To298.Box273.QInterval	72d6721b3c00ee67d4e040a29960494919c126313a3c20e93fe2ce0eb4ca1d41	equivalent
CompactCells267To298.Box273.lambdInterval	5cd4f5e5d990cfdacc221ca6c90531709e92865bcac2ca8daacfb31799162e4f	equivalent
CompactCells267To298.Box273.compactCandidate_not_isWeightedPerimeterMinimizer	556068bdaf6218472995546d5c66db6c0d7b68ae9278bd5b757141403eba5345	equivalent
CompactCells267To298.Box274.QInterval	efc61faaa2ac46de6147413296b599a8fc74cd09e96f23c062cb8c3e3eb83372	equivalent
CompactCells267To298.Box274.lambdInterval	8c6d9619151aba6d08818b60e13aad259240105136d329a4188947439e3e7dd0	equivalent
CompactCells267To298.Box274.compactCandidate_not_isWeightedPerimeterMinimizer	4b7758f897978a1ec76f7acb1a565cc9087ce7d191c8932e10569394a816544e	equivalent
CompactCells267To298.Box275.QInterval	6d8283bd2090eff210b68b79122d36e763d03cb0e35905ada41c91fcacf6f2ce3	equivalent
CompactCells267To298.Box275.lambdInterval	17038b2e925788c291604774d10173f483d6274e36a3781c0065c2ee83289259	equivalent
CompactCells267To298.Box275.compactCandidate_not_isWeightedPerimeterMinimizer	b79bc3e5231577945702eea3fb0402de2daac421f202b8b78a3bdce963e128d7	equivalent
CompactCells267To298.Box276.QInterval	55a23a1f8758cd52e35bd80da99e007704fed75d08825c351aa2995c24fa74c4	equivalent
CompactCells267To298.Box276.lambdInterval	aa4cf09ce24cde5f85274fc920451061231d32756c45e6f46d0c857bbbb38323	equivalent
CompactCells267To298.Box276.compactCandidate_not_isWeightedPerimeterMinimizer	47f7434c6e8d120d3d61350806144024c21d4691c8372b69bfebbf2c374eae66	equivalent
CompactCells267To298.Box277.QInterval	9ca5f152c93a5a63ac17360955c908769ed4ccf90faa1ea2c72966845ee55404	equivalent
CompactCells267To298.Box277.lambdInterval	aa202b663066773b50275e6016e1c001e0a67c76b7deb77f4b01cc1a21e7b98f	equivalent
CompactCells267To298.Box277.compactCandidate_not_isWeightedPerimeterMinimizer	e332e958ef67ecb873e3cb32480b7927ff53211cfa5feb97c9d55e6436e0976e	equivalent
CompactCells267To298.Box278.QInterval	19e5cdba9253c8f88abb00e3bf1dcf11427a64caf49a14b3c9ace6b1d8401ac8	equivalent
CompactCells267To298.Box278.lambdInterval	2b4df54ca29036e6e248cadb01e5218c87b92ecc42d39303690177e6c3488e82	equivalent
CompactCells267To298.Box278.compactCandidate_not_isWeightedPerimeterMinimizer	abb6db2fba90fdffc1fb66939236d19a650fbcc3f88ad1bbd4d8c3dd81169246d	equivalent
CompactCells267To298.Box279.QInterval	4886d845046d0a72e812fcd21f0d2f9e2bfff6bb2251e3c43a45f8367090d84b5	equivalent
CompactCells267To298.Box279.lambdInterval	96cd393fe4b2a719255f80900faaf0e988bd50190e9c68c10c8d9a49daca829f	equivalent

Declaration	Source hash	Audit relation
CompactCells267To298.Box279.compactCandidate_not_isWeightedPerimeterMinimizer	7fed697dbdf7f7a71b7942ec1f9265f06b8990043eb6537b0b3ed2fe4d2f9efd	equivalent
CompactCells267To298.Box280.QInterval	a7e9a15279f93102cc0da7ef664c9abf526348548476dc869a1dcf2a74c16d58	equivalent
CompactCells267To298.Box280.lambdaInterval	78d53ad11ec05642026f38b372e2d5a0c8ff4c7bef34995b473c8f00c93934cb	equivalent
CompactCells267To298.Box280.compactCandidate_not_isWeightedPerimeterMinimizer	14d57910c3c248f208966b2a510d07bc778aff1fb2b0c196ff27fdc4e478c540	equivalent
CompactCells267To298.Box281.QInterval	2e87c9fe086e51d7eaab0143097a3a0552bf8395192c535fb730c05c2e6d6c42	equivalent
CompactCells267To298.Box281.lambdaInterval	11c79bb7c30e1fecbfc6213aa637b95786186050212df6dca0061033914871ae	equivalent
CompactCells267To298.Box281.compactCandidate_not_isWeightedPerimeterMinimizer	0ef84a28c2d67637e1421911f4b502d66a9865e2c17a4f9177218d59acf2f5a7	equivalent
CompactCells267To298.Box282.QInterval	ea7fc6241256c5b3eae060243019a6f68a2eaab268737eeb89194faae3605083	equivalent
CompactCells267To298.Box282.lambdaInterval	3d44dd72edc6deafeb8c660bfe31c1f2f00c5ce44ecdf2c716fd9e5c7d4e89cb	equivalent
CompactCells267To298.Box282.compactCandidate_not_isWeightedPerimeterMinimizer	b3c3fbdb4c81f0538ea35c6ca504769c17bf6e3499f75578ebb2780d33eb060e	equivalent
CompactCells267To298.Box283.QInterval	78d71b82a71168b6e7d6ae267787793e310979640c9920b56bb15e0ebd1de4d8	equivalent
CompactCells267To298.Box283.lambdaInterval	658e5bf66e2fb78fedc3af3c15b44c36c64d9489c4f52d824d80ffaebda399f0	equivalent
CompactCells267To298.Box283.compactCandidate_not_isWeightedPerimeterMinimizer	0fb81e1c469e98951ff0965ff01b247c2bfaea9debfdcf821a84bd70da76da2b	equivalent
CompactCells267To298.Box284.QInterval	c4060d58be7709d6f53a6508c2e83df2c429a02d13516900da43f6558261539e	equivalent
CompactCells267To298.Box284.lambdaInterval	d44dc2cdf338f6e5febda6e9d7e494d108e5d36bbd993f09ab0498794eded2eb	equivalent
CompactCells267To298.Box284.compactCandidate_not_isWeightedPerimeterMinimizer	4299ef6831a32dc6c8e2617631b89724561e5d15740153b3a15eeec6c65e3a7c	equivalent
CompactCells267To298.Box285.QInterval	e503386ea6918099fa1dec8921832a16fe6e36ab02162127d76a57787f0f26ca	equivalent
CompactCells267To298.Box285.lambdaInterval	bfcfc848b4f2bd0b7312818301d43792237f9f5d6231f6865ef64b37d36cdf35	equivalent
CompactCells267To298.Box285.compactCandidate_not_isWeightedPerimeterMinimizer	35020c603db4bafa3a610fcf958a9bb87850615086c914cd1d6cb47e965f48a0	equivalent
CompactCells267To298.Box286.QInterval	162cdf5287eef36a2602636e10a00ebce5efd9fca3128076c0cfb37b25209cf5	equivalent
CompactCells267To298.Box286.lambdaInterval	d42be64d0855eb24dc77b7b34eefef1457458e1ea96f0a9f0890c9707e8a182d	equivalent
CompactCells267To298.Box286.compactCandidate_not_isWeightedPerimeterMinimizer	b093bf799cbebf6e42850a894cae8f3977a21651864c077d1e8f0c89445a970b	equivalent

Declaration	Source hash	Audit relation
CompactCells267To298.Box287.QInterval	fa65e4c2a04682349e548e83572339931d996d83b98950a9453657cc74d06095	equivalent
CompactCells267To298.Box287.lambdaInterval	561c6b12bfff7bff5118cb8879bfb45ac48ca04d031b1cf978456a2a0943ea8bd	equivalent
CompactCells267To298.Box287.compactCandidate_not_isWeightedPerimeterMinimizer	22825096914b16d968c337a561d471459684fdeeb1e41b494e6d27e2fa77d070	equivalent
CompactCells267To298.Box288.QInterval	065ad8fb0b32bad24fa1f52af0071c82635eafb84ec7d0064e16f087e3678b87	equivalent
CompactCells267To298.Box288.lambdaInterval	c1155ae89e020b59351678347ca4405ccf83db807f099e98e5d5a2fc5e87fbd6	equivalent
CompactCells267To298.Box288.compactCandidate_not_isWeightedPerimeterMinimizer	e6b724dc8621b8a4698dc09420f9e0cda2433f4d94e44ca3773752e1b9d2abd6	equivalent
CompactCells267To298.Box289.QInterval	0125107f5901e5134de0fd7513e881e203bc8ac0fbb11350dce87cb4204bdacc	equivalent
CompactCells267To298.Box289.lambdaInterval	0a5721986f028c864d14d9dade4946590c5972a5fcb83e52d3f4173f46cf4f86	equivalent
CompactCells267To298.Box289.compactCandidate_not_isWeightedPerimeterMinimizer	ad1c7e5f7e06558740ce22610827a575ff7c78910b2679e5c4be7fa67d5f07f8	equivalent
CompactCells267To298.Box290.QInterval	c54bd2ea22b166eba7ce64be6a6b375b33d88d5df3f2d67e14fd5d13821583b7	equivalent
CompactCells267To298.Box290.lambdaInterval	03f13342fb2f93a84b02cf50ae1554ec61ae601b6f86843e325898b54c3b8561	equivalent
CompactCells267To298.Box290.compactCandidate_not_isWeightedPerimeterMinimizer	5dee263a2a32e7f718519eb202324c7a94ba79f21e71b4e37aa0b15b25ccc70a	equivalent
CompactCells267To298.Box291.QInterval	7a67cea69bd2c9dc3c60927bd5cc66114df4f7bd2e91a6f4736a9c4cb9641008	equivalent
CompactCells267To298.Box291.lambdaInterval	ab30391d54411b079fa6a479f69ee05ad062f09357c238c7db3c5466c320564c	equivalent
CompactCells267To298.Box291.compactCandidate_not_isWeightedPerimeterMinimizer	ff721490c99aa53371530f972348bdf2314af8cb09e6bde35b761b98f77bc5	equivalent
CompactCells267To298.Box292.QInterval	d63f45ded7481248acfdaddcdc15472235ea80322586323d59327ca9f6b859e1	equivalent
CompactCells267To298.Box292.lambdaInterval	e5aece0493a49fed0d209c666c5c685db906dfcd8287fffb5827e326f85fceb12	equivalent
CompactCells267To298.Box292.compactCandidate_not_isWeightedPerimeterMinimizer	d5c6beeebf2655cf336ed5b098b04f27b5a4523fbedf22d28ad246a873776f04	equivalent
CompactCells267To298.Box293.QInterval	bf644b7e43e45ffdb68415cb696a73185b3f8f9e321425be0de538da71046d51	equivalent
CompactCells267To298.Box293.lambdaInterval	dc2ede481e715ff4320d928fc2724eb43dd2f81fcc5fb055bedd89f253a73bef	equivalent
CompactCells267To298.Box293.compactCandidate_not_isWeightedPerimeterMinimizer	e0496e000f31d02b3f6ef84f5c76daf625cd63a8593d6a18a8e86d6cc1d8a70a	equivalent
CompactCells267To298.Box294.QInterval	f2f131a58b4fdb98b6bd1a997888472e6f910bc82fb06f416abe8cd9ef153d94	equivalent
CompactCells267To298.Box294.lambdaInterval	069c5577f7bc03beaa3fbcbb1df2e5f7c8baa4789e3fa6c41a720007a0c41e31	equivalent

Declaration	Source hash	Audit relation
CompactCells267To298.Box294.compactCandidate_not_isWeightedPerimeterMinimizer	22d1dc5bbef7fffb92de02b8c148934a95f13fa3301c2006bd5734c627ba2d324	equivalent
CompactCells267To298.Box295.QInterval	648ebc4fed0d481e4d1ad58a7b2b0945541a6614a521ad1ac07c22c47a34c21d	equivalent
CompactCells267To298.Box295.lambdaInterval	54d9be9313c958e95f33b542189ebf8096128a66aba1b1cf8349329b8426db3b	equivalent
CompactCells267To298.Box295.compactCandidate_not_isWeightedPerimeterMinimizer	7901d02fa9e602555ecc279141406393d3a656b9e266fc36b27a3a7cda8c6dd8	equivalent
CompactCells267To298.Box296.QInterval	d3b4d144489054a7fccda4f6386da471cb2e291c185c505a56f45d87e239e57c	equivalent
CompactCells267To298.Box296.lambdaInterval	be0c2d3c3edd7ff5cdd967090a7714381255496049da93f6653c4aea173330d0	equivalent
CompactCells267To298.Box296.compactCandidate_not_isWeightedPerimeterMinimizer	44f53fab7e5ac740cbd4560326a0af4d6354c1c5865811043e61212ef31ad006	equivalent
CompactCells267To298.Box297.QInterval	e59dca2244e94f97d297b8d9847a9479a3331a62d97fa44cce4f48b05fd4e2c2	equivalent
CompactCells267To298.Box297.lambdaInterval	b95e5d2a2b492908ff08031a0c4b9f76afb1c40b37d09cdfdb9e5db4175d27b7	equivalent
CompactCells267To298.Box297.compactCandidate_not_isWeightedPerimeterMinimizer	bb9c29612aad64bdd631f54ec5a089b0e2e0eb3f51adc84056d56ed00cfe5ef4	equivalent
CompactCells267To298.Box298.QInterval	835de753ae0ec3df0dfe83525a259b7b75178ae8c21405d4cf1e7bf0429a5fec	equivalent
CompactCells267To298.Box298.lambdaInterval	328da4f75ff24eec9e5a6c86ce2e37c38ddf7a0406a00a8b261cef13733e19c6	equivalent
CompactCells267To298.Box298.compactCandidate_not_isWeightedPerimeterMinimizer	2666e6a74147ea59e15fac3c19f1ed56ccbfb827e6ed4df4d09e4330214471d5	equivalent
CompactCells267To298.bboxes267To298_candidate_not_isWeightedPerimeterMinimizer	0a72a1f202f6857175a7e61464061e13c05cbabcb155f735a89c1d320184c206	equivalent
CompactCells267To298.bboxes0To298_candidate_not_isWeightedPerimeterMinimizer	5d8fa6e5a14d1b11a879f314abdba46a8285eac2ed228c0d25f54818fb503228	equivalent
CompactCells299To330.Box299.QInterval	60a01de3e0ae1d37543f957590473300f16fab594809397d957e75d797f7e43a	equivalent
CompactCells299To330.Box299.lambdaInterval	ce6719ad5916cefba426a14a88aeae8d758ba20c3f34cae9b260de948de3076b	equivalent
CompactCells299To330.Box299.compactCandidate_not_isWeightedPerimeterMinimizer	de6ce82646aee8cb785a56e4e5b4c0001e7c99d1839744bce87ada71dd35a263	equivalent
CompactCells299To330.Box300.QInterval	093a068d9624fe1909b6105b425bc12aeb8ad703ab5e0b56a76f1e80d59d5dbe	equivalent
CompactCells299To330.Box300.lambdaInterval	6016250cdcab69e402878317f98fa6d9c2d3d2b4bbd1be194fb4496f74db84f6	equivalent
CompactCells299To330.Box300.compactCandidate_not_isWeightedPerimeterMinimizer	aeb02067e2c65ab20bd78f83c165ce69462d5844fdf0438f188b434787db5f32	equivalent
CompactCells299To330.Box301.QInterval	e3a0976cefee777beac7f7108561ad04507e129aaf0aecd138262822d770f5d2	equivalent

Declaration	Source hash	Audit relation
CompactCells299To330.Box301.lambdaiInterval	7634a0f7a2a4952e7a5afce0d6cc110bc62d661f99d0c27cfdd63e99b6599ff3	equivalent
CompactCells299To330.Box301.compactCandidate_not_isWeightedPerimeterMinimizer	f681616d975d01fc3996c6080d58b4fb03dd68b078f291a104009d3686d2dfdd	equivalent
CompactCells299To330.Box302.QInterval	5be1a33e76e9d0dd588cd07873570a2397da5a4c131ce435216e65f611aa671a	equivalent
CompactCells299To330.Box302.lambdaiInterval	1d739300a1e88a57435f8b9c334b8e4dcdb6124996e1de939a012cabd149a917	equivalent
CompactCells299To330.Box302.compactCandidate_not_isWeightedPerimeterMinimizer	0914b1d1f7537f4bb6cf6d980510f8fca535edbacc6d0a5422e229d436ea0aad	equivalent
CompactCells299To330.Box303.QInterval	1b23266c1ed0a8d11f5522e84aab68b5b1b66de54c39d45f21f88c7033c51c1d	equivalent
CompactCells299To330.Box303.lambdaiInterval	0a47825fdda09e2ba866a869dc0a027b5c1850de48e8067bb7e841d66cd64932	equivalent
CompactCells299To330.Box303.compactCandidate_not_isWeightedPerimeterMinimizer	a741fac72fb6878ce6f474f7a86aa497e4976ee21c7c702adedeb67ae23185ea	equivalent
CompactCells299To330.Box304.QInterval	1e73d71ccc4e64479972c1838882731373117c4f66828ceffba26527c5c2d83d	equivalent
CompactCells299To330.Box304.lambdaiInterval	e88652e692eee9ffadb359b8d1b6af60598b082288e4beb8742aec8f7568a2fd	equivalent
CompactCells299To330.Box304.compactCandidate_not_isWeightedPerimeterMinimizer	4404c8f32f88e149b61341c2b405bbe926404fdd38c3482f995a2e71526a563c	equivalent
CompactCells299To330.Box305.QInterval	eadd69ca00ec958a870281401075c2eb8913c02b8e0a9e558d8eb5e3b15b53ee	equivalent
CompactCells299To330.Box305.lambdaiInterval	662cd8e379f32a7f523e2337d8e7b18e703c8ef3db07961296acf1679c184fea	equivalent
CompactCells299To330.Box305.compactCandidate_not_isWeightedPerimeterMinimizer	bd9b4ca27c373a93c29df6abc20d4ee1ef6851e745f1de49ee89437aaffeee4c	equivalent
CompactCells299To330.Box306.QInterval	a5e56432c086dd5dfcf2eb59b2a19ed8ae0427751fc484327d1bc2db5c6d9825	equivalent
CompactCells299To330.Box306.lambdaiInterval	47e90d00cd9c51ed50a8e3efced5da4a6b9c54c5cc7d1573fab8734bb57e7d78	equivalent
CompactCells299To330.Box306.compactCandidate_not_isWeightedPerimeterMinimizer	40e641815fd963e4f4f3b6cf1f0a134362110c3da2ee0ee28d9429c7651df81b	equivalent
CompactCells299To330.Box307.QInterval	c7920a6e19c8ffa099f9009d8b5d19c4d0188c9b5f488b1dfd7d0142cdd01e91	equivalent
CompactCells299To330.Box307.lambdaiInterval	c036a9aacd3d56e021bf7a89779c88cb42162fd9d7b0c83d2a6ee2a3a470e33c	equivalent
CompactCells299To330.Box307.compactCandidate_not_isWeightedPerimeterMinimizer	c5070139c2049e07b0604327a3dc991d43df55c758ec1150b95a2987fd2bc907	equivalent
CompactCells299To330.Box308.QInterval	a2a2ac5b16f739fa099643b377d3d8ddae985e89e39eccc78eb359ec43ae36ce	equivalent
CompactCells299To330.Box308.lambdaiInterval	471ee94af67e6f6c75ab30845457002c78555e345a87682b2d900f20d4b4d2d7	equivalent

Declaration	Source hash	Audit relation
CompactCells299To330.Box308.compactCandidate_not_isWeightedPerimeterMinimizer	bf785e172028fecccccebd887e3c50b3c6334aee744613768579aa7cce755020f	equivalent
CompactCells299To330.Box309.QInterval	ff7d87ddf9d6fe28fc04491d9fdf8fa5d95c50851cdaffff156e4a25c728c9c44	equivalent
CompactCells299To330.Box309.lambdaInterval	1709e7fd764913424c382984d10577ef09b81f63da14bd9be9c44d477cd7a37c	equivalent
CompactCells299To330.Box309.compactCandidate_not_isWeightedPerimeterMinimizer	0623b98063c1ae22482541248ee5b3ec748e260d5c9fde476c782934bdcae56d1	equivalent
CompactCells299To330.Box310.QInterval	60e2aaf49f6f157ed5d42b56f947aa85267ac6f3f0daa52eca4f0c9ab98e7d3c	equivalent
CompactCells299To330.Box310.lambdaInterval	7c090060de5c098c6bbadadae372696ed13cb70b72491fe2e18e0ad7b43a11b9	equivalent
CompactCells299To330.Box310.compactCandidate_not_isWeightedPerimeterMinimizer	e4cee0d40ff2dde6076ace420cb833a364bcbbe4709328a30b51602f95168b61	equivalent
CompactCells299To330.Box311.QInterval	de171df2059489abe14740364d80921e4cb7153ce8c9e403a2c4e18568058fd4	equivalent
CompactCells299To330.Box311.lambdaInterval	e6c44839c95b06decc4fdb9e28c6c7ad812cab151a417d3d65e94845ea4613df	equivalent
CompactCells299To330.Box311.compactCandidate_not_isWeightedPerimeterMinimizer	9a19e6ca24ec32ac93b3b15bd2c271dbc78dc0d6a7a68624f410e8eacee46bc5	equivalent
CompactCells299To330.Box312.QInterval	79db48d54ce71d6ce1c9dfac286f20cd658a2bd4fe83a5f87c5d1d76689be6e2	equivalent
CompactCells299To330.Box312.lambdaInterval	da081a0b30999c000ed929680355bc8f8cf229fbdaa70e15748717cf8b9b2ae6	equivalent
CompactCells299To330.Box312.compactCandidate_not_isWeightedPerimeterMinimizer	6ca026e9a949237dad099598116bdc2b398faa2b3a5085173e939855f3449e4a	equivalent
CompactCells299To330.Box313.QInterval	1ea562ff35001054d45edd713f1f1b097c08208776a45ea02d405c52c38ba802	equivalent
CompactCells299To330.Box313.lambdaInterval	d70d0f838643111559dc0eb5b8c22bc2b792c3e9f548c1bc4dbb7c8ab1e35df6	equivalent
CompactCells299To330.Box313.compactCandidate_not_isWeightedPerimeterMinimizer	3454b69eca246862bf6f6d38ade211f343e3005985759f2539c2afbc8b402cd4	equivalent
CompactCells299To330.Box314.QInterval	7fbf392fe51dbd784ad2b12de24561412f38540add2d960bedf20dfbfd95f7b4	equivalent
CompactCells299To330.Box314.lambdaInterval	f447ce19739b5788ee5973f13cfd72b5d938d2f4a65acbb76a35f44d614b52ee	equivalent
CompactCells299To330.Box314.compactCandidate_not_isWeightedPerimeterMinimizer	6771c062d2de745036ea7ca06a86465d861604819965c1b7465a1bfe4617c7e1	equivalent
CompactCells299To330.Box315.QInterval	fa74df8f12ae343316417c25682d7a77755c11e6a9d8fe5688038f4b1e8f4bf4	equivalent
CompactCells299To330.Box315.lambdaInterval	8b469cba1866a80f2fbb1301244486a5d74cdd05377c520e2de394607c9810f8	equivalent
CompactCells299To330.Box315.compactCandidate_not_isWeightedPerimeterMinimizer	86b588a8899d8261831f407b5aa67197b4cf960aca6df3af75355cb73692b95d	equivalent

Declaration	Source hash	Audit relation
CompactCells299To330.Box316.QInterval	fa4a8222124d3b36f78c7ced0bc5de8c0074e6683d5d608a676ee786d8dd8a53	equivalent
CompactCells299To330.Box316.lambdaInterval	f5b7dc739ef938ccf321f001184c634f4cabff6d015194dc046f0e27085dbd73	equivalent
CompactCells299To330.Box316.compactCandidate_not_isWeightedPerimeterMinimizer	196f58197df05abc657e3abc676ccac31d880b1eba6ddf4093cfbe5123399c4d	equivalent
CompactCells299To330.Box317.QInterval	558950596f069c8b28e24c6db7ffab9960ed232148da5dc11996ee9e18e65986	equivalent
CompactCells299To330.Box317.lambdaInterval	f80e41c5ab217355f4f49899ac0074cfcfcf3a38c844b2b0d5125c7fc7c2dc87	equivalent
CompactCells299To330.Box317.compactCandidate_not_isWeightedPerimeterMinimizer	832f7d2b62c3c49a328f032bb1a44b1ab458ce58429d7333ecf0764532c8c2d2	equivalent
CompactCells299To330.Box318.QInterval	b3c715d0db88b36dfe4dc5610a8692a3b529797288600f0901447533346885e3	equivalent
CompactCells299To330.Box318.lambdaInterval	74131d3d2825d63733c68e350cda3ae3db9f3c0988026a47d3e31e684a4b1dba	equivalent
CompactCells299To330.Box318.compactCandidate_not_isWeightedPerimeterMinimizer	7c2de8d0bc07f4d46fbadde73ca829140a01b71e5d4e17fc951e1c20e68dbd94	equivalent
CompactCells299To330.Box319.QInterval	d492a36ef976deae55c0731bddae249be931ddc4e9a4affad7aff7432819ff	equivalent
CompactCells299To330.Box319.lambdaInterval	c354f77a640083e809b48578a34d852f97e0f34c2ebd0bbfdde22c10e203be2d	equivalent
CompactCells299To330.Box319.compactCandidate_not_isWeightedPerimeterMinimizer	13c94645ec9b78b59fb44e5282ab48e076476549b0b621ddb19d45495619e36a	equivalent
CompactCells299To330.Box320.QInterval	26ef9d45efd456553b2bb379629583f8bc21e9a9e3d3091826e1f52aa0883aa8	equivalent
CompactCells299To330.Box320.lambdaInterval	5ed2d92228785f2ceaba2bc5ac102235c9ba186999b5f3a74c3dfff3817682a	equivalent
CompactCells299To330.Box320.compactCandidate_not_isWeightedPerimeterMinimizer	22b20362858bc0fdc2b60bf044ac98601c28c4044aa76c8318033419b478b133	equivalent
CompactCells299To330.Box321.QInterval	d599041e9d6abfd611b6e9358ebc9f0fc9e6392850eab42d49ab87cd4977ebe0	equivalent
CompactCells299To330.Box321.lambdaInterval	110bc6ce0b5eb6e28462aab842c9a08d6ea3b8c75ea24753623559bfc5f59a0e	equivalent
CompactCells299To330.Box321.compactCandidate_not_isWeightedPerimeterMinimizer	2cd41effad71e4e95fb4d1daaa489ff3ba8ce64972cb01af2f3daa7cb3c49ecc	equivalent
CompactCells299To330.Box322.QInterval	ae3b7b2c25f5a09f966fd4d114b5b5d1f2bf1c2d0d73e75a4969db775757f795	equivalent
CompactCells299To330.Box322.lambdaInterval	dc148397a42504a2d0dc9aac1a084ca67f525fa7c7cb41dcdff8fcee1a396bd57	equivalent
CompactCells299To330.Box322.compactCandidate_not_isWeightedPerimeterMinimizer	9f1d6d334392bac53398f217c8fc2e8f88b7cfebf02e5c56390ca1d48c4afe07	equivalent
CompactCells299To330.Box323.QInterval	8850df4faf4202b2a5d3e80da3071c19f76882839aa609c4d16bff2469967243	equivalent
CompactCells299To330.Box323.lambdaInterval	600608e413a570cb5af3c9bd6dc353444990bfe375e05e4f64ec713b75723533	equivalent

Declaration	Source hash	Audit relation
CompactCells299To330.Box323.compactCandidate_not_isWeightedPerimeterMinimizer	4ae9a5f724b6921f83886af154a54b6aaf2c920629b9057bca5491a9b435ecf	equivalent
CompactCells299To330.Box324.QInterval	efc0d066ac4002f98a538997b8bc1945db8eb3f0f6746ca3d7071d542ba8e74	equivalent
CompactCells299To330.Box324.lambdaInterval	ff8302adcf83db37525abe956df193c0b18807594c7978dcf41ed1a59fc6d3eb	equivalent
CompactCells299To330.Box324.compactCandidate_not_isWeightedPerimeterMinimizer	cbd7f1a4c2ce3ffa24fdb8c94b67d1e3d1c0193d7a2fd2b900ac019a3d9c5cf9	equivalent
CompactCells299To330.Box325.QInterval	62ce2e8685161ad607ae852f7e2a09b659e61b5e5a08477d066864f439c70c47	equivalent
CompactCells299To330.Box325.lambdaInterval	edca0f7d880380a044dd0b727f1350fe38e6a51a6c2e34cad180ffa80dad8949	equivalent
CompactCells299To330.Box325.compactCandidate_not_isWeightedPerimeterMinimizer	c744e529f1043c25c7c769b7fe62104c6d413574f2896c59513b467fb2cf3cde	equivalent
CompactCells299To330.Box326.QInterval	8dca2876bc33abef5629905788350d1a053658625708f7bc0d43fb8a5e7d6140	equivalent
CompactCells299To330.Box326.lambdaInterval	9ca000d6b4a7de9d5b03a3407b8d86a2b00237014f6ffcf69529abd9bbb0146f	equivalent
CompactCells299To330.Box326.compactCandidate_not_isWeightedPerimeterMinimizer	7a538145310daa6db31eac2887c6d68058880fb44dd634095030d2860f666607	equivalent
CompactCells299To330.Box327.QInterval	547b56150aad2c9288589655d2ae576da44927b941c6d3faea70e2ad7e20fc0	equivalent
CompactCells299To330.Box327.lambdaInterval	0eefd5a64a4d0b873c576bb308991fffe255f5700013fe5d350a1f14d57cccdc	equivalent
CompactCells299To330.Box327.compactCandidate_not_isWeightedPerimeterMinimizer	c29ac17dc70d9e4d0d3d1432ca7b2d8bbb1c0266d8d270bd6dec9614d8f25717	equivalent
CompactCells299To330.Box328.QInterval	21a4ccf64983cbf05eb343d468041feaa1f0af519dcaa3bda03c9c558e15e8ae	equivalent
CompactCells299To330.Box328.lambdaInterval	1c16f729f3e7dcce84ee47024d1ab9c9cc0f7f9d58d188c3c6aa3f4b0ddc8c3c	equivalent
CompactCells299To330.Box328.compactCandidate_not_isWeightedPerimeterMinimizer	98b5b21d14a372129416132e40f7f7b30e37c2827ad0216bb1836cb73f427e8c	equivalent
CompactCells299To330.Box329.QInterval	48305d08f9c70a0501e45c8ca2eafc02f630dfbbdd7cbafffa5586405d56a40e	equivalent
CompactCells299To330.Box329.lambdaInterval	5068c37f038a010d14ba985773c05a207e9da75503c98ee6bef18063ab539a28	equivalent
CompactCells299To330.Box329.compactCandidate_not_isWeightedPerimeterMinimizer	c1907667be89f76fb37e8ed2d55c1a74a4818cc961944248b40400e85eef3221	equivalent
CompactCells299To330.Box330.QInterval	02dce17e64469058e8e81936db3fceda66b8ba35c301374b001424fce5307719	equivalent
CompactCells299To330.Box330.lambdaInterval	879222679bea52ba5a249f4fbd992b99402316247d65dca69f48307069600e7f	equivalent
CompactCells299To330.Box330.compactCandidate_not_isWeightedPerimeterMinimizer	1f6d37ce580ae9f79c6dcd10ab55fc703a1ee04a3e1fc576a05f0208f233a36d	equivalent

Declaration	Source hash	Audit relation
CompactCells299To330.bboxes299To330_candidate_not_isWeightedPerimeterMinimizer	7e98912c86e9fba0851957a22b6bc522e999cd679c540db59464ffa8a98e3a15	equivalent
CompactCells299To330.bboxes0To330_candidate_not_isWeightedPerimeterMinimizer	6a14b6cb363aec1a44ce2c486c92df6d405a30fcee652a23429e375da8809fac	equivalent
CompactCells2To10.Box2.QInterval	9f76101b52fe8ff78f20b4093a715ed7960e954f3f42a0ad2bd3ea183779db92	equivalent
CompactCells2To10.Box2.lambdaInterval	c53ed8f607b2a172720b438c38c6ed7a906d00946eda7222edbdeb4c79dd09bf	equivalent
CompactCells2To10.Box2.compactCandidate_not_isWeightedPerimeterMinimizer	6422b8de2bb2f94e12194734c7b60eed8d82e51c6c7414b3c8f42ede8d08dc8a	equivalent
CompactCells2To10.Box3.QInterval	cc3a62b8e72cc6cc216aa391e6dcfbc5d8f8b54a5e4db46639b23eb0d19a79a9	equivalent
CompactCells2To10.Box3.lambdaInterval	bf6e718ced211284e9c083d4b4fa9ccfe489a79410f9800ec5f685d4181bc2de	equivalent
CompactCells2To10.Box3.compactCandidate_not_isWeightedPerimeterMinimizer	b0e96674236f38844c8a847b7abab48ab5958f89638af334f84d84ecf2f78611	equivalent
CompactCells2To10.Box4.QInterval	9409921c87fd845077b6307abf8caabe89f7ac70027b39327283c61d00e8e024	equivalent
CompactCells2To10.Box4.lambdaInterval	68ee44e074d072d2070d86ee58844e2a241fa9951963546cc437b4abdde96397	equivalent
CompactCells2To10.Box4.compactCandidate_not_isWeightedPerimeterMinimizer	483d91c7efa65943915d0f5863c4cc1458986b888eb425855746571e8848c5ac	equivalent
CompactCells2To10.Box5.QInterval	7406a9905a7abf7f64e360b837ec5db014e67f7fb717a80794c5ad9f479d6af7	equivalent
CompactCells2To10.Box5.lambdaInterval	c15f79510699151720ba5274f43dfff6126bd2e2e732134545df3ca44e4d9f3f	equivalent
CompactCells2To10.Box5.compactCandidate_not_isWeightedPerimeterMinimizer	e5fc5f9c9f09c1677693075b9586334123089b2625722f0bdb75e334cb70f80b	equivalent
CompactCells2To10.Box6.QInterval	96f255a674a295dbe30690be85590e9713fc2b6d81200e6ded70a6abcf548902	equivalent
CompactCells2To10.Box6.lambdaInterval	05880725d08583816e515d816f39c12929ae2b3b9192fd9dba9b19a2843c44ff	equivalent
CompactCells2To10.Box6.compactCandidate_not_isWeightedPerimeterMinimizer	dbb9357698938c492ad278c41a0baaac4bd5758a0404f7d257830c06deb13662	equivalent
CompactCells2To10.Box7.QInterval	9b68870b5eb188684790db3c614475b9193521118f2776d515db9e9979ee35d1	equivalent
CompactCells2To10.Box7.lambdaInterval	dfa020bf522e90c29a884cd2bf11e585f3ec6974bfb964b69370d681d7ab4797	equivalent
CompactCells2To10.Box7.compactCandidate_not_isWeightedPerimeterMinimizer	0da3ed013f1ae59a43b34800e1cce47d435862e1e4664b83f3b3a6ad011adb9b	equivalent
CompactCells2To10.Box8.QInterval	e343cf12296a2d502d68aea19a384fa8309a136d0061d192a6586b152d781d5a	equivalent
CompactCells2To10.Box8.lambdaInterval	0e243e5f496ad80341e77752cb24bde29a052e28051e6ff54d9e7fbcc541b87d	equivalent

Declaration	Source hash	Audit relation
CompactCells2To10.Box8.compactCandidate_not_isWeightedPerimeterMinimizer	213e896af43316fe86ef8076df8f55c23afcab78918b3b72c2e7fe6f8285d642	equivalent
CompactCells2To10.Box9.QInterval	a51ed0621f47f23b08218cfce9e0f0ec72279010948538fe5cb726dcf5f956bd	equivalent
CompactCells2To10.Box9.lambdaInterval	c3094e13b21ef653246445000d5faacec3502ab92d8144de1088bb8d9def7523	equivalent
CompactCells2To10.Box9.compactCandidate_not_isWeightedPerimeterMinimizer	f7b7e3311f48346b700c1b7a1c3c9203b359179faa5c7bfc02c862921692d974	equivalent
CompactCells2To10.Box10.QInterval	001d1502508992bdf3a391f08d9401dd77f79c1ba2b5c1272f89f97e653cbf22	equivalent
CompactCells2To10.Box10.lambdaInterval	ea9e70a906c23fd72f3094dadfcb44d3165b9410a4f182b3e4cc1d7e7f02d440	equivalent
CompactCells2To10.Box10.compactCandidate_not_isWeightedPerimeterMinimizer	be835c46a58d63f773b69c88aa4f679b954e8e6c91d8738c232d4b9cc8279ba3	equivalent
CompactCells2To10.bboxes0To10_candidate_not_isWeightedPerimeterMinimize r	aa677d90a7f590f6abedc4ef1586d8b735dda78f9f506acb42fc8461ef9403f	equivalent
CompactCells331To362.Box331.QInterval	dcd686951aafd7f107c353aa472ac41feaba4e19098807cc4724fa8f7260d3e7	equivalent
CompactCells331To362.Box331.lambdaInterval	2bbecad084bec1acedeec48bc030c09244d6762d4285b692d238b46ee7b1af5b	equivalent
CompactCells331To362.Box331.compactCandidate_not_isWeightedPerimeterMinimizer	e530b00caedef3bb13b8bcc4eb00d4caab969cb9839a82a56db044900fc2c872	equivalent
CompactCells331To362.Box332.QInterval	aee4a3d17f5186a742d00b65e3da84773bd070bb0e4ee873f5b903db9bd850ce	equivalent
CompactCells331To362.Box332.lambdaInterval	7204df1dfd8c676a5cb380e2a008886ac119cec8f270d772ecde4c4bf68daaeb	equivalent
CompactCells331To362.Box332.compactCandidate_not_isWeightedPerimeterMinimizer	f5249aaa64688d03d7c14c064fbfb7211f445e8e85906105954fd94d0c528dc0	equivalent
CompactCells331To362.Box333.QInterval	bae3f3d03e178ea0b4f7b103777cca4e77361bdb78319315639870f35a291a07	equivalent
CompactCells331To362.Box333.lambdaInterval	9d078338e8a134870abf83886b526e2af76eb5a47296e17a68f68305d5a8f489	equivalent
CompactCells331To362.Box333.compactCandidate_not_isWeightedPerimeterMinimizer	e32af367b61b1e6127fcbfcd608695956f957f4211c9072ab02a796120aec13	equivalent
CompactCells331To362.Box334.QInterval	26fc7fa01162b96e6872b613c309e950e91a85c1117e248740291fc323bbe058	equivalent
CompactCells331To362.Box334.lambdaInterval	39953dd5dd5166106d89e7149370365f9126ca1f8a0f0c65482d66d516a09d6c	equivalent
CompactCells331To362.Box334.compactCandidate_not_isWeightedPerimeterMinimizer	71c82c8ced2a8358b0954e9f7e2cc687da75e36e3137980337037de0c6e49350	equivalent
CompactCells331To362.Box335.QInterval	341f103d9b48926621331c3f1c937d3ad4e485ca95578d42bcc8d870a13d91de	equivalent
CompactCells331To362.Box335.lambdaInterval	daa53595d838912d582f1af16931ada55ba56867301b540fba6cce5e77739618	equivalent

Declaration	Source hash	Audit relation
CompactCells331To362.Box335.compactCandidate_not_isWeightedPerimeterMinimizer	9e0799cee69427e73d87cdcb1126b6ca6e37400c401a45747da059f53ba8dfd	equivalent
CompactCells331To362.Box336.QInterval	cd521c797f15e8d746f7fb3cbd109edfb7fee4e9f4024b83bd45102a11716c03	equivalent
CompactCells331To362.Box336.lambdaInterval	ddc849933829a99c71bf12916b5a4d500ed368b5e36d3a23d7b9a1d47d60db4c	equivalent
CompactCells331To362.Box336.compactCandidate_not_isWeightedPerimeterMinimizer	5ab840cae8bc981f4c0eb709c0c7dc680bbe475e4aed9364a8ed7b6ad1925184	equivalent
CompactCells331To362.Box337.QInterval	74ae386847f3b723bb828ef8f4e78c32cdc700b276b12ce89f5ed38ced829918	equivalent
CompactCells331To362.Box337.lambdaInterval	58e79ac11d8c2838c22be993fbb5aac291108838aaff2dcf7d73a0884af059a4	equivalent
CompactCells331To362.Box337.compactCandidate_not_isWeightedPerimeterMinimizer	930d7718c997d804ff2997e9cfb18eb2a69f2e0965a4c0936f823595305e4230	equivalent
CompactCells331To362.Box338.QInterval	6c26cfb5c56dbeff53aa1a4d02e74badf28ed4ec7949a6a08b61a7b893412561	equivalent
CompactCells331To362.Box338.lambdaInterval	84b899c3c03952df968578cbfbf632116046a1d9f738e5486eaca2c184612ead	equivalent
CompactCells331To362.Box338.compactCandidate_not_isWeightedPerimeterMinimizer	59c1613a682a6d060eccb3ab5112778e315e0a11914109e8175784d5fe54e833	equivalent
CompactCells331To362.Box339.QInterval	4204fae7058e5bba978c95299080c9315a6a5910b3d76d94153e7367a4c04395	equivalent
CompactCells331To362.Box339.lambdaInterval	2470f1a39fdaea9c5728360504cf42a800a26e6bda765aac545a2d542d3b02f4	equivalent
CompactCells331To362.Box339.compactCandidate_not_isWeightedPerimeterMinimizer	1cec017be4ea3cb52900eae3b04d1a328112b5ec5f0f6b2e57a0ed9569a484a	equivalent
CompactCells331To362.Box340.QInterval	a8102a1b59eedcb7202462fa7a054ec3dee5792592350c40533718c7b3058ce	equivalent
CompactCells331To362.Box340.lambdaInterval	4ccaad73c302dc522dbc5d93e53a7636209ce87bd85fa65a9b88ff6ee31ea3c8	equivalent
CompactCells331To362.Box340.compactCandidate_not_isWeightedPerimeterMinimizer	b0e6d54cbea40150086e961e07943871b6da2d0747f0096476c4b27786f4b58c	equivalent
CompactCells331To362.Box341.QInterval	b78164502d0081c13458042687c90f2524f6adc17cf841a7abf64fb99ed4c4c7	equivalent
CompactCells331To362.Box341.lambdaInterval	0be67ae5fd3c37c28cb14c884262a3fa8eeae0b10f52830c7f995bd1330cea52	equivalent
CompactCells331To362.Box341.compactCandidate_not_isWeightedPerimeterMinimizer	cca9207874c8e156af23975810fec9cdcfe31b4e1ef5af64e73ab5b937e2d79	equivalent
CompactCells331To362.Box342.QInterval	c935a7a2de43dec37d55b13ebc223735230c9745b3c738173f360b232cc8a91a	equivalent
CompactCells331To362.Box342.lambdaInterval	c1459832a1ea01d5d8b250b4c8c5e6eb74c82c81526dc31ea9a3275d845ac15d	equivalent
CompactCells331To362.Box342.compactCandidate_not_isWeightedPerimeterMinimizer	cff7e150c381be6772b2f3197b84bd8558b1d2119b0f24c0dc3beb36d8e24c63	equivalent

Declaration	Source hash	Audit relation
CompactCells331To362.Box343.QInterval	03325ef9fb26a21660b022c873979616a254f97213c6cf7d51a00d6954748d53	equivalent
CompactCells331To362.Box343.lambdaInterval	776c752f757d4c2ca13019b0efbdc3009491c27886fb780f42346ca10caa560d	equivalent
CompactCells331To362.Box343.compactCandidate_not_isWeightedPerimeterMinimizer	897dbf408688a9a6eb9372396239a08e46fb9c0790e5e926e649daf2e7665bd6	equivalent
CompactCells331To362.Box344.QInterval	6f8e26032db429958e7961a976a101343912b593b3b6bbe22812708dbee84289	equivalent
CompactCells331To362.Box344.lambdaInterval	9eea941bf858141fe3329066cdad75962bb057416eecd9b445e7fb2293c31d4	equivalent
CompactCells331To362.Box344.compactCandidate_not_isWeightedPerimeterMinimizer	d3c7d5c933f8f261825bdb58cee9cf0b593df88fd5066db60abdd46f333cb3ef	equivalent
CompactCells331To362.Box345.QInterval	1552d75085568486f1f335e98786617614beb342d08945949fd872745e096667	equivalent
CompactCells331To362.Box345.lambdaInterval	01292adcd1009e290eac1f833375ae605880f8372caf7e9739ee4a47532dea55	equivalent
CompactCells331To362.Box345.compactCandidate_not_isWeightedPerimeterMinimizer	217f18b976027157a71e427a6a50c650488afa607aff56fd936757b7c1ea9222	equivalent
CompactCells331To362.Box346.QInterval	ccf57ac538f8ba5d65c8724a657f15b0cf90e8aa7e5848d49c043a047ca31482	equivalent
CompactCells331To362.Box346.lambdaInterval	ac9b58f5f5c07cfc9939c488ff6b711a4578d1935f0c7ffee083c67069c4e528	equivalent
CompactCells331To362.Box346.compactCandidate_not_isWeightedPerimeterMinimizer	1d8e0921a226eef1b854a1f715b86aab82099b75014da98092da92c6fa2d544d	equivalent
CompactCells331To362.Box347.QInterval	87fe0f8a624495547bb65d0f4bf6e3dc168054a47696c8b2f54c8d8c30004176	equivalent
CompactCells331To362.Box347.lambdaInterval	39eca9c2d1ae0bfc091084650f32e892e31b3ef9549ea848c2f840cf48f3e9ad	equivalent
CompactCells331To362.Box347.compactCandidate_not_isWeightedPerimeterMinimizer	f114d260ea2c104c3a6f5120501f022a99233bb07be9ede0b3f12434f41ce246	equivalent
CompactCells331To362.Box348.QInterval	6cc795ce3261bba43824830efc8f3069a459fb7aa1ea68de64cd709a05eb907a	equivalent
CompactCells331To362.Box348.lambdaInterval	c401e595684a31d654999212ca694774836ab32f29cb8a4eeab3bbc97a186ba7	equivalent
CompactCells331To362.Box348.compactCandidate_not_isWeightedPerimeterMinimizer	75af9165cb335f0b247187348f65c805017c9cba92f2d3c3d9ddebc15bf7f3c4	equivalent
CompactCells331To362.Box349.QInterval	45e63b77e72093938621d292cbaf024d5896da3bf7de8ed4d0315d2dc9099420	equivalent
CompactCells331To362.Box349.lambdaInterval	cf23c9577a0c59b7c211572024798cb9b72d220b23107ed22700ad8c4261128d	equivalent
CompactCells331To362.Box349.compactCandidate_not_isWeightedPerimeterMinimizer	a0b31e5eb3bad0f95af02da925d60073a09865cee7a097d49fadbd4d62d28cf3	equivalent
CompactCells331To362.Box350.QInterval	fe9b02fce21fbd3d27c7114122bbeb05ffdf3f2fedb6a6aa28ebd49470d60988f	equivalent
CompactCells331To362.Box350.lambdaInterval	9ea3149c6b57457c52d11896ff5c3ee079beab4bcd518f3813539c0e1c69419	equivalent

Declaration	Source hash	Audit relation
CompactCells331To362.Box350.compactCandidate_not_isWeightedPerimeterMinimizer	6c6dafc18fa01991065d64b82360131fbb07d7bd8107864f87889cc491349b7b	equivalent
CompactCells331To362.Box351.QInterval	0a76023843ee278e15b0e472a4321f8d927aceedbe036cc217e1775a44659f99	equivalent
CompactCells331To362.Box351.lambdaInterval	e6c7fc040e57c23af77814e96a78878e5461c13f0844f73d1962a594e3efeb1b	equivalent
CompactCells331To362.Box351.compactCandidate_not_isWeightedPerimeterMinimizer	efdfef5af19776adb30180860014fd086dd4625eb5396ea4938c3c33819520228	equivalent
CompactCells331To362.Box352.QInterval	2c2acf2542e45bb425f5a32d78bc85bbbdea042ab1497e1626d4a65873aacbf6	equivalent
CompactCells331To362.Box352.lambdaInterval	54292194e3ccc5be06e2b9c19392f2abf2d32fcae973851bae31a2543c1b0656	equivalent
CompactCells331To362.Box352.compactCandidate_not_isWeightedPerimeterMinimizer	74275ed76a2f53818e650c9f74eddf5ea627c5b0599dc83b361617024d7f6b87	equivalent
CompactCells331To362.Box353.QInterval	7ce69d63d4c20275084396a1506ec02d1884c635b3bef95a19d29836ae028756	equivalent
CompactCells331To362.Box353.lambdaInterval	2f38ac9e124e12e39f48791789b4782d7123eb5e421d49d245719c775fd62b50	equivalent
CompactCells331To362.Box353.compactCandidate_not_isWeightedPerimeterMinimizer	c5de22dc0c6c0fb7ca10e320cb8dcd7387c45d149f5f0555086bfed11fbc4c35	equivalent
CompactCells331To362.Box354.QInterval	ac92608c41795599933b5bc92820cf46c961de8fdd3469fb1027b50992b914d9	equivalent
CompactCells331To362.Box354.lambdaInterval	0b7142d87131d0e7e48dd885a1e97dcef65c4cd69f5067fcfe98b2c3c366f278	equivalent
CompactCells331To362.Box354.compactCandidate_not_isWeightedPerimeterMinimizer	be250a765eee04a3ab55ab2e9cccff72cf3fe622bb1af45c153da02c7850d6e1	equivalent
CompactCells331To362.Box355.QInterval	a2c289155b8b1e26518ccf265962850b668f074edd9b5a744e6eae801c4a3928	equivalent
CompactCells331To362.Box355.lambdaInterval	86e53a86ea7fa538e9084bdb87af4062a2d42e8eed0fe83c51edf942bf659c1e	equivalent
CompactCells331To362.Box355.compactCandidate_not_isWeightedPerimeterMinimizer	42b4b76bf7de4ff44491e69c5002f7b8f73e08fea3253e66e2eb9a5507529d54	equivalent
CompactCells331To362.Box356.QInterval	448e92c42fe82d0d81b7fff06ce02cebbabced3d68f157c162f19a2eadc5052f	equivalent
CompactCells331To362.Box356.lambdaInterval	8c885c8b8d994bbbf93d658e2e6a5692584d4d5d07ded7405ea7d00057037ff4	equivalent
CompactCells331To362.Box356.compactCandidate_not_isWeightedPerimeterMinimizer	94031cd835c3f3deef1ba68ef9c854b336f8ca0f4c322047878659c8d422a0c6	equivalent
CompactCells331To362.Box357.QInterval	b77d60d77c94bc05d2743c12523ffa3a0c9c16cbad8542803934f43fcdec4732	equivalent
CompactCells331To362.Box357.lambdaInterval	508fa6f63234bb79f815d3755ddbd25f207ab6ae80fd05df66fa6038bc41e628	equivalent
CompactCells331To362.Box357.compactCandidate_not_isWeightedPerimeterMinimizer	401fb2690f668e1fe4c158ddca75c7366add80d497965e5ea5aebc750ad492cb	equivalent

Declaration	Source hash	Audit relation
CompactCells331To362.Box358.QInterval	4744300896da49de1add85ce3ea500a5613511e5a03913f7d04d2b135d1e3d95	equivalent
CompactCells331To362.Box358.lambdaiInterval	fe8d3aa936e634726a777cb03345b4a32f090fac6956e77e3e16965868fa46c6	equivalent
CompactCells331To362.Box358.compactCandidate_not_isWeightedPerimeterMinimizer	de2f02fa53b1afa8b59d9314bbf5ec76c918fa988730ea5f834dc751fdb44e53	equivalent
CompactCells331To362.Box359.QInterval	9924deddfefe1e90c5cd0b256bc14d4dfd5ca4d98b415b7c4e0e9aca6bccd804	equivalent
CompactCells331To362.Box359.lambdaiInterval	61c91dcbbbac201b8d230eead38ce21f992ce8ec2e12d99162b766c78f72ce83	equivalent
CompactCells331To362.Box359.compactCandidate_not_isWeightedPerimeterMinimizer	09b66f405ce7bb606f1d8b0a19637bc04b9b03f0c64ca1d273de5cc7f31b2922	equivalent
CompactCells331To362.Box360.QInterval	7b7aaf5992425332f6100cf04033e08e93e3f802b21cdbc07d13087e24dcc3cc	equivalent
CompactCells331To362.Box360.lambdaiInterval	4a1ca2a9e75d32522ecea6ccb1b56c68e2f88433930e9913f1eb41be4c56060f	equivalent
CompactCells331To362.Box360.compactCandidate_not_isWeightedPerimeterMinimizer	7cb1a6d714a60a13cb4c63a8b671c3a919c40747e6285787da9e4261f7e34e43	equivalent
CompactCells331To362.Box361.QInterval	493687b0c553ac59ff4c7638fd6736d034421d7b61a577cd7ac0be64a83ed6c4	equivalent
CompactCells331To362.Box361.lambdaiInterval	e663afc5d4bdd885992b58d3c3f804e3a2021bac6e3e990a799ce64b7049c8f3	equivalent
CompactCells331To362.Box361.compactCandidate_not_isWeightedPerimeterMinimizer	8a377c80535649868a783ffa4aec3a0bd566e4c1ef7651e5a12a12eb7483e869	equivalent
CompactCells331To362.Box362.QInterval	6fa0d777c135c8d087b1f11528c6ec103564b17a44026a834faf1ac36d663b68	equivalent
CompactCells331To362.Box362.lambdaiInterval	4b74818de65f764b4bf2a7ab320f2c3f1e10f17e347a37d0aa5f2e21274c9a06	equivalent
CompactCells331To362.Box362.compactCandidate_not_isWeightedPerimeterMinimizer	dc832c687c7332b25e7c7e3f26ccad38a622cfa43ca26c0027fc926495313c3	equivalent
CompactCells331To362.bboxes331To362_candidate_not_isWeightedPerimeterMinimizer	a679d85532a5511eae40fa13aa30ed8386ca711c43a0d8cba7e12d1fd1f8a7	equivalent
CompactCells331To362.bboxes0To362_candidate_not_isWeightedPerimeterMinimizer	bf72333aa6bc2f95060c0f54daac3bc1e92f6ca45c9b41617ba871071eae0a	equivalent
CompactCells363To394.Box363.QInterval	b7226dd73de0d66b264ff14cc6fb35349906e017b5e5afb1d2540500ad357a95	equivalent
CompactCells363To394.Box363.lambdaiInterval	883ab4ad0b008b3cfb7da24f8c27ec97c8ab3aa48de16268d2a81122498a5722	equivalent
CompactCells363To394.Box363.compactCandidate_not_isWeightedPerimeterMinimizer	957b698b8812fe43b275500f0d46c255db47a74717fe9d296880c073aa8a94ad	equivalent
CompactCells363To394.Box364.QInterval	4768147bde8447307afa04d90346380e6ae24affbf799a75995b188c9773b9db	equivalent
CompactCells363To394.Box364.lambdaiInterval	70486ccbe374e7cf384ad5ab35b6b4f97a2f7d05d1c73e627504bf519c496942	equivalent

Declaration	Source hash	Audit relation
CompactCells363To394.Box364.compactCandidate_not_isWeightedPerimeterMinimizer	efab64bbc23a3b3ab49b11e324eb12d77d44294c7d1dc786ef88d34444695d7c	equivalent
CompactCells363To394.Box365.QInterval	18dc1942d0d076d173f8a47794f075ee80fd8e50a1ce864ec06d001d99a8c761	equivalent
CompactCells363To394.Box365.lambdaInterval	50a1f6663d3102e6aa331056e8d1c959fca95f136d4781dbfc549e23477f600a	equivalent
CompactCells363To394.Box365.compactCandidate_not_isWeightedPerimeterMinimizer	51b7be19caeb1536cdb6023014d897f00bf985e657ce568f2c8b0b4a3af84517	equivalent
CompactCells363To394.Box366.QInterval	0ab31ce4dfdf6c438a2d522f4a581f34ec041b61a31946f5a577adf75c5e4a81	equivalent
CompactCells363To394.Box366.lambdaInterval	98b10a6bec78fa43e18834fa1e967a3868fb3dc68592114d4f1ab94177df036e	equivalent
CompactCells363To394.Box366.compactCandidate_not_isWeightedPerimeterMinimizer	f8b718cb37a51ac2d1f782a047bf4ff654ea82da138f6d49d58c79766cd392ba	equivalent
CompactCells363To394.Box367.QInterval	d251007366e6dab3afd297587099a51b22a8d505aaa331e4bd65bbc0c1510600	equivalent
CompactCells363To394.Box367.lambdaInterval	36f30dc96a63e125c50645f2736989eacb9b5d6bd91033e7432f46d58634e946	equivalent
CompactCells363To394.Box367.compactCandidate_not_isWeightedPerimeterMinimizer	30a428b6028f2eecba079280ad1bb0e94457dd4c05fc5e282b115b113b0dde86	equivalent
CompactCells363To394.Box368.QInterval	b8021a7ec94679a0a256ea1cddbfb87ed0c378c49579677fca02672687b6342ed	equivalent
CompactCells363To394.Box368.lambdaInterval	6fdf39b0edcdc03a142e42f6f0f504d34cdffc16d2ce85c083532042e31d836	equivalent
CompactCells363To394.Box368.compactCandidate_not_isWeightedPerimeterMinimizer	39e14bc771d597f80685f15ac40fb4b47348146ac73b4e17325d24aad6edd4de	equivalent
CompactCells363To394.Box369.QInterval	aa4cfa5c2655c6a0414ced55329dcc727e95d2045276f515fc71a2fc7fad48a9	equivalent
CompactCells363To394.Box369.lambdaInterval	9096b15eb07790daadda7dcba44c5b7852ed2433295c4c674f2c2c2fa09c2def	equivalent
CompactCells363To394.Box369.compactCandidate_not_isWeightedPerimeterMinimizer	4b1875ba56a4484c9eaab0a4c832cdd85cc98e39b41160a5c2638e125140a84b	equivalent
CompactCells363To394.Box370.QInterval	676c3bd28873144d3c9287dc577fdec74a872a806bc03f51951d38d23835afa8	equivalent
CompactCells363To394.Box370.lambdaInterval	f5653f9893cfc354c9a675416c8486cd6d895297a4cce0968ef75b8dadf1f36e	equivalent
CompactCells363To394.Box370.compactCandidate_not_isWeightedPerimeterMinimizer	e4c6981880fac8807da307d8acd53ad6be709c5df6f548c7b1ed9373571a0e31	equivalent
CompactCells363To394.Box371.QInterval	c20ebc5709bd0e6b0d2964d132c92ab3cf5fdceaeb0c17d72447b917a71d6943	equivalent
CompactCells363To394.Box371.lambdaInterval	76612ffcad624d71c47831509cf5c38be8d79dcbfcfc996008428344470e6a4b	equivalent
CompactCells363To394.Box371.compactCandidate_not_isWeightedPerimeterMinimizer	9425116f1f48722f924983451bd89e55454a5aa6e1144e8d980a5f6636556de0	equivalent

Declaration	Source hash	Audit relation
CompactCells363To394.Box372.QInterval	9558439b18844edf2124ac0f66bb05d840d0c9d5a90cc104a4835b9141a1da15	equivalent
CompactCells363To394.Box372.lambdaiInterval	47cd732e1afc4ed6a8ed9b268ecb262d139b0fa43730a0ec4ced8fbca2da680c	equivalent
CompactCells363To394.Box372.compactCandidate_not_isWeightedPerimeterMinimizer	8bdcaad5d6565a58aaa409aa65d6c6d1ed1e279a512dd56ae84f1c4987307b1e	equivalent
CompactCells363To394.Box373.QInterval	edbce8d676c332ce9ab2cc44ddeb93c75e0957e67a5e597b37ac27dd14ffbaaa	equivalent
CompactCells363To394.Box373.lambdaiInterval	656b780aa27f83e0e0b125abbd0bfcde7287a6b6511909ab77a7588b6a6ee8c9	equivalent
CompactCells363To394.Box373.compactCandidate_not_isWeightedPerimeterMinimizer	ef7485b3c7b019b5cfaac85001a9c9844e90dc158cb06d1820cc4c88d73359e9	equivalent
CompactCells363To394.Box374.QInterval	d08198086ed733ce66d1f4d2d8a5b54ba231e1bc24e10e20988dbad4c8ab49a5	equivalent
CompactCells363To394.Box374.lambdaiInterval	62dac308684fb127ea6ed333d0aab88325342bd13060c03831394c3aa85864a7	equivalent
CompactCells363To394.Box374.compactCandidate_not_isWeightedPerimeterMinimizer	75a7646955a202ef1e2cdbc9d5ec9cc107bfc00fd465cba0b0262574d3bf60df	equivalent
CompactCells363To394.Box375.QInterval	69ed5a7455b8679687efc6f24b9dbb2932ed84a1263f8dec683139e885046102	equivalent
CompactCells363To394.Box375.lambdaiInterval	d3cd21662c6aed792f6765a5f6a49d2b0b18c164389764291808f34adbdb9aa6	equivalent
CompactCells363To394.Box375.compactCandidate_not_isWeightedPerimeterMinimizer	02fa9d3cde03db0f9adf6ba9865451391086eaf600bc05ab563cc19c73031456	equivalent
CompactCells363To394.Box376.QInterval	c80f54d6c5be78ea30692ec741dad47521ad6e2d711b93d4a2f4953bdc00596e	equivalent
CompactCells363To394.Box376.lambdaiInterval	b512d7df3359c2be892cbe06c66878d0715f9e74eb62ad59b346386153823e2b	equivalent
CompactCells363To394.Box376.compactCandidate_not_isWeightedPerimeterMinimizer	f258a58d595d52ce5c004bf0d88499e0f59c23eba95e708295825feed3529a5f	equivalent
CompactCells363To394.Box377.QInterval	b5a3b7701121b569e1d5597914ec445fe9745b2ad79e0338d638beb50933367c	equivalent
CompactCells363To394.Box377.lambdaiInterval	423ad85bb7dfa21aaacff575cac599a10c3a66602e01ee0d0be38f60639555b	equivalent
CompactCells363To394.Box377.compactCandidate_not_isWeightedPerimeterMinimizer	6c89312b2851be71902e7b86489bc697413b086d054ac58a1ba5e914d7637551	equivalent
CompactCells363To394.Box378.QInterval	fc5d87fd5ed24b68c7c06ee2dd639ac209829c348d55c59529c46ccb8b621177	equivalent
CompactCells363To394.Box378.lambdaiInterval	d6ae335f79d137f0e250020e9ae0a7115b7fc473c27f4de87aa1b75d648943a6	equivalent
CompactCells363To394.Box378.compactCandidate_not_isWeightedPerimeterMinimizer	b527dc1ba17dc7fceec8209d9e377ab0cbc3452b7efc331e121d61a3d1ecabb2	equivalent
CompactCells363To394.Box379.QInterval	6b400d9a3806ebefd9f7e1b2becae17cbc936e034ffd091ba2b62a1ecce1cefb	equivalent
CompactCells363To394.Box379.lambdaiInterval	9c290c816d6f5f48337237607787e3bf3cc8df1e3b320e6c44039111ad657c7f	equivalent

Declaration	Source hash	Audit relation
CompactCells363To394.Box379.compactCandidate_not_isWeightedPerimeterMinimizer	ae6b2d0c09420e389f2d38b20ee2d7eb73e3bef338c4f3d460aed45442908d52	equivalent
CompactCells363To394.Box380.QInterval	04d10d6568845d980636151a3f8ca7fc4bfec0ac41538b3680689768f44c02b0	equivalent
CompactCells363To394.Box380.lambdaInterval	510a2c504f7fbb8b425d71cde5a0b9c97f077fd6352df1c89d6b9cf172312604	equivalent
CompactCells363To394.Box380.compactCandidate_not_isWeightedPerimeterMinimizer	a69d5a582fce10d2369c381b5c42d15cb910981c4d02a06da8c1ea2917ac2294	equivalent
CompactCells363To394.Box381.QInterval	868584aa79dc91ec69a15c6b146cca1983f54a59e1c25a3ca7ac6fe836dfad8e	equivalent
CompactCells363To394.Box381.lambdaInterval	dd5505a0aa98fa750024b2c4e66a628f49e5e4957f6e941674b39bbad8401428	equivalent
CompactCells363To394.Box381.compactCandidate_not_isWeightedPerimeterMinimizer	6afe6ee7e6fa909c4e5a9c119483497a89a96a85142662e4da632e307921461a	equivalent
CompactCells363To394.Box382.QInterval	40a7084a00c6b6f30bfb026e57dfa3583815ba5d96872618ab276937cb742193	equivalent
CompactCells363To394.Box382.lambdaInterval	6841dbfc9773fca5df89d5f8b5372380ac3b8fbb38b3c516b2ac1a6ca0c8addc	equivalent
CompactCells363To394.Box382.compactCandidate_not_isWeightedPerimeterMinimizer	82755ec7055aa5596c79aa8176604542928ab066cb39a45ec101f67f240acc19	equivalent
CompactCells363To394.Box383.QInterval	a035de3f2cee2a0298a4cb47dc5222c09a191fc6338fb28713e87fa649ec475b	equivalent
CompactCells363To394.Box383.lambdaInterval	ab99c902df77524952ce6cd6793e0caf25ec7d1ae80850717296b366e279f224	equivalent
CompactCells363To394.Box383.compactCandidate_not_isWeightedPerimeterMinimizer	bbbd8d6eeefc2dbc2bba984a519068881593aad2a085a8bf7266c65b1ba8d4b4	equivalent
CompactCells363To394.Box384.QInterval	6f673219c7d38ab6ab5403b6578bc706aed46fe0ea778b2354fd9ba223deafc9	equivalent
CompactCells363To394.Box384.lambdaInterval	500d0d181b8a24f119ce108b08654c172c110ec2bde26beb35412454a45ac88a	equivalent
CompactCells363To394.Box384.compactCandidate_not_isWeightedPerimeterMinimizer	e10a16884ba0ae475cf84031982835157afc3691e90efe2cf5722ccfa5472c6d	equivalent
CompactCells363To394.Box385.QInterval	2ef21fb1b246e24a3a1daa7d8c41b2e9fc9570c9b63a0bda830ff34a90a89c93	equivalent
CompactCells363To394.Box385.lambdaInterval	5ec6879d21f7cb9b5e7f8bed4e5447cc54f736aabffa53332f9cf1aed46c8c43	equivalent
CompactCells363To394.Box385.compactCandidate_not_isWeightedPerimeterMinimizer	bf0fbcb3e326cf407d727557f1d4ac2c4f06138e4332782560e68f6ca2db708d	equivalent
CompactCells363To394.Box386.QInterval	07056a578ae196a8db197d56deb93c85623530d2bc984de1f22e3e9f06d43d11	equivalent
CompactCells363To394.Box386.lambdaInterval	3819d708c70ebb75faacc336c6d14ae07534a179ec560eab15f53af4807032f2	equivalent
CompactCells363To394.Box386.compactCandidate_not_isWeightedPerimeterMinimizer	93f8cc09f0731b97486ea6a8de62e1e647845ff6b8735bab4bfb7238c8da2aa0	equivalent

Declaration	Source hash	Audit relation
CompactCells363To394.Box387.QInterval	af65eb5fc4a449a2791fffed3dd0bc18512fc96dbb888978bb2c3142c6da7f3d	equivalent
CompactCells363To394.Box387.lambdaInterval	af58ebdd4c390315df5f533232016b0b77980da7b3541a1ca72304b690244074	equivalent
CompactCells363To394.Box387.compactCandidate_not_isWeightedPerimeterMinimizer	02ee2f181a1032de485b06a3dd2a8fbe597be0612c8cc19542c6df54215407e6	equivalent
CompactCells363To394.Box388.QInterval	d7e285041e493dcedb2d10e143171a8bf27d55be662e285d6c6299be5e2b0c45	equivalent
CompactCells363To394.Box388.lambdaInterval	568b72cfa3f099d07215eb5cc665b1adfded8d13477066b598e5f29908fdf8151	equivalent
CompactCells363To394.Box388.compactCandidate_not_isWeightedPerimeterMinimizer	fac447f25bcd9903ee9e6f18def65345a0709f8606bdc454b7833541baecf4b0	equivalent
CompactCells363To394.Box389.QInterval	692aae7765fd31cf346d2890372c7ed1b45b781f8c88983e91d72301d473e1ef	equivalent
CompactCells363To394.Box389.lambdaInterval	5ae4a1a81ae4247892bdc6e0a147aa065d11e78e2140dbc5d2076ca54776c374	equivalent
CompactCells363To394.Box389.compactCandidate_not_isWeightedPerimeterMinimizer	14d1c457c9b761afd747f763acac989e0dab7458d208f346ece8d9b1b4fb7dc	equivalent
CompactCells363To394.Box390.QInterval	717c95916ef1e64191fe2946ebcb7a49949a86c7af0e0034de0816a4043cdf02	equivalent
CompactCells363To394.Box390.lambdaInterval	24ec9dcfba68955e9f6da9156c919fa72199d31cb47c40ef2035042720d99801	equivalent
CompactCells363To394.Box390.compactCandidate_not_isWeightedPerimeterMinimizer	7a25fa47668a9cded7ea0974dfba5aa424f786b177669ca22efb2a07773f6d37	equivalent
CompactCells363To394.Box391.QInterval	f030b0d1ab92f800ba7375e937f97d2c9d918864b8d71a67365d264379fc5e33	equivalent
CompactCells363To394.Box391.lambdaInterval	bf473bce6ac972acc7cb9088d92b491f855072c7a09f7603f53417b171694d27	equivalent
CompactCells363To394.Box391.compactCandidate_not_isWeightedPerimeterMinimizer	0ad94d0cf6255c408da1cdfcd38a7444a5827f27443b02e7f9aedbbed0b748fa	equivalent
CompactCells363To394.Box392.QInterval	5bda4def7352bffda088d1b357afc87cc66c2c691260bb755cc6380352500515	equivalent
CompactCells363To394.Box392.lambdaInterval	5c6986486289f9f6b9702a68c7f158715f206a6b07f078eb698148477d1df987	equivalent
CompactCells363To394.Box392.compactCandidate_not_isWeightedPerimeterMinimizer	1c2f0098c42f5b9f2c891997afbfa38170c81e8c32ed53a713cf6738e8a38c12	equivalent
CompactCells363To394.Box393.QInterval	bb1d7c351986f377e18431b2bfe5eecb64a04ebd86830a3abc3e293bd5a5f23c	equivalent
CompactCells363To394.Box393.lambdaInterval	6012891d468f94280251df363509c3e620c3b8c65fd4bb55faa0009aeed757e6	equivalent
CompactCells363To394.Box393.compactCandidate_not_isWeightedPerimeterMinimizer	cc06db6a8777b5bd56272cd991c372fbbda94dffdd8364abf1393efecc11dd2c4	equivalent
CompactCells363To394.Box394.QInterval	9b7248c928ab8e06ae31b0b0d42d1d39e7d32d0e6402134ee14ca02049cce957	equivalent
CompactCells363To394.Box394.lambdaInterval	0f9a5b6964e35a0bd90ae503d81457a6595d0d810ee0563faa7fb68b74e91df7	equivalent

Declaration	Source hash	Audit relation
CompactCells363To394.Box394.compactCandidate_not_isWeightedPerimeterMinimizer	abefc3987ea094dda8352a3838aced3c7653e88c831ba9c6292c9540fc82acdd	equivalent
CompactCells363To394.bboxes363To394_candidate_not_isWeightedPerimeterMinimizer	0fa450abdd7678b5a8610e464a7b15b0e82e6eb86c06d02f425f47fbdde54dc9	equivalent
CompactCells363To394.bboxes0To394_candidate_not_isWeightedPerimeterMinimizer	294384e6ee34e0c65f14626c29e1b34677a087948e642ad024bf21993373c5ff	equivalent
CompactCells395To426.Box395.QInterval	a07269d89ccbc06a8ab4e544b848f43cce5bbf0ad4bcf7800a65737592835bc7	equivalent
CompactCells395To426.Box395.lambdaInterval	85d6da50494bfff10b74464750c03d2b51887f0d5eea78aeff34e68525845d0	equivalent
CompactCells395To426.Box395.compactCandidate_not_isWeightedPerimeterMinimizer	e4cf85d9af9a72be9abf64846566a6a8c174f19b0b375a773f8f1018c0144f4e	equivalent
CompactCells395To426.Box396.QInterval	0c89fd06d9c74aacf54bc4af0fc1ea2e6be6d2dbed8f56e21c7e7cc7a974a49e	equivalent
CompactCells395To426.Box396.lambdaInterval	cfcc806918e8e4c6856601eb5b7ae832f31d7dd8b50997fd239773c689664f4f	equivalent
CompactCells395To426.Box396.compactCandidate_not_isWeightedPerimeterMinimizer	435f66d3ad22af20a26819b9c5e81ddb221fe4eb365eb651b41c66fdea3eea69	equivalent
CompactCells395To426.Box397.QInterval	833a432822de64c72459fd85336a6344c48d6ebab024fe2d25042c99a0e68530	equivalent
CompactCells395To426.Box397.lambdaInterval	74b1a6f9cedf57ad2c305a8c8e590e99877e87f21c118c608c9a1e6578f42ae3	equivalent
CompactCells395To426.Box397.compactCandidate_not_isWeightedPerimeterMinimizer	364260882799b1b279ca1aa98a3dff9ec088c0dbd12c80c41d56356332c8dfc7	equivalent
CompactCells395To426.Box398.QInterval	d8d708bb139753e571f00e71bf9785aff9d1fb64eee4e53d9edd81c0dd04d4e8	equivalent
CompactCells395To426.Box398.lambdaInterval	fc59c1975add3c13656949f03ebce560df1a2d2b870170e74545ac5dc4238bef	equivalent
CompactCells395To426.Box398.compactCandidate_not_isWeightedPerimeterMinimizer	bd33c189dd3087e0e727384510a1b5ffbddbf8467bdf376bc290f7c21ba5651c	equivalent
CompactCells395To426.Box399.QInterval	b583d1e271c441b743bf104535be43f49bb8806597c2c8475e692fd6410b72b6	equivalent
CompactCells395To426.Box399.lambdaInterval	a5f5708abd11faa79d3f88c2e202c11bfe1db3a6aaa1b343a478349e455c4cc9	equivalent
CompactCells395To426.Box399.compactCandidate_not_isWeightedPerimeterMinimizer	bd519cd934131912563b4449a70b481b5cfcd5e6072ff8111eb4c9b6c15bb4d5	equivalent
CompactCells395To426.Box400.QInterval	26b2bdf3b6713685dc70941fa01549da45d8c8ca9e9cbc895d2b3d7f1ed97fc7	equivalent
CompactCells395To426.Box400.lambdaInterval	4e1967ef6f737040be102e6c4c50517e77303939da92f1615596ad271818e0e1	equivalent
CompactCells395To426.Box400.compactCandidate_not_isWeightedPerimeterMinimizer	a81b23042eff76b0543941dca89ac0970d84c878fb7fbf1f80abea839e4e3c23	equivalent
CompactCells395To426.Box401.QInterval	49deb1ac50290e3ceef1ce6795cea481f0b6fdd4432a35ff9c1162224f957c4f	equivalent

Declaration	Source hash	Audit relation
CompactCells395To426.Box401.lambdaiInterval	b1c3eab5be677a9a4c6d14d0f673b5141b8826cba24605849ebabbdf8c75cd2e	equivalent
CompactCells395To426.Box401.compactCandidate_not_isWeightedPerimeterMinimizer	a82f3ee3816cfed113de94cc7004d9dc5569d379c63538c143d75c0dd121a0b5	equivalent
CompactCells395To426.Box402.QInterval	09dd10f843ac72962ed73585318eeb5012cc09ea4541dbd137b9d0bbca9bd5f3	equivalent
CompactCells395To426.Box402.lambdaiInterval	a110988d10fb1fd7958262ed20d5be17ace8b84493cbc5d16089ece25258269	equivalent
CompactCells395To426.Box402.compactCandidate_not_isWeightedPerimeterMinimizer	b14ddbe3fd8b5ab22ff30bb01e2dbecfce8d0d08e6280e4257afed6753f144ea	equivalent
CompactCells395To426.Box403.QInterval	aca4f2ec1bc3784718d25ed4e45a96a7d0765abf49e9d0d509fc62b22f78d3cc	equivalent
CompactCells395To426.Box403.lambdaiInterval	a3a8fd8e5c171915cabf6004e26e0ecbcce2d328999b2884dd3ce5f284aebbd0	equivalent
CompactCells395To426.Box403.compactCandidate_not_isWeightedPerimeterMinimizer	0f546ce5d2b38d9590b13a57deb03237f2a92347e587fefade974da47c0f29a2	equivalent
CompactCells395To426.Box404.QInterval	bc212241ff36711cf32445e761a8f41da9c8b0017b32972bc0f9aab715637e6e	equivalent
CompactCells395To426.Box404.lambdaiInterval	3c7c8bca35858f6ac3a4598cf5d059fd709f0815cf128cac2297b6889e0e037a	equivalent
CompactCells395To426.Box404.compactCandidate_not_isWeightedPerimeterMinimizer	6b14e125ca4ce1c2d0130e40003318484c10e35423cc9601a71f44406ab8110d	equivalent
CompactCells395To426.Box405.QInterval	ae70945355d84943dce73a27b803b0bd1fe8312d7ffb72763c7ad6a9d26d4722	equivalent
CompactCells395To426.Box405.lambdaiInterval	311307310acb2069c3c6006242a662a4ec347baa07619555662a727b8d32b046	equivalent
CompactCells395To426.Box405.compactCandidate_not_isWeightedPerimeterMinimizer	8ec0842240b2de90532ab90001e8f707c5d5694e6f00c35592ceb8aa444d70cb	equivalent
CompactCells395To426.Box406.QInterval	27c26acd944d37400cd794d6e543fca44c851b746f19a261919ac0ca3d288bba	equivalent
CompactCells395To426.Box406.lambdaiInterval	25cca2a9468ed00ccfdbd9cb27c5fa0fb2ab23a9a5fb5ababc9714e58263a076	equivalent
CompactCells395To426.Box406.compactCandidate_not_isWeightedPerimeterMinimizer	bc20cd598716752f718bd4c5fb3856ca47f6796078d0e5dce8838fef81cd4aa7	equivalent
CompactCells395To426.Box407.QInterval	790f07753642c72ab2be270f3e546c12d2a972af9fb44783fad5614677014c1a	equivalent
CompactCells395To426.Box407.lambdaiInterval	b8b8ab3318da9679cc5f8d4bc94c73234cf59a13204028fbc8970b197ee5ab1	equivalent
CompactCells395To426.Box407.compactCandidate_not_isWeightedPerimeterMinimizer	1be58a004b3a602d5ce7b10871dcb9fa7f5a92d0e41f26b3d88966c1b764584c	equivalent
CompactCells395To426.Box408.QInterval	1e516c4b9644b8c754c8a43dd2237357701b2fdb04f3b96209f0ae4581c82067	equivalent
CompactCells395To426.Box408.lambdaiInterval	1353ac525d7cc7517a375e03dcd3599326023ed63072617eec29de8d7a944b98	equivalent

Declaration	Source hash	Audit relation
CompactCells395To426.Box408.compactCandidate_not_isWeightedPerimeterMinimizer	e93e45f866baa3bdb8586b36832f582382bb900508c61b97d2f7151adb69f5b	equivalent
CompactCells395To426.Box409.QInterval	779e974d0c79f03d6bc9b324c56124251b1513157105e6ccad80b22a930631f9	equivalent
CompactCells395To426.Box409.lambdaInterval	190a12f3126dce561d8b0c740db80801446e18f44c225e5343b133cefeef4e4	equivalent
CompactCells395To426.Box409.compactCandidate_not_isWeightedPerimeterMinimizer	9dc958982a4d48eb69adc819dfd49f2c7c3962576ab4136a44fff07b9f5b322e	equivalent
CompactCells395To426.Box410.QInterval	0c6aad63a19d621dfdfaacc9ce4e345890bc26960e39839356a10b62a46554ad	equivalent
CompactCells395To426.Box410.lambdaInterval	074bd2bbc46f0525d876f329522b6b7eb0bfe800b75754fba7f5250ca98a3673	equivalent
CompactCells395To426.Box410.compactCandidate_not_isWeightedPerimeterMinimizer	2131af0b8dcf50df90bb191d382e45037858254c6471feccd8baf7be7b0af490	equivalent
CompactCells395To426.Box411.QInterval	348a4fe534af1394f9a14e56434b01609377384f441ca03bd869a7f31e9a019b	equivalent
CompactCells395To426.Box411.lambdaInterval	de30d8ba556b053cd4b5b1f888b15c4ecac44574c6faae0ac09fb064d6ed4a9b	equivalent
CompactCells395To426.Box411.compactCandidate_not_isWeightedPerimeterMinimizer	77f91a70aa00a9c9844edebdd7af09f274291b408877c948c58a325cbbe42dec	equivalent
CompactCells395To426.Box412.QInterval	97234dd7246d2729d381c0fa61407ba5fb3f14703d78b5975300492f95d80f9a	equivalent
CompactCells395To426.Box412.lambdaInterval	c3424ee2e15e2295298387941571843f65164133e36d093ed37781e6bec52c60	equivalent
CompactCells395To426.Box412.compactCandidate_not_isWeightedPerimeterMinimizer	7ad4e21cb7f5aed0b5a60d7f3df7e6f05cd8e84444d01dbcc4e8cb3516ccdc53	equivalent
CompactCells395To426.Box413.QInterval	5755336573eb2b9ccf0100e6b85529f69295b6ead54a242749e89b0e131b6264	equivalent
CompactCells395To426.Box413.lambdaInterval	72f970e95a87fe92d0553d0aded1b5feefd006001992d82e06faba2606d17c4b	equivalent
CompactCells395To426.Box413.compactCandidate_not_isWeightedPerimeterMinimizer	9eb576b8114f84a548c1e2c4b6a7247e5d55ff18a7df028a4baf5de3be44a431	equivalent
CompactCells395To426.Box414.QInterval	c80fffff751783cdbcc307011169df236e7cb1c6fb27a479edda689b0169737e	equivalent
CompactCells395To426.Box414.lambdaInterval	233ab33b16d00375289c31549ef39afb779acbd20e15e8f60f6cdb2298e0c8db	equivalent
CompactCells395To426.Box414.compactCandidate_not_isWeightedPerimeterMinimizer	6269108dcd6cc97be34bce5697f6c7711f34f36a471107b3948a4d49a4a33e4b	equivalent
CompactCells395To426.Box415.QInterval	8dd14d33f18720f1133f6f176785d78cffc2b5b91b463dc4ca78692ddf3bd829	equivalent
CompactCells395To426.Box415.lambdaInterval	832eb00b35573156e89b9ca0566088a2636d3e760b687e37160226f435161531	equivalent
CompactCells395To426.Box415.compactCandidate_not_isWeightedPerimeterMinimizer	8b652a56b6c9eb10aab98bd5e4c8d5cbbb2d7c6341225bf5577b2f591cf13aa4	equivalent

Declaration	Source hash	Audit relation
CompactCells395To426.Box416.QInterval	8ea9275220a858f1d2cba9fec55aeb8e46917adc836f0b4718b9b16a9512d3d8	equivalent
CompactCells395To426.Box416.lambdaInterval	33c38fa9c2725aad2a0db5140a3910e5b30a87006abec133ab648c75ce087e26	equivalent
CompactCells395To426.Box416.compactCandidate_not_isWeightedPerimeterMinimizer	d476e4f976d93f2b190410a90e325c2c29b248ab4013b63fe1eb43c2bba53c16	equivalent
CompactCells395To426.Box417.QInterval	d08e07924cb3fe63c9b626e808eb004fee83e3f00f0cedcc97274d8dc0b81516	equivalent
CompactCells395To426.Box417.lambdaInterval	7c3b65f899ece19da04e1432a7b22c267b0f3bc727b498194b1987fee7c5fca7	equivalent
CompactCells395To426.Box417.compactCandidate_not_isWeightedPerimeterMinimizer	11742fd6cff0bf9629ededc6a75c32df9bd694342d1a2d0a354ad034f706c980	equivalent
CompactCells395To426.Box418.QInterval	91891a9255fbdfc2ad26289c261422781b959fd627fb9d663346f2a857031fcd	equivalent
CompactCells395To426.Box418.lambdaInterval	5f57414ee580781597f582886a31f1e2c792c965ec81b82ad5ad493ea592ff2e	equivalent
CompactCells395To426.Box418.compactCandidate_not_isWeightedPerimeterMinimizer	6f72181f3b2fe905015b03a177f8764f2f0c332b4345e8e85db06bc374b7269f	equivalent
CompactCells395To426.Box419.QInterval	2784f59fa5df2198a021d67d794bfbbcc3f2b516576a66cb1f1a082c991defc0	equivalent
CompactCells395To426.Box419.lambdaInterval	d73a759562b86345f1f1eedb6be6cdfcc189a1c0534c1bdd5df0bd29295d0914	equivalent
CompactCells395To426.Box419.compactCandidate_not_isWeightedPerimeterMinimizer	714608c5f87663c5d98006185635fddcd293c225be9f77b998331c82f5b2b61b6	equivalent
CompactCells395To426.Box420.QInterval	570c0dd9009a2175e4a5c4eb7717b050a5ded628aedd2af64c8b684dfe977676	equivalent
CompactCells395To426.Box420.lambdaInterval	0155516ec33c21d4fca7e8cc09ac5eadeec15ce657921e4f207a2ab5713da0d8	equivalent
CompactCells395To426.Box420.compactCandidate_not_isWeightedPerimeterMinimizer	e8b0e9a09596a2ef34e7e2eb59d4e781bbe676b1a9632aa72ce1eac18bf8f345	equivalent
CompactCells395To426.Box421.QInterval	91958e6b45f6f816e17f7e22247d8ba168717fd629ea5d1a1766bf2e6d3fcbb5	equivalent
CompactCells395To426.Box421.lambdaInterval	6ad91299211379276c2a72a3d7fc63b247f5b0466a2f1b1f4521b0b025af3b60	equivalent
CompactCells395To426.Box421.compactCandidate_not_isWeightedPerimeterMinimizer	3e6ec76c53ff767fc1ff99ca0c8170a86bd312f24697076e02a81f15e9219eee	equivalent
CompactCells395To426.Box422.QInterval	2e98bc82c95bdd4c03882eac25dc14fdb2422a9985b97eff16d0231d12cebec	equivalent
CompactCells395To426.Box422.lambdaInterval	8210296a52c8589510f6c9f6156bbbc12b1629f5dcd68e9df91db58d3cad309b	equivalent
CompactCells395To426.Box422.compactCandidate_not_isWeightedPerimeterMinimizer	70f6cc54f9b06cd1cba9c4cd7adadb6f52dcb10c411f08d3142a3edfa18a0615	equivalent
CompactCells395To426.Box423.QInterval	8560443e78bd86c885f9a5d77987ebd9cc86039c5719fb87d316b53ec4716025	equivalent
CompactCells395To426.Box423.lambdaInterval	401ed80f1a6066b0543ca8ff1cb3dee4649d9738446aa5d8bf96c37c534423d9	equivalent

Declaration	Source hash	Audit relation
CompactCells395To426.Box423.compactCandidate_not_isWeightedPerimeterMinimizer	2941bf86af3abdc20552b4c9a8ba95d5aedfaea2de65c8c0cbd65c7cc22ecfc3	equivalent
CompactCells395To426.Box424.QInterval	1b4109758033e3edb15b7951040486563947c6b068f8d1bd28f7ae70df14e1f5	equivalent
CompactCells395To426.Box424.lambdaInterval	4616dda428cfbc6c9f0406627c28c65d28b87049b3d8effbfa99e9ee6ad463d	equivalent
CompactCells395To426.Box424.compactCandidate_not_isWeightedPerimeterMinimizer	eddc62afadc076da0add5436816e17487a30274239809206263a33aa85fc7f0a	equivalent
CompactCells395To426.Box425.QInterval	0cdd4c938f7cff53ae11a92a0c9d8a6c929828b239ce8af8fb692e13d7fcff86	equivalent
CompactCells395To426.Box425.lambdaInterval	31096686c6961e1bfe04201a95c76518561d70d522b66596b367acda81863370	equivalent
CompactCells395To426.Box425.compactCandidate_not_isWeightedPerimeterMinimizer	4e4b541e2cc84962e2c99e46abc086ede99fcbec0357baff7edd3c68e743be58	equivalent
CompactCells395To426.Box426.QInterval	b865b07abe4f1ec83cbf719933dae6af3f5550d80a604a308f0a684240f60c7e	equivalent
CompactCells395To426.Box426.lambdaInterval	d8d991dc2b9d79e7df5891fd526db28894f7c60d9c798dfada16a77efc7d8212	equivalent
CompactCells395To426.Box426.compactCandidate_not_isWeightedPerimeterMinimizer	6ba7c0eaf9512f92514d313c0a9b16b4cafb9a058b3019d6f7bc68447c416ecc	equivalent
CompactCells395To426.bboxes395To426_candidate_not_isWeightedPerimeterMinimizer	13bb594ee3c9c8e40e879a62e7361315bc33319360f7b2001d62b02a93f644f3	equivalent
CompactCells395To426.bboxes0To426_candidate_not_isWeightedPerimeterMinimizer	5de91fccaf4a735282810a47d096ff485f65d3c560f267891c93be715b4f98cf	equivalent
CompactCells427To458.Box427.QInterval	0e9df8ef2b7d1a5c5d20b73c2455c1fbdfd5196e664401f42331dd7b95f395b6	equivalent
CompactCells427To458.Box427.lambdaInterval	8b4ee7994f7bf1a4e705f6b2b3993d368a2bd844883c85ad9447428ab1aa4739	equivalent
CompactCells427To458.Box427.compactCandidate_not_isWeightedPerimeterMinimizer	bb116500c4da23be7d02c7a61784002be42000db10e11ca53b91790f89c90fc2	equivalent
CompactCells427To458.Box428.QInterval	4f716ac1bb8ce46a42520187991bc106ccf23efcaf0eda843ecd6beadcdd7141	equivalent
CompactCells427To458.Box428.lambdaInterval	79bf934830606759fcd44b3451e35bcb5ac3c16a9b8e2a2e49190442fe99c972	equivalent
CompactCells427To458.Box428.compactCandidate_not_isWeightedPerimeterMinimizer	7edb4e8f28fa33fd7f72eba789663dd53b137c2c8459b5ced0ad2d89edb68f3b	equivalent
CompactCells427To458.Box429.QInterval	9448df56ad8c7013fc379034bbc78f04dc5a12b6dd408a7545effd253cd2b43f	equivalent
CompactCells427To458.Box429.lambdaInterval	caf9f31f57c6cea1a492911badb658c58c02127562828408df42f567c7eb016d	equivalent
CompactCells427To458.Box429.compactCandidate_not_isWeightedPerimeterMinimizer	dc9d2ca52756f73c671920eacbf418ef31201995bacdf68cf2d1458834daa7cd	equivalent
CompactCells427To458.Box430.QInterval	710dc01ca031c594e0ef2b68b94d540b99d0783b69fe7357eddbc18a444731	equivalent

Declaration	Source hash	Audit relation
CompactCells427To458.Box430.lambdInterval	365caf48f0ef4d77d0f9bad7a30e4161145b137ef40c25cf3faff5e6418fac42	equivalent
CompactCells427To458.Box430.compactCandidate_not_isWeightedPerimeterMinimizer	60881ba86aaf6391f3bb94b1c22251081417692e3a10770dba9ac471ab8b4fe6	equivalent
CompactCells427To458.Box431.QInterval	5f772c15ec9c435b92c0bfd8b50025861edeb80e46d392fc80859655ea5b86ca	equivalent
CompactCells427To458.Box431.lambdInterval	dbc445ca317000f55d89ff1a3b57173166a89a44b7c2e336a85fe12e448e3443	equivalent
CompactCells427To458.Box431.compactCandidate_not_isWeightedPerimeterMinimizer	479f3da88b52a6a2aac2ed5915ebe7114957893b5cfd2d95192536f0b5875e55	equivalent
CompactCells427To458.Box432.QInterval	a26413da62df34429d04aea8f4fd694716748307eade9e5b051fa5946554c8b	equivalent
CompactCells427To458.Box432.lambdInterval	18eda012bbf2ef7a465bbfc67c121e5ed6b1d22990cb50ab7b571e14feceb2bc	equivalent
CompactCells427To458.Box432.compactCandidate_not_isWeightedPerimeterMinimizer	3a5eea04393f2450013c37daa18cb9c2044638aff9aa9b02f71cab7ad5c9446e	equivalent
CompactCells427To458.Box433.QInterval	6d6e51d98ac9e9d6bc20fd23c9a9491775f620d6e7c31e567fc5b8fca2cb58ad	equivalent
CompactCells427To458.Box433.lambdInterval	a9cc061a078e293969bc9f972b3c31eff2db7194526c25da06126f0acf8855c2	equivalent
CompactCells427To458.Box433.compactCandidate_not_isWeightedPerimeterMinimizer	ba3dbf59026ae186e7f81f1408e309dddab71a68cbe379cd1a3ee053f9bce2a7	equivalent
CompactCells427To458.Box434.QInterval	2d6defe00bfe232a9ff22eb23924d8937a4a049a6ba72fd11df76aa9a1728e4f	equivalent
CompactCells427To458.Box434.lambdInterval	d7014dda0955ce04ce4ee9326d0392306002b9fd7352e212473dd4f78eb5e393	equivalent
CompactCells427To458.Box434.compactCandidate_not_isWeightedPerimeterMinimizer	94507cb96c4a312676d3e1581649106d953d58fda852773a22c417933aec5157	equivalent
CompactCells427To458.Box435.QInterval	638464a9bc9421e5b4faf29dd7923736f0691a9d0ff9cd0a9af9fc8cd7af8383	equivalent
CompactCells427To458.Box435.lambdInterval	e18267b330abcd7bdeb6323ff401dfc8512b38e3f5ed80d417a774354acfb27	equivalent
CompactCells427To458.Box435.compactCandidate_not_isWeightedPerimeterMinimizer	4059fc14676b5130128eb3d893b59ce5a040f142883e131794457e6667ec90a9	equivalent
CompactCells427To458.Box436.QInterval	af0c47461962aa7b44eb2dac337e0fb7dda cca8ea8cc910ebd3f490715354a5f	equivalent
CompactCells427To458.Box436.lambdInterval	887f116158fae5abb0dc007641030233d3d54d5e5eeef39129aea2dddc18440	equivalent
CompactCells427To458.Box436.compactCandidate_not_isWeightedPerimeterMinimizer	92d18db1d15b49d5154953758f849ab8c6d18851be3549980e5adf3b786e27ae	equivalent
CompactCells427To458.Box437.QInterval	70918866a2dae0c81d6a98f9364c35294b5757381f06b64c3eb3674e330c587e	equivalent
CompactCells427To458.Box437.lambdInterval	398ed393093b873fe2f94b3ca51e49dd7e5064993da2bbcd33d1c16af1b11725	equivalent

Declaration	Source hash	Audit relation
CompactCells427To458.Box437.compactCandidate_not_isWeightedPerimeterMinimizer	280368221eb83d09085f6c01fe4755692a13bf40a6bc7ad0e1b5e5e6e52040c1	equivalent
CompactCells427To458.Box438.QInterval	b92ef4555e796cf37abe62d0c50ff19825b77c94f25753e0a259d70c00d8d233	equivalent
CompactCells427To458.Box438.lambdaInterval	8af46ee39be29a55af5cd7d542252e5a2cfe748d3713a7e3e3c4879f4cd7707d	equivalent
CompactCells427To458.Box438.compactCandidate_not_isWeightedPerimeterMinimizer	83243af559d2af9a29596a4cfec7b241cb2e6702db4fc11436d25623e6c8ef7f	equivalent
CompactCells427To458.Box439.QInterval	e356e5b74d51d2f4b55dcffa9dbce138931276cc28bd9f5c67db704146021a20	equivalent
CompactCells427To458.Box439.lambdaInterval	9052faaa15f39815b7cae066e6352ecd5108496b2fed30435246e8f76563f869	equivalent
CompactCells427To458.Box439.compactCandidate_not_isWeightedPerimeterMinimizer	4e492c23481cebedfb776f73496b881e156a2a228eef8f5594448c1c13b2ec89	equivalent
CompactCells427To458.Box440.QInterval	b2518b467896a9d7ddb4770152c44adc40f748444f18bba607a23ade680111a3	equivalent
CompactCells427To458.Box440.lambdaInterval	4626770ee2f112c38fb7847bc207dfbeb806c4dadd7c9897773cab9439df83bb	equivalent
CompactCells427To458.Box440.compactCandidate_not_isWeightedPerimeterMinimizer	af64ac6996c4ce38d5c8ab69e8b464f9175037bad243a50ff3732ce1f9b304e4	equivalent
CompactCells427To458.Box441.QInterval	63e70f36efd591250c0be03135bd06a62ed88d3413caa25f5e9e4fce5e529694	equivalent
CompactCells427To458.Box441.lambdaInterval	eff9aee0b657b0e6a090e7b498a85b0abc86344a0f6d86ec50d275ed000058db	equivalent
CompactCells427To458.Box441.compactCandidate_not_isWeightedPerimeterMinimizer	25ad56acab767eee7dbff902f54a0ed6daa96dd6eaaba39b02adc478680ec477	equivalent
CompactCells427To458.Box442.QInterval	974155c2417053da71a40cb6c2032295d53b6cd41186f3d5b2b6c6c69a452041	equivalent
CompactCells427To458.Box442.lambdaInterval	b4cc3496f38b9316403de5200e0bd89b6e69f3d23f8b4cb33b6dc10b297764f2	equivalent
CompactCells427To458.Box442.compactCandidate_not_isWeightedPerimeterMinimizer	ef4b15cb54c2233aac5f3f0ceb8ee639bb47b829b58578c4d983ec8b93c5b699	equivalent
CompactCells427To458.Box443.QInterval	7f3e3ee47b0d28b83778acb9e505d505b8f5b6e1229143dcb05b857b19e5f0dd	equivalent
CompactCells427To458.Box443.lambdaInterval	700f06d46b06461ffc1aa1476a1d41b64248236fd4511a1efd08483d99f9f5a8	equivalent
CompactCells427To458.Box443.compactCandidate_not_isWeightedPerimeterMinimizer	5185c9ff7b04a21acabdc0441376f43a69d2e89920dc6e1a66b599a6a50255a	equivalent
CompactCells427To458.Box444.QInterval	a993f3aeca34e3ab69fa239b0ae1b754d5020a3fbe9475c2c8f7540332956a37	equivalent
CompactCells427To458.Box444.lambdaInterval	b9a7961fa8a4b943379dc9338ca8c526c8182c7d175a93b475e08211cbdeda7f	equivalent
CompactCells427To458.Box444.compactCandidate_not_isWeightedPerimeterMinimizer	dee48ee6adf2d06f4fdc12a56f0e2dfc8a50797400b4d52bad319568a807d5f	equivalent

Declaration	Source hash	Audit relation
CompactCells427To458.Box445.QInterval	407c6ba4f34c30cea10ce0955f8266c118df225b65f2c5f3e0a07f1f72a47f29	equivalent
CompactCells427To458.Box445.lambdaiInterval	3f968474f948b31812d70b9738a6f26bcfcfe0d52f10f7872f4ead8edc3271e9	equivalent
CompactCells427To458.Box445.compactCandidate_not_isWeightedPerimeterMinimizer	a5c944c8f102211211a8b689f35ac9b8f269d050e10efb30df396d7c6244e142	equivalent
CompactCells427To458.Box446.QInterval	f7055f7c00786c25b8f266391c4582707f6b4177e05dd54d9ea861590fd2c84e	equivalent
CompactCells427To458.Box446.lambdaiInterval	dc7b4a560457b8bce3fa3c85785a7d38266c0e6e3e38c3846086d443f3d540b8	equivalent
CompactCells427To458.Box446.compactCandidate_not_isWeightedPerimeterMinimizer	d53ec82cd205065363f1f5714cc4b9f481345ef0db2de8fc53d2e5a7efd489bd	equivalent
CompactCells427To458.Box447.QInterval	3439052cdeb04fddf64ef638ac216fdce4ed06aaeca16522511eef9d6b1bb59a	equivalent
CompactCells427To458.Box447.lambdaiInterval	3790d0ea9396ae6dcfbfbf94a2a9c0051800af05ecec6891e8c2e9db61b314d	equivalent
CompactCells427To458.Box447.compactCandidate_not_isWeightedPerimeterMinimizer	75e7d98a84fe185312cc60b432b548dffa67bc1fa5eef3ed87d9edd6ecdeed06	equivalent
CompactCells427To458.Box448.QInterval	faa7c36ee0514a669ecaa51a7ba73b7d7d4a976b946ebcdb40a3e43a50115bcb	equivalent
CompactCells427To458.Box448.lambdaiInterval	2ca1f76d698652c7290f8a652799e7d646927447a4fa17cac15a30093836d3f0	equivalent
CompactCells427To458.Box448.compactCandidate_not_isWeightedPerimeterMinimizer	b21d90b5cdf89bd463394fa8de9b42156d8939e3b7086aa50689e757caaeb361	equivalent
CompactCells427To458.Box449.QInterval	6ef0145098d69f8e7f3e67c1b3704b3ce338b8e4c96c6da27d77c65120bd2a1d	equivalent
CompactCells427To458.Box449.lambdaiInterval	f974c6528b02c0f360388595e44d4811b4b31ed9b9dee7234c7141aca7739b7d	equivalent
CompactCells427To458.Box449.compactCandidate_not_isWeightedPerimeterMinimizer	e89c4d6dd1edba920b8ed14fcf7d133bcfd8f71923264e506e4917e306c6d8ea	equivalent
CompactCells427To458.Box450.QInterval	e41f3dbca8c94bf75c3fcb107fdaa0079167ee4346dd8ef24509c19eca2c4076	equivalent
CompactCells427To458.Box450.lambdaiInterval	a5b21d3b9bc34e9434e4607e81c0118b6833afea051cfe2e61ebf0c95b352ec6	equivalent
CompactCells427To458.Box450.compactCandidate_not_isWeightedPerimeterMinimizer	d38c143e3a5715121d0855a71e4f7b410b514df0c27b470b1e79fd0a5c75eedd	equivalent
CompactCells427To458.Box451.QInterval	d47040773846753f77adff080188dd895b56832bf2b6abf1af85f74a07c2ad0a	equivalent
CompactCells427To458.Box451.lambdaiInterval	3895e213a5a297e39107e4e73e332607a30676b33f5b5f1e0e0424c56f601ff5	equivalent
CompactCells427To458.Box451.compactCandidate_not_isWeightedPerimeterMinimizer	d38161b30986b4fab10f454e35cf4d76e289be0e0c04e8775cb7c2d2eae5a34a	equivalent
CompactCells427To458.Box452.QInterval	8e529d1cac05c0b85171982d10a6f14f3268197274099b6a25a64fecf4708b22	equivalent
CompactCells427To458.Box452.lambdaiInterval	209655edc6d6161a1b6f630e3809f9ff0c4b8c65d44360878771f69a133b6657	equivalent

Declaration	Source hash	Audit relation
CompactCells427To458.Box452.compactCandidate_not_isWeightedPerimeterMinimizer	f697758635874a79d91a9dac2b269ba4b4d49a99e230a3d097b2c1c507f51dd4	equivalent
CompactCells427To458.Box453.QInterval	0ec1df2a5f528fc9d5c16e00cda6d357497fbb7006b3a058132277ed28f59778	equivalent
CompactCells427To458.Box453.lambdaInterval	5c02e38903b9bab519d29e6b148cc11192f388940ca18edbd0d136544177adca	equivalent
CompactCells427To458.Box453.compactCandidate_not_isWeightedPerimeterMinimizer	6373249b9bee0a82397c0d428def9b4aadf125929bc8a3cea28c987fae0107f0	equivalent
CompactCells427To458.Box454.QInterval	b1f96df69e8007da8dd6b2dd4c6dd397adf97f85584b8af1c96e894a3291cf9a	equivalent
CompactCells427To458.Box454.lambdaInterval	2fd105e8e7afe3cbca7ceff9b1288c68118d68372454c0a0b4b2096abbc84c4b	equivalent
CompactCells427To458.Box454.compactCandidate_not_isWeightedPerimeterMinimizer	311448ebbeb847cbcf6c6499d161cc68daff0f3bf1e36a215a46a92fc56848b8	equivalent
CompactCells427To458.Box455.QInterval	75a2bbf229dd7b76b42d8154d7a4b7f1711aa7d5894dcb3d22898e5fd7b07176	equivalent
CompactCells427To458.Box455.lambdaInterval	b061a93caedbb00c5215d0e394abc4af5db5cb018e577124005a915cf1e8b9d6	equivalent
CompactCells427To458.Box455.compactCandidate_not_isWeightedPerimeterMinimizer	6ebd77cc03481e09529d7755e2818f28178ec57b726fa0eca5b793795640b6	equivalent
CompactCells427To458.Box456.QInterval	0a4c02709d8a7dff169c634c3e00a4e057c8f141d35c3bc036c7fabcd663f654	equivalent
CompactCells427To458.Box456.lambdaInterval	f1ac27a4cebd6de06e3e4c04f1048b9df86965029538f4ea326e8ed53c92a6a4	equivalent
CompactCells427To458.Box456.compactCandidate_not_isWeightedPerimeterMinimizer	c5570aa038b42c106008f451f7518e57b7e806b296172ef03dc6f5fb6fdee935	equivalent
CompactCells427To458.Box457.QInterval	eea109537146e2804958175c67574162e059fa681f422295604c75087e451e7a	equivalent
CompactCells427To458.Box457.lambdaInterval	e78a898129c07c485ad013ecf5d2820adced5f7d5fad593640be3150adbd9f19	equivalent
CompactCells427To458.Box457.compactCandidate_not_isWeightedPerimeterMinimizer	a2e785cd3792484caf08dcb3a914c6d6cc069549dd8b52522ea3d3117f88f800	equivalent
CompactCells427To458.Box458.QInterval	c8f8c431d7f703224c983d9c3d75f859069e0e556e75a95917b328392e91ab4f	equivalent
CompactCells427To458.Box458.lambdaInterval	5e307d9a2b43ecb19e46fb387e1d337a789f943be51dab2d58552855ca5b1131	equivalent
CompactCells427To458.Box458.compactCandidate_not_isWeightedPerimeterMinimizer	76f8360c6e70c576aab39db72683e0b7adf063e7b8780fdef4ae512f2e6f3c95	equivalent
CompactCells427To458.boxes427To458_candidate_not_isWeightedPerimeterMinimizer	e0b16fdb42feea7420ec6803dd722182023c63369007c60b5a5af45d6cd63111	equivalent
CompactCells427To458.boxes0To458_candidate_not_isWeightedPerimeterMinimizer	8b2d1248408db49b206aedc59cf59834bd17891d05545d10c80cf286db38b007	equivalent
CompactCells43To74.Box43.QInterval	1261c4cca7331e376975e91d01aa4235a2fc5f9aa2e296a768142f2d201290ad	equivalent

Declaration	Source hash	Audit relation
CompactCells43To74.Box43.lambdainterval	8215442a970a3993db087e9cdf8869e8858f3ae02ef1389430f6fc78dd0a5323	equivalent
CompactCells43To74.Box43.compactCandidate_not_isWeightedPerimeterMinimizer	93a8b5dc4819f8e247b51400e398c1829db6f00906524147fbe86d2cf4a78029	equivalent
CompactCells43To74.Box44.QInterval	5a37a9ec3887b9b9ee1109d3a0e2b08b378601789d12ec40145776d71de99d5a	equivalent
CompactCells43To74.Box44.lambdainterval	2269a45ad233ec4bc3ecbb0b9df9c2e11a6be80d69bf08bede0f58ccc53c93ac	equivalent
CompactCells43To74.Box44.compactCandidate_not_isWeightedPerimeterMinimizer	b11aca50a996c5cbecad6fffb891e25b3bb4abfc930640eb5bcc784e79d7a0dd7	equivalent
CompactCells43To74.Box45.QInterval	6f530d66d303c5a180a75d56b846027f530212f3ff6e21b0926be2d0116bac2d	equivalent
CompactCells43To74.Box45.lambdainterval	ce1e80758a0109d5014d744c9466ee424fa51903ffed8014ac3b8d22dd847f89	equivalent
CompactCells43To74.Box45.compactCandidate_not_isWeightedPerimeterMinimizer	68108078406295819cb9167c1a8804b604bb3c24a5aa5157bbbbcbb2238273a6f	equivalent
CompactCells43To74.Box46.QInterval	054926e64b6485910c93cf7780878aa4275da6bb6298cf73f99ca3cf09a9264d	equivalent
CompactCells43To74.Box46.lambdainterval	821d6320896fd36d17a00ce87e0d6cf3ff61425f33d943a197d59fe7b98a8261	equivalent
CompactCells43To74.Box46.compactCandidate_not_isWeightedPerimeterMinimizer	b27f88f8b3ced2c42dadf47c62075d01fb654bc72c941b36360728f69fb4d787	equivalent
CompactCells43To74.Box47.QInterval	db6d2c9a9d1faf028f60601fdc7b8ecdcc7f4ae2f1447400bc49d9ce09da149b	equivalent
CompactCells43To74.Box47.lambdainterval	a60a4b435f0f684f763b9ca4942d9b9b265ffd541dd8d51923198b04a6f2964b	equivalent
CompactCells43To74.Box47.compactCandidate_not_isWeightedPerimeterMinimizer	84067a2d7c1f1d595eb8e54f5c05784e298f1e5f0e0a4784d1e8bf32c0ae6fea	equivalent
CompactCells43To74.Box48.QInterval	750f930199f8966fac74c1ea044c007fef7a5c28e025918e4a73ae1b0ef0a589	equivalent
CompactCells43To74.Box48.lambdainterval	034fe2df7ecc445f3fdb260977927988fb431cc8073045b74191b6e5e90f3eb3	equivalent
CompactCells43To74.Box48.compactCandidate_not_isWeightedPerimeterMinimizer	b8dd5c7a93b372cac5bcedf26e3b9b7e7613a6ef1ec4db01fe4c267c99b2803e	equivalent
CompactCells43To74.Box49.QInterval	7e4cb71a771371b9db2b0cac87ff66c07aa4bac604c4bebf9a95e4ec91d7765c1	equivalent
CompactCells43To74.Box49.lambdainterval	8d283497644b4c96b730d05fa685566b45ec1e82eb42c27791d7a9a45be40102	equivalent
CompactCells43To74.Box49.compactCandidate_not_isWeightedPerimeterMinimizer	31e2836bc3784629860194745bea9201cdb6da0362cbb41295c5cbc75dc59e9b	equivalent
CompactCells43To74.Box50.QInterval	e0a7441cb4db45e013e3d26056e2c043d8cf06cc845f01c01d0d22522dbb21fb	equivalent
CompactCells43To74.Box50.lambdainterval	2f7d38def8f3a0b9b3706ba0feeabf17e9d754bf9e2afffaa2c4583de17bf7df	equivalent

Declaration	Source hash	Audit relation
CompactCells43To74.Box50.compactCandidate_not_isWeightedPerimeterMinimizer	5b209ccdf2c3da828cea2cd1599e92af848180fc6d5ea16dd5d43fbfcfb776ef	equivalent
CompactCells43To74.Box51.QInterval	b60ba65e206ab0d8865adf83a48f822bde47e408396c732fc1b9a4418aab8b58	equivalent
CompactCells43To74.Box51.lambdaInterval	190db8017cbab52d839ebe7b8bef8b980e56aea470005e75e8ac709170babbf2	equivalent
CompactCells43To74.Box51.compactCandidate_not_isWeightedPerimeterMinimizer	7e04f05f6c2bd72a59443702d0513160b19369b00718e5ab59bfe47b3cf12cb4	equivalent
CompactCells43To74.Box52.QInterval	ff0748893af35af5c3e407d91cbe510f3a5d497eb2dc00479341f55705d99984	equivalent
CompactCells43To74.Box52.lambdaInterval	20f9f399cbe1454f15fdffa1c4581b041ebed3ae191909d6902bd456ab475d8c	equivalent
CompactCells43To74.Box52.compactCandidate_not_isWeightedPerimeterMinimizer	90bead8888ae6073ceae83cff6e3439f20339ea64bd0fd6c22a002c8fe4f52f2	equivalent
CompactCells43To74.Box53.QInterval	2eb7e97b40ec3fdf9e2ee54ae7d3dc4dfa0cdcaab2b975a8b4940383a3555bfa	equivalent
CompactCells43To74.Box53.lambdaInterval	c69df8e821fce9ba00a17f4d4658f12d6975dcccfa9856c03294f9ceb1fe84feb	equivalent
CompactCells43To74.Box53.compactCandidate_not_isWeightedPerimeterMinimizer	c2e75679a360817cfe6cd0356678e9c02309c720fbfad93b47f2211427814d1d	equivalent
CompactCells43To74.Box54.QInterval	3f800bcc30972a8960053c11f89c77c7bd308706a741368c5838896166ef6ccd	equivalent
CompactCells43To74.Box54.lambdaInterval	17ade2c5bfab035a898562db430d6476c3a70704da3db5e734c9f1ac9feedf4	equivalent
CompactCells43To74.Box54.compactCandidate_not_isWeightedPerimeterMinimizer	162f8fd98724fdea4d7bc759d1b6a461187136cba9c8a804d6c9e389389777c9	equivalent
CompactCells43To74.Box55.QInterval	dbcd8856fd5c4f59b7a1e8c0b5b8b45bd1db55eca4a0a11f05f58b1b65a2dd4b	equivalent
CompactCells43To74.Box55.lambdaInterval	3798f1499ed03ef25fcd2298e043614655e286ba896f6ec96d7632fb8763a604	equivalent
CompactCells43To74.Box55.compactCandidate_not_isWeightedPerimeterMinimizer	b7952daf857d84e9e5a877af0114eec3e214c7b2b42cc316dc564808ad9a88ee	equivalent
CompactCells43To74.Box56.QInterval	2e894b3e7bdef46531b7e13f73c0fa353f7269feeb84fdcfcebb0a01053cac35f	equivalent
CompactCells43To74.Box56.lambdaInterval	c65c9906ade2b4181b7f26de7c8be76c1c225c44e9446ec0d93af3ceacdb101d	equivalent
CompactCells43To74.Box56.compactCandidate_not_isWeightedPerimeterMinimizer	b8ed6290280439a9f2bb3cfc34f6560666919637295fb1c047727943f6b00caa	equivalent
CompactCells43To74.Box57.QInterval	ef44b3ede9464aa6498133463c0924f2cf058102b8c161e749fce481b98e8db0	equivalent
CompactCells43To74.Box57.lambdaInterval	0398ef385c6368b3ac260ff973fd7e7967d322abb331c335bc3f47e00ce0017c	equivalent
CompactCells43To74.Box57.compactCandidate_not_isWeightedPerimeterMinimizer	fba5dbfe07d5da2defd58110b07a2643f57f7bc873b89880b567d063751e89ec	equivalent

Declaration	Source hash	Audit relation
CompactCells43To74.Box58.QInterval	1d500fce47d738929acf7ef6bca6a2b2526b953821f742cb5282b0aed4050b02	equivalent
CompactCells43To74.Box58.lambdainterval	a2f2dd96505db5f0e9ca7b5e00822710cdfdbec738fb9e8de57d79a62dcb6bc9	equivalent
CompactCells43To74.Box58.compactCandidate_not_isWeightedPerimeterMinimizer	06a0dd0b2763606f9f07a0a16dc8f26c54cbd75180a8b16561759a0e15387cfa	equivalent
CompactCells43To74.Box59.QInterval	017b30b75ffbe3cc96599c327f79a2ae7efea074098612a54ac97a6c53023ced	equivalent
CompactCells43To74.Box59.lambdainterval	554c099a192b11604086e0464ddcdcc9091701383fc905ab704d15d4665dd2ae	equivalent
CompactCells43To74.Box59.compactCandidate_not_isWeightedPerimeterMinimizer	2b096ce162c52d6bf33996cc1eaa864ad87a550052eb581d4708de281dfcb239	equivalent
CompactCells43To74.Box60.QInterval	6e9b27174bc6603e1570675cca1456cd897e3ee6c8d82dac9063b1d8bdb264ac	equivalent
CompactCells43To74.Box60.lambdainterval	d650ab34e2c33e07980bc64c6757250b4c041a8c4a968360bf3859fce1b018d5	equivalent
CompactCells43To74.Box60.compactCandidate_not_isWeightedPerimeterMinimizer	f968d849cb0946380b3b679a11b784cbee15e48c6bb6a8fa67aff8260acf0120	equivalent
CompactCells43To74.Box61.QInterval	bafb7f457f6d60f70f9f3ad225baa2d4e6339d0a58914e26c4d0ed0b123ecf86	equivalent
CompactCells43To74.Box61.lambdainterval	e92146d71092e7e3d020b2e0f71d94e4e70a7006294056ff173d474092711571	equivalent
CompactCells43To74.Box61.compactCandidate_not_isWeightedPerimeterMinimizer	9d695cc5ea35c76f56936cd49e29d58365cd9b98547847f4f5ba66b83dae7bdf	equivalent
CompactCells43To74.Box62.QInterval	939eaecdc896f91acd80f040486a1813e406fc34c7f287d7780528431e1fb76a	equivalent
CompactCells43To74.Box62.lambdainterval	81da428a5ea380b8cb68d7ebdfa498dd43e6d848e9ed30ce02473b23f7706dbc	equivalent
CompactCells43To74.Box62.compactCandidate_not_isWeightedPerimeterMinimizer	20e395eb241db6752c4ea4c3212f40e253fc74a309b439c65c2be675bf6d826f	equivalent
CompactCells43To74.Box63.QInterval	28485561aa8d972dae4b5de78fd04f94179170518da30e81773b0de692bcf228	equivalent
CompactCells43To74.Box63.lambdainterval	0118296a2b484821c9a3fb78996e889366e5465bcfb2b9c1e71d373517860221	equivalent
CompactCells43To74.Box63.compactCandidate_not_isWeightedPerimeterMinimizer	8f8b8bbd38f41f2b290508cdb32b3f0c32afb616577423af8dfd1c0aa7ab2dc5	equivalent
CompactCells43To74.Box64.QInterval	a0cdf8fd4871eeba0c54606199b032b7640b04758209d448f61406f031dc8c74	equivalent
CompactCells43To74.Box64.lambdainterval	74a754f5af46cbd0b4f960c96936065242b108c8f13d81a9a266cc9bb3f2e890	equivalent
CompactCells43To74.Box64.compactCandidate_not_isWeightedPerimeterMinimizer	aab883fa654eba4f8b669295129dd99a35ec60b994847a9b864fd47b733a7d55	equivalent
CompactCells43To74.Box65.QInterval	d9456ab76afa02b7c8dded329f9319e796c58e693e24e72460d1d397750e5d25	equivalent
CompactCells43To74.Box65.lambdainterval	169f990120e2469839e3b73ecaf0c4cd73a76d62ecdd2f5cb0a369a70b8b6fce	equivalent

Declaration	Source hash	Audit relation
CompactCells43To74.Box65.compactCandidate_not_isWeightedPerimeterMinimizer	c83a46ee79e17481a4c90a3506a41dd05412369d3ecc452acad5be105a94c6a5	equivalent
CompactCells43To74.Box66.QInterval	3d5508ffcf5317a29e2673fb991b1362b8c2a9a5734fc3c47883381fcb10989	equivalent
CompactCells43To74.Box66.lambdaInterval	d64945d3aa5cf4af984621f496b1e699e8b52588273aad2b310405c43eaa8075	equivalent
CompactCells43To74.Box66.compactCandidate_not_isWeightedPerimeterMinimizer	7140181c83713ef747c8f97381d2d6fe776a3a5d145f3af340f88dd248d2f85d	equivalent
CompactCells43To74.Box67.QInterval	b18225b9c9b927c915ac29c0c02ce45ec712a103eebb9fe7d3abcf6f72551dee	equivalent
CompactCells43To74.Box67.lambdaInterval	c254c3531067eda5c6f75e7cd8b3fe8901d956596677f380ec2adba6e6099561	equivalent
CompactCells43To74.Box67.compactCandidate_not_isWeightedPerimeterMinimizer	f5cedf091e430ad9df75a58be99b500676626e4037dcf6efc9b3ff706027095d	equivalent
CompactCells43To74.Box68.QInterval	8de528a2707ff93ab2cf1a16b1ad625b37e3a95a41cd55e270afbeffc332957a	equivalent
CompactCells43To74.Box68.lambdaInterval	f1e87e6e9e893b824fba763ac353cc14232909730664fbd8512f489cd1bc6214	equivalent
CompactCells43To74.Box68.compactCandidate_not_isWeightedPerimeterMinimizer	8334d4a03a1198e262496f5c679f5059bfe1166fc8d65659a5878dcffac52838	equivalent
CompactCells43To74.Box69.QInterval	e8e015800831a7111f2ea90d6d3f211628cf72460ed89029ea0c51fdf811eb20	equivalent
CompactCells43To74.Box69.lambdaInterval	f7aeba115f34d4969a43003d3b91de4831a958687e6379a49ff3746f018b6398	equivalent
CompactCells43To74.Box69.compactCandidate_not_isWeightedPerimeterMinimizer	e3e53e58a9a59b117b157509112a7c12fb1a26cd3452ace0bf9d1981a0b2ee3e	equivalent
CompactCells43To74.Box70.QInterval	62eed18f2fc14cc2b639f3b76a5eb5b1af60f226485d3b81d86ca6f4f4a4833f	equivalent
CompactCells43To74.Box70.lambdaInterval	e461d0d5f94dd6022abf0786dee48bde8e7df5f24c794c2c8bd703d08ac28b8a	equivalent
CompactCells43To74.Box70.compactCandidate_not_isWeightedPerimeterMinimizer	bd46be24bc96458bc1bc8c7e4860c0bac3914e33ee098eebf0b3d69dce102dc9	equivalent
CompactCells43To74.Box71.QInterval	889c78e50c0c964641848be9825bd7bfe4bdf54aabd4ce965f878c18fb4aa80b	equivalent
CompactCells43To74.Box71.lambdaInterval	3daef6dc9626deeb3a9f04a6160787bb8a826c74efbba618807ce3fbbd1ae7d	equivalent
CompactCells43To74.Box71.compactCandidate_not_isWeightedPerimeterMinimizer	734006c5743e342f3d4c0926f3f36c4330b6ce847890c386a861978595366a31	equivalent
CompactCells43To74.Box72.QInterval	1e264414824aced85fb66dc7c5907c2a425516488a3eb0af63d91d3ca64823a6	equivalent
CompactCells43To74.Box72.lambdaInterval	b3fb98adbacb9bd932d24a983baeaf83dc50787be8baca21e9d7fcd8c8cdc140	equivalent
CompactCells43To74.Box72.compactCandidate_not_isWeightedPerimeterMinimizer	7b0e735575c318af1c7f299548fe38520af7e686b8d7be54ae35cdf3942e2939	equivalent

Declaration	Source hash	Audit relation
CompactCells43To74.Box73.QInterval	849cd971d41b79e7af538b93841f205560e cae5852307f2fdb267873e66add85	equivalent
CompactCells43To74.Box73.lambdainterval	1bfe66e1d2738e27fda346c5133f3152a59 25a2f481c5c3941c7e64a6a6a9fc8	equivalent
CompactCells43To74.Box73.compactCandidate_not_isWeightedPerimeterMinimizer	4858c8837cd7dd0e5c6966437dff8d9edf2 f8da84e2a652f25244411e50df5cd	equivalent
CompactCells43To74.Box74.QInterval	f8994da3096dfd2bb7f31929a0f8752ba40 900cd88353714911bdc2745037210	equivalent
CompactCells43To74.Box74.lambdainterval	9da36d762f052853d2d0b09c51cfdefa3e3 f77e8db61eeb89ca36f9777d05220	equivalent
CompactCells43To74.Box74.compactCandidate_not_isWeightedPerimeterMinimizer	6c7282268fff20e2126edbf0979123c528 5108017283f643bad7c1e4ff505db	equivalent
CompactCells43To74.bboxes43To74_candidate_not_isWeightedPerimeterMinimizer	899c5a82ace73846d50118707c3d3dbbe69 807b75da9128da1fca8d919e346cc	equivalent
CompactCells43To74.bboxes0To74_candidate_not_isWeightedPerimeterMinimizer	19a57bf71e6f4e645925e362a438f84e571 c445db421bd1aec567a18e07ae39	equivalent
CompactCells459To490.Box459.QInterval	52903f640dde5a600fb0677517b747db2c0 a6377f3cd5deb5c7dfe1065235a38	equivalent
CompactCells459To490.Box459.lambdainterval	3e7a349fad65708b04976ee3d08e361538b 9ee6abbfbfdff716a7fd5520c71c0	equivalent
CompactCells459To490.Box459.compactCandidate_not_isWeightedPerimeterMinimizer	f1edb2f7e2270167f49bcdd7ee0edf0b17f 2709b6d5de4044f967551fcd99775	equivalent
CompactCells459To490.Box460.QInterval	0297e1805b602c6474e5867108b72643826 fd9f89fd13db0558bbe41e9992b0b	equivalent
CompactCells459To490.Box460.lambdainterval	9b3e576178fab309d7c6e3b399ec23d8ab3 9a1553e3dea2e7140f48383b1276f	equivalent
CompactCells459To490.Box460.compactCandidate_not_isWeightedPerimeterMinimizer	23dd8ed191c476b8b84d811d62ab2a1670a 4c057a0549572415a33c502f0ba74	equivalent
CompactCells459To490.Box461.QInterval	6646be10f46e75ba667b713b87efaffc23f 1cdd1551f9f31bc6a76d167a00127	equivalent
CompactCells459To490.Box461.lambdainterval	c21cfc67f7a38508d9e6655239aefcfa45e fc0cb0cf541187cf6059a0a77a618	equivalent
CompactCells459To490.Box461.compactCandidate_not_isWeightedPerimeterMinimizer	5d41577405a6740c2d19b83fbd50ab5d9d5 b21d64c25d0f32b8731739c696fe4	equivalent
CompactCells459To490.Box462.QInterval	9e79ad8649e8c087447088939b02af133b2 f2dbd8e0a3788df3a978c114f4ff2	equivalent
CompactCells459To490.Box462.lambdainterval	98889d1c37e26e8121e62a3ebe67f4a0327 f648458eb6457c88105d81219d7b0	equivalent
CompactCells459To490.Box462.compactCandidate_not_isWeightedPerimeterMinimizer	b20308c9b89c64bd576cd17382d29a39abc 0de9919e9560e6837916cac719169	equivalent
CompactCells459To490.Box463.QInterval	2876c9cbacc9f5d4f055c2131741ea53af2 dde2d956642124f75d586805480eb	equivalent
CompactCells459To490.Box463.lambdainterval	dba89f39dca7f6f3819b3969e69b9f74520 b549801381e69b9311ef1d1af1698	equivalent

Declaration	Source hash	Audit relation
CompactCells459To490.Box463.compactCandidate_not_isWeightedPerimeterMinimizer	f6d7574bb3e557cacf12bf9878c9fae9bb37e3cf3675874566e81b068e683d20	equivalent
CompactCells459To490.Box464.QInterval	a37abf85657a48490f5b881fba25e4c8363797bf6b07944b4fe311e34ce01253	equivalent
CompactCells459To490.Box464.lambdaInterval	5130293d01706c05629e59711e503cd41a4c3c87d10a758908bd0a29b3d87672	equivalent
CompactCells459To490.Box464.compactCandidate_not_isWeightedPerimeterMinimizer	85d39014566fd29e77bc49ce86443989137d1e0a523082736f00d3e42a734e64	equivalent
CompactCells459To490.Box465.QInterval	fe64da221eb70144157ae58729138a17492bdf8e08d25fb1452c698c08038dbf	equivalent
CompactCells459To490.Box465.lambdaInterval	badd90e18c0712c3cefb8e86590c403b8d0e818585f8bf35cbb7690bc599d7db	equivalent
CompactCells459To490.Box465.compactCandidate_not_isWeightedPerimeterMinimizer	9110d08478142c0552b2d5363a01e94591e4cc178fc9a904ecdd9f4678288c37	equivalent
CompactCells459To490.Box466.QInterval	27c8f653eb15aa41351fa8fab002f53ede a7ff5865ae89948e29606747c4522	equivalent
CompactCells459To490.Box466.lambdaInterval	75def372e993534c0f76982872c6156a62e beefaa3ce03804a96dca400ab89a9	equivalent
CompactCells459To490.Box466.compactCandidate_not_isWeightedPerimeterMinimizer	6374fb5ca080788862365d896665b687bc950d97a154d4f9e69be5a110c2c8b7	equivalent
CompactCells459To490.Box467.QInterval	b2de94ccdb890e82c1b7cba8b24c67a0f9969d1f4ebc7e88cd1b6af68d287cb1	equivalent
CompactCells459To490.Box467.lambdaInterval	44d8db37c655cc1ec36d9d2996b8e537dc283587d51ce9a5a556b8b879abd685	equivalent
CompactCells459To490.Box467.compactCandidate_not_isWeightedPerimeterMinimizer	923813d1157aeb9e48b1e74a189496d93d25983749615308fab552c171a7dc28	equivalent
CompactCells459To490.Box468.QInterval	d257ce2c1a1b0d6efcd2f8281723938acdf77e9b40bce77a3fe1e4ec961da305	equivalent
CompactCells459To490.Box468.lambdaInterval	ff04b5cea4902cf2851600d208b7049e64972229d41b3be2804d61a7ab9641f4	equivalent
CompactCells459To490.Box468.compactCandidate_not_isWeightedPerimeterMinimizer	724e38bdd4b63da6d033480519b6253d45e0f088fdb056cf8d701d96e90cbefe	equivalent
CompactCells459To490.Box469.QInterval	ffe9c1286d2b4a30b5c4c18aaed980749809baff619bec3163b2312b0611d076	equivalent
CompactCells459To490.Box469.lambdaInterval	133145e7c56bf94f99ece4ff4682fe2412c713e3eeee07259acad141ca47a9cb	equivalent
CompactCells459To490.Box469.compactCandidate_not_isWeightedPerimeterMinimizer	bba03dfd0fe777b9e149449e37818195dad5bb36c2c6cd5222727ed9340a7f8	equivalent
CompactCells459To490.Box470.QInterval	bead64a31bf243002ad726e1ea9101b0915f37e55a5cb4c0411b806b339af2d	equivalent
CompactCells459To490.Box470.lambdaInterval	4a449a5be23af67d0eb92c26bbdc02f2a3c4897478551e652ad517e483a1aa85	equivalent
CompactCells459To490.Box470.compactCandidate_not_isWeightedPerimeterMinimizer	c09ce9b4e58ecce21d0dbd1d2cd93a629f3d89f5c5fe2f0d2a680ff54f92737a	equivalent

Declaration	Source hash	Audit relation
CompactCells459To490.Box471.QInterval	af18b86305c66c18d27034ba9e5772ded07397bfc201e36eb2b76f287af8774f	equivalent
CompactCells459To490.Box471.lambdaiInterval	a1b01d8edf62e7ec36c2019e2b97369960cb56f5720ad828b93e0232e252c324	equivalent
CompactCells459To490.Box471.compactCandidate_not_isWeightedPerimeterMinimizer	c09b2d249f76fe0f95be4ee46b68b37ed8177cd3c889c3b860f0f5c6bd03cc7c	equivalent
CompactCells459To490.Box472.QInterval	f452253de8736cf53da8f41a8e6cf61734fead0150b70be479ecd84ff2cbf2ff	equivalent
CompactCells459To490.Box472.lambdaiInterval	9a2e53945947682f2d9fa718d4569300c2fb82400451099cdd180493a503a206	equivalent
CompactCells459To490.Box472.compactCandidate_not_isWeightedPerimeterMinimizer	2cc18fb92b49cc7cb1acb45eefd34698bb95741142965318091bd1b6bb36eb59	equivalent
CompactCells459To490.Box473.QInterval	6222d9c0f76b01798e712273c51da4eb1c9905cac716e7e4cfc12505e91146ef	equivalent
CompactCells459To490.Box473.lambdaiInterval	8a06cc2f2db289505bf8ed7ee4031248135536c7d08780ec38a92989d006b1e8	equivalent
CompactCells459To490.Box473.compactCandidate_not_isWeightedPerimeterMinimizer	d30738899c0c63b8bbf49b316f6ac8bf979ecb227a551dc6b26d7eda190c0a5e	equivalent
CompactCells459To490.Box474.QInterval	ec837aa8822f3c9ffea8bccbb92249bb560f3eea2855fdc246547f62d3277590	equivalent
CompactCells459To490.Box474.lambdaiInterval	1972f4dc925f75d5742a38807f0a21ff1861e55177c3952ba7eb202e2c11dd14	equivalent
CompactCells459To490.Box474.compactCandidate_not_isWeightedPerimeterMinimizer	4660ffe10835d581853c89880b73bf9436eb6f592c5245eab7fe59703e72383c	equivalent
CompactCells459To490.Box475.QInterval	a58a1c64bc48b12cb70c33f30105dbf302b6aa99aa74d9ddc9ec66d636350404	equivalent
CompactCells459To490.Box475.lambdaiInterval	c22d1eda5370f7c63975e4f336344a03fd712692202f3af314ef4a13cc36c09c	equivalent
CompactCells459To490.Box475.compactCandidate_not_isWeightedPerimeterMinimizer	7f445e8920ddaa125452fdd91df94b3441d072cd077b226d84cc218eb33e2b31	equivalent
CompactCells459To490.Box476.QInterval	bf797dccdc85d275ad116773da45e10674f5beab5b49f3ac52e8fb2c17b2b242	equivalent
CompactCells459To490.Box476.lambdaiInterval	d483af74702ed14c5a56ef1293152ea847b58c5ffcd816d16952d022759d002c	equivalent
CompactCells459To490.Box476.compactCandidate_not_isWeightedPerimeterMinimizer	743ef470b1692776f996df7a525a67591abe3171e5cd8d55e89f1b50d9baf688	equivalent
CompactCells459To490.Box477.QInterval	6c4ed6288553acc83f856fd48c6e4a74d8a5b18ea9848533368fb8f133d433f9	equivalent
CompactCells459To490.Box477.lambdaiInterval	8e0174967449fe210117220d6635b122487e9eee2bddeea3b89f60a7535746df	equivalent
CompactCells459To490.Box477.compactCandidate_not_isWeightedPerimeterMinimizer	01144bc5f70910e93fd69ca3df771f9e3e4fd00f84312d5403a06afeb1735595	equivalent
CompactCells459To490.Box478.QInterval	b23efa1efef0d8ca461742cfc7d48805571e1782eda1420c2547bcf35f31ddb4	equivalent
CompactCells459To490.Box478.lambdaiInterval	031b07fa25aea43782c3ba64d54029cac915ccbec328cfde7c825edbd0c1225ac	equivalent

Declaration	Source hash	Audit relation
CompactCells459To490.Box478.compactCandidate_not_isWeightedPerimeterMinimizer	002fb7722d9e1d136ff9028c3a2e70e756cf93cf57e9383141885d2df39ffc94	equivalent
CompactCells459To490.Box479.QInterval	20ac7fc0be6aca2e75325d2d44d92406fae6ae59045eb03d5bae2838057e382c	equivalent
CompactCells459To490.Box479.lambdaInterval	d6bb3f6a64cb8c72173181376ac02411ee4f0b59b8bcb5e011f74e64def4438d	equivalent
CompactCells459To490.Box479.compactCandidate_not_isWeightedPerimeterMinimizer	5545964dcd9fe0460cb6f7276cdf7fa7d2a001dd75d9cfcd9db86209237db414	equivalent
CompactCells459To490.Box480.QInterval	68e9bfa0c9862156d1546c20097d5cb2b516b7517abbd58e99f290958177c251	equivalent
CompactCells459To490.Box480.lambdaInterval	bd8939df406a4bc00f7c89d04b707f33fe86e3abf086ec44494377d6ab4f5e73	equivalent
CompactCells459To490.Box480.compactCandidate_not_isWeightedPerimeterMinimizer	f50d214f505a1f19bc6f9c8eb1d0d3d2b59b176716ba8c81b987da8927840148	equivalent
CompactCells459To490.Box481.QInterval	1d96fc44ead5d1ca5097df070986e83549eb3f636e6ba2fa48dc15003c227090	equivalent
CompactCells459To490.Box481.lambdaInterval	c38b42afd6c97f1448aa2692415438744a5ddd9095972970819fe88f0e34922a	equivalent
CompactCells459To490.Box481.compactCandidate_not_isWeightedPerimeterMinimizer	f3ba628c4b5c95f66fe98ae45a75c4e73749aeb9349ff3a5147000d4efc1c047	equivalent
CompactCells459To490.Box482.QInterval	cb7db6cbf108b65e24ff1fa948e925077c82517b10edbb38ff21de037749d57f	equivalent
CompactCells459To490.Box482.lambdaInterval	de00879da4524ceee005a17d7ad561ca3b64f41f27e973aa55846a77b7fd54a2	equivalent
CompactCells459To490.Box482.compactCandidate_not_isWeightedPerimeterMinimizer	d52a8bd2c44a312891e7434a45448388645c5f7bc425a536790184ad5a076ade	equivalent
CompactCells459To490.Box483.QInterval	0502ded01a935960b0135f2365ff4d0bc6e6f0adc048b8c605815893a0ec28f4	equivalent
CompactCells459To490.Box483.lambdaInterval	f02d65b5fbd9b2ad82b459cc8350fe59115ce820c84ce9d774c914ed5aec20f5	equivalent
CompactCells459To490.Box483.compactCandidate_not_isWeightedPerimeterMinimizer	63b4f87afacd1ca1a5b7e4f079401f38360588d1f5928514fc53662131ab0dd8	equivalent
CompactCells459To490.Box484.QInterval	7defff93e844207663ffa465ecbe176e38ad4d5beb04173b57f32b6ba7239ab1	equivalent
CompactCells459To490.Box484.lambdaInterval	622241801ed83a2ebbe4b93421c9c9c898d01f649a636f3bc83353302f6e51cb	equivalent
CompactCells459To490.Box484.compactCandidate_not_isWeightedPerimeterMinimizer	ca014f69ed5aa5cc0484029671d38a86029db475a3cb1964e82709128c1b5fc7	equivalent
CompactCells459To490.Box485.QInterval	0e00d2a02ad916c9a0e9f13a6f520efeb430d189ad1131243053dd8567b7e361	equivalent
CompactCells459To490.Box485.lambdaInterval	ddba1978970f34fcfac14ed53c5a175d95841b7c3f5a564e0437fd86b52c8486	equivalent
CompactCells459To490.Box485.compactCandidate_not_isWeightedPerimeterMinimizer	23101b72fe58aadcb92088d4e9176d75da3e4dd43e88218b273d9be5ced2cb42	equivalent

Declaration	Source hash	Audit relation
CompactCells459To490.Box486.QInterval	902ceb24f71125e9e58c0f09aa2fbd9257d4bbecfe20969c0d8e6760912d6533	equivalent
CompactCells459To490.Box486.lambdaInterval	b8592803abdc728953adbf5dbbf538fe52243ca4818db67eb8fa70e2c1f38cd1	equivalent
CompactCells459To490.Box486.compactCandidate_not_isWeightedPerimeterMinimizer	e99daeda5d3ae53e2ba0ac34e4f244ec418442f4fcbba9a8b3f86d8c2e01e0b0	equivalent
CompactCells459To490.Box487.QInterval	ff616a4569f54097d0db24f54013882c98d6ec79c6acac1991a1c999d2672300	equivalent
CompactCells459To490.Box487.lambdaInterval	7c56859922394c5dd1c552b0f645fe1b00c5d0cf1aac9a948a7d1c9766eff09f	equivalent
CompactCells459To490.Box487.compactCandidate_not_isWeightedPerimeterMinimizer	a8f893f0c6e302ebe9630b11f2b12422b3f2e03d5a6a80d944a1ef7ec2db758e	equivalent
CompactCells459To490.Box488.QInterval	564da7988817fc7de5e6af1888e14699577a994c6112c0d2a307c1d52b226697	equivalent
CompactCells459To490.Box488.lambdaInterval	aeb9d7b61d75c3efb63cd0f99c32300805992e332be749f0bf49d8e3d2bba1c5	equivalent
CompactCells459To490.Box488.compactCandidate_not_isWeightedPerimeterMinimizer	fa2f9c1db165d170f2cf5156dc9715399a4c41a0ccf0273f1d3cdc712efb133f	equivalent
CompactCells459To490.Box489.QInterval	54a91f82fc9b82d557d64bc5d5f7dee91ab08db14580d89f634b4813e8473627	equivalent
CompactCells459To490.Box489.lambdaInterval	5732d0670f554cfead118291b56e1d324e53f52f2d5fcdb907b6da9f596f3120	equivalent
CompactCells459To490.Box489.compactCandidate_not_isWeightedPerimeterMinimizer	d8f4fde36411c4a38f50c1e02d378bb9eed0a43bcc3d391ae20450413682662c	equivalent
CompactCells459To490.Box490.QInterval	495ac24da6cccc019535c9c9f047e5b7a4ad24a4fa3eba2acefee5bce42d9f21	equivalent
CompactCells459To490.Box490.lambdaInterval	2bb97661d89fa0278602d94f860c965baa7fc88965f5b4bb43d324b8daf4cbfc	equivalent
CompactCells459To490.Box490.compactCandidate_not_isWeightedPerimeterMinimizer	559db61d5e7ab4c17f2ac2619d5781d5f6f5915e546d17f87f0f4a1875a528bf	equivalent
CompactCells459To490.bboxes459To490_candidate_not_isWeightedPerimeterMinimizer	27ad82bf73b6abb8762fc45ee2a255aca2971ddc7f458b820ac0c4925e9acffa	equivalent
CompactCells459To490.bboxes0To490_candidate_not_isWeightedPerimeterMinimizer	33fb8ab16ea6825a544fc1a3086d20f4e603f22327858bceaaa87909ad993a65	equivalent
CompactCells491To522.Box491.QInterval	2caa802180db4525643828346dee0e4bd65fa2016fa8519f099c6d043356b387	equivalent
CompactCells491To522.Box491.lambdaInterval	839fa7b16604f10ac6a21ba1b27dcacbe3033f471d3a31b1509f83f4a5e22635	equivalent
CompactCells491To522.Box491.compactCandidate_not_isWeightedPerimeterMinimizer	ac4202f31f5fa7c6e1ddf753ad18d772a874e5b36be4fd767c18f384ea4145d9	equivalent
CompactCells491To522.Box492.QInterval	fd03fd73b6ca656d38ab3b871c9016b7a835468df316f1a74dd5e7aafca6be12	equivalent
CompactCells491To522.Box492.lambdaInterval	b2a2ad79309c343d27605a6ef939ea03e56c6a40b52d868af022dd5143539b66	equivalent

Declaration	Source hash	Audit relation
CompactCells491To522.Box492.compactCandidate_not_isWeightedPerimeterMinimizer	b6a33122f6153fb29fbbc91a56e7677328841412030fcd0a8e9c7342cbc28bc3	equivalent
CompactCells491To522.Box493.QInterval	046a3c16c611b329d19343dc720e5813a54a5682ff83df97e872b1a420f2bb5d	equivalent
CompactCells491To522.Box493.lambdaInterval	9c07ab1d79712ad223669c1c203c0e77874ee44327e49b7e1277264ecb1327cc	equivalent
CompactCells491To522.Box493.compactCandidate_not_isWeightedPerimeterMinimizer	9f64f6980a0842652eed2985becf145e24eb9001e9e7cece31f418c2d4a79fce	equivalent
CompactCells491To522.Box494.QInterval	ac7896d932f851bb4c9f790ca3b1a3f33ceb6292709aaf4d269382cb7b1308f	equivalent
CompactCells491To522.Box494.lambdaInterval	07fcb9048731018f75610de192b83f4e35d3901e8477ab0f79c240c04a4df856	equivalent
CompactCells491To522.Box494.compactCandidate_not_isWeightedPerimeterMinimizer	ff10f7236006880e4111940890847cb129ed99329ad6ab35b600271293bf9641	equivalent
CompactCells491To522.Box495.QInterval	e654407911c7603390e8973119d7b5699896f3a0ec2980a48a0649a9aae9aecd	equivalent
CompactCells491To522.Box495.lambdaInterval	db23d39a3c491193111f118f2cea54ab8c78a9b8c96a3c0c4d8cfc7d870b2dd7	equivalent
CompactCells491To522.Box495.compactCandidate_not_isWeightedPerimeterMinimizer	4532be52cce096892adf1684a0c5fdec75725345cae32354adad0ba2dfe5ede	equivalent
CompactCells491To522.Box496.QInterval	872a17bc5fd8bd9dc2ff572e77e21019d7de85beb2645b6cee9a00e0675a93fd	equivalent
CompactCells491To522.Box496.lambdaInterval	34dbfd054b1c8c3b148ed7b3671245f89d35ab035b68440e20c4a9a8f8c82f6f	equivalent
CompactCells491To522.Box496.compactCandidate_not_isWeightedPerimeterMinimizer	3835a9a2c1a163793740d7a314c8901a2bc48c35941679f03cf6ad9adab93926	equivalent
CompactCells491To522.Box497.QInterval	66ff02264e117ac4799aabeba756cb204b829815e6ed33fe17573f58c4820e52	equivalent
CompactCells491To522.Box497.lambdaInterval	b59fdde634a83fb3d48d52481fec2f729ee36718e7e97986caa330f82869452	equivalent
CompactCells491To522.Box497.compactCandidate_not_isWeightedPerimeterMinimizer	d3de44f0d2185da2f9b86dd4ec07a85835e5cdfdf679ee057ca5258abe6dd925	equivalent
CompactCells491To522.Box498.QInterval	4a5d83fd1db3c5d2751333497211b1dea1a58e8dd2e0e349fb404f62af3e7049	equivalent
CompactCells491To522.Box498.lambdaInterval	3591ab8602f4a8c49722864b2cc3898af59fded834033ea3713269fe0cc67740	equivalent
CompactCells491To522.Box498.compactCandidate_not_isWeightedPerimeterMinimizer	5629f07b23483ce030d8bfa0a2a2b1477a98d4dfa71d95a6091c279103261979	equivalent
CompactCells491To522.Box499.QInterval	4abf24c49ba56f2e7e2e1afddd607e06c0e715b8e476680e6c5ea2eb8cc485cf	equivalent
CompactCells491To522.Box499.lambdaInterval	a3585c4e1ca33f9e5d6b3d1a3e6a6e0a3fafcee407c69fc01d60e2d7b5a8f02a	equivalent
CompactCells491To522.Box499.compactCandidate_not_isWeightedPerimeterMinimizer	f93133a5091653445c502a8e1eb789628d53a7b60f0bbac907cde35fa490379d	equivalent

Declaration	Source hash	Audit relation
CompactCells491To522.Box500.QInterval	9a389f78e6ec4372d1ea956d424437bcac1e650b2592c3a250dc0f6fc08f0b15	equivalent
CompactCells491To522.Box500.lambdInterval	ead17e4c7e5aa3c32ac6da502013e470ef5ccb3c1a4acbd146f8cd8bc84274da1	equivalent
CompactCells491To522.Box500.compactCandidate_not_isWeightedPerimeterMinimizer	63d188611ebc9c4806818b3ff83330ca1c0953557d5240826ffeeba01f93c4b0	equivalent
CompactCells491To522.Box501.QInterval	58f13e67a08e5054e8f6c9811b9863295c4f90af55185e407c99a0aec5454ce7	equivalent
CompactCells491To522.Box501.lambdInterval	eb9d9fd84b2efcdd8ff96b1fb744dae5912be00f4bc0f5a9dc97066f1c7072c2	equivalent
CompactCells491To522.Box501.compactCandidate_not_isWeightedPerimeterMinimizer	804031c0a8104a9e59a03719e66574e185a6eb4a31272ccd9a77a2c60e436f18	equivalent
CompactCells491To522.Box502.QInterval	0a35a65427c1cccf8b5dc6106ce69cb0075f8f72d0c215fe0da1944134299795	equivalent
CompactCells491To522.Box502.lambdInterval	ad5f0bc82d3de7c5639cb1a7235ebb7b16702c070ec925dcdae2d561651407c0	equivalent
CompactCells491To522.Box502.compactCandidate_not_isWeightedPerimeterMinimizer	85ac2a95b34598c0d93ddc41b4ffd0fe21434b061341bdea16e67b7553136ea8	equivalent
CompactCells491To522.Box503.QInterval	506de77ab469aeb67478ccda50acb2036f513550a5d24de9ff5e4fe4a5dbc334	equivalent
CompactCells491To522.Box503.lambdInterval	03e2cc212c8857748cdd0b7a90de493f46e1fa39b5b07c45cba31ba15fea959c	equivalent
CompactCells491To522.Box503.compactCandidate_not_isWeightedPerimeterMinimizer	309d7eb400d841a7daf4dd3cc65d2594255359b9ce1cdedf8522a3b0a730d9a1	equivalent
CompactCells491To522.Box504.QInterval	b345dc55c2acc75847ba567c23671e5d6f58c2ba00391c789b9bd469b5e60f9b	equivalent
CompactCells491To522.Box504.lambdInterval	43749bab63740d72e8ab4c251ba27754f8d7a89197da9d0af4fc28e9cccd971a	equivalent
CompactCells491To522.Box504.compactCandidate_not_isWeightedPerimeterMinimizer	0fca3809e36cf565d2c13355bb14677018177be01a9c948dfa14e7a87c517afb	equivalent
CompactCells491To522.Box505.QInterval	60c29608c31f0149dc5730def007efbc36b4e9c064976f888a7fd3a38c77ef7f	equivalent
CompactCells491To522.Box505.lambdInterval	3e3f53594ab3ef30dc202df5ea625169e1b1e808bcd08c8f64a6789e5c37c66d	equivalent
CompactCells491To522.Box505.compactCandidate_not_isWeightedPerimeterMinimizer	464e466d8213abbbe65973ea0ce35e7123efc81fc0d78938459b7f0dfe50abc2	equivalent
CompactCells491To522.Box506.QInterval	5add6483d020c447da6703c4a66d62a6f2aa59da63e30594e55e161fbcbc13e1	equivalent
CompactCells491To522.Box506.lambdInterval	52692a38757a2f591b9207a09a3dbd177ebc6e28984c0ff94c91bb1b90beab5b	equivalent
CompactCells491To522.Box506.compactCandidate_not_isWeightedPerimeterMinimizer	55ce169726d539fb4ead5893ab2f8cb15d65062f3d1854012716f7d9d74584ea	equivalent
CompactCells491To522.Box507.QInterval	86c631d49138bbc457999e1515cbec1cf561d34b500c2e467d352fa142fce423	equivalent
CompactCells491To522.Box507.lambdInterval	e6e8f6344ddfc8aa732eae634c5ad14c24582147b5cfdb7323d95b5d6c34961	equivalent

Declaration	Source hash	Audit relation
CompactCells491To522.Box507.compactCandidate_not_isWeightedPerimeterMinimizer	8a790c53416fb4c9a924c87aca35c7ebdc0c5c930584b44b0752df50c16c6ea8	equivalent
CompactCells491To522.Box508.QInterval	40b1de33df560728a6384315941b3ab0141ff37c94a1b6d7939ad048e5540858	equivalent
CompactCells491To522.Box508.lambdaInterval	8056fd16a2444a6060c3680a0d8dec83666bac2ed959a8847a85feb1cd8c279d	equivalent
CompactCells491To522.Box508.compactCandidate_not_isWeightedPerimeterMinimizer	56c5fb76c1b6565ccab831bd6f6416333b0265a6df73fd5e84e1bea6e58a2486	equivalent
CompactCells491To522.Box509.QInterval	d2f319bb1468583b7ea0b15ffe78369ec6c88ca3e885957e407b904af1b0e879	equivalent
CompactCells491To522.Box509.lambdaInterval	0c6c379481d7da337310d93d6f254d516c4b4ef83e230a8cd7ce9502e54c7bb2	equivalent
CompactCells491To522.Box509.compactCandidate_not_isWeightedPerimeterMinimizer	30a9da5d1c2110d814072fcac7d66606b9f0508ccd4801b1c794d565ea12c445	equivalent
CompactCells491To522.Box510.QInterval	a6b9c529b40d22213c9f243f725e5a3aaeba2ee0e093edceb88c14d8c6678861	equivalent
CompactCells491To522.Box510.lambdaInterval	fddfe9527f8bc92282b6748126e51ccdbfe15c5db89b39de4b0bb3720445dcdb	equivalent
CompactCells491To522.Box510.compactCandidate_not_isWeightedPerimeterMinimizer	5cf68642652e93ecce889b87aef101ec8bc33fccf0ab6a784fc7d8efce874c74	equivalent
CompactCells491To522.Box511.QInterval	a12628c5718985f0076bb25e9a86b6a5ac8cc56acf51958dbc9b66378f1140c	equivalent
CompactCells491To522.Box511.lambdaInterval	4fffc674af01b0964497fc7069e50a1423349767d0c4d2bd7c01a31981d63330c	equivalent
CompactCells491To522.Box511.compactCandidate_not_isWeightedPerimeterMinimizer	7c8c6b86fd6dcdf54ef14111d6510724fa6d6f63a77638b75d608d0abc4e78a9	equivalent
CompactCells491To522.Box512.QInterval	26d4a08b21e2a86ff64d6c7d7532144a6b411726330f78fb4d963712642cad0e	equivalent
CompactCells491To522.Box512.lambdaInterval	151dac92c7917491b58659816d638a024972a97f9082244322d2bca219921217	equivalent
CompactCells491To522.Box512.compactCandidate_not_isWeightedPerimeterMinimizer	e2948e4453db05175c56a2eb6bf0f43722556d7c990d68942351b917264540d9	equivalent
CompactCells491To522.Box513.QInterval	0642c5b8ee59fd81f0fb38c2d0e5d3a2f530ac51c19aaed58449abd18bc108e0	equivalent
CompactCells491To522.Box513.lambdaInterval	3de63310ff71711ef8d46ceed93a52d3127a88d12364984cd3e1e55bc4e0aced	equivalent
CompactCells491To522.Box513.compactCandidate_not_isWeightedPerimeterMinimizer	fa536e63ef246cdec10a02d5d2c4b1467b4c86dc37ce065982a54deea0ad4033	equivalent
CompactCells491To522.Box514.QInterval	d0d17083d32f93c946f9471341f39b1ff5f49c895051289657d57f5dbac8bdec	equivalent
CompactCells491To522.Box514.lambdaInterval	6a282e1b162b68b99618593f8b673838e693a3b89311a1b6dc95e8e4a5ac5e80	equivalent
CompactCells491To522.Box514.compactCandidate_not_isWeightedPerimeterMinimizer	c6c52c39666a9c934e92e42001e235e25791f25a35317c36089e6f501b9d810d	equivalent

Declaration	Source hash	Audit relation
CompactCells491To522.Box515.QInterval	2461fb5f08b2041335564dbe4ae8df56c3b17e0fd01f21cdce4af0a8b879b0d1	equivalent
CompactCells491To522.Box515.lambdaiInterval	efdee65c029d838f8eb279b5b133e8be2b71792e8c5d30b9b5e6b1edef4df07a	equivalent
CompactCells491To522.Box515.compactCandidate_not_isWeightedPerimeterMinimizer	76d9ea707a099440b814084e72ef82dd9a0d577d10048936938cc778ba1bdac9	equivalent
CompactCells491To522.Box516.QInterval	16094adfc729caecef3cf807861d5795d16821c2aabe7fbe736161d13a311db0	equivalent
CompactCells491To522.Box516.lambdaiInterval	2551144c48e07476e57353193c53e0814fe34d98b9954ba59efe19aef677824a	equivalent
CompactCells491To522.Box516.compactCandidate_not_isWeightedPerimeterMinimizer	f73f37d2abd4bcd7246ced201dabad7ed8fb56233f0a073bb0de3303ac45244	equivalent
CompactCells491To522.Box517.QInterval	d4ba99887fe13e0b741d505d092a1c34d304f3195370f9c5ee057edb3775b4cf	equivalent
CompactCells491To522.Box517.lambdaiInterval	1e21b162d13f5050cabce5066ca725c62b88b711f2cae8f65f2154238be2a526	equivalent
CompactCells491To522.Box517.compactCandidate_not_isWeightedPerimeterMinimizer	e3c41ecae6e9322cfc6b66d3ffedca9f3d5f818cc985484e8bd9700c72b727b	equivalent
CompactCells491To522.Box518.QInterval	44bea5e4034d311c54d4c0973d2ff1a491e005ff1342bff3e808709227b44dc9	equivalent
CompactCells491To522.Box518.lambdaiInterval	4265787d49a047dda2ea530e7085eb66561d0e6c018cfb1af32da099557b8c1d	equivalent
CompactCells491To522.Box518.compactCandidate_not_isWeightedPerimeterMinimizer	7c96c7a4cd3e87e955c1b49363e8f20e172e78e50b5b02c797ef7bed2b44014a	equivalent
CompactCells491To522.Box519.QInterval	ac096f39d01e31b9de0421fd16f39b69103957c5e7e1539a7becf564af7c840b	equivalent
CompactCells491To522.Box519.lambdaiInterval	34f23b7694e2c369f30cc0a603017b6043a40afaadd6b58e9760c7d6a7c66201	equivalent
CompactCells491To522.Box519.compactCandidate_not_isWeightedPerimeterMinimizer	2bc12ed36dae2a32cdfd922f4e4c5ffe8adb77ff3f810609485df5ff8d68272a	equivalent
CompactCells491To522.Box520.QInterval	d6f6f70cbaedef22c98422eb8f56976c19c6f7edd04ad4d367ff1e2025ae3b87	equivalent
CompactCells491To522.Box520.lambdaiInterval	2c64024d16f4c7da10b53b50dee24b3b704176018f4e7d468fd01a9e86f21a5a	equivalent
CompactCells491To522.Box520.compactCandidate_not_isWeightedPerimeterMinimizer	4b6ee823012c1296d201162c2516e7307a794f5554ca9f3a1a7d3344b328a4f7	equivalent
CompactCells491To522.Box521.QInterval	a513b4fa03e98276206e6da426ee7d860bf3eb3ce96fc8123c187f5d5d9737a	equivalent
CompactCells491To522.Box521.lambdaiInterval	a0cc9c5d2f98e1e1d08f4e7fe774c96692dfc5fb13148ddb2e3ef1e700aee81	equivalent
CompactCells491To522.Box521.compactCandidate_not_isWeightedPerimeterMinimizer	d2b4c84a92ac35f8a151b2a3f8daeebb57f662fa945f9f269ec796bef4f10f36	equivalent
CompactCells491To522.Box522.QInterval	807a5fc228b8e856ed363d33085f118bd22955567a1635933a1b4f26a60af0c5	equivalent
CompactCells491To522.Box522.lambdaiInterval	051dd504923c14faa3f1079396305980eaa8eed564cd70f981ed4183f031ca1	equivalent

Declaration	Source hash	Audit relation
CompactCells491To522.Box522.compactCandidate_not_isWeightedPerimeterMinimizer	26d358f0ed666e8b52112629d6f56e22b4d60cbe6fa08c9ccd706cc6b8076b82	equivalent
CompactCells491To522.bboxes491To522_candidate_not_isWeightedPerimeterMinimizer	519a5a87e0161528802dcf4bb2eeb37fef001868a1a13132b239753a9bcd9fb	equivalent
CompactCells491To522.bboxes0To522_candidate_not_isWeightedPerimeterMinimizer	bc47b736499c0e8fe846f5b70579af4c1cb5a61b2f667e3f4ca4ed5ba96ea1be	equivalent
CompactCells523To554.Box523.QInterval	491ad0c954a58e1d93f4960aa0caa317b81c572669a7ff39c2b86a9c6e2d782f	equivalent
CompactCells523To554.Box523.lambdaInterval	749b37b26a68ac274412454ba05acc4b7a453cb704eeffe1bf2e2fdabab60c6f	equivalent
CompactCells523To554.Box523.compactCandidate_not_isWeightedPerimeterMinimizer	f5543b354075724095ff656effcf47c4b264a111745db07a8a6b065b36d991b3	equivalent
CompactCells523To554.Box524.QInterval	ef2154b2be4689316df39112cf03270df7394ca158b922439e842a6622b1ae00	equivalent
CompactCells523To554.Box524.lambdaInterval	8c8bd6fe9b029c83bb3b0a1dbe9d6e35af6fa80ab9af501373bf6686e3d29d0d	equivalent
CompactCells523To554.Box524.compactCandidate_not_isWeightedPerimeterMinimizer	3ac558b3ff73ca184f84d29b48fd93cb9cc8fc1303dee6feecedda465f3e910	equivalent
CompactCells523To554.Box525.QInterval	d45aa2d92173761f68df21e4ed2c129deca8a64a6ebb82cc5046128477bb2c95	equivalent
CompactCells523To554.Box525.lambdaInterval	2ffdd19c5ff2105207cb59676f89af03a58386f1be97817f05063cfc1bcf09ea	equivalent
CompactCells523To554.Box525.compactCandidate_not_isWeightedPerimeterMinimizer	e07d39712cf76f673a880ea6a5aca8fd7fd440b5edf2f256a147bbd909d517f	equivalent
CompactCells523To554.Box526.QInterval	f554546977bc59de940485a81d3fec6a15cf67ac5b5bafdac7489f7d14fbb67	equivalent
CompactCells523To554.Box526.lambdaInterval	aa0fbab013b5207891939b9e9a8378ad6d84836ce85d776c0fffb3fdf325c5b13	equivalent
CompactCells523To554.Box526.compactCandidate_not_isWeightedPerimeterMinimizer	287b9cc5d4d55655c8e98609ec7ba10fedf1ffffe4337746002648fa3f722f3dd	equivalent
CompactCells523To554.Box527.QInterval	2e2a7ee0cfffeda90fd8ece61a419762cc1b8f5741124e9d65029777b29620de5	equivalent
CompactCells523To554.Box527.lambdaInterval	290b39b1acb766549d534c68f8190ef1df8dae0865a1675ca2ed0fde3980dfd1	equivalent
CompactCells523To554.Box527.compactCandidate_not_isWeightedPerimeterMinimizer	8735e25ae951e366240b621c61c38c3015696ed1e8db9304d209c2925bf8ce57	equivalent
CompactCells523To554.Box528.QInterval	62a3904f9cf7d36d188b3e32d4a5d8012b038395cbeac6054afc2ae1b751298b	equivalent
CompactCells523To554.Box528.lambdaInterval	731df43aa4889377175aa20e4e2004d6d482130f10de83fdc45258c6632a5779	equivalent
CompactCells523To554.Box528.compactCandidate_not_isWeightedPerimeterMinimizer	03c526a0cd987ef8a1e8ead8fe65d9a8bf9fe324917a12f267ec4c30c4617ca8	equivalent
CompactCells523To554.Box529.QInterval	191bbe5b5ba16aa54b3f36ced64c4aeebbf26b13102f020c0f91952afd7de2a8	equivalent

Declaration	Source hash	Audit relation
CompactCells523To554.Box529.lambdaiInterval	5839d6a96c2d2ce2e9e53db31ec696bfdae bfc5942929bd4511c65be02cc75af	equivalent
CompactCells523To554.Box529.compactCandidate_not_isWeightedPerimeterMinimizer	9c32af118924d4315094bfcf8066649647a 34c5c41cdd442afe95f1848ff147f	equivalent
CompactCells523To554.Box530.QInterval	237df7f577ffa2ea8180caa48f184534c5c 792b97bac13a90e1a81501d0bf8b9	equivalent
CompactCells523To554.Box530.lambdaiInterval	f68f75fc63e507f5310517bef58927743ce 49c76e5e37b5aa36c1c2ecb8c89ef	equivalent
CompactCells523To554.Box530.compactCandidate_not_isWeightedPerimeterMinimizer	1dd1df67e9406a54d13c9a4199c5bb283d7 a8ca330a99ab11f7a36f4f7532607	equivalent
CompactCells523To554.Box531.QInterval	f7c533ff6cc52e54171b4de760dbd3fd873 75ca37b50cdf9a5cdaef0ebc4ae40	equivalent
CompactCells523To554.Box531.lambdaiInterval	83247397af41217377de1aaa4b9fa58a6d4 356244f6fc9c1a5e94eb3b93d3fe5	equivalent
CompactCells523To554.Box531.compactCandidate_not_isWeightedPerimeterMinimizer	1d95640c82ae63efbe2e51e6b082cbba059 6849ffd4c5e644e1fe5138f122f1c	equivalent
CompactCells523To554.Box532.QInterval	14732dadb294741d67fb90c92a9b4ca9483 6901f84d61b140a9ed0724d90abf9	equivalent
CompactCells523To554.Box532.lambdaiInterval	9c0b5d51c35ec7a20cf95dbba92c7ff12ee 7025d30509b4db494b8c21b1fac52	equivalent
CompactCells523To554.Box532.compactCandidate_not_isWeightedPerimeterMinimizer	498941a89413bb72e660c846941f6254156 dad2d5bd8d5e6751cbc0185b9e6e8	equivalent
CompactCells523To554.Box533.QInterval	7c25cc10392b4d45d9f92e84a74b6edcde8 7fe558473d13137187ae0f8767bde	equivalent
CompactCells523To554.Box533.lambdaiInterval	a6aa2b6f236a3650e84dd6ef52de0689eba ef78d3d7f33806f277df5a80433a0	equivalent
CompactCells523To554.Box533.compactCandidate_not_isWeightedPerimeterMinimizer	a00c3db76568afd4a466eb5de8f028502cb dd7ad980e6f8cef08456889f31500	equivalent
CompactCells523To554.Box534.QInterval	ab466863750b3f2e669c347e9691b60f6b4 d29933db7e2bbb791b2be48c841d3	equivalent
CompactCells523To554.Box534.lambdaiInterval	0948827390378bde12998918e42b3cfc42d 4a7ae9779a66bd2b4e2875fe7b48f	equivalent
CompactCells523To554.Box534.compactCandidate_not_isWeightedPerimeterMinimizer	dccd254836f9daff39e0b1b8f3865425fcf fc4aaae069b9d056eeb15f4d40b69	equivalent
CompactCells523To554.Box535.QInterval	596d377e4cc474702c98b615fe5c42bdbb4 7ea4d0ecaa35293e36eba005082cc	equivalent
CompactCells523To554.Box535.lambdaiInterval	759fb57df18fa9f245043b3297d3819bdf2 1f1bb9c811fb2c7767e1be19b0eed	equivalent
CompactCells523To554.Box535.compactCandidate_not_isWeightedPerimeterMinimizer	d23281126a737df8ab45b7f3df1b1e1b021 25f329f16e9293a8b2a938e7100a8	equivalent
CompactCells523To554.Box536.QInterval	7f3f161da8d5fd9d0208a604f562c0057f7 2469df99f58fd83ebbf8070c81061	equivalent
CompactCells523To554.Box536.lambdaiInterval	9839d68bb2cdf9d4b598dfbad3b7e611b7c db8481473dfa6fc4e976c567e1214	equivalent

Declaration	Source hash	Audit relation
CompactCells523To554.Box536.compactCandidate_not_isWeightedPerimeterMinimizer	35b8a03bdf1b53205ab28a3cee82d1cc2ef192b6d0bf253f582b99ca7322a13a	equivalent
CompactCells523To554.Box537.QInterval	c319ac4535f4ba85e879356c25065c1675b08de4930ad9e6985af50cd358e1f7	equivalent
CompactCells523To554.Box537.lambdaInterval	2ad4bece67db09e997b7a61f77544efb23dd5812574037ff2bb7b7d63c3daae1	equivalent
CompactCells523To554.Box537.compactCandidate_not_isWeightedPerimeterMinimizer	b7d735c5ca50de0f9d84c7b7b6a8943ffb157d92424ab81548d88511098e211c	equivalent
CompactCells523To554.Box538.QInterval	8606b13d6dbf89d5d9327dea11322a6e2434745be4685282b258777182e9808c	equivalent
CompactCells523To554.Box538.lambdaInterval	e993d68629b7442f7de22a3b4287d6982f0c828e56cc394e354bca4b2c34409d	equivalent
CompactCells523To554.Box538.compactCandidate_not_isWeightedPerimeterMinimizer	e9d8709c6a4ff80da906505d30addf6e480d6aeada78f87e2492f082feb79aaa	equivalent
CompactCells523To554.Box539.QInterval	fc0b023bf0ee51446cb8b1bd31eb6898da7080a09b6bd42ddd0abc6e80174672	equivalent
CompactCells523To554.Box539.lambdaInterval	cf017cd56090e1f47c7914654afd524d9f2f97dbabee36fbdcf345b8709f0e7a	equivalent
CompactCells523To554.Box539.compactCandidate_not_isWeightedPerimeterMinimizer	92589a07bac32083de8ea61b609bdbadf19dba8f1de4d46c25cfdc06b85d61d2	equivalent
CompactCells523To554.Box540.QInterval	a87209bb1a157b5350403a06c6d45b9215b5abc4d197a3051efcb01669020fbf	equivalent
CompactCells523To554.Box540.lambdaInterval	76bc951e10c788f3922695c0d0bafd3bf5d60a17a166f7bd2be95c831a35e4e7	equivalent
CompactCells523To554.Box540.compactCandidate_not_isWeightedPerimeterMinimizer	23df5ead564a4e962654236cd5e5e094b1cd74b4c4ced661ab70f540efa56608	equivalent
CompactCells523To554.Box541.QInterval	0508a4f3fbc8b856de0d7d57aa1cf639ca3edbac4271ac3cd4ce9fa67883354a	equivalent
CompactCells523To554.Box541.lambdaInterval	8a4534ad20dcf0c030ccf54ad3597135154f09fda67cefe7f3c4cd4a380fad44	equivalent
CompactCells523To554.Box541.compactCandidate_not_isWeightedPerimeterMinimizer	69838fdcf6365f7255ad0bb8890fd640e18110a955286086a368e50d18299d4	equivalent
CompactCells523To554.Box542.QInterval	87c52696b14a5c79fd58671fbfbaa531f3e9e1ed2a5044a5772f0bb51182ea7e	equivalent
CompactCells523To554.Box542.lambdaInterval	cc90f290b652ce9e24edfe1a19cc015797cd7a55c8bccaf8323cb3a3af7dfae3	equivalent
CompactCells523To554.Box542.compactCandidate_not_isWeightedPerimeterMinimizer	bc35d864bb9ec802fae94f1bdb9376152a3febe834b0bf5c6c75eccaa6b89a42	equivalent
CompactCells523To554.Box543.QInterval	1c2a04be20f1fb98308e0bf565d56f4fe7132c398ebe7c6f4a4342c5fb6fe56c	equivalent
CompactCells523To554.Box543.lambdaInterval	fcf554e9f5cf60dc22be62b3533630204259375a962a479a26ed42e9628b4e9f	equivalent
CompactCells523To554.Box543.compactCandidate_not_isWeightedPerimeterMinimizer	940a1ab80bb7cccfb8ec7e357f4ce9fa81daf48f6c065bd3b61fcae0d63fe95f	equivalent

Declaration	Source hash	Audit relation
CompactCells523To554.Box544.QInterval	a93cc435ea21f9759fad8ccf8b2d39c85a385af22c5212294169086a288da456	equivalent
CompactCells523To554.Box544.lambdaInterval	2a40fce765afe1ef4030c15aa6bdc03dd94c6bc1691f64e224de0c1f1890c254	equivalent
CompactCells523To554.Box544.compactCandidate_not_isWeightedPerimeterMinimizer	44480c8f4c604ca53ee549faa1f5229be4c33d3fd0dec1a315571c07cfe5e58c	equivalent
CompactCells523To554.Box545.QInterval	7cadab5643baa4f38116d21a9fed25f66f561c024048e762c26190eac66d86fc	equivalent
CompactCells523To554.Box545.lambdaInterval	c9e1e9a322612696cae72e19df732e4037c052be77d087c8dc53e879cc1d351e	equivalent
CompactCells523To554.Box545.compactCandidate_not_isWeightedPerimeterMinimizer	e037839ca75136ebfefe0e5adf2e4420d214cc7662baae90caf8b6d8af9817d2	equivalent
CompactCells523To554.Box546.QInterval	0ac28d867e21188fb2d84bdd6e90d07c456cb7ae3cf475f335f3b5ffb81eed17	equivalent
CompactCells523To554.Box546.lambdaInterval	a769e3b2913d0dd984f5b2149129539c1708989722896630258439bbb695b759	equivalent
CompactCells523To554.Box546.compactCandidate_not_isWeightedPerimeterMinimizer	7007e4f7cc8806aa89aa27fa3b9b2885b5bb5a7a355ba32c4bb57296560867c2	equivalent
CompactCells523To554.Box547.QInterval	42295d7683e2c8399a0560922aa028dca2891406ff4e53c287aca21afa1708b0	equivalent
CompactCells523To554.Box547.lambdaInterval	0121c63ba2e181370cfc486a6f485653d9da5c7649cabfb12efdcc4e51e143c1	equivalent
CompactCells523To554.Box547.compactCandidate_not_isWeightedPerimeterMinimizer	340fe769b94abd0172bd72193685dea6b872143b3f17f933df6bbccd1942b3dd	equivalent
CompactCells523To554.Box548.QInterval	a3cc49035b0658228f310feb485ce7ab59c97618bcfda9c512ee42c5aefa7c7c	equivalent
CompactCells523To554.Box548.lambdaInterval	637c2feb9a3f1747a4a58cf49c4bbe3f6417c12741869c7892099319b70ef811	equivalent
CompactCells523To554.Box548.compactCandidate_not_isWeightedPerimeterMinimizer	96daaddfd520cfd965088ca940986a0153446d078fd008baea4d790f3a9e445c	equivalent
CompactCells523To554.Box549.QInterval	a3a2da63eb00fb9f464995ef10a17337d19ed1ae002277be02e8d9081c5faf21	equivalent
CompactCells523To554.Box549.lambdaInterval	a52c63c374ac6e6ba12f77900f28474643f0a6bb360c0f65bdfbeb6875067d61	equivalent
CompactCells523To554.Box549.compactCandidate_not_isWeightedPerimeterMinimizer	a91141c652b0bb4e29448dc3de1bada620831dfab2e1b610d658f7d488abc56f	equivalent
CompactCells523To554.Box550.QInterval	bbc099caa2eac28ef598ae1461a40a56b00d7114c4ae416b59d50a945fcf3873	equivalent
CompactCells523To554.Box550.lambdaInterval	392331f024cf15271f3321a546849670f7349a37e8aed6061c73185f40a28858	equivalent
CompactCells523To554.Box550.compactCandidate_not_isWeightedPerimeterMinimizer	0103dd1564cdf03058cd8ce7f20ec29aae9364e5291815d5c7a8d92cd9049742	equivalent
CompactCells523To554.Box551.QInterval	551751d8219c0b8aa707cf812a632d489f749e856a6e5bb10fe145913f771c26	equivalent
CompactCells523To554.Box551.lambdaInterval	20ecffaa8f8c194fff7a64d3b2c1c59e6d460ecea36d80d41a0f2b479a50285a	equivalent

Declaration	Source hash	Audit relation
CompactCells523To554.Box551.compactCandidate_not_isWeightedPerimeterMinimizer	56fb7bcdcb839f714b10bce1c62598b9252a553704c46fe18391656ea7a98acf	equivalent
CompactCells523To554.Box552.QInterval	8479308034df9ec213cafc1d19e6090b95ab6aa708ed17df12224e03231eb3a7f	equivalent
CompactCells523To554.Box552.lambdaInterval	3e7a851bebee38f641e41f787c13675343e4d081a9068e8a344f5b33965b6b6a	equivalent
CompactCells523To554.Box552.compactCandidate_not_isWeightedPerimeterMinimizer	3bab700b2f1d9e74f83cb8da301a73a531bc194f73b673a8bd90ac2be5580e2b	equivalent
CompactCells523To554.Box553.QInterval	d359dca3221598bf9d338112b156142f6fe809bf76b8c0f9acd970e21a5eaf85	equivalent
CompactCells523To554.Box553.lambdaInterval	f86870775c9e10b86e7af76db986c429a53072650747519fe6d7b082322c5b09	equivalent
CompactCells523To554.Box553.compactCandidate_not_isWeightedPerimeterMinimizer	64fde315ead0a76ca76a7b5bb5ae905573b444a4f7bc70f9c4b9ba6783fff2dd	equivalent
CompactCells523To554.Box554.QInterval	1de5339ad93fda53e0b32fd45e7466f706a92836be5459ec1bfb1091f0346372	equivalent
CompactCells523To554.Box554.lambdaInterval	99294e6ee935279fe992ec61dae99d5c5b437af4394b01f9f24c434a581495a9	equivalent
CompactCells523To554.Box554.compactCandidate_not_isWeightedPerimeterMinimizer	4258c2f5442d9897495c1f8051e43663c5f9e52f0efa3e508148c4aed2d066cd	equivalent
CompactCells523To554.bboxes523To554_candidate_not_isWeightedPerimeterMinimizer	6f287b6ceb7989f7dc1265f35f2c2c2c4a5d1c0db2babcdc1fa6457c20ab15d8	equivalent
CompactCells523To554.bboxes0To554_candidate_not_isWeightedPerimeterMinimizer	a7689b4b3d8bc3beaa1b72b84f2f8c1147de017e3e7919dbd671c8ff9d0854b5	equivalent
CompactCells555To586.Box555.QInterval	d404dac7b9282781354c4a6e85a233b1dc5f66c1e9a1eec0de36a13a8c57bc9c	equivalent
CompactCells555To586.Box555.lambdaInterval	5d4fefc977bf40262990560fe33d06adfcc565cff63c5756af78621ed7b10c	equivalent
CompactCells555To586.Box555.compactCandidate_not_isWeightedPerimeterMinimizer	e4fa023ec42768a04ec999b66236a00f7a799f8cc90bcc68b7de2bca5f32230c	equivalent
CompactCells555To586.Box556.QInterval	5cd5db36bac943521617da54b9522ad0a4d579ae0f1e41eba477c5be18d210e5	equivalent
CompactCells555To586.Box556.lambdaInterval	37ab701a70c472504b2871eae245f8028f3393b1798f4d071e16efc5b2e717b	equivalent
CompactCells555To586.Box556.compactCandidate_not_isWeightedPerimeterMinimizer	897e448cc1b48d6f1754fcab48bba189e0091b729d184635af6f7ce74578224c	equivalent
CompactCells555To586.Box557.QInterval	bacaf8920d281513f3b4c8e024c01187351fbd968d3ec836d62b49fa97dd596d	equivalent
CompactCells555To586.Box557.lambdaInterval	921e58143f1b7c9d8d14e72b4e89a702a6605f7bc886e238ff6623896ea004d8	equivalent
CompactCells555To586.Box557.compactCandidate_not_isWeightedPerimeterMinimizer	e1299e5432c990437588c21477f85e813d9cec98c5d495387ba9cb89989edd83	equivalent
CompactCells555To586.Box558.QInterval	8358c3c26511cbbd3a3957344459928474e208724d63197085afb378cb605ee6	equivalent

Declaration	Source hash	Audit relation
CompactCells555To586.Box558.lambdaInterval	009293198687a1a3c084947a1586cb91ef84ebda7da546247031a78b96b181bc	equivalent
CompactCells555To586.Box558.compactCandidate_not_isWeightedPerimeterMinimizer	2b2a5a3435a90e613c6e88bc394a47d139b4e454287d2d20c2878ea402a69657	equivalent
CompactCells555To586.Box559.QInterval	aecc6134782b14aeba67e68b35fa79c7c104d7470b305aec99945543d8e3d84d	equivalent
CompactCells555To586.Box559.lambdaInterval	882a9170aab04fc3dafed0ed0aa8e332813e0bb199ec619153bd16f7de295f2d	equivalent
CompactCells555To586.Box559.compactCandidate_not_isWeightedPerimeterMinimizer	377c8e905a6c84790b2b9d330c610dfb52680bfd219e9d5b7cf8c5ee00f9c067	equivalent
CompactCells555To586.Box560.QInterval	e6c1a090f9506653a8f8a8ff14fd9e5cdb1a28f3f6bb4872593f82fe855d3453	equivalent
CompactCells555To586.Box560.lambdaInterval	040c07287418b3f009533e7d1b448bffff58bbea8132903177863f28dd896c001	equivalent
CompactCells555To586.Box560.compactCandidate_not_isWeightedPerimeterMinimizer	5ef6d55630f4fcdcbf4abfec2e64f20aca8e8ffc0a2c840ece4bc251cb2e92b1	equivalent
CompactCells555To586.Box561.QInterval	72ef4ef789a95d9b4d844d9295a4f5e9efdaf2c4854e8bac23307e9e6cd72089	equivalent
CompactCells555To586.Box561.lambdaInterval	b7d7e9f7f26dd470d8713cd887546cddc37a9779a65fdb45d64faf71cb1bc774	equivalent
CompactCells555To586.Box561.compactCandidate_not_isWeightedPerimeterMinimizer	c99ae54d7b530872505eef19c3fb557b7b38b058fd4d5e35f558585f4a691994	equivalent
CompactCells555To586.Box562.QInterval	118e9458ad33a5dc4f32950d3b4db71a01f26a82977b03c1e2827c88823c797b	equivalent
CompactCells555To586.Box562.lambdaInterval	a58ba6874b8c5ec3d77be516293b14e6cc52906903c0114f46dd0a8bff805e0c	equivalent
CompactCells555To586.Box562.compactCandidate_not_isWeightedPerimeterMinimizer	2a5625ca8638989a9dcd335a7e24298d27f2b9a266a0596bc6c326cbd1276b81	equivalent
CompactCells555To586.Box563.QInterval	071a99335484ad08e4897b19646afb5496eb78ab17d2a55fd00e008d799ab9ff	equivalent
CompactCells555To586.Box563.lambdaInterval	0982089e75d6ae768d188477d7b20c5a9fcbd3da029c0d0410d877282e0d0394	equivalent
CompactCells555To586.Box563.compactCandidate_not_isWeightedPerimeterMinimizer	ec78db004e071147bc18031cd806d08473a3824ed53e5e9d1ac67215577bb6f9	equivalent
CompactCells555To586.Box564.QInterval	6502df884b4de2bb9ee9c82a8ce4588cf71dc240b4e287513eb942fd6b616ce4	equivalent
CompactCells555To586.Box564.lambdaInterval	3a94a485742d7e7bd081ab48fd86dcc097d06338988d1589a0bcf43c91033f31	equivalent
CompactCells555To586.Box564.compactCandidate_not_isWeightedPerimeterMinimizer	d696c4d3b873cfaa3670a0eb2f4f8cfbf606ac63fdd0f2d461b5843cf8652b6f	equivalent
CompactCells555To586.Box565.QInterval	a12ec0bc9652d16efa21afd0a5f912e64683797a3d832bdacf7068ba04b976c3	equivalent
CompactCells555To586.Box565.lambdaInterval	f3c992e473cd2bcc8e4cb8704efa21740fe727ed587ac40fb7530f8547dc607f	equivalent

Declaration	Source hash	Audit relation
CompactCells555To586.Box565.compactCandidate_not_isWeightedPerimeterMinimizer	842702280c7c307a8dfbcd57e4912152a63a9732cfae8847e471cf3ed7df7a35	equivalent
CompactCells555To586.Box566.QInterval	bc56cd04e97eccd4e3c7fbfc729ceae706eb5f89f8a475fc2fb0ddf216d5553a	equivalent
CompactCells555To586.Box566.lambdaInterval	c7d29c38f8e640a9f088a0ea67b35ced4b69b1b2e5ad7d2ca9ba2427c3390c0d	equivalent
CompactCells555To586.Box566.compactCandidate_not_isWeightedPerimeterMinimizer	8e079bd29013a5be2e2716b19765e89dc01316c825ce4b38aa50ce0625f86e2f	equivalent
CompactCells555To586.Box567.QInterval	8359b78af999ec598985229a1444aaaa020529e4a12931e47ae14750ed047f78	equivalent
CompactCells555To586.Box567.lambdaInterval	918936c6d2b5c5c3c621519bb34b8ce398bf14885ac4f9740904fd97532e61f9	equivalent
CompactCells555To586.Box567.compactCandidate_not_isWeightedPerimeterMinimizer	51447a0466681869a4a7b3a65a5843cd25c9c69a8288991d142c1fe137c9974b	equivalent
CompactCells555To586.Box568.QInterval	8770152fb901fdb8068e4ca66027a172f75e6603444129dda2d7135a719269ba	equivalent
CompactCells555To586.Box568.lambdaInterval	88e7fc81f0aa9c9bb2e9be0aef4fcc14f355a9e2d894509a36764be5966b29f5	equivalent
CompactCells555To586.Box568.compactCandidate_not_isWeightedPerimeterMinimizer	200302c045c4198ae508b0dd077dbd588648a34c9921444f219ed77789b6caa5	equivalent
CompactCells555To586.Box569.QInterval	a8c241b6b408677a0764b2d33a488b707ef0ce03ca1fa1ff25511968d8874982	equivalent
CompactCells555To586.Box569.lambdaInterval	ab4dd9fc8ac681d406e156cf6500f057de5bfbecd50fecbf2345469e11b2faae	equivalent
CompactCells555To586.Box569.compactCandidate_not_isWeightedPerimeterMinimizer	7686f29da535f523a12f492e019cf090dc884310e9a8ae0042dbe2b1a33f2823	equivalent
CompactCells555To586.Box570.QInterval	6a134a5ca33b27be64f827c2cb2ad2052244e8c0c43316e4c9a533104ehead98	equivalent
CompactCells555To586.Box570.lambdaInterval	e1054a9c883cf5569128a49dd7fb294c8209191a3a4e6a1587aab7e8d3c48e5d	equivalent
CompactCells555To586.Box570.compactCandidate_not_isWeightedPerimeterMinimizer	55479c5d395cc6959440a1dbe46253957eb7dc32c828363e49945e4af5373d92	equivalent
CompactCells555To586.Box571.QInterval	a81ca6412589d60a4d6e01c96574cfe6cad5b9ac2d5772e22ecef01bb7516c60	equivalent
CompactCells555To586.Box571.lambdaInterval	1de55d384f9a6cc9f8a7095b06a5d3ee301015901d5d73fec13d48eab7bf0077	equivalent
CompactCells555To586.Box571.compactCandidate_not_isWeightedPerimeterMinimizer	325d59a769fd79a99c9d1faba68da21ad76d1f9567615cd37d8e726f299af71a	equivalent
CompactCells555To586.Box572.QInterval	261964c29fa6b14759f8dd70675a4d16e191b718e9c2bf5c99a4a91c56576cf3	equivalent
CompactCells555To586.Box572.lambdaInterval	d4741683f38a269acce93b96a2d0560496d59a02f677e876445ba68c49093f35	equivalent
CompactCells555To586.Box572.compactCandidate_not_isWeightedPerimeterMinimizer	b2c6403fb29f07c85baad4b28d799aac1f3a05d0e4287df47a5aa3aba51e652c	equivalent

Declaration	Source hash	Audit relation
CompactCells555To586.Box573.QInterval	63fb9da3597ba2046d86904c2519973446d6b1d04ff0a3566aa49a79e331ffef	equivalent
CompactCells555To586.Box573.lambdaiInterval	fb418b6f5c09435e8e0ee9b959f0000ceb3658f968c73175421135005d51952d	equivalent
CompactCells555To586.Box573.compactCandidate_not_isWeightedPerimeterMinimizer	3cea8a6a7ed6ff5f8e08b29401a96feed77a94781970436c5b324cadaf1e9fde	equivalent
CompactCells555To586.Box574.QInterval	5110b1f64ccdc7f6c4bcf21964813722d93d3ab5212b39111938f749f863e63f	equivalent
CompactCells555To586.Box574.lambdaiInterval	c13bfe9892bad72848416c23b80104f8a193491011d2c976532e8d74e9b464bd	equivalent
CompactCells555To586.Box574.compactCandidate_not_isWeightedPerimeterMinimizer	479715e2f06c9258d2bba5c0fa648f7b7c61686056f220c3ab7d0e1dcf735ceb	equivalent
CompactCells555To586.Box575.QInterval	8bfdde00716c5f23a33798a2a9f08378e32c4bf533643f29e555b74732f3c0b8	equivalent
CompactCells555To586.Box575.lambdaiInterval	408b01e87343a7cb6f9c12692603402a1f1199cab62d2b114f2972dc3a42b422	equivalent
CompactCells555To586.Box575.compactCandidate_not_isWeightedPerimeterMinimizer	01024732efaebf5eeb94e7faa24bb5c1f108e80d21ce7538d665b9872862b057	equivalent
CompactCells555To586.Box576.QInterval	0c484b45991e080e2e5e205104cb1bd3463a10d76bd4c13087fb03cf1311fcd9	equivalent
CompactCells555To586.Box576.lambdaiInterval	e01439800a478cbd260adfcc6785ab28d1f2b48596c955044100997c705c63c9	equivalent
CompactCells555To586.Box576.compactCandidate_not_isWeightedPerimeterMinimizer	052cdeaa21c430f113b500887170fd97f6b4473ef82c8ad0681a1a50e38ff433	equivalent
CompactCells555To586.Box577.QInterval	323b04d30dab82c63362a0ff357bd4cf40ac85fd04755c3a5de2f6195c936825	equivalent
CompactCells555To586.Box577.lambdaiInterval	7b1ff7b4f51c6c7bf8cc64dd07fb311d86c312458b5594230721af4578d75990	equivalent
CompactCells555To586.Box577.compactCandidate_not_isWeightedPerimeterMinimizer	36234d4952f9b2edcd0bffa5332ee59ffa2831298a6a565b32d0f2594ec3c4cd	equivalent
CompactCells555To586.Box578.QInterval	653b1499c3418f5d04610a8f1ca7cc3216105048c616c8f80007c904a8b385af	equivalent
CompactCells555To586.Box578.lambdaiInterval	b323c5ba6e7e2a9c52b17665047cae64de3cee8a789eac402db4ac4e44c32f55	equivalent
CompactCells555To586.Box578.compactCandidate_not_isWeightedPerimeterMinimizer	93052f94ab80999790975d930e937e996645a0c32ff58c33bc07a50588403f46	equivalent
CompactCells555To586.Box579.QInterval	5e83bf2b343105736c86c80158d1894f8ef496dc826631751aea71591785606e	equivalent
CompactCells555To586.Box579.lambdaiInterval	8af94576a7a1002fdccc9d9416847ea90a773c62c0c91ba869b5ec2ac2d514f1	equivalent
CompactCells555To586.Box579.compactCandidate_not_isWeightedPerimeterMinimizer	526aac93a2f28ff09204d0ec4afcadf07fd5fdf2cd7dbdd0b1b611d4896c8127	equivalent
CompactCells555To586.Box580.QInterval	fe20773ce2d5b180d00214959ba0db56d3e4452b9ee429425cd3bfff07f49017	equivalent
CompactCells555To586.Box580.lambdaiInterval	7ad7bbf2ba5544230a66005ec11f47ee10a08229741c3dd349c41fb4e3da6623	equivalent

Declaration	Source hash	Audit relation
CompactCells555To586.Box580.compactCandidate_not_isWeightedPerimeterMinimizer	6eb87e101a3cd54db561529b9b515d870f75a3df5ff34ee64534a93d01ecbd64	equivalent
CompactCells555To586.Box581.QInterval	121822657782ef0f050d9c4ad0e2fbefb71fd92634fda171e66d19b7bc4503d1	equivalent
CompactCells555To586.Box581.lambdaInterval	5657852017e5f6a1f995cb73fd6b1fd7c8dd698c5666695ac9e6b408c1f408cd	equivalent
CompactCells555To586.Box581.compactCandidate_not_isWeightedPerimeterMinimizer	046b5b35df1eb0d291f70d71f594afe12641cc7c3733a5dde1c32d7564166647	equivalent
CompactCells555To586.Box582.QInterval	561da3c1bef639dd86d84fa97220b22d1e74128cecb7a0a47b3e879f37653bd0	equivalent
CompactCells555To586.Box582.lambdaInterval	299dd326fc02fa405796a7b4064fc1a6694ca25455656c7c6b5472aba21cccf4	equivalent
CompactCells555To586.Box582.compactCandidate_not_isWeightedPerimeterMinimizer	8a57cac0b5af64cbf08bd7ee28d68763c793e21cadf28715ba134ea0a6ee34cf	equivalent
CompactCells555To586.Box583.QInterval	c88f2bb90f5fb504f4a042ac1edbc944cddb5fffd2db616afadcbf0326673a320	equivalent
CompactCells555To586.Box583.lambdaInterval	57c1c7d85838d8e8786d06722b91fb52dedcf2cc4e10f3651fae00fa6b87e80e	equivalent
CompactCells555To586.Box583.compactCandidate_not_isWeightedPerimeterMinimizer	c94f13489c1b5c1a51eeceb02477b152cd44d55e69a07f0ea8584c68e7424d9b	equivalent
CompactCells555To586.Box584.QInterval	06230eae9f07220b5e6a7cab94281b017df054cf464a0625f1eb4ede34450495	equivalent
CompactCells555To586.Box584.lambdaInterval	e482cd20cd4cb3a46c8fa53a96254c378db78d4d3db9ea056641dbe7a64ee19f	equivalent
CompactCells555To586.Box584.compactCandidate_not_isWeightedPerimeterMinimizer	38ece1403b8193262d5d06be2879701d4aa856390ad8ef316664783902346b2f	equivalent
CompactCells555To586.Box585.QInterval	540fb1107287be34e78a077a950b0b6abcef408f00498d511477d944be9dca4a	equivalent
CompactCells555To586.Box585.lambdaInterval	46036b8de5452eedb5fd96b6dbddc022901efaaa0e6060ebabf916a31ca9c4a4	equivalent
CompactCells555To586.Box585.compactCandidate_not_isWeightedPerimeterMinimizer	b5a177149d26bafbd6934e68017c2d221209a1827d3059e01a73930c29834761	equivalent
CompactCells555To586.Box586.QInterval	3281f4362b90534c9081afd43e94435a11e8dc33aa697621a5b412a3e1138117	equivalent
CompactCells555To586.Box586.lambdaInterval	90965c1b91578ee3bc5e4fda8e273b9cf9012aceec513ef66c55a3bd41d527d6	equivalent
CompactCells555To586.Box586.compactCandidate_not_isWeightedPerimeterMinimizer	b4a13aea8701da4fa1cffbb5a17c692f4398c3dbadcb26608fbd8b53a40e30	equivalent
CompactCells555To586.boxes555To586_candidate_not_isWeightedPerimeterMinimizer	43695f63fc0a7b371c8a41b7e8ded42abf49092fed1430f50caa71594c86021e	equivalent
CompactCells555To586.boxes0To586_candidate_not_isWeightedPerimeterMinimizer	4e0355d670f6fa8e166f437507c48e7ef4f5bdcaf99f36e6d5c963c0cd59f92d	equivalent
CompactCells587To618.Box587.QInterval	51e87baeb57fe905a844fb478576697997eb53f500caa70ec1a7c26dad058b4b	equivalent

Declaration	Source hash	Audit relation
CompactCells587To618.Box587.lambdInterval	6fb0699ad85a6683bf79e474d5d2bb8dc143fdae3d19dfed3935c56329676643	equivalent
CompactCells587To618.Box587.compactCandidate_not_isWeightedPerimeterMinimizer	452728d366df445942690325a3c490e0c3b29d6494e041c7798e45a9cdc5263c	equivalent
CompactCells587To618.Box588.QInterval	8b74605a4578fb491980b4133a0b2f0ea3311e505ae4595e9c6fcd2882a63d96	equivalent
CompactCells587To618.Box588.lambdInterval	accda278b063b66bb630c5eda8db9c67818b01d174df99626ef9d735d2a03cbd	equivalent
CompactCells587To618.Box588.compactCandidate_not_isWeightedPerimeterMinimizer	fd0a6cbe448ee48a0b516582f21ee5558d30b405c8e380fec6222d328540115c	equivalent
CompactCells587To618.Box589.QInterval	4449f1c79d2fa3bb4ccacfc325ae4ea6fed204a3a61049c5d5bd0bc36c9555b	equivalent
CompactCells587To618.Box589.lambdInterval	056d629663e9d02e20cd51a7379f2a588747e02fd1e014fc1b9d33133c6b27df	equivalent
CompactCells587To618.Box589.compactCandidate_not_isWeightedPerimeterMinimizer	5e426c81eafcca9561096af2d9c00603d018c8388754e6e70f264fb007f99c7b	equivalent
CompactCells587To618.Box590.QInterval	d1e4b19f93979c2cdfa7ae1eb205d26719481503f05786dc71fc6b82a7f61848	equivalent
CompactCells587To618.Box590.lambdInterval	b1979d11727affc8992cdfe67a75cee2d1520c12e352eb19dfb420c7323e2428	equivalent
CompactCells587To618.Box590.compactCandidate_not_isWeightedPerimeterMinimizer	43ce0004d829282ecbflabebfc0f800daa24827a5dfd57c7376c7ea92cd85699	equivalent
CompactCells587To618.Box591.QInterval	8519ca41acf61c091860883cbe2591cb79a1f844afebda190d1ebc4ad329db34	equivalent
CompactCells587To618.Box591.lambdInterval	ee2a712a43c7df31a7657f92e8780de38dec000471017cf8c4ada83205430831	equivalent
CompactCells587To618.Box591.compactCandidate_not_isWeightedPerimeterMinimizer	2e258d0a56e6e5c5228dd0552a004dac32a25920f2c178547e5c47913ecfaf01	equivalent
CompactCells587To618.Box592.QInterval	88c07788c313bc2145b469244280664f223efbe66e5c03b2739775dc8739b0a4	equivalent
CompactCells587To618.Box592.lambdInterval	4ee459fc1f08337ea2c00eafce607854afa7055d51b29162a6db8777b91f552b	equivalent
CompactCells587To618.Box592.compactCandidate_not_isWeightedPerimeterMinimizer	46ec382876cb7c9a51386a5ace333c294d4fe34091183e8e567fb030df32616	equivalent
CompactCells587To618.Box593.QInterval	3e9a268b445586e518435e01ef8683e2241bbddf7a7c6e9980e22516b587206d8	equivalent
CompactCells587To618.Box593.lambdInterval	f5ec1a2570bc8c5f101ec414b9470e99801295a69337650a2a6d01e270a9357c	equivalent
CompactCells587To618.Box593.compactCandidate_not_isWeightedPerimeterMinimizer	fb37b8e21312665c94c73e82aed5f6a019f7594444c5cf6a18bb5f54c31cb8eb	equivalent
CompactCells587To618.Box594.QInterval	eff90a32c96c2192fc22f0cdfc83ce187b9b2c8919697d389b4e28493e3be6e0	equivalent
CompactCells587To618.Box594.lambdInterval	e2b08004fed1476d2766912308804e8c566a0ee91234de488c2c6ecfa273b647	equivalent

Declaration	Source hash	Audit relation
CompactCells587To618.Box594.compactCandidate_not_isWeightedPerimeterMinimizer	73cbfcbba715b4859f10574217bf2eff444b6c80bcb3e2d3bd36399d1cab30c2	equivalent
CompactCells587To618.Box595.QInterval	19b3cac8932b5041013920f91145ad25189b66aeea998117f3eb13ca0ce16a4e	equivalent
CompactCells587To618.Box595.lambdaInterval	a87106007054015c8dbbe16ae2e86da72dd638c74766d243e2c19a5a48d8659b	equivalent
CompactCells587To618.Box595.compactCandidate_not_isWeightedPerimeterMinimizer	161e2aedcd1ba58010d8f6ebf7d81f044187cde1457eb0ca1e030d7f15fd4a02	equivalent
CompactCells587To618.Box596.QInterval	472b151a1777192edc29a0349c7bb55186014d51f6f9fb7558b1919f87254921	equivalent
CompactCells587To618.Box596.lambdaInterval	d3895eb746d0044a2d40eb7e1c66e6c49962f6c3bd7b64950ca4e737549ce1a7	equivalent
CompactCells587To618.Box596.compactCandidate_not_isWeightedPerimeterMinimizer	8126d0731f0835ed89f0adcee759c58873cac29c21f81a741460d9006c8bf3a1	equivalent
CompactCells587To618.Box597.QInterval	d16329386b8ddb9829c576d5d46026db6d2621f6a0be0a712d92d015abcb8878	equivalent
CompactCells587To618.Box597.lambdaInterval	0765f4bae4bfae3463fbee6f015cc499de8a4b7ae5d3da97d2bcb8c1ae56c98	equivalent
CompactCells587To618.Box597.compactCandidate_not_isWeightedPerimeterMinimizer	19ce9affd1fdf5c1544327e223f0e350cda118bc2600762fcc37faf271ba115	equivalent
CompactCells587To618.Box598.QInterval	6aa6160f1b60e6a57560cc92216fd7e1de1bed3dbcedde9e6a91110edf47cced	equivalent
CompactCells587To618.Box598.lambdaInterval	5f7e7a6ac2e0080ce7f97a1c8d6faeb5e663f1f3b35292b2e79b79aec6b73783	equivalent
CompactCells587To618.Box598.compactCandidate_not_isWeightedPerimeterMinimizer	46ddc24696327c4acb610d495dbb119dfeb7b43cdc3e093f6456d0be0e5a254	equivalent
CompactCells587To618.Box599.QInterval	88f6c9b91034f54140f36600dda12f3a089947b08a79e3d393cc10619ebbb84d	equivalent
CompactCells587To618.Box599.lambdaInterval	663569c7c74b2c665950a099967c5e41a8eb82c73efbf8344f72adf5342d556b	equivalent
CompactCells587To618.Box599.compactCandidate_not_isWeightedPerimeterMinimizer	5d747951a3f98ab0007d69f702eeaf0330e69999ae60fe16287a9206de2f073a	equivalent
CompactCells587To618.Box600.QInterval	23a641bbaa8905d66b5435bc96fdb79453ab653831a2e5d61eed59887581278d	equivalent
CompactCells587To618.Box600.lambdaInterval	793323a75c0a341948552ad69525b7814996dccb7ba3efe413c21acb9825fd41	equivalent
CompactCells587To618.Box600.compactCandidate_not_isWeightedPerimeterMinimizer	7d2f812f0d18a0903f82f737f6b5f30c420f22538a121d3a44d5977760976e6d	equivalent
CompactCells587To618.Box601.QInterval	887ccb7db444fc77b777ef0558ea29d5aff2ae9ca1a83ddd1839297047aa7a9c	equivalent
CompactCells587To618.Box601.lambdaInterval	19fc83b18bf55be1df1aa1392751d00127f54882e6b4fc8fda03df0d9a000937	equivalent
CompactCells587To618.Box601.compactCandidate_not_isWeightedPerimeterMinimizer	a3638ece5064167ae649acdbf88642da6da0ab7c6decdb2b072ac9466673bae8b	equivalent

Declaration	Source hash	Audit relation
CompactCells587To618.Box602.QInterval	8cbb8d1e142b631d3c6de3fd50145ef90ff378141204629a12071e4eecdf3b8c	equivalent
CompactCells587To618.Box602.lambdaiInterval	f23f3f3fe05afd2eb531332a204eef881554e611b6abde25d3d3f87139c45d23	equivalent
CompactCells587To618.Box602.compactCandidate_not_isWeightedPerimeterMinimizer	66807cdbee988c70ab5dd5ab9b9d1b4d9c9a3437a4c552be16a8ae3341784a7c	equivalent
CompactCells587To618.Box603.QInterval	e1896cc15dd91218f81eb245807195110fb4d76593f9f582cfa810dc64b8d643	equivalent
CompactCells587To618.Box603.lambdaiInterval	ded1e4d36e660bf03f6cc3f616c4b179b42e11e364f6348c8e4d05c67470d1bf	equivalent
CompactCells587To618.Box603.compactCandidate_not_isWeightedPerimeterMinimizer	b0b337338b129c218ebcd0af7448a6681ec7fbe9d76570d160ce155c475b13c0	equivalent
CompactCells587To618.Box604.QInterval	d42c9a68406b3079cf80ac139bcdaff807b92272944dc16e3d3e458095c559b7	equivalent
CompactCells587To618.Box604.lambdaiInterval	6b9398f1dcb187d275c8eb5a104f8554a081ce17060b456e32f57a3a48824e25	equivalent
CompactCells587To618.Box604.compactCandidate_not_isWeightedPerimeterMinimizer	576f9519357783f305077fa8fadb2212b930aae6ef98bfcdc535989680eed70	equivalent
CompactCells587To618.Box605.QInterval	0ba5b97c62eb4f56c21f3ac93a5db4725219c030c147667e4e9fb7f0c207ceae	equivalent
CompactCells587To618.Box605.lambdaiInterval	1bdac8456874faf8b52233a3b1cfaa9628338c16acef609c6b0e2795f3f50e2e	equivalent
CompactCells587To618.Box605.compactCandidate_not_isWeightedPerimeterMinimizer	c3f85213b8f8bb77ec70b40ed3efda237c900573164d003b65387005f4ebf18d	equivalent
CompactCells587To618.Box606.QInterval	d2c4594148d0d6bd3ad01b13037c0ede0199c22db728e792e8bfbed7ac484288	equivalent
CompactCells587To618.Box606.lambdaiInterval	63c8a0a304b4c8c79594f91119d20658c0f90e2fb6373ca0d49e0497ada430a0	equivalent
CompactCells587To618.Box606.compactCandidate_not_isWeightedPerimeterMinimizer	a4d7513eaf40fc883d922324d60b5a12ea2df0b94c6ba7c067bac3b73601fbaf	equivalent
CompactCells587To618.Box607.QInterval	bf81d1121305e2326d16fbf17bfbfc65f0b1da6669659fed651461f611d2020b	equivalent
CompactCells587To618.Box607.lambdaiInterval	e384cab53e51cd43f0efaf1be2a486a8e2d4de8b76f4acd38ad1ae2358b44c96	equivalent
CompactCells587To618.Box607.compactCandidate_not_isWeightedPerimeterMinimizer	c518add820b462eee03450ffe8874df7af848eac4e0451b3f242b7c263289782	equivalent
CompactCells587To618.Box608.QInterval	21f189af3608778a5c1807949a28544047f7ce14286d8cbc963156828b56cdce	equivalent
CompactCells587To618.Box608.lambdaiInterval	3f3483f5c08cc6e900c1c0c9d774ec60937ffaca822f2211d19f455790de2570	equivalent
CompactCells587To618.Box608.compactCandidate_not_isWeightedPerimeterMinimizer	03db74da999670f34bcfe5d2b1ddb7ce7e13dcc23c55c7daf5e561aa3c75a27c	equivalent
CompactCells587To618.Box609.QInterval	6131a5867d64f591031e860d1ac8e05c1a0f68d9372bb6f1ae22f83811a99ef5	equivalent
CompactCells587To618.Box609.lambdaiInterval	7593b61b49aa4835ff77e3b153a2cca89964c3cec84b6c3af136712e6c710d7d	equivalent

Declaration	Source hash	Audit relation
CompactCells587To618.Box609.compactCandidate_not_isWeightedPerimeterMinimizer	ccf3355b718ffff534a6b95126283d4e17da9eb7fac75a0e645e311c7703de8a7	equivalent
CompactCells587To618.Box610.QInterval	b9c07de58bd66263441c7ba83aacd1a2c47b98b9490c82a1159b8958cb7b59f0	equivalent
CompactCells587To618.Box610.lambdaInterval	33e848075094bd127c96102af96b95c789c40221a726a87a05f775746721ffa1	equivalent
CompactCells587To618.Box610.compactCandidate_not_isWeightedPerimeterMinimizer	850c745712988e3e9d1c416b640da134714d049e4d0c9246e893896b1a88f88d	equivalent
CompactCells587To618.Box611.QInterval	c757ecb867ce4eaba32d95c8b42b8af72d5e748533307e77daf176c136a61ef4	equivalent
CompactCells587To618.Box611.lambdaInterval	4a5863e7f9eda4b116e709875b74544570fee1086b8f47d3244550dee9dd0472	equivalent
CompactCells587To618.Box611.compactCandidate_not_isWeightedPerimeterMinimizer	a01e4ed0dcb7390a1e63e0f7685d1df7bc817ebeaa66063c625f0d8cc79c9c28	equivalent
CompactCells587To618.Box612.QInterval	6b5cba28da9b689840eff1ad275027a78745bda776cdc7bb5854c727512b78f9	equivalent
CompactCells587To618.Box612.lambdaInterval	fe858b335d3e33db8004d5f61a6639010156149a06fddd199432b72f821c3479	equivalent
CompactCells587To618.Box612.compactCandidate_not_isWeightedPerimeterMinimizer	da7a96f5410a52a69dc44765818c54756e8d9da2580a437269a6db794e215611	equivalent
CompactCells587To618.Box613.QInterval	79acb5153a01fff3299691bc12ead38b4b532a8ba74a841e2ca1fbd03337a2f9	equivalent
CompactCells587To618.Box613.lambdaInterval	c9f136fcab3c5e61d112384ca1b75c815455daf807a00cc27620b53b7d8b2082	equivalent
CompactCells587To618.Box613.compactCandidate_not_isWeightedPerimeterMinimizer	2a4868aaf8bdca2ac2f272fbbd07d318f637802f2c39e91e5d348ec1e5832dc0	equivalent
CompactCells587To618.Box614.QInterval	47ef075fceba10aa89aab73f129152fe2cb145602672dccc8f648732641c4dd1	equivalent
CompactCells587To618.Box614.lambdaInterval	09bc0c67255c1cd590290941ffb76d3e4551960267f7d1859b83aaec20e5f576	equivalent
CompactCells587To618.Box614.compactCandidate_not_isWeightedPerimeterMinimizer	a4bc9f4f27360a3eb92ef60896346d722392d263dc618027003e41a5efc0770c	equivalent
CompactCells587To618.Box615.QInterval	a0940c903cca59865d936fad457f428a0ac0c7f749f79e5510c8ab1da1f1631d	equivalent
CompactCells587To618.Box615.lambdaInterval	aa10a192581e788aa992af5a2535fb48c122d23182f08ee7a9d3d4ed14c7449d	equivalent
CompactCells587To618.Box615.compactCandidate_not_isWeightedPerimeterMinimizer	77a7fc8088172d51a9fd8d782beefea98feab20fc47b23c2b5f23288632f06d7	equivalent
CompactCells587To618.Box616.QInterval	819eac5638c9ca46df706f2f4c9a7f1fa7359f6005655699f6efd536b749e2e1	equivalent
CompactCells587To618.Box616.lambdaInterval	462b522b0811e26ee54b515bce1d18a04c8ed8ddd4224f7431abdf76863aebda	equivalent
CompactCells587To618.Box616.compactCandidate_not_isWeightedPerimeterMinimizer	197d7bc1302ced824df2293aad169fb90ac3bb0a257382acd27839543823193d	equivalent

Declaration	Source hash	Audit relation
CompactCells587To618.Box617.QInterval	480af36011f9605f622e668e6d3389a920cf24d8e9a056599ce7cb775072a7ea	equivalent
CompactCells587To618.Box617.lambdaInterval	aa8a25f7bfe2c4c3677192733df60f7f2684ebe2b3def8ec94782819cad06f7a	equivalent
CompactCells587To618.Box617.compactCandidate_not_isWeightedPerimeterMinimizer	c7a7eb32973c675cffa0fbdbd6fdc6761dcec33e11eae0bbcb46ba45b2308233	equivalent
CompactCells587To618.Box618.QInterval	e8f90712e7ab8985588c5300ba4b7d5a920267da45a8f9877b94413bd240e528	equivalent
CompactCells587To618.Box618.lambdaInterval	c710df8e0089f2fcd73638bd11e942bb4af962bd9717cf5bcbffa4af6fb9c130	equivalent
CompactCells587To618.Box618.compactCandidate_not_isWeightedPerimeterMinimizer	a152fcf362ae519ffe64e781590e8fdb589a289bf4a7c255c5091509309a710f	equivalent
CompactCells587To618.bboxes587To618_candidate_not_isWeightedPerimeterMinimizer	e54a337d7aa2a41fff24ac9f1348db5c1e5a6af3e85a0afeea734d85fa44ad74	equivalent
CompactCells587To618.bboxes0To618_candidate_not_isWeightedPerimeterMinimizer	1b0ed147674dac6a2c6fd87897d076a463b3e3fa9269908743a14a02ecad21f6	equivalent
CompactCells619To650.Box619.QInterval	f4b72ad9f2a29644a981fc5b7e10053fc6e5442b57ea3ff8d52fcdc720e1f4cc	equivalent
CompactCells619To650.Box619.lambdaInterval	5ce9c50a18bf313adeb9592dfb6aa84d657cb7bf93e7a37a8e1c4073e844d75f	equivalent
CompactCells619To650.Box619.compactCandidate_not_isWeightedPerimeterMinimizer	4b8f90918c2dd537317a9c64eaacfa69493ef71bc0c2bde39b6eebb143f2b9fb	equivalent
CompactCells619To650.Box620.QInterval	54b0d4189e1c19ca210e1ac5050c69bdd577ad497e813ac3b0097dde5f4a632a	equivalent
CompactCells619To650.Box620.lambdaInterval	04f0d820cf8c74490a3f420c6cc5e9b084d139a99dcbaa89f54b1241c56f4cba	equivalent
CompactCells619To650.Box620.compactCandidate_not_isWeightedPerimeterMinimizer	1dfee46cacf261bdb2ed9aeb1f7c86e845e1042a262ff351ac23c0f0afe40b22	equivalent
CompactCells619To650.Box621.QInterval	2aa42ac8c0aa03e11cb62c3741adf11fcd364864b410ed9b2f10c6fbc84d8ee0	equivalent
CompactCells619To650.Box621.lambdaInterval	b3a4eca4f7cbde959cc4bdd027d605f1bcb a5abfafcfb903303562ef159494cc	equivalent
CompactCells619To650.Box621.compactCandidate_not_isWeightedPerimeterMinimizer	0726427ad7668b0b82b8136cf46b26b496e953e29afc3b1b9c29555a6377f6fb	equivalent
CompactCells619To650.Box622.QInterval	1a84551a04bd6233d8ad8bf9b7571fabf3861e512c2a09cd4a170fa412786aa2	equivalent
CompactCells619To650.Box622.lambdaInterval	fa6bf3309348df64d1c957ea06d02913bbcc8cd5ad13f35396d80165575e542	equivalent
CompactCells619To650.Box622.compactCandidate_not_isWeightedPerimeterMinimizer	706a43fba3752c1db02274a38d4f3ce451f2b60ea6b8f86d8fde5923dcaa041a	equivalent
CompactCells619To650.Box623.QInterval	cc0ce746ac859306bee99521720588b5e7322accfd86e91940f9bddaa132ada	equivalent
CompactCells619To650.Box623.lambdaInterval	e72c41d4edf5739db09cfd245ffd0b067858b799febe3a878636e9d3e1c77864	equivalent

Declaration	Source hash	Audit relation
CompactCells619To650.Box623.compactCandidate_not_isWeightedPerimeterMinimizer	7d0f8dbfc104d7aa8d9c34829680ce6f82870228c76b09b9589e06a6629879f5	equivalent
CompactCells619To650.Box624.QInterval	6a8a4be74c11a9dc9ebe81a8b940e7747f0d0fe0f083e2fe5205a1e2fc260d6e	equivalent
CompactCells619To650.Box624.lambdaInterval	f7745369d2c81af69bde4189dd976357f0814607501e3b80fc0d893d8f11e56a	equivalent
CompactCells619To650.Box624.compactCandidate_not_isWeightedPerimeterMinimizer	0a060b05e14daabb4fefce5b131e341f1de19fee77457d534ec30c38b06ff0d	equivalent
CompactCells619To650.Box625.QInterval	5468d7ea6957ffffc042ce1c7d85f5c5de0a4ae8b4085958c25b19fb8a0b90b2	equivalent
CompactCells619To650.Box625.lambdaInterval	f10bbf286802e53e2b9e8720a83344f2064b13135b902a35f0e4359d4dc1f089	equivalent
CompactCells619To650.Box625.compactCandidate_not_isWeightedPerimeterMinimizer	5d41f76d69ae1573cfb4e9697e14d63a25616e005de3d685f246a317f3c6ddfe	equivalent
CompactCells619To650.Box626.QInterval	000a101ee28c0334c74b2c980d6afe4633d36ceabfa689dd0b754d73d48ed6da	equivalent
CompactCells619To650.Box626.lambdaInterval	dcdfdf239a70d5069c7186d89c0f6fbc34923e97faea3329a7a70c876cd477b25	equivalent
CompactCells619To650.Box626.compactCandidate_not_isWeightedPerimeterMinimizer	6f6a334461ba1af4e800bcb721fb452ff1ee6344a74fe7468ff0297560c34212	equivalent
CompactCells619To650.Box627.QInterval	b75e4a98087e1b5b9d6291a43415a93ba0674d45741d5cf376dbdadd39f654bc	equivalent
CompactCells619To650.Box627.lambdaInterval	0affe955d5e3a5f7ae327b0d94e22a62040041e6a3dbc4b03c45769037c0c6ef	equivalent
CompactCells619To650.Box627.compactCandidate_not_isWeightedPerimeterMinimizer	fcff20813f5db5f4f3692715517415bf6e32afcf39148dcd80feea4e75465c68	equivalent
CompactCells619To650.Box628.QInterval	5d3350b54121860a0e24726d906cd042b97d1d8cad3fd9bea62be1b7819f2bab	equivalent
CompactCells619To650.Box628.lambdaInterval	409cae742a59a417ac731a3cfa6ff5edcf1015277446ea28c6b73fe075b92e67	equivalent
CompactCells619To650.Box628.compactCandidate_not_isWeightedPerimeterMinimizer	265b957d728bb4a850094909e3bc003d7d66e1ffe9bed0505aa825195e6be86e	equivalent
CompactCells619To650.Box629.QInterval	bc5532b78c047c151d4dc01653b737d788d0b5ca0a74cbc790841ef5e684e7f1	equivalent
CompactCells619To650.Box629.lambdaInterval	36082152be777ea58fbf8acc7c996bfd1cf3d9afa3006369a43a1576630fd98	equivalent
CompactCells619To650.Box629.compactCandidate_not_isWeightedPerimeterMinimizer	948b31fcb593713eff83717c35bdcaf1cb16e67b6ffd5573a4550362fb8d9d12	equivalent
CompactCells619To650.Box630.QInterval	2b82b7fb12a98cb55f64edb97ef79f048496c32b85cd811e78604d231f16c5d2	equivalent
CompactCells619To650.Box630.lambdaInterval	91117903a50a9d20479c802bc030f89ef13e1173351b924d46d40c868dec5bcb	equivalent
CompactCells619To650.Box630.compactCandidate_not_isWeightedPerimeterMinimizer	9508821bc73c98d0e8ba11c48ebdcbc021c47282c2f36ec11ece2366ee050d5b	equivalent

Declaration	Source hash	Audit relation
CompactCells619To650.Box631.QInterval	99278cf344b3c80d6ea155ee3c7132cf9ca84ce3412faf22803bcc26078d2229	equivalent
CompactCells619To650.Box631.lambdaInterval	6db4ebfd8389c3eca7df0021c8a5499ead880a5d6b7b42bdbb00bd19c953fcc56	equivalent
CompactCells619To650.Box631.compactCandidate_not_isWeightedPerimeterMinimizer	905f026a9956e70f1185413ed9bc72f401b962af11d3d95a2b833f58e36e26ab	equivalent
CompactCells619To650.Box632.QInterval	864b8903af9763211243fc456a94a5102927a7734ba912aaef26927374491c48	equivalent
CompactCells619To650.Box632.lambdaInterval	4486add855295c746606ee3f3b96a1dc397c9b0fcf10110567fc36fb9cbcf895	equivalent
CompactCells619To650.Box632.compactCandidate_not_isWeightedPerimeterMinimizer	4337600041e7bdf2d98ef369b9b78de557631a707d5eb9e24f2d8d8cbb810360	equivalent
CompactCells619To650.Box633.QInterval	0f39e6d1a9c59c41ff45f953eaac5afffa96a7068284073abe168155b6f37b7b	equivalent
CompactCells619To650.Box633.lambdaInterval	9c17e342492b1537761d495279b3eee1cb174463fa3cff7224fdccc74a06fd62	equivalent
CompactCells619To650.Box633.compactCandidate_not_isWeightedPerimeterMinimizer	dc304d4bb5920df3192e28bf1d4fa6e92b53d997e7462772539a9652b287379f	equivalent
CompactCells619To650.Box634.QInterval	9b33867212c87ec0c6259631734a32e3311356816f7f1a8ea5f62936dbd993f0	equivalent
CompactCells619To650.Box634.lambdaInterval	a3783b296de7383808d22d23dda520fb190e45e968ceafde25ba856ee631960b	equivalent
CompactCells619To650.Box634.compactCandidate_not_isWeightedPerimeterMinimizer	42eb79c7892d552bd0e2c1215c07a58656642d110870a50111a38730cfa96c9a	equivalent
CompactCells619To650.Box635.QInterval	0003096bf18867ed75354ef0109b893f9708875a6ee9309bcb3fddf128a100695	equivalent
CompactCells619To650.Box635.lambdaInterval	05bf2d62826763cc9e4c95c8256d3a618e7a80fbae380f9ab95379f66f0d2a20	equivalent
CompactCells619To650.Box635.compactCandidate_not_isWeightedPerimeterMinimizer	2af542f8a25cfd957c3564653f52604ffc91b21669efd4bdd5c6969014c6615b	equivalent
CompactCells619To650.Box636.QInterval	07fb2c8c1f6be68df6e0f4ff8f6c3278bee833dc0f892382def11487ac1038a8	equivalent
CompactCells619To650.Box636.lambdaInterval	d372c6213436801297d0510efe40641e640ae88cd857f7f0a38a797047cc5c41	equivalent
CompactCells619To650.Box636.compactCandidate_not_isWeightedPerimeterMinimizer	0bcb142269fbc19bddb5df6cbcd6c8e6a7ca0e9725a287a24980656423fe0757	equivalent
CompactCells619To650.Box637.QInterval	94f323d71101240cb4e603a312fa4946b94b9718a1757c9fcd197ba377149833	equivalent
CompactCells619To650.Box637.lambdaInterval	671ff074503c4b7313fa138591894ec195e8f47ad3e919ed76ed54a3d57b72d5	equivalent
CompactCells619To650.Box637.compactCandidate_not_isWeightedPerimeterMinimizer	684404c0730ad9ac014af7635641055e0ec66db3c27439872b12a2f6ef159fb4	equivalent
CompactCells619To650.Box638.QInterval	80d8a6db18a98004cf5ff82625e82adf9e841c1dff44e8199b8467c46c041cfd	equivalent
CompactCells619To650.Box638.lambdaInterval	be0608a4e36f6d08236628aa35d2472446608edd1fc2c0b62b691213bcd6698a	equivalent

Declaration	Source hash	Audit relation
CompactCells619To650.Box638.compactCandidate_not_isWeightedPerimeterMinimizer	d8c72b86f8379dea18fb49d66d60fe815bb0c00ff27566a2d7702f5bc5772dd92	equivalent
CompactCells619To650.Box639.QInterval	109288fe5bbb495230d510c0f240183d289900ff2f8ec7086f5c7d7a8fbbda68	equivalent
CompactCells619To650.Box639.lambdaInterval	7b43e3e7e93740962b5eabe9bfcf3552170e10457dfeed5055eff92abc213e58	equivalent
CompactCells619To650.Box639.compactCandidate_not_isWeightedPerimeterMinimizer	84be3434241b218fee142c4da3b0e1039d40cc2846a0e38354f5ec9f14874b44	equivalent
CompactCells619To650.Box640.QInterval	9ca487ec058f6845d778dcf304278665a54ac3aacbcdcf0c7fa2b881ca3c8104	equivalent
CompactCells619To650.Box640.lambdaInterval	847b28a283d04441a0332ef38569191f67e11dc87c5f317ebb350a3ce46f53df	equivalent
CompactCells619To650.Box640.compactCandidate_not_isWeightedPerimeterMinimizer	e1cb38a51b2cb1b25dac45b8c04ff7db33c62b6b935f08963d97ac0a85cb0b4d	equivalent
CompactCells619To650.Box641.QInterval	9c83663a09e6ffe83675a36c947f44cf06ea9bbb4ee2ab6846d48612b1ef57ec	equivalent
CompactCells619To650.Box641.lambdaInterval	9fcf2ed15e31ed9859122abca423de8f353ea4f15c7bdab7d73f69dd0c4c432e	equivalent
CompactCells619To650.Box641.compactCandidate_not_isWeightedPerimeterMinimizer	b5cbfc51ea524441bc0bf4c267b6ac0a9777648ee5a415453b318c104f8c77cc	equivalent
CompactCells619To650.Box642.QInterval	ab4d955af1fb30d93f9b33d44fc6b6d9e2c4ceb7895a9b62b2dd4c25ccbb51ca	equivalent
CompactCells619To650.Box642.lambdaInterval	f24d3234c4b95f416a645aff529ee807b6a9b007aeb267163435f23e17590b85	equivalent
CompactCells619To650.Box642.compactCandidate_not_isWeightedPerimeterMinimizer	6d20144c346de9ab3412af879a6c3f45a9891b9ff721a331bd18894b5fc28eb3	equivalent
CompactCells619To650.Box643.QInterval	8aa23957d6d03fb4dcbf8d4039ceaa860ad8e8b2bb05c581b6fee0b05cdd336e	equivalent
CompactCells619To650.Box643.lambdaInterval	0f79c14a71c3645dc51ca42b97b98f3c6e99025f2f502fdb0ddcbf61de65dcda	equivalent
CompactCells619To650.Box643.compactCandidate_not_isWeightedPerimeterMinimizer	a3f2c6156d607b58541516cc421f7ac174535b609f4d76541e82ae0b3318a988	equivalent
CompactCells619To650.Box644.QInterval	cc46409e3220dbf7ab0b6c5e142bf69b118d30d66d3d828378d021b775e8c98f	equivalent
CompactCells619To650.Box644.lambdaInterval	9842e29ebb0c967de995c5ca994ba08047e67329204d6e6dfa48d32daa74afe7	equivalent
CompactCells619To650.Box644.compactCandidate_not_isWeightedPerimeterMinimizer	23bacedce984d240aa88dc49d08b6529b9ad8e1fe432e5802c92ef9c761c553c	equivalent
CompactCells619To650.Box645.QInterval	7e41c2e44bbd7a58480e0b460cdc02d28a5d7933e1d310af37b40ae0989faf07	equivalent
CompactCells619To650.Box645.lambdaInterval	92048706d846c0131900e8c48fa1b815054f0a023b92b7890f78380ef67f6e15	equivalent
CompactCells619To650.Box645.compactCandidate_not_isWeightedPerimeterMinimizer	a32d5adc4c300e2480db81d36186e3bdf838e72aef7aae766d3a468a5b0018a7	equivalent

Declaration	Source hash	Audit relation
CompactCells619To650.Box646.QInterval	9745deb9de0762556ababd5c3224829ee9b09ac72c5ffc04237d3772ae85abe0	equivalent
CompactCells619To650.Box646.lambdaiInterval	583be25a6f72ce5e16e811938d1469b93375aa719de0270584ec2cf107b13a6c	equivalent
CompactCells619To650.Box646.compactCandidate_not_isWeightedPerimeterMinimizer	8eb64b04b1953a5c0710c1d9a6e0fe0318e32edde0f8d1db12cb6d27176b2289	equivalent
CompactCells619To650.Box647.QInterval	e92af27724002ba0030911fccd5ba2871f74b9d57d4fbabde267394c0929ec7f	equivalent
CompactCells619To650.Box647.lambdaiInterval	d209658116134865596dc9fedb0bc10cdb12d4dc43da49ff94ae925ac fcd2021	equivalent
CompactCells619To650.Box647.compactCandidate_not_isWeightedPerimeterMinimizer	5739bd52b2f20a7d08bf1c6b163631821a8ebbd97547d9f609b2908f34697e56	equivalent
CompactCells619To650.Box648.QInterval	0c6908d8a6fb682d06cc7f72998576d3089c248afde9b78ab0198dd3bfbf568b	equivalent
CompactCells619To650.Box648.lambdaiInterval	68132f6d8acfb1c662a4d9b5db5e9b09bd33321649d34cb42213810b128fde5e	equivalent
CompactCells619To650.Box648.compactCandidate_not_isWeightedPerimeterMinimizer	2a6b10d5a146088d344446564092e67865a62c3d25440449986570097ec01a33	equivalent
CompactCells619To650.Box649.QInterval	76477ce06bf23d67d88dfe079ff8906e0e08f288dc1ee73b782d07dd6915acc3	equivalent
CompactCells619To650.Box649.lambdaiInterval	773f1095fd6e13a478443c91b13344415fb3ce06eaad63c16d0c9f0b07e202bb	equivalent
CompactCells619To650.Box649.compactCandidate_not_isWeightedPerimeterMinimizer	ae77bc071662fb386b65001f70b79d5406ddce7086844c84e03cc9974bd70e3b	equivalent
CompactCells619To650.Box650.QInterval	0d8be187cfd768b6a9c2e0299c212f35763e62520bc5e160d28d5ac0ebca6ab8	equivalent
CompactCells619To650.Box650.lambdaiInterval	11359763f4d554fd72a30aab25d0305b9c8169b17da0213a3b99032840e5024d	equivalent
CompactCells619To650.Box650.compactCandidate_not_isWeightedPerimeterMinimizer	ac7aba840171e73866684958227c5a799a4c977863a5716f659e0e2cd6caf344	equivalent
CompactCells619To650.bboxes619To650_candidate_not_isWeightedPerimeterMinimizer	5eb50b7567e60b792c061c6c2e09e28ed4625c4b11d54d88867571339d816fc1	equivalent
CompactCells619To650.bboxes0To650_candidate_not_isWeightedPerimeterMinimizer	d489bb1a51ec984a7ad9bfab6fc2ab9d52490501a0db592e09cd6d04704be4a6	equivalent
CompactCells651To682.Box651.QInterval	accf9d76568ca504be806575938912d00d6686bf81dcf46e42adf68c5b80c666	equivalent
CompactCells651To682.Box651.lambdaiInterval	6b6e89cd328ce235bc30c403d835c86deb4bd59fc9ed3c0dee56ec8a2e225b16	equivalent
CompactCells651To682.Box651.compactCandidate_not_isWeightedPerimeterMinimizer	a157be5451024043a7a1844a23a9ca1b2cf3ea192d2b60be94d98b4e6839eaa6	equivalent
CompactCells651To682.Box652.QInterval	257cac110f763deabc7c98d10d35b303855b6260a930d4df41d08138b1861a28	equivalent
CompactCells651To682.Box652.lambdaiInterval	4696bdb06200077aa0f4d602aca7cb4ad0600c274cc66da39a182c76d2649ecd	equivalent

Declaration	Source hash	Audit relation
CompactCells651To682.Box652.compactCandidate_not_isWeightedPerimeterMinimizer	9fedaaee8f155d7e775e44ff559e090d4b08e2bdcc22ff51a7ce5b847abe9aac	equivalent
CompactCells651To682.Box653.QInterval	516030603c34aaa21b481d13b0fc11752369bdcc7df8649c3e5c662a59473436	equivalent
CompactCells651To682.Box653.lambdaInterval	9f499a1179edd6f47df50c8c6f9e14e2e570e7d3f685a1d2c9257ffd7667197d	equivalent
CompactCells651To682.Box653.compactCandidate_not_isWeightedPerimeterMinimizer	c6908b89688414d04ab3e100a4711458c4b3878a27f3c1ad69dd9188670ce9f0	equivalent
CompactCells651To682.Box654.QInterval	e7b05fd31bd2b6d180a04ba4b9cb8145ed534f40eb33482b94abb91e3f1b7101	equivalent
CompactCells651To682.Box654.lambdaInterval	40e30995733206809441f4f5cc50fa45e4465cc9810103d8bee5b362a17b3f6d	equivalent
CompactCells651To682.Box654.compactCandidate_not_isWeightedPerimeterMinimizer	7eb88be9b0eb09d4abc6f4348b57538111c39b8828bce7997d1fa17b010b7914	equivalent
CompactCells651To682.Box655.QInterval	4ab2c235f6d4d726d6d23f60fa2a397984f205cd1b62e729f1366f38348db564	equivalent
CompactCells651To682.Box655.lambdaInterval	09e0b3e465a1bb9dc9fc5df67416382163be6d3ed1745aaad660c564415375c5	equivalent
CompactCells651To682.Box655.compactCandidate_not_isWeightedPerimeterMinimizer	5e8a405175ae85e3a279faeab232c0b05b47283fd795ae10978a7c03277e07f9	equivalent
CompactCells651To682.Box656.QInterval	d574bed3c026e53e6d076b6c8da34df5ec9346aa8a24f8013d51717ad3203f99	equivalent
CompactCells651To682.Box656.lambdaInterval	cc4e4d33ea7535d36e25cd95933c981e29fce3c7adb7341501eaf7fe82663b6	equivalent
CompactCells651To682.Box656.compactCandidate_not_isWeightedPerimeterMinimizer	d0460bf2be435d608cefd4c8e4717a4ee8bde593e86ae473b345ad69df3b358	equivalent
CompactCells651To682.Box657.QInterval	a08bb7ee15ff1a42948cef1e33a3ea996447b826f3d91b0cd0037220e8fa0e56	equivalent
CompactCells651To682.Box657.lambdaInterval	29c9ee185eec43d6a1c38bfa9a983be776157e79d6bda2b5cbfe4c407cfd65da	equivalent
CompactCells651To682.Box657.compactCandidate_not_isWeightedPerimeterMinimizer	a41d1e024250e4649b4b856ac0f9addb332d7611da390461138c628574ffb18a	equivalent
CompactCells651To682.Box658.QInterval	5742575876ac1290a48593f21cfd978fb513725bce57873aef4b84434cbcd06a	equivalent
CompactCells651To682.Box658.lambdaInterval	c6f86e0b4eec0f9e64ac402d84be1c89faa495163ce53198d9a624d946309c9e	equivalent
CompactCells651To682.Box658.compactCandidate_not_isWeightedPerimeterMinimizer	7f827a9aba7655a4c3ddbaf938b7a11cf93a6236c5b1f62a05e5c9502e6e256	equivalent
CompactCells651To682.Box659.QInterval	ca0b4145ceda421fd9cdd526f66f309113fdeeb5c3f06059f3cc6eb792a1ab0	equivalent
CompactCells651To682.Box659.lambdaInterval	08c472eed1dbc62d4633905c2136dc78b0b28a1969a4515091647016dcdfa0a	equivalent
CompactCells651To682.Box659.compactCandidate_not_isWeightedPerimeterMinimizer	6ac090c5b9beb5dfd293c177ad105711100b6ecdfe839450d34137b55fe72d85	equivalent

Declaration	Source hash	Audit relation
CompactCells651To682.Box660.QInterval	0e0236dbdc18ebabf0bbc4efa32838a6ed40aadbb7c7a6f5e77ad37f2cfd4674	equivalent
CompactCells651To682.Box660.lambdaiInterval	adc825d097d624396ea75c2699ea5aac2060d85720121ef0c fb78435829313a5	equivalent
CompactCells651To682.Box660.compactCandidate_not_isWeightedPerimeterMinimizer	7d68bf90118abfacc46e00cff0c3ca653b217db2d80bab6ca89d12a93eeab9ac	equivalent
CompactCells651To682.Box661.QInterval	c56e5d8f946b57465849ab75bff0a499b34a9ec8101e147ba32f506e573366f8	equivalent
CompactCells651To682.Box661.lambdaiInterval	4709c0f9153b7d6c00c5113cbc5dc284a067bea69c748d4371693766d5f87eab	equivalent
CompactCells651To682.Box661.compactCandidate_not_isWeightedPerimeterMinimizer	2993ab1f81c8a111ab48e3bfa8d9b43893d03418ffe9e15256c5106a25ae58af	equivalent
CompactCells651To682.Box662.QInterval	10f544dbf0e6f79a92108e5980df4d097c30efc153f69ac876da6be675bd67be	equivalent
CompactCells651To682.Box662.lambdaiInterval	01327020b992ae071b4731566b89afcc265950cb7fdb58c4bc848ef60615c1fb	equivalent
CompactCells651To682.Box662.compactCandidate_not_isWeightedPerimeterMinimizer	e044d14b76e06769a2ae34c94b6ff3318d8b63808b2de461cc6ac004b02f4642	equivalent
CompactCells651To682.Box663.QInterval	13216958b4fea9c2ce957d844ad4546f3d23b3e471e81b4b265139e59c9b13a9	equivalent
CompactCells651To682.Box663.lambdaiInterval	01a96da1bcd167e5f13ac9cc68cac968b4d987157714a0416d6909ee3ea76dc6	equivalent
CompactCells651To682.Box663.compactCandidate_not_isWeightedPerimeterMinimizer	72078e45f1cd3f786ee3d2ac495560fc8d2296d5e42bffffe28f8dc86718c89a3	equivalent
CompactCells651To682.Box664.QInterval	caf8d8e35bceae73106f0f4958738d32b9e5fda1349e6f2f20b9194c4c43d778	equivalent
CompactCells651To682.Box664.lambdaiInterval	c35a00dff7d3e25762d2f1c91ed89fc140d164638e3a34d27c6553ebfb6fd802	equivalent
CompactCells651To682.Box664.compactCandidate_not_isWeightedPerimeterMinimizer	4403a18937ab22644e7104ff170a7dd6d4cbc199dc7ddc0110c0105170ec82b2	equivalent
CompactCells651To682.Box665.QInterval	d0809ba99e9d3100af54b350a9f1bbafbf28b121fba121c8f3ed0e34551a9b02	equivalent
CompactCells651To682.Box665.lambdaiInterval	ee7e6d470c43626709721168b7e454e2981f0d86a003f5b9071dec2bfd50b8ab	equivalent
CompactCells651To682.Box665.compactCandidate_not_isWeightedPerimeterMinimizer	8e669281de3837d1821722bdf699840a75fdd45c011970e912ad81e29626c222	equivalent
CompactCells651To682.Box666.QInterval	bc1ab998fb522b839342637402ad8ff103fb5f96f3308f8ed0252c391cb66472	equivalent
CompactCells651To682.Box666.lambdaiInterval	f3d6f1486d30c25065e4c840179e5a897e5db92f66daa39b32e30db60849f697	equivalent
CompactCells651To682.Box666.compactCandidate_not_isWeightedPerimeterMinimizer	870e5ae5ecdea3eeca8a80aa37bf8bed12f4f35f5fa46ef64c03303d33f6bf88	equivalent
CompactCells651To682.Box667.QInterval	50e16366331a554523cac38dcd793375549b915b7996e7b6c8b873281b7bbc05	equivalent
CompactCells651To682.Box667.lambdaiInterval	aa4096ca0e8ef220b91d071eed21af4bfd3a282629ce854b53ff931c2e151063	equivalent

Declaration	Source hash	Audit relation
CompactCells651To682.Box667.compactCandidate_not_isWeightedPerimeterMinimizer	179c1c16f10b603b4d404b7c05a73d7d76b536de6c24e2e1d6f5791aac6c1426	equivalent
CompactCells651To682.Box668.QInterval	bad93032d8318319c304d48c17e89838e966fffaee4dda747597bcd68378634f	equivalent
CompactCells651To682.Box668.lambdaInterval	56cb563492867150ba2000e1a88206f53ac9b2d521043e7a76b9724f7047ee8c	equivalent
CompactCells651To682.Box668.compactCandidate_not_isWeightedPerimeterMinimizer	521704d08333735c75b52b854b23509ea535d2ba9a5cb3ebc8a4748aa06ae393	equivalent
CompactCells651To682.Box669.QInterval	642fc169149df41e62e2bf03c823821cd9bcd692a2a29c2abdbd083b5226cdbb	equivalent
CompactCells651To682.Box669.lambdaInterval	da7a299a17751d33d2df140d2936d2684c9a135e8e520d785610c4d26cc68954	equivalent
CompactCells651To682.Box669.compactCandidate_not_isWeightedPerimeterMinimizer	37df9e6d25902d9f2449c07f4692759e10f3dad8c86ef9633a91731befca315d	equivalent
CompactCells651To682.Box670.QInterval	e09ce9807966726b755c5deb077b03eadcf8aa49414981221fc71f2b1965eace	equivalent
CompactCells651To682.Box670.lambdaInterval	b205a4c28ac6a359a447d1c1829417f7fa5764204601f84978d5bd7b4b66f6ce	equivalent
CompactCells651To682.Box670.compactCandidate_not_isWeightedPerimeterMinimizer	6bf9d7c76b45f8d8aed34b4459a773dd70816034933fc26f0a0e6a3a7ad3d62e	equivalent
CompactCells651To682.Box671.QInterval	e4cd3e8b783a8551a2d6316ade3e31f6d8833a433fc3ae42e9f9dcb22918fa09	equivalent
CompactCells651To682.Box671.lambdaInterval	4a6b0d899a1023e2ade54f4b47c32804fa82227d77cc55ca329066964f223c71	equivalent
CompactCells651To682.Box671.compactCandidate_not_isWeightedPerimeterMinimizer	7e0ef7f9adcaa1b6f0b6f644ec2269e09f46a252f660012a7397897d0883ccce	equivalent
CompactCells651To682.Box672.QInterval	1fe9ea7da9b75cbfbc039163c51e211ae4b7d3e26f76c9f3b46f85d6ab36b55d	equivalent
CompactCells651To682.Box672.lambdaInterval	b4c4cc9d9efed0e1b23f6950839652e73bb6c2c00e89f13157ffa3570b2c656d	equivalent
CompactCells651To682.Box672.compactCandidate_not_isWeightedPerimeterMinimizer	d5e6860c4b318587bdb6691c358fa24465ed670808ea62fbd5ac0072bf6c4e27	equivalent
CompactCells651To682.Box673.QInterval	15395cfac007b579cdcccd90bb6e7dedf39cebc8fe15498368a4a0d33865894	equivalent
CompactCells651To682.Box673.lambdaInterval	a0312129995064d554302b80f622932108c5ef37873784bbea30c98544819de1	equivalent
CompactCells651To682.Box673.compactCandidate_not_isWeightedPerimeterMinimizer	0b6b652add1ee55db7cc9fba17b3309489bde09ff47f505305d2c65ba3a3ed1d	equivalent
CompactCells651To682.Box674.QInterval	b16af38b207605f1eb9a78b2d73b32d23dc628b9cdb5fc09474349d50255803	equivalent
CompactCells651To682.Box674.lambdaInterval	338709e6ba21be66a117278fa40809d2cdc467540e0ee4d82010be4fd3136ec2	equivalent
CompactCells651To682.Box674.compactCandidate_not_isWeightedPerimeterMinimizer	b7eea96e21da4272d6c7b918d3bc8cd3df3bb1c7ee3e99a19dcfa56b8e0ae4bd	equivalent

Declaration	Source hash	Audit relation
CompactCells651To682.Box675.QInterval	4aa609b89be60aacdcaa0816cca772989ab a1c5d3d143bd13c0a02e5c8a17260	equivalent
CompactCells651To682.Box675.lambdaInterval	2df050d0d3ce23a0cca898e2c9995ab4b3c b0a34c2d6fc205b2ecc898787bd0a	equivalent
CompactCells651To682.Box675.compactCandidate_not_isWeightedPerimeterMinimizer	6832bce798a26c78a9ddd84a3dc1c46cf31 c6f8b1e62a813e8ba5b261cb13245	equivalent
CompactCells651To682.Box676.QInterval	75b1b1fcbcb004e5eb39210775c64442fe78 a6e5c8dd185c3a32f54ccf6b40633	equivalent
CompactCells651To682.Box676.lambdaInterval	ccc2224d6d7a65b83d43b562e3f64b4846d 4e71aba5a471ed585fd84c30ec1e6	equivalent
CompactCells651To682.Box676.compactCandidate_not_isWeightedPerimeterMinimizer	abc639c3fcb4d7c9f2a454b5980d363e5a4 a619c17c35fcc3660f2238a2260b8	equivalent
CompactCells651To682.Box677.QInterval	094dbb8f83b87fa0f218b0e9e749fa79253 8e24db5008e2ccbb9612d7cb140aa	equivalent
CompactCells651To682.Box677.lambdaInterval	48d95719761b83c8a675f01dc6065e269e2 287aba6f2cc59e29c1dd27f878481	equivalent
CompactCells651To682.Box677.compactCandidate_not_isWeightedPerimeterMinimizer	958bfd01bcdcd4d8c666f2bc32008eaa383d 440c18decbb99c45aa8931ed21a358	equivalent
CompactCells651To682.Box678.QInterval	0a04531380f7f0dca80e33d3d7a890f7199 0d70c2bdd0f88dae90c8d4727017d	equivalent
CompactCells651To682.Box678.lambdaInterval	f346c91b5e98191c33d015e7d3c23eff2ba 486859b1c0d916d476c9b4af28701	equivalent
CompactCells651To682.Box678.compactCandidate_not_isWeightedPerimeterMinimizer	0d584c7e2cdc2194d34108c0e75e015744c 88e71fcfa82b9e691ce3bc6c6146b	equivalent
CompactCells651To682.Box679.QInterval	636afba493041ccb3560630eea6406a2f5d b1e5080157c7d48e5f00419ed72e3	equivalent
CompactCells651To682.Box679.lambdaInterval	25a6a5ce74338b0652b9ea837b1285fe18c 06fd934a699781ddd26876bc26ea9	equivalent
CompactCells651To682.Box679.compactCandidate_not_isWeightedPerimeterMinimizer	bd3addba362a118bb05e6ccdbcbcf2a10 b05761240de58b1d30c026884a953	equivalent
CompactCells651To682.Box680.QInterval	cd3a25a7cc37c6b9ac145d14b9b121e1fa9 e718dc0f3194e26e1410d459fb4ce	equivalent
CompactCells651To682.Box680.lambdaInterval	511c8e9df1c9d928d6bdfddf6a09c3d1850 7777a5cd7b8c890b3b4465b6b5e67	equivalent
CompactCells651To682.Box680.compactCandidate_not_isWeightedPerimeterMinimizer	6a14910f8de953dca167eea0e014e4a1fbb 14465f7f040098a30a69d65c7d99c	equivalent
CompactCells651To682.Box681.QInterval	92c5e750a6437ff2b1106c0ded9f6ff5282 31c6960643b54714f79ba0ed92c32	equivalent
CompactCells651To682.Box681.lambdaInterval	e8d04ea0a262e567d744d5c437070af7c12 7410270616c0403a75c225f2ff58f	equivalent
CompactCells651To682.Box681.compactCandidate_not_isWeightedPerimeterMinimizer	1112e0361a19ac74ea8239ffd8a3456c731 e5ef21f978f76051fc54594044f2c	equivalent
CompactCells651To682.Box682.QInterval	296c8be0ce897e3e380048cd7adcd993461 554fc3a0b5b07bd45075f68c044e1	equivalent
CompactCells651To682.Box682.lambdaInterval	505ab63bb5341d3ee96541f8a2c86bf3b21 4276b98297bd43abf7fe3fca3bf3d	equivalent

Declaration	Source hash	Audit relation
CompactCells651To682.Box682.compactCandidate_not_isWeightedPerimeterMinimizer	504c0f23f54843b4e895d299f48f9cd75cc55cd47160d35eaaffacc32da33dbf	equivalent
CompactCells651To682.bboxes651To682_candidate_not_isWeightedPerimeterMinimizer	5a366bf166f5e4e29cefed51f25ee61d0bbd6e759f2f065810a08813ff1b3335	equivalent
CompactCells651To682.bboxes0To682_candidate_not_isWeightedPerimeterMinimizer	9d862510e06e53f6fb69dafaf725e55e6af9a3abecd59353c1a4b54caa23156d	equivalent
CompactCells683To714.Box683.QInterval	a39e5b7b02385e8e1329341ed8bebae8278ea95bb2f868bc54593b84e0f5ffd4	equivalent
CompactCells683To714.Box683.lambdaInterval	1f414d00e10757abfab91e6a25b9eab0463057600f484a50191beb99ab646737	equivalent
CompactCells683To714.Box683.compactCandidate_not_isWeightedPerimeterMinimizer	688deda50b41e95f884186c4e6aea45b772c21a06d04ee4a6c1ddab1c1d6c9e4	equivalent
CompactCells683To714.Box684.QInterval	3f49c67ad08ecbcee30200cd7de1d0c8bba10b6d30189907ee0410304565642c	equivalent
CompactCells683To714.Box684.lambdaInterval	edda46a97b8475ba5f97642b277d051328291d590cda49e107749073bf6499ef	equivalent
CompactCells683To714.Box684.compactCandidate_not_isWeightedPerimeterMinimizer	26c8cbcd7665c5a7d5e935a6a8daf114bf7b0970be352ca5a949b828d5719179	equivalent
CompactCells683To714.Box685.QInterval	bf4c29a01ba36f42de11cfb6795feb50e92de7493a79f812fb8e675baafba4f7	equivalent
CompactCells683To714.Box685.lambdaInterval	c7fafa4944a2eb29bbc074ba6cbdfef165ce1f16b4dc8ab028f62fb68e2fbc0d6	equivalent
CompactCells683To714.Box685.compactCandidate_not_isWeightedPerimeterMinimizer	0e42e112d30b24f357c2ddd7c436794a9e95c8f350b1e8225fd5105c42fb12c7	equivalent
CompactCells683To714.Box686.QInterval	a7d3572875010a7fe2bf55110db04c854652a69817615be78d8e0ef61c4691b3	equivalent
CompactCells683To714.Box686.lambdaInterval	d777fc413b90f59852904ebc7bfe7f332463cef8192cb78cb1c24459e8f1d3c3	equivalent
CompactCells683To714.Box686.compactCandidate_not_isWeightedPerimeterMinimizer	81d501717df88b960bcd1cc924c8ef01a0985634606c3fae5ab45bb7b8e54745	equivalent
CompactCells683To714.Box687.QInterval	073760db5cc7f5ae82f1629e24939245e2afa40a235deafea9dad108b1aa362	equivalent
CompactCells683To714.Box687.lambdaInterval	8ed1178c8627eabe55ad61efe491cfaa76fdccaa9837898c68ba64799e9e1cc	equivalent
CompactCells683To714.Box687.compactCandidate_not_isWeightedPerimeterMinimizer	f5a82843458f196266d667f24af52e7d3a1b8e29053b4265b727f69d06cf374b	equivalent
CompactCells683To714.Box688.QInterval	ffcd995de249ecee184ecdbbc67a2fa66371d1ca0ee39eb456f6fdd50a14ea0e6	equivalent
CompactCells683To714.Box688.lambdaInterval	4bbc0c94163ec989d7fc3294c861ab69db9a1dcfa7668e1fb975e235515bcf54	equivalent
CompactCells683To714.Box688.compactCandidate_not_isWeightedPerimeterMinimizer	2e7ad6ef40bc82a78a7f2480104c4bd5c6840a3ca02e2b047ce9805121d9bae0	equivalent
CompactCells683To714.Box689.QInterval	b5368317791f41f9c730bd78d9ba0e408ecbee46d313f071c51131de054897d6	equivalent

Declaration	Source hash	Audit relation
CompactCells683To714.Box689.lambdainterval	a35c7aa3ff6e7c6b54b2a762a34b9a333284ca26d45f7dfc7f99c40d8fca43b7	equivalent
CompactCells683To714.Box689.compactCandidate_not_isWeightedPerimeterMinimizer	77fb3c759a73d9779003a5a15f1183af37c1aa49e332c6f24148e36cca7daebb	equivalent
CompactCells683To714.Box690.QInterval	fab9744634a30134f4ad3f74c9c3e96c940fa58424e0615aba3656f01af6ce7d	equivalent
CompactCells683To714.Box690.lambdainterval	acd15d465906f2850237dbd40843f188c93193fbc0305107ea16c012dab78ec1	equivalent
CompactCells683To714.Box690.compactCandidate_not_isWeightedPerimeterMinimizer	d1329c392d64a6b1b93ceb5cbb7c8ec344d4f98e996cab84e74ac1ca663500d1	equivalent
CompactCells683To714.Box691.QInterval	368d18a75424fa6cc662358a732b9a698be90e91e525c196e974b57672c474a2	equivalent
CompactCells683To714.Box691.lambdainterval	3216db827395363dc1a40ea8cb583257cb37e967081fa872816341cbec880af3	equivalent
CompactCells683To714.Box691.compactCandidate_not_isWeightedPerimeterMinimizer	932f460f57b064bc1c128be646215470e3e56624d0a1a8061712704a01c1dcbf	equivalent
CompactCells683To714.Box692.QInterval	0145bdc11a4d86772e748646eb7db4508db1406037a9356812740d1acf91a477	equivalent
CompactCells683To714.Box692.lambdainterval	5cbb89069abfcadacb2add327cca9128dfdcd7d9bb3dc3f8fec1a84f149b2bb9f	equivalent
CompactCells683To714.Box692.compactCandidate_not_isWeightedPerimeterMinimizer	55e2346a721dc4d322fc6041ab3bdb7dcd7588c36ca7cb5fc50aca779137a055	equivalent
CompactCells683To714.Box693.QInterval	f0f9c31817be380f590c4170af353b4ad05aa07367cf7ee9c54de39fa7e816bb	equivalent
CompactCells683To714.Box693.lambdainterval	807f90062102076a864eff7bccce55b4123099a2c718e4416a8408d929030aaa	equivalent
CompactCells683To714.Box693.compactCandidate_not_isWeightedPerimeterMinimizer	bd294a8d8811d82d49add1b6d8ac455a2b5d954e3dc99de22b67d91e249f44ec	equivalent
CompactCells683To714.Box694.QInterval	384f47269e242d5ca092662478e8e50d89895d34f8d1d63823929fa4ca2b047b	equivalent
CompactCells683To714.Box694.lambdainterval	c42253cf1a2e45b2fa6137ac2837bcf0549201f688ef279555a8a99e065fcff	equivalent
CompactCells683To714.Box694.compactCandidate_not_isWeightedPerimeterMinimizer	47939b984acb4aec464475333b7d2a2c56189cd945f384b7c61b2afdc99cc22	equivalent
CompactCells683To714.Box695.QInterval	72db47c7b475156eda566188d552456dbd4a6729128c63ef832992a1322d7d16	equivalent
CompactCells683To714.Box695.lambdainterval	5979751e75d3a5ae7cebe9c60cc702247f69ae3e2a1ae4dc8fa8a924d4ac6f41	equivalent
CompactCells683To714.Box695.compactCandidate_not_isWeightedPerimeterMinimizer	9bb1469317d3f8762c628aac7f4289cd2004ee2d89274df12905f93cc1dcb700	equivalent
CompactCells683To714.Box696.QInterval	c327eb79c86b18d0bd6439dfcd99f886ba78d8ff9217b64ef6f3d413873faf9a	equivalent
CompactCells683To714.Box696.lambdainterval	6cc3b2509c5dc1ba9d32df20e2e9af8322d478c8f14a59be1b0b834096943a56	equivalent

Declaration	Source hash	Audit relation
CompactCells683To714.Box696.compactCandidate_not_isWeightedPerimeterMinimizer	8a39b4408e784d89f75836371ca7e7d3f65e5fb9eb0dd75757d8403b09a8a8e4	equivalent
CompactCells683To714.Box697.QInterval	cec1dbe17a83b6ee183548c02f1e8a24072c1960c8f019d76dd8d6542967e075	equivalent
CompactCells683To714.Box697.lambdaInterval	e5e94fd2e53ddb4159c55e0036bece01fa6087bce51cd61ab0d2da878ebaf683	equivalent
CompactCells683To714.Box697.compactCandidate_not_isWeightedPerimeterMinimizer	17bafdf1bdd7867cf476abc68aa3a83b884570d404ba07a0494708af5373c9c4	equivalent
CompactCells683To714.Box698.QInterval	5e2b3b6d8d5976737a9b1551e919d49f11f6e49954dd10e904ffc8f6a2981b04	equivalent
CompactCells683To714.Box698.lambdaInterval	03498102a7b8f71cf87ad62071d5a7a796a33df67421240c59ed0c04cde815d9	equivalent
CompactCells683To714.Box698.compactCandidate_not_isWeightedPerimeterMinimizer	c24e6269abc54b1af0f664b636f5b5e5e8428ba3aabdffaff0a15ccaa383d2eee	equivalent
CompactCells683To714.Box699.QInterval	7779a49211bbeb51a2010a08f414641246c93fe9a7e5fef510853c574988da33	equivalent
CompactCells683To714.Box699.lambdaInterval	9ec34c80b863d1a27b1240680fa26ff9cc35249702422b52975329fa65b7d0d0	equivalent
CompactCells683To714.Box699.compactCandidate_not_isWeightedPerimeterMinimizer	e01e8a3407e6e3ae506b1b7d9734d7f39bdba4460934adb901442331ba440187	equivalent
CompactCells683To714.Box700.QInterval	fd96fef4bbed04d0ae3230f969a8e4e5f9eb96e47326924e5d36691ac84ac8fd	equivalent
CompactCells683To714.Box700.lambdaInterval	057a064370a535d0e62783d51fcc4dbfdb13dc27e548ba6c719975265e3e7318	equivalent
CompactCells683To714.Box700.compactCandidate_not_isWeightedPerimeterMinimizer	9258d906eaa0dbd951a2c08c21ec93253885d7fe5847151ba4b955f0a63d1b24	equivalent
CompactCells683To714.Box701.QInterval	a6a9876d4bc46e5581f51ab9dbd2bea36b10b8059cda6e95bbaedc95a9fce78e	equivalent
CompactCells683To714.Box701.lambdaInterval	798b9edc6eae6972757c3673a28cbab13fda29b7f91c2d8ce55d2c34fa9e3c35	equivalent
CompactCells683To714.Box701.compactCandidate_not_isWeightedPerimeterMinimizer	50d71f1df995522321235e95ed3a4b7ab5491604e6733c4254f96a1cf98828e0	equivalent
CompactCells683To714.Box702.QInterval	6785de17e0aef1c4ed8c61ce4f24e1c415eb57ac4b8770af7067686065aebedb	equivalent
CompactCells683To714.Box702.lambdaInterval	8ce01cc6a678ffa58a62c2a3acdac75da593215c07dd63b1e5d49b3d82bf2597	equivalent
CompactCells683To714.Box702.compactCandidate_not_isWeightedPerimeterMinimizer	a37784f78e762a385282ee23c467430b34943fdf11710a916d6f4face25d9ec0	equivalent
CompactCells683To714.Box703.QInterval	b4d5596f373707b5a09e940e549ed470e6b1ae53b9e8a7c9390a868329d41931	equivalent
CompactCells683To714.Box703.lambdaInterval	7893b6c1cce50e13ea9c4612a29473c2f1628a90583087f7c2c5492d2c73b8ea	equivalent
CompactCells683To714.Box703.compactCandidate_not_isWeightedPerimeterMinimizer	27f5d1c8205c3afdd5ebefd4544504ac555a74abe7e6c7a3b7232a1d8812db9b	equivalent

Declaration	Source hash	Audit relation
CompactCells683To714.Box704.QInterval	2603ebe4aa13968a8def83108610f6300567480c43bbdd54eca819fbce3c15a9	equivalent
CompactCells683To714.Box704.lambdaiInterval	1815d664ca1423257a66a1f0de647f2ec80c92931e4fa84934fb5e5da57b487b	equivalent
CompactCells683To714.Box704.compactCandidate_not_isWeightedPerimeterMinimizer	0cfd8cee58d5e1a80cd794f00668f91dfffb80c0011bfd2368e8f804c940365f1	equivalent
CompactCells683To714.Box705.QInterval	afbef6a9b077730085ac7c226b367ce74a31f9b380b5dd3c51d1ce7a67030049	equivalent
CompactCells683To714.Box705.lambdaiInterval	191c75e90267f82258a29e03f603f9885a514d2bf0d532c34457ea56d643d28f	equivalent
CompactCells683To714.Box705.compactCandidate_not_isWeightedPerimeterMinimizer	5a2e7082af711e04e120fda46e7a1065d6b14128283ba1234778b7188601db	equivalent
CompactCells683To714.Box706.QInterval	461165445a08cc8e26c5c946c1e842c3a8d20125fb80201bf74355d57e80accd	equivalent
CompactCells683To714.Box706.lambdaiInterval	447baba8ae6f3adc6441035a10cb68518358dff8ecfc33fa70f4dfaa72e1048a	equivalent
CompactCells683To714.Box706.compactCandidate_not_isWeightedPerimeterMinimizer	d1c39d73d2716435be585b24942dbbc58eaf0755385b9a1d25f67e3642e7fca5	equivalent
CompactCells683To714.Box707.QInterval	af0d046579192a897e2450b0c6d30d67add0117d95968bd9cd57f528ec272b	equivalent
CompactCells683To714.Box707.lambdaiInterval	da58cd3f6ed0d713c6c21277a43a6850bf49c7d07422db94fe5be3a09760301b	equivalent
CompactCells683To714.Box707.compactCandidate_not_isWeightedPerimeterMinimizer	15c50b37bb32cccfc229e97d38fef5de8bcab379fffb70369fdcf17c5bc03069e	equivalent
CompactCells683To714.Box708.QInterval	c9a0e561a0b3e8ac45e9fd143f6afd7008e37110f0b94efa45adbb7f39915dab	equivalent
CompactCells683To714.Box708.lambdaiInterval	b1ce36c8342bda0210f10048d6bf07b5c83e88a6366abf905617d8e4f79429df	equivalent
CompactCells683To714.Box708.compactCandidate_not_isWeightedPerimeterMinimizer	d2e0d22c4046bace97d182f8055c01105fca1a7a2459303c783f52f07d000911	equivalent
CompactCells683To714.Box709.QInterval	bf51e473fbc3e56aa2987edaf9c8e460028cb58a430fa9b7330605fbedb7b8f3	equivalent
CompactCells683To714.Box709.lambdaiInterval	356238f3f5d0dce5c7b71adf1ab4b978486e99e71d5052024e298f9874d040e6	equivalent
CompactCells683To714.Box709.compactCandidate_not_isWeightedPerimeterMinimizer	35dfadd694a90f144ef3a3074eabf4f6917bf4398fced2c512db68fa748f5972	equivalent
CompactCells683To714.Box710.QInterval	47c9ee4e9f91901bdbc79406327b866aaa0df5ff4db6d7e7acae645290057a31	equivalent
CompactCells683To714.Box710.lambdaiInterval	dc625c52502ed5be9e4c4bde329c2d49ab69205e80e8aa21de9e6ca462b86431	equivalent
CompactCells683To714.Box710.compactCandidate_not_isWeightedPerimeterMinimizer	7617718d3a13971f55c908813f579808c65848234a2b914026d621e1a8b0d88f	equivalent
CompactCells683To714.Box711.QInterval	97df9181f090f53868077e8cc17e5662cb079a1f2994a71b2913687866bc5668	equivalent
CompactCells683To714.Box711.lambdaiInterval	2b5445f273301412a391c9720d5bfa7e6b70b1abe98f770c39019f0bf0918e03	equivalent

Declaration	Source hash	Audit relation
CompactCells683To714.Box711.compactCandidate_not_isWeightedPerimeterMinimizer	7fb894038b59becd6ae5d97b8109defb025ab95fbd04db1ca3ed27ad55ef77a5	equivalent
CompactCells683To714.Box712.QInterval	e226150f474f24596ad4525c4cc5cb82e0eb1b35c7c3259bd8c9f13c81a2cf04	equivalent
CompactCells683To714.Box712.lambdaInterval	28e7477f7dc6f394537d47948b26f7f452b5e7bf8434b29a259f8ed8164676ca	equivalent
CompactCells683To714.Box712.compactCandidate_not_isWeightedPerimeterMinimizer	858ad903e49ce5e1d7c7a79a33c0d4eed4a6859ce7b67cddbba4041e1fb4cb1b	equivalent
CompactCells683To714.Box713.QInterval	4d2d5081ec943331e2b95de359189b488ce1a4876c8bf48e5dcbf12e356fabb6	equivalent
CompactCells683To714.Box713.lambdaInterval	e73a525bead2ca23d13d661140953d986430c1598bf6827ad9dcdab7016b7de3	equivalent
CompactCells683To714.Box713.compactCandidate_not_isWeightedPerimeterMinimizer	9a01fe7e4a78c9c317260140dbcc8fe14a64582ac159db99c5315ee2de4eb842	equivalent
CompactCells683To714.Box714.QInterval	11aaf7fe2b77452c5ac84d6cacb663d388dffea08a632509653b51baeb3380cd	equivalent
CompactCells683To714.Box714.lambdaInterval	102e0d855c5add8cb3c9b0ebf84d94c47bef8ee00aeb0fa64e1c0ed0a2ba9673	equivalent
CompactCells683To714.Box714.compactCandidate_not_isWeightedPerimeterMinimizer	824fe73f266d3caba8be23410920479f6ee09e7a715c496a29246388e0eea6c9	equivalent
CompactCells683To714.bboxes683To714_candidate_not_isWeightedPerimeterMinimizer	80d47d6dfb0a92dabe4b702e8a677ac3856eb3df280bd93c228441c5aee536b5	equivalent
CompactCells683To714.bboxes0To714_candidate_not_isWeightedPerimeterMinimizer	bf173ebf38135bf1764ec474da71caedfaa9ef90f181f1dbb6694cce522f5324	equivalent
CompactCells715To746.Box715.QInterval	35b868ac3d6583f682c05e44499424702fe917489cf506832040ea6681f45fc6	equivalent
CompactCells715To746.Box715.lambdaInterval	b68996a4f9417c2da9eebae039f42f8a7f61501be6f08b2ddcfa812474485f4d	equivalent
CompactCells715To746.Box715.compactCandidate_not_isWeightedPerimeterMinimizer	98de2911113af9aa4801219ed82e5a12a8e5155aaf0f8536b5085183de9b8599	equivalent
CompactCells715To746.Box716.QInterval	b59674e5a274c39eb588388567b5a90ad3b07749651075c1af5f61e15b0edac7	equivalent
CompactCells715To746.Box716.lambdaInterval	376469b1e7d464f921d233774c86cabe8238d8766b46645ad5730c038cfbcef2	equivalent
CompactCells715To746.Box716.compactCandidate_not_isWeightedPerimeterMinimizer	3d4d9ab416d28e68e2c7a848f9778d38275c694dea42e9a665d6db6f5e8f2514	equivalent
CompactCells715To746.Box717.QInterval	c02396a32d8b5193683fa71198c23947e7cee9377ccf20c0f0904d929d3176cb	equivalent
CompactCells715To746.Box717.lambdaInterval	635d47f830aab13c1a487f86b1802188479eb3702a6c0cc21033bd181ed75678	equivalent
CompactCells715To746.Box717.compactCandidate_not_isWeightedPerimeterMinimizer	6ae7d30a4415fc4d0cbcd87d18cb33bde409750e686a1dccf2022a70922da1e8	equivalent
CompactCells715To746.Box718.QInterval	dec4cd7c2f2be606231912259040476267ae52ba62765c4f34273ddf706ddc4a	equivalent

Declaration	Source hash	Audit relation
CompactCells715To746.Box718.lambdaInterval	60353a1d9881437f47719f03d31fafd835b6ef7f0d273fcf85d193c97842cf61	equivalent
CompactCells715To746.Box718.compactCandidate_not_isWeightedPerimeterMinimizer	b684946b9b6ed214891916a8de3e1be53fde85011305f2d275ffb77479098c44	equivalent
CompactCells715To746.Box719.QInterval	86facec1f7cc58b15195c4f0074b43d67490cda41d355b116c08bd5bfd4c724d	equivalent
CompactCells715To746.Box719.lambdaInterval	61aea9ec9adb1fa481f009cababfe548aeeb11c00c159a904392a9723d00141f	equivalent
CompactCells715To746.Box719.compactCandidate_not_isWeightedPerimeterMinimizer	b6d8e3c32cb038ce858b4e273753bcad2d81614401852f58b168620d32ec1ff3	equivalent
CompactCells715To746.Box720.QInterval	9d4a5e6006b0388bc02482ff0ede40971f6f331e8323ccf529c389162c77f81e	equivalent
CompactCells715To746.Box720.lambdaInterval	5b4e8d21b02e46bacb60d9b557753c28724c6406cdd7834cdffa8c1bc931fac6	equivalent
CompactCells715To746.Box720.compactCandidate_not_isWeightedPerimeterMinimizer	3a2f67f6ece5b190609855376c454a121f06ec1dac547e6420b9c01e614f67f8	equivalent
CompactCells715To746.Box721.QInterval	57df846bde058e8d7884c81613e4097204d2633a90818d0f979c8c79c3208376	equivalent
CompactCells715To746.Box721.lambdaInterval	e9714680d46119910025871eccc8b34936baef1e44c5c165ddcb3a2d1b2b2c58	equivalent
CompactCells715To746.Box721.compactCandidate_not_isWeightedPerimeterMinimizer	52444ebe3bfd8bd72a0991ee861dca3eb5bc204b421bfa376d1abcf0a5ab373	equivalent
CompactCells715To746.Box722.QInterval	69123a7610f394eeceace2110856a942ea0078005e741d08f4271bed7238d28f	equivalent
CompactCells715To746.Box722.lambdaInterval	f717d0ad7703eadd74d7c722be16c00e277abff8db04d961418c674dc2eb2880	equivalent
CompactCells715To746.Box722.compactCandidate_not_isWeightedPerimeterMinimizer	889650b5eedfc4c1e63b9c78e52a057d409b7b0b27add1f88db0f1490363c8a9	equivalent
CompactCells715To746.Box723.QInterval	24da470ce3573b3ed4b35f9127e9bb0ac3773b61cbd335c5d69eac9b5891c7d6	equivalent
CompactCells715To746.Box723.lambdaInterval	731835651c3e7b14fb67016ffbc40344a52f72bca4c44f5eb63e6a06b9a606d2	equivalent
CompactCells715To746.Box723.compactCandidate_not_isWeightedPerimeterMinimizer	e5e8021a6068b2ef396c1b7a3ad21edd4393d4a60be535cecdfec8f19ce6c792	equivalent
CompactCells715To746.Box724.QInterval	9242880f264139fc1e44bd3c1f5101a41ec352c150c29560bd5b89ca8739a079	equivalent
CompactCells715To746.Box724.lambdaInterval	ae71bf6598b3508253787fac3733b6371ca7c57bc031cd8d191351b35833e90c	equivalent
CompactCells715To746.Box724.compactCandidate_not_isWeightedPerimeterMinimizer	8064799105d7375f662eb17fc157d212ebc8802777eaa75b93dc2934966be76	equivalent
CompactCells715To746.Box725.QInterval	c831e9d4ccab2e5dee1a484c86ea48ada6537cd5b51dccb8a62e81d485eb5516	equivalent
CompactCells715To746.Box725.lambdaInterval	71b89b99e82a86b1d8470cd021d6f71e8c3784b5cd8d3d0029c4662dd5dfc537	equivalent

Declaration	Source hash	Audit relation
CompactCells715To746.Box725.compactCandidate_not_isWeightedPerimeterMinimizer	c6f2e58868336d86476e5198a30649545109f2832f06966746f6658b6ab4732e	equivalent
CompactCells715To746.Box726.QInterval	c9117af16b0bdf45647477cc40751d00234d812d74fdd877c13e5296e074067e	equivalent
CompactCells715To746.Box726.lambdaInterval	bcfe654a912270f48a67ea3ab7b0cfe9216da72bf62066a9fbb2314b6b72e9c	equivalent
CompactCells715To746.Box726.compactCandidate_not_isWeightedPerimeterMinimizer	89c5dd302237bf70fcb591e0e4b72e40e93eb1a939d22dfe2c4610ea748877fc	equivalent
CompactCells715To746.Box727.QInterval	dfbacfc63d2aa837069d10878a641ee1e88e728c7681ab859303dc13d926efd8	equivalent
CompactCells715To746.Box727.lambdaInterval	f76e6a2c6459e8182ce7b994873461146243755f076d24038d0328f0444f69e3	equivalent
CompactCells715To746.Box727.compactCandidate_not_isWeightedPerimeterMinimizer	b2a56f096063a2a1d85f3d6778c478dcff6a455170eb4df2331b726294a7b7f4	equivalent
CompactCells715To746.Box728.QInterval	9add877e71d0ed7374ca0030dbe07a0ef17948e83b8bd74de81a44430b56721a	equivalent
CompactCells715To746.Box728.lambdaInterval	a0f8872977ad1128d88ba2a0b3018126ce138a3e4ac7f58fb58112ef46b0f924	equivalent
CompactCells715To746.Box728.compactCandidate_not_isWeightedPerimeterMinimizer	0de7ea7514a8d7399c68a593fef1c67cf603ac605f464e6673cefc1ff3630142	equivalent
CompactCells715To746.Box729.QInterval	28a0905df37f39dba6e802808ba69143330cc8ced1871e76452cf3130c2fc72e	equivalent
CompactCells715To746.Box729.lambdaInterval	5325f4927489b5b4017a7f9b9cb244666b866fd6cde52e5baee80f86bdf1bfaa	equivalent
CompactCells715To746.Box729.compactCandidate_not_isWeightedPerimeterMinimizer	b1b0a469ebb2bfd0325d93e6563a04d1f9b0764e23a1ffca49230cb4912c716b	equivalent
CompactCells715To746.Box730.QInterval	1cc5da0e2d0e5642744a8d18d7a694d9b74e691a1aea499f23b5f9978d39ac71	equivalent
CompactCells715To746.Box730.lambdaInterval	b3d0242d2c7e1b33d9a97f53f486d3d5993dee453282f350bd8c5965f51ab54a	equivalent
CompactCells715To746.Box730.compactCandidate_not_isWeightedPerimeterMinimizer	74ef036574beb04fd5eb055f38d70705b7db7ca6038d8cff2b33b55d30d641ad	equivalent
CompactCells715To746.Box731.QInterval	474b6d2dac4d5733a052b43ebd7106ebc03e3e4457376ab00671f49044a86c3d	equivalent
CompactCells715To746.Box731.lambdaInterval	f19bd0416f0779cb5ddb73f19eb78a67fd55226d5883180e206eae5767badd22	equivalent
CompactCells715To746.Box731.compactCandidate_not_isWeightedPerimeterMinimizer	8eec25a1cbe358ca6ea060754117d7eb4483df553f7fb834555eb23c10383262	equivalent
CompactCells715To746.Box732.QInterval	05f73d22238c9c0bf5a8321ce82a1bb6c41ed14f6b12878fa6e7eb99019f0d61	equivalent
CompactCells715To746.Box732.lambdaInterval	232cf174751539e905a95ce73a978eca606a496df6a2a7a0b348f8d15348bc5e	equivalent
CompactCells715To746.Box732.compactCandidate_not_isWeightedPerimeterMinimizer	7015dd21a7a237d5e14d6730333095873f3670ed5d1f1c040f095855bf275bc2	equivalent

Declaration	Source hash	Audit relation
CompactCells715To746.Box733.QInterval	ca6d1599c27c5c9681841cbf9f35e73eb58b89df7d5f4f9c5abfc5b6aa411185	equivalent
CompactCells715To746.Box733.lambdInterval	187c425fd1c4c13196540d1be69804315dc5d78a0b63d1e14203ecf9b3ea4ef9	equivalent
CompactCells715To746.Box733.compactCandidate_not_isWeightedPerimeterMinimizer	74ebc3e9dd446769ad0d483670049095383436dcdd997956470999fee0281f21	equivalent
CompactCells715To746.Box734.QInterval	4f5a3473c894c6e5cb08109c721e9d58a4934dae2674e75cf6949238bb7afd6a	equivalent
CompactCells715To746.Box734.lambdInterval	be41fcd15dfdf59e45b438fcb6eb7ab78273b2a2a68202e8c4af4938ec2cf11d	equivalent
CompactCells715To746.Box734.compactCandidate_not_isWeightedPerimeterMinimizer	ffd114f3836bac102c9fe893afe94e6798d570154d2f432c63972498a33c3390	equivalent
CompactCells715To746.Box735.QInterval	e1238f27b9fdcd24a39172374dce375c42049b051db2cd9f2993d5861379bf09	equivalent
CompactCells715To746.Box735.lambdInterval	a4f9d0cf1e8374e2dcecc9647c09c00b843f52d456b5e9c83c82fa95555318be	equivalent
CompactCells715To746.Box735.compactCandidate_not_isWeightedPerimeterMinimizer	1ae2b9c5e0dc7e9e1ee11046b78b3adf123aac13ad5992894f45fe59a7d0d0a9	equivalent
CompactCells715To746.Box736.QInterval	f95ad5a6f0c2408da875ac7274d22f22587cc0c36a5e94cc51c3f66d4cdfd4cd	equivalent
CompactCells715To746.Box736.lambdInterval	e34ec184dff0ff7622dbce2911d9663ba37957e5cdb544b57562b3be62c49a9	equivalent
CompactCells715To746.Box736.compactCandidate_not_isWeightedPerimeterMinimizer	d98554abc398183588939e59f3c7f6cf4058a5eab1e673a81055a5a2bf249aa9	equivalent
CompactCells715To746.Box737.QInterval	403b9d86d0918f23f831e9394f9b96b41c5987d968d1ace796658c16bc90879f	equivalent
CompactCells715To746.Box737.lambdInterval	89272ed3bff65fe8e9eab34f02044ab5074ff82b27c6357da13a7890031110b1	equivalent
CompactCells715To746.Box737.compactCandidate_not_isWeightedPerimeterMinimizer	ffb34c8bb9ef9b17dc9c0260ebb99cd212ec5bd3f3e3981b59d60c2ddd41022	equivalent
CompactCells715To746.Box738.QInterval	e6af23c4267f8d91386ef44c63c09055eabf3841ad1157dc176cb6fed4fc805	equivalent
CompactCells715To746.Box738.lambdInterval	db3d9aeac2d7a35d325b9b3737f7be73c4b084a2e2ec4a22bc8b2519c01cafc8a	equivalent
CompactCells715To746.Box738.compactCandidate_not_isWeightedPerimeterMinimizer	e847735b23992a38afa1513b2e62adbb59cd573690d5b621b598f88cf06c8ec1	equivalent
CompactCells715To746.Box739.QInterval	8fb70649323fb034765b20f3c5ee58025a02dbe17e1ce778022849eee80c6037	equivalent
CompactCells715To746.Box739.lambdInterval	439a5e6f056512f8a7675449b9b2e39924bb3576218fffb2e332233fa628f0b1	equivalent
CompactCells715To746.Box739.compactCandidate_not_isWeightedPerimeterMinimizer	653ac4ab44f7e7b96099367721d1aceb9d2fde64b34f67e80a3a85d2258207fc	equivalent
CompactCells715To746.Box740.QInterval	4303af5caa048328f56782def4c2961f7aa9cb8f86cf54937206fd1c1b6b674	equivalent
CompactCells715To746.Box740.lambdInterval	1c289b35606e628dd42e780e2b3fecac8c4f1897232708f78c4e5f38087f4f38	equivalent

Declaration	Source hash	Audit relation
CompactCells715To746.Box740.compactCandidate_not_isWeightedPerimeterMinimizer	097f19917d139ab3308cda53bdf4f97b9a694997e57965ca6ae3362c675b3e4d	equivalent
CompactCells715To746.Box741.QInterval	1802cef1ba7cdee3cc9b02ecdcea72b34fa74a970839d257fb5a347463fd8aaf	equivalent
CompactCells715To746.Box741.lambdaInterval	99c7590a1f6e14a027ae01f0059d68f27e8a7a1b3579055adf3669814a11a0c4	equivalent
CompactCells715To746.Box741.compactCandidate_not_isWeightedPerimeterMinimizer	19b3dce51937e45ac7f941bb4c1132857bca15379545bf223260b5cd3836f380	equivalent
CompactCells715To746.Box742.QInterval	8a31d42262cd8382b4d45b3bf9134cd568ecc2f2415682140681f23a40fc54aa	equivalent
CompactCells715To746.Box742.lambdaInterval	0bc05d4bd4578915c51d35fecdd04cc653fa460d9ab3b675182f4e52067f1b4ab	equivalent
CompactCells715To746.Box742.compactCandidate_not_isWeightedPerimeterMinimizer	ce2f8c12b79cd3f586582e96620b87ffa3e39d7f158363eb116f45c789659d	equivalent
CompactCells715To746.Box743.QInterval	ce46dc11c51f93bf8968219868c2d70364e61394f1a27021a9e7842c8ae8b0e3	equivalent
CompactCells715To746.Box743.lambdaInterval	a912dfc7cc7b3276e8bc6366e7b51466cba1f69c674ba95eb9f61e02903b140f	equivalent
CompactCells715To746.Box743.compactCandidate_not_isWeightedPerimeterMinimizer	7b55a07ea3ceed3b257fac0a4c43bb96b3ee064800fafd5e2f6f9dfbf0e24d96	equivalent
CompactCells715To746.Box744.QInterval	ca5446140a9006c4a6191cd0a11e7b23c521a362bc7b7fd5b547cb78336de9485	equivalent
CompactCells715To746.Box744.lambdaInterval	3a4239aa001cfdad039cbdad9e6aa224e5ede466b968788f153e15f415d6ae6d	equivalent
CompactCells715To746.Box744.compactCandidate_not_isWeightedPerimeterMinimizer	7e579dadb4cf3151263eab21c8b0ca203b7dbc486e88fcbd33e3d15b334d69a6	equivalent
CompactCells715To746.Box745.QInterval	a5538bfeadcd81695ef2a7c4a6c767386530324ba3fa569742303028401d2fd0	equivalent
CompactCells715To746.Box745.lambdaInterval	cf4289ef8ef7732de28f97321980de760fdaf3379bc24387d907cc3b8c4606de	equivalent
CompactCells715To746.Box745.compactCandidate_not_isWeightedPerimeterMinimizer	4939e6ebbb056d22eb681bfc3b8b13a24c5fd0d7312a6d50e16c6a3d9b10e9f6	equivalent
CompactCells715To746.Box746.QInterval	65ceb33be902159a694de8be913481668ded828abc32248d79078b704e81dba4	equivalent
CompactCells715To746.Box746.lambdaInterval	93520e93fa90026157a0390ed399b78569281bd7c94c2ea3ace9ed9d4f20cae4	equivalent
CompactCells715To746.Box746.compactCandidate_not_isWeightedPerimeterMinimizer	963f8153955418456e8fa50bb4a1e8e069474da7c5e4e186c8d3e0ecd6bcde7a	equivalent
CompactCells715To746.boxes715To746_candidate_not_isWeightedPerimeterMinimizer	1189909df99af468a91429737d960f11737a8b6b672f112f37836588563f27bc	equivalent
CompactCells715To746.boxes0To746_candidate_not_isWeightedPerimeterMinimizer	10f01d242597312468da9eef2c4697c3396ee4c042720726dd2d194461b7d655	equivalent
CompactCells747To778.Box747.QInterval	9528ba8cc68c8940fe51f873ac86edde382a3d97f58915d58baf26e3ea56b68c	equivalent

Declaration	Source hash	Audit relation
CompactCells747To778.Box747.lambdaiInterval	2bae94d2c94f94594c868d120b7dd57aa4b acbbcaeebbcbdb395caea1df66597d	equivalent
CompactCells747To778.Box747.compactCandidate_not_isWeightedPerimeterMinimizer	f2d0bae9a195cfb77c9cf198b7da74a200e 8c6bcf45697d00aac137e8c6661ff	equivalent
CompactCells747To778.Box748.QInterval	decb7b67a0307bdb432ccb6db2cc7dcfd15 fc2bbd867aa105c2d4c454dc990ba	equivalent
CompactCells747To778.Box748.lambdaiInterval	c6401d51042fd838fd0d88f8898d307ebaa 3ae7534a9272f718e3d7e8ad3c74c	equivalent
CompactCells747To778.Box748.compactCandidate_not_isWeightedPerimeterMinimizer	70c4b47edbd8096022064da0e94247542c9 9f3655c307d2330892c04f13055e3	equivalent
CompactCells747To778.Box749.QInterval	0216a778cc78b29fedfb8a1e541f82036d3 a57b8c6d9580c3c0db5ff24d0a807	equivalent
CompactCells747To778.Box749.lambdaiInterval	1d6a98546003802c7e5cce1905e3829c492 bed30d70fa6fb79143e7f41fb653e	equivalent
CompactCells747To778.Box749.compactCandidate_not_isWeightedPerimeterMinimizer	e1d891e1a4dc858fb2ed74c37886b65a9c6 c7f13ed115bdfd7388a7b1c5d6ea9	equivalent
CompactCells747To778.Box750.QInterval	a5baea9c72db66c9aa9665ee1dfe4d0eedc 7943fcdda83efba130e3622d59e34	equivalent
CompactCells747To778.Box750.lambdaiInterval	abf0aa5e7a02f051cbbc8f96f56d2e066b2 240ddc20256b1659e23402d52ad65	equivalent
CompactCells747To778.Box750.compactCandidate_not_isWeightedPerimeterMinimizer	76565b629e403ccafb2737c75e9b1023864 10b169d001aa29801f4421515fa73	equivalent
CompactCells747To778.Box751.QInterval	573cc79142032f1002bce1320478104d191 7d412df4ee0abb90ff4cb2db011ed	equivalent
CompactCells747To778.Box751.lambdaiInterval	d4dd7446bcb4a18861c192b89c894dca06f 4e515f0a8d43f4071cb2e6dd5747a	equivalent
CompactCells747To778.Box751.compactCandidate_not_isWeightedPerimeterMinimizer	ee945760de07fb4ffe8055085f4120f59ac 5dad18a1f5d42388b8085fe7c3551	equivalent
CompactCells747To778.Box752.QInterval	bfe7af7d8d1705f0ed35350e396c31cd803 b6deafa8d71121bae42f3e4838657	equivalent
CompactCells747To778.Box752.lambdaiInterval	557a72ccbbcf42b10bad1fab706920388cc e8d6f02076fdb23cc8111d36ee76	equivalent
CompactCells747To778.Box752.compactCandidate_not_isWeightedPerimeterMinimizer	7f1b494bbb4ef3509cd66027e77bb09347e f8daca446edd70b92748e1c21e077	equivalent
CompactCells747To778.Box753.QInterval	2d7b28872bef5fcebff8d9e90dece714573 cdb32b86da495bf42e35ace43f951	equivalent
CompactCells747To778.Box753.lambdaiInterval	b7a7ce2d6a70e60bd7e5d3f22371432c62c 33f27abb20f7812c9c220fe263c17	equivalent
CompactCells747To778.Box753.compactCandidate_not_isWeightedPerimeterMinimizer	bb998d2d6727c8b294846f13ce6c60b8fff 732008eeb31b0aba965ea8dce9668	equivalent
CompactCells747To778.Box754.QInterval	4c90d8bd6b8d7ca549792d936e616ae5b05 59b3097cec5050e3d4476bd348168	equivalent
CompactCells747To778.Box754.lambdaiInterval	8543621764f67646bead79e1fde6d784eaf 10399305f235376db788d3e52853f	equivalent

Declaration	Source hash	Audit relation
CompactCells747To778.Box754.compactCandidate_not_isWeightedPerimeterMinimizer	663e6477c9d9e4f57dd74511aa70c820fba818190d57f1b60d4279fcf7cafe62	equivalent
CompactCells747To778.Box755.QInterval	734fca154d1dfc3f1438f48f771ff9adfe6fb691efb9153180177c0b1aa3ea60	equivalent
CompactCells747To778.Box755.lambdaInterval	fa1841abd7846f13c32a01abdd24a22458c22d4ffa89fafdfd16d5cf9170180a	equivalent
CompactCells747To778.Box755.compactCandidate_not_isWeightedPerimeterMinimizer	c8e28afab8750ee6c3b0644484453cedef746cd3b4eccfaac14f3bfe55f7663d	equivalent
CompactCells747To778.Box756.QInterval	207da29d30f9b036467ca6d99cf6f1c7c310874d5254bb38677ec3c6368abbf0	equivalent
CompactCells747To778.Box756.lambdaInterval	967f28135edca2f7afc6aea8db703151d4679c7e17c4d8b5d81ec2f165bd5532	equivalent
CompactCells747To778.Box756.compactCandidate_not_isWeightedPerimeterMinimizer	12ec0e4124be940d3a6ce407bf3695dab89f9a03f94d04bfc0c66be83e5d1a1e	equivalent
CompactCells747To778.Box757.QInterval	f55af1640956d21de1510a8eb5319381ca394125f6a1e974131ec6c32934a803	equivalent
CompactCells747To778.Box757.lambdaInterval	bc41a6aed737d0e387b5b98a56dbce7423033bc475576260bd50061ba423c9c2	equivalent
CompactCells747To778.Box757.compactCandidate_not_isWeightedPerimeterMinimizer	8a4ff5a0ee720666fc80f86defe13aaaa28828d19f45bd3be5f8d8122f989577	equivalent
CompactCells747To778.Box758.QInterval	bb695c095cc04d9668869faca51f1a6fab9a4a116c9f418686aae3007c2df9	equivalent
CompactCells747To778.Box758.lambdaInterval	1b284190499e93b638d772ea5461b9cd20a5a8768c8e61f1da1fc6faa04d731c	equivalent
CompactCells747To778.Box758.compactCandidate_not_isWeightedPerimeterMinimizer	f9eab568ff90a32f04b22de0fc0f2a7bc5bb8c9df5a4939e09df3d3bdf7d9289	equivalent
CompactCells747To778.Box759.QInterval	0036048e64e61af0f60954b3bfb1d4a1035f50fed3dd73a6ddcebc62430ff75d	equivalent
CompactCells747To778.Box759.lambdaInterval	94115b2a7f6f057bb1bbe345c44f44ef41ff0858b0003fb319e0c13762a5440b	equivalent
CompactCells747To778.Box759.compactCandidate_not_isWeightedPerimeterMinimizer	8e359b032948e0bd322d17995ce218d783ed7b909a67ccab22c3e93f67b0ef76	equivalent
CompactCells747To778.Box760.QInterval	45f051b8a787d05987c9977bd0233e5fe8bd4da943f6f44d46032ec4616142d4	equivalent
CompactCells747To778.Box760.lambdaInterval	db3b94ef65e5a25dd91f8c5322c64a64913ea75aaa6a2545d14f720682828c3e	equivalent
CompactCells747To778.Box760.compactCandidate_not_isWeightedPerimeterMinimizer	ac5272e796c5f2d554487b607159d0247b619e38d01db0606f91fef728fd62bf	equivalent
CompactCells747To778.Box761.QInterval	b78b67fc5244ff354ed0aaaf8cda4f35a86e6e2bc7e5ff350df724cbb16a4c2f	equivalent
CompactCells747To778.Box761.lambdaInterval	bbd1609ece8489a707c9090b4d472a907dfa7d8db25700e03ad8f6245ad2ff59	equivalent
CompactCells747To778.Box761.compactCandidate_not_isWeightedPerimeterMinimizer	a85aa4fe7070f4c159ff6d60f5742380ffb6ab3ad412a5ff8197215d2f987612	equivalent

Declaration	Source hash	Audit relation
CompactCells747To778.Box762.QInterval	4bbb08d05e33c79cbc59f8ed908a0d2b420170584c71e5c7f7408d58ebd10b23	equivalent
CompactCells747To778.Box762.lambdaInterval	31805b446adb8ba6141c24ec58434567ece878cb4fe1b31d55866146f3528895	equivalent
CompactCells747To778.Box762.compactCandidate_not_isWeightedPerimeterMinimizer	4141dde4222af05bfaa19cd9eeb1eead8e720ff23a89e878c0111bc122b4e22c	equivalent
CompactCells747To778.Box763.QInterval	c35c69c98198f31b3198bbb52063f1428bc87b011f9fa0cf59dd581d3d34e18a	equivalent
CompactCells747To778.Box763.lambdaInterval	6ff5f49e466f1e227f55cb144453f753ae498d63305de3b7a2285f1958dcebf0	equivalent
CompactCells747To778.Box763.compactCandidate_not_isWeightedPerimeterMinimizer	1029524dc87fd9bbd59b3bbebb5678ebe68675efc8d6c69988d0deadcc77455d	equivalent
CompactCells747To778.Box764.QInterval	bdb e11ac1d6101e96ac42925e8b37a07b5b582abb0f46c41684ddc06d8cc265b	equivalent
CompactCells747To778.Box764.lambdaInterval	e72d62dbc528e5b37ba04fe78d1f86bd93f4967959e5cc2b211d03ad422992d4	equivalent
CompactCells747To778.Box764.compactCandidate_not_isWeightedPerimeterMinimizer	0d34022c4b0a7300944579d49b5fadb64bf790772b1ded821a0e4e56d34b82e8	equivalent
CompactCells747To778.Box765.QInterval	fa528822efe6678784707e39b8a0488e0297f9313ee1850dd2b9dbb20e2ef5a5	equivalent
CompactCells747To778.Box765.lambdaInterval	70a6e5ac406f156d425ab77755b42a462b3515b8994d23efa01e843040a166a2	equivalent
CompactCells747To778.Box765.compactCandidate_not_isWeightedPerimeterMinimizer	636f6a7328c62e72ac81c2c5f548d85eecfa6a5f2b05766ff8ec977ddf03d05	equivalent
CompactCells747To778.Box766.QInterval	261f69c281980d55c82d21fea53eb7478b10b0ce7a54536d09c5ed2e8568e69a	equivalent
CompactCells747To778.Box766.lambdaInterval	3f985445d8cc8937a71ffaed2233bb38b1de0fd3cc7d843c9e4efd15ef7797ab	equivalent
CompactCells747To778.Box766.compactCandidate_not_isWeightedPerimeterMinimizer	6c1cee18d41b3fd5c80390353d9c9b9a93a822a90568304e6e854f10ae37a5e3	equivalent
CompactCells747To778.Box767.QInterval	3e8426aeb e234bce4a965d923ae6402c566331b0839f78265649fd5fb3b513f6	equivalent
CompactCells747To778.Box767.lambdaInterval	920d686317dfd53d0ea46ace4a4b5cdd7e49e6bc1d4775e18427a5694f953b8a	equivalent
CompactCells747To778.Box767.compactCandidate_not_isWeightedPerimeterMinimizer	c280219e3a883720034547696007830cebfb56977f1dba6f6620478d0220378f	equivalent
CompactCells747To778.Box768.QInterval	ce58f14592ae9294558b2f569b134265f0d104b91c00b17eaaafed7ad3245aa	equivalent
CompactCells747To778.Box768.lambdaInterval	90ebba86c3a5cb23fbe1b9d2ef12668870c4fece121b55be6f0a5815cafb4534	equivalent
CompactCells747To778.Box768.compactCandidate_not_isWeightedPerimeterMinimizer	38e219ab7240827d65ed9bcec8d0fde5ef277a0c077b616acb2c22969d2b33fc	equivalent
CompactCells747To778.Box769.QInterval	d80c3dd1d8cc4d739f41a6e67911f62983d59e68f95b5490b63bf22a437ea460	equivalent
CompactCells747To778.Box769.lambdaInterval	90e70f402f22ff0a78710f1e7e6199c7602cddecc47a88c64c7c80896bb5139b	equivalent

Declaration	Source hash	Audit relation
CompactCells747To778.Box769.compactCandidate_not_isWeightedPerimeterMinimizer	2bfc107947099f2fc235f5e2f3234fb7d31f35b0c90d3b3d703eed0a7914ca02	equivalent
CompactCells747To778.Box770.QInterval	ac818adc2c387c119294220da638c01f7f1949608075974486ab8f04ac45c7f3	equivalent
CompactCells747To778.Box770.lambdaInterval	4432ec3495a8affdfaac4524877e14f952d88671a99fd9835c5cd82c232b76d8	equivalent
CompactCells747To778.Box770.compactCandidate_not_isWeightedPerimeterMinimizer	ff5b3e3efb6681afe10001156408a31ea434c32acead5be2bff82fe07e3c1c2a	equivalent
CompactCells747To778.Box771.QInterval	5522bd63e78124243bb63f9c79df39887c95f37eb4e4202eafdb3eaa44408f07	equivalent
CompactCells747To778.Box771.lambdaInterval	578e1e06f9729bc46a2c5de306ce2b3959494fa3c1c79cf3a3abb8b36be01ae8	equivalent
CompactCells747To778.Box771.compactCandidate_not_isWeightedPerimeterMinimizer	02907ed66e87165aedce0aef14dd0fd932feb2f2469a4c9cae48b126bc1ecd65	equivalent
CompactCells747To778.Box772.QInterval	c350a5329e96360d238d41cb31bdc932be7c083e4b9c4fd154cd694ef199870	equivalent
CompactCells747To778.Box772.lambdaInterval	aa8e87c5b9a4888143fae7d9b308089dba8e74166d4dc83fb82c068c8d2e1abe	equivalent
CompactCells747To778.Box772.compactCandidate_not_isWeightedPerimeterMinimizer	0e540fa3f4dff5fb97e2d36f5a297394a28fddd1c772bafef718e046c82a7f07	equivalent
CompactCells747To778.Box773.QInterval	921ff95c5f1eba157cec52df4ee34405b8f25258d6551970d7346a2d9977ddaf	equivalent
CompactCells747To778.Box773.lambdaInterval	c812b29c354d1ebaea4b0be8671a1274ae2e9e8aa1571d5bbde47e7fadf7740f	equivalent
CompactCells747To778.Box773.compactCandidate_not_isWeightedPerimeterMinimizer	3ef389a9f73cb4813b7a3013cf26502a5c56d1c78b557e9f551bd4824e6f09ff	equivalent
CompactCells747To778.Box774.QInterval	9b96b3ff4376804030e9cf793bcc06d5ef1829e7b0ea477da5dfdc339c14ed06	equivalent
CompactCells747To778.Box774.lambdaInterval	47bf6b2beb2178512ab860b3414dca4250822b23b0a433316d9dd623b86599d8	equivalent
CompactCells747To778.Box774.compactCandidate_not_isWeightedPerimeterMinimizer	89e55a16142eb4ba72e5c4cd68e6b75fd5dec64e2271afe7f9e15a4bec94bc8e	equivalent
CompactCells747To778.Box775.QInterval	f21ba99aafd36970ebfa2e0085e4b96ed0ca6e959625bdeb949ade91ba2eb102	equivalent
CompactCells747To778.Box775.lambdaInterval	64ae2e62f551dbbf65a1f1bf3920c6d7ab9c1cf630c57641c4188a7d97f1bb9c	equivalent
CompactCells747To778.Box775.compactCandidate_not_isWeightedPerimeterMinimizer	43fbf3edc5cf3505e908e718f1493e03c925b326f0ca8f91bb2637bd1aaac273	equivalent
CompactCells747To778.Box776.QInterval	0414c453d80a75d2f1abcc44b29bc1a1877d02d3b04da7e70a5cf9d08941f595	equivalent
CompactCells747To778.Box776.lambdaInterval	38a3409dca522b9a0530f78baeb293b95463f3a7e1a04b7753bc3a49fae9099e	equivalent
CompactCells747To778.Box776.compactCandidate_not_isWeightedPerimeterMinimizer	cbbfba2fd5b90fecb37a1d3bdee05bd8ea401b0588d890fbc52b81890fb5a2ed	equivalent

Declaration	Source hash	Audit relation
CompactCells747To778.Box777.QInterval	e74000afb70ef034562837cc94a146c0cd58fc7a7a27c149c82b2cfd6e3c0f	equivalent
CompactCells747To778.Box777.lambdaInterval	2cae130236c8ef4d10dec41da95aade8598160ca3999295a039d36eb3e9218e3	equivalent
CompactCells747To778.Box777.compactCandidate_not_isWeightedPerimeterMinimizer	fbf89579fba31e56540b5005aeba02c175dcbe2ee7e7ce62c028d3ad7df6a8e3	equivalent
CompactCells747To778.Box778.QInterval	bd001c0079cc452c820d4f30a87590f71cf1a17f464341e7d4cc5f2a801b0bee	equivalent
CompactCells747To778.Box778.lambdaInterval	107b379b7d3f4b73cd5e0c58568c3ec6e08741b17afdd44f6d1836d03750c2c0	equivalent
CompactCells747To778.Box778.compactCandidate_not_isWeightedPerimeterMinimizer	a93edd7bb6f265a20d621d2cca0def30103d0f8604b1b7a2cd9a437300dae730	equivalent
CompactCells747To778.bboxes747To778_candidate_not_isWeightedPerimeterMinimizer	88cd35cedccf4ca857edac090c27fc142ad622d3b277c38a6f5a21601f23d1c0	equivalent
CompactCells747To778.bboxes0To778_candidate_not_isWeightedPerimeterMinimizer	89260661d6f382750b7663e0fcbb1a1f311d22dc0f16491baffa94d97c0bcc14	equivalent
CompactCells75To106.Box75.QInterval	fd26d809bf758506b1df3de0411c2bff2cbf59e533ece162dc328a5871de099a	equivalent
CompactCells75To106.Box75.lambdaInterval	b7c912126cad4302a9d52ca0ee70753f6c3e6cafe86cd8bc55838300d96850ad	equivalent
CompactCells75To106.Box75.compactCandidate_not_isWeightedPerimeterMinimizer	20b670a377ee5d780649296de359989a7d2a6e97412b86839c55de2fe22902fc	equivalent
CompactCells75To106.Box76.QInterval	609c222e12dba255947b462f4b3571ca2b583018a4967f6356b938c3df3ce2af	equivalent
CompactCells75To106.Box76.lambdaInterval	af9f420f83f052c648407f463ed0fb4b81a070334cd0728be33c5a2b8407c5e0	equivalent
CompactCells75To106.Box76.compactCandidate_not_isWeightedPerimeterMinimizer	e541834560f054b1b493e17548dc4d4cf6b250a5ae45326316386d09eaa3c0b7	equivalent
CompactCells75To106.Box77.QInterval	8d235171185a2d78af604ac7f1763344a0bc9d660a5a69220601c2b3cbe44681	equivalent
CompactCells75To106.Box77.lambdaInterval	e8d845f3a51faf13e9c9f6ab69a1e3ea145d6278695d26539c38d69641a12654	equivalent
CompactCells75To106.Box77.compactCandidate_not_isWeightedPerimeterMinimizer	2d96c5a64510ff3d69acf4a5b755cdcaffb3797d68e3a45529b1c173362abe95	equivalent
CompactCells75To106.Box78.QInterval	0466f075f5de220f27b8f8e770490852422d14d152a6f3ff1694d1759f6a96cf	equivalent
CompactCells75To106.Box78.lambdaInterval	90c9a9d45e0b532dfb8322a9936cdb46a22c1ae0ead561823cb6a698e9b59f60	equivalent
CompactCells75To106.Box78.compactCandidate_not_isWeightedPerimeterMinimizer	e3bb6fd795f2e896c6059824186dd88e67f293c463e05c46e5fe48df27c964fb	equivalent
CompactCells75To106.Box79.QInterval	0f6be7a6803f0f35e59b5e280e7dfe8bd217f973093071ddafd1d7502c897f86	equivalent
CompactCells75To106.Box79.lambdaInterval	9b519bc5014de061ba05c308933a1300bf747457a8377d5ec5667362fdb8e09b	equivalent

Declaration	Source hash	Audit relation
CompactCells75To106.Box79.compactCandidate_not_isWeightedPerimeterMinimizer	03d179a89f209353e1f48b848f72be4c2db50261f3165a6b9a3e9baed7983fdf	equivalent
CompactCells75To106.Box80.QInterval	5ff2fc026969991ef9229deadedb9d4a00dd0c4f5fd495b8a0f3ec4d2bad574	equivalent
CompactCells75To106.Box80.lambdaInterval	5c6349bb8347e8ab10aab01950cc36605a66a0217a94bb246f2dcc65cd10ddf6	equivalent
CompactCells75To106.Box80.compactCandidate_not_isWeightedPerimeterMinimizer	47b6ff5df2f77c27e008d78d04500c62d9073cd2924f1d22c2679a6a2731040b	equivalent
CompactCells75To106.Box81.QInterval	071a5e8464a2af9c69b1a3ac78dccbe7557298a8cb3ffce534703b15b618389c	equivalent
CompactCells75To106.Box81.lambdaInterval	6aeb76b92868542091d755b8579cae53b5e0203e2966c8397171ab1266f8bdd8	equivalent
CompactCells75To106.Box81.compactCandidate_not_isWeightedPerimeterMinimizer	5ff5fa935cb0be5bd8f1b185839656cd6f41c225cf8793e42b823433ad74cb0c	equivalent
CompactCells75To106.Box82.QInterval	a5e06805262101292152ac4e3b38cfe1df95d7862e8ecbe6645b7aeeb73e5fee	equivalent
CompactCells75To106.Box82.lambdaInterval	9792b5499142134f271d94f6b4445f78e605e9cc61cd6252cf16cbf9efd7cb18	equivalent
CompactCells75To106.Box82.compactCandidate_not_isWeightedPerimeterMinimizer	1bb1a2acc2fa64eb9ce4eca67c4031367542c31d322423bf5cef1ab386bc2009	equivalent
CompactCells75To106.Box83.QInterval	52878858ee8493390c442fa97b85f7e14816e17553f4aaf3710125c193c037ef	equivalent
CompactCells75To106.Box83.lambdaInterval	a8542f2e10a52c1fb4afe08b79cabafbb228eda65d75219c45857760bbfa19ce	equivalent
CompactCells75To106.Box83.compactCandidate_not_isWeightedPerimeterMinimizer	158a5e77fc848d58d05f7e2a5b3d7e372cd0542ed6fbd0e4f40c3c6f34c8cac2	equivalent
CompactCells75To106.Box84.QInterval	01950c27d3074af280aed25f9d2952a775824b4018c4ee09930e174453239e7d	equivalent
CompactCells75To106.Box84.lambdaInterval	a01bd4a426b8e9a79caa4b173d99cc58bed8358b5201224f6ba2064b80dc6a37	equivalent
CompactCells75To106.Box84.compactCandidate_not_isWeightedPerimeterMinimizer	9992241401a05e5965c210099d1b524af8d03a3bb46f05219161387654878086	equivalent
CompactCells75To106.Box85.QInterval	f89eaa1876105aaeccc90726c399f0f7117559f72b786c9c5e12ffd2b3a7fe4	equivalent
CompactCells75To106.Box85.lambdaInterval	2394452abd939a9a068ad6756620c9da1a3be0ed855bbb7877cdacb615d2b103	equivalent
CompactCells75To106.Box85.compactCandidate_not_isWeightedPerimeterMinimizer	ae7d7543c0c599b907447f95e6e892ac18402cc54122ec0e9f483581859aed94	equivalent
CompactCells75To106.Box86.QInterval	f691ec6624e21bd412cc9df91ee8870d00cdadc9d8502728af96acb4da1ce3e7	equivalent
CompactCells75To106.Box86.lambdaInterval	b9afd4d63e602875d68b8ff398ef4aaf000646ce65eef3b95e85bad6a18fa583	equivalent
CompactCells75To106.Box86.compactCandidate_not_isWeightedPerimeterMinimizer	fdb54ce8adeeaf107b58abb7cb49dc7308095e011addaef81c194261368beed5	equivalent

Declaration	Source hash	Audit relation
CompactCells75To106.Box87.QInterval	3bb5ceb5e174e3040db3ba6211dff6b4869eb9f2d5cbc336b40562badf830b1	equivalent
CompactCells75To106.Box87.lambdainterval	2d05cb02033f465fd5087b35c06daa724b6982eeaa1534b3f1f9edaecd4304f8	equivalent
CompactCells75To106.Box87.compactCandidate_not_isWeightedPerimeterMinimizer	e17a83550a69de9efe7c3cdee2a0aedd35e5ab35145cd5e417a1ae4adaadeb7e	equivalent
CompactCells75To106.Box88.QInterval	e74b7833ee9f66083566c8bca62b4e9ec28962e8570d0fda64e14bfa61c1f4df	equivalent
CompactCells75To106.Box88.lambdainterval	eef429fe8b424b9d0dbff7770422c8bdd576f4f0cd3f242c3a5d6caf351eb0e	equivalent
CompactCells75To106.Box88.compactCandidate_not_isWeightedPerimeterMinimizer	b4954bb0ed5184acddc2e612a6e8c0219759fb2c8fe890e35b41adb9897a35a9	equivalent
CompactCells75To106.Box89.QInterval	49ea057f6068b923678f0bf637c4e1d7f0959fbd01d97b3e9dcc84d531be9ec8	equivalent
CompactCells75To106.Box89.lambdainterval	a18a5794a6710ecd44c727d545fac7789f7fafa43e5cc8189df0049fbc3df3de	equivalent
CompactCells75To106.Box89.compactCandidate_not_isWeightedPerimeterMinimizer	ee071cbfba03bfde533f3889d2c589bf61a832e200383fc35c27a9ad85cfa947	equivalent
CompactCells75To106.Box90.QInterval	c89b26b7cb3f3551ab50077c95d331feb9ab63dfe45180fd43180e14547eaa0d	equivalent
CompactCells75To106.Box90.lambdainterval	313e6671225c16db14ba2f53259aa829925e1ebee6048c16d4ad1b1551eca9b6	equivalent
CompactCells75To106.Box90.compactCandidate_not_isWeightedPerimeterMinimizer	c03dbd7a7a7fcafe16206c03c6eb022f5a5d1d0e36b9c856e3b3c4f962202b35	equivalent
CompactCells75To106.Box91.QInterval	aa884b881753d2a7c0d1725d276f2fe3ac9d4d95b469e53876fa69ab31b2ef0d	equivalent
CompactCells75To106.Box91.lambdainterval	daf9c4ff13143c1721ef44b1999f110c76d62817105b6da972193b02dc416f39	equivalent
CompactCells75To106.Box91.compactCandidate_not_isWeightedPerimeterMinimizer	34306eed505f7811ca7e8bb2910dd4a958eb63d5387adba2ec5607f90aed25f2	equivalent
CompactCells75To106.Box92.QInterval	3d65ce750c63050a7a39e0f223cdea14381a083dfac5f7ec01d15aeca99cd0b3	equivalent
CompactCells75To106.Box92.lambdainterval	82b3195c62b3e6e3e51bef16c8221e251a3a7d3c2ac9ec9b75e95a7577b0f6c2	equivalent
CompactCells75To106.Box92.compactCandidate_not_isWeightedPerimeterMinimizer	94e60313659aae4aedc71e25d9131c58bd043c80a3dd7fb5f617350853c93508	equivalent
CompactCells75To106.Box93.QInterval	aae3fefad9104d021468131cfbcb6e6487cd881748cc95d905d87105bf65050	equivalent
CompactCells75To106.Box93.lambdainterval	2d201c7c4be769b354692492e0ec55f2f23993d497f929e52285e049e8bd3175	equivalent
CompactCells75To106.Box93.compactCandidate_not_isWeightedPerimeterMinimizer	af43d8267201072c4853c3c6c1aef16446c76258ba5ebb7e7d40b82c72851389	equivalent
CompactCells75To106.Box94.QInterval	d7dc7480a618e21323665653ad6e92ab6f64dbc1e90f5db5e6da62b0df43c53b	equivalent
CompactCells75To106.Box94.lambdainterval	64d10b3cc1e82aa5ce26339647b7e9442729cc5c238f91d564bb1f263228d404	equivalent

Declaration	Source hash	Audit relation
CompactCells75To106.Box94.compactCandidate_not_isWeightedPerimeterMinimizer	0dfaceac154e90547ab5b58c81503a83803abf7f0d09e61d219e6d7c5e7af75a	equivalent
CompactCells75To106.Box95.QInterval	49e3d05150325285f8258bb0842c4cb193c5e9fc486c7d0cebcfb52553a37dae	equivalent
CompactCells75To106.Box95.lambdaInterval	d0ec1eca90265273d63c0dce5bfff7233a2d7cf57fe934975ff4c151db94a267f	equivalent
CompactCells75To106.Box95.compactCandidate_not_isWeightedPerimeterMinimizer	3593cbd4afa44eef75c984cd7372695d38713004fc0c2bd29564d433f854c2b9	equivalent
CompactCells75To106.Box96.QInterval	f9f6e5cb5dd0c4c1eabbab1e473ecc81d1d7a41e8aa5c92ebc12575ef3401bef	equivalent
CompactCells75To106.Box96.lambdaInterval	b61e9ff13f114c977da2691f67bc6ecf0d3d4a12b8c02da61c8239a3e39772c	equivalent
CompactCells75To106.Box96.compactCandidate_not_isWeightedPerimeterMinimizer	d71f0f6ad9f1f2e718d27e3db90237a80bf867589a67fc463e37eebc35dad230	equivalent
CompactCells75To106.Box97.QInterval	ecca3e67fd159ab73d5884635ccc97d43817aabdcaccb576d86303dd5aab0bc0	equivalent
CompactCells75To106.Box97.lambdaInterval	c083bad9b07bb71df65ee2917be7f91f27ff6ea1c22af8f5575d537ec71dca92	equivalent
CompactCells75To106.Box97.compactCandidate_not_isWeightedPerimeterMinimizer	f16649cbb3a0634a22c49a59994fbbedb44cf2b886e311bf9482e00aaac4b1a2	equivalent
CompactCells75To106.Box98.QInterval	44632bf5d93aa737ad2425bb468b06a7d683368e37dc6a44aaedbdc8a8d08271	equivalent
CompactCells75To106.Box98.lambdaInterval	a54e16b298ddf1e9d45be856a13991bb5a329769b40c49efcdbe1b23397749e0	equivalent
CompactCells75To106.Box98.compactCandidate_not_isWeightedPerimeterMinimizer	170e8243a6cf8e54eba99195472d1df971881aff0b33f7e49bff43ebbd293702	equivalent
CompactCells75To106.Box99.QInterval	820b3a6cf2009b96a7ce8f4e8ea6440e3a5be83039d9bc8e5724bce95c2f7469	equivalent
CompactCells75To106.Box99.lambdaInterval	babb8eb5a744daf05023119d90fbcf3d88995bc2c0b978076673d1c43d06bfee	equivalent
CompactCells75To106.Box99.compactCandidate_not_isWeightedPerimeterMinimizer	b31ec91256744cb24152a309d47a311ab0ad264ad12bb1f4d7926daf944380de	equivalent
CompactCells75To106.Box100.QInterval	c0665446c4b09cd4dccd49cebad9e8e9d7718dbc f59d679a2c299d362adf9d3f	equivalent
CompactCells75To106.Box100.lambdaInterval	ae1e10a5c3ecee6ecf90a0eeeb280e45967436caa2217d92c6a785ad3895a80	equivalent
CompactCells75To106.Box100.compactCandidate_not_isWeightedPerimeterMinimizer	1635e986627b7096cf280d7051636e1055bc0de1c7a16746a5d984482785d5ee	equivalent
CompactCells75To106.Box101.QInterval	79e96c7c6fe4ef48eea3530a9c78b6d1213bb5dbf9d255e09c83f33f114fb43b	equivalent
CompactCells75To106.Box101.lambdaInterval	153502c597f3de90d22f3dd008b1575d50ff81c50d0e458977d5ca8bad3a9f8f	equivalent
CompactCells75To106.Box101.compactCandidate_not_isWeightedPerimeterMinimizer	f9da92af0d33ca39b60f7274e5ded2063b869a5ebcf54ed22c524f2a3daf424e	equivalent

Declaration	Source hash	Audit relation
CompactCells75To106.Box102.QInterval	c9f4c678097594de1ef28352d3276c31ea555f3c82645e811e8a5a60f1196c0a	equivalent
CompactCells75To106.Box102.lambaInterval	dd70a687f1cde29eb5167cd60f62d6d76216348365f53d83297a45ba72d71fe4	equivalent
CompactCells75To106.Box102.compactCandidate_not_isWeightedPerimeterMinimizer	266fe654a0be25dec662dfdc013091fdb2b4a5ae914bba2859dc1a2058cc8ded	equivalent
CompactCells75To106.Box103.QInterval	4250225c62f762dceec61bca992f3f35df5a3ea073a13807e3024ad13c18255c	equivalent
CompactCells75To106.Box103.lambaInterval	c88f4bd77b6be6d3d11687148ca79c4d40017c2f290497f154b40a3d3dab0d16	equivalent
CompactCells75To106.Box103.compactCandidate_not_isWeightedPerimeterMinimizer	5c7ad453925d87153ee8edaf204656d23ea2e3cf538a5bd96656158b06eb53f9	equivalent
CompactCells75To106.Box104.QInterval	5ffe3d992c83a6f351676aa9fc5cd73710f7991dd2b94a6025dcf3d50a0503ed	equivalent
CompactCells75To106.Box104.lambaInterval	4ace804745fea95dc4db1d12e0ab32cc3528db5ca13112a8fdb2fdbb01fb1676	equivalent
CompactCells75To106.Box104.compactCandidate_not_isWeightedPerimeterMinimizer	7ef30d6cbe602d3b58fc250f639493e024df79f8d82aba4cd88a919e421c6518	equivalent
CompactCells75To106.Box105.QInterval	a4d6303c0ac7f7dd5308aaeb2ebfb75a1c89c879e2e6fb6750395084a507d92b	equivalent
CompactCells75To106.Box105.lambaInterval	4cf2a3880bd7b50323133abd994a7a75c23b81b62fe83d2065620d6d5bf00b11	equivalent
CompactCells75To106.Box105.compactCandidate_not_isWeightedPerimeterMinimizer	e7899334c29d7b874d9946e09ddf779f06960203fb33a89ca21d24860afb6534	equivalent
CompactCells75To106.Box106.QInterval	2fcbd7355933982c510e0b3b7d3304442e3d57338fc0ea5e5c89edde9166a7ea	equivalent
CompactCells75To106.Box106.lambaInterval	ef88cb5feba6caf62154ccc6b775cca2abe1f937ecf0918479145e02153b8e5a	equivalent
CompactCells75To106.Box106.compactCandidate_not_isWeightedPerimeterMinimizer	c1a4a22e313edd96babb6284930a2cf555a8c9d3632bf1586e5a1253d08d9adf	equivalent
CompactCells75To106.bboxes75To106_candidate_not_isWeightedPerimeterMinimizer	08b5f9cad78f29935aa4fef58417683c89c5deb79b1d08ac9db0225fdb5545d2	equivalent
CompactCells75To106.bboxes0To106_candidate_not_isWeightedPerimeterMinimizer	1f26f7f057597424051b4b5292c617a1c9b7c66674c2de9f226def6336f15a73	equivalent
CompactCells779To810.Box779.QInterval	030b21694da625662bd3de78ae9333a25c86224293eb04453bdd6419c7adf4bc	equivalent
CompactCells779To810.Box779.lambaInterval	f6cb4b161539aad39d91b5bed2f30a9742277ec3c2c256a018ce65ea53bf7024	equivalent
CompactCells779To810.Box779.compactCandidate_not_isWeightedPerimeterMinimizer	c60432de58f336849b93b812e304146e3e45fe7c83151324acd47485755e2a58	equivalent
CompactCells779To810.Box780.QInterval	f28fa31810c2dbd3e0e58c6554dd46f71e256912a635b0405d53df0a22cdeea1	equivalent
CompactCells779To810.Box780.lambaInterval	1e0207a34224874ea77d0b0ef841421a43f84509f71e7f9df6c2dfa989f3e2fd	equivalent

Declaration	Source hash	Audit relation
CompactCells779To810.Box780.compactCandidate_not_isWeightedPerimeterMinimizer	08a5a44f382c92ba63ff4cde6a90be7bae243e7cca2020eb4e02c40a396117a1	equivalent
CompactCells779To810.Box781.QInterval	45c150b78406546fd4c8e78e074f6f09d1f9c7be91822a61b371e15d233baa0c	equivalent
CompactCells779To810.Box781.lambdaInterval	bb6b3a27440de100afce593dd3d7e8593d8c1bbcd50078d168d7adafc20ba5dd	equivalent
CompactCells779To810.Box781.compactCandidate_not_isWeightedPerimeterMinimizer	edc17a0a2fedfa3ac14d5d1e7bc96a4ebfb2a3f3413e6547e1475435c38c51c5	equivalent
CompactCells779To810.Box782.QInterval	4d11cd92b8d9d158383409f92a817f53776a156057d04b3125107c8fedc955d3	equivalent
CompactCells779To810.Box782.lambdaInterval	77aeae6897491b19021abda41e5835738dd8ced75a4c91ef86f786cb32e4838c	equivalent
CompactCells779To810.Box782.compactCandidate_not_isWeightedPerimeterMinimizer	df2af7c648dcd8d6bf2bc15652a284358efbbf5e7acebc82a5827ebfd2cce6b6	equivalent
CompactCells779To810.Box783.QInterval	321babcead80f5418646b51daccf0ea69cbcb75b873e0608c433df914d918071	equivalent
CompactCells779To810.Box783.lambdaInterval	d2108bcfb2969d3c1bb4d01fe3def1b5fcf7e7253201f2d0e6cf372323ae137a	equivalent
CompactCells779To810.Box783.compactCandidate_not_isWeightedPerimeterMinimizer	118958e40e77e235fe90e65c2210fe742818e8b0064d2f39252f28d71276613e	equivalent
CompactCells779To810.Box784.QInterval	4ebcf9ef044c934a57ddebba49918e8596dd2fe52e58ce298c2d1544c73513d5e	equivalent
CompactCells779To810.Box784.lambdaInterval	1d12fe7b2bac91622a30799394db4babcb6a9566a1f92918e9c169a6b2fad5886	equivalent
CompactCells779To810.Box784.compactCandidate_not_isWeightedPerimeterMinimizer	01c3b0609e6aacff5a21fa182f6c63117a726922e3718d13ce157a704c2967ce	equivalent
CompactCells779To810.Box785.QInterval	52bd634adbb3f4834569415b582a285fcd6d17d880bc8e16f3ac29986b7edb72	equivalent
CompactCells779To810.Box785.lambdaInterval	0c4b81277c9458a9278b70fb4659211053a239c8d998d69b3adb8db93084f4f9	equivalent
CompactCells779To810.Box785.compactCandidate_not_isWeightedPerimeterMinimizer	2d190814b9b60540c912b0192bffbb0e6eb10a381e2cb0f64288ce56bdc0dfc4	equivalent
CompactCells779To810.Box786.QInterval	1b27d7523d0d11459e7ce111eac4f27ea3fb774cf6723c5cd8c4a281ae7bdd5b	equivalent
CompactCells779To810.Box786.lambdaInterval	6227e0b4f748afc63a79ce9601827e1a3066ffa00a1c284be700d4f69e6cfa6b	equivalent
CompactCells779To810.Box786.compactCandidate_not_isWeightedPerimeterMinimizer	8f8c81fc20d841168087edb4567714db44a50ae62cc63d3a66fdd0ffd31ce287	equivalent
CompactCells779To810.Box787.QInterval	649b5e7087256855a9eb1cfa25dfffea5d43b28eb85159a8d77d34059a5ead6d	equivalent
CompactCells779To810.Box787.lambdaInterval	45b2c4d6bcfd25d038d0671563ecc99d22801edea4e5c43909ff1e43fda0ee8a	equivalent
CompactCells779To810.Box787.compactCandidate_not_isWeightedPerimeterMinimizer	366fb18235f35438954d87149736f7194e786292bb8cd6867d0511bc5a9ea15f	equivalent

Declaration	Source hash	Audit relation
CompactCells779To810.Box788.QInterval	9ce280bc542f3f71bae835dce32a8d1b95a dc1553ee9285151f50ea43f2d2d13	equivalent
CompactCells779To810.Box788.lambdaInterval	365b8af8cd84ccf3789a1b0ad0cc044b05b 264a30021148a3852fd7cf64507ee	equivalent
CompactCells779To810.Box788.compactCandidate_not_isWeightedPerimeterMinimizer	4c6978db1fe8fff76162dcadab29de31a05 1f27b5db0955db69afd70b96d5e7a	equivalent
CompactCells779To810.Box789.QInterval	9506ce56a7cdd60daf365d51c962f151b03 6cc2fe7f2df7fb618b0cce8213d4b	equivalent
CompactCells779To810.Box789.lambdaInterval	5e75daf3301a0e9850805d85692b59f6cd3 455cb2b075c84d4255b7ccaa6a27f	equivalent
CompactCells779To810.Box789.compactCandidate_not_isWeightedPerimeterMinimizer	ff16facbe237e66b719f1e7f53326aec7bd 74b46fd51ef527b317ba86868311d	equivalent
CompactCells779To810.Box790.QInterval	e76f5a28aa7c273abd2e99c4a7cefeb3e4f 34b9325a2230a0c3317c26dbd0579	equivalent
CompactCells779To810.Box790.lambdaInterval	fd13bf7d3a5397a700d5c7fee32bb756953 21b27f8be37f209bf46f4c8e95439	equivalent
CompactCells779To810.Box790.compactCandidate_not_isWeightedPerimeterMinimizer	a2a1cf94d4809f14b97ce5cab377427609d af01620fc26f1e102c61fac9dfcc9	equivalent
CompactCells779To810.Box791.QInterval	8236c8ca56f099209eb1f9752c25d213e3e 8068b8157c05ec8622e38f0095bac	equivalent
CompactCells779To810.Box791.lambdaInterval	5462d6e22fe1bfc642820fbc48de5cfff100 4cddbdc539dfd3555bfa239e431af	equivalent
CompactCells779To810.Box791.compactCandidate_not_isWeightedPerimeterMinimizer	44b2c2ad904e55cd55cbe5aaabe03a898ec ba13d8762ec314ad9a0b10658f0c7	equivalent
CompactCells779To810.Box792.QInterval	6e928fb641fb9650c1c50e45053b1b3d3b5 8655ef16fb1878efb1d6d121332e6	equivalent
CompactCells779To810.Box792.lambdaInterval	583b03efc90251024f22fe1513c518491b2 d092a4d8df7378be6d906864d6b7e	equivalent
CompactCells779To810.Box792.compactCandidate_not_isWeightedPerimeterMinimizer	158f0b493585e264639ef46046044c27832 73e533f8322849495bf6b39945013	equivalent
CompactCells779To810.Box793.QInterval	0eb0b5474791c5e8caba77fcee2b40432b21 75e16cf27c9e22704f901f1aa08ad	equivalent
CompactCells779To810.Box793.lambdaInterval	4bcd5d50572f94cdffd5b69b1a7275900df 83f9b02fc0a2a5782cda4864637fc	equivalent
CompactCells779To810.Box793.compactCandidate_not_isWeightedPerimeterMinimizer	2c4f864242ef57b0956a44fa0729b3bccdf ce41b2fcaae45f396e769f8f17dcf	equivalent
CompactCells779To810.Box794.QInterval	975b03a71c7e69c77a396f2d4991629aa02 3371283390b5e6f5cba33ec9c2f3f	equivalent
CompactCells779To810.Box794.lambdaInterval	678ea445087918ec53cca4b81b822ec833b 6546721aac8c1e2c8fc15ce56f752	equivalent
CompactCells779To810.Box794.compactCandidate_not_isWeightedPerimeterMinimizer	23ddf15ed9e6fe80f87cc843a9b02c16dbd 70204e35a140fff4d3b864f901b99	equivalent
CompactCells779To810.Box795.QInterval	d7f9ebae8f13afd97215e3d90643da063aa a870a764a6b354acd198063fe8f2c	equivalent
CompactCells779To810.Box795.lambdaInterval	2ba3b46cdf71cb48e88365324f32b515b26 99cd2fb619c6216424b9981bc2b33	equivalent

Declaration	Source hash	Audit relation
CompactCells779To810.Box795.compactCandidate_not_isWeightedPerimeterMinimizer	6445d0d9c1eceb556604a0f148434607a08c97e5955c812bded442720ff02d0e	equivalent
CompactCells779To810.Box796.QInterval	6598794ca535fa993a41880fa033db0668d59c0a31b84a2788b70c2de8ad2403	equivalent
CompactCells779To810.Box796.lambdaInterval	bbef2e243f4aab89aacbcddece73a20f51dd340b0e3dd83dca37e2f56b63c7ac	equivalent
CompactCells779To810.Box796.compactCandidate_not_isWeightedPerimeterMinimizer	22a39ca33038c4cf6184a4bcfe1b5805ac5ee6e2f5df0a56261c5f4f65b3d2d1	equivalent
CompactCells779To810.Box797.QInterval	bb52862b48bc7eaf933899c5a4a531e9531f81157df2a5f118d57a3f9f6d885c	equivalent
CompactCells779To810.Box797.lambdaInterval	ae582791fdb82c5e53ec7a6ca5172e23cb37b8d6b00744db93df94f84b0d1f01	equivalent
CompactCells779To810.Box797.compactCandidate_not_isWeightedPerimeterMinimizer	5694207737dd1aa1f261dcba72b59f918980a0e16ceeee6534ef92e33aabfec5	equivalent
CompactCells779To810.Box798.QInterval	96528bafd0acd524a5e5c92ef68c50d23595eb7f5ad806b47ed71343fe63d50e	equivalent
CompactCells779To810.Box798.lambdaInterval	cf652e86fd35d529b6c94c4c3704664741989b6b38c956a5d66327a0d0764883	equivalent
CompactCells779To810.Box798.compactCandidate_not_isWeightedPerimeterMinimizer	584a534585e76896b23479d66cc336190426e328223134df5eccb4b22e7a17f6	equivalent
CompactCells779To810.Box799.QInterval	80c6e6814a7e80d26171efc7e4a612aed349eef6ddf42f4220fa69240411130c	equivalent
CompactCells779To810.Box799.lambdaInterval	6dfabcd257632744b7c29a40a076b218e0c948ffcd40ee967b68f19c0d2289f3	equivalent
CompactCells779To810.Box799.compactCandidate_not_isWeightedPerimeterMinimizer	eae3ea4a96654b0b204566731434de2c2a184c4af09f150ed4560293880fb0b3	equivalent
CompactCells779To810.Box800.QInterval	0d42d0de563c01eb97334c6987c13d05ef2133f659cf87e514888de9bf5f1f4c	equivalent
CompactCells779To810.Box800.lambdaInterval	e466e7054c467bc81481035844c665dadf3a4e73fc79de60eab5587315cd925c	equivalent
CompactCells779To810.Box800.compactCandidate_not_isWeightedPerimeterMinimizer	fad0a26b065bca720365b2cd574bd53f7ebca032a42f68c19c60de74c71f852e	equivalent
CompactCells779To810.Box801.QInterval	8487a4e018e87b80d4d511734bb4140273dda378cc618bc9d87795f4ea4fbf30	equivalent
CompactCells779To810.Box801.lambdaInterval	a45033974ac1e9a473a9b04f420190b30ffb56976c32443ba4905ded465c8de5	equivalent
CompactCells779To810.Box801.compactCandidate_not_isWeightedPerimeterMinimizer	5bcb2377c16938368f0088da76f7a2d0f6be665c50470f272e78807b64c1ca74	equivalent
CompactCells779To810.Box802.QInterval	02ffbd987af00760491826eed0b31b7806f9bd6d19512aa8839fd45cc06f8232	equivalent
CompactCells779To810.Box802.lambdaInterval	ae9bd7d7833a95b9ff2b397b21c1e2332d67013b5dd80e69123984f0cf7d6560	equivalent
CompactCells779To810.Box802.compactCandidate_not_isWeightedPerimeterMinimizer	b73ae84368799047e227a7c846c53b42280fca926071d086938441ea879ab3cd	equivalent

Declaration	Source hash	Audit relation
CompactCells779To810.Box803.QInterval	af69038a49fc4cbd5305925e0c840b2100affb65b316c5e6064222d3d194c8dd	equivalent
CompactCells779To810.Box803.lambdaiInterval	0bfd81def38a08e142f33e29d74c677d47644d7362f9e44df37af4fa9cb19524	equivalent
CompactCells779To810.Box803.compactCandidate_not_isWeightedPerimeterMinimizer	3db68670802354e526e17293c93b61046ee614908bc87e72c4a8b1ba9c76f313	equivalent
CompactCells779To810.Box804.QInterval	b1211f5d2dd1f9be7339a98853247ec9a79d412a22a1a634e858e0bd8a84e098	equivalent
CompactCells779To810.Box804.lambdaiInterval	904db87f84959da74e7cb5e4a30277d063e364861acc370ad9f5823845f9420a	equivalent
CompactCells779To810.Box804.compactCandidate_not_isWeightedPerimeterMinimizer	0a1656fd74a6990cf94958748c985a601548396e603fa710b05ee3dbbb684633	equivalent
CompactCells779To810.Box805.QInterval	376ef41c67bc8b22f864736d77ab627879e5f9eca3bc8dfe92f97269f9ed8d83	equivalent
CompactCells779To810.Box805.lambdaiInterval	53ac64e602309c7f8f1945e6cd4c51c84892743a90f1a9236b1d9a903aafac51	equivalent
CompactCells779To810.Box805.compactCandidate_not_isWeightedPerimeterMinimizer	1ee5987c8fba7d5f0e82d74e8879dc26a2f96b8e7808c8e16d5d03d2a6d28187	equivalent
CompactCells779To810.Box806.QInterval	c8028b131ab6a6bef98bd400c0ee1624ad69705f6d726f9b40b3783752e46654	equivalent
CompactCells779To810.Box806.lambdaiInterval	47dd7fa76d95fe220134568959915aa4b1c0a0dca295872328b7fc7b7d1cd458	equivalent
CompactCells779To810.Box806.compactCandidate_not_isWeightedPerimeterMinimizer	08327b38f0ee173538feadfba9941eb635d99674888037e350306ac5f9bb4c65	equivalent
CompactCells779To810.Box807.QInterval	30d8bb05e3eb5198f2fe0198b1b7cc0198d73cfd335edadb557009efd0feb5d	equivalent
CompactCells779To810.Box807.lambdaiInterval	73dcf0fb9ed1d9e40dcc32dbafb4cd448fadb92bf98c986d9f4ea71517b0d937	equivalent
CompactCells779To810.Box807.compactCandidate_not_isWeightedPerimeterMinimizer	e1b8714926920eeb32e378a299fa416204dfaf08347a0294a7689a50c30bd06f	equivalent
CompactCells779To810.Box808.QInterval	41be0ffacc833bcd3b5f4c0b9c42c7aafa371200f6032a0a916e508a77b05f5a	equivalent
CompactCells779To810.Box808.lambdaiInterval	50f1907f9d597dfb1c7ba555c9891b2da1537645a34a701b77bb7699dcbfb7ae	equivalent
CompactCells779To810.Box808.compactCandidate_not_isWeightedPerimeterMinimizer	71ad92769487fd728eb3afa46a2f7fc231d46b690df8b217d2c2177f2ac39cf8	equivalent
CompactCells779To810.Box809.QInterval	cad7c7249cfe8d710838d8dba57e9a99b5d8aa242a259c1139b33a46fbf97ac2	equivalent
CompactCells779To810.Box809.lambdaiInterval	98a83b79e48e2fc88e501c56223fb495348316c74b11a99246b8e8f9b6ea6285	equivalent
CompactCells779To810.Box809.compactCandidate_not_isWeightedPerimeterMinimizer	d871fa88e8954db6c9509a4da7d419d381a2ed46dc2e9906c4c84242b38f0298	equivalent
CompactCells779To810.Box810.QInterval	3606648055d638330186f2522a1a23d858e262e1cdc4fb5169d9d79c64f2211a	equivalent
CompactCells779To810.Box810.lambdaiInterval	5075d57c76cbd6740a7fe722b72e8316fa932b46f88f5359fc3b42a88ca97069	equivalent

Declaration	Source hash	Audit relation
CompactCells779To810.Box810.compactCandidate_not_isWeightedPerimeterMinimizer	4462618a4a659a646640a92baec7ff9b626ba3014c183f0e1d5d4cd12b15bf74	equivalent
CompactCells779To810.bboxes779To810_candidate_not_isWeightedPerimeterMinimizer	6d302b492c1b278b9cfde0f5e9251db498a14fd684b60801c60068e9a9851f24	equivalent
CompactCells779To810.bboxes0To810_candidate_not_isWeightedPerimeterMinimizer	0efbcc8cdc5dcd419becefa317f2a16040543d75bd08e7962d419d8b3fef4e3e	equivalent
CompactCells811To842.Box811.QInterval	3045f38851d63c713d4815f50c5b636b81b911e9c5676ea19c677ad4a3e09b3c	equivalent
CompactCells811To842.Box811.lambdaInterval	86a1603a41bd49a0eb9f664be8089282215ad9b8f06ca0054c88e6531369fe2a	equivalent
CompactCells811To842.Box811.compactCandidate_not_isWeightedPerimeterMinimizer	1204fdb4b1cc140bd028982768a985c2a3d9aa30a94e86711f752b3f61008caa	equivalent
CompactCells811To842.Box812.QInterval	ac58ad390318b034aac65bf5def56818feca51e403f0d18a86fd28be86526f32	equivalent
CompactCells811To842.Box812.lambdaInterval	1693a45b257e137e324b168c040e7a4eee792e2ced7998b8fcdd5177fb57737a	equivalent
CompactCells811To842.Box812.compactCandidate_not_isWeightedPerimeterMinimizer	b1f3d4823fe0dc21dfcecaae664bb42ae980f0c9be0fe1ddc7a250d81018ea4b	equivalent
CompactCells811To842.Box813.QInterval	91b72e5affe220f7376b10f981bca395d4ca801f29ad0e50be27e84fba063f43	equivalent
CompactCells811To842.Box813.lambdaInterval	f7469d229151f3fd77892564e718e78a85b9a5752208067b44dd895acf466710	equivalent
CompactCells811To842.Box813.compactCandidate_not_isWeightedPerimeterMinimizer	879851b516772f2d4bb38790b76a51148046c4383a2a66b5ae8b44c04d88a792	equivalent
CompactCells811To842.Box814.QInterval	be9ad95a2960f938c56625ae2555857e734452fd465f62893dd40156ad6ebeec	equivalent
CompactCells811To842.Box814.lambdaInterval	39a99e22162f1684b95466e7dd0f684859cf8cf9ee6d7fffc6db9e55ab427c622	equivalent
CompactCells811To842.Box814.compactCandidate_not_isWeightedPerimeterMinimizer	5f81d059de8183ae1b23d0999fd0eb9e4d1ea0bc8c761816a0a7e5af5bb7a6a3	equivalent
CompactCells811To842.Box815.QInterval	e20476a98237b8bb57f90e28b6418f4dcd966c6edd45732cc3c19001c71c5bc1	equivalent
CompactCells811To842.Box815.lambdaInterval	1cf9dd5dfc3a1fd085d4cca759d32b4f05b11ef3062844292a106a845ca5f53c	equivalent
CompactCells811To842.Box815.compactCandidate_not_isWeightedPerimeterMinimizer	1ceaa6bf36adbb1317673f7c27e45d26a3643edd7ca3d0299abebfe881e69745	equivalent
CompactCells811To842.Box816.QInterval	1ecc755e3744487ded8a22215a01f0cf8824fe3d3849fe8dad579b1fa0b258b1	equivalent
CompactCells811To842.Box816.lambdaInterval	4ceacb81a3870ed170c854171731cc517d8a5832130bf4e45a503c2c9a69037c	equivalent
CompactCells811To842.Box816.compactCandidate_not_isWeightedPerimeterMinimizer	228be73079307d56f982e7090095120aec8174cf807378500997cd4e03857dfc	equivalent
CompactCells811To842.Box817.QInterval	75ce7d42084266f7f252af1245e7fe72fea701361c377e76ea09fa89ffe341ba	equivalent

Declaration	Source hash	Audit relation
CompactCells811To842.Box817.lambdaiInterval	0f6934ff7631d8d5fcdba8ff170836ace831cf442da668540233002dd903f33e	equivalent
CompactCells811To842.Box817.compactCandidate_not_isWeightedPerimeterMinimizer	02899abd2d7e4d66c1acc00fc98a4c87b11e897a64f87a79817521dfac86dca3	equivalent
CompactCells811To842.Box818.QInterval	e7c7c782a545ba449b2fc666512395f9b510db98b628630e013961a1c3ebcb60	equivalent
CompactCells811To842.Box818.lambdaiInterval	cf000ba7569b6b3bb1c081776bcd2f87d15870c470a038139029db4d79d1a0c	equivalent
CompactCells811To842.Box818.compactCandidate_not_isWeightedPerimeterMinimizer	73bb5b7a4954a863185b95b93922a19e7f09c8cafa1ea264ec6964c899c4a99b	equivalent
CompactCells811To842.Box819.QInterval	95b7fbd73fa88a1096f9ba282cce441cb4a1adfa8fe8160bf92ad54344ea64f	equivalent
CompactCells811To842.Box819.lambdaiInterval	56fb635dd19fcca435beb4c81d20c87964f2b7031ba49f549baeba51c28223d2	equivalent
CompactCells811To842.Box819.compactCandidate_not_isWeightedPerimeterMinimizer	bada667f223f4aec53bbfc97082e4524b42e8660c86473f0d74a9c8f8f771bbc	equivalent
CompactCells811To842.Box820.QInterval	f766cf3b10b9ed1da2b98b1232b8e07f497dfb821291b4a575bd46eb84f7b24c	equivalent
CompactCells811To842.Box820.lambdaiInterval	1e98ae73dc5c2813fca9a2873e9ba07d51dbc8981ff5c3ffbfe2a95694bd30e3	equivalent
CompactCells811To842.Box820.compactCandidate_not_isWeightedPerimeterMinimizer	4f80f488e7eb785815ec70b0145f3a146c952e90f661c481e4b278c21aae29d6	equivalent
CompactCells811To842.Box821.QInterval	d3c4bb62977607f54b28ed1e1f53903fe0c664e506f22ddf342f68bd64d7ed87	equivalent
CompactCells811To842.Box821.lambdaiInterval	aa4ba61e2c6ccba73eee0951ef76e0fa1ddb5d49ca9814d1205c17dbef96cf0d	equivalent
CompactCells811To842.Box821.compactCandidate_not_isWeightedPerimeterMinimizer	bcf52ab2875635c33ba5423bbde03c23cbda5d4d0f232ce890c4213b7ee6df67	equivalent
CompactCells811To842.Box822.QInterval	876cb97672270d738d09bf0f3059e55c781165743aea96dd4bd5979f7a22e41e	equivalent
CompactCells811To842.Box822.lambdaiInterval	058e531c37687ef17afc96b43013a171d63d45bca6a59c341ddec2ae3b94be0	equivalent
CompactCells811To842.Box822.compactCandidate_not_isWeightedPerimeterMinimizer	cc4c289cd59b3280d97ec9d75a51992ae5152dd29ad403533dadd083eb62a4ce	equivalent
CompactCells811To842.Box823.QInterval	009d255eee25777fc590a5528caaecf3c29b26a2512be9f3f0af54268618b1ed	equivalent
CompactCells811To842.Box823.lambdaiInterval	e48d07bd4c222c031ae737f208d6b748e86490859d073286f15ef4e9b02e7154	equivalent
CompactCells811To842.Box823.compactCandidate_not_isWeightedPerimeterMinimizer	ff6a14090f797bcf941de04240ceb6493938245945be49b82f690f80f96ce3d8	equivalent
CompactCells811To842.Box824.QInterval	dad15f53209f0e79d784cc6c0c2005c4169d9e0179eaeaf0660b5258d562ccea	equivalent
CompactCells811To842.Box824.lambdaiInterval	e5c3036430d8dcd4eadb2d3fa96cccea321dce43565792a7fc40cbcbda964f00	equivalent

Declaration	Source hash	Audit relation
CompactCells811To842.Box824.compactCandidate_not_isWeightedPerimeterMinimizer	e7410193392e0ef22f7e5e993e7bf74f6260d60f585b1d0516a02ab7413bf53d	equivalent
CompactCells811To842.Box825.QInterval	c2f550309f955a9dd8a724a0fa9d199d3eb182847a8c83f46033dc4d50b0522e	equivalent
CompactCells811To842.Box825.lambdaInterval	5d62e37df7f16a93e459de262b9ecf8fce13fb41f795dc426d159ba3807ca689	equivalent
CompactCells811To842.Box825.compactCandidate_not_isWeightedPerimeterMinimizer	3ce7252d349b347a384e43ba61f4ad538ff2b0231d38b349f71e751a3ef1f29	equivalent
CompactCells811To842.Box826.QInterval	b2e9ba55462b6a4e9e36f59a6a318afc91ebd4a1477f7874e6cb1cd4695ea168	equivalent
CompactCells811To842.Box826.lambdaInterval	5328f98531ced615a766f082d9940483b8a47c5d28d6fbda69f5afebfe29d906	equivalent
CompactCells811To842.Box826.compactCandidate_not_isWeightedPerimeterMinimizer	bfcdb3b0c975ac1f9cb54416a47a465ceea0218660c3eae7aeaa96fc59c004d4	equivalent
CompactCells811To842.Box827.QInterval	9ae8f02b77b8ca44c93f591fe5d10bea07cfdc3c84c57500024090b916ccbfb6	equivalent
CompactCells811To842.Box827.lambdaInterval	99ed607504b6b1948c139c002c530b21295995ce6d7bdeb792df7eef0a8d2ef1	equivalent
CompactCells811To842.Box827.compactCandidate_not_isWeightedPerimeterMinimizer	2b52b0b33176aa57596c4f30dddf0c316a848673b1c7fb524e282ee35da156e2	equivalent
CompactCells811To842.Box828.QInterval	492054d70d2da0b153d0f661cd94fa3d583a3c13b9649ec88a841171bda2d039	equivalent
CompactCells811To842.Box828.lambdaInterval	13101fa6fece40393834d5369deb90f0db43d0dc4ca5e12ea1926ed847143f51	equivalent
CompactCells811To842.Box828.compactCandidate_not_isWeightedPerimeterMinimizer	b7785630a541599df44e7aaf1ce0629ce4a3283d12f6019bd11005b2effc8653	equivalent
CompactCells811To842.Box829.QInterval	dd0cb984911fc12c7c28c1e79a1c66094b5358f7b4b70e1900cc0c8ef1a20833	equivalent
CompactCells811To842.Box829.lambdaInterval	18bac0d0cb29e8cde8ef0ba48d7b75f29a26e300f9d77e51ef98f423fd9e7942	equivalent
CompactCells811To842.Box829.compactCandidate_not_isWeightedPerimeterMinimizer	ed81ebf5279c26fa8ae02681ef5994d1b5f7d05b45e4ab35b3ae82f9970d1279	equivalent
CompactCells811To842.Box830.QInterval	39c3002a6adc387e28357a30e5283b70778800f2f7dfa715d146fff630a90cab	equivalent
CompactCells811To842.Box830.lambdaInterval	4ed7c8bf4559989b6c9e27584960318fc30650fdffc21e134dcac72898242809d	equivalent
CompactCells811To842.Box830.compactCandidate_not_isWeightedPerimeterMinimizer	9ef076b84f2e3386e7033a41f43fcbdfd7ee195807cf1e750a5748325ca37ebe	equivalent
CompactCells811To842.Box831.QInterval	8cc5f56765a81eb13d42b65fa4d9ed1d1e30b2f5451c92b4f530a39e533f9359	equivalent
CompactCells811To842.Box831.lambdaInterval	d4bbb6c4608d87427485047d5785ecf4654bc81097cb0193442b1ccb7d6f1637	equivalent
CompactCells811To842.Box831.compactCandidate_not_isWeightedPerimeterMinimizer	14f1c7edc8f39f447695552762245372d493797aa928808b45657cb5a9cba6fe	equivalent

Declaration	Source hash	Audit relation
CompactCells811To842.Box832.QInterval	2bd5b521f7d6cae00a09bbebd9b1652c0998e34afa4d60f3d633fd32114026d	equivalent
CompactCells811To842.Box832.lambdaiInterval	611e59ce941ed51c8111b0ba6a997b3254e3f603c5285b594f807103762cd8f8	equivalent
CompactCells811To842.Box832.compactCandidate_not_isWeightedPerimeterMinimizer	5973a1fa61b62d6fe2efde32c65848ad7d3cd4010c73a1a3338f4c10c2f11f0f	equivalent
CompactCells811To842.Box833.QInterval	72deb7628de2aaff2f09c7212c575f2fdf40cbc32829516105223640cb713057	equivalent
CompactCells811To842.Box833.lambdaiInterval	356d8d00ea928b637056069290f2082d3ee18d9807dc709df39ac796a4ab263b	equivalent
CompactCells811To842.Box833.compactCandidate_not_isWeightedPerimeterMinimizer	133b78eee2121f690db2cd99e591dcbccdd76f10c03cd09fdb477fd74d6f9b34	equivalent
CompactCells811To842.Box834.QInterval	c59d909e0a9794958757d3c5b606eceda6cacadf5ab1ce7ea845b3c32118882b	equivalent
CompactCells811To842.Box834.lambdaiInterval	f2e8bf28675b6129313e474598bcad7746bc4291425ab8ab0917d81cea362d43	equivalent
CompactCells811To842.Box834.compactCandidate_not_isWeightedPerimeterMinimizer	06f97f463551d150187bf3a83bed3068d85dbf517a8a341ef1b8217447532ac9	equivalent
CompactCells811To842.Box835.QInterval	a7ff3589da78f41c3814866cf53d8bc38f51457dba6abba0d07db8211c124260	equivalent
CompactCells811To842.Box835.lambdaiInterval	a1d4531407ed025200eed3b7833f172bbbd86ec0a59a7fe23653e0baa8f7bb1b	equivalent
CompactCells811To842.Box835.compactCandidate_not_isWeightedPerimeterMinimizer	94eae1e8f4b05575ead3112f1f99a32e12d27c0c4f83867740efbfcc61e39670	equivalent
CompactCells811To842.Box836.QInterval	bd6daa7ca6076971a47e34491a83b21dcc7eed6d186cc2033150340a209856fb	equivalent
CompactCells811To842.Box836.lambdaiInterval	0de7e30a4dcbe219dbc471fb72f37bdae30088730a737221744bff4cfd31cb5e	equivalent
CompactCells811To842.Box836.compactCandidate_not_isWeightedPerimeterMinimizer	d616ba313701903dc32726a81e3a9c961bd9f2c0ed6142d3cf95125805546ddd	equivalent
CompactCells811To842.Box837.QInterval	2a6efe476bdccacfe4620c3b6cbdafdb29fae9f5898792bb640a1daf858a5198	equivalent
CompactCells811To842.Box837.lambdaiInterval	fcf03f266b73eb62da739cddbca9e014624379594b846f4d72a4f0b2e73f8cb5	equivalent
CompactCells811To842.Box837.compactCandidate_not_isWeightedPerimeterMinimizer	a89dd22a69e3075dcb638de4ad751341caa52aca84ed92bde841648dd6d9bc58	equivalent
CompactCells811To842.Box838.QInterval	aae3fb3dabe9fe7bf40adc65377d57156070f297ef41971a97777e630f0aa041	equivalent
CompactCells811To842.Box838.lambdaiInterval	189ae894f60ddce88f189553fba387fa125e8481f6fea9c5e1e65078ecc93359	equivalent
CompactCells811To842.Box838.compactCandidate_not_isWeightedPerimeterMinimizer	7929f6359afe1fe704e86bdd1385def4891a3966161ae5f03371f1d821ba84bc	equivalent
CompactCells811To842.Box839.QInterval	21006b48c2caa5edbccd1fe0cfcfcfc0c1b2f529d9db0826eacfa969eb72fcb7	equivalent
CompactCells811To842.Box839.lambdaiInterval	5977aa0570ba37e5138383ba01cfee2ce51d9e6b7a2a5e2bc72f98c41343098b	equivalent

Declaration	Source hash	Audit relation
CompactCells811To842.Box839.compactCandidate_not_isWeightedPerimeterMinimizer	81a4efaaec4418ac782b063c0db488aa6ced32b7628bdba95ce086998861f6e0	equivalent
CompactCells811To842.Box840.QInterval	32498dab0e5c64a2c1b2b807e7dae2359f822eff10eb315a2fd303b66c33e5d4	equivalent
CompactCells811To842.Box840.lambdaInterval	57fecc89b502aad146708ec65cdf51594c91d59e59be89fe5b228a9efa67d	equivalent
CompactCells811To842.Box840.compactCandidate_not_isWeightedPerimeterMinimizer	83f096d882d904f48843ae481edd6995d99adc49e0aad4ad694f7f8530f6119a	equivalent
CompactCells811To842.Box841.QInterval	736292e3941b939d6b6bd47b95d74c1f69bad8350bcc00e193e6577a1e0d3162	equivalent
CompactCells811To842.Box841.lambdaInterval	1c486245a96e0c98bcbe07a366ef7da4f23e463909b941198574b795f9ae8b02	equivalent
CompactCells811To842.Box841.compactCandidate_not_isWeightedPerimeterMinimizer	246fc943e2b9d29a7644e089b649865225dc36039ed6574e04de3be8aedd38af	equivalent
CompactCells811To842.Box842.QInterval	b3f008f3abaa2bca5f94d9ca797dde89ca0510e98f0d50653d768ec0874c0133	equivalent
CompactCells811To842.Box842.lambdaInterval	354cb9f8f5f1ba8327133507f991070092c07553755f047d916a0a85775d9be3	equivalent
CompactCells811To842.Box842.compactCandidate_not_isWeightedPerimeterMinimizer	4e0a9eb8a79c706773531f73e775910995df0f43936f7863620a8b32e7a327d7	equivalent
CompactCells811To842.bboxes811To842_candidate_not_isWeightedPerimeterMinimizer	08441dcc5f6d56e615ab70aeb751ac7b1c0f12eca84e0828d45c03eb9e079	equivalent
CompactCells811To842.bboxes0To842_candidate_not_isWeightedPerimeterMinimizer	afa957a7194efca701a71a44e3242dff3f6dd069fa9f2518e01fc40db9ad7226	equivalent
CompactCells843To874.Box843.QInterval	73027f83fdfebba89e54647c089b5d667d1c23e5b9edb87cf9985c547b1fb1ff	equivalent
CompactCells843To874.Box843.lambdaInterval	35301ddde0ccddcbac03086c3656b0e345f78c711fac1d0b1ec0c5c07d819574	equivalent
CompactCells843To874.Box843.compactCandidate_not_isWeightedPerimeterMinimizer	eb0a9f1fa2e7125603cd4dbc5996f65f794c81b2cdfd2de18ffbac84438c9a41	equivalent
CompactCells843To874.Box844.QInterval	5d021b54f147b724be79fa3e12fc265839f61c39bb88520acf15bcc1a44bab39	equivalent
CompactCells843To874.Box844.lambdaInterval	9a0f7a542ac526a6f7b56bcbe41f8610d8062d081e3cf480e2ce56820a26db4f	equivalent
CompactCells843To874.Box844.compactCandidate_not_isWeightedPerimeterMinimizer	87da79dc340514139a936865393cb79b73a8a65bb9dde5dcec42a996338fca92	equivalent
CompactCells843To874.Box845.QInterval	5362f23b96823bcc7e978106159564b360b1c0fafbc6b0e0eb1d7c6fcbb88b6b	equivalent
CompactCells843To874.Box845.lambdaInterval	3d6bde973fac6cf9ee97b827cc0f4afbb21862f1bd93f7323cc312b097c9dca7	equivalent
CompactCells843To874.Box845.compactCandidate_not_isWeightedPerimeterMinimizer	28751f4f6c5c5920f436235348431b11dc3facfef9dd2f2b1fc8002e9215b3fb	equivalent
CompactCells843To874.Box846.QInterval	162571ecfea62d96c4d53b3f85f818635b21493bd1f42d4d20ebb0ffd34a80da	equivalent

Declaration	Source hash	Audit relation
CompactCells843To874.Box846.lambdaInterval	c08126240a209b1be3448b5f9db1de01e03db8521cf7063250bca3c2c19903fb	equivalent
CompactCells843To874.Box846.compactCandidate_not_isWeightedPerimeterMinimizer	aeed4e8b856a0ef3640ef8db520a138a537c84a9a8d3e52f5ce53221f3fba4ec	equivalent
CompactCells843To874.Box847.QInterval	8f9367847d0f03bb851567d13b72253dd04725ef7256c7839cb47b12c246d5b2	equivalent
CompactCells843To874.Box847.lambdaInterval	e5f3e483b83dadcf8825ffd71ff130509ff1731af71de6228b6d3b713351e6ec	equivalent
CompactCells843To874.Box847.compactCandidate_not_isWeightedPerimeterMinimizer	3033434fe1d969d329e0af343954caf099904244ad590308be9b09062a999252	equivalent
CompactCells843To874.Box848.QInterval	2c961de4e25aaea5117f45fb6a517198f80d9b985edc96c280af716ee3005632	equivalent
CompactCells843To874.Box848.lambdaInterval	8da1edc35d517d81abaf943c14aabb6ecc96630b152038058d1fd55fa967fced	equivalent
CompactCells843To874.Box848.compactCandidate_not_isWeightedPerimeterMinimizer	168e7b8aeb7d98fff804e57e886f54cbf00d7cee940de31cffec415c2ff5f6e5	equivalent
CompactCells843To874.Box849.QInterval	8e2fe3cd98552d6ca4fbd213ef741b4723579d74659bf604daaf9e5314b44168	equivalent
CompactCells843To874.Box849.lambdaInterval	c56a5caf760b159b7286689f790ddf32b1256da0d6bbc7ee5dadaa35d78e81d5	equivalent
CompactCells843To874.Box849.compactCandidate_not_isWeightedPerimeterMinimizer	048c1a3c51d39fe99fbabc0e069620a1acfb907f4abd19490d199c00093fc3e	equivalent
CompactCells843To874.Box850.QInterval	374cad997d7b0732a9e5620c4df7d63566f8626b72621982fd9c8b62bdf4598f	equivalent
CompactCells843To874.Box850.lambdaInterval	4652389aa81b5066cbd3ea0942f4d82759a855d67307b44fbad48d938d43b563	equivalent
CompactCells843To874.Box850.compactCandidate_not_isWeightedPerimeterMinimizer	98d0da05bb85dab7882409b4ae3d9011a986739d581062b53c36662102963423	equivalent
CompactCells843To874.Box851.QInterval	8df47ab4a3a975b5782e02d3e803283fe43644027709cd7702f6a45f79ecd472	equivalent
CompactCells843To874.Box851.lambdaInterval	d0a3812cf83fd09c2904aa9af7ba1e320ab27887fd34005f81e43e0cbbbe8d87	equivalent
CompactCells843To874.Box851.compactCandidate_not_isWeightedPerimeterMinimizer	7fe42261a33b018160d3b2208f4bd8be773dfedcb73e62a59d75172eb98a6664	equivalent
CompactCells843To874.Box852.QInterval	e81122202d5390490eee1cb51362e635507ec080391d9eb9ac67088a5d744221	equivalent
CompactCells843To874.Box852.lambdaInterval	77f9a01f0a028727767d5a321dd4576e686f3c8da2ae1b01b0321b74badb206e	equivalent
CompactCells843To874.Box852.compactCandidate_not_isWeightedPerimeterMinimizer	9c4f3d24ee2e4a5eda09130b35790aef1930b52b6f915ec798f103652b453946	equivalent
CompactCells843To874.Box853.QInterval	ef83bcc84baa40a471ef7717acc7cd0892a89ef1312f6b4a6a317a3ceb68a35	equivalent
CompactCells843To874.Box853.lambdaInterval	9fbc16b8f8e023457b231b9341910acb06beb71e3b22ecbf416fb97adb98e4a	equivalent

Declaration	Source hash	Audit relation
CompactCells843To874.Box853.compactCandidate_not_isWeightedPerimeterMinimizer	38a6f709743f1781b3edf72c2871d4d1cd2eaf7732507c23115408d10bc6a514	equivalent
CompactCells843To874.Box854.QInterval	f11c3298cb11ad83190186f9277bb5d94ea2e5ebc8db6440d9f07171a007571c	equivalent
CompactCells843To874.Box854.lambdaInterval	89cf548284f6c2dfb1fb8d8aab5d85de575cd53ad40bb4c2949817c4b68c0d6d	equivalent
CompactCells843To874.Box854.compactCandidate_not_isWeightedPerimeterMinimizer	f7af64907806429aae17b5e796e33bd9a44d1f45375a41b008e125000830693	equivalent
CompactCells843To874.Box855.QInterval	8ec971460f9d3c7f071b81f4fd6ecf607dd6a642933b863463b84963d175af79	equivalent
CompactCells843To874.Box855.lambdaInterval	3045e81a81385f4f2845480a2bdb391dd063eeb3eda476b6b8f0a03ac6a57a77	equivalent
CompactCells843To874.Box855.compactCandidate_not_isWeightedPerimeterMinimizer	27d7dfbacb70439a351e05fbf49f49870a93c4de71c0cdbfde2bcced4e83a32a	equivalent
CompactCells843To874.Box856.QInterval	a1536090a23ab0baff7c47bc55b270957d9c7f9362d2ec4fdfa64c26a61e1b4e	equivalent
CompactCells843To874.Box856.lambdaInterval	183edd1867da706986e8432cd7c94779e0737c4d770e90b5312a8fcf2f7c36df	equivalent
CompactCells843To874.Box856.compactCandidate_not_isWeightedPerimeterMinimizer	573ee2731053a2211d404713279ab48070739dece15f8543f6f69ff7fe94959b	equivalent
CompactCells843To874.Box857.QInterval	72d59c7a40723d269bfb48632dad612a4bba665aa980504358726f7120c570a	equivalent
CompactCells843To874.Box857.lambdaInterval	042162d72819b2c74d47f71d714d3af171adb8378c04db75b7d68f8dd60eb86	equivalent
CompactCells843To874.Box857.compactCandidate_not_isWeightedPerimeterMinimizer	566bdaae5f028be4613561d29c3fc99f57af2bccd6a63579c39697b5e256e1af	equivalent
CompactCells843To874.Box858.QInterval	8c66cc3841cd4e23e74fa74bf400cb66be2b497f00b389a5e62ee84dabff3679	equivalent
CompactCells843To874.Box858.lambdaInterval	d8b7c1213cdf30ec4feb9053098b51c250377bfb6a6dd8e900eb83a4273251ab	equivalent
CompactCells843To874.Box858.compactCandidate_not_isWeightedPerimeterMinimizer	d8d9daca079de400378d49ba247d694ab7e792e164359379b6808b9880145012	equivalent
CompactCells843To874.Box859.QInterval	613bf234d2a0ddbcb4b47bc20f7032639d8383c372222f1f1e8ca16e25f7fe7	equivalent
CompactCells843To874.Box859.lambdaInterval	0ccce00c8feee5a1d6eb188a8d01b8283fa580e0c5efbacdfcd753bf830a5056	equivalent
CompactCells843To874.Box859.compactCandidate_not_isWeightedPerimeterMinimizer	b88293c7f0e10dd18432778ff6ecb0bc866a37e280812970c24ee56b9e678a75	equivalent
CompactCells843To874.Box860.QInterval	60265da9e2fffb01c7d4f1679b0a8fc5948ed83ff4c7154c78a0bc05bb8d6f0aa	equivalent
CompactCells843To874.Box860.lambdaInterval	e62675a63c5177fd91d336f37d138b03acede20397a2baba7f347e5728ba7893	equivalent
CompactCells843To874.Box860.compactCandidate_not_isWeightedPerimeterMinimizer	919498d03b684dd355012db32a488547855a517647ffdf60b52bd7ea38a4794c	equivalent

Declaration	Source hash	Audit relation
CompactCells843To874.Box861.QInterval	eac1c80e433f7be3794b009a85dbb39c98ebd7aa230f94cab271ce65ccf9b4a8	equivalent
CompactCells843To874.Box861.lambdaInterval	3a193425c6a163dbcb7af2c5af47aff80b0ed68edeacc2f4033fab3732e70810	equivalent
CompactCells843To874.Box861.compactCandidate_not_isWeightedPerimeterMinimizer	4ad470a33b745362bcc293e070b9590659eb7d0aaaa7a7a30219aceb173fe5ff	equivalent
CompactCells843To874.Box862.QInterval	ddd64e2a193f5067fabb1e3ef5667ce0204032c81f3628c03e50ecaf7b9d04c4	equivalent
CompactCells843To874.Box862.lambdaInterval	8ac3a59f2f9ff9cae4a002cdc481ee5c68475bd9593c37f8bed4f55e570095e2	equivalent
CompactCells843To874.Box862.compactCandidate_not_isWeightedPerimeterMinimizer	dbf893434247acd0c19cfa9ff29eff203e0b5bc17842325de70cb344791d4fbfe	equivalent
CompactCells843To874.Box863.QInterval	414d63878ffdcc9fc90c17eb74659f7712577855ddef2630d8f48b560540a10d	equivalent
CompactCells843To874.Box863.lambdaInterval	b5008a27e9322c55a1ad66014d4ac770f5e04079be34779cf82cc3630f87d3a6	equivalent
CompactCells843To874.Box863.compactCandidate_not_isWeightedPerimeterMinimizer	c550170a34e325f71192a3cdbcdd0f50f48adc48395f3e996f6e9a41ee8c8db4	equivalent
CompactCells843To874.Box864.QInterval	460604015958e415f0256242270bbea46b90e9901d295c89ce0ca60643c53ca	equivalent
CompactCells843To874.Box864.lambdaInterval	8f60f3b1e4f10926c46ad6e6e5b1947c2573ee750a09318ff829d3d7a466dcd9	equivalent
CompactCells843To874.Box864.compactCandidate_not_isWeightedPerimeterMinimizer	e6e3e398865daf5b5f5f9991ba23c8ea63ce1a50eee1702b6e1a6654a26dfbc3	equivalent
CompactCells843To874.Box865.QInterval	b3a8d8355645ebf1acc070afad3436bac15084bfa5eef391c4db3ab525559e86	equivalent
CompactCells843To874.Box865.lambdaInterval	2e96b1af74fc69a4e100ec0da4cd20c414aa13a0c9472115f21cf261b4fa49c8	equivalent
CompactCells843To874.Box865.compactCandidate_not_isWeightedPerimeterMinimizer	984164a38042c515b260f1707c7b80053e158554e516b060aff41602cab0c11	equivalent
CompactCells843To874.Box866.QInterval	9104a4fe453bceca980db2fbfce3fdff2213ca02126ab61286cc881b37c22ed9	equivalent
CompactCells843To874.Box866.lambdaInterval	d55c10680d6eb36857440531519f2371bfb3f344cc4d1d5bb55c4162d7be5d53	equivalent
CompactCells843To874.Box866.compactCandidate_not_isWeightedPerimeterMinimizer	d597403ab4530fa1099b85b7044df86413c139626deb9af2096eab7c2792c55d	equivalent
CompactCells843To874.Box867.QInterval	03d16d557bed08c2294215615f61f6e4183e4407ee47fa460438187f555e078f	equivalent
CompactCells843To874.Box867.lambdaInterval	7db1c9a474d893e49c14c0b69c71dd61a237ea756db9acb50c77294a503115e9	equivalent
CompactCells843To874.Box867.compactCandidate_not_isWeightedPerimeterMinimizer	48e8e85d85e3a220b467fb6f4a150045406fac286cdbeea8ba458edd89fc90b6	equivalent
CompactCells843To874.Box868.QInterval	da1526d03f6cb5ce84c8e78e222a01af720d3353abe3f3f83bb44d43e467706e	equivalent
CompactCells843To874.Box868.lambdaInterval	cfbc12dffff3dc84f5ba29a82f590127309836e4b27a11eace235c27035642a20	equivalent

Declaration	Source hash	Audit relation
CompactCells843To874.Box868.compactCandidate_not_isWeightedPerimeterMinimizer	2430987e7d793fe196463d8854249a79c704eb56248e7e0296eb62a9f46a0db0	equivalent
CompactCells843To874.Box869.QInterval	f67af56a7b07097dcc5272e0c8a00769ddd675413d855302dc78dc0e54789582	equivalent
CompactCells843To874.Box869.lambdaInterval	9210a6d207225116a30267d388886dbc9f92d82ca6e5b094744c71cc545edec0	equivalent
CompactCells843To874.Box869.compactCandidate_not_isWeightedPerimeterMinimizer	a8cd50c892da095b330c50030cb7a87afd43444f2e2543bd3a91a30e706b322c	equivalent
CompactCells843To874.Box870.QInterval	eb18a1b8a7dbe96b858a902e57ecd7747b4999160fd582bc86f2148d4940421a	equivalent
CompactCells843To874.Box870.lambdaInterval	e6c24f28d59b4b0e5af8c3f5a75a8b60a5773cd6de2d9b9430fccf260c77599b	equivalent
CompactCells843To874.Box870.compactCandidate_not_isWeightedPerimeterMinimizer	fd5fbe4784311c8259a8045a24673993f37e81cf5865d28b01dceb1023aee44f	equivalent
CompactCells843To874.Box871.QInterval	f6a73ca1dbf159a4839434fb241450c2279e3ef11bbc9faf7950d8660831f17	equivalent
CompactCells843To874.Box871.lambdaInterval	2fbdeca582129bcfdb848c6b1ffbb202a5a641a932dfcc0a3e8ebf52ae235f5f	equivalent
CompactCells843To874.Box871.compactCandidate_not_isWeightedPerimeterMinimizer	8e806ce9ce6affabe24569fcd1cb1313b12108ea41a878ce26e8c24fc83b94b	equivalent
CompactCells843To874.Box872.QInterval	75667d110c5d939bffa603dc3b3f7e11e5578b4e4229fc639e78413589ccf847	equivalent
CompactCells843To874.Box872.lambdaInterval	e444069e1afbdbdd2ff1d5935a10900bb80a5caf822339aface7b74d89c27dfe	equivalent
CompactCells843To874.Box872.compactCandidate_not_isWeightedPerimeterMinimizer	0829a54a5305f4bd81f81da8f0327e26bd21bcd6e7177cb6efb973efbf9a6029	equivalent
CompactCells843To874.Box873.QInterval	27097b7e2030bb3ceb590af4c8392a0d995155816303ab07cbc1a5174c0ac136	equivalent
CompactCells843To874.Box873.lambdaInterval	329e4529c65f2a1e0fff704233295e59823083a6d81753a94028e509cffa9b2d2	equivalent
CompactCells843To874.Box873.compactCandidate_not_isWeightedPerimeterMinimizer	cc3c9b416c6f1aa2571820be50239bc018413f22e1ddcab6dbdabaa101ff34e2	equivalent
CompactCells843To874.Box874.QInterval	a07764ba8bf465713420556c323aa1191b37a8001de195890034df32e2cbacda	equivalent
CompactCells843To874.Box874.lambdaInterval	a87c53c2cbdcbd1d0b6d89390a4161d8db7dd0d578485bfbe5f303471762be3a	equivalent
CompactCells843To874.Box874.compactCandidate_not_isWeightedPerimeterMinimizer	a289d590a8d90628e83629a1d79f9043764897936f1f0b7294c1d1f7846304f7	equivalent
CompactCells843To874.bboxes843To874_candidate_not_isWeightedPerimeterMinimizer	643b0dd434b1a9bf8b054c51df5bf687f08c1c45fd5ed0bb6827fa9add52fcd1	equivalent
CompactCells843To874.bboxes0To874_candidate_not_isWeightedPerimeterMinimizer	4ba78cbde422e099596af8f9c96fc3b3f64fee508a002cd547513696dd1abdef	equivalent
CompactCells875To906.Box875.QInterval	61c6ff96c582d732d3c307e6bb0804f4cef92dcde6e9dc731bf3b66ae0e58132	equivalent

Declaration	Source hash	Audit relation
CompactCells875To906.Box875.lambdaiInterval	0cbbcc7cc12681b824679eeddf65500cb82eaae48cad608f3749530e8dca4ab5	equivalent
CompactCells875To906.Box875.compactCandidate_not_isWeightedPerimeterMinimizer	c61266d58af7ddeb408148948ce67df7d9ff633497009837f657946a4404eef8	equivalent
CompactCells875To906.Box876.QInterval	992be4bb521792a19766fe73859fb98876a2ed7664960608720924980f222221	equivalent
CompactCells875To906.Box876.lambdaiInterval	3cf811c71b714a5e309d35a19fab9e5554ca784ae199b390667f517ea5beb99e	equivalent
CompactCells875To906.Box876.compactCandidate_not_isWeightedPerimeterMinimizer	f19e18baa9047357569c0b2c18e747f5b03a49d979328926db12042b5add5f3a	equivalent
CompactCells875To906.Box877.QInterval	9ee9417d8905a00bf42decf40964f0d57a1fd16939e920416543e96bf050090a	equivalent
CompactCells875To906.Box877.lambdaiInterval	ca27777ab21a1109d1092e5b916378d9597775383b5a4ff51aee2e46190d2943	equivalent
CompactCells875To906.Box877.compactCandidate_not_isWeightedPerimeterMinimizer	aabccdaf77ec98564e5aa9ccd28f754990c43f315362409d8109419a1613a2a2	equivalent
CompactCells875To906.Box878.QInterval	5c52636d74c7960c7bd28dd7f655016ce66280bebc3b079f89e4b49691e9cde3	equivalent
CompactCells875To906.Box878.lambdaiInterval	a1974f719f4aea7ec2ef7aaea8d30e1469b5074983187c0909ae6a57554525bf	equivalent
CompactCells875To906.Box878.compactCandidate_not_isWeightedPerimeterMinimizer	f1cf25c7f1b8740edca5ac6d33946df53da881d9350ba79980b5d4905417aa62	equivalent
CompactCells875To906.Box879.QInterval	146c6d78201775c262a92939686f776d15e57f92c179f73412e6267f1d303421	equivalent
CompactCells875To906.Box879.lambdaiInterval	bca1e8654ad870f63d2ff179201f9518c5434571d0bfe577afe986d4cd08a5e	equivalent
CompactCells875To906.Box879.compactCandidate_not_isWeightedPerimeterMinimizer	a58292b9492ff66463ebba9c2b7e9962ac07cd7442157b98ecb99c254f3cbe65	equivalent
CompactCells875To906.Box880.QInterval	54cf6c40faa21039d42914366221b9b3afcadb87c8f265a4fd0afbfb2c2744b8e	equivalent
CompactCells875To906.Box880.lambdaiInterval	0990a5279d278fb80c3758913cffce95aca05c07cfd4901a86c7f5c4cb0756c4	equivalent
CompactCells875To906.Box880.compactCandidate_not_isWeightedPerimeterMinimizer	29a0c02fbaffb7e30be8f12c97f36f2faf48535d4ed0b3df7cae36c160fa75b6	equivalent
CompactCells875To906.Box881.QInterval	4092d89ed4f6ded9c4a36e14dd4af233129040ef9856e98f9d90983d6436d52e	equivalent
CompactCells875To906.Box881.lambdaiInterval	36614dffeff3d157db4ac74669337ad9001e7108038194797926d21e75db79eb	equivalent
CompactCells875To906.Box881.compactCandidate_not_isWeightedPerimeterMinimizer	543f2e9b36c92dea612700e8911099736e2e74247478487230d20910188203c8	equivalent
CompactCells875To906.Box882.QInterval	32bfcb803c9011fdafce2210d38b6edf539b0f9025b2e6a14b270c49f3df6809	equivalent
CompactCells875To906.Box882.lambdaiInterval	5da3e05732f46ec608d2847e2bef57e00db98bc3a4ff8e814440dd5e35844bd6	equivalent

Declaration	Source hash	Audit relation
CompactCells875To906.Box882.compactCandidate_not_isWeightedPerimeterMinimizer	42479505985eaa91a09bf12e176763289402802fa987417a2de25314a391f63b	equivalent
CompactCells875To906.Box883.QInterval	ae6fce74a8f3894c9a8c98d5cc6f8e5330be5eb5384a46af43a0b52f52851816	equivalent
CompactCells875To906.Box883.lambdaInterval	0626dfb23454c4f473a5884ca6e1c9e44524929ab80e4a6ea63f9cf65f252e1b	equivalent
CompactCells875To906.Box883.compactCandidate_not_isWeightedPerimeterMinimizer	cc8668aac5afa9430687063063d42c32938fb5a67524649a30ef40ee4f5647c3	equivalent
CompactCells875To906.Box884.QInterval	5d1483f34bc38642ade0faea324a90bb8c9c5f9f96ad066234a36b52ca7adf3b	equivalent
CompactCells875To906.Box884.lambdaInterval	5b7014e1995b175ad9f95a87b915a6d759e173b425ba2b4a050bf207627816ab	equivalent
CompactCells875To906.Box884.compactCandidate_not_isWeightedPerimeterMinimizer	b9a015ca20c4206761f960c6e50133d44de5d6d2ab66624b3e972fe7d0c8c72c	equivalent
CompactCells875To906.Box885.QInterval	221711f1878608dab2fd56546d047c75a562a7bcc08271c4599befd51a006204	equivalent
CompactCells875To906.Box885.lambdaInterval	a3fe4eb6cfa5ac7494e6ef49df3187d95ef3b8a5cafba35456501e316ed822c2	equivalent
CompactCells875To906.Box885.compactCandidate_not_isWeightedPerimeterMinimizer	11c8d05c0d035f7330c580fa7582b87b0345fb2f8c0df49f1e2473757e5bb752	equivalent
CompactCells875To906.Box886.QInterval	00e19eb192b202ef60b0aa3bf9f554193ad960a2bd8b844e9cd85407a6ba3ea2	equivalent
CompactCells875To906.Box886.lambdaInterval	566702cf2214da88868f8761e5f0159e8680ed100aa392c52d8fc54cbfa02bff	equivalent
CompactCells875To906.Box886.compactCandidate_not_isWeightedPerimeterMinimizer	f37f150451c5a663ad4c7470c38f2aea1a04c9852cb2dff0ec97ea0eb2186a5a	equivalent
CompactCells875To906.Box887.QInterval	fcfa9708c0d582fa75ec01994317ba8b9c341e5f5023ed77d8c5d3458c62ab3e	equivalent
CompactCells875To906.Box887.lambdaInterval	b5473c109a54ff25cb39e3be0d5eee5bb451ab00541d8a4c2b8737bc640330f1	equivalent
CompactCells875To906.Box887.compactCandidate_not_isWeightedPerimeterMinimizer	eb56d5fc69859c2c2dcc3df434f1ad8de8a7e8c1b73b236be9ed9ac329e383fc	equivalent
CompactCells875To906.Box888.QInterval	8dd8d8eea7c3156e194bac4eaae7cf8abce7df3740ddf614f015407571bc57c4	equivalent
CompactCells875To906.Box888.lambdaInterval	e02ee195a437903123bbf74c75c74d6b9ade79681e2dac87aff78b7f6b97fca3	equivalent
CompactCells875To906.Box888.compactCandidate_not_isWeightedPerimeterMinimizer	86bffb6ded964592310e47f43effe09bc50d1fce22d65c3f861fbb1cf583375c	equivalent
CompactCells875To906.Box889.QInterval	cd227fd60334a6d6837e470f790e7d0d381fb324ce39cf241158b709e1aa8eef	equivalent
CompactCells875To906.Box889.lambdaInterval	c2cca6d9c95127c8c09be96eb2fc636a16afcb84846480d37ba9e258dff71221	equivalent
CompactCells875To906.Box889.compactCandidate_not_isWeightedPerimeterMinimizer	b1df2cff9bc21ec0443aaff5974d84deb2e6eed390d0623d5c690eae9f772f0c	equivalent

Declaration	Source hash	Audit relation
CompactCells875To906.Box890.QInterval	cc1679b2b9d9c9e4cf3b0eb5f0f0be42974689b1573587c97ba12d3d1912e381	equivalent
CompactCells875To906.Box890.lambdaiInterval	f1f6e900c3c67f334b08f764dcb82d95b1e1ac2413de7897f15c2e86c14ef6ca	equivalent
CompactCells875To906.Box890.compactCandidate_not_isWeightedPerimeterMinimizer	d06239b55fc7ce0438f6ac04e67352edda35417311775f505fb2dae170c5191d	equivalent
CompactCells875To906.Box891.QInterval	e46a34ccc17f8c4a2a41ee3e7279322247df849d2da5730aece78f6704837254	equivalent
CompactCells875To906.Box891.lambdaiInterval	bfb44f25c4e9ec9824c4349397f9c3ae8eb44266cbf556c320760aa9d8b36310	equivalent
CompactCells875To906.Box891.compactCandidate_not_isWeightedPerimeterMinimizer	244e2bc05a5c7ec1a19d54d7030079f341c0f6c0c0fcf9fc88fbca3eef02b1c1	equivalent
CompactCells875To906.Box892.QInterval	9da55871861c7fd4c96ce65d84ceaf1e80b040c6164198955924a6d983adf7b9	equivalent
CompactCells875To906.Box892.lambdaiInterval	0f6a44cea1516075dda95e913c948f778554ba47f61f7acc921a1b811b6cc3c0	equivalent
CompactCells875To906.Box892.compactCandidate_not_isWeightedPerimeterMinimizer	60c3d1ccf848e28562540f2285211506be161539e9dad1603e310002d2c3cd6c	equivalent
CompactCells875To906.Box893.QInterval	bbdc6c608828b513e0bd34aa5e75c60c40e82e29ba28a2cc4e4803e050050911	equivalent
CompactCells875To906.Box893.lambdaiInterval	a9265193b41f429f2e5c197c73bcf8a0e2a11119bfff9f074473f52d172551f18	equivalent
CompactCells875To906.Box893.compactCandidate_not_isWeightedPerimeterMinimizer	4e3a8c536855e39dc1a936d95fbd44c9d6d1b154ccc4f024973483e3ce9d2abd	equivalent
CompactCells875To906.Box894.QInterval	184ff302210ce09dcfa2d823ca1310cb9bf48cafe224057534b75964ae00a0ad	equivalent
CompactCells875To906.Box894.lambdaiInterval	e9791fee73eab42a648d2babd68a7ea43e51e077c76c123835b1e4af64a0842d	equivalent
CompactCells875To906.Box894.compactCandidate_not_isWeightedPerimeterMinimizer	889602a000a98c2849fbd6a107736dd92e59e77a557bc652dfd9eef760098c1	equivalent
CompactCells875To906.Box895.QInterval	648545b94d9746b69f3e43028c38347958d202b7a068a082ba0a43e2808891f7	equivalent
CompactCells875To906.Box895.lambdaiInterval	f822dd4823d0a40be845217c78da87117dc7199e50d3f8196c85adbfa49b2796	equivalent
CompactCells875To906.Box895.compactCandidate_not_isWeightedPerimeterMinimizer	6ffd357955028b263fb83733f2ec4b0ca60f8c2ba4aa61f183b5515f40db97c8	equivalent
CompactCells875To906.Box896.QInterval	c575a50d5167d0ff064184954fc07b969511af1031298c88066db6bc9128fbbd	equivalent
CompactCells875To906.Box896.lambdaiInterval	eb6acb2494c0c964c990daf62cd1a3730e1db733cf691de3a440ca87a92e5b61	equivalent
CompactCells875To906.Box896.compactCandidate_not_isWeightedPerimeterMinimizer	58727b123979570e1b4a927fa4a92641e3663c74fdc3104407eaf3c7fc83d596	equivalent
CompactCells875To906.Box897.QInterval	f9792561e381a9746b66904fa4278924d6028e89481cb0dc663178a6080edd7a	equivalent
CompactCells875To906.Box897.lambdaiInterval	dbaa4dfe9ac49ebaa1c7438532a769083e70b5fced1050a5eebf614a262c2177	equivalent

Declaration	Source hash	Audit relation
CompactCells875To906.Box897.compactCandidate_not_isWeightedPerimeterMinimizer	0f13619b653532cd58dac68563aac6b0e101eddeed570779b33a7f59cdb4cc27	equivalent
CompactCells875To906.Box898.QInterval	3d41c2823fc955dc122c61881baf2b0cc40c2d7f7eabe3f279bc2b068d6ec184	equivalent
CompactCells875To906.Box898.lambdaInterval	cf28d13a936f3aef74cdaf461d06f1746178568c798e52eafc0bb864009caa68	equivalent
CompactCells875To906.Box898.compactCandidate_not_isWeightedPerimeterMinimizer	cf1c655d06c5f50fd090abd3dc1ee54d716a08086c2cb7536d7d9a4d53ae2ce0	equivalent
CompactCells875To906.Box899.QInterval	867f63fb9ec8f07214cdc8cbc31c24d94efb543e7de72bcf683c69ad87084b24	equivalent
CompactCells875To906.Box899.lambdaInterval	3c2eaec2c353de79d4a8e5c4a7c34621bab803d690d7d66d3cec3d77fe3760e	equivalent
CompactCells875To906.Box899.compactCandidate_not_isWeightedPerimeterMinimizer	d8e783b5012de085047b5193d76abd65b35dc2bc0c00ee768077688d635e7f2a	equivalent
CompactCells875To906.Box900.QInterval	7f1b220ff74eedace58fbbe74eee6f76f65ff3fe42a33b3cd5072e5e6dad731a	equivalent
CompactCells875To906.Box900.lambdaInterval	a0ac2d9448e3af4e557e326fd7e59a050c88d060b10a55d2878aa473401a4295	equivalent
CompactCells875To906.Box900.compactCandidate_not_isWeightedPerimeterMinimizer	a9d3b0d95e34d46b171ce1d36581e196498eca2866eea83f322cba63657c2cc0	equivalent
CompactCells875To906.Box901.QInterval	d6ea8a7a57d4544b9920394604e23499a84ff3e789dcc6da56d330d6ba52a5cf	equivalent
CompactCells875To906.Box901.lambdaInterval	ceaf34f791b7ac79827fd72f7309bd1012be6bb7b31631c66d5ec00d29355ab7	equivalent
CompactCells875To906.Box901.compactCandidate_not_isWeightedPerimeterMinimizer	11aa4f12e4c08ab11f1cb11af216b5152cab64320be57c755f36b4cf6a2bf161	equivalent
CompactCells875To906.Box902.QInterval	98ce15654f44c34842c5e828c3a6989bf4a5184d081f6d4993b3477a4555ac5c	equivalent
CompactCells875To906.Box902.lambdaInterval	be28895e7a4db9014907838c1197f64bf06e54531c06536d2513911ab0a9dc57	equivalent
CompactCells875To906.Box902.compactCandidate_not_isWeightedPerimeterMinimizer	9baea41df623769ab9ee0d10a64c823ee689aa9fd361d44b3cf99f54ce2df4ca	equivalent
CompactCells875To906.Box903.QInterval	f7395c0b739dfdc8b2e66301664bff6940e0a2c2a66460631bd15f4bf14cc14e	equivalent
CompactCells875To906.Box903.lambdaInterval	2f29a98f0d11f7093cf7176e1c62970ab7fa0bc4ce0b4511449bb3f2f80be72b	equivalent
CompactCells875To906.Box903.compactCandidate_not_isWeightedPerimeterMinimizer	e7986187dd8d2ee7299973752ae23d1b8773f3cc25fa602b726be4fd1d627dc1	equivalent
CompactCells875To906.Box904.QInterval	0bf7784c122a95fe45d5811cdd4692a40bbebfc1f5522193a3a0a713432dfe4	equivalent
CompactCells875To906.Box904.lambdaInterval	a86708692342190554c97207b5ac208e5c0e93f253ac3a09c14fcdb8aea824b4	equivalent
CompactCells875To906.Box904.compactCandidate_not_isWeightedPerimeterMinimizer	3de40ad6c9459fcd8927a9413124db5f10a7f7e31a549a5f920a925ee94d7ae4	equivalent

Declaration	Source hash	Audit relation
CompactCells875To906.Box905.QInterval	ed68c944831f1deb6c75a4992ce58a21a2a30e22e4e4a9f0c1e203a94972737b	equivalent
CompactCells875To906.Box905.lambdaiInterval	9dd4ce0a5755a2afb5883d5105fc78368b67cb0a89a81af7c58b731271f16981	equivalent
CompactCells875To906.Box905.compactCandidate_not_isWeightedPerimeterMinimizer	72f54e5d00550a1e1346d34d868c131c98760c1584d8f76416801deb07f692ab	equivalent
CompactCells875To906.Box906.QInterval	f68c1a240795ba7ce53d7acc8f756ac8abe06ac9574bc265791dea359aa02e67	equivalent
CompactCells875To906.Box906.lambdaiInterval	e2941732cf39d1cf47df5f68c113592d7e32525f845511d90852dce5bca50442	equivalent
CompactCells875To906.Box906.compactCandidate_not_isWeightedPerimeterMinimizer	af59d0790dca28dda64f3c33ec77fbe507e6ed8439d4ed44c1182b7e1f789e15	equivalent
CompactCells875To906.bboxes875To906_candidate_not_isWeightedPerimeterMinimizer	3a00d7bf7a2f4a36f1d6e38f3f47398e6f6f6ecbabde4b4f590cd28db1d4ccfb	equivalent
CompactCells875To906.bboxes0To906_candidate_not_isWeightedPerimeterMinimizer	5753c755fc4c59462748908ef1312df1292b55cb8422639d41c8061a02975611	equivalent
CompactCells907To938.Box907.QInterval	3679bb304174f4ed9e94ec630e96457fafdc0e6c0b79501abda961342cfba6dc	equivalent
CompactCells907To938.Box907.lambdaiInterval	813925f120d54e95296835166b9f43eec3e25688e62b7dcc81e2878972053c36	equivalent
CompactCells907To938.Box907.compactCandidate_not_isWeightedPerimeterMinimizer	116ff4f601c39fe9c522594bfbe768f7eabc04e671981553e4b27f5ca98cc6d7	equivalent
CompactCells907To938.Box908.QInterval	61bd60855127f650786a2e886a492cbc06102fed6b2d0d5decd3e2d893e27d25	equivalent
CompactCells907To938.Box908.lambdaiInterval	f7c4847bff21c478b194a0813714fde7ddf965a57c88fba1f00f4b99f119eb12	equivalent
CompactCells907To938.Box908.compactCandidate_not_isWeightedPerimeterMinimizer	6e7a1fa2f46ed0d314668de67a8ed77e7c9e4b10d6bfabc883c80ce3a1fa4ebc	equivalent
CompactCells907To938.Box909.QInterval	82342a459fcc62120c7d52d36f4ac170e770c5eef9aaaf24d62374ef7ecd3655	equivalent
CompactCells907To938.Box909.lambdaiInterval	dd829ea128471f8eff5be16e6e5830a35f214158bfb6bedb3d6681668015d7a6	equivalent
CompactCells907To938.Box909.compactCandidate_not_isWeightedPerimeterMinimizer	ebbb308bd5bef97b4be5c51b19b7770e21460f0f9876e52b365944a1a79f21d1	equivalent
CompactCells907To938.Box910.QInterval	b1eb5a94d83e542e8f68623a7ffb67907683ff1c68e2a0456ef3049fc17953d9	equivalent
CompactCells907To938.Box910.lambdaiInterval	bb28fc4fa0ce1dd737cc0daed7fc2b09453abc8753b92f14d0642ee5095759f6	equivalent
CompactCells907To938.Box910.compactCandidate_not_isWeightedPerimeterMinimizer	6c627f6b4f522d2d4bb9bb6d25dd04c5490eedaf5ac441a09a5005b8e190cb97	equivalent
CompactCells907To938.Box911.QInterval	28e1c65bfc018b032776695d6c2bdc084e9b3032b69570dab242b87acd8c93ae	equivalent
CompactCells907To938.Box911.lambdaiInterval	a94b12952234c8c3cafda67063bc4fff014fb33e63e1384820c94caf22af7979	equivalent

Declaration	Source hash	Audit relation
CompactCells907To938.Box911.compactCandidate_not_isWeightedPerimeterMinimizer	ea0ae78cd9f60b445e1d7681619c86642d3a83a7922d0b394d2eb75d92966113	equivalent
CompactCells907To938.Box912.QInterval	5eb490200acf3c2ecd5642ec75a573bcddf c4ab1b94374554617bdd48d3ce117	equivalent
CompactCells907To938.Box912.lambdaInterval	6774ad39cc33f3304ce5115bcdcd53c00e4b2ebaa5ecd0038ad63251d0cf0797	equivalent
CompactCells907To938.Box912.compactCandidate_not_isWeightedPerimeterMinimizer	daa50fcd4623176416268bb243e29ecbf2bc0ba929706d4d14276e970299c261	equivalent
CompactCells907To938.Box913.QInterval	a2210a61e8e50ba36f16725fb97525ca400c3a0ddf4d5db91e3f5dd84a6f4115	equivalent
CompactCells907To938.Box913.lambdaInterval	989e121361a51dfc5c67b5f558ba1d699bc0d5c5893ec9ccd003d2a2b21279cc	equivalent
CompactCells907To938.Box913.compactCandidate_not_isWeightedPerimeterMinimizer	75f24a3e71772b6ef9ea93bbc91b2eab92f5c0ab109be9aae5e3a942040c0a11	equivalent
CompactCells907To938.Box914.QInterval	4d754fed995a007ff5c6c118e87e271483b780dea48942a60a2ba78e96057e71	equivalent
CompactCells907To938.Box914.lambdaInterval	2ef5a809210112a1b1985b18a7734a7d3c00d59811b0bf3775170fc47960b8cc	equivalent
CompactCells907To938.Box914.compactCandidate_not_isWeightedPerimeterMinimizer	6ce92252dac73e6863661b9e273e3900e07bbd2a96f47912cb8ea421f4551efd	equivalent
CompactCells907To938.Box915.QInterval	b81c990a1b1b958ec26b0878711986d579a41c30bc0d40893c659d00959917ef	equivalent
CompactCells907To938.Box915.lambdaInterval	d25d2e8662953d47bfa5efd9959dd0d8673d0c3db71f2bd599bcd7d2d4041a02	equivalent
CompactCells907To938.Box915.compactCandidate_not_isWeightedPerimeterMinimizer	780030b52affe6e5c224001465cd448631aab938d24334ae4fce8a3dade057b9	equivalent
CompactCells907To938.Box916.QInterval	6c0b0d790d1615b88bc6615fc10d2557a76f20db5f1485a8f5115ccc05b8d2ff	equivalent
CompactCells907To938.Box916.lambdaInterval	a560f0cd8e79514d13e25043a3cc6c90beebe35794db7d8652a2b6b940e469d2	equivalent
CompactCells907To938.Box916.compactCandidate_not_isWeightedPerimeterMinimizer	9601b0df5b9250e82190a8b67de6675652ea2b89cf23b1933ed56f670dfb18e5	equivalent
CompactCells907To938.Box917.QInterval	130126da4c6b3762c5ec3f67d711b2cca05484fc01e16b5c5f4d7bc4e714a1b1	equivalent
CompactCells907To938.Box917.lambdaInterval	7d4423cfd7f25d3aa57aa0fbf840b45aa0ea161e335a6432d106bc4975b4daa0	equivalent
CompactCells907To938.Box917.compactCandidate_not_isWeightedPerimeterMinimizer	2f545d5c024977287313a45bf633688eb1af4de82ce1f59e599b413529932909	equivalent
CompactCells907To938.Box918.QInterval	1cb93746b358933a22905cd0ac5c142b482bd3b5d967cc82cddb8cc36e0d414f	equivalent
CompactCells907To938.Box918.lambdaInterval	1225d67f6e362caa80e35fade2302bab1835f0db9b9084fdfbfd30154a44b14c2	equivalent
CompactCells907To938.Box918.compactCandidate_not_isWeightedPerimeterMinimizer	e9fa38b292e334aed1dd4a0774c563dc03a40cc711501441d0353542d38cda76	equivalent

Declaration	Source hash	Audit relation
CompactCells907To938.Box919.QInterval	18a342079ef688034a8c36631a484dd1fd0b6e5184e2852c9a841b8a2321e018	equivalent
CompactCells907To938.Box919.lambdaInterval	9ccb4a9e3aa97096188be9ba4979fc8adc0f308022d6d74e62965425e7032333	equivalent
CompactCells907To938.Box919.compactCandidate_not_isWeightedPerimeterMinimizer	78d1be5e6342ba9f327b440a37d60a9edd2e667dd710dd0f482d94161fb30597	equivalent
CompactCells907To938.Box920.QInterval	9d9c54c20e5787f98290324a8bee30407ee72a30428145f60e4deb5a22083bbd	equivalent
CompactCells907To938.Box920.lambdaInterval	9ee0e163e77a6c6dc2e67b28cf88b5c3c80336a25d7139ab087a938cfe2043db	equivalent
CompactCells907To938.Box920.compactCandidate_not_isWeightedPerimeterMinimizer	b56742e94e342150bf1d3308489c86870c53f5940b2784d9d43f6c34ccd480a4	equivalent
CompactCells907To938.Box921.QInterval	de5e0cac35d9abd80f76ce63b20930dc03e1517ae547dbf76c4c1e85a74ed471	equivalent
CompactCells907To938.Box921.lambdaInterval	35e5ff8e01c46158cea644aa63ba3c1eac20d27ab174786c3945ed5ca25646a5	equivalent
CompactCells907To938.Box921.compactCandidate_not_isWeightedPerimeterMinimizer	53105efeb550db993f54dc2b717ae27707a020971d673a8d4a1676b47368030d	equivalent
CompactCells907To938.Box922.QInterval	3f9d40fa88344177d0baf6ab75c4f5641b71327690038c9edee37479289d2aea	equivalent
CompactCells907To938.Box922.lambdaInterval	1104361ca62d10b43989b8aea491682be83de3ddd313fc03c8f2635ae0120037	equivalent
CompactCells907To938.Box922.compactCandidate_not_isWeightedPerimeterMinimizer	752e1a4b435b04352fb7019e543247961454f3f7cc566a8c8a9b8056965da291	equivalent
CompactCells907To938.Box923.QInterval	3fc8bafb36805b6f6b286a373d959f198335793a74265f0d2b5ff11502795956	equivalent
CompactCells907To938.Box923.lambdaInterval	affb217374d14336871903575c8cccc0c5d7e11722e45eb605741180f83624c8	equivalent
CompactCells907To938.Box923.compactCandidate_not_isWeightedPerimeterMinimizer	8d3ebf12365f75fbf4ad0fd2ecdb7a8387737dde9cd4994b3c94d1684cc9505c	equivalent
CompactCells907To938.Box924.QInterval	9ef70b86343856fd12fc542227227051f9c1f8178e5befe37ca2030eba0cfd34	equivalent
CompactCells907To938.Box924.lambdaInterval	b51eaf2c1315da6f03e9a9902937ff5b64927765c5d165b36daf735213f7a8e1	equivalent
CompactCells907To938.Box924.compactCandidate_not_isWeightedPerimeterMinimizer	d4fac9fd266d4a64737ab6eb5f7f961356afb0f1f86e2f82c6dc1e120777d157	equivalent
CompactCells907To938.Box925.QInterval	29085f80f90070ba2942dcf7b5de0f1ce1ddc012af80f2b07618f87394d175fb	equivalent
CompactCells907To938.Box925.lambdaInterval	ce69963b6806baed68f48ee50e8b2e64e5985e4754c010a7aff0a666586cad4e	equivalent
CompactCells907To938.Box925.compactCandidate_not_isWeightedPerimeterMinimizer	500c066a82aa1f962d6d4d20b3403ab1aa03c30addf15a07f3e258acf982443f	equivalent
CompactCells907To938.Box926.QInterval	4d85cf36091f22f023aad8cc5b3b83b5d0db18280a65d329594424d03f583cd8	equivalent
CompactCells907To938.Box926.lambdaInterval	6d82c399416ab3c010288f5556ef4ca62735e3d0d743bc3848fef334b3a28c22	equivalent

Declaration	Source hash	Audit relation
CompactCells907To938.Box926.compactCandidate_not_isWeightedPerimeterMinimizer	de6e520ddf4bf9bead76e30871b86d29be2485b65519ce9fd71510b52f67e734	equivalent
CompactCells907To938.Box927.QInterval	b26ba705e5f90a3074680b2e9212680751c6dcb66eb7d5197e03d3cf55cb1ff2	equivalent
CompactCells907To938.Box927.lambdaInterval	ae2747400b701d0bd4a0fd990a368b9f6c982c7781f9eaaeff018e23f57ea7a	equivalent
CompactCells907To938.Box927.compactCandidate_not_isWeightedPerimeterMinimizer	d1351647ff01fb53fb2422ecb7156fbac55ec0c943391319af371ce4077a43a6	equivalent
CompactCells907To938.Box928.QInterval	912c22769481d66c0f8bd6aede9153dccb618018474b46cb5028896c934f4b49	equivalent
CompactCells907To938.Box928.lambdaInterval	39cec87ebc133e92b70b103d22dbde64609f7d06ef3cd229b2f5b6657763968a	equivalent
CompactCells907To938.Box928.compactCandidate_not_isWeightedPerimeterMinimizer	245cc0aaeaba300a63251eeb0da76bcfe76f37b495b1d4b7f89712c422c22cec	equivalent
CompactCells907To938.Box929.QInterval	ba4455f88d0dc3fcfdaa4da2e0087094e8aac2163d4ef5be45110a563ef0d817	equivalent
CompactCells907To938.Box929.lambdaInterval	c7c3cec3c5b267ea22a39c9d6c56684fb8a22af9c0a70d16e305380f49ae7aa6	equivalent
CompactCells907To938.Box929.compactCandidate_not_isWeightedPerimeterMinimizer	898aef66fb7edae58087ac58bd18a7ff854d7dd7d96cc9002113bc3d86d752f0	equivalent
CompactCells907To938.Box930.QInterval	463ba8c175b2da2c6563cd1481f991fe382ee798149564ed82a9c17b662f1046	equivalent
CompactCells907To938.Box930.lambdaInterval	6b3f64b201257ba068d5a2c6abc56d7e4c4315493ad501dc02bc9c8fc6094180	equivalent
CompactCells907To938.Box930.compactCandidate_not_isWeightedPerimeterMinimizer	ddbfeb3e4c815b85d9a6c212ab56bd536b38c1b4b1e2e0a0c7b77b6b7c90507d	equivalent
CompactCells907To938.Box931.QInterval	36d9f412cb3f5e4d4230b35a8d31dc65ed6e8aa47d06fced778cb6428012c834	equivalent
CompactCells907To938.Box931.lambdaInterval	3494b25653522e2707a9707bd703996abc2d17068d5e971540a4c1a68ad9c1b3	equivalent
CompactCells907To938.Box931.compactCandidate_not_isWeightedPerimeterMinimizer	50cde685ba6c7db4886992da0859b9a01d72098223db9c5bc51d681ce6a2935e	equivalent
CompactCells907To938.Box932.QInterval	5649937582d583466054b8b067239032f8cfee8717ce6c8dcdaa4e33e380e2f3	equivalent
CompactCells907To938.Box932.lambdaInterval	9663dba0fd2954731573235d0d18bd1d27f6749e0f334823bd36cb1f452ca900	equivalent
CompactCells907To938.Box932.compactCandidate_not_isWeightedPerimeterMinimizer	3e4a0f7b57c0cba9f294d97bc83355f9dcc46d9d8be9d15002ce4056d17c65ab	equivalent
CompactCells907To938.Box933.QInterval	101edadc29c3103dd0753e593a2ed9aba5b234609c63d3a82de1d62b9cabcdc4	equivalent
CompactCells907To938.Box933.lambdaInterval	87f45b5f7919bbcdc93a50ff2718e6243cbe9f1832da2b23b75188fa0ac697f9	equivalent
CompactCells907To938.Box933.compactCandidate_not_isWeightedPerimeterMinimizer	2c421c456f2fd900cde1ec7c971d5cb1aa835cbd3acdaa600fede4e8f4b1ad79	equivalent

Declaration	Source hash	Audit relation
CompactCells907To938.Box934.QInterval	82df24e42432f7c5ef8f72c858529955803aad13d900f497202e9cb6199183d0	equivalent
CompactCells907To938.Box934.lambdaiInterval	6e6c707a2ee92d722eb42f3c3b9b04dc2d8a6e6df06a0de23fabf0ceee4e3676	equivalent
CompactCells907To938.Box934.compactCandidate_not_isWeightedPerimeterMinimizer	99c9d59c95e4a52aec06ecf0fa1806f1c5a00b81545521794061402fc7827eba	equivalent
CompactCells907To938.Box935.QInterval	f52228bf8e2ae1a82f0b9f1d62089dc19f026b9672cc6c6e188d2480e9e2cb33	equivalent
CompactCells907To938.Box935.lambdaiInterval	78e37cb3dc49dd087d04107eb9454c595f9293d5c8d365c86a1eb4fe4b19f318	equivalent
CompactCells907To938.Box935.compactCandidate_not_isWeightedPerimeterMinimizer	63a689e96ee0cc8f618487117ee5608a1bf aad3897d784cc77e066394a6abf87	equivalent
CompactCells907To938.Box936.QInterval	f28e4df8f0edfd20afaff0c28d2f3ec1146fbd6794ef68f43db8021a861d928a	equivalent
CompactCells907To938.Box936.lambdaiInterval	4805aaa87e6b8579227980d377834839c1890d8ece3408cbd5c68e5ad56b8ccb	equivalent
CompactCells907To938.Box936.compactCandidate_not_isWeightedPerimeterMinimizer	7810bd0cab92e55e2cbcfe3ae7516a3f513f26e7d184360331df91aaff7c9269	equivalent
CompactCells907To938.Box937.QInterval	08b1c916fcfe4e780c1da3c676ad851d8c989bd6d3132362bcceb495c7949ddd	equivalent
CompactCells907To938.Box937.lambdaiInterval	cd92c9c5aa4265e7a451a6a8c53f4754376e32055f0f99ccfb58faf4bccdf4cf	equivalent
CompactCells907To938.Box937.compactCandidate_not_isWeightedPerimeterMinimizer	f5031cc47b0dfb52585494598131a07ced3cdd54862ffee68ade583b273cefc9	equivalent
CompactCells907To938.Box938.QInterval	3e55fa120b1cac0e45c454f4412fa1e308bcd91cc4814b2d1da0d8fe23ba66ea	equivalent
CompactCells907To938.Box938.lambdaiInterval	c951fc0025362e2a6c384696a83918cbbd6ca3b24a07e09b059662769859de8e	equivalent
CompactCells907To938.Box938.compactCandidate_not_isWeightedPerimeterMinimizer	0d125c9148ea77632aead845519c5c2c2f80023b7c7fc78597e65900c1322ab2	equivalent
CompactCells907To938.bboxes907To938_candidate_not_isWeightedPerimeterMinimizer	d50e2da16fc128de2b119a6f1801159d987dac732f0bcac88a2e23434b85234f	equivalent
CompactCells907To938.bboxes0To938_candidate_not_isWeightedPerimeterMinimizer	6a29df51ff0443617f7318e6c98da0ff006accbf8a78bdc713b91776d7bbb50e	equivalent
CompactCells939To970.Box939.QInterval	001551f1d2e67c3334b8d62160b4cce997dfc97574700940ccc67cf23f4893e7	equivalent
CompactCells939To970.Box939.lambdaiInterval	e1be114d267cd6bb786bc5796d4bbb84c148356ed21f1c72b991ce79d0428935	equivalent
CompactCells939To970.Box939.compactCandidate_not_isWeightedPerimeterMinimizer	90867974fbb7b19ac688f8fef80e3ddeef4d6b3a29ed6ed72185335a850474d	equivalent
CompactCells939To970.Box940.QInterval	51be92a5f5739c61d67b6693e5cef5994bdf3e3ca27d4dc01a74f2523f9d14d6	equivalent
CompactCells939To970.Box940.lambdaiInterval	fba3bcaef36c9fb1cee63d9a893572e2601e210f7b89cdf7c64e6e983de141a	equivalent

Declaration	Source hash	Audit relation
CompactCells939To970.Box940.compactCandidate_not_isWeightedPerimeterMinimizer	02e1fcae3263fbd1b489ca9ca95d990c9e204ca1ebc926e54c20d541acb3c205	equivalent
CompactCells939To970.Box941.QInterval	d2e2faedb79babb318a12e553866998b9bd44b28d72e7b3db927d135147d965	equivalent
CompactCells939To970.Box941.lambdaInterval	4ecbf43297148aec23183275ab1a2738b46ffb447e0786507fed9251db5fd850	equivalent
CompactCells939To970.Box941.compactCandidate_not_isWeightedPerimeterMinimizer	14fbc987fe46fd4528b17c3df64e1ac26153998d2f6ce9b8c9e6646280c4eaa6	equivalent
CompactCells939To970.Box942.QInterval	4d2b23448596a4cdf8427776ab77be2f3730700dd4971c0a5aade60555bc6875	equivalent
CompactCells939To970.Box942.lambdaInterval	ccd13b81dc55012761673e568cd1932c39d070ff621e4c0b37ff867df92d7f9f	equivalent
CompactCells939To970.Box942.compactCandidate_not_isWeightedPerimeterMinimizer	b2bf6eb0d0a040313ce04976ffa380f09290e5fb2a3a88381dcf5a4480a4e0c7	equivalent
CompactCells939To970.Box943.QInterval	647392fb99a8401e3897edd444842a16d6ab8aa08dc5dbeac5d3915da1b73ff1	equivalent
CompactCells939To970.Box943.lambdaInterval	8e4cee3e55ee12bc199ea3cc862309d93c15f5e757aa3f7f161a959eb2da3696	equivalent
CompactCells939To970.Box943.compactCandidate_not_isWeightedPerimeterMinimizer	483219d65941114eb194922ef4b9c2adb2854c4d656bafca4ab60526139f8a18	equivalent
CompactCells939To970.Box944.QInterval	8d96fe34e5f0edbb34be47fe39a4c34f84671bea17ba1868cf3dcb1cd81619c8	equivalent
CompactCells939To970.Box944.lambdaInterval	79712f56722b1cfc45e669ea6cc331eb582b2a343deab405991036b7f6029ba6	equivalent
CompactCells939To970.Box944.compactCandidate_not_isWeightedPerimeterMinimizer	66c97cef0fea350de7952bf10e1d649041d6b482ff4615e15e64089d20623f37	equivalent
CompactCells939To970.Box945.QInterval	39c9818c7fc164ff7fa13fedd76bd0cc4e3f8b8affcecd0d5aeb9e092dad22a4f	equivalent
CompactCells939To970.Box945.lambdaInterval	3aa7ce69c4608f155016a7b90bb209038d19b06416a0e03f6ce5b1dea81137d3	equivalent
CompactCells939To970.Box945.compactCandidate_not_isWeightedPerimeterMinimizer	4a31bccd69dabd8e4cdcb0d3f3f949a575cfb7de7d5d67f5139433393162433d	equivalent
CompactCells939To970.Box946.QInterval	6f6d2d9bd66787e4336a6524e4b9a50171566483a467ecde5703f41823b2c652	equivalent
CompactCells939To970.Box946.lambdaInterval	5d2864e736c877144f8b149547dd85d3e7981a650ea796fec1b0e57cd4c20add	equivalent
CompactCells939To970.Box946.compactCandidate_not_isWeightedPerimeterMinimizer	11c5727732a96fc3017c6fb1662464d3cfb845ed3bf3bf6f47eb21c961facfc8	equivalent
CompactCells939To970.Box947.QInterval	6c9740964607ffd645969d8a22d772fbd917ff7523fff5c54b18fe5688670219	equivalent
CompactCells939To970.Box947.lambdaInterval	ad2dc0781ae6e5f84e8409e76247e9f715c793b1f2acea4b1c925f1f8147f2d0	equivalent
CompactCells939To970.Box947.compactCandidate_not_isWeightedPerimeterMinimizer	8fa0beb51c5062fab7b94f6643ea5383c475c83d24b55c308ef20ae4eb7044b3	equivalent

Declaration	Source hash	Audit relation
CompactCells939To970.Box948.QInterval	72edaf5cb8b3bcd6d6a2a72bdc969b9801e83b5c540272c1f97ca244d5aa11da	equivalent
CompactCells939To970.Box948.lambdaInterval	7a65539863ba2b023ffe834c397ac8c7c29ff5e05fcdae2bbdb8a16235ec88bc	equivalent
CompactCells939To970.Box948.compactCandidate_not_isWeightedPerimeterMinimizer	295288ad8becd39973d5603f626a74190ff8ee1dfb74a6281e6beeb36f9fba64	equivalent
CompactCells939To970.Box949.QInterval	f4673233b7197d9448fd460ada053c5ed1f5581b6d310bec50f2ffc092cc4b3	equivalent
CompactCells939To970.Box949.lambdaInterval	f88eca34c69bf05b4335cf93672005b742c2147a9ffc804acbf3a33b87ffae83	equivalent
CompactCells939To970.Box949.compactCandidate_not_isWeightedPerimeterMinimizer	8168a73fae98902d81c5c797931e99dfa131885bd1ab8ca84e95f9a6c1e529ef	equivalent
CompactCells939To970.Box950.QInterval	59745501dcf4c4ca9f33aabab5c1a76511fb5a3f64641bd3518714c4b3750d3a	equivalent
CompactCells939To970.Box950.lambdaInterval	3968f4141082814090b5927dc91616ad804df25af27a6f2876536d16c7aaedfe	equivalent
CompactCells939To970.Box950.compactCandidate_not_isWeightedPerimeterMinimizer	44a2547e4bf71e7ba2d0ea2379d8cee623d77d6d708f241a967a1ca607ba6342	equivalent
CompactCells939To970.Box951.QInterval	7d2a7a427c9bb811dd573907e6471a1becc2ee98dbe53a88d989f80ddb27eee2	equivalent
CompactCells939To970.Box951.lambdaInterval	e3d5f64c507bb83b2c78c968453dd1005bd3c595ac0d0c8e6b98822bd78da7b8	equivalent
CompactCells939To970.Box951.compactCandidate_not_isWeightedPerimeterMinimizer	6eb95f929016db11cdafe4774be9af10815c2a5993780bc0542757dd03bacbb7	equivalent
CompactCells939To970.Box952.QInterval	a74e2dce97be889926d6695f449c258524c037a44f6da70ccf9427946a5872fe	equivalent
CompactCells939To970.Box952.lambdaInterval	279994e908d36a8a6359df80a5ef644ed076d6531ce8472513237d2553df38c	equivalent
CompactCells939To970.Box952.compactCandidate_not_isWeightedPerimeterMinimizer	ce20aac007d42946918bfc7663a1f5a599092ce5b6e52c5e0fad814362d44124	equivalent
CompactCells939To970.Box953.QInterval	71c25972d677e6a73dcd41fbe0e1b1439c2314b56207afd7c68877de132a3dcc	equivalent
CompactCells939To970.Box953.lambdaInterval	dacf49a0951e03ee3874c25a1102b6b9fb2cc4e0a7fc1c4aaf755df685cbef95	equivalent
CompactCells939To970.Box953.compactCandidate_not_isWeightedPerimeterMinimizer	15c7db81e91a78be20cbe8ecf55f5b15aedda0899b4af94d1b19c02348fd0e26	equivalent
CompactCells939To970.Box954.QInterval	b8a5cd1bb3d8f5054a7d6b694abd1b7c52b49124b35b130a5f38af4082db7040	equivalent
CompactCells939To970.Box954.lambdaInterval	cc9e5c2aa198454a3ca76dfeaf302d481b3497bf30bbc5d77d3549210b2ae481	equivalent
CompactCells939To970.Box954.compactCandidate_not_isWeightedPerimeterMinimizer	9512514c37bfd7998c440b8a0f79da719f9b0facfe19f019c18820b50e650a39	equivalent
CompactCells939To970.Box955.QInterval	7b71c305315660a119157267661996973e977a0a6852fcae4483a6f895f75103	equivalent
CompactCells939To970.Box955.lambdaInterval	a955e341d4a7e1193229e1f7a1000622d96d2a1c14ce1a8c588f04511c00f19c	equivalent

Declaration	Source hash	Audit relation
CompactCells939To970.Box955.compactCandidate_not_isWeightedPerimeterMinimizer	2e86d6b72f642b102270c4e5779b3912a3a366c6c28c9e676eef479db0e96250	equivalent
CompactCells939To970.Box956.QInterval	2d37ac41c2bd3036ee0219d3833275e18032d262598ff2492666a64c40042281	equivalent
CompactCells939To970.Box956.lambdaInterval	253ba1721d93b37feb78a0df4cfc73edc9038aeebb0e18261d251e3881249b21	equivalent
CompactCells939To970.Box956.compactCandidate_not_isWeightedPerimeterMinimizer	49cac853b450278ee05551b48f9bf5e9d655b59828171594212d2e143b04e024	equivalent
CompactCells939To970.Box957.QInterval	7f83f9ceb9934114f6e70354a4d66147a7ac3eaccf580a020437b476cc470c74	equivalent
CompactCells939To970.Box957.lambdaInterval	75c0bd40d4414ba856506993c036034ebf3916d9ba78660d1241eb3aadd91ee7	equivalent
CompactCells939To970.Box957.compactCandidate_not_isWeightedPerimeterMinimizer	59e30a24e9affba8bc160dfe032b19d0aaa9a924291d7920fc84fb8ef855cfe8	equivalent
CompactCells939To970.Box958.QInterval	dcb85d0d316736328556747f0d2760a2c5d90f4209154f7903f0d041679e960c	equivalent
CompactCells939To970.Box958.lambdaInterval	3ed2a70d0c43bef6ad482156972e2719967d8266ce497412d3b5c2156e22f175	equivalent
CompactCells939To970.Box958.compactCandidate_not_isWeightedPerimeterMinimizer	7a8c071c35c36827cd36b2451b7be8a3425874afb724a0afcedf44814b1d937f	equivalent
CompactCells939To970.Box959.QInterval	b5b66440ab498eb7e1657223135b5189a61d005020a424ede66d79a9081fbb2	equivalent
CompactCells939To970.Box959.lambdaInterval	5ecdd5a7ca52cfd61fdd5db23888381b10d9ec68734dce5d882d7fcb5db3a3e0	equivalent
CompactCells939To970.Box959.compactCandidate_not_isWeightedPerimeterMinimizer	08f9aaa06d88d34ed8018f4311bc616a7013288a9f71c42fad1942600fd05ff	equivalent
CompactCells939To970.Box960.QInterval	15cae777e0493978ac4e468d6d13779dbdfea15a26efa3b284f4c4690c94bc40	equivalent
CompactCells939To970.Box960.lambdaInterval	9e5465e434e02eb70588462206d3808680b823b5b54d96afd27d71bb769ae14f	equivalent
CompactCells939To970.Box960.compactCandidate_not_isWeightedPerimeterMinimizer	45c1b4724b9012e973fb5dd21ac50e22c16a21ff78543dc028dab6ef2f48d409	equivalent
CompactCells939To970.Box961.QInterval	6ef04336cde86dec53423c9bf64475221fdbad20cd94897572368a43d5f62f0	equivalent
CompactCells939To970.Box961.lambdaInterval	69a4573e90c9342badd123fba48eec9fd5aebda6bbcca80a298b29fecbbbe72a	equivalent
CompactCells939To970.Box961.compactCandidate_not_isWeightedPerimeterMinimizer	b355eceed03a37f85459be84628256e5a0066f644eda66d8578760be947b7749	equivalent
CompactCells939To970.Box962.QInterval	703fc4d2a3b9e2939e58cff7bba5848f952a04d0a02ff4f5835bf2852112793e	equivalent
CompactCells939To970.Box962.lambdaInterval	8c0fe57367a47b6f2514a6fd8fe0c6ee04a4d9a6024ebaa01d4d6855d0e84fb9	equivalent
CompactCells939To970.Box962.compactCandidate_not_isWeightedPerimeterMinimizer	caedcc85019999ddd2552d7934827b374244fc6626f904e5144af701c2429f57	equivalent

Declaration	Source hash	Audit relation
CompactCells939To970.Box963.QInterval	a7195164dab555cd72c50173b20f21376f89ed2e1e28d088361ecb6cdc6f615d0	equivalent
CompactCells939To970.Box963.lambdInterval	363aeca8917eef76a87527444ab015d84dd436036beb9e7796e8989c9faf47bb	equivalent
CompactCells939To970.Box963.compactCandidate_not_isWeightedPerimeterMinimizer	97d4cdd356a3d2ec0d9c493ebb085275d7e806ec73a315cf32e997bb0b6ee33e	equivalent
CompactCells939To970.Box964.QInterval	b67b07cb928b9809995be20a3093e9a385f5308b2341db9c1d7c6dda18ae233d	equivalent
CompactCells939To970.Box964.lambdInterval	ac5f9043bec1a6600a889bd641b4db2c0d29379bfdc9511db543166c2d8e8b5	equivalent
CompactCells939To970.Box964.compactCandidate_not_isWeightedPerimeterMinimizer	989e1015c8e128b481f8300fbe25d4a9af44ce90132748d052d3932f4c937773	equivalent
CompactCells939To970.Box965.QInterval	063763ea1858ac47afac13b73da87351d67493b4069637cfb8d5b5f208ae356b	equivalent
CompactCells939To970.Box965.lambdInterval	1809dcce69cf07c9b70d7ba677cfcfc69632d62e44a1a62964c1be0e244749d7	equivalent
CompactCells939To970.Box965.compactCandidate_not_isWeightedPerimeterMinimizer	0d3734b75eaf21025161f51207122d73d0a6b0a6430c337a6b08bc14fb15049f	equivalent
CompactCells939To970.Box966.QInterval	20e243e17fb53ca8724ff39020928921406e821c382becd483e66855e1a0a264	equivalent
CompactCells939To970.Box966.lambdInterval	a46a26e46459a45242cbad7ce0d1de44807953c83d4f2897c72df2450a66d15b	equivalent
CompactCells939To970.Box966.compactCandidate_not_isWeightedPerimeterMinimizer	212de0748844646b1092eb8edc13fc8e6a1e749d72bb5ebdc5ff933bb00262b3	equivalent
CompactCells939To970.Box967.QInterval	67f58cd7a5f892b413ba6410c5002c1e4586601bf4d0713217bac46243e5d942	equivalent
CompactCells939To970.Box967.lambdInterval	b24dc3937ee8e181ba653869996cb4dcb877dc7734637e3b763d5c05a97a2a60	equivalent
CompactCells939To970.Box967.compactCandidate_not_isWeightedPerimeterMinimizer	177b0771ea3793bd0bf9b258bc27a409650aef5ac1769eb8be7079f04dfc42e1	equivalent
CompactCells939To970.Box968.QInterval	949aee2ce935aa5f442e529bec71b4f5660c14dbc66507add583b9dcd7d52ce8	equivalent
CompactCells939To970.Box968.lambdInterval	8ea1acdec8f741bff9117a13183f68f3eb70a1cf6a6c43198e4d6f4c68d1e6eb	equivalent
CompactCells939To970.Box968.compactCandidate_not_isWeightedPerimeterMinimizer	f419262ae3d80161b71c023b895bee7b7c0894aeb99751a9eb043a78666e592d	equivalent
CompactCells939To970.Box969.QInterval	bd95fac79dc4eb1b18d971e4ac5705ba87c2163bac9c41d1ee24611a02bcbae8	equivalent
CompactCells939To970.Box969.lambdInterval	8bcacf4065bfddb103429b8b7ddd46b1a2cfff967e567a83672c412d849a08625	equivalent
CompactCells939To970.Box969.compactCandidate_not_isWeightedPerimeterMinimizer	1b22c0ce0b437b5a315bbe663d30e599283bd8bc64f1d110e59c852293cb2651	equivalent
CompactCells939To970.Box970.QInterval	a649d47ecf96ed1939d927b98bf488202be0baaec7bce6c9431c850d03607c7d	equivalent
CompactCells939To970.Box970.lambdInterval	eb33100df28428bd0e24e60fb63a0e7ef3bf65457736ef77cb5af70f5282c357	equivalent

Declaration	Source hash	Audit relation
CompactCells939To970.Box970.compactCandidate_not_isWeightedPerimeterMinimizer	21c7a75bda7b6785099a1007db7149201d832fa8d44ef5d8610d27fea71e7942	equivalent
CompactCells939To970.bboxes939To970_candidate_not_isWeightedPerimeterMinimizer	0348ee46b2efa245082e5e1469eaafe26785fa2d504c1d8568302ce8e0b1bcfc	equivalent
CompactCells939To970.bboxes0To970_candidate_not_isWeightedPerimeterMinimizer	1af9623ff32ed1306032f7c9c33fa085ce585dc883973c02aff47312f4f9cd4e	equivalent
CompactCells971To1002.Box971.QInterval	2aa9b78703ee2ae1ccc4976a059d3c39afa2156794f1073e57f8f48f2f9bde42	equivalent
CompactCells971To1002.Box971.lambdInterval	84494f706e6d3c3f3971ae1dfce3e717255960b7f5daa6bd0453c9a0f6dc78a9	equivalent
CompactCells971To1002.Box971.compactCandidate_not_isWeightedPerimeterMinimizer	766f7bd7a11e25a20728254db8a8c193e321ce62a3406a610163e72913da85d5	equivalent
CompactCells971To1002.Box972.QInterval	6f5b3e53fde24d06c6c7617be50dd43fb62b547b430fbafbc8d57f50fa1052886	equivalent
CompactCells971To1002.Box972.lambdInterval	caf3675876523740ec08e8b2b26a943d05ec578089736807c901c7d2a05dd986	equivalent
CompactCells971To1002.Box972.compactCandidate_not_isWeightedPerimeterMinimizer	c07fd47b1c05e8f30a1a29a76ba98530b22ca1b963467ca4390945766365df93	equivalent
CompactCells971To1002.Box973.QInterval	194370388974df48653b9ff02197caf4841624aa4ed5c084e8d89d7e0f3f13c6	equivalent
CompactCells971To1002.Box973.lambdInterval	445d2554d1af182a751557e40590eb3635f7d9fa37605d6958930ea6d02171a4	equivalent
CompactCells971To1002.Box973.compactCandidate_not_isWeightedPerimeterMinimizer	b0138e5bb6f9afc30514e00fe0c44507604b068e1bb02252a21c62c721ff7d92	equivalent
CompactCells971To1002.Box974.QInterval	474b15a8343d35edb75dfe7e596dd1794bc09b770840a0af32d8c6f2157806cd	equivalent
CompactCells971To1002.Box974.lambdInterval	aad8cfadc5a8c51297d085ac86e579ba9f8488d91c7ae155ddc501f0ecd449d7	equivalent
CompactCells971To1002.Box974.compactCandidate_not_isWeightedPerimeterMinimizer	7cf2403908688fd8ed07d21a26aa086b500b7b846704af44b85d4506cb9b0301	equivalent
CompactCells971To1002.Box975.QInterval	e71ee2d00babd46c24f3586670c218f11a306b922a73804196a52b1ab0e58a30	equivalent
CompactCells971To1002.Box975.lambdInterval	7d6a06a9dff46d17752bad0b090e91142a4ad511c05d0fc0b6d00b8bcc0d0ac8	equivalent
CompactCells971To1002.Box975.compactCandidate_not_isWeightedPerimeterMinimizer	2f4b3099efd876e83ba00797cc1f91a1a940d838cfbfff45f09806a4f6a3f7c48	equivalent
CompactCells971To1002.Box976.QInterval	179ed42344f21d4da7b992abc9f9ea013777af3ea4b061e7b0b5410d829de506	equivalent
CompactCells971To1002.Box976.lambdInterval	543b035279693e9a75818a4485be4cbc886c1cd6b60eade416bf464f5fe5ea33	equivalent
CompactCells971To1002.Box976.compactCandidate_not_isWeightedPerimeterMinimizer	f0c10fc1179791f07d367ad69e6305b55044951e2169d3dfead69074a0411ae6	equivalent
CompactCells971To1002.Box977.QInterval	fb76097d90597826820f3b5d2cf2893c9406df74cb7ab8720fb6d561e5ce80b4	equivalent

Declaration	Source hash	Audit relation
CompactCells971To1002.Box977.lambd Interval	87ef25f2b89d3c04d391b76ff38b52d3668 0bbcdad46455ac1e3f2ddc6aa2e5c	equivalent
CompactCells971To1002.Box977.compac tCandidate_not_isWeightedPerimeterM inimizer	da6f7c3c3e47b6cb9841c5868eb62bb8a52 2ecc6e66bfa28b9b5238662184127	equivalent
CompactCells971To1002.Box978.QInter val	c4ce4eda1935c6326b97fdf1d2e6bee8bcb 2009e82e4b17c638b21a4612ba1bb	equivalent
CompactCells971To1002.Box978.lambd Interval	ccc4f8d1d7506caae5208a030a5bf27a98b c837e30bc287274c2152f7fbdf968	equivalent
CompactCells971To1002.Box978.compac tCandidate_not_isWeightedPerimeterM inimizer	bb4783b85dd269cf6d98e9661a60784c8a3 16498500575239be77ec976a25925	equivalent
CompactCells971To1002.Box979.QInter val	84391e9fe79be1812519f804ebc2f5099e8 499514c064480be6c9537b3bfa952	equivalent
CompactCells971To1002.Box979.lambd Interval	e6116ca7af8ebca4c54dccbc909dc92686 0a23820815194844d0c3ec1804ca7	equivalent
CompactCells971To1002.Box979.compac tCandidate_not_isWeightedPerimeterM inimizer	30fde344598469e10fb736aef664c3a3842 78ba3a09ed611e0f0bcf31d5f06c1	equivalent
CompactCells971To1002.Box980.QInter val	f3e61b554add802d8c566398d8a9faf2bdc 59645b37d2a935b2c9cb0e858b6bc	equivalent
CompactCells971To1002.Box980.lambd Interval	c020e89903fc66b60648b39d358f0431beb 6d3afcd5310d896b4c2b008dc46d6	equivalent
CompactCells971To1002.Box980.compac tCandidate_not_isWeightedPerimeterM inimizer	e005d916aab6876f14597d7ffe297a20df3 ddc3e6f425a38c2c20ffaf375a2d6	equivalent
CompactCells971To1002.Box981.QInter val	8561a0a735df59f771163c6df11cbfb5f10 2e79739473435064f6d8867762131	equivalent
CompactCells971To1002.Box981.lambd Interval	337d32c0be5ddf8c52a11705ba1a723fa8a b1047b42b1bf227046f86cd6c900d	equivalent
CompactCells971To1002.Box981.compac tCandidate_not_isWeightedPerimeterM inimizer	90cd1fb3288c49fa202ec5b26e8da7f420a 39bad62b2120f4b0fe50b0134b0b4	equivalent
CompactCells971To1002.Box982.QInter val	050ff192720c314c1be8222551936c557be c8f2767c08ab6d0489e93f13ee655	equivalent
CompactCells971To1002.Box982.lambd Interval	09dec889acbdd86611d5f3ca8843f95d17e fbc871bc7e092195a17bd66fb9fec	equivalent
CompactCells971To1002.Box982.compac tCandidate_not_isWeightedPerimeterM inimizer	b37c61eaf742bb194d4c1728946ee36db5a 6effda8835adb5f84c6c98db258a	equivalent
CompactCells971To1002.Box983.QInter val	0b4aba11b4674bc44c815b2824e7118b3d0 039da57d557131117796086949987	equivalent
CompactCells971To1002.Box983.lambd Interval	a0dfe45d261ac0ed5331595ccd4412bc0bd 805ece44fbc02efdfcdfc5591c2e	equivalent
CompactCells971To1002.Box983.compac tCandidate_not_isWeightedPerimeterM inimizer	da4590fa2bb38fdb9bd5891b43f394d180a 024f6ad3974735d6ad7233c7dbdb2	equivalent
CompactCells971To1002.Box984.QInter val	f19abe0731c614be929984583bee6213385 84dab39b930f4acbf2100d9f61000	equivalent
CompactCells971To1002.Box984.lambd Interval	1563449729c9eedd420413dbb6d24599906 d9ad197c2eacffe4079faeb459c19	equivalent

Declaration	Source hash	Audit relation
CompactCells971To1002.Box984.compactCandidate_not_isWeightedPerimeterMinimizer	bef4194ddb910fe0c2be39189c98184c7397fa858b2f31f20b8d17eb39005760	equivalent
CompactCells971To1002.Box985.QInterval	a23b8526641ac8330967f5b388429beb559b292f9ddfe4721b2f15dbf86a77c2	equivalent
CompactCells971To1002.Box985.lambdInterval	d06387a55e3afeb1db5c5ac354360c4a68c7e60a0b0fc6a983a2905b0a9fa9a0	equivalent
CompactCells971To1002.Box985.compactCandidate_not_isWeightedPerimeterMinimizer	b63c67d2aa8264d40439fa3780a2ddaefd6dda0488ad278051b0f5c41858662b	equivalent
CompactCells971To1002.Box986.QInterval	2d4c69eb541f0c1585f74a8ed7fff83349aab99c262481675296e25f4599f783	equivalent
CompactCells971To1002.Box986.lambdInterval	aa4f8ff3405a54e48336d55c7efe862ab34f93512fb8154ee300965aa1f02dbf	equivalent
CompactCells971To1002.Box986.compactCandidate_not_isWeightedPerimeterMinimizer	eea06aa43ce2e1bf72a53f425246c1766ab76789e673ba038d53d98dc37c2a47	equivalent
CompactCells971To1002.Box987.QInterval	17e62a2489cc568dd63340b430445f18034cb70a6469bef19a3c053ef2d43862	equivalent
CompactCells971To1002.Box987.lambdInterval	c1475c81450568cab343f232bce2903aa8fc5cc21e3d7bb20ee11a01f29b4310	equivalent
CompactCells971To1002.Box987.compactCandidate_not_isWeightedPerimeterMinimizer	df1df9accdc2ca426dbc0a855457aa0b9e083ac59e1a8e5dc0e8b1afeb1f45a8	equivalent
CompactCells971To1002.Box988.QInterval	9495f2eac2ffbeaffac27f5f14cad04ec9acafb483052e1438bf720e338c2557	equivalent
CompactCells971To1002.Box988.lambdInterval	00b17bb03298c615ef588f9ade7e87482b7df941c54f750db61b5589c54fbeb0	equivalent
CompactCells971To1002.Box988.compactCandidate_not_isWeightedPerimeterMinimizer	4cf68ebaec74889993b0163ec9366bbd3b183f420e1fdaf68ecb1505f1afc9c7	equivalent
CompactCells971To1002.Box989.QInterval	42fe43ee65a8c6c15677003b90152b84990dfd5e4640f435b2ea7efff8236f11	equivalent
CompactCells971To1002.Box989.lambdInterval	b34e868ecce2dc7eb60239c2681f686c0a55e31ab11a8b3e75939beb2f16a390	equivalent
CompactCells971To1002.Box989.compactCandidate_not_isWeightedPerimeterMinimizer	89c072b87d86d260114abc8d3e197fc44aa346388d1e8882dce481b3c255438b	equivalent
CompactCells971To1002.Box990.QInterval	07247b410490db6a057d000f3fe0838725deb12b72c4cd11431746b2df7d8ed3	equivalent
CompactCells971To1002.Box990.lambdInterval	4b363cf47197213dc212dc1d518958ffe1e0e90e6a65776ea32dc27af5e90bd4	equivalent
CompactCells971To1002.Box990.compactCandidate_not_isWeightedPerimeterMinimizer	48e8b62b068e364ac8a7bb3ea454b72caa8e20832afd54019505243b137dac7a	equivalent
CompactCells971To1002.Box991.QInterval	2ceec609a6deb07e9e673a27af6f758e534a7b4d8e57e45cce44c7759071473a	equivalent
CompactCells971To1002.Box991.lambdInterval	d1af12d7bad7c1808aeb632a3d5d95007fe a872d9bb57e929ee381e76e03c1b6	equivalent
CompactCells971To1002.Box991.compactCandidate_not_isWeightedPerimeterMinimizer	8fd720216d6774ce21beb8a3a75c6d735645eccc13ea834ef2e20c42816dff81	equivalent

Declaration	Source hash	Audit relation
CompactCells971To1002.Box992.QInterval	1c01c4dc856d249300c927ceb21c0d8e42cf93185d99529e32cde02448d80a4f	equivalent
CompactCells971To1002.Box992.lambdaiInterval	7a3920dcc082fff02504bd72759de5391c3935a45972faf1377ca728a45fbe6f	equivalent
CompactCells971To1002.Box992.compactCandidate_not_isWeightedPerimeterMinimizer	aaf2e8cab5e6d918c9956d5d80d07e7bb4aecf98c74a850f2919e89e6d0ec40b	equivalent
CompactCells971To1002.Box993.QInterval	9c102773e53ea1fe6d01bfc3158ab130bff084d3df84438eb0f1e6681fe2d49d	equivalent
CompactCells971To1002.Box993.lambdaiInterval	9fd4d2f3cdae1bfd79846ea2ff4bf631cc5653351ccd752cc83c2494fdb8d74f	equivalent
CompactCells971To1002.Box993.compactCandidate_not_isWeightedPerimeterMinimizer	5071fbd44b5c6c4f4d0f985ce2d852c2e5f6e63b86a6fa3b17686024144987f8	equivalent
CompactCells971To1002.Box994.QInterval	0ca634bab30b7c7f93426d175e27ee344b5ef837b478037da3a050d767675846	equivalent
CompactCells971To1002.Box994.lambdaiInterval	56111e600904b0700f2141e7440c8a118862aa90e3e09d80118252e64e7f613c	equivalent
CompactCells971To1002.Box994.compactCandidate_not_isWeightedPerimeterMinimizer	4b2a3d8171f514bfdeaa30287a7f1631a7e55bccaed20b656cee4eac5448b874	equivalent
CompactCells971To1002.Box995.QInterval	4ade3f22e47219884fcbc1c5a167f8fbf7a418edd840b83443de76688d6aa563	equivalent
CompactCells971To1002.Box995.lambdaiInterval	84250a50f6b4eb3d42ac07bbc59eded7d44bc84e5564706f6d193dee3a34b275	equivalent
CompactCells971To1002.Box995.compactCandidate_not_isWeightedPerimeterMinimizer	966c4d3f28f99be4c2ae68d12bf74948d9d9dc3372b5c1cdf5ebe98037cad8e	equivalent
CompactCells971To1002.Box996.QInterval	be7e2003ee52330c4e448cda32ce8ae97ff64d1cf1dc0509b61583a4f326ffc9	equivalent
CompactCells971To1002.Box996.lambdaiInterval	31c12ca87905e0d762b1cabf68df4cdf36afa903fe45bd7ebb4075b852ba5ccf	equivalent
CompactCells971To1002.Box996.compactCandidate_not_isWeightedPerimeterMinimizer	f988c508e1ab8702b9a9d4178993f77596350d7f7f3b521eb298d945d340ef59	equivalent
CompactCells971To1002.Box997.QInterval	37958bf6eea5cf1eac561eb9d9c78a1fff26a59a0e75ac2ba1667e704564de38	equivalent
CompactCells971To1002.Box997.lambdaiInterval	ce4794cf31e6777aa554c8c6f67a2e2c4681519149a196d1d1874e2fa51e2479	equivalent
CompactCells971To1002.Box997.compactCandidate_not_isWeightedPerimeterMinimizer	779f545d991a9921a4d141fad5af8a27a650f15eb33d3c51b50b179845b34757	equivalent
CompactCells971To1002.Box998.QInterval	94b1a31340830ca038838cf39b649eff96f8ea7a77ae9d62ebf0166a362ff811	equivalent
CompactCells971To1002.Box998.lambdaiInterval	55a2aad66437d8d53723e5f8ac83bd9e479421a5dbda898df33080f87c67e75e	equivalent
CompactCells971To1002.Box998.compactCandidate_not_isWeightedPerimeterMinimizer	addc1922bfe832e90b48b972e75a17a7c86e42f33cf901543273d8a857493fcf	equivalent
CompactCells971To1002.Box999.QInterval	ee4fd86044a0e109e594df26c339dd7e700297b1bb14d8b2b36e3947d6085263	equivalent
CompactCells971To1002.Box999.lambdaiInterval	db541f7d1ea4c2043e6e6cf4e146f846bc6ad3acf409205be9956b165f0a16b	equivalent

Declaration	Source hash	Audit relation
CompactCells971To1002.Box999.compactCandidate_not_isWeightedPerimeterMinimizer	b1b79732342f6f8d7c717d01d545de2505fd333cc424e2bb0e456e1ab7a710d5	equivalent
CompactCells971To1002.Box1000.QInterval	5c9b403518856be5fa968df392fb27236807355eafcc642f05ec8d50af0787f1	equivalent
CompactCells971To1002.Box1000.lambdaInterval	b0267289acb827f6c3c15f55346cc603b095c3505104719b2da21e03db7ab4b5	equivalent
CompactCells971To1002.Box1000.compactCandidate_not_isWeightedPerimeterMinimizer	5f04bf2359addb365b97378c307ed5238a296df2ec6da2652fb7699bbc4f3a68	equivalent
CompactCells971To1002.Box1001.QInterval	92a3ddce95258445948d478bde8b43715586d390d3f64cc9d082e83d90cad013	equivalent
CompactCells971To1002.Box1001.lambdaInterval	59d3c1caf14f47535685f495fc2436b4b958683976cab889ecb313ce09ab6320	equivalent
CompactCells971To1002.Box1001.compactCandidate_not_isWeightedPerimeterMinimizer	838dff7d593286f068954e6a71d40293d304777cc2d30d3e54491b55f5bc2b8b	equivalent
CompactCells971To1002.Box1002.QInterval	56f6bf61216e8ed6339ac911af6291a81dc5e6751715de0c2424cc1f2da8f72d	equivalent
CompactCells971To1002.Box1002.lambdaInterval	5be8db7c10e9512edb133db0480af0803fb5fd60978b454a9ac3973078f3cd68	equivalent
CompactCells971To1002.Box1002.compactCandidate_not_isWeightedPerimeterMinimizer	6c6e6244b7f8187eb1689d1866e69fa29d1695051ac7dbcb92becfc1fc7b11a9	equivalent
CompactCells971To1002.bboxes971To1002_candidate_not_isWeightedPerimeterMinimizer	c05223a8d6bbcb47aea3d4e5a213a08f37849f2c2b7fdd88a452aaf00283a2d	equivalent
CompactCells971To1002.bboxes0To1002_candidate_not_isWeightedPerimeterMinimizer	38783d955697a0e6460ee6c7af025c6d1a3c7dea04ada888cedc4de8a629d94d	equivalent

MiddleFaceCell[0-9]*.lean

Reviewed all 452 indexed entries: Brick, brick, lamInterval and the candidate boundary for each of the 113 bricks 0..112. Exact rational endpoints form the closed union $[102001/100000, 25781/25000]$ with all 112 internal seams present. Every brick records ordered density bounds, affine centers/slopes and the two cross-coordinate couplings, plus positive error radii. All 113 CellCertificate proofs instantiate the same complete angular-branch, opposite-face, negative-K3 and negative-G schema, and their checkedConditions proofs apply generic certificate soundness. The candidate boundaries shift density to the centered coordinate and invoke the same all-curvature geometric exclusion; they add no stationary-curvature or root premise. Generic common-zero existence uses the continuous incidence projections and triangular IVTs with residual order $(-F, C4, E, C3)$. Both author-selected end bricks and both end candidate boundaries were inspected in full and agree; nogenerated batch modules provide the advertised exact interval assembly.

Declaration	Source hash	Audit relation
MiddleFaceCell0.Brick	927e7b2d26353aa0f6b5291a55da180383cdca4c1a118bd1f2c2be6131ea029e	equivalent
MiddleFaceCell0.brick	6826922bc65ff716ea9f2f5179745288974bf844804e2d773c508e15617304c9	equivalent
MiddleFaceCell0.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell0.candidate_not_isWeightedPerimeterMinimizer	c186745e2baa4f5a1e70e59dbcdeb62622e696b9a3b2aa4cb9d30703f720f43c	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell11.Brick	37064f1906aa6a9d0382f4665218a3044d4d6522e77c02dc1748a2643f500442	equivalent
MiddleFaceCell11.brick	74b9198c97ef1af9e41670b35587391ece70b5c463f069a6305d2686f97177dd	equivalent
MiddleFaceCell11.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell11.candidate_not_isWeightedPerimeterMinimizer	cbd80e0421530d63f847a7447cca8727b99a1484929a8487f1581662fac379d1	equivalent
MiddleFaceCell10.Brick	dfb2d9943dec214e92a4f54ca07b8ea4b313c45b3c52a4b1e1a1ed5322ca36ef	equivalent
MiddleFaceCell10.brick	875797fb1eb5a25b516cc242ad6452f8dc0d9ffd04c37c93b8ae744dc331eb80	equivalent
MiddleFaceCell10.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell10.candidate_not_isWeightedPerimeterMinimizer	85428bd03e21ea81d78e51732ed52f9172565ff417edcc184f9cd3e8079033f7	equivalent
MiddleFaceCell100.Brick	a8171f87dee8e25cbb24a75f6def9d25dbe def73d4381471a3e5f96b83099bdb	equivalent
MiddleFaceCell100.brick	dd7a2d1a2b25dce173aaf5738da4a042b57c7f4eefc8988166f7e09b5a45fed5	equivalent
MiddleFaceCell100.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell100.candidate_not_isWeightedPerimeterMinimizer	c939bfd952b2a5b36c2046e60ce73214cb624656ddc380aeed7edd15ef70028	equivalent
MiddleFaceCell101.Brick	a5e1a929882a03e8fa552c7b425f51e229813864b71386437fbf0b7091cca9a8	equivalent
MiddleFaceCell101.brick	3238c37297b6232bc1e903b223e435821fb12195fb23e61a7382337ff7a5a1f7	equivalent
MiddleFaceCell101.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell101.candidate_not_isWeightedPerimeterMinimizer	3c99b634f26ee22bb60a6f4407b3c5485cf22924ab3c86b7c66e9592cf66cae6	equivalent
MiddleFaceCell102.Brick	7bd71285ffdc2dd9c94e6b306d9064d04945740418b700069df9e24de8504ed7	equivalent
MiddleFaceCell102.brick	6421cbee537a9e9d50cc381bd9bf7661a631e8b8ed2eedceb02c698b199f898	equivalent
MiddleFaceCell102.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell102.candidate_not_isWeightedPerimeterMinimizer	bb704f431ef2c10d4c3dc586f8847f50d0885b82298b08083e9ce862eb573268	equivalent
MiddleFaceCell103.Brick	92b99d25c250dfae849fda7d9d4545daa3e043a47bfe4784c239c4e0ac67c341	equivalent
MiddleFaceCell103.brick	929f857e44917331990f739d186c69005bc b3c5567605331a2fe5e6012c791a3	equivalent
MiddleFaceCell103.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell103.candidate_not_isWeightedPerimeterMinimizer	01f97e8fc75e086938b09d68722fb18acad663b37d518995b4e4d9fb448b1271	equivalent
MiddleFaceCell104.Brick	07b8a51ca0ec9a8d73c6241e43546f23bdbd26da8f3714ae0113574663693611	equivalent
MiddleFaceCell104.brick	4317f9d345c970d16c5bf7aad1819eeb62321b686692a83ab0eaa63600fba26c	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell104.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell104.candidate_not_isWeightedPerimeterMinimizer	cbbe180cb50850c25226cf7c1f9b089a8f2fe005f3b8d5ba28ec09293fda25e1	equivalent
MiddleFaceCell105.Brick	90bf41a259cdb75de53fb0bebe41f5bdb2538e068e70568bba7e18438ba0a7ff	equivalent
MiddleFaceCell105.brick	d34ddb8552e287fe2496e9457a6c103e944ffac22b1a2903ccbb910ac0f1eb3d	equivalent
MiddleFaceCell105.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell105.candidate_not_isWeightedPerimeterMinimizer	3855bede4d6c20c5fa449fd12225215d82c66aabf5dc2c518077388835b97275	equivalent
MiddleFaceCell106.Brick	43657c08113da4763a323445e0a8d23e9cab2995c04ca900b53e38cbba68cf0d	equivalent
MiddleFaceCell106.brick	dcdce058b9e3fe7f3b9f273bcbe70dbabd4a36d2d60e42959b7eb682d4ba3538	equivalent
MiddleFaceCell106.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell106.candidate_not_isWeightedPerimeterMinimizer	c4323cd7dcb81dd23bbb13e7a6da6db369314dc092c81930c824f1c12bfe05f0	equivalent
MiddleFaceCell107.Brick	fb97f7a9d17ed29f207a2bc58f493e65b7756f82f628a6504ff0176944fd563	equivalent
MiddleFaceCell107.brick	d79e9086a17cc0f1e5ba7ce1e0aeca7cfdd288120b9ccd9ac07eedd01db6a1ed	equivalent
MiddleFaceCell107.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell107.candidate_not_isWeightedPerimeterMinimizer	f3c170ddcf0f5f71486c4a7d455025f384a3d60752b417e48d9a09a1c7501bae	equivalent
MiddleFaceCell108.Brick	7a12777f272bea6b574c86aca5e0e027b839357478e71accad6ce6ecc655ed66	equivalent
MiddleFaceCell108.brick	dd865f669b8c5fccaafeb5146a05d4166797f26e8683d15f71fa7e3972bbddf6	equivalent
MiddleFaceCell108.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell108.candidate_not_isWeightedPerimeterMinimizer	e2399880ca42d06b51edebcdcc1162a133006306e9370f7c007b070be55e30f5	equivalent
MiddleFaceCell109.Brick	eff349c87e09b79fb97fc3a49f54b7478d67e316bf96fbf8c5451dca7fcb1400	equivalent
MiddleFaceCell109.brick	8e34bdd669070a698fa44b2d021dfebd9bd639a51e391cb53bd55912d658200f	equivalent
MiddleFaceCell109.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell109.candidate_not_isWeightedPerimeterMinimizer	875f3a258995fcc9566c1c6752f674ac21c416e158f0c088b0726d5313d63ef1	equivalent
MiddleFaceCell111.Brick	422f1140c77856bb80f87f2a668e15a32e1496e36e3c5d80735b28ca996bf036	equivalent
MiddleFaceCell111.brick	8d4d270ab7e7beff9107c59c176795ebf6b29af4cd78ed650d9e95debee763ce	equivalent
MiddleFaceCell111.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell111.candidate_not_isWeightedPerimeterMinimizer	bba989a84f2e693a08430344e5d59fcf650f736d817781053f3bcfba22f6f409	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell110.Brick	364e5651eb375637678c7401201c01287e60f9480217622bab46937282a3a73f	equivalent
MiddleFaceCell110.brick	2b973783bf5f5189781c989d58a5453d575ed1c5bf9732f9cf08e5db07628f77	equivalent
MiddleFaceCell110.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell110.candidate_not_isWeightedPerimeterMinimizer	349927598530c27e4d8f9890c5881cf9ee7e149fa20350a9a24837a42cc3f6cd	equivalent
MiddleFaceCell111.Brick	a0572ab587465cbd7939ef4bc9a5812128d7791b2582595b0979307269e8e2f4	equivalent
MiddleFaceCell111.brick	170ec15586f29cd18b465787aa85613f84c4f0cca1d8bab0ddeb17ddfb29d250	equivalent
MiddleFaceCell111.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell111.candidate_not_isWeightedPerimeterMinimizer	fc0fdb0e6d11b157a69766d9e95c0dde52d140ec249b874e8fb65a9f0d6cd579	equivalent
MiddleFaceCell112.Brick	1cefdf019326f37bcd23c4746a1f17260594ae1bd43fa54e77bbc59f229fb244c	equivalent
MiddleFaceCell112.brick	13fb7924d17e89d141f66fb85a48c1b9a6678969b8b75487fa8596207f305a5d	equivalent
MiddleFaceCell112.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell112.candidate_not_isWeightedPerimeterMinimizer	9d838a95d2ecd15a5c3a7b7579ec34eee2f0903e2a1cf65b4baf655d19534b4	equivalent
MiddleFaceCell112.Brick	20f5a3ac6954cc2a35235d046bc1afd1741df16e34eb1f0043b84128fb5fd7be	equivalent
MiddleFaceCell112.brick	645b0c50d437bc380f3e2db27b2241f847129518715cf148baad24cab7075a6	equivalent
MiddleFaceCell112.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell112.candidate_not_isWeightedPerimeterMinimizer	1261c9b50b52efdb6406a4fc5ca1c13bf40bba1520cfe0b4412b490d6c6698c9	equivalent
MiddleFaceCell113.Brick	01ecd47a9f84702695d2ab20463f5f6627b03642a407bd1be1654cc40cec7a27	equivalent
MiddleFaceCell113.brick	e068d913393aca117e9e68d3e1724ff00858fe7a3ae39cbd46590a079b9d6e60	equivalent
MiddleFaceCell113.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell113.candidate_not_isWeightedPerimeterMinimizer	63dcae663f848cca6593ae1ba84ba472422d8456046cfa86d039eee3cbce501	equivalent
MiddleFaceCell114.Brick	4b59b7712d40357b392a8bc1c9a38f091c5e88101e52319e23414ab54774979f	equivalent
MiddleFaceCell114.brick	3de956ca214dfd5cccd0799485e19d7fe9e92a3a639eb0adf86181891f74ff5d	equivalent
MiddleFaceCell114.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell114.candidate_not_isWeightedPerimeterMinimizer	5d72385bd489b20beb035d250ff87892f0c53643833a6171d4986d04a38bfde8	equivalent
MiddleFaceCell115.Brick	ab0196e6107213171313caebf38aabcc031fe291d6b066d8fff4d4fc5bf05994	equivalent
MiddleFaceCell115.brick	2034a2b13d7fff9a7fccf162e2c764fcc76926eb4e3793a46fcce5c0d415d65b	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell15.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell15.candidate_not_isWeightedPerimeterMinimizer	1ca64a20a3b67b4ace3477b29aea5b919693419135610bb1e9fbd03ba3faf125	equivalent
MiddleFaceCell16.Brick	ad4fc4e603a493f6adec4e71203eb84c0cc86d059ad4e496d5b16eb28e307178	equivalent
MiddleFaceCell16.brick	86e79a68a77907460d5ac610273fe45b85c112f814f81be4b71fb84d7214a548	equivalent
MiddleFaceCell16.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell16.candidate_not_isWeightedPerimeterMinimizer	fcaceeb4a8a1b27254a145e54ffe85d4902383a36e2630bf100580f45c2473b6	equivalent
MiddleFaceCell17.Brick	c0d813432e1fe40bffea0ab691cbd1ae0295160b22e492478dff5757390b33c4	equivalent
MiddleFaceCell17.brick	ad60b7a243e71d3581c0de53832c27f60174ec98af830acc003d60666d558d99	equivalent
MiddleFaceCell17.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell17.candidate_not_isWeightedPerimeterMinimizer	74f39636ed37da4a1c62c28cd1ae81d47d3e8c392e547791369a33627eafdb6e	equivalent
MiddleFaceCell18.Brick	d1f79827df04ecff873ff6923fbde15cbe8690196354c628b60f23899c657573	equivalent
MiddleFaceCell18.brick	9dc58f5b65bee40fc92533a497e8a65e558a6c35b50d3bd4244b2310fe605ae1	equivalent
MiddleFaceCell18.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell18.candidate_not_isWeightedPerimeterMinimizer	310e9dad68db89ce9f7ad60a1883d1e073f172a1f657a4829ff1d35a50509358	equivalent
MiddleFaceCell19.Brick	79bd62fd038bb44e1184712d2033563246bba03597c4586af6df9b3a5edf5c5d	equivalent
MiddleFaceCell19.brick	2d1dfe4ac74d7072ae8c067afa29ec8a6f115fcdccc6ad9f53d441fc81f8b986	equivalent
MiddleFaceCell19.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell19.candidate_not_isWeightedPerimeterMinimizer	a8d4c101ed34f7defea68b7756a9f1130822565b72fc50a30faa5428924c899e	equivalent
MiddleFaceCell12.Brick	011a73b220738cb17074df2b18f45ebe2ab8d59d1e0642e2f61e8ec4bbff55b2	equivalent
MiddleFaceCell12.brick	a170dc2fe5cef8659c28f7bac53ffa921a3bed2ce822770eda493d548da6b184	equivalent
MiddleFaceCell12.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell12.candidate_not_isWeightedPerimeterMinimizer	2c79e479b97dcf55c49fc1e2183763dc5eccffa0759c249045ed8f55fd8e71f	equivalent
MiddleFaceCell20.Brick	049a6f985b14968233b76aa3dc3241eb50ddcc4ea443d1d737d573ab3af0bb77	equivalent
MiddleFaceCell20.brick	47b194ac9325aeb229e86dc2bd50fe3f24c32e4887600506f152c39f177206b4	equivalent
MiddleFaceCell20.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell20.candidate_not_isWeightedPerimeterMinimizer	4b2bd45881d855e5f66887d56eadc7166d8461cc0983298a6496b6551cbc60c6	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell21.Brick	90ebb1927a9afa9794678bd9c0887ab198ac9eeb40af36e3e80e62c809eb2378	equivalent
MiddleFaceCell21.brick	7add5151efd595536d0e2f9d261dc6c749877ed7f719bb8cf07206afca4f78ea	equivalent
MiddleFaceCell21.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell21.candidate_not_isWeightedPerimeterMinimizer	dcbb0c95c3ff640f798d4a4548b38fe23481688b19b2d2590794d3903e030c0d	equivalent
MiddleFaceCell22.Brick	4999b65af88e2a6bb34c6e2e5b81abebb41e5a58d487e76595642b47be5fef4e	equivalent
MiddleFaceCell22.brick	b6794f19d2ad69e4db918a47ef4af5a85b1103a9174a147d14629e2874ef00c5	equivalent
MiddleFaceCell22.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell22.candidate_not_isWeightedPerimeterMinimizer	3f2eac28da23af7125633d0fc0818813a9ba150dc472fb62c664dc4557c9c6cc	equivalent
MiddleFaceCell23.Brick	f16d4ecaa36b559fa5cc11e83d02bc4e9c06bec9df35f1be22d168d58d4a2e16	equivalent
MiddleFaceCell23.brick	c93b38920cf23da2708aa0b11c82efb6b493353fe89ceeaaf62b590517851184	equivalent
MiddleFaceCell23.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell23.candidate_not_isWeightedPerimeterMinimizer	2c9ae9591b3f94474f5ce07961806ab51a5397725e0f4903e1c8b40dd0fcb6b9	equivalent
MiddleFaceCell24.Brick	5a8c83462766a16f507824d8f0cd14a5ba4eedbe88c68797cc0694632c17754d	equivalent
MiddleFaceCell24.brick	72da8f7d24ba964179fc11b84b13e8ffd0875de498efc2239f897b1e85c97e00	equivalent
MiddleFaceCell24.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell24.candidate_not_isWeightedPerimeterMinimizer	7993f31b20f5091a307bccc814d02ea4a8ef25fa04ff8f6ba5c6fedd5be99f7c	equivalent
MiddleFaceCell25.Brick	08591daa72f0dfa9205ac13c3b19e046ffa0dc133c96da376267ef83b9fa56d6	equivalent
MiddleFaceCell25.brick	df943767dcac7319b08845e3e7174c9d315d0e4da6e7ed51e929fb0fd2e40e4a	equivalent
MiddleFaceCell25.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell25.candidate_not_isWeightedPerimeterMinimizer	fd0c9e7c06fe3db2a20f03a2f551b97d3025cf5b074a07647cb9e45d584ca311	equivalent
MiddleFaceCell26.Brick	cfae21b626c40d28ba07b5dfce0fb6a3d1bd70c2558e86462a71f450a8b3c111	equivalent
MiddleFaceCell26.brick	25d98e6750e3cb05331116592870fb40841e9b7905885f36b1778a2312a703b	equivalent
MiddleFaceCell26.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell26.candidate_not_isWeightedPerimeterMinimizer	2189db9af68806069e8a36da892f06b984488c240770a6000e024c02e514184a	equivalent
MiddleFaceCell27.Brick	045e2e7fcb3bcf6b90169d7b4491e0733e48c078f4db9b95c13492e5f0a49ac7	equivalent
MiddleFaceCell27.brick	70e0712d29560bc034491be4133c1efb3e4a4e900e1f7d4041e104cc136b0bd6	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell127.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell127.candidate_not_isWeightedPerimeterMinimizer	d103527cd78c17fbd3bb014125faf1ed069668dc574f0ef1713856da0836c2ac	equivalent
MiddleFaceCell128.Brick	47425b6995b04404bc0f959aaf7a0aa326c6d60e63cb2020fd96dbbd113e3a70	equivalent
MiddleFaceCell128.brick	8d0fa6a129fb0893c39672d3c88502f1fd433accde985e8b720a6d61860e3896	equivalent
MiddleFaceCell128.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell128.candidate_not_isWeightedPerimeterMinimizer	f6850a423fa365710137e49ede10fec12de1e3e7a507c28a121cc2db107a427d	equivalent
MiddleFaceCell129.Brick	730baff5b0ee3434e50b9971440701aeb0f604f541f7ff59d7d24dbf151d199c	equivalent
MiddleFaceCell129.brick	15f28419af91b0a5276012e31d35d7b2257516a8467659235fec0dc68aebcc	equivalent
MiddleFaceCell129.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell129.candidate_not_isWeightedPerimeterMinimizer	4ba81e0957f1e1644ab8c0e9eeeb5577af35fcf820abf1328a44cd5c188baf60	equivalent
MiddleFaceCell13.Brick	3c982b71c68a0f3d59d23f747d50174f73c4e56a59a9d8d627e7ec9582237f13	equivalent
MiddleFaceCell13.brick	7e5f321d57734cd54642b4c9a74e698411a75699495ce0d1322082caabf79b7	equivalent
MiddleFaceCell13.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell13.candidate_not_isWeightedPerimeterMinimizer	bdc1fb8217b716abe1637d56c691debde3c9e0a58ac781fd83f70193653367de	equivalent
MiddleFaceCell130.Brick	34c60a0a002d690e3908aa790ca9e292c86dbf37022bb7411e97d74b4b37c113	equivalent
MiddleFaceCell130.brick	fb3e2254ed16a1230d7bc1d93df46b0c315735872f2c3e42130456a2935a7db4	equivalent
MiddleFaceCell130.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell130.candidate_not_isWeightedPerimeterMinimizer	ace3ab9d3255b54ba8f67b856f01dbecd3259b6aed66ab342ea9a852cfa148bb	equivalent
MiddleFaceCell131.Brick	22ebbfe17f3adc5b7c39368091701a2353cc132bf6094f9cc6873c4da10940bf	equivalent
MiddleFaceCell131.brick	75e470ca5d19cdc39769ded4aadbd6daa1cb188ad8446171a7135e4e90ba6762	equivalent
MiddleFaceCell131.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell131.candidate_not_isWeightedPerimeterMinimizer	3f63b972acb1f4943f46a4d8cbcf4cfc06cc82862fea5d7c988258da5bd6581	equivalent
MiddleFaceCell132.Brick	7b184f7e370273cb649a2d23456c9a9efe9d0d676ca64784298e783f48038484	equivalent
MiddleFaceCell132.brick	3638094e0418ffed437ca8977e42b6ac4eb44158cca8b82278b4fd8d80b1cedc	equivalent
MiddleFaceCell132.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell132.candidate_not_isWeightedPerimeterMinimizer	83fa3b157c41d8ebe8daee20ed6cf1979b4038133028e8b39a70c4972d1d1522	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell133.Brick	e4eb49d5987ca3dbb972f96e24fb122e61fa6da1f5a27858e81577e192df6101	equivalent
MiddleFaceCell133.brick	1715f8a6f5a6faf227ec0b759e1911d3f458ec5b6c06360342a056c30de4b6a8	equivalent
MiddleFaceCell133.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell133.candidate_not_isWeightedPerimeterMinimizer	993c7ef257d7790e832e8a2d443c53218f112b59b6a87b84387ecf9dfc56b960	equivalent
MiddleFaceCell134.Brick	600111299cf800a5af5507411d009491aecb343e14c0fa7d359596e57fad5296	equivalent
MiddleFaceCell134.brick	ebc09de22cda073ec8aae961a1c3c74e4bcd61f837f93d9c30387d35fd78d6d1	equivalent
MiddleFaceCell134.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell134.candidate_not_isWeightedPerimeterMinimizer	ba88fe90e268803835e68ed9cf41af33c2048f5ac7fb4830063e2e511db50768	equivalent
MiddleFaceCell135.Brick	df71c07137cc5af5c4d9a83aa9856d7ba5a101a0dfe0e21681f944e22c416268	equivalent
MiddleFaceCell135.brick	9675d1896bfc496d644a8c98209a6242faead44fdd2f05e6188c8c43d555620c	equivalent
MiddleFaceCell135.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell135.candidate_not_isWeightedPerimeterMinimizer	b6da973a892aedf8afcd9ed65beafe8c719fa5d43e1acbb0777bb083b865a38e3	equivalent
MiddleFaceCell136.Brick	3e4dfd8587f83c8099531bf55c217014637dc59b3cd5849abdb8cbf2dd4e183c	equivalent
MiddleFaceCell136.brick	ca7bd4588d8afc48231b0c7548c0529eb064abeefcc154d1488a1c982af279d	equivalent
MiddleFaceCell136.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell136.candidate_not_isWeightedPerimeterMinimizer	fbba412a79446178749cf0ab49fac9a34d76fd66beedbb9c4e192a9c87960804	equivalent
MiddleFaceCell137.Brick	efe1921541ef7d27fbcc7a0ce94a60ddd6cc6b27eea924e91cb9a51dea293ee2	equivalent
MiddleFaceCell137.brick	80b726a309f53af37c3906c8f094aa67cf1cca606c030c9833542bf26fb020c9	equivalent
MiddleFaceCell137.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell137.candidate_not_isWeightedPerimeterMinimizer	ca2f927cc91214d12c9b048fd17f834624ba8974459b45179a3c52d335484c1c	equivalent
MiddleFaceCell138.Brick	e98d854f1bd927500ee2821c38b9d65184ab91938777a6ba664c0b696dd04b3d	equivalent
MiddleFaceCell138.brick	695cb02809bd74ad06040f534f715c882a8a65cada5fb66e07d5ba85e096f09a	equivalent
MiddleFaceCell138.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell138.candidate_not_isWeightedPerimeterMinimizer	98f1e7151b39b252d1496c132272cfaa23201496f712c2358fc7bbf1c699b6f9	equivalent
MiddleFaceCell139.Brick	6ce801d52ad8adfa03d7efac7d5f517d3a048c3703428dec26f7d3525d55ac2	equivalent
MiddleFaceCell139.brick	e77f251a6565b4bcf9a5e8987401386d2a526fc1a574a71e95c0e164be6bfd18	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell39.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell39.candidate_not_isWeightedPerimeterMinimizer	11154e3d5d8e2a2adc8664892701753d3a3f49e7538dc6be34ab8ab7f2aed49e	equivalent
MiddleFaceCell4.Brick	48cd60fb0640964dc38d625c1eb0182c48e15d3595ef7bb0e0c6cebf94ead9f5	equivalent
MiddleFaceCell4.brick	76cfc2f29f682f43b63aba2f6c4f71dda700f9c2d080e2953d644d5320ba90e2	equivalent
MiddleFaceCell4.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell4.candidate_not_isWeightedPerimeterMinimizer	f580efaf916338aa54ebb6014efe4aaf0309e7918f40d6f23e93cec375dfed58	equivalent
MiddleFaceCell40.Brick	0d5a5466e6bd582d980c772ee1a718a398bbf3df14c2b8dbb52ee40fbcd0c391	equivalent
MiddleFaceCell40.brick	dc48ab98a7807fff971491e811b2e69bc49ded91c13867142be584b4d068ac04	equivalent
MiddleFaceCell40.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell40.candidate_not_isWeightedPerimeterMinimizer	abb19e3c8efe82a1547c9dc36773b9af383b97eeefad59e703d13ad54005817e	equivalent
MiddleFaceCell41.Brick	4bdad4719d39d9a98a7298d2c42560e620e90066535c6b965736c709257d2ddc	equivalent
MiddleFaceCell41.brick	bb95231040a422364cab97b8c6bad59479e1ff3634f4e116d63725f2b15184fb	equivalent
MiddleFaceCell41.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell41.candidate_not_isWeightedPerimeterMinimizer	5620720074b49e865939ba17ba360a82773ef2dd1bd3828d6ab4ab0c4b7b7e33	equivalent
MiddleFaceCell42.Brick	29bf5af8895817fcaf29a0a5314d691781eb1ad105d35ab8febee15fa74ca477	equivalent
MiddleFaceCell42.brick	7a9c56b7db11cad9037c9e7626ba2ff81e5363f7e943efe7b6d744c881bcd2f2	equivalent
MiddleFaceCell42.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell42.candidate_not_isWeightedPerimeterMinimizer	068e43d5aa65631d3e8af92c77aff5f4525b7e9f34c8e11314bf09c062fcb700	equivalent
MiddleFaceCell43.Brick	c39ab82e10945b205921be5276ec9aa0a8b34c4f46260057c5bda2953be92a50	equivalent
MiddleFaceCell43.brick	b125d7317a4df21e4cd5c594a4e895e1278257c61ddd0ad91ccd159f49cd6eef	equivalent
MiddleFaceCell43.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell43.candidate_not_isWeightedPerimeterMinimizer	36490d10389560d7d507fe1efe6727b77967ab1e60ade75d23d7302efd157f24	equivalent
MiddleFaceCell44.Brick	2de2668313ab5c07ad11effca4120357c634750ebf03f33d29261d2c6e798b92	equivalent
MiddleFaceCell44.brick	5d0ee609c8e4e5264bc03fe7cf6290e3552d0896e304e420ad3ffeea1f6cc80a	equivalent
MiddleFaceCell44.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell44.candidate_not_isWeightedPerimeterMinimizer	1bdee456e4b24086876480ccc10a47ac35d9d46eceee2ccce9478ed3a4a627fc	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell45.Brick	6a83cd0fd0879451c629b6fc20ff71078cc27c407b6de8a1fb35fb793a9b775f	equivalent
MiddleFaceCell45.brick	fb1c811187719487e22860d60a443b4ba65c5163b553801a2133ec09518995b8	equivalent
MiddleFaceCell45.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell45.candidate_not_isWeightedPerimeterMinimizer	72e61f315f257f14446b71cca19f4a37b00e18103be742db82982e0c5a0d2ae1	equivalent
MiddleFaceCell46.Brick	5280a023d19d033c8b19b594c67cc6f94119edd398ec4715bd4e513d57e2eb83	equivalent
MiddleFaceCell46.brick	a8bfc2a85d0c07f49a0b0645539e666572f88d911bc9baa61ac366332453b0bc	equivalent
MiddleFaceCell46.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell46.candidate_not_isWeightedPerimeterMinimizer	29087dd65e20e2108e1203971e3a6f6ff4f5df419fce019e29f220d201804173	equivalent
MiddleFaceCell47.Brick	63f7920068833f41c2dccc5b46afbe3dd793a98e71f15b9b00d3d44ea5f6010	equivalent
MiddleFaceCell47.brick	4ec8e0c7c1133c4007e3b85ea8f90928a9c621861c061d97139e12a750a731bd	equivalent
MiddleFaceCell47.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell47.candidate_not_isWeightedPerimeterMinimizer	32ffa8c9c70d9af3f8a0781dd43e96d3f1bb8147b1490c7368c8bde57b25bcb7	equivalent
MiddleFaceCell48.Brick	01366ad92590a79c3eab14cbabdc5a8866a5528e479a608c11d69b0ab46896fb	equivalent
MiddleFaceCell48.brick	dd8fb665d1f533527d1c42a91b475f6ace059911279f1369c02c4db3dae68ad6	equivalent
MiddleFaceCell48.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell48.candidate_not_isWeightedPerimeterMinimizer	0d2f461c89e5827365ea86b81b5d9bcf755e80325caf1af6a1c73d0eebf2b556	equivalent
MiddleFaceCell49.Brick	7459c849bdb8249ef9145590e4f25d650163ab067eca7b8302e575f09d87fa39	equivalent
MiddleFaceCell49.brick	3fab3c59bb3e9a2be3ef27fd40976374e5e8fb78b79ba0c1ee074a333b896be0	equivalent
MiddleFaceCell49.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell49.candidate_not_isWeightedPerimeterMinimizer	76313ad42a6d2185ba5bda9d348e7ecef55e41db83dbdc3a3c20150467e34045	equivalent
MiddleFaceCell15.Brick	9ff337826da5e83fda8f5e259812b9a6fb0d324165fdc97d31c356dde6b37c1b	equivalent
MiddleFaceCell15.brick	f8f178042e6ea085477d36184ae9edfd4f3087b36c6818c69791a66ca7604b72	equivalent
MiddleFaceCell15.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell15.candidate_not_isWeightedPerimeterMinimizer	203ecd80ba5622ebae38198c8d5dd3f62dde91a84f786f85b1d41f673f431b6d	equivalent
MiddleFaceCell150.Brick	757fc72813d427c4b786712da7a49ba4e4b2a061f23dedc0f3b4834322b0ae33	equivalent
MiddleFaceCell150.brick	bb164d35ab272eabda268f960c4ec1d4b3e8d493de5f4104d3a6f423e5037593	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell50.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell50.candidate_not_isWeightedPerimeterMinimizer	a0d1963f0fe12d432ac674dd5ffe6f064ee1ef4e4ae4104e24a5564404f447e5	equivalent
MiddleFaceCell51.Brick	9384d7973d7f928352271500593bb86d4967fde6523028cc361c872a435d06a1	equivalent
MiddleFaceCell51.brick	507c6242616e4e0598ef22db33665a8866238274e6be3608d1a3df81da2e2b35	equivalent
MiddleFaceCell51.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell51.candidate_not_isWeightedPerimeterMinimizer	6f2e0bab5951b6f24bf00d086621baabdc85d1be2c07e01023dbb27b6ea7e2a	equivalent
MiddleFaceCell52.Brick	53cb46e23454e7fdbbaa2e488355264235f540cee6716663e79c0c5a878ca73c	equivalent
MiddleFaceCell52.brick	75a90bebb10d7271d32c5557fcf371c93767dec1f9f8fb5632d22e679f0071	equivalent
MiddleFaceCell52.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell52.candidate_not_isWeightedPerimeterMinimizer	eb36a0c715aa59eaf2add9fbd1c23cc8978af55f1fab1ccab93dab08257ef5e	equivalent
MiddleFaceCell53.Brick	1b372d246b1ec8a516e87941fe764d9027c5d8aa8731c9c7466bdf5fa6c99c7e	equivalent
MiddleFaceCell53.brick	34cca59419dede788ed44d5f18c049ba33c05eb8193f2c1c26309a762802ccac	equivalent
MiddleFaceCell53.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell53.candidate_not_isWeightedPerimeterMinimizer	028aa7a1e8a7043a0a9875776a74c9565e6f8b6c086c51cdf6d8c5f3a02622b8	equivalent
MiddleFaceCell54.Brick	019b5e384b555dc6fdc2ea9210dda85105cd997c274c66864bdb7f65d97ad590	equivalent
MiddleFaceCell54.brick	50e037c1361e8b98b2190c4cb74a93ed535c6c4632df3cc34e8a69d849cc6fef	equivalent
MiddleFaceCell54.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell54.candidate_not_isWeightedPerimeterMinimizer	5c3c33c392db59a4dbd25f45ce48afb44af17496ec33d42719eaa5165740983	equivalent
MiddleFaceCell55.Brick	5eb8433f72def7696c21715fb4bddcb5436bf6740e36ded4599742bced105f43	equivalent
MiddleFaceCell55.brick	b2498e18c442cdedfdb3e1e5a2ce9515d035b5930a3bedb2b84f389be015c4b	equivalent
MiddleFaceCell55.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell55.candidate_not_isWeightedPerimeterMinimizer	96e4d8ca1cc25d7c02936550ba9b0af6050acddb0e55da78e6ec622d445090f1	equivalent
MiddleFaceCell56.Brick	bc7932632cb06e7872a745a8bda2ffe4ff695439775f570f7930977d5f5f2770	equivalent
MiddleFaceCell56.brick	2256baccd0a5c1c85c9d448ece0597699de49c994e7a1ef0558f1005589a482a	equivalent
MiddleFaceCell56.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell56.candidate_not_isWeightedPerimeterMinimizer	9b6d7728072f7fdc6d0b4db27b0aca2a2ddc8e5fed3611a09507220660196fbc	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell57.Brick	fa632f1bf02ed6f70b702ef7e883c71f7b3090964739360f5665765c9c393075	equivalent
MiddleFaceCell57.brick	2a522ec8f1a02d5846061b2bd89947208f4d68af82c71b4dd024c5900c71341c	equivalent
MiddleFaceCell57.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell57.candidate_not_isWeightedPerimeterMinimizer	376f02a02f821a6f6a31dc563c2e0f6fc2bbeb4f8d828bd99d138b1b140fe303	equivalent
MiddleFaceCell58.Brick	e0c6301d9398a187c1684d4ce31c7e2a663acdfe08456275a865b7b3102287c	equivalent
MiddleFaceCell58.brick	08c3e019272f8327acc0e6150a36491eeb6d3577997a3b6c5897a91f2d40ec40	equivalent
MiddleFaceCell58.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell58.candidate_not_isWeightedPerimeterMinimizer	6ed293f5a6565943f6e0ddc509395dbc9f6eadc61aa0d70336488a1457b3cf9f	equivalent
MiddleFaceCell59.Brick	93366befc77fc4532e58c8c07c684528c95703ed3969c09c3af97a3dd978fd8c	equivalent
MiddleFaceCell59.brick	9af0f4fbddb434df63f8b118938422c037813b7992cc4efc53c4896ec73bb517	equivalent
MiddleFaceCell59.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell59.candidate_not_isWeightedPerimeterMinimizer	ca09e9c4b920e14d9b0055a64681e496d518dc550f2e2ad38ffe3ce991c54c3e	equivalent
MiddleFaceCell6.Brick	2c83651390c0d47851e2a70832b426de5fcd21f0486225e842bd3aa3a8f99e4	equivalent
MiddleFaceCell6.brick	c1935f798b415ad91f8996c1b753f4461b2f3344ae0b8108a63450aa93488c9a	equivalent
MiddleFaceCell6.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell6.candidate_not_isWeightedPerimeterMinimizer	8dd35623c9fc58a188684e0ffba352cc8f372e6f4e6dc8b3b707ab38a32e3532	equivalent
MiddleFaceCell60.Brick	57f78186ade7afa2a689f64476c530065a66da6a66a9ebc8ac590c571a57c353	equivalent
MiddleFaceCell60.brick	49f55084cc0a3ef783c0d928e40479443bee26669f254368f995e998c8f2dcb4	equivalent
MiddleFaceCell60.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell60.candidate_not_isWeightedPerimeterMinimizer	17b0c72d3f748f7ba74579f27875c40429209b4533fc2be18bd07b085790e06f	equivalent
MiddleFaceCell61.Brick	a21e3ead542cbf21c18e37ec15a35f5d1f4312c14f7730d1274c3faa71190557	equivalent
MiddleFaceCell61.brick	44a41ae2a19bf7d12322f907c746b316213e2d91d3dbf772087644d10150f696	equivalent
MiddleFaceCell61.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell61.candidate_not_isWeightedPerimeterMinimizer	59e39c5c4c9d75442163963805af636e4bf2c0eeb8885a92e2856a1e5edcc105	equivalent
MiddleFaceCell62.Brick	ae5b229b0d739a07b2a21fc80fbe78bbf956c5673e002614b179e9a787c2bb14	equivalent
MiddleFaceCell62.brick	893eda77ff3ace51482a9c8ff9fe9ee7d2e2357abd1376f4cc06bf04130fd508	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell62.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell62.candidate_not_isWeightedPerimeterMinimizer	311ecacd555e3dfd79b0cafadbb2999b9e3863e61caac9a27d64b0cb18edc4428	equivalent
MiddleFaceCell63.Brick	acc685650a48f4be5d2fda8f2f236ef5ec5eb2d7d69f84b899750d8bc9108219	equivalent
MiddleFaceCell63.brick	8848cf2f984c8cb8295c39cca2de587f59c33c72704ae7495304cd6daff63571	equivalent
MiddleFaceCell63.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell63.candidate_not_isWeightedPerimeterMinimizer	0bd6efff31944c4c935815b0ca4c5630cfa3d7968d8a445f43d01ba427a50d721	equivalent
MiddleFaceCell64.Brick	6b3f307ed54f914c6ce2b1071f94dbf36a74b3ae65bcc82f18f6f8df3d4665bb	equivalent
MiddleFaceCell64.brick	02964721a0844905e5a998ee8d930ea6a21b40404ef4869cc032803782a9aef8	equivalent
MiddleFaceCell64.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell64.candidate_not_isWeightedPerimeterMinimizer	3a7ff9a8302c41ed4b3ac35692f785b3ba35492d2e906a1c8fc75a74fa4fa126	equivalent
MiddleFaceCell65.Brick	f0d6c18d179e7246079064b43e887ec300f6e6e19ea96eb2ab89e446383d7d19	equivalent
MiddleFaceCell65.brick	2c2d9ee4dbcd9e63cee8c8cbdd3dfbbec3307843aec5ad346c90615aaa0362a9	equivalent
MiddleFaceCell65.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell65.candidate_not_isWeightedPerimeterMinimizer	3fbe8040d7a6f2069799ff79e7aa8e9ed2b76b5f190e6fad37dd0bedc5d3c898	equivalent
MiddleFaceCell66.Brick	4e929fcb1272110a43d05e51a429753db9e2aa3a7afdde7c61b635f28f68ec70	equivalent
MiddleFaceCell66.brick	7d0e3aa6d2b535f398d12fe40f99af27f146291a9a8697d042b4cf5aca0711fe	equivalent
MiddleFaceCell66.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell66.candidate_not_isWeightedPerimeterMinimizer	59fef6b88141739652a2fd3810164fbc9627974e5678a392c22a281deee9ae11	equivalent
MiddleFaceCell67.Brick	5c6c9bf60ced5754e62e13bec820feecf58c055e90cf8be3d49cb661580cc85	equivalent
MiddleFaceCell67.brick	6dafdf0b9af15ad7508f097d54ca9c462578b548a10482aed6f2c99db3905a0	equivalent
MiddleFaceCell67.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell67.candidate_not_isWeightedPerimeterMinimizer	2432d56a46c435b4344119b9fc92429eee8a81ca54d36c3a8a74a8e594c52c11	equivalent
MiddleFaceCell68.Brick	14d630bf52f4319721b460e79158d608626c195698693badec4784cb95033179	equivalent
MiddleFaceCell68.brick	a156d2a7fa4a1efb1f5890c59ac72adabd452c668c3f04ffe075732812483d14	equivalent
MiddleFaceCell68.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell68.candidate_not_isWeightedPerimeterMinimizer	1f91a9ddad52e966b8337d2f8e4fb7b264d403d2659d4249c1e2cb1e13b717f7	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell69.Brick	f3d90eb5d1befd4b2921fdef318326e423d864868283d67cb14a7b9e35d762cf	equivalent
MiddleFaceCell69.brick	fa73967f9b53a4fcddfcf8c2268107278daa93a7c35046c3233f8a8206f37c81	equivalent
MiddleFaceCell69.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell69.candidate_not_isWeightedPerimeterMinimizer	2cfdbfd5c3a32b738a4e5a0b1496f80d9fc6fe995018441421b741712d35a87a	equivalent
MiddleFaceCell7.Brick	a21b216b1cd9d16284f53df6bd94b490c6b6caa2a11b8d8b1d39855f250c9531	equivalent
MiddleFaceCell7.brick	40b0173b10d63d32bd765f8e28397d57eaf473d9fd834431a37251824234526b	equivalent
MiddleFaceCell7.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell7.candidate_not_isWeightedPerimeterMinimizer	4434b695c4fb5f10cd2218845feb07df257c58a0cd30c28943a08e7dbe523dd	equivalent
MiddleFaceCell70.Brick	9690f836b7dd451b9296f7cfae7d07a4c0d5f470d494ae52aa661ff35a58d38a	equivalent
MiddleFaceCell70.brick	61f4fd22eb5d96ff733096e5989f28bd4768b5c36e8970e2c3ed8d759f66a035	equivalent
MiddleFaceCell70.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell70.candidate_not_isWeightedPerimeterMinimizer	2263b741b2069b51601b284e3ef952a054671cbb5daf224d60a7047cb3658482	equivalent
MiddleFaceCell71.Brick	6437a07026d594aab523cf1d51f33441df96b6b9ac5af25132350846468eb767	equivalent
MiddleFaceCell71.brick	072d9d8d624c2a2d5c698a163db512a8cb1ab5efd669c3258bc47ed5361244f4	equivalent
MiddleFaceCell71.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell71.candidate_not_isWeightedPerimeterMinimizer	bf33b0a72a5fc4f7cc099260d839869a60a4466fd82f4ec39ef8a6d5421c416b	equivalent
MiddleFaceCell72.Brick	d917658e3e305283f973bef5d7ccdfd340b32cf03033c32a776511e20bea10be	equivalent
MiddleFaceCell72.brick	ce2e84ff21f8558f6dde915afaf766a80abf404ac1ae8271443493df29d82f4e	equivalent
MiddleFaceCell72.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell72.candidate_not_isWeightedPerimeterMinimizer	a79d2bac93d9c1e432858003aafcd2a9e4c655dc43d3b89d177b016976e3f987	equivalent
MiddleFaceCell73.Brick	1ce244173af7f7b3692a20829f115be2c11f338ece5a8cc01179b918837bf4dd	equivalent
MiddleFaceCell73.brick	1ba7c7ae7d53474a5ca9b4b72790c16197bed85fb1d28b29e12352327f011d06	equivalent
MiddleFaceCell73.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell73.candidate_not_isWeightedPerimeterMinimizer	3531ba96d91b6ff96ed27296ff32fa125c037ca13e6aa8779a4fd865a7f86ced	equivalent
MiddleFaceCell74.Brick	d6e02804be6c550f98e48a2b10b3ce2c44b481d6ede506c43aee8937b229160c	equivalent
MiddleFaceCell74.brick	672d336bf508cef45beea04d784a1adf379f226a79182d5fd8daba1d1583ac7a	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell174.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell174.candidate_not_isWeightedPerimeterMinimizer	74c5223a1d7bd048fbc29e479d022778b1dd9b8a95c03bf8df544264c0dab7b	equivalent
MiddleFaceCell175.Brick	6d7a8c3bd56af88c3ac137a2ca4a9a31a793c2a3ff02a19268f5b04d01143587	equivalent
MiddleFaceCell175.brick	c9ea8c315e795c0f620936e446292bf33b5ba7c00231eb80f90376651edaa74	equivalent
MiddleFaceCell175.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell175.candidate_not_isWeightedPerimeterMinimizer	572604102a270e038fcb33cc32bbd988228223b87181039ef8ff923d28a7da66	equivalent
MiddleFaceCell176.Brick	de78088c38e40ee33dbea20bbdb174a823674170d4cde2dcd76193a582a617f	equivalent
MiddleFaceCell176.brick	d9df6035cdf8745fe1d6b40482437dee6307251ce63011665e0ffbc90d91950f	equivalent
MiddleFaceCell176.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell176.candidate_not_isWeightedPerimeterMinimizer	85936dbafb78201c9b4435a5a9629ffaab693545c045a4571ad4d22f3c1da36c	equivalent
MiddleFaceCell177.Brick	ad2cff64744e9cc38316842db67072d50d3475f2ea8e0f85f790cc6310dca1a4	equivalent
MiddleFaceCell177.brick	7af9c7d378463801e055d50ca1173ddfaabf781ab041d17473fc4151b411a2a7	equivalent
MiddleFaceCell177.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell177.candidate_not_isWeightedPerimeterMinimizer	143e8c199e91c818b6dd41c233f831915991114c5e792db039d2c23e1510dfb7	equivalent
MiddleFaceCell178.Brick	4a4361ebefb331a6d9e656d3b13bdfb8c1689bc47a03ab5d99a6667e66e8ef79	equivalent
MiddleFaceCell178.brick	ba18ab673976f8dce8f1fa61f0aaae4d0c792585ac357cdfa229d8a026944118	equivalent
MiddleFaceCell178.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell178.candidate_not_isWeightedPerimeterMinimizer	7d32b14f172a072c1f81e09b8198ce9ca054526eb85b7081f58a202d9a55d606	equivalent
MiddleFaceCell179.Brick	e06715b99fec3ff6fd1569ad58cc754ed7dce78a4cd93407fef33c6ff66f694	equivalent
MiddleFaceCell179.brick	05b3715d4d1e52a3d4b49c73018c3d6844baed464f967adc9b083fba5fe1247c	equivalent
MiddleFaceCell179.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell179.candidate_not_isWeightedPerimeterMinimizer	6cc0c763d029a2411b2b4b4273e670060f1ccb0ccc619a39c0f872d8aea8139b	equivalent
MiddleFaceCell18.Brick	c1bd829ce56335f7e21ed7dbedd56b9cd7ade514d87668b57269b6c3ff22304a	equivalent
MiddleFaceCell18.brick	615e5073546f89e29cce77d7b7428fe8372b3b277471bbb077692c616fe40a95	equivalent
MiddleFaceCell18.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell18.candidate_not_isWeightedPerimeterMinimizer	a46e2f11d26c8841fd9af98cef1778f32897543897e1ef81c0ac324e5675c610	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell80.Brick	71e2309537720e847c9a6c72c4757a0261f648fff89f6278c7e52fe1c21566aa	equivalent
MiddleFaceCell80.brick	560385cca95d01232ddb9cd033540d339e48b4ca681fc658ed45cc4b5be90739	equivalent
MiddleFaceCell80.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell80.candidate_not_isWeightedPerimeterMinimizer	1570f8f2e91871b795ab0901244f47648fd0d28572228aa7fdd47f888a483125	equivalent
MiddleFaceCell81.Brick	70d98ed149a553d35636fc79a3ebeb3cc82f6789828acc336eed4e42f995db2	equivalent
MiddleFaceCell81.brick	2884468804590f9fe03bf7006b82261b45b96c1a5c896b5a03917dc9a95b566d	equivalent
MiddleFaceCell81.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell81.candidate_not_isWeightedPerimeterMinimizer	f5557f592400d59db127bc685a091e7ef1a18482a26c8ec31f57726abc106f20	equivalent
MiddleFaceCell82.Brick	08cc483ee9f34fd43de5983ec3e372ac8feb4e39aceba7597dc67e47f129ee8e	equivalent
MiddleFaceCell82.brick	de42acde7caba71c256dc8337d1165502ad5b6b84439309329820bf7aa04d2e3	equivalent
MiddleFaceCell82.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell82.candidate_not_isWeightedPerimeterMinimizer	10acd939e2fc5525f8337623d0f1bf2460ed4746e5756849c9fd336eabfa6329	equivalent
MiddleFaceCell83.Brick	9650ea6442abf5e43c8ca08371f29586e25d18368b53ca610562b57ba2c0f52e	equivalent
MiddleFaceCell83.brick	62f8bbc3d45be2b13d1d935f7149bdc83805f74acf477c07dfb0b5e6838758f2	equivalent
MiddleFaceCell83.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell83.candidate_not_isWeightedPerimeterMinimizer	666d81da2d0064ab8100666129b2ebea74fbd964cfd2fb5f9376f32a1048d99e	equivalent
MiddleFaceCell84.Brick	d34e8c6620b44cbf04fbcd8b440d3b9ae90a70cafe2ecbc7721da6e271a6e603	equivalent
MiddleFaceCell84.brick	05af87b3578cb7840d1bd95184be863c619a618db671ea9c370ceb2342f147e7	equivalent
MiddleFaceCell84.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell84.candidate_not_isWeightedPerimeterMinimizer	ba359568852f017ef26961a4dd130650067528029288a7ed1c08b8c3ba9386d4	equivalent
MiddleFaceCell85.Brick	be024ce60ca137be0e5fdc9363fe54718e704c4eff09e8bae759ff882a7a15d1	equivalent
MiddleFaceCell85.brick	6642369ddc479d24bfdd9974ef5360aa3943fb1fbfe5891cfef20f9b208065eb	equivalent
MiddleFaceCell85.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell85.candidate_not_isWeightedPerimeterMinimizer	a6247b6da305574c9ddee7bdd9a1cbc5e48a20e811c6305962499e42b7521679	equivalent
MiddleFaceCell86.Brick	10ab4f94ecbef3c8c3567ffd6557093c16f5787dabaddf2935b5c822187668c2	equivalent
MiddleFaceCell86.brick	97f3fa8d817bd40b1b0a001113cedda05d1caeee401240447c3cd8951274f496	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell86.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell86.candidate_not_isWeightedPerimeterMinimizer	dcc22c8bd8491f3c0735b3ac99fee72450e415fd312e9c10faf89345a30c7751	equivalent
MiddleFaceCell87.Brick	cbc129c276b8619e093d753c3d53b663219b8d3900469455b42811bbba02dad3	equivalent
MiddleFaceCell87.brick	590c31ec2907d440625fd3defdc16af9124c517ca7b2138b01ef1c771042a2d3	equivalent
MiddleFaceCell87.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell87.candidate_not_isWeightedPerimeterMinimizer	1983f6b5d120f11b9df46d4a05bebfef3acf07084eb9b691c349d23f756d82a8	equivalent
MiddleFaceCell88.Brick	d756c619ba11b2fb174f79fe7c028e274d0503ddd7e8f5e036910e91ab7a074f	equivalent
MiddleFaceCell88.brick	3bc79fef75c4cc2bfb54caadebfe4d0a6d32c32d710ab85135791650552c6b99	equivalent
MiddleFaceCell88.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell88.candidate_not_isWeightedPerimeterMinimizer	ee0dc2ae864acfa75f53a16c34b57c7809fcf24f923f77009f6e9c6fcdc5a6b5	equivalent
MiddleFaceCell89.Brick	9995d8f0735e89d4d22371dc5db0e6b45c79d84622112064caf49cdcfc9c5b9c8	equivalent
MiddleFaceCell89.brick	7131e68fd89095f47f2f0651a01b47ee86608d86dac8b9478ca7dd379e7ef991	equivalent
MiddleFaceCell89.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell89.candidate_not_isWeightedPerimeterMinimizer	c539115f908ea3c854f13e214af2c379cdac74ec4091c5662325c2a1f6bd89e1	equivalent
MiddleFaceCell19.Brick	c90ecb97bcca2b0b1ece9bea1a892e190d84007307c237cb7b84d7ec3b0e6024	equivalent
MiddleFaceCell19.brick	206210c28923a9253ad7d8345344ea0655b058110c31c53b5e5a6a5d8b053598	equivalent
MiddleFaceCell19.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell19.candidate_not_isWeightedPerimeterMinimizer	7213ad2339ef4769866fbe9f0a71501c0da780c17ab8006f5a6c40b4447551d7	equivalent
MiddleFaceCell190.Brick	2eb37468b09417053141932c23a09dab269faeccd192307cd73e740a7d43ef17	equivalent
MiddleFaceCell190.brick	bd32ab36c2769524155ac509d94107635d10cba0428d34fecca4b171de849095	equivalent
MiddleFaceCell190.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell190.candidate_not_isWeightedPerimeterMinimizer	dafff59d6235378af6b0880a575280e1069d461a36206afbd1e26956655cc951	equivalent
MiddleFaceCell191.Brick	34f1e9ad7cfd4b476da7cae8a83a0da382284c646ba2a2383b0f7fecfda5e4f7	equivalent
MiddleFaceCell191.brick	380b05899d820354e486b3ad6b1ae2f996e222ba447488be0433f974c5b6fcc6	equivalent
MiddleFaceCell191.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell191.candidate_not_isWeightedPerimeterMinimizer	325b487ffb7001998ca4bbd5c8bf369dadbe1c7c09bf25a0b629b46aeb7470f7	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell192.Brick	c8f88e0ab8be959c8cab6ee6276482346f10092933a6346711dbadcea88fe1e4	equivalent
MiddleFaceCell192.brick	14dc866f267ee2076b6a8a01f8d73b289d5a6d681617580be877e76c2a69cfd5	equivalent
MiddleFaceCell192.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell192.candidate_not_isWeightedPerimeterMinimizer	3ff009b02820a6a83bdccc24bf4132cae807fc772620b71528e22367e6b67133	equivalent
MiddleFaceCell193.Brick	29ae0c7679f7842b43aa4f39871c3d6c0e4b9568b52f48639e3cd76e786a9012	equivalent
MiddleFaceCell193.brick	c7a8d47c0018dcbe08fb8f3444931f223d4a38b493275594a3f606489af28d0d	equivalent
MiddleFaceCell193.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell193.candidate_not_isWeightedPerimeterMinimizer	9dc1a3f1a04048f01e8fa76f7ae1f00a783f959fac32923af716a43938dd5445	equivalent
MiddleFaceCell194.Brick	54d0814c2f91e4d1b0530355ac75bf4a3673e1cc78f850e9f036515e6a1f3d8e	equivalent
MiddleFaceCell194.brick	7de96974533b4ef5b836e6f752ed9cd536c4ef41caf6b6c4180f38eee4a07e8e	equivalent
MiddleFaceCell194.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell194.candidate_not_isWeightedPerimeterMinimizer	6b0ec5a16b25bf5b9133990825e7851d57dad283c6c222688a828f74d3543221	equivalent
MiddleFaceCell195.Brick	4f6254c1e4db993da672c61044c530a142aaf24876f6ea438ea84b38faa5cf6e	equivalent
MiddleFaceCell195.brick	7cb31dfff056fbeaffd924fd8063457668a9747d3d11d80cbca01889a6677b16	equivalent
MiddleFaceCell195.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell195.candidate_not_isWeightedPerimeterMinimizer	7684dcd90ea13b1172127269b6966efdab365d74371f7e711935242634ffdb66	equivalent
MiddleFaceCell196.Brick	3b9bdf29101617b1e9bf8633e4fef5397c4e6d7dcb133f4405048fe4d9d1ee66	equivalent
MiddleFaceCell196.brick	c45e2ed57bcd6d49ebd45c3d1c752697b337f0bfc1f270fbb3ef5f4075b06c3	equivalent
MiddleFaceCell196.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell196.candidate_not_isWeightedPerimeterMinimizer	c1dabc8c1fc8310ad63535b0bfb9632f22b6b0b2d012ae504038d0ba6ab3f3d0	equivalent
MiddleFaceCell197.Brick	06ac3b95089ed503287244914cfd52c32983846693cf6dad05ea3fb3e37c6c2c	equivalent
MiddleFaceCell197.brick	88f1cf09b3253d9a97de813dec4a629284bcbcf83b60cb6ffe8d5b08399853fc	equivalent
MiddleFaceCell197.lamInterval	f66ba63b466bb0baa42078e66a7ef00780edcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell197.candidate_not_isWeightedPerimeterMinimizer	37f1de9565ef09275e9666cfa7f7f137646aab315b65b6ccfb9240f14e9e33f0	equivalent
MiddleFaceCell198.Brick	1d489afd608a3fac58e5ca1d1e56b02b8581cf9957f1be0d06901a170a224cad	equivalent
MiddleFaceCell198.brick	99e4e8b1cbde397c76a9f6c8a98f1b474d643ccf0c2e41c084f19b7050116bfa	equivalent

Declaration	Source hash	Audit relation
MiddleFaceCell198.lamInterval	f66ba63b466bb0baa42078e66a7ef00780e dcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell198.candidate_not_isWe ightedPerimeterMinimizer	6ecc93ce4c3ed951951af788cf1a0886858 fa768f5dd4cbc5b54051e12d31ad0	equivalent
MiddleFaceCell199.Brick	69acff0df68b5b3f545e8743f26e335af57 0d16f7cb007df1bee399f705fa30a	equivalent
MiddleFaceCell199.brick	42e1b8daf31e6b66339c1badb8b202b6790 36c5ef2ad3af3b48805d477ca6ed1	equivalent
MiddleFaceCell199.lamInterval	f66ba63b466bb0baa42078e66a7ef00780e dcfd66cb63df195880be1a2d957b6	equivalent
MiddleFaceCell199.candidate_not_isWe ightedPerimeterMinimizer	4be83c4c3331b21c491ec69bafa0f988c57 b160f2eb2242b52dabb540a89611e	equivalent